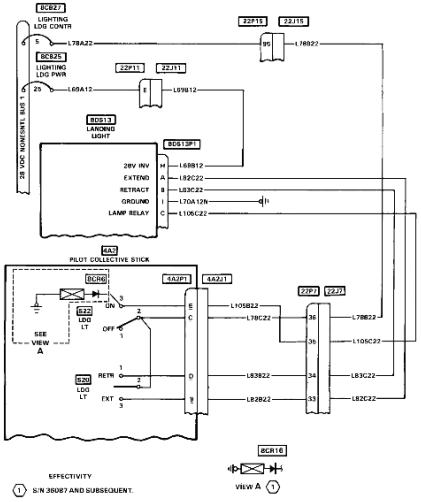
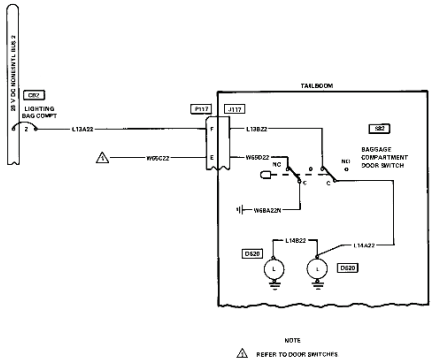


ICN-412-A-980000-B-97499-00187-A-001-01

Figure 1. Landing Lights (Sheet 1 of 1)

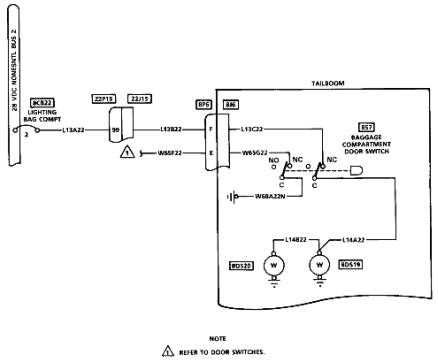
Intentionally blank page



ICN-412-A-980000-B-97499-00188-A-001-01

Figure 1. Baggage Compartment Lights (Sheet 1 of 1)

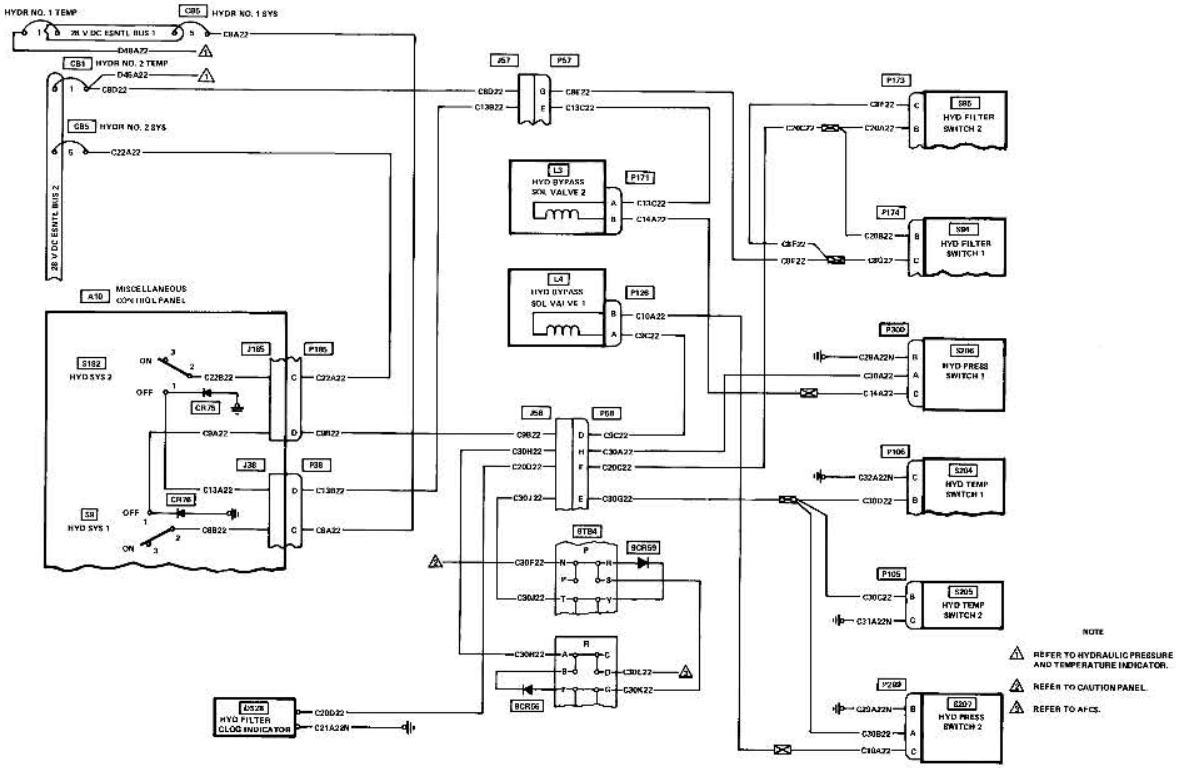
Intentionally blank page



ICN-412-A-980000-B-97499-00189-A-001-01

Figure 1. Baggage Compartment Lights (Sheet 1 of 1)

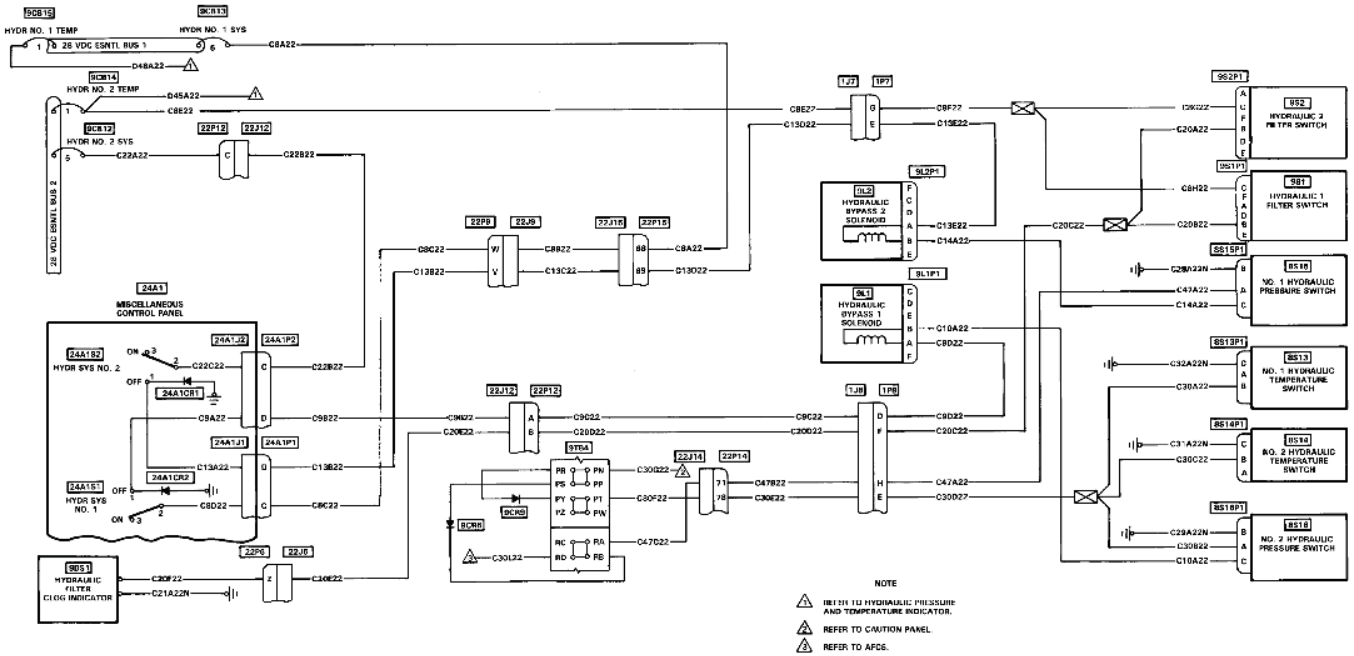
Intentionally blank page



ICN-412-A-980000-B-97499-00190-A-001-01

Figure 1. Hydraulic Systems (Sheet 1 of 1)

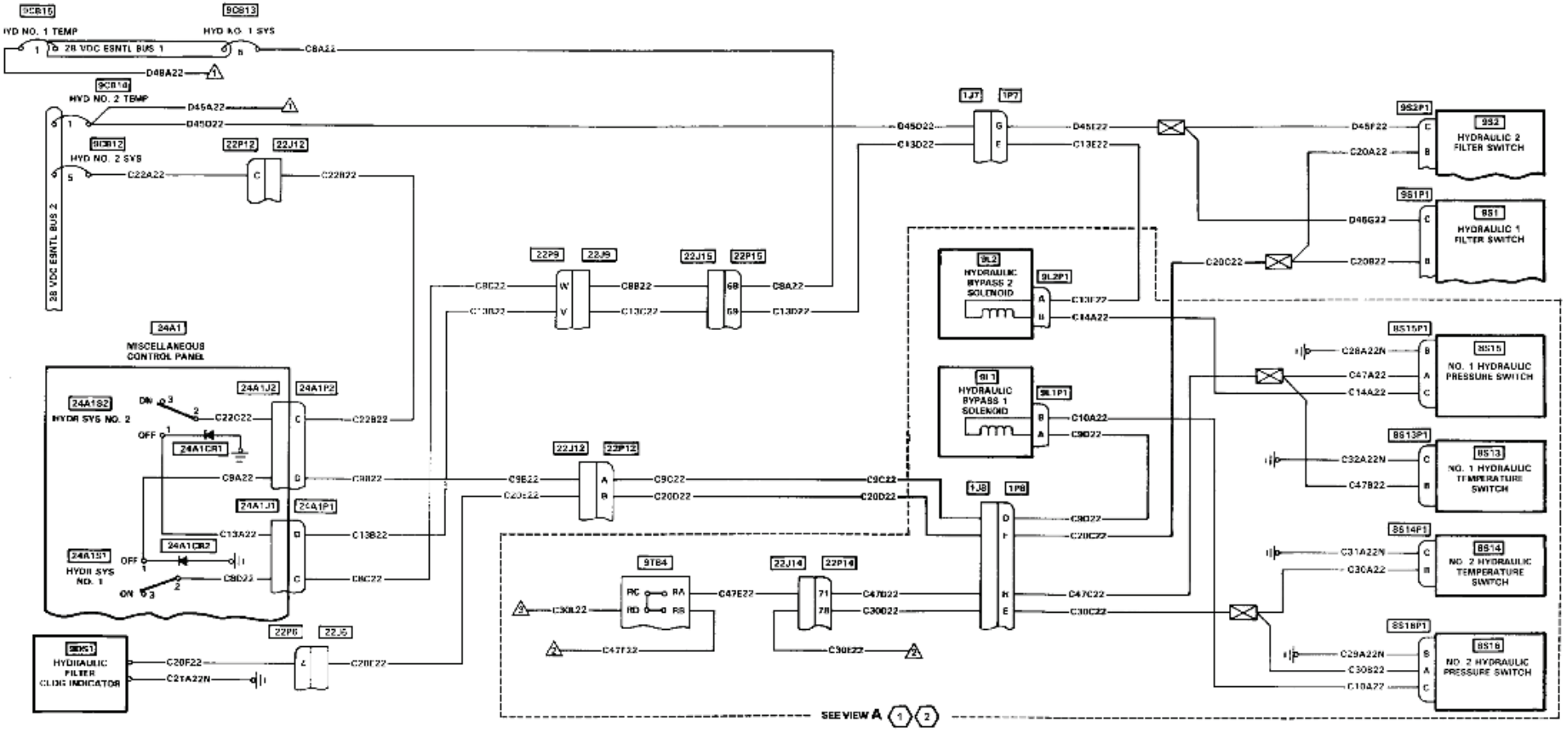
Intentionally blank page



ICN-412-A-980000-B-97499-00191-A-001-01

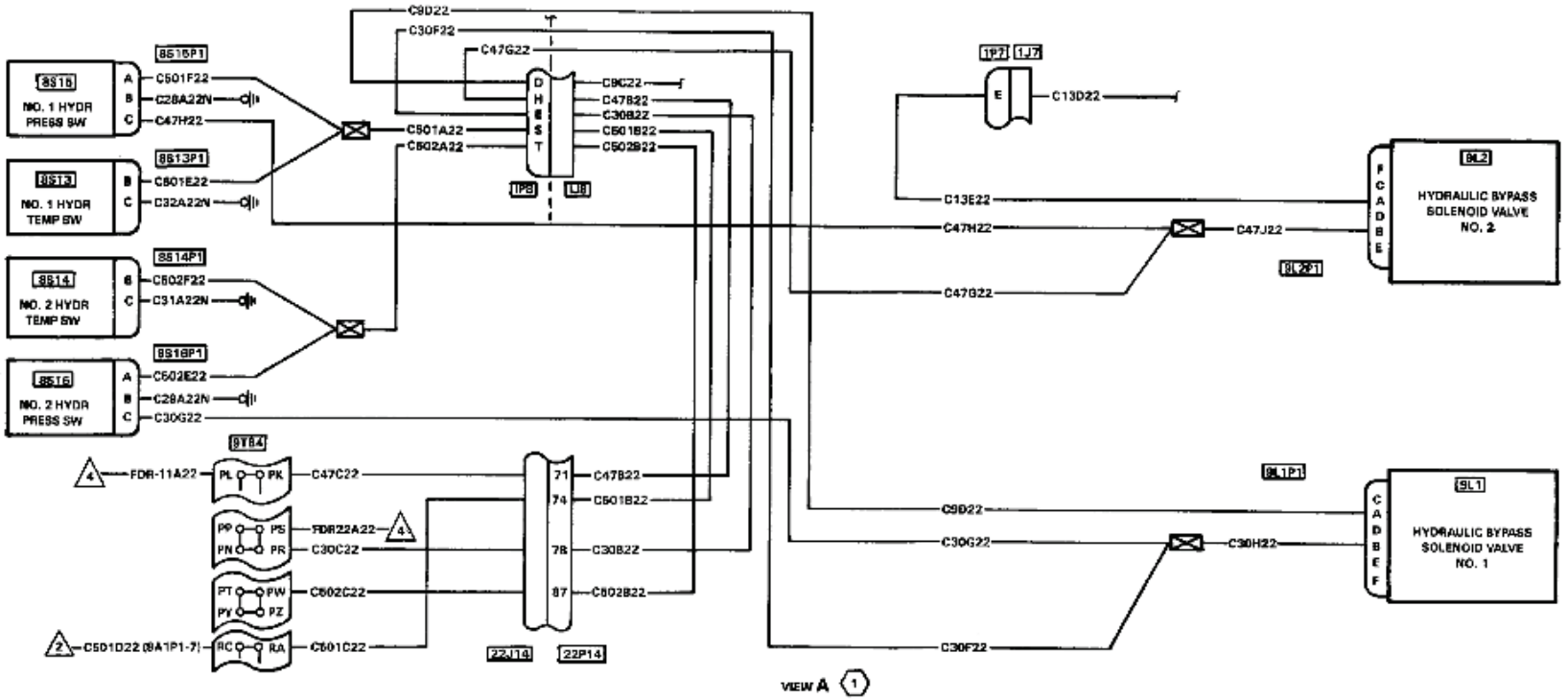
Figure 1. Hydraulic Systems (Sheet 1 of 1)

Intentionally blank page



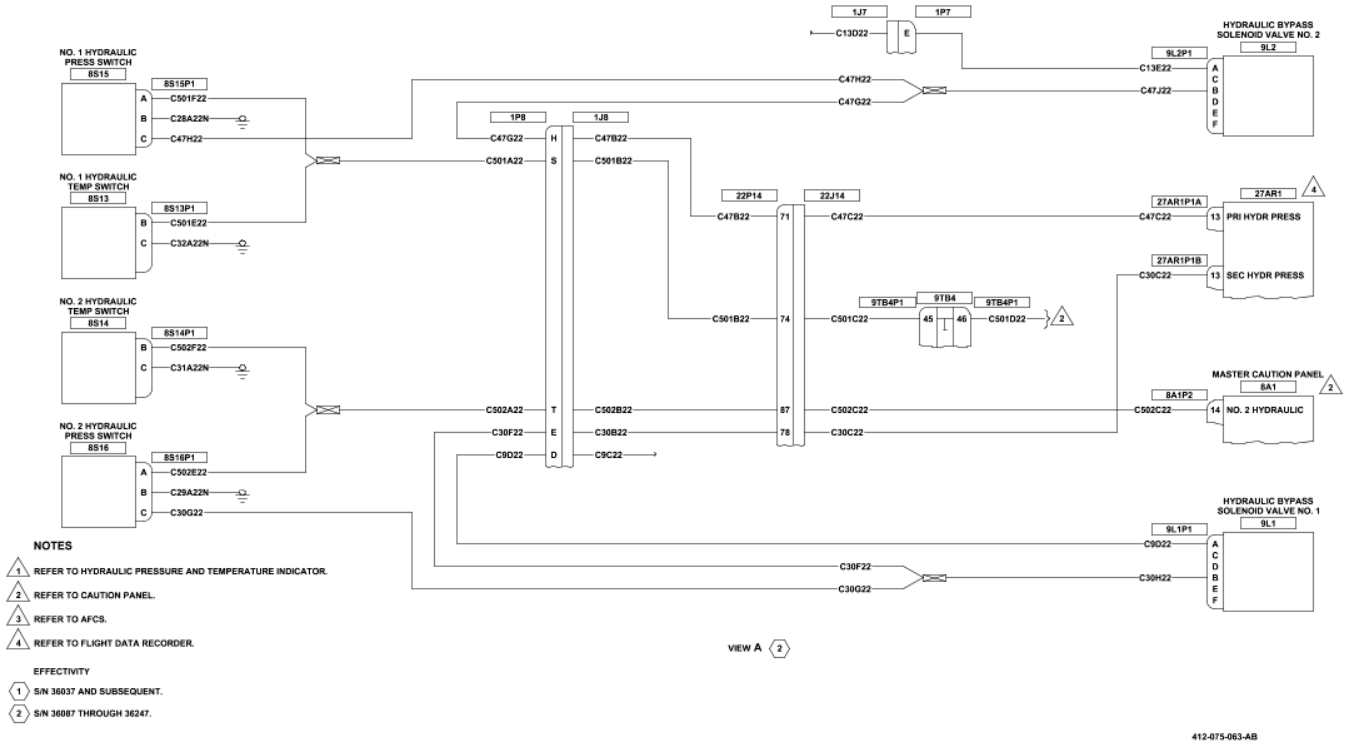
ICN-412-A-980000-B-97499-00192-A-001-01

Figure 1. Hydraulic Systems (Sheet 1 of 3)



ICN-412-A-980000-B-97499-00193-A-001-01

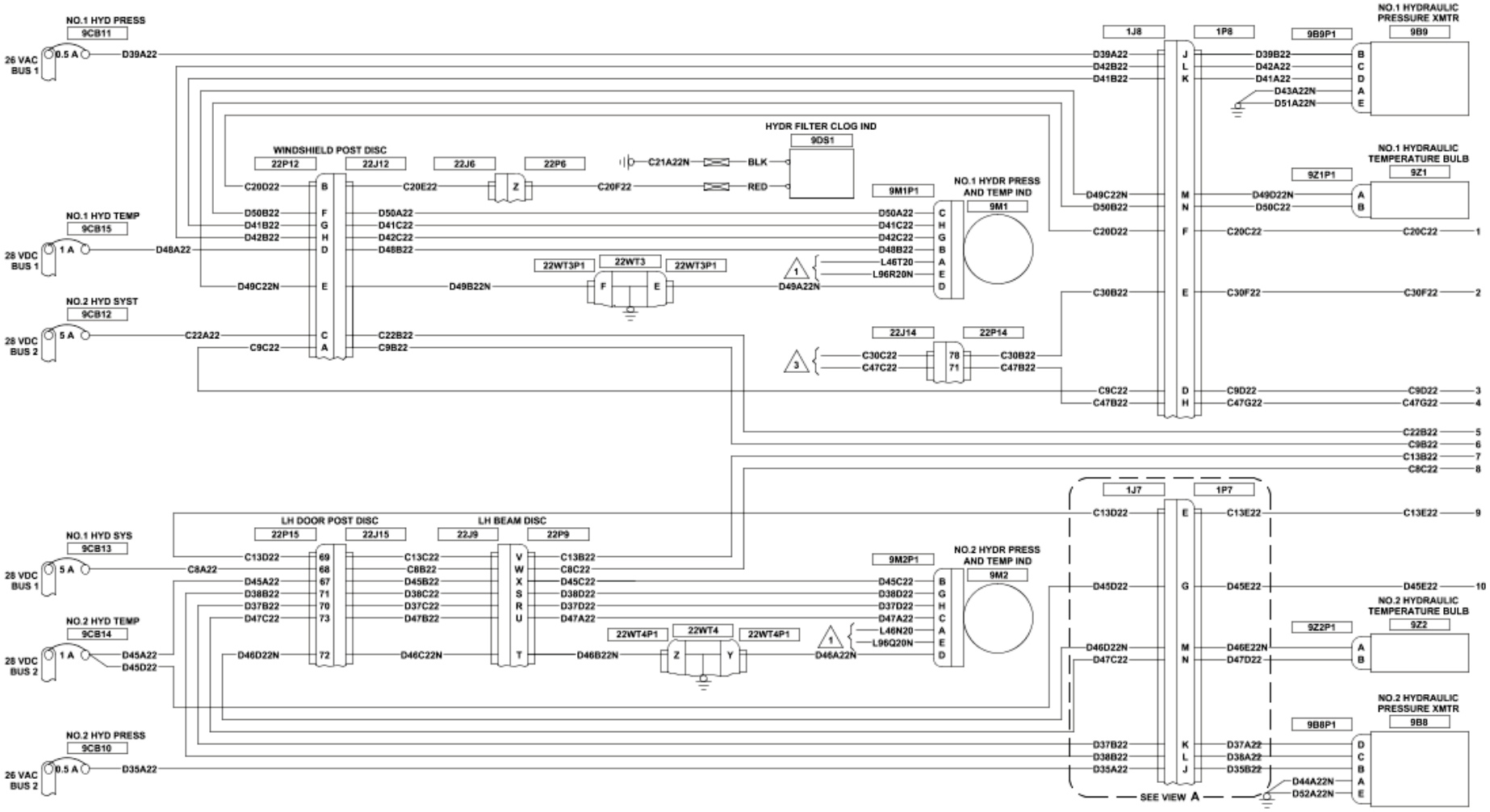
Figure 1. Hydraulic Systems (Sheet 2 of 3)



ICN-412-A-980000-B-97499-00194-A-001-01

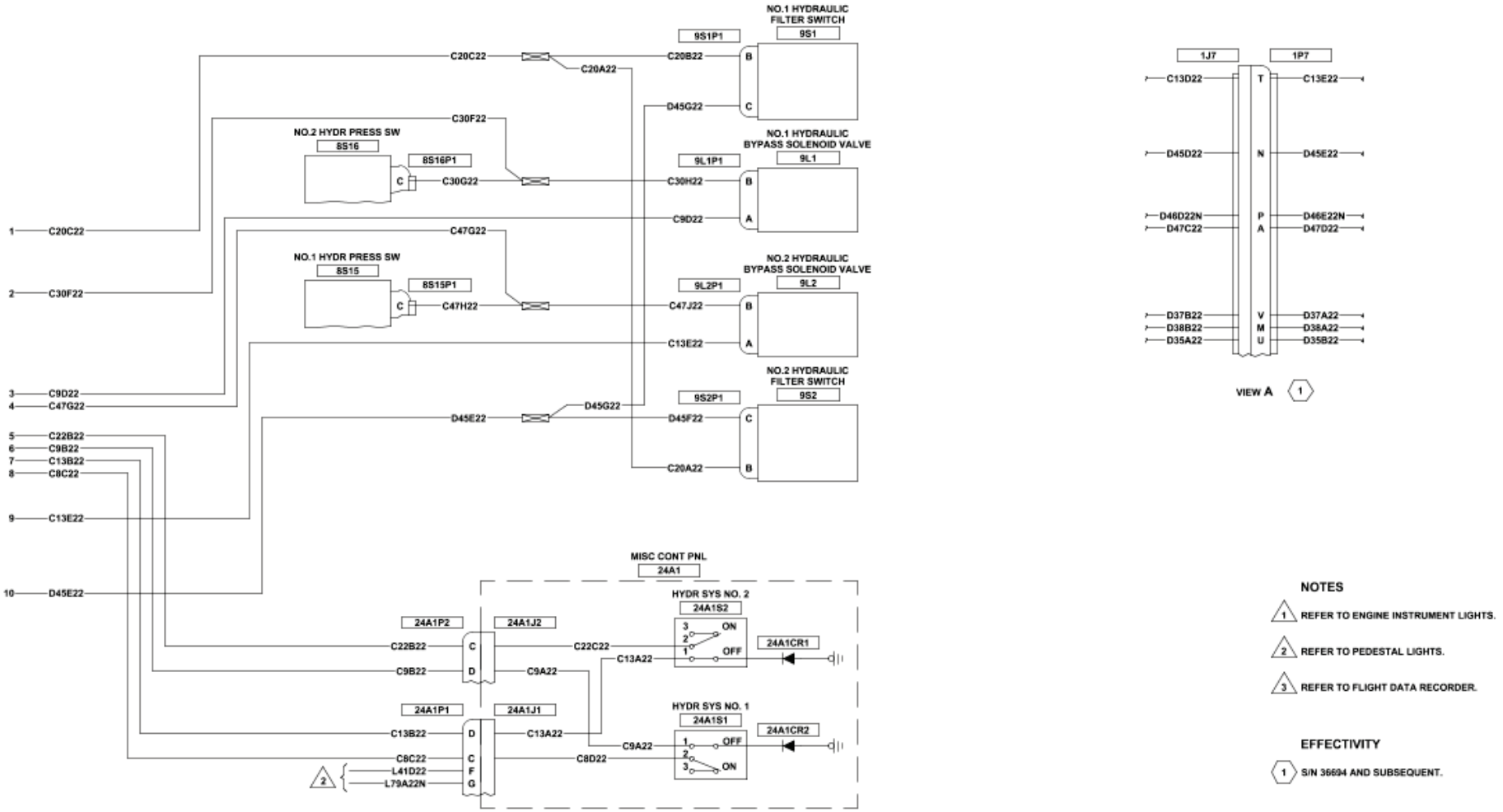
Figure 1. Hydraulic Systems (Sheet 3 of 3)

Intentionally blank page



ICN-412-A-980000-B-97499-00195-A-001-01

Figure 1. Hydraulic Systems (Sheet 1 of 2)



NOTES

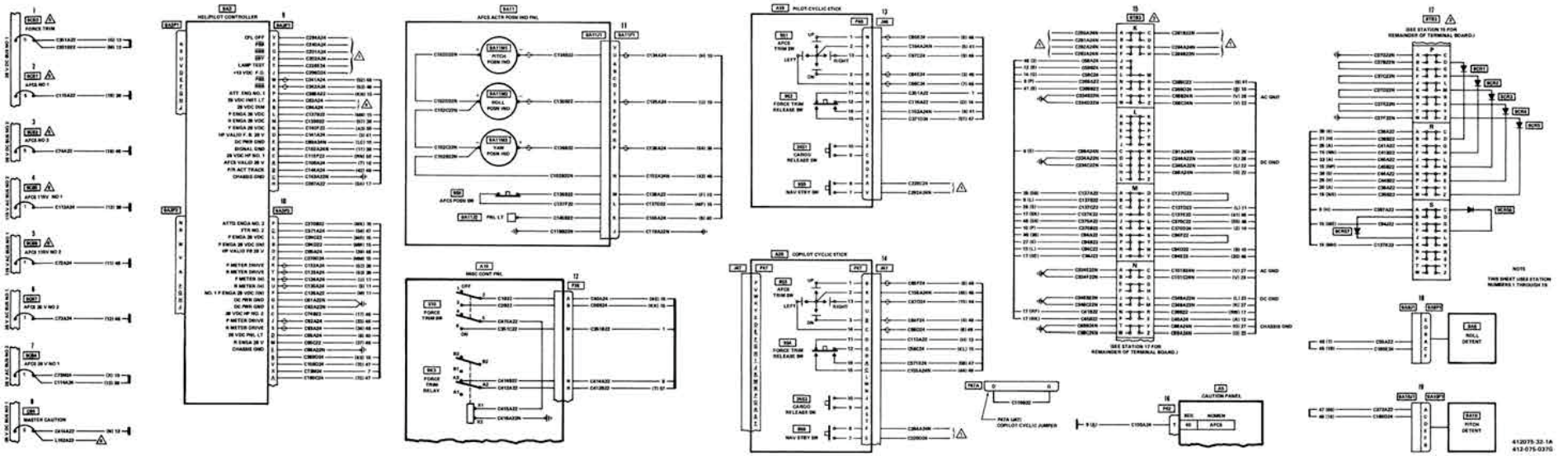
1. REFER TO ENGINE INSTRUMENT LIGHTS.
2. REFER TO PEDESTAL LIGHTS.
3. REFER TO FLIGHT DATA RECORDER.

EFFECTIVITY

1. S/N 36694 AND SUBSEQUENT.

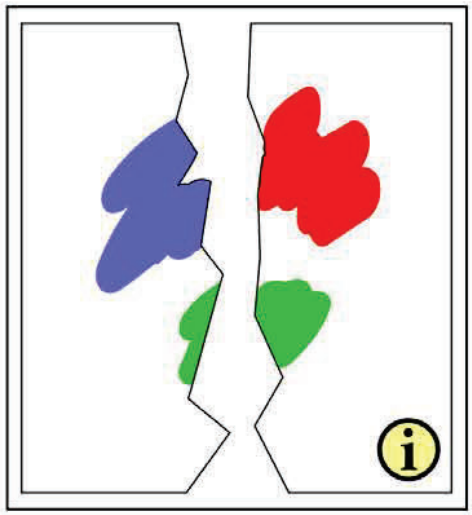
ICN-412-A-980000-B-97499-00196-A-001-01

Figure 1. Hydraulic Systems (Sheet 2 of 2)



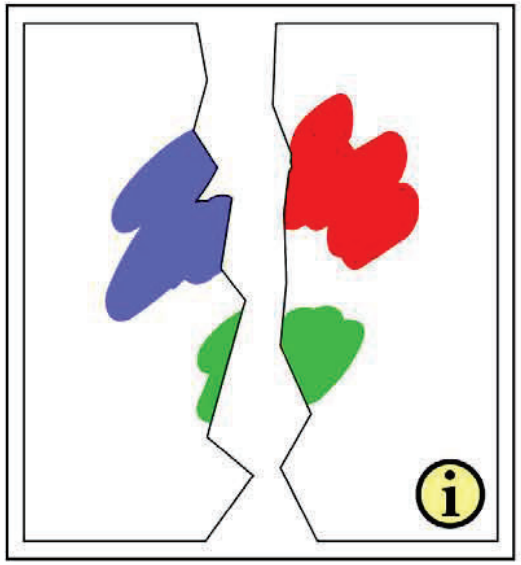
ICN-412-A-980000-B-97499-00197-A-001-01

Figure 1. AFCS (Sheet 1 of 5)



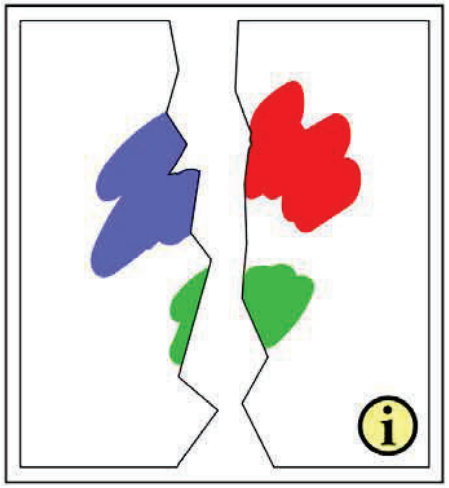
ICN-412-A-980000-B-97499-00198-A-001-01

Figure 1. AFCS (Sheet 2 of 5)



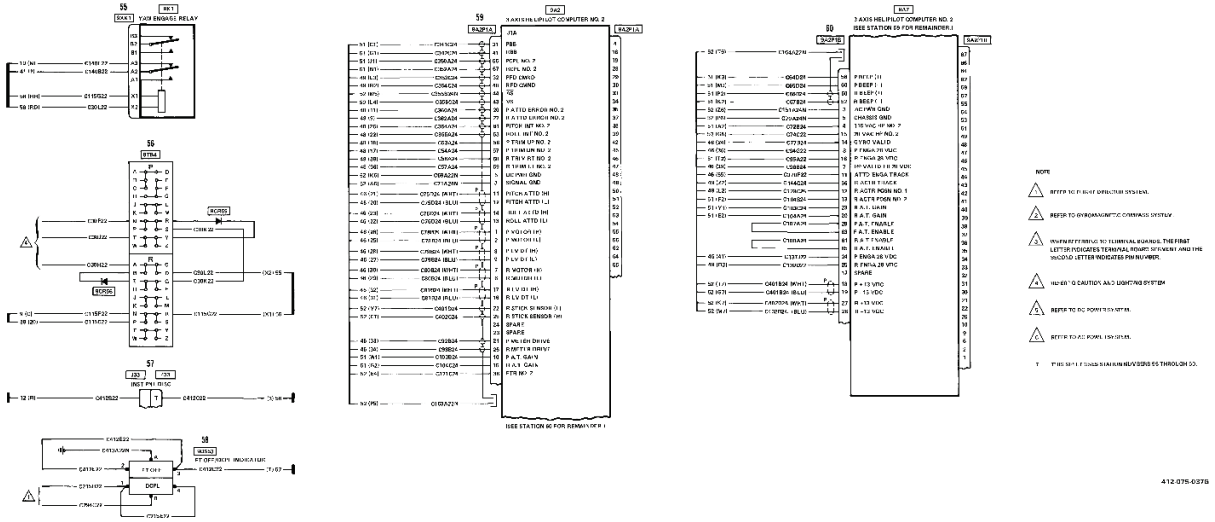
ICN-412-A-980000-B-97499-00199-A-001-01

Figure 1. AFCS (Sheet 3 of 5)



ICN-412-A-980000-B-97499-00200-A-001-01

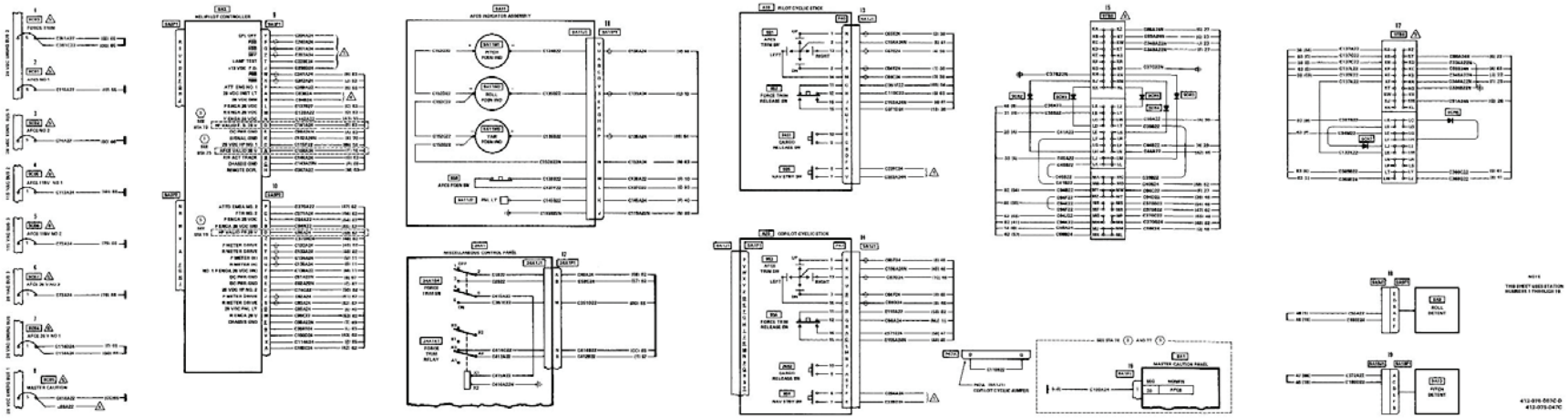
Figure 1. AFCS (Sheet 4 of 5)



ICN-412-A-980000-B-97499-00201-A-001-01

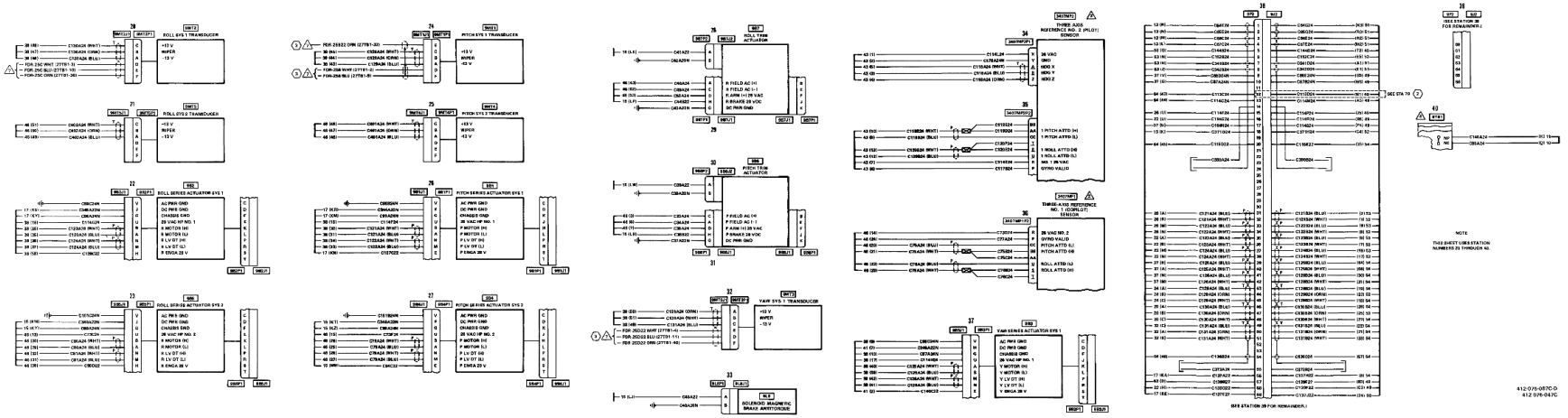
Figure 1. AFCS (Sheet 5 of 5)

Intentionally blank page



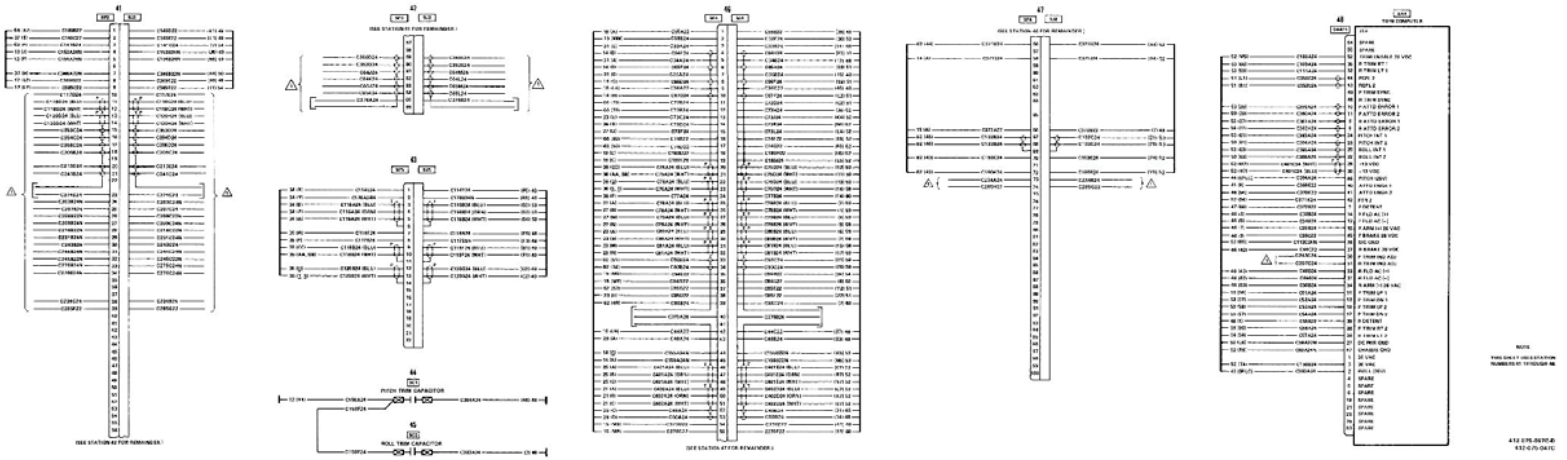
ICN-412-A-980000-B-97499-00202-A-001-01

Figure 1. AFCS (Sheet 1 of 8)



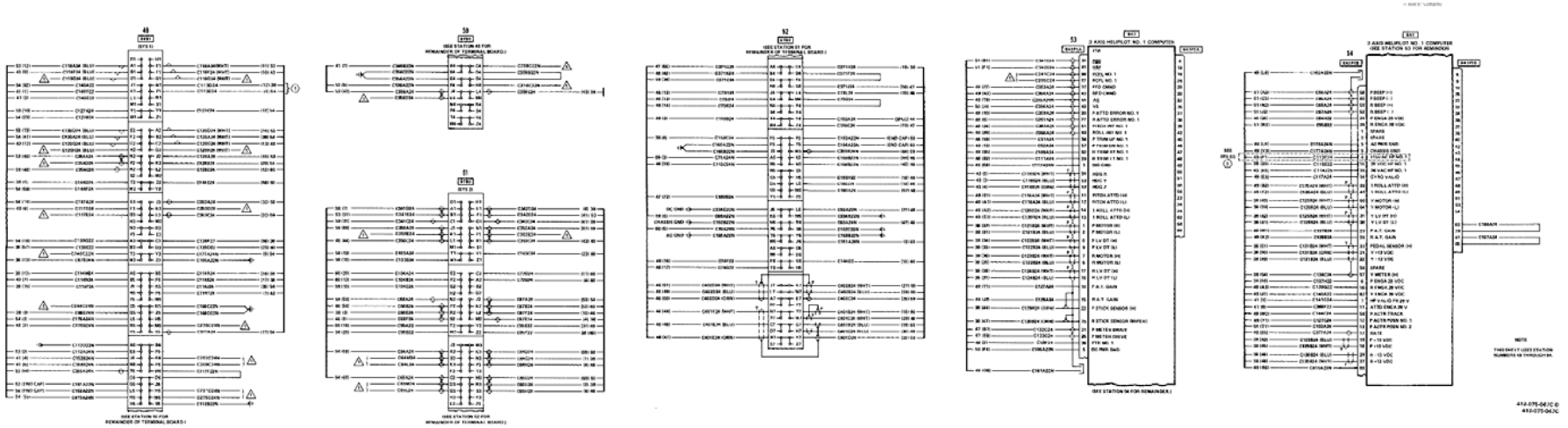
ICN-412-A-980000-B-97499-00203-A-001-01

Figure 1. AFCS (Sheet 2 of 8)



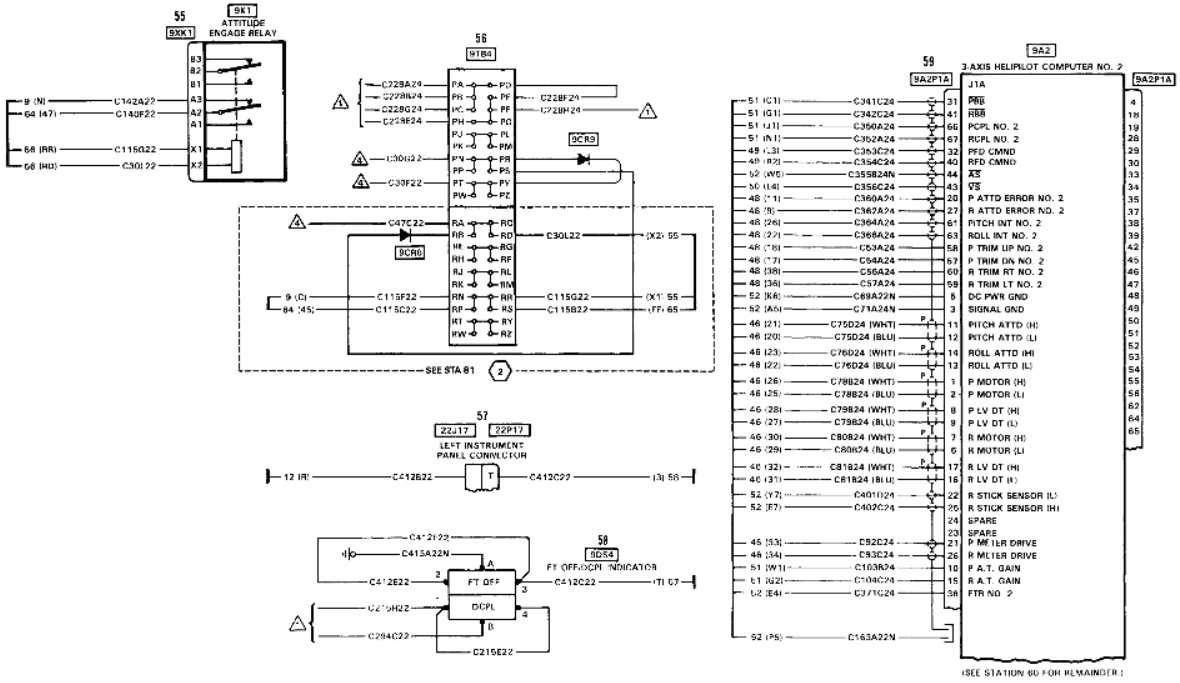
ICN-412-A-980000-B-97499-00204-A-001-01

Figure 1. AFCS (Sheet 3 of 8)



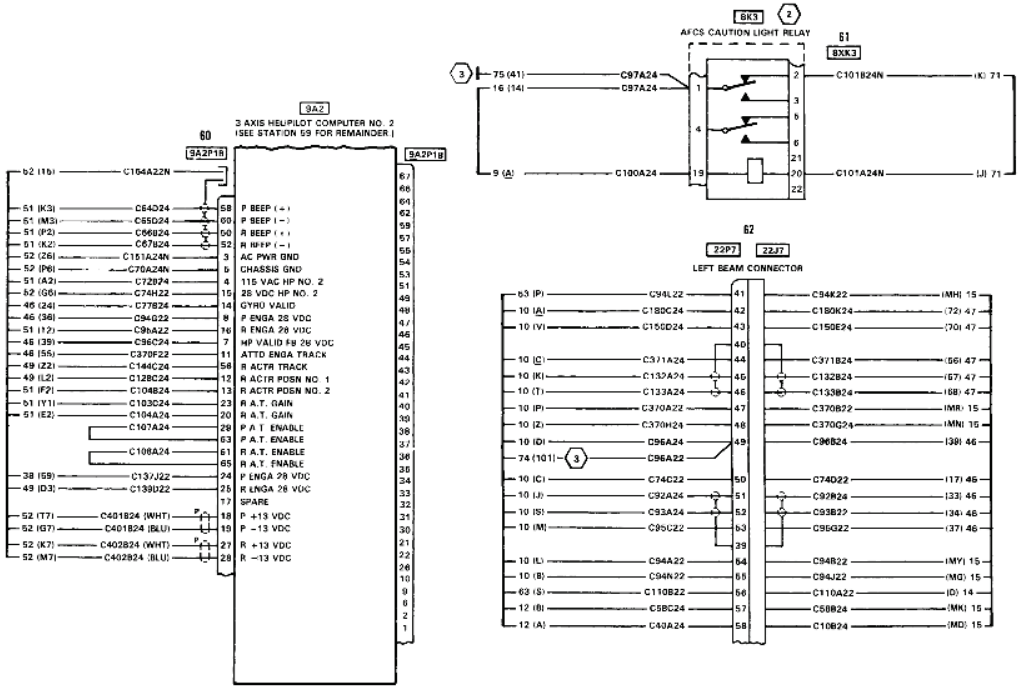
ICN-412-A-980000-B-97499-00205-A-001-01

Figure 1. AFCS (Sheet 4 of 8)



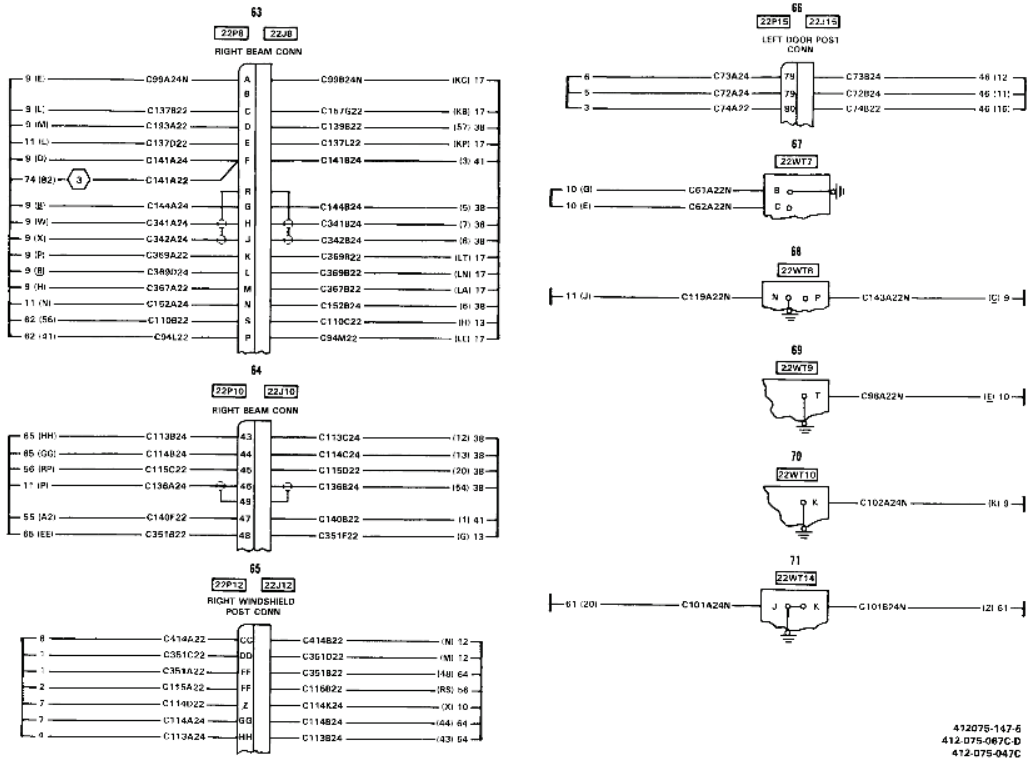
ICN-412-A-980000-B-97499-00206-A-001-01

Figure 1. AFCS (Sheet 5 of 8)



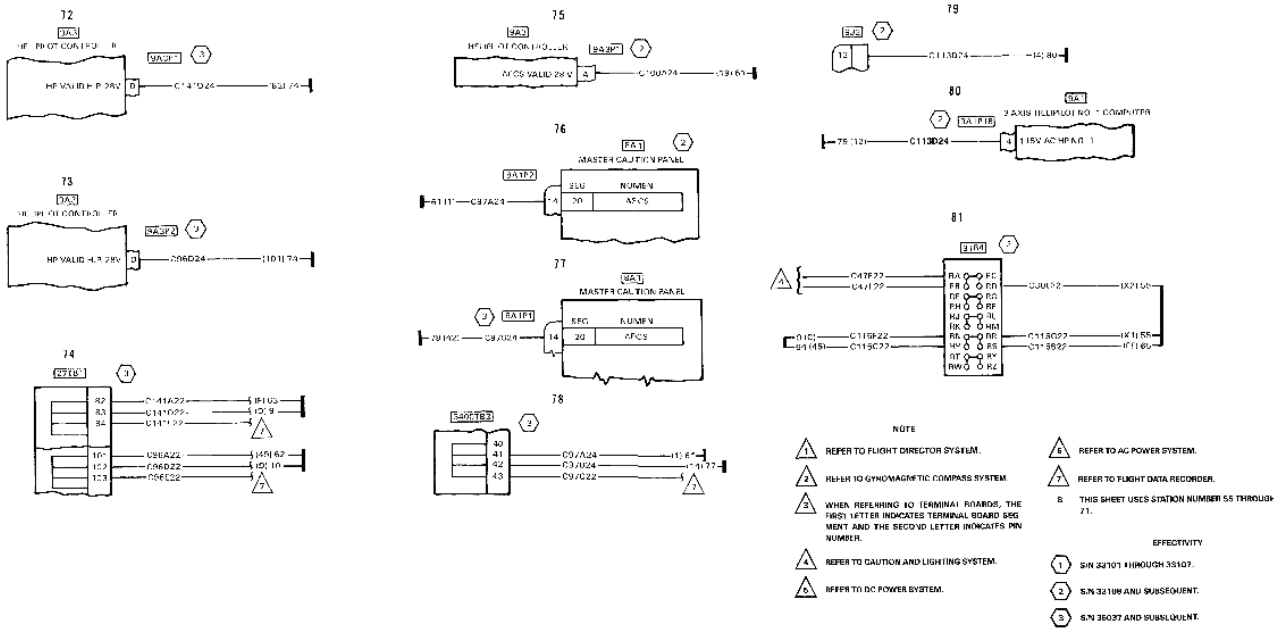
ICN-412-A-980000-B-97499-00207-A-001-01

Figure 1. AFCS (Sheet 6 of 8)



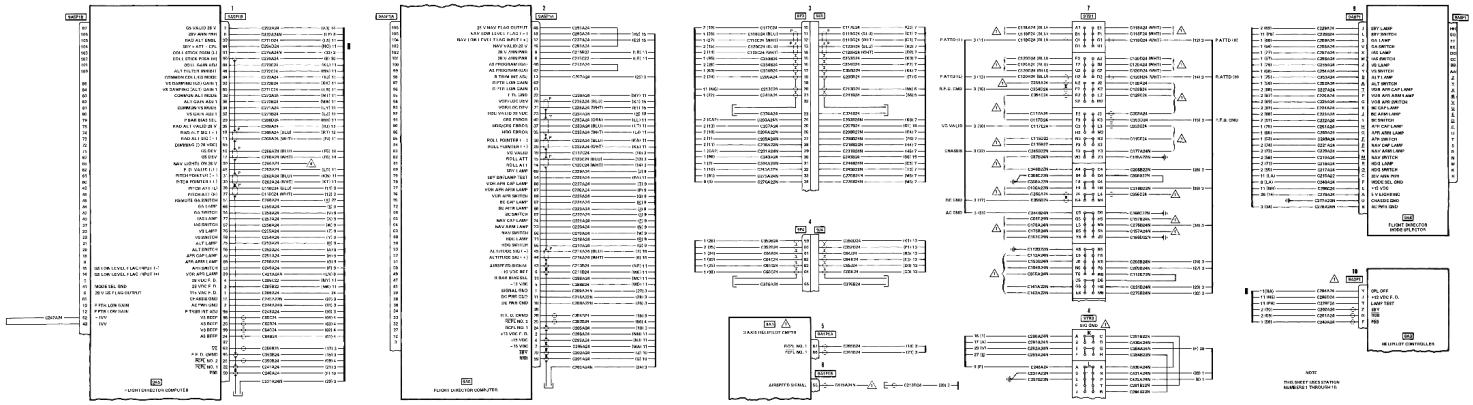
ICN-412-A-980000-B-97499-00208-A-001-01

Figure 1. AFCS (Sheet 7 of 8)



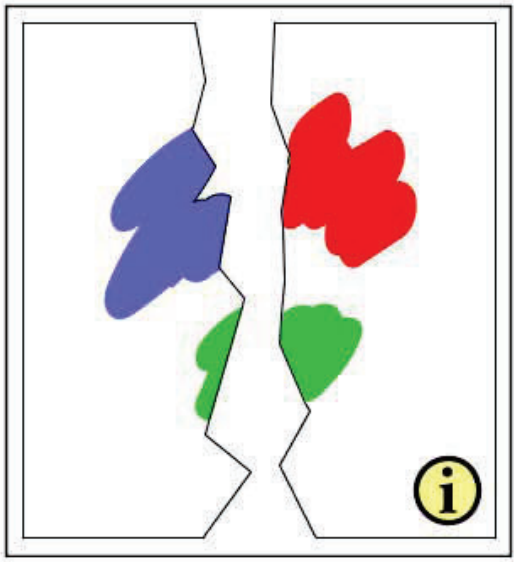
ICN-412-A-980000-B-97499-00209-A-001-01

Figure 1. AFCS (Sheet 8 of 8)



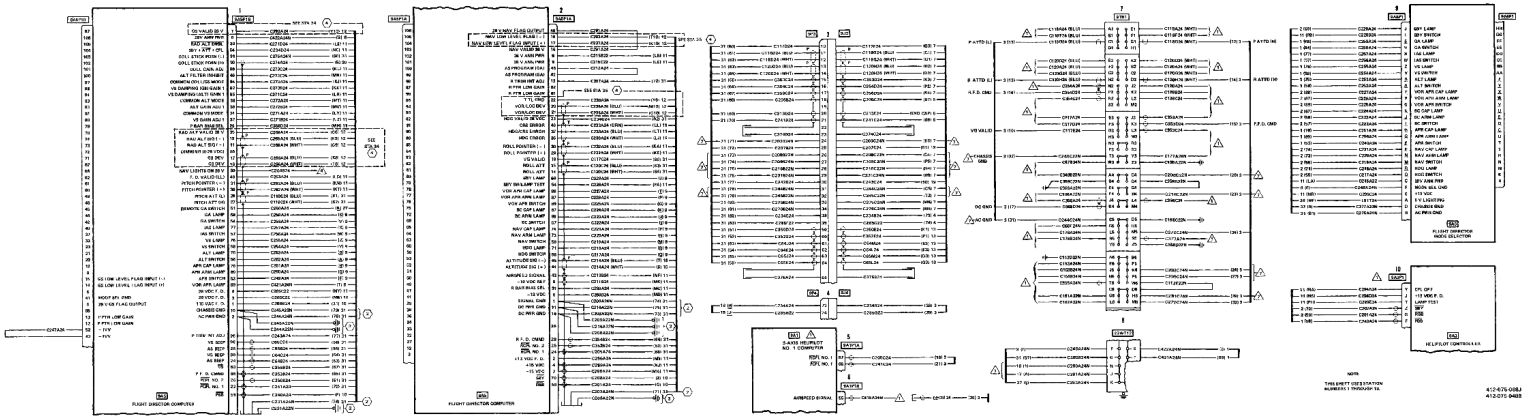
ICN-412-A-980000-B-97499-00210-A-001-01

Figure 1. Flight director system (Sheet 1 of 2)



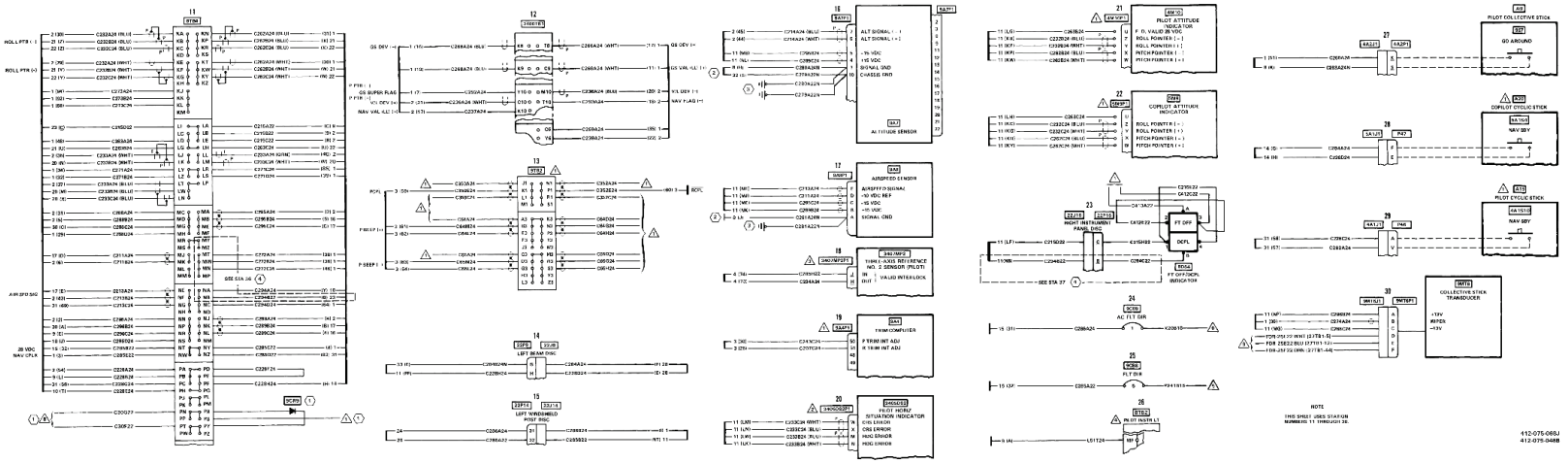
ICN-412-A-980000-B-97499-00211-A-001-01

Figure 1. Flight director system (Sheet 2 of 2)



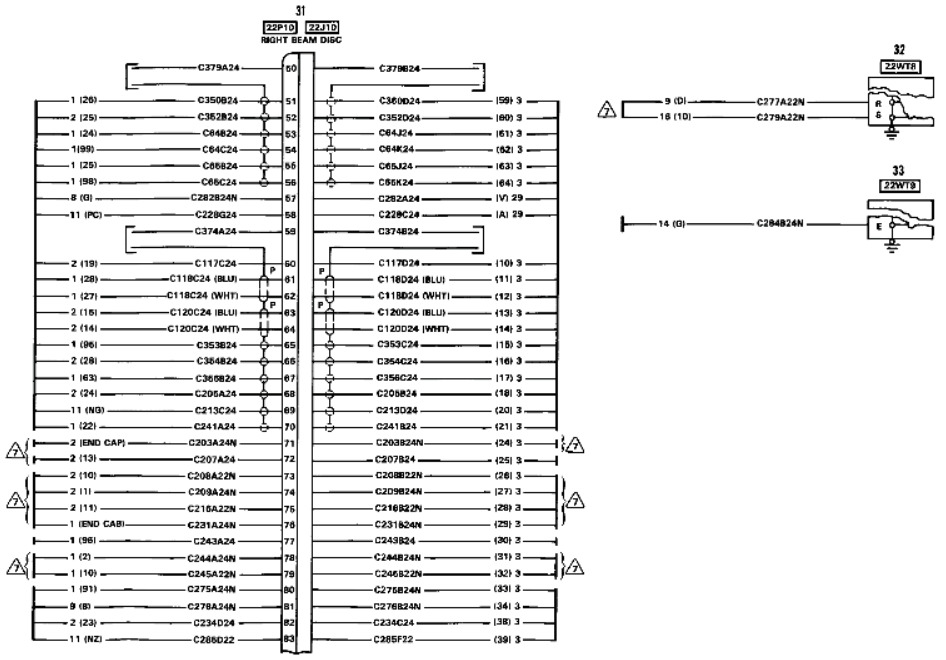
ICN-412-A-980000-B-97499-00212-A-001-01

Figure 1. Flight Director System (Sheet 1 of 4)



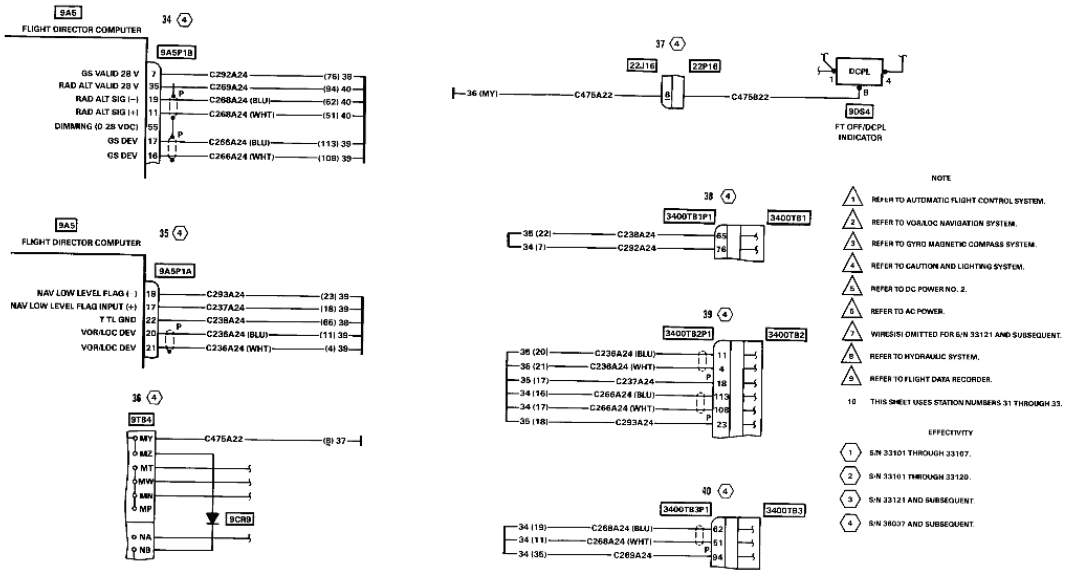
ICN-412-A-980000-B-97499-00213-A-001-01

Figure 1. Flight Director System (Sheet 2 of 4)



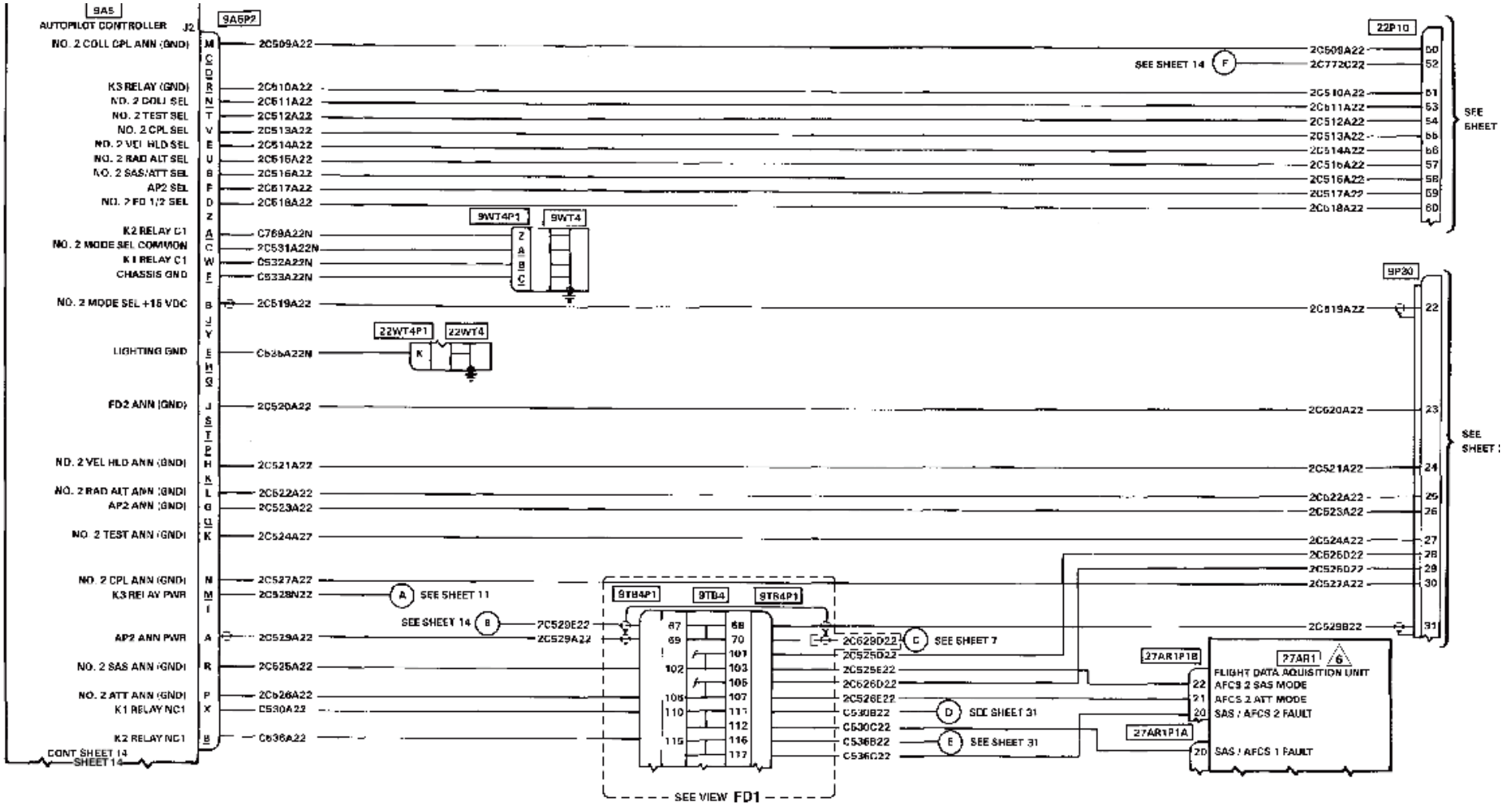
ICN-412-A-980000-B-97499-00214-A-001-01

Figure 1. Flight Director System (Sheet 3 of 4)



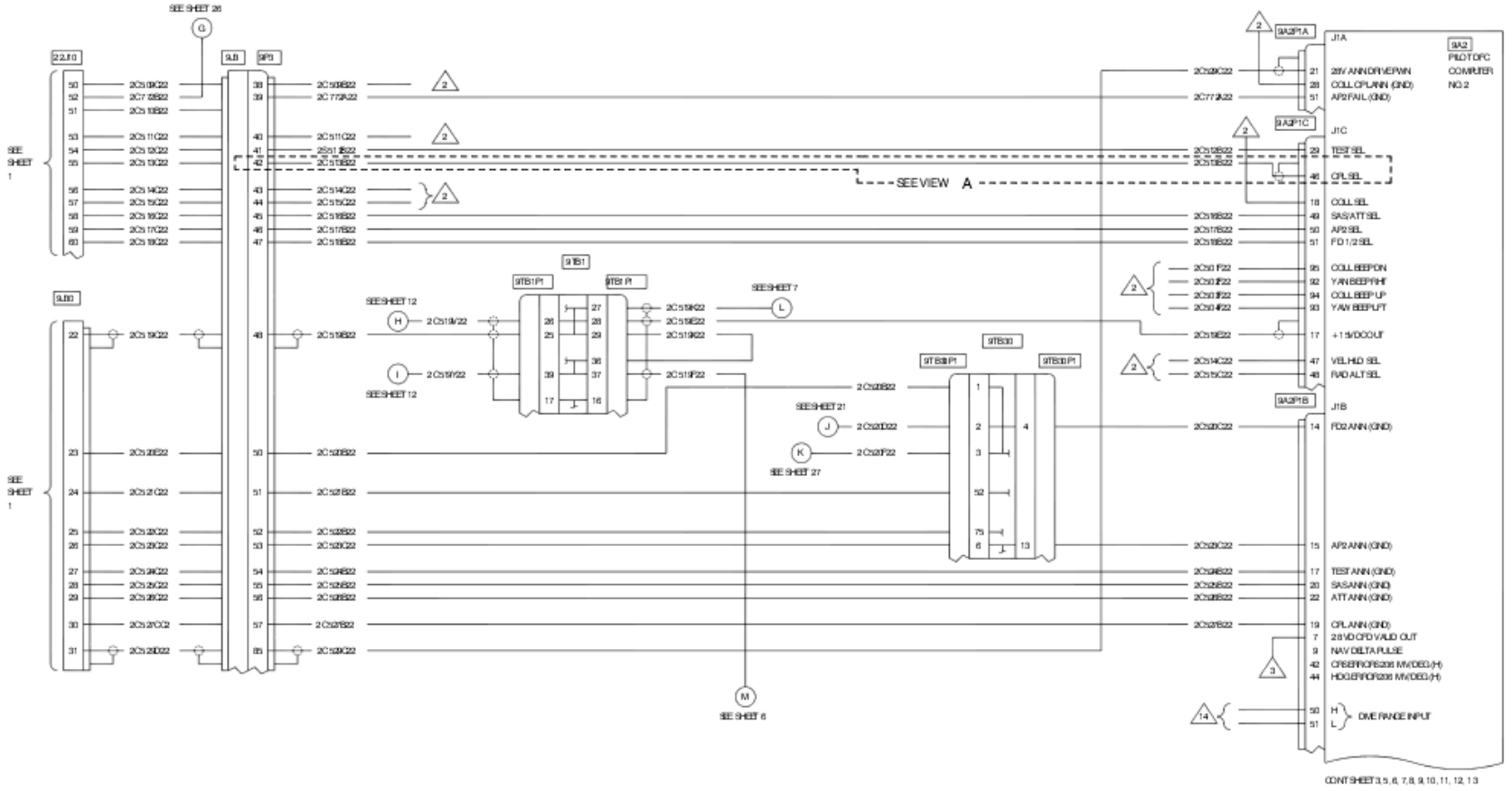
ICN-412-A-980000-B-97499-00215-A-001-01

Figure 1. Flight Director System (Sheet 4 of 4)



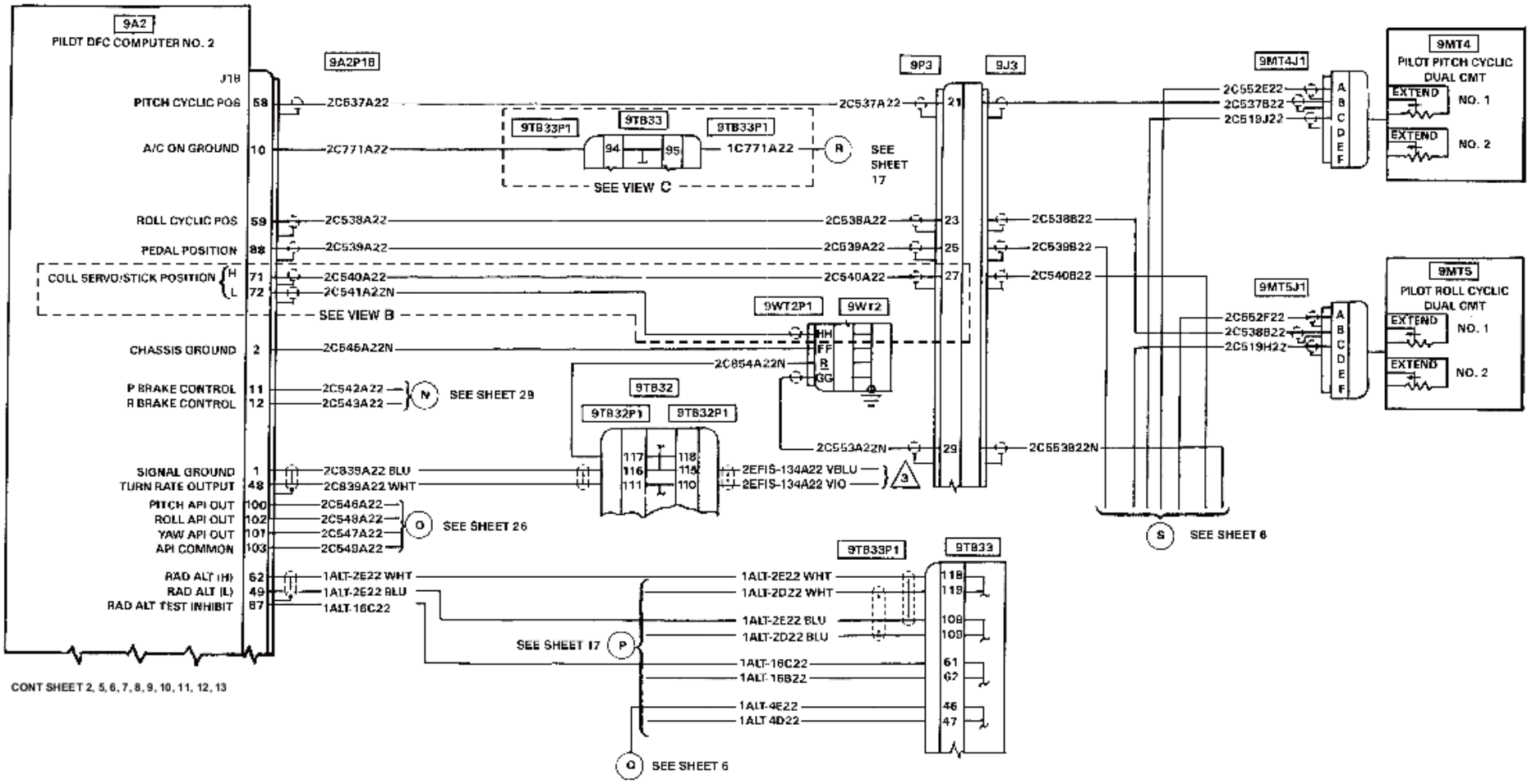
ICN-412-A-980000-B-97499-00216-A-001-01

Figure 1. Digital Automatic Flight Control Systems (Sheet 1 of 37)



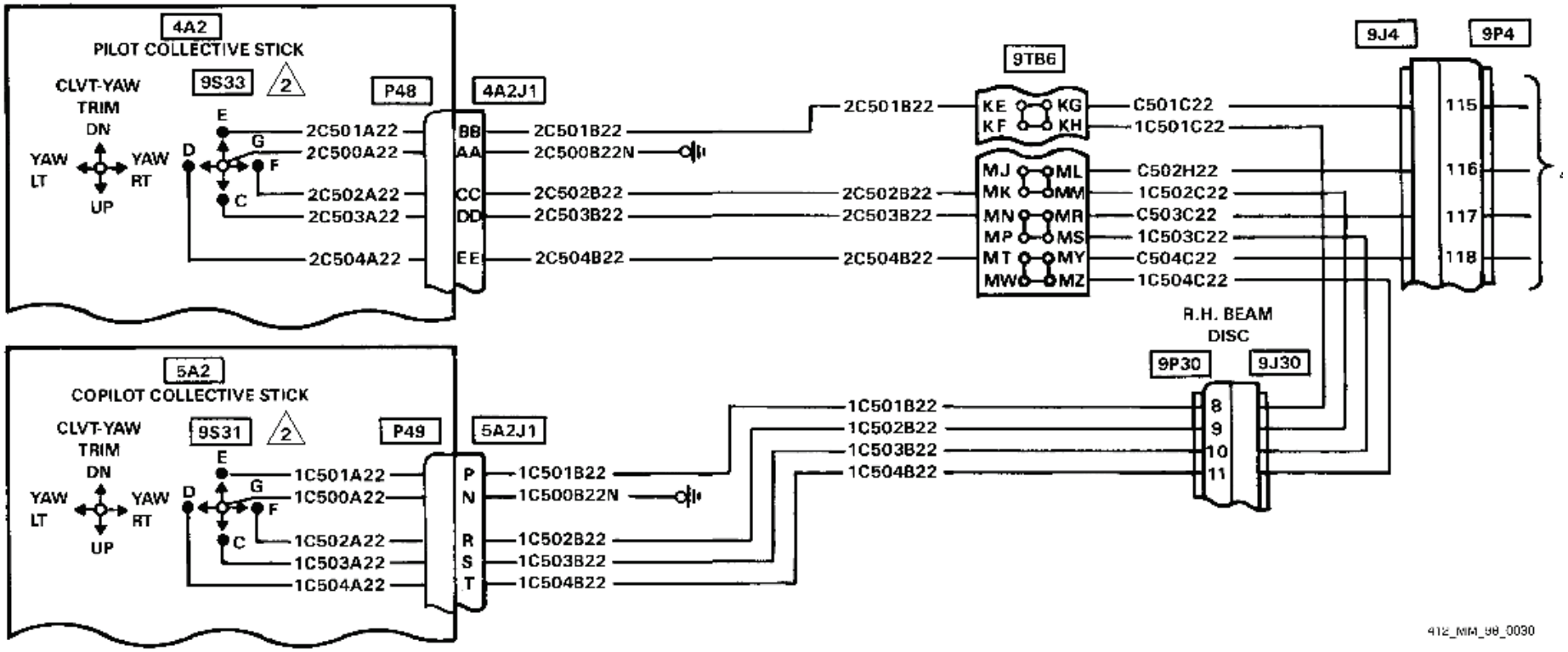
ICN-412-A-980000-B-97499-00217-A-001-01

Figure 1. Digital Automatic Flight Control Systems (Sheet 2 of 37)



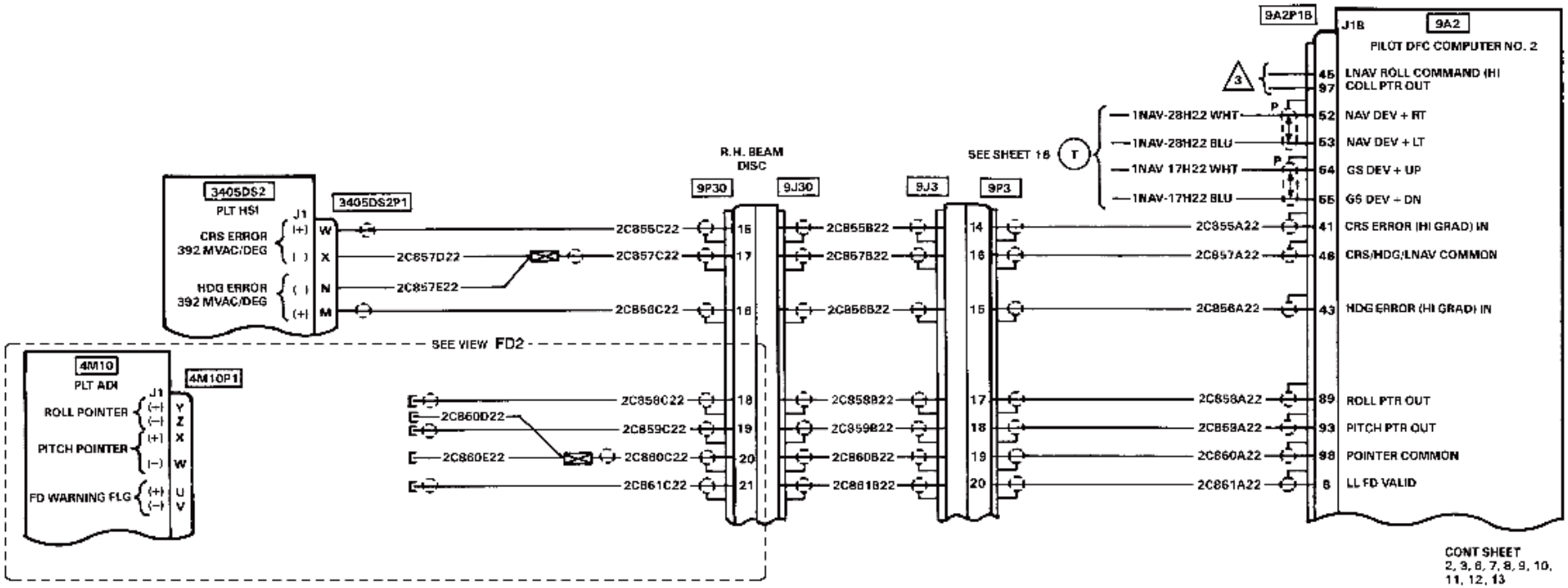
ICN-412-A-980000-B-97499-00218-A-001-01

Figure 1. Digital Automatic Flight Control Systems (Sheet 3 of 37)



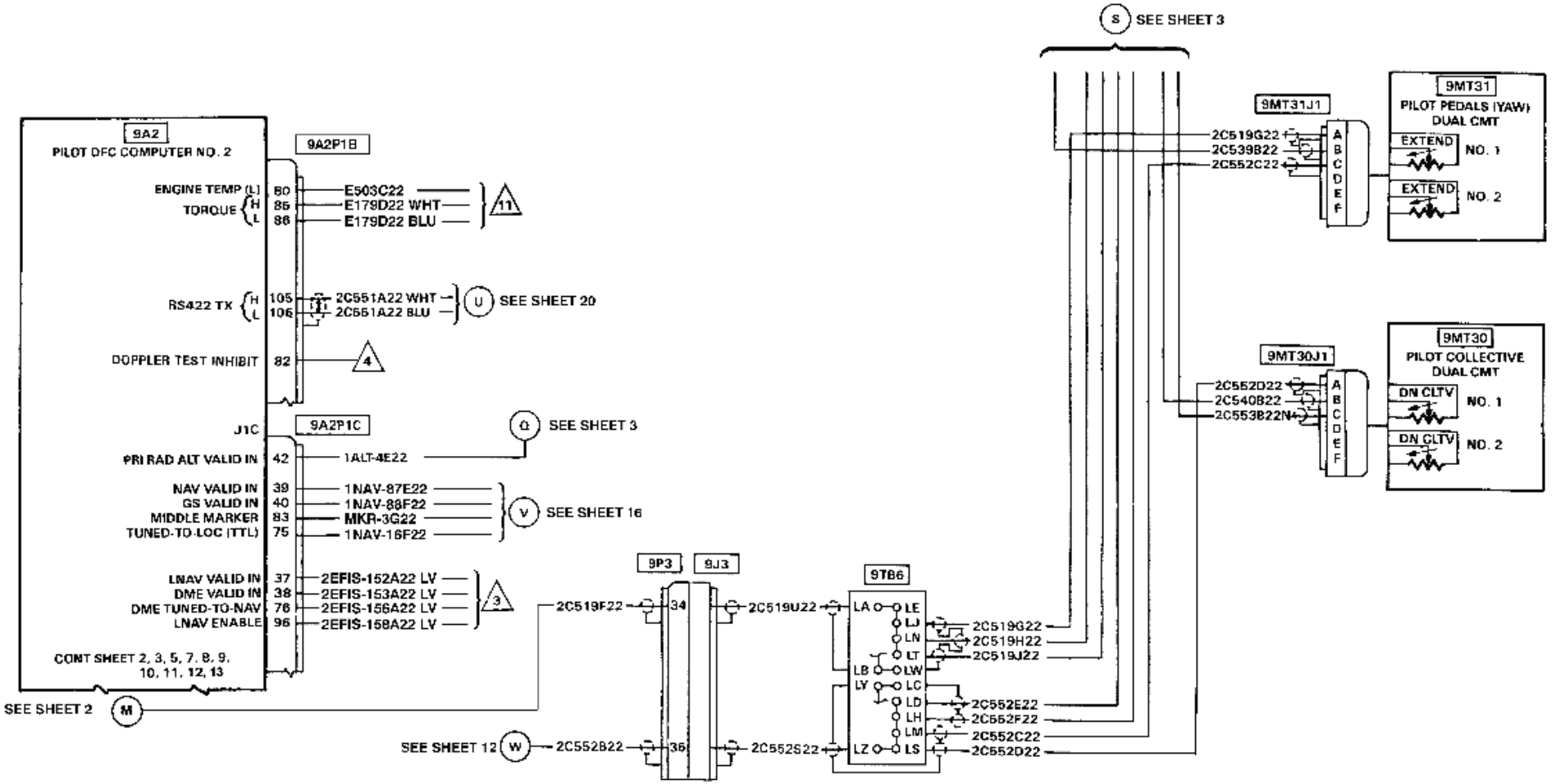
ICN-412-A-980000-B-97499-00219-A-001-01

Figure 1. Digital Automatic Flight Control Systems (Sheet 4 of 37)



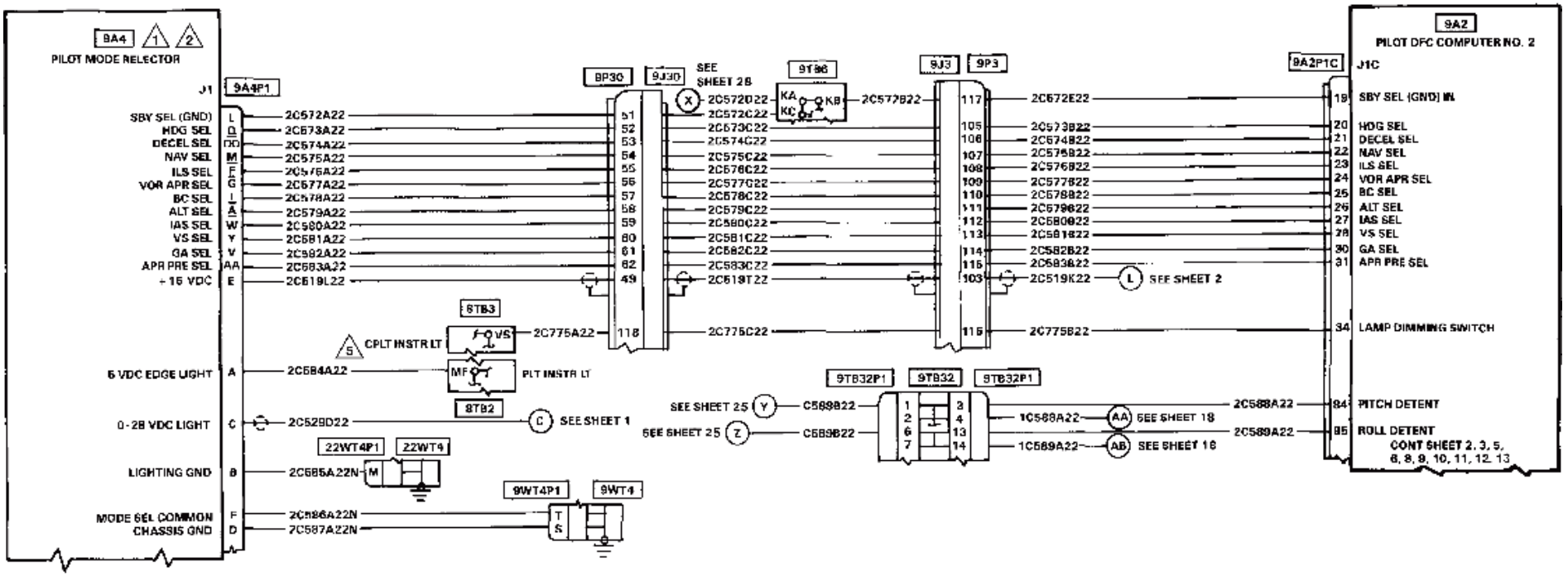
ICN-412-A-980000-B-97499-00220-A-001-01

Figure 1. Digital Automatic Flight Control Systems (Sheet 5 of 37)



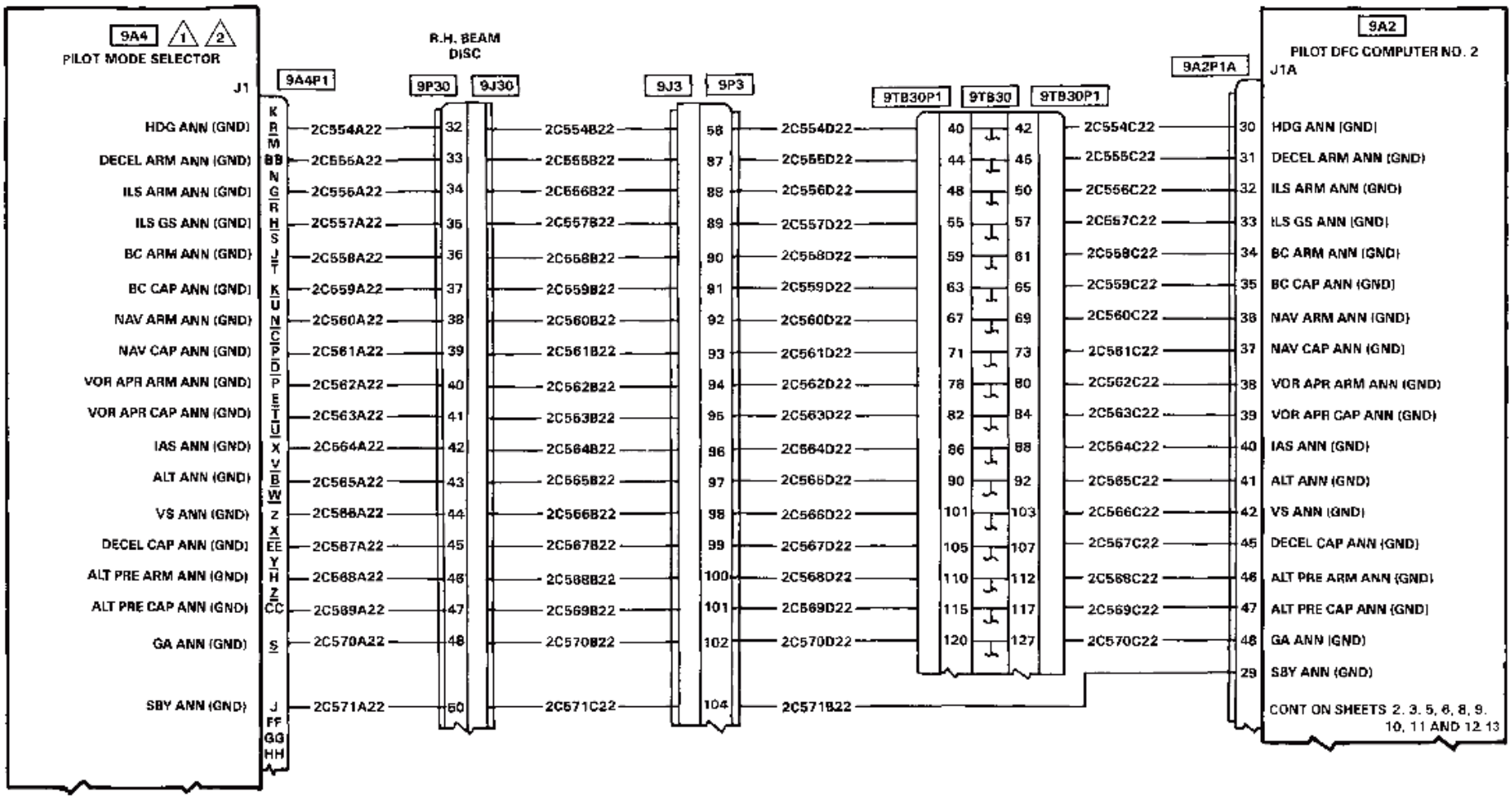
ICN-412-A-980000-B-97499-00221-A-001-01

Figure 1. Digital Automatic Flight Control Systems (Sheet 6 of 37)



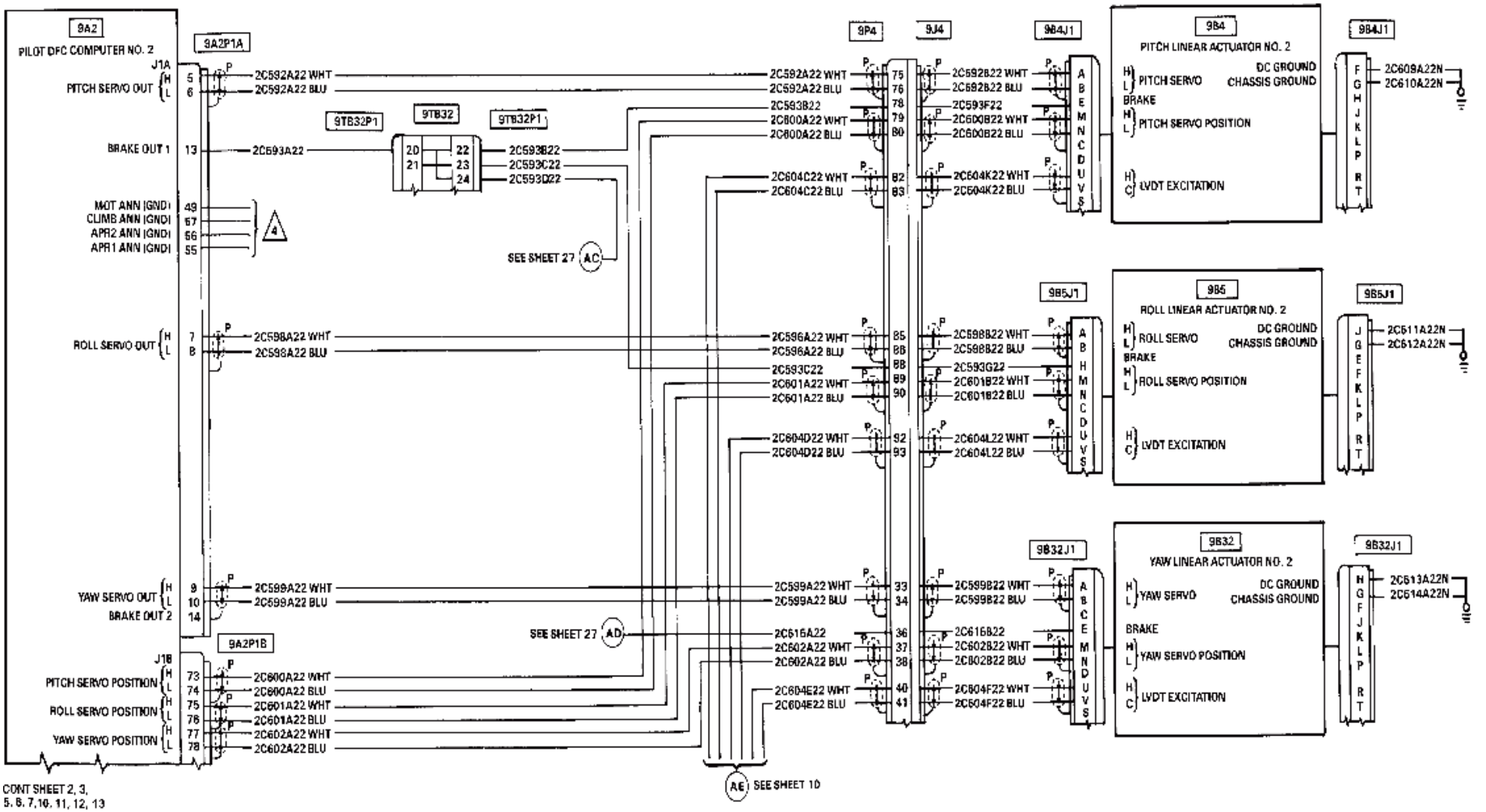
ICN-412-A-980000-B-97499-00222-A-001-01

Figure 1. Digital Automatic Flight Control Systems (Sheet 7 of 37)



ICN-412-A-980000-B-97499-00223-A-001-01

Figure 1. Digital Automatic Flight Control Systems (Sheet 8 of 37)



ICN-412-A-980000-B-97499-00224-A-001-01

Figure 1. Digital Automatic Flight Control Systems (Sheet 9 of 37)

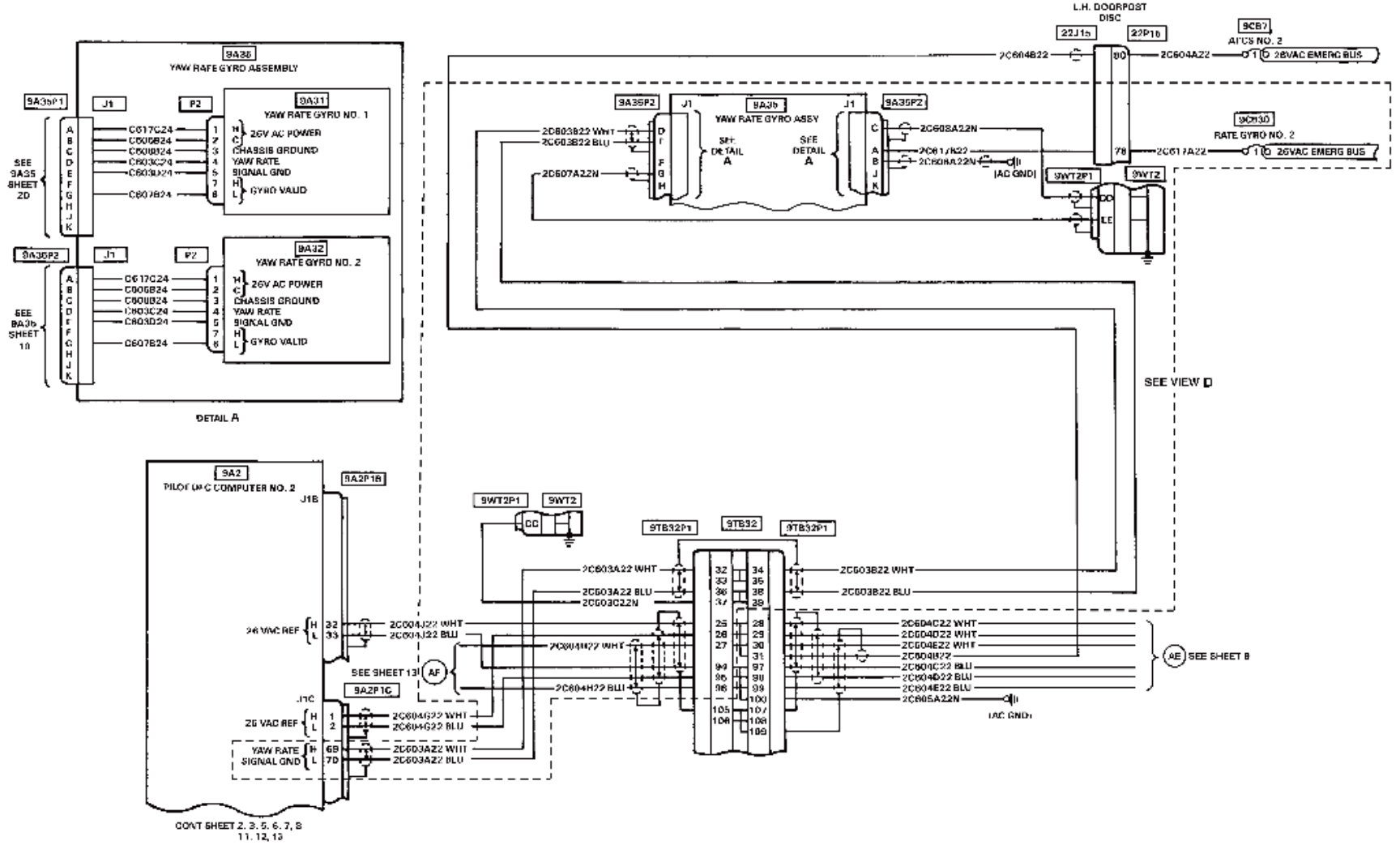
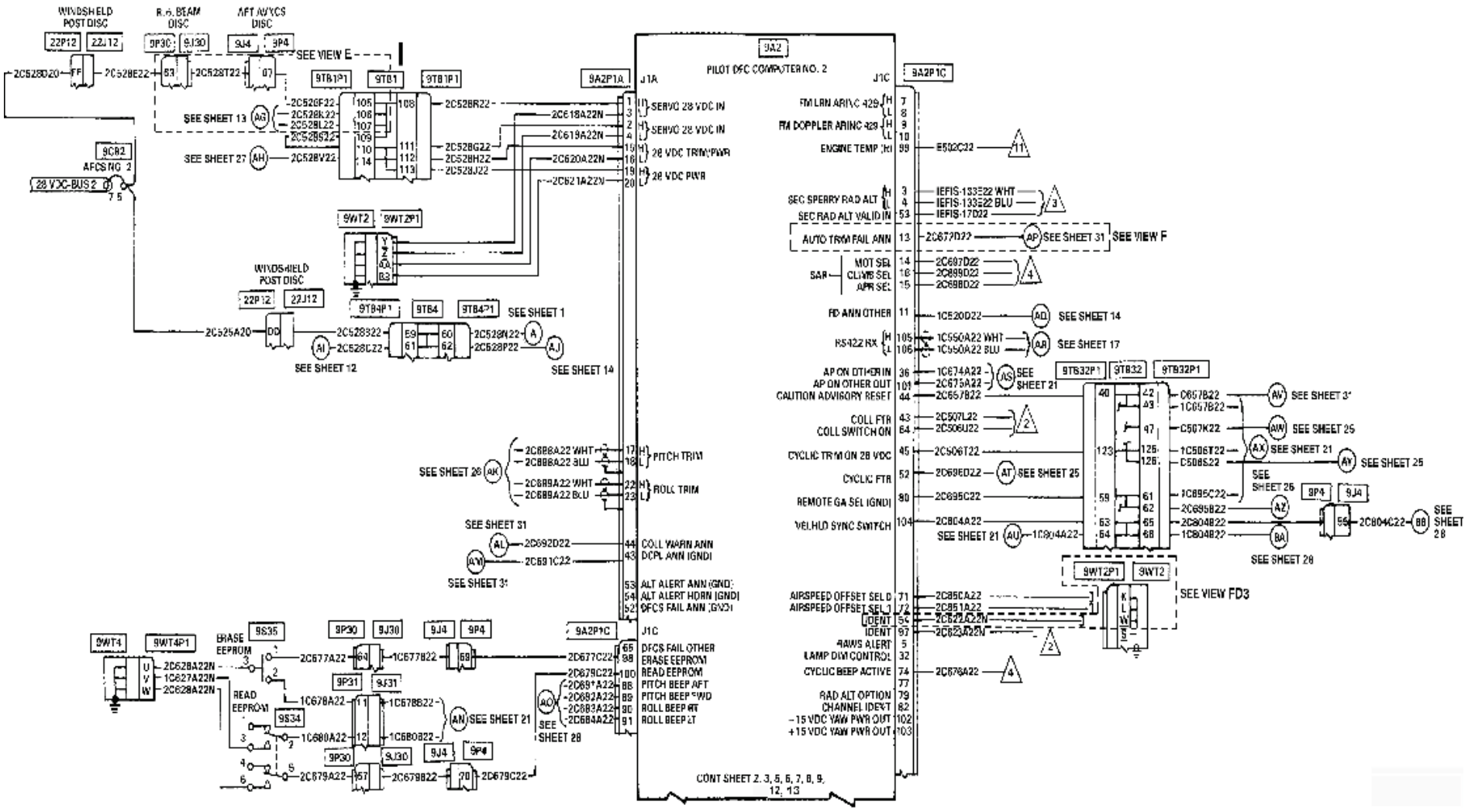
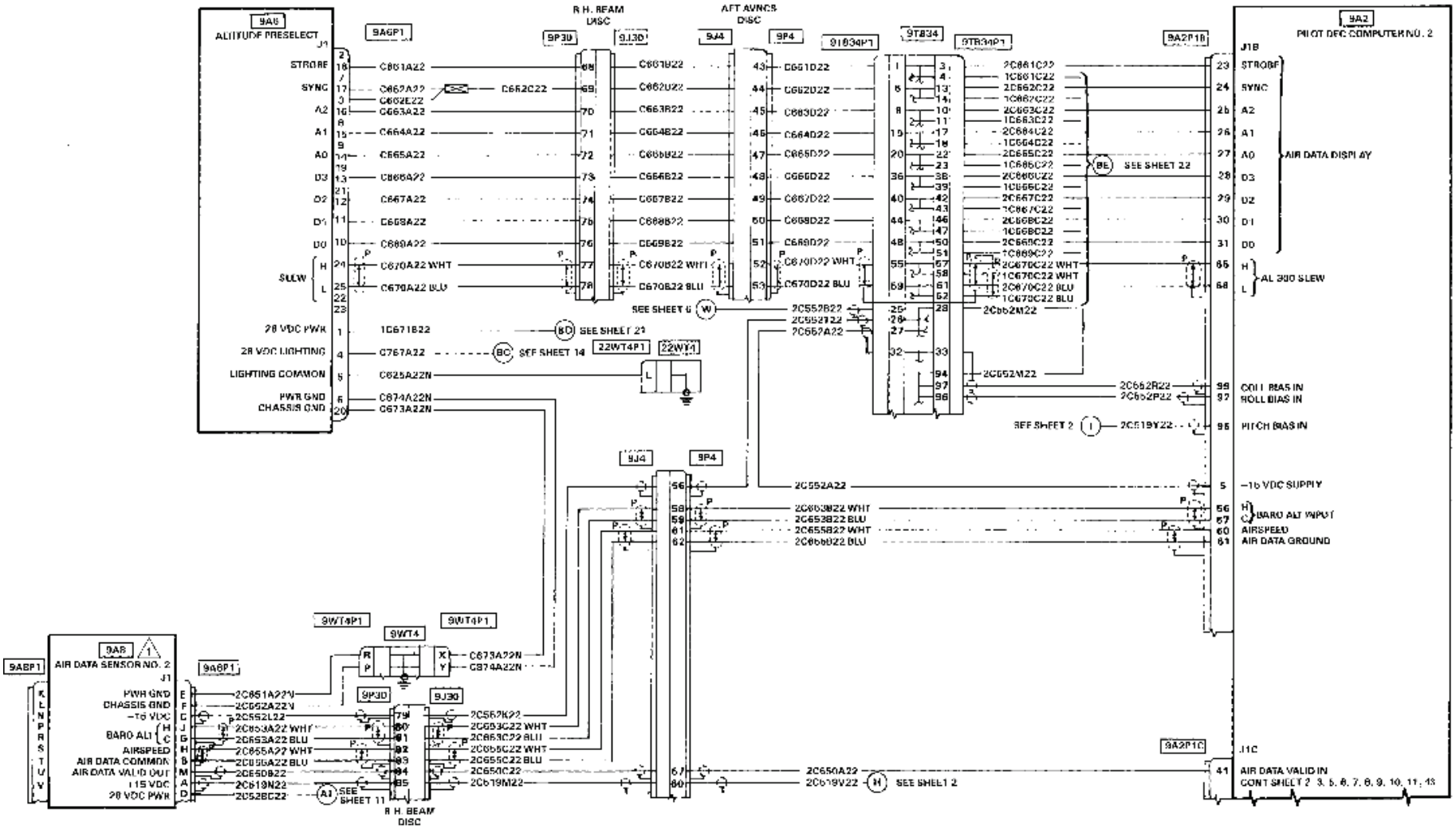


Figure 1. Digital Automatic Flight Control Systems (Sheet 10 of 37)



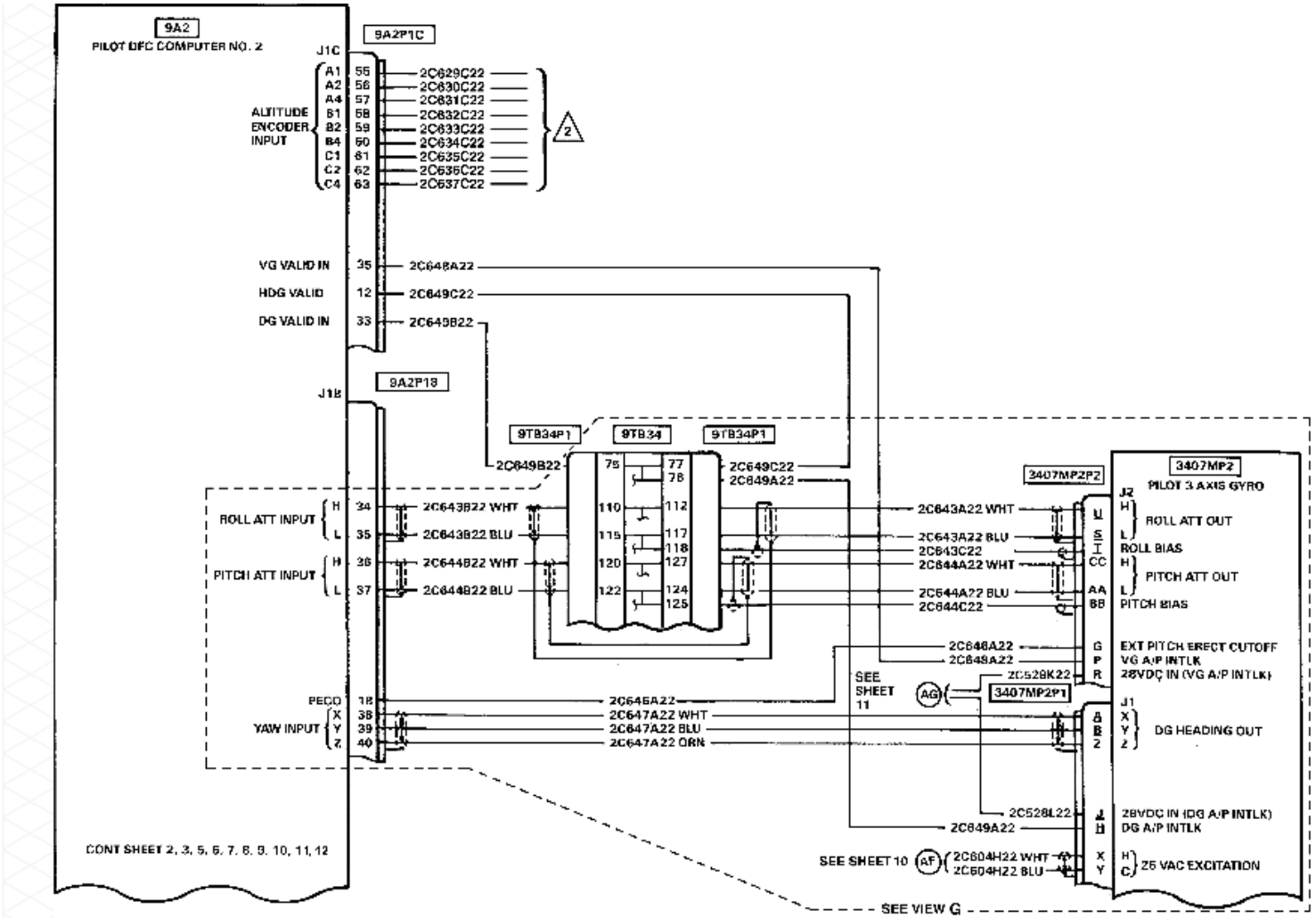
ICN-412-A-980000-B-97499-00226-A-001-01

Figure 1. Digital Automatic Flight Control Systems (Sheet 11 of 37)



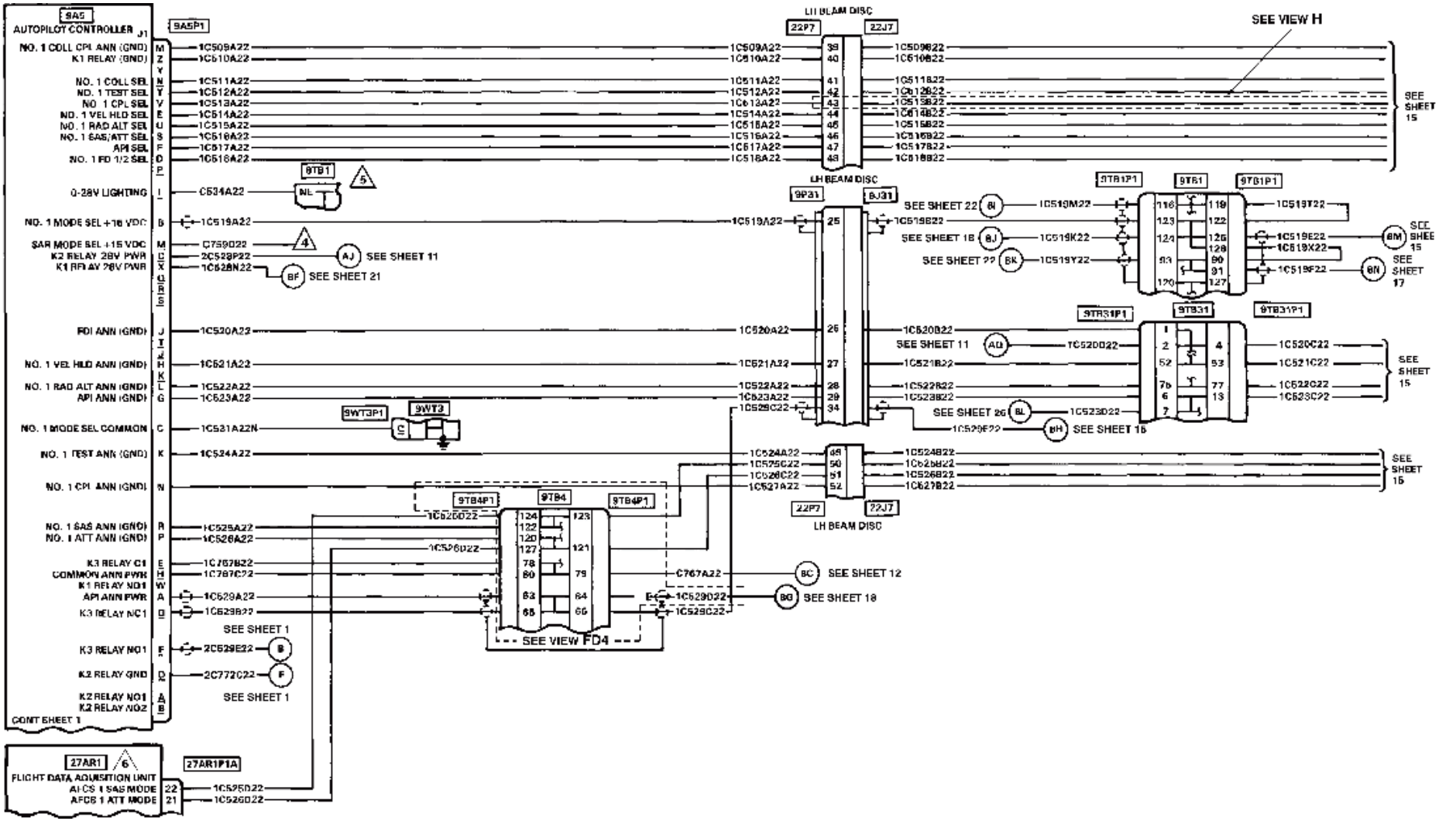
ICN-412-A-980000-B-97499-00227-A-001-01

Figure 1. Digital Automatic Flight Control Systems (Sheet 12 of 37)



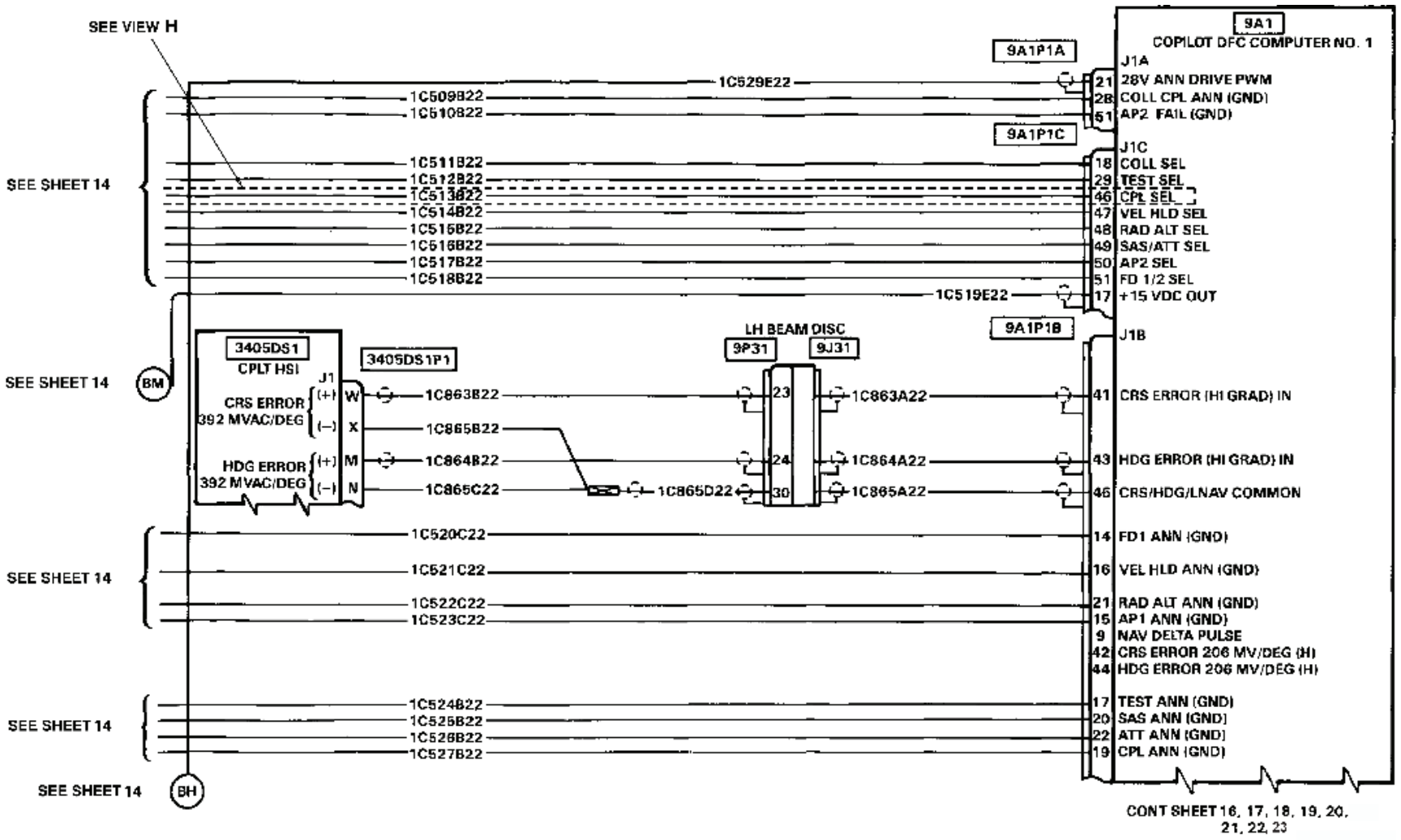
ICN-412-A-980000-B-97499-00228-A-001-01

Figure 1. Digital Automatic Flight Control Systems (Sheet 13 of 37)



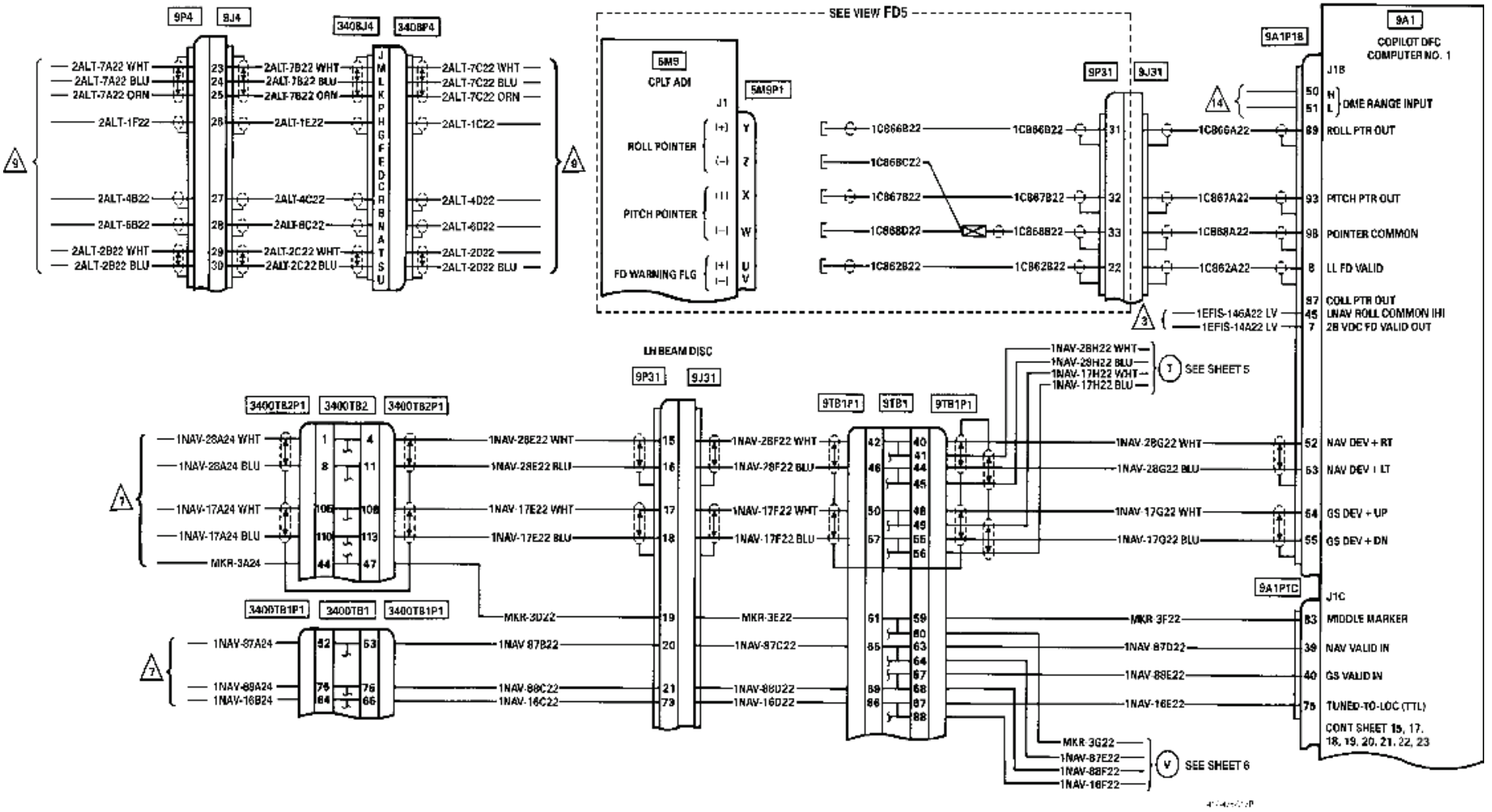
ICN-412-A-980000-B-97499-00229-A-001-01

Figure 1. Digital Automatic Flight Control Systems (Sheet 14 of 37)



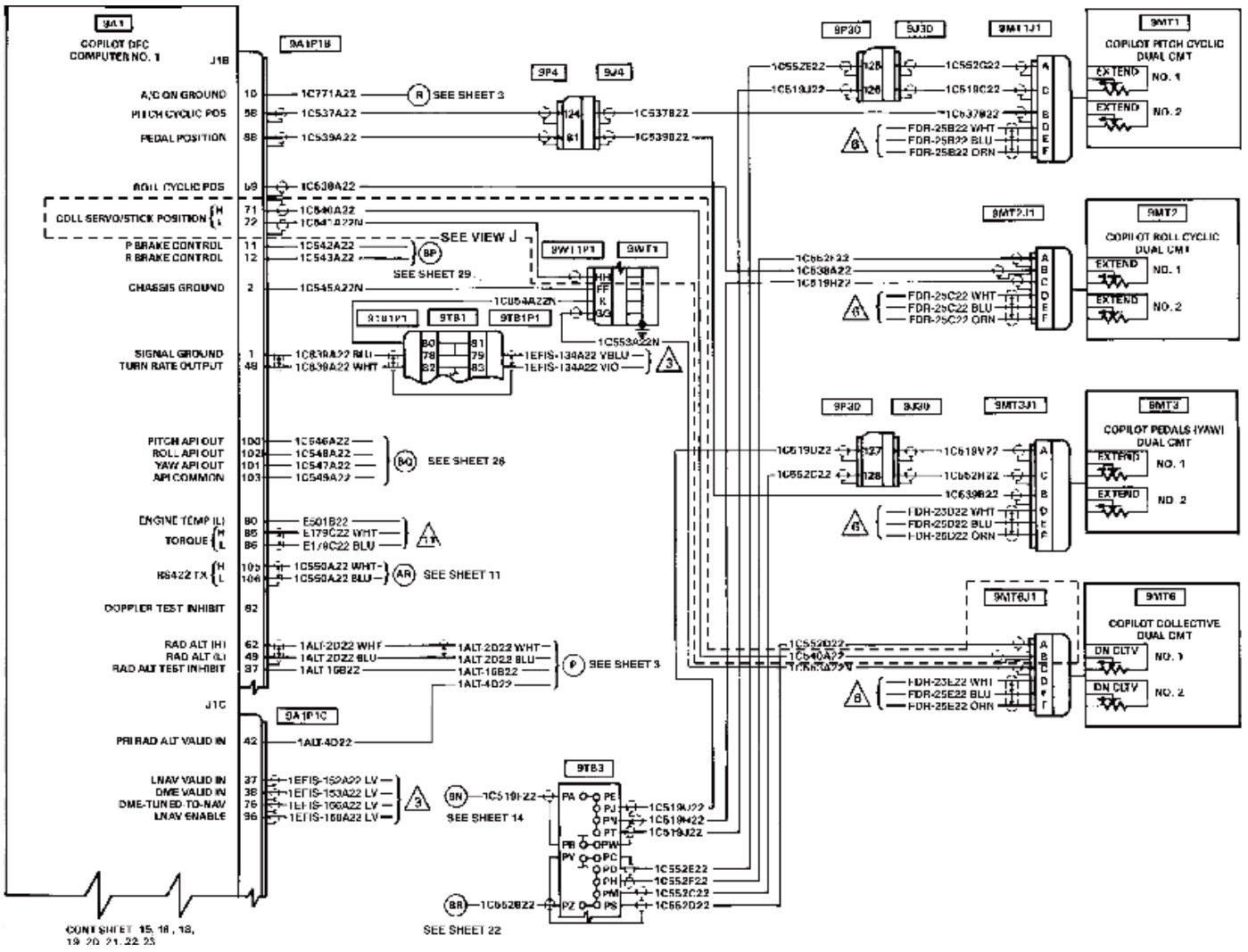
ICN-412-A-980000-B-97499-00230-A-001-01

Figure 1. Digital Automatic Flight Control Systems (Sheet 15 of 37)



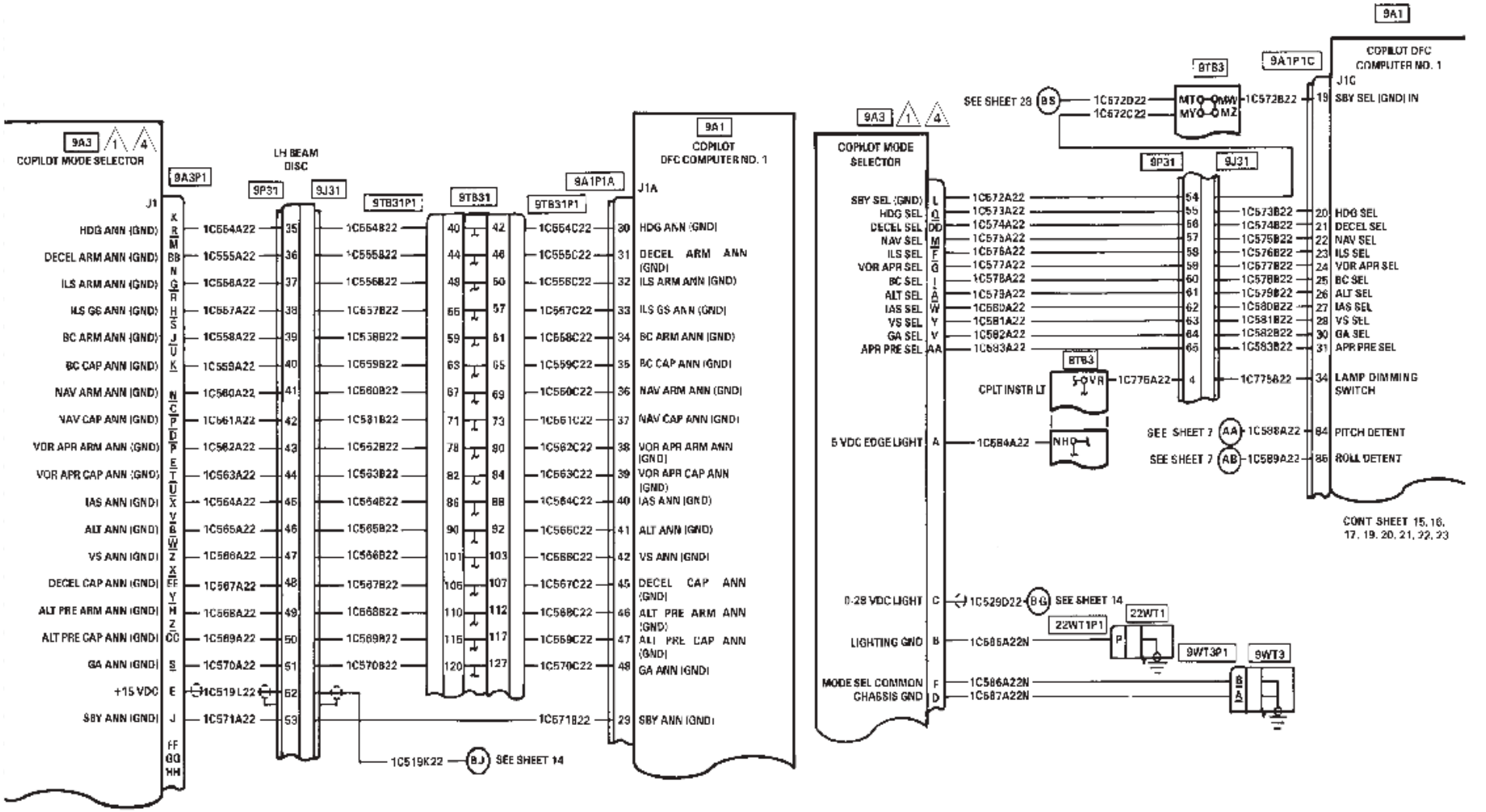
ICN-412-A-980000-B-97499-00231-A-001-01

Figure 1. Digital Automatic Flight Control Systems (Sheet 16 of 37)



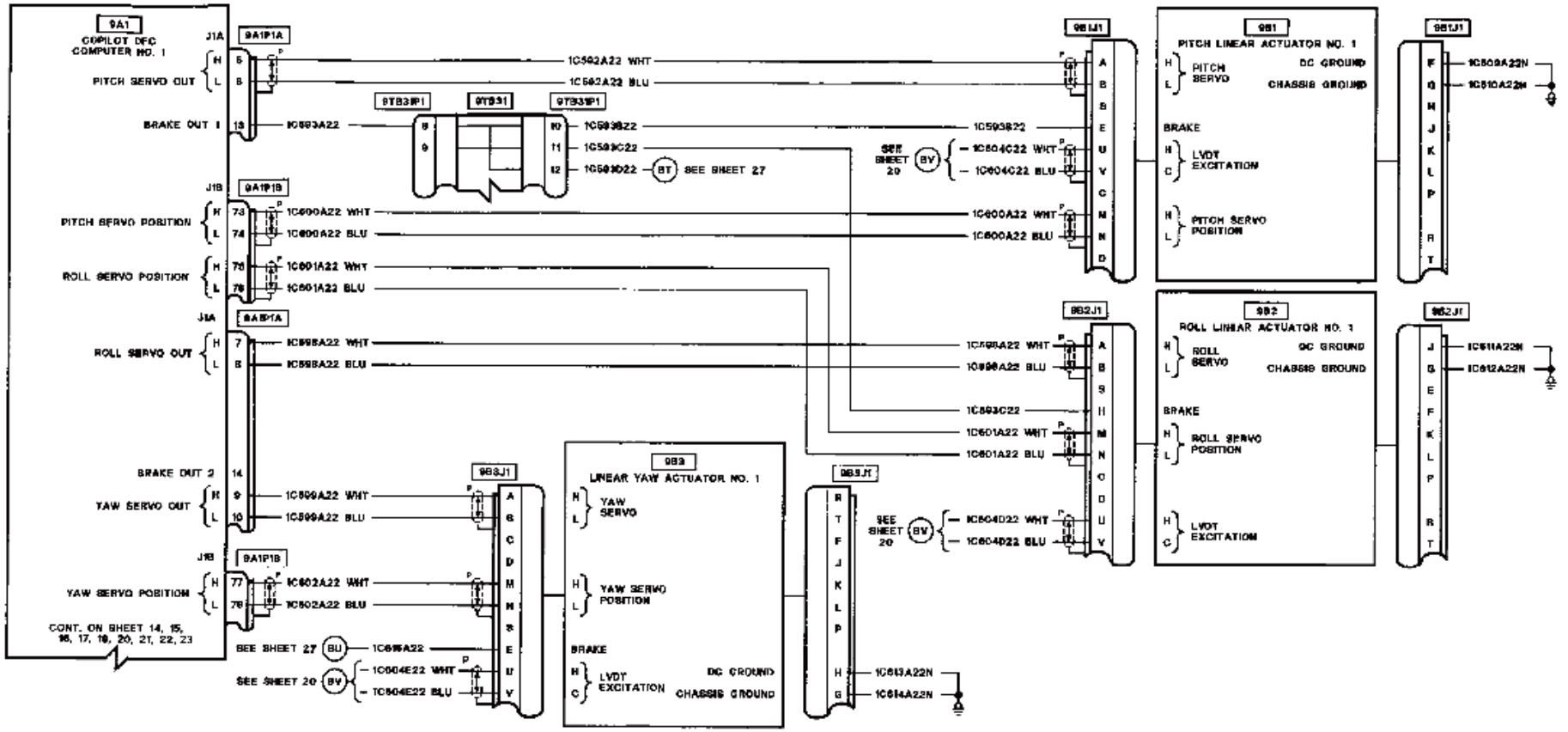
ICN-412-A-980000-B-97499-00232-A-001-01

Figure 1. Digital Automatic Flight Control Systems (Sheet 17 of 37)



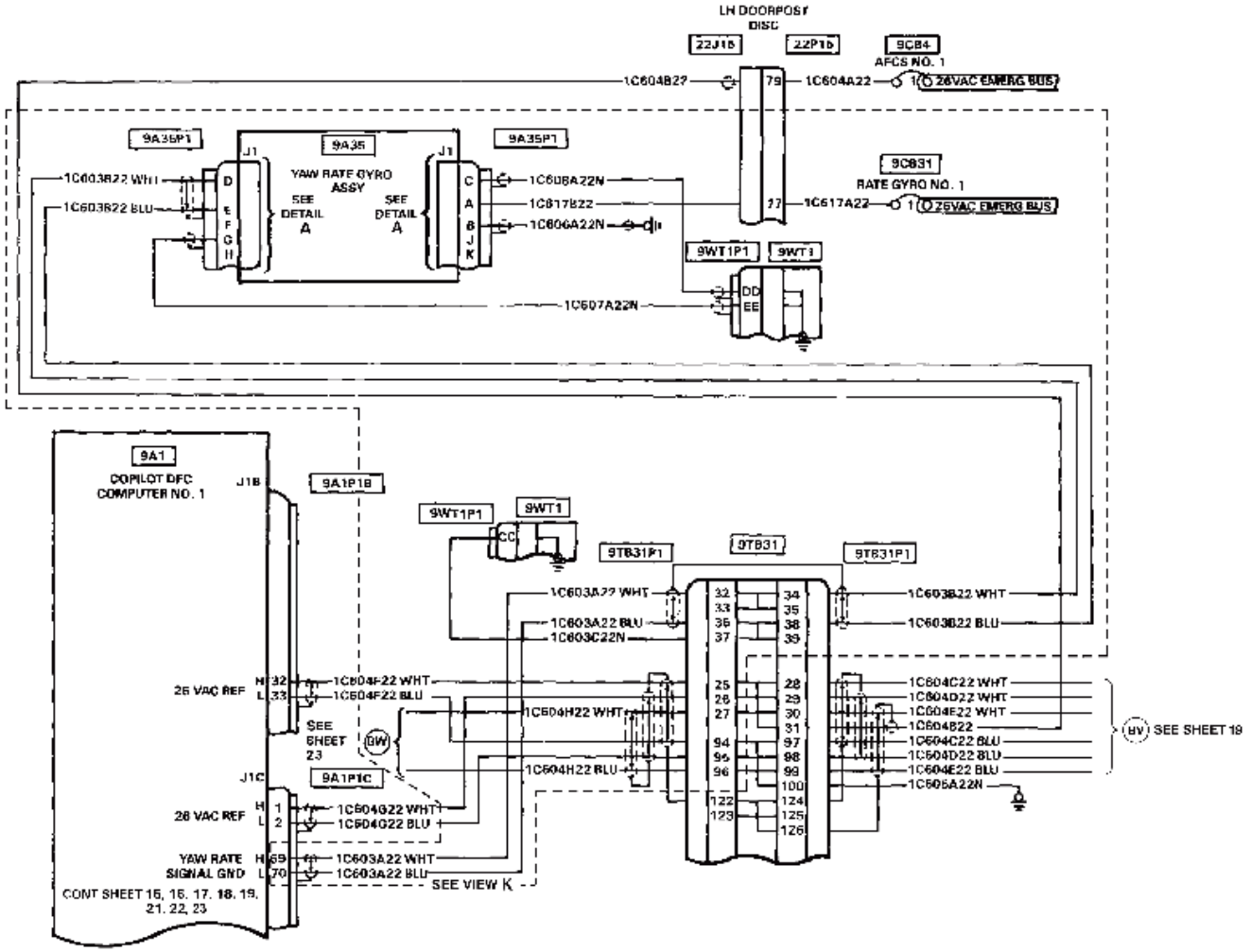
ICN-412-A-980000-B-97499-00233-A-001-01

Figure 1. Digital Automatic Flight Control Systems (Sheet 18 of 37)



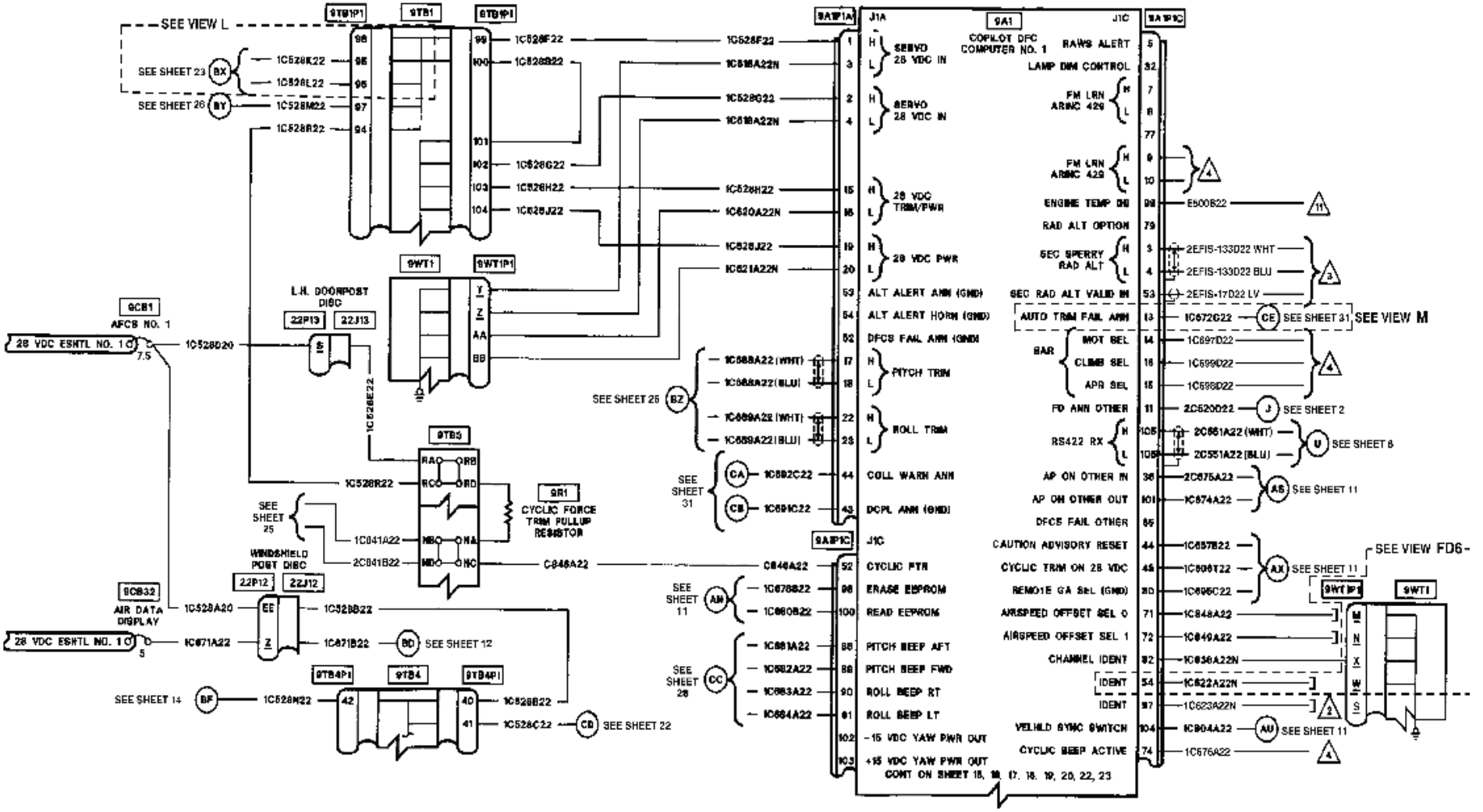
ICN-412-A-980000-B-97499-00234-A-001-01

Figure 1. Digital Automatic Flight Control Systems (Sheet 19 of 37)



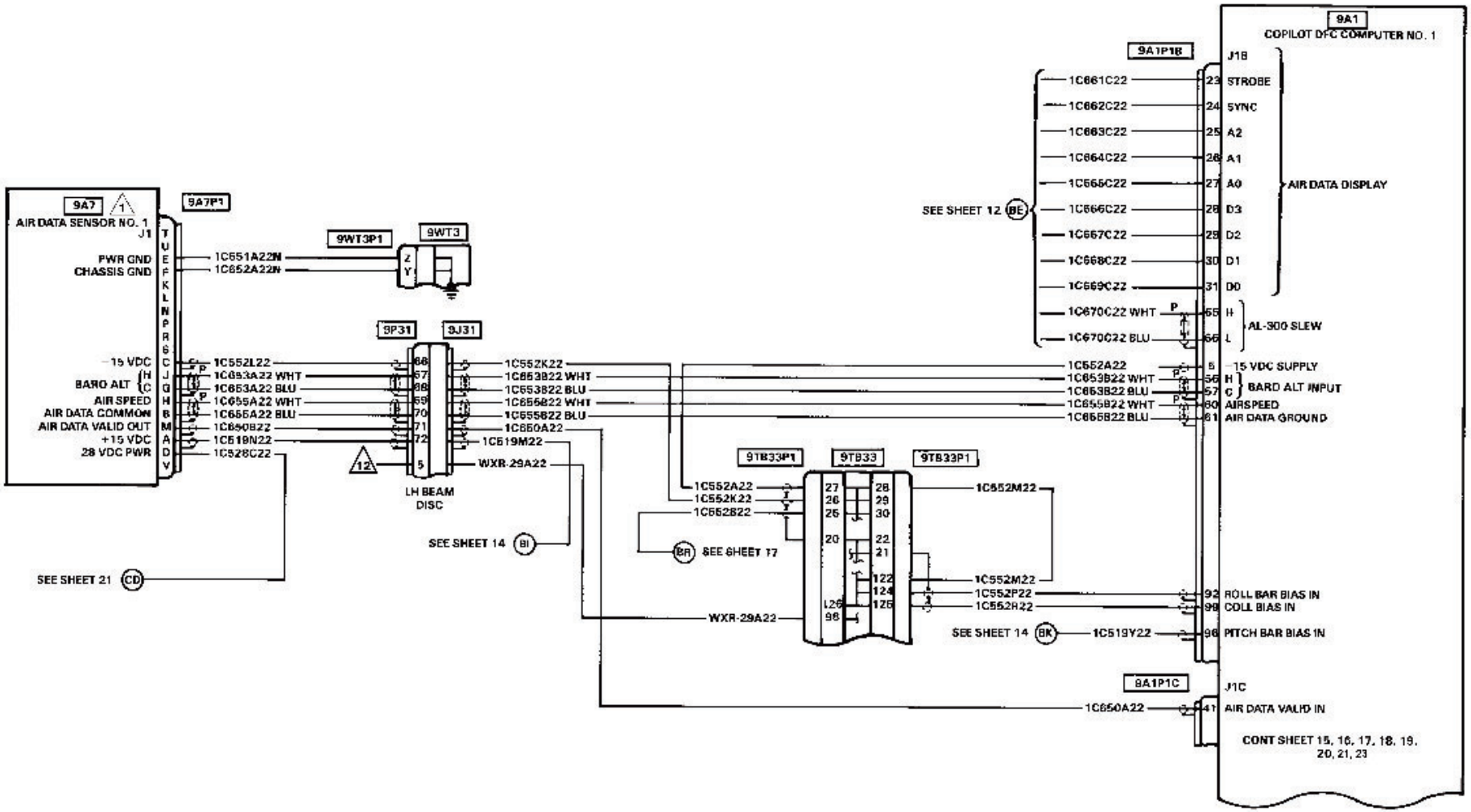
ICN-412-A-980000-B-97499-00235-A-001-01

Figure 1. Digital Automatic Flight Control Systems (Sheet 20 of 37)



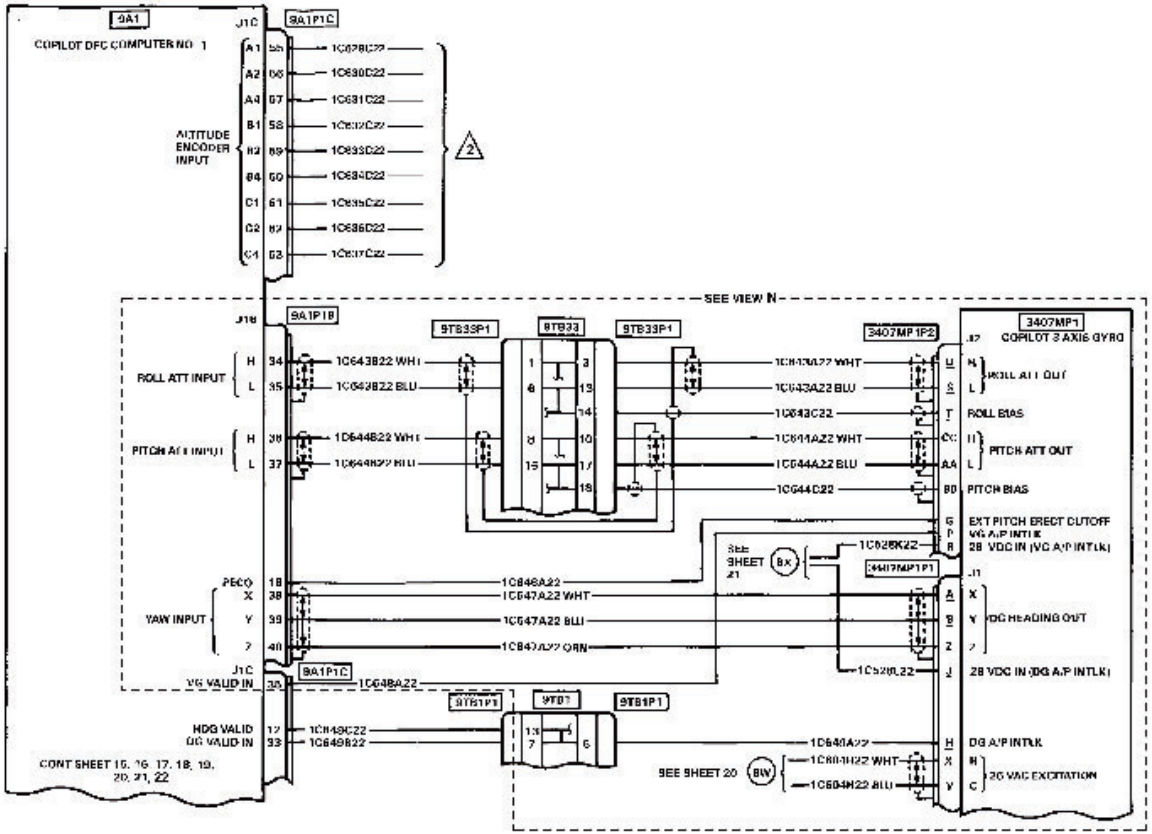
ICN-412-A-980000-B-97499-00236-A-001-01

Figure 1. Digital Automatic Flight Control Systems (Sheet 21 of 37)



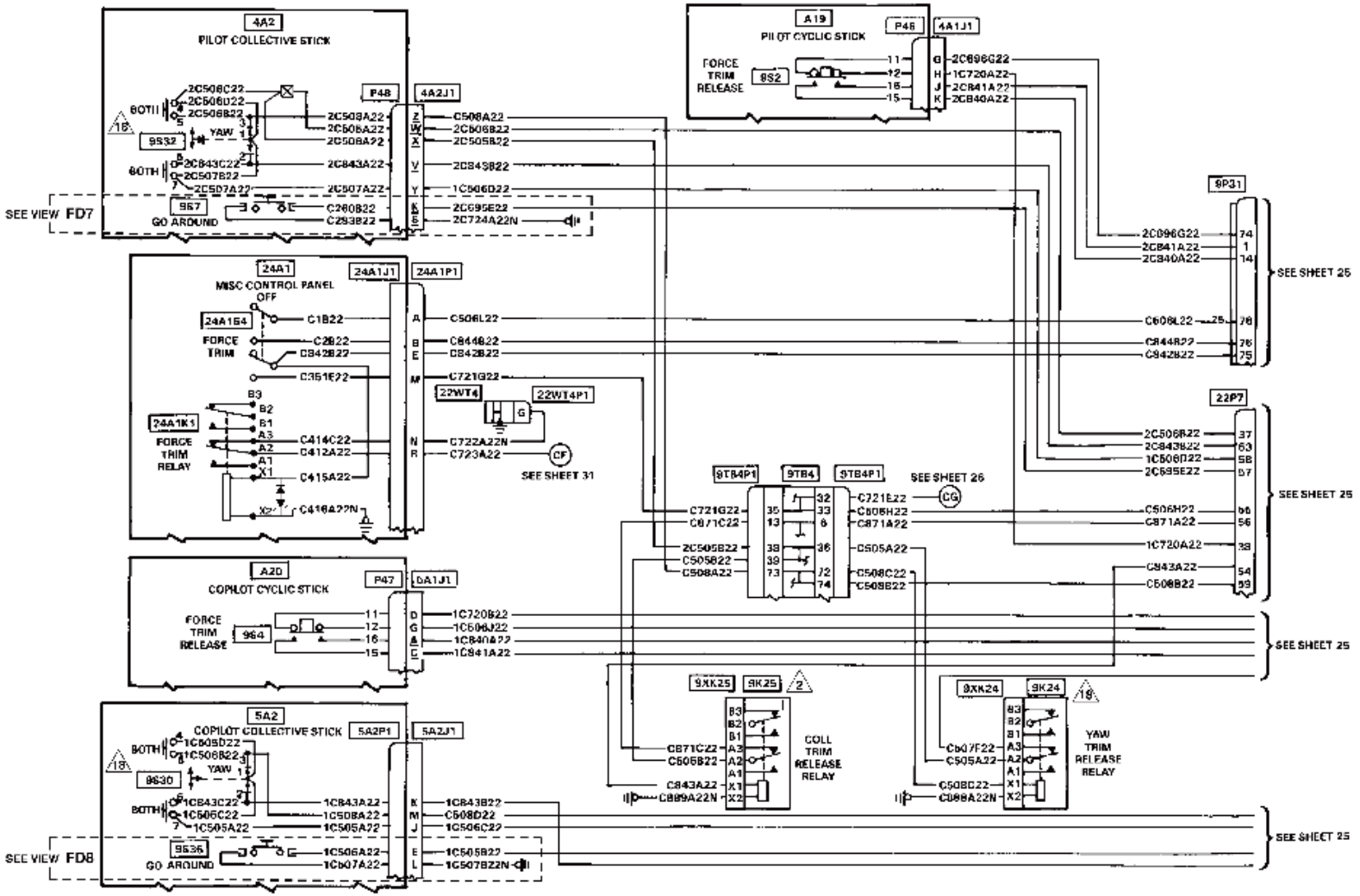
ICN-412-A-980000-B-97499-00237-A-001-01

Figure 1. Digital Automatic Flight Control Systems (Sheet 22 of 37)



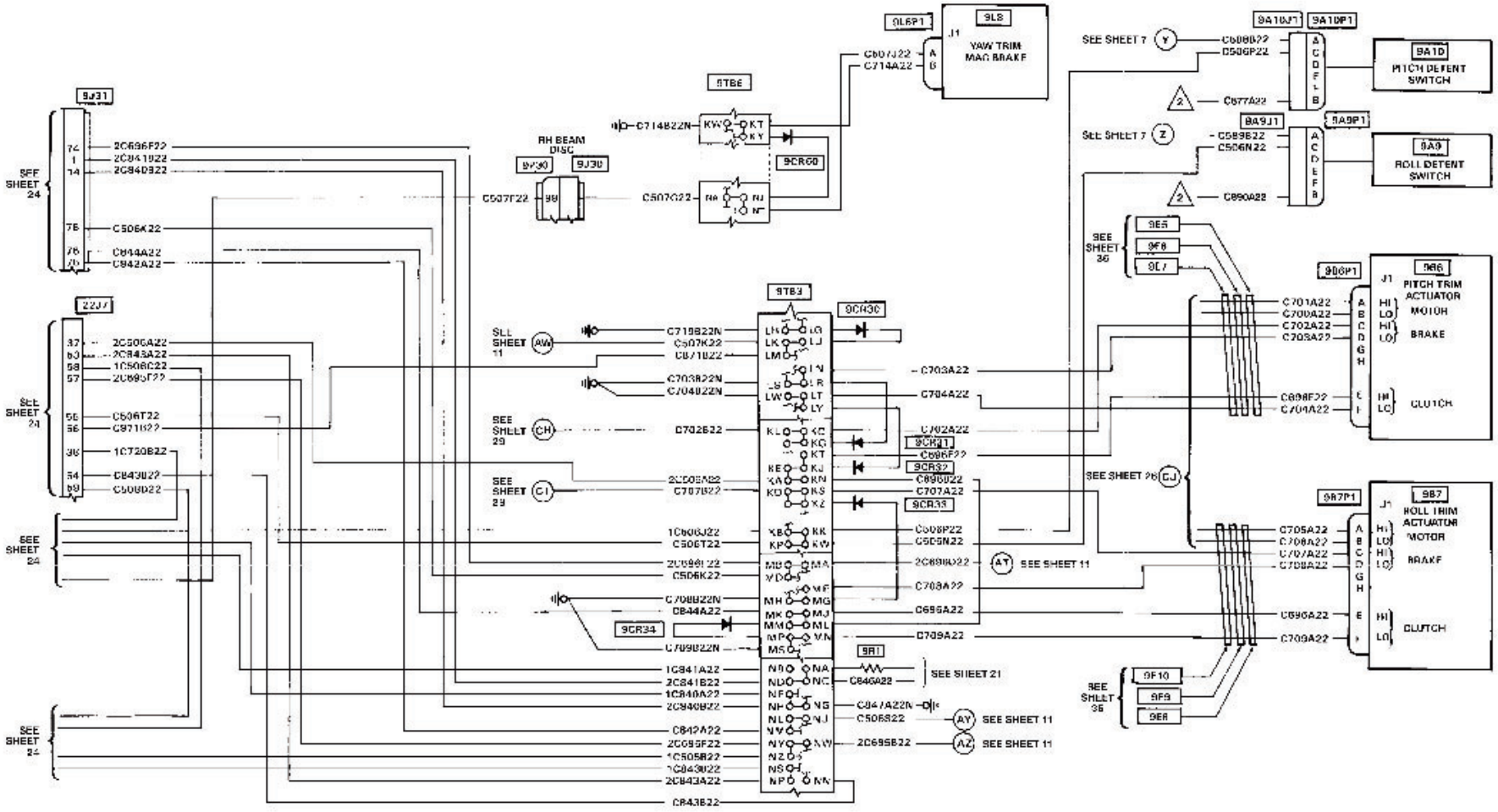
ICN-412-A-980000-B-97499-00238-A-001-01

Figure 1. Digital Automatic Flight Control Systems (Sheet 23 of 37)



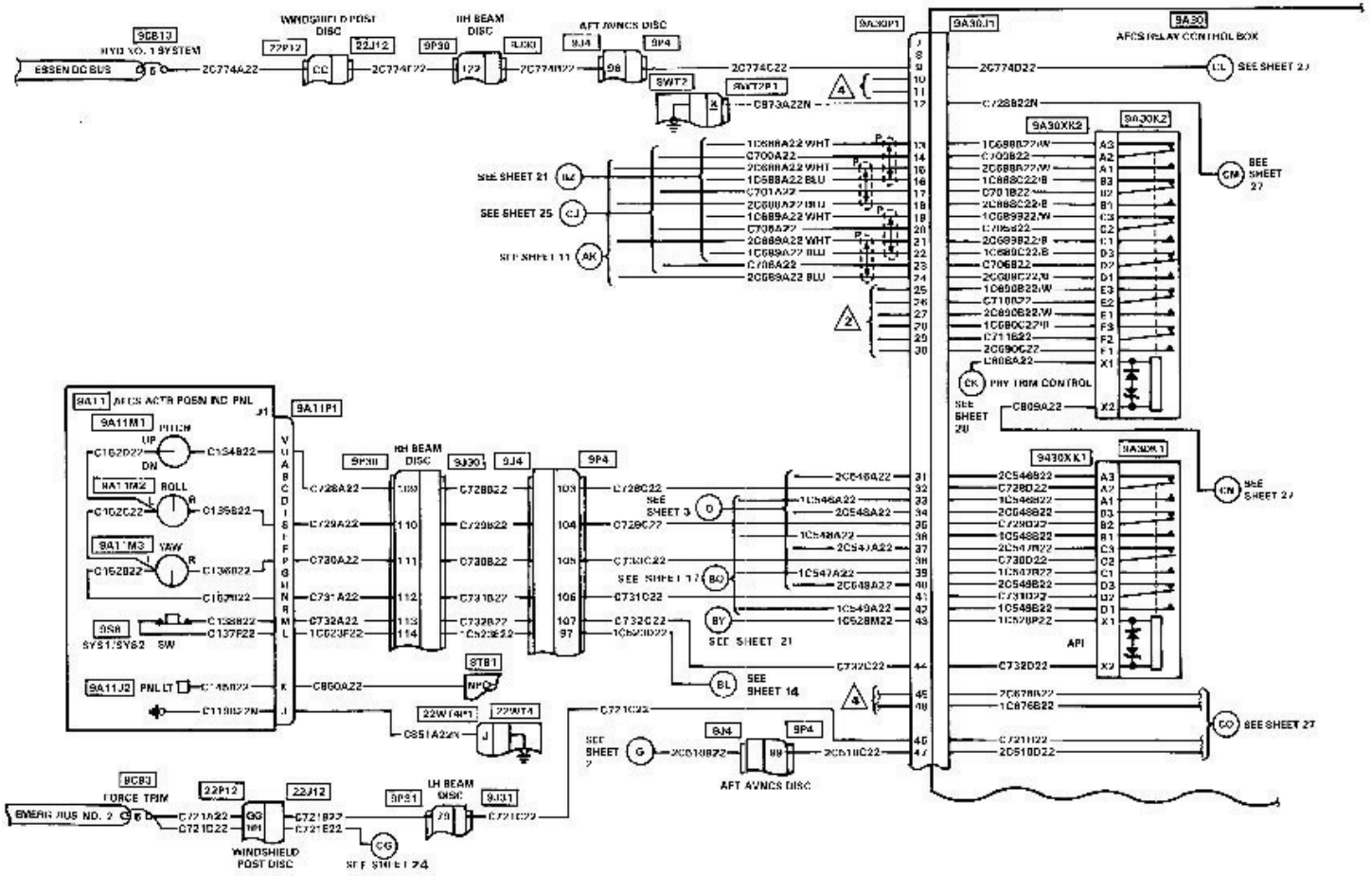
ICN-412-A-980000-B-97499-00239-A-001-01

Figure 1. Digital Automatic Flight Control Systems (Sheet 24 of 37)



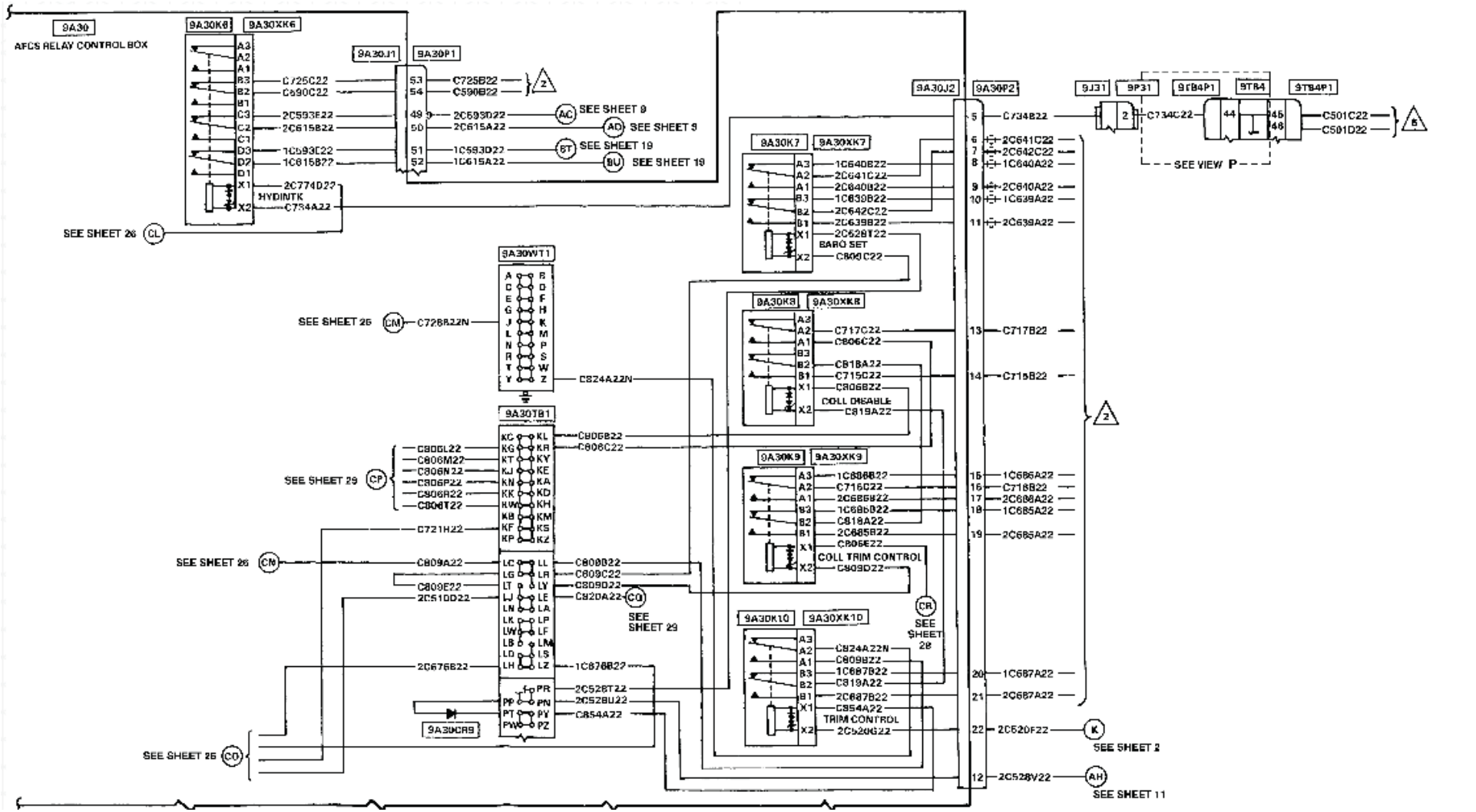
ICN-412-A-980000-B-97499-00240-A-001-01

Figure 1. Digital Automatic Flight Control Systems (Sheet 25 of 37)



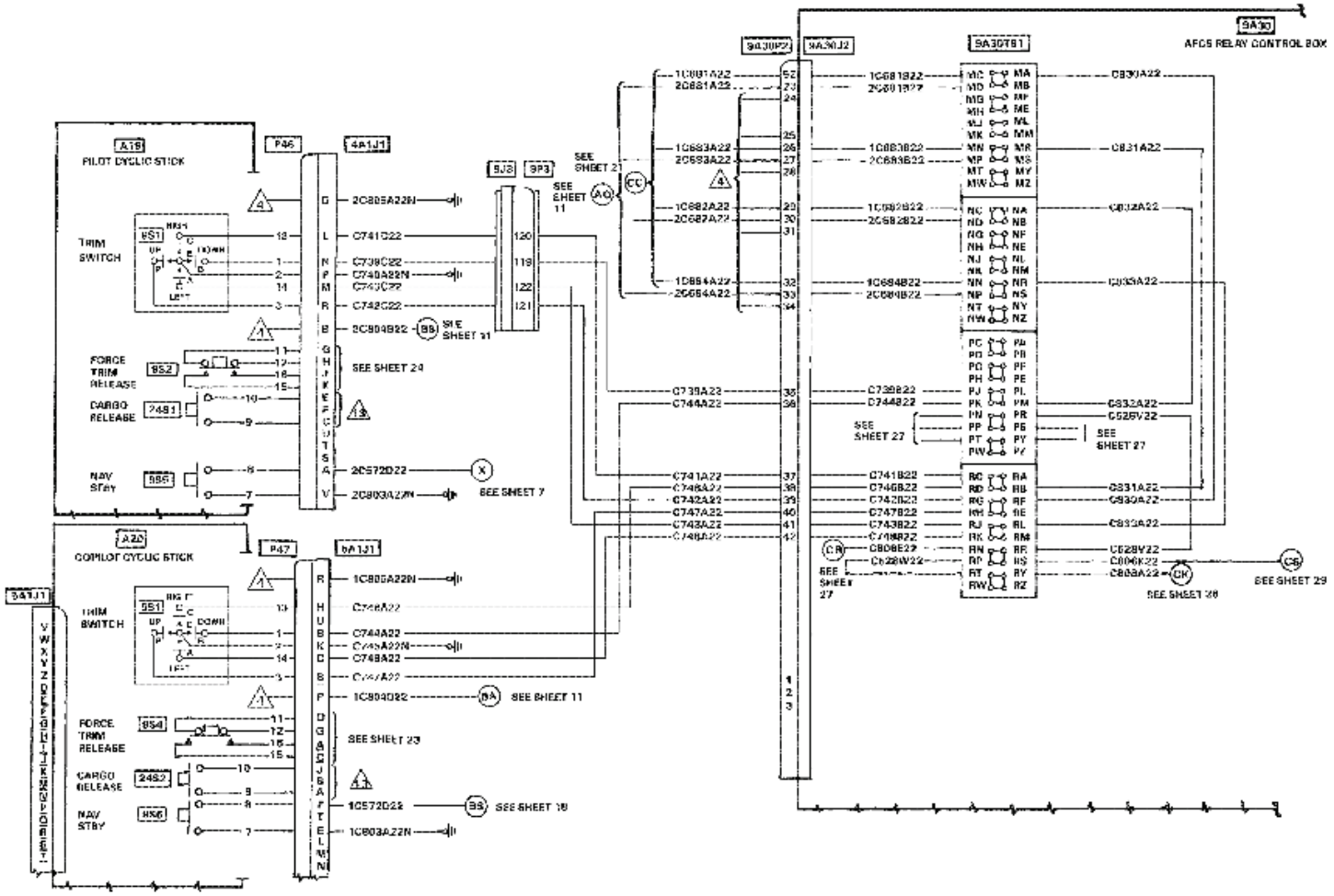
ICN-412-A-980000-B-97499-00241-A-001-01

Figure 1. Digital Automatic Flight Control Systems (Sheet 26 of 37)



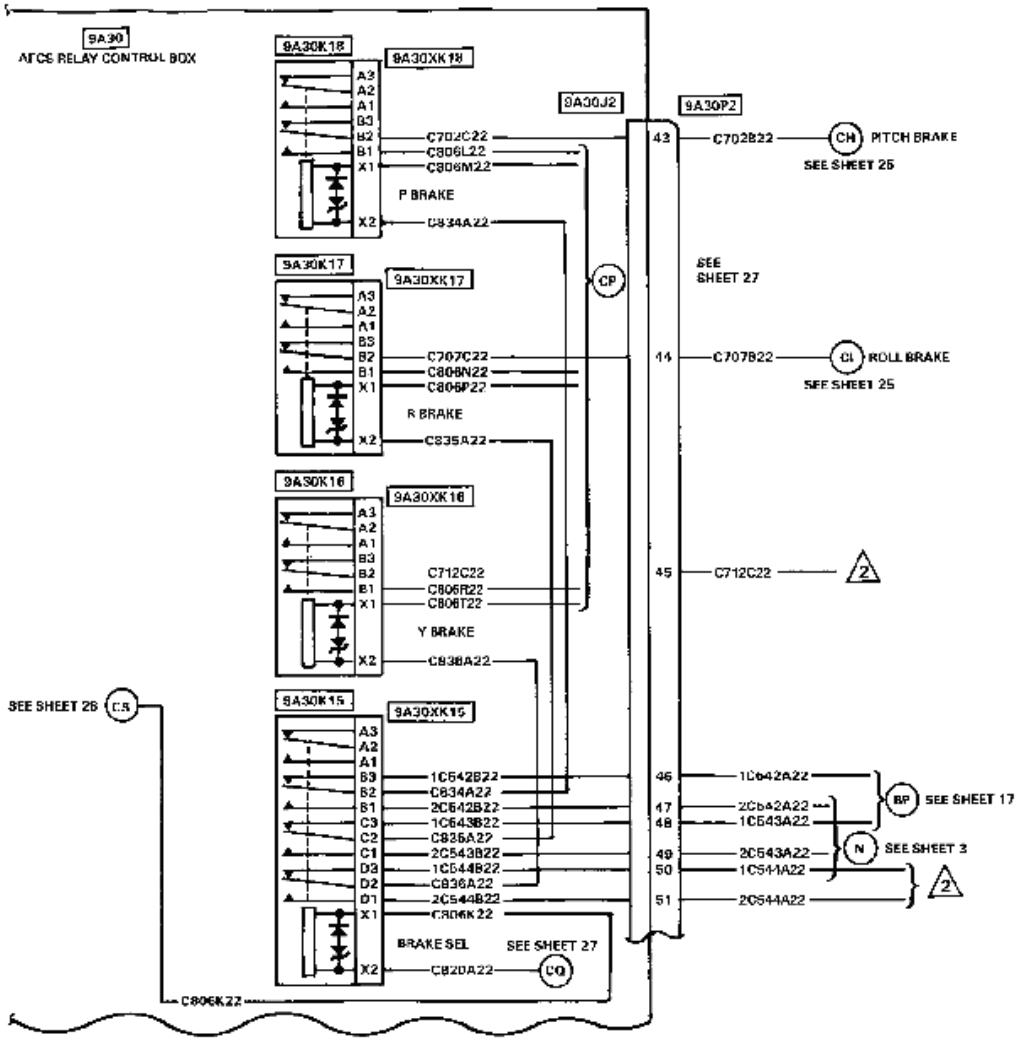
ICN-412-A-980000-B-97499-00242-A-001-01

Figure 1. Digital Automatic Flight Control Systems (Sheet 27 of 37)



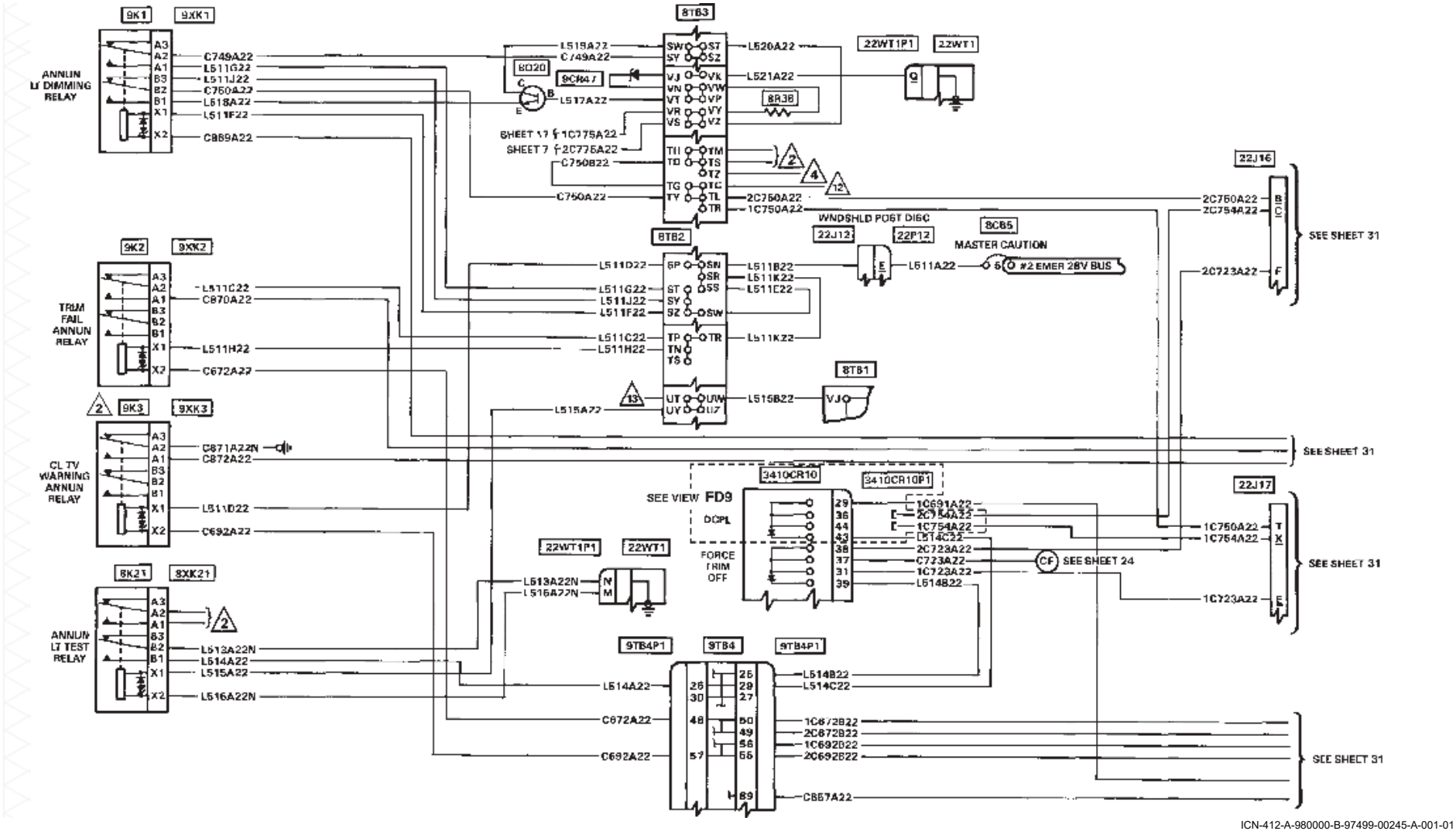
ICN-412-A-980000-B-97499-00243-A-001-01

Figure 1. Digital Automatic Flight Control Systems (Sheet 28 of 37)



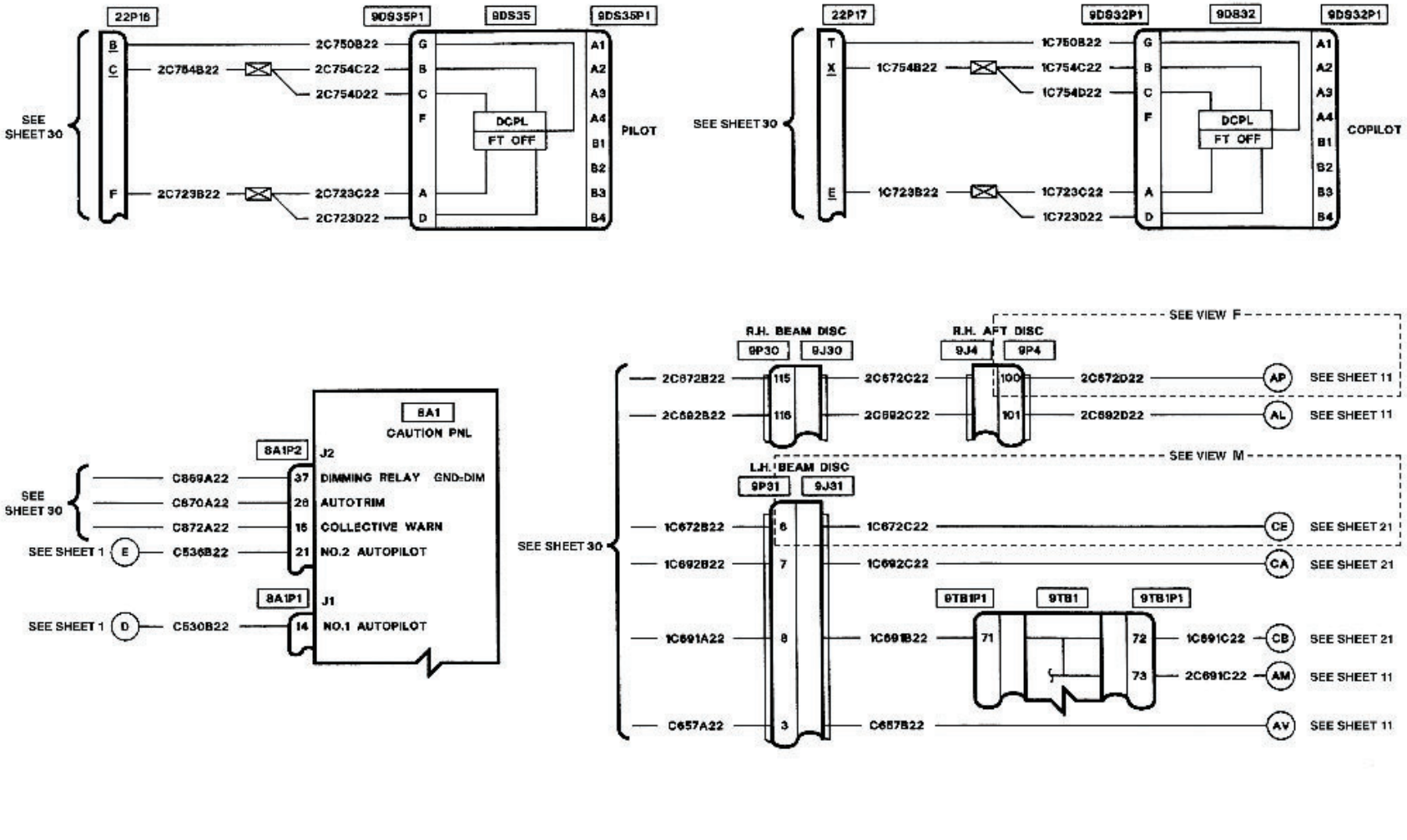
ICN-412-A-980000-B-97499-00244-A-001-01

Figure 1. Digital Automatic Flight Control Systems (Sheet 29 of 37)



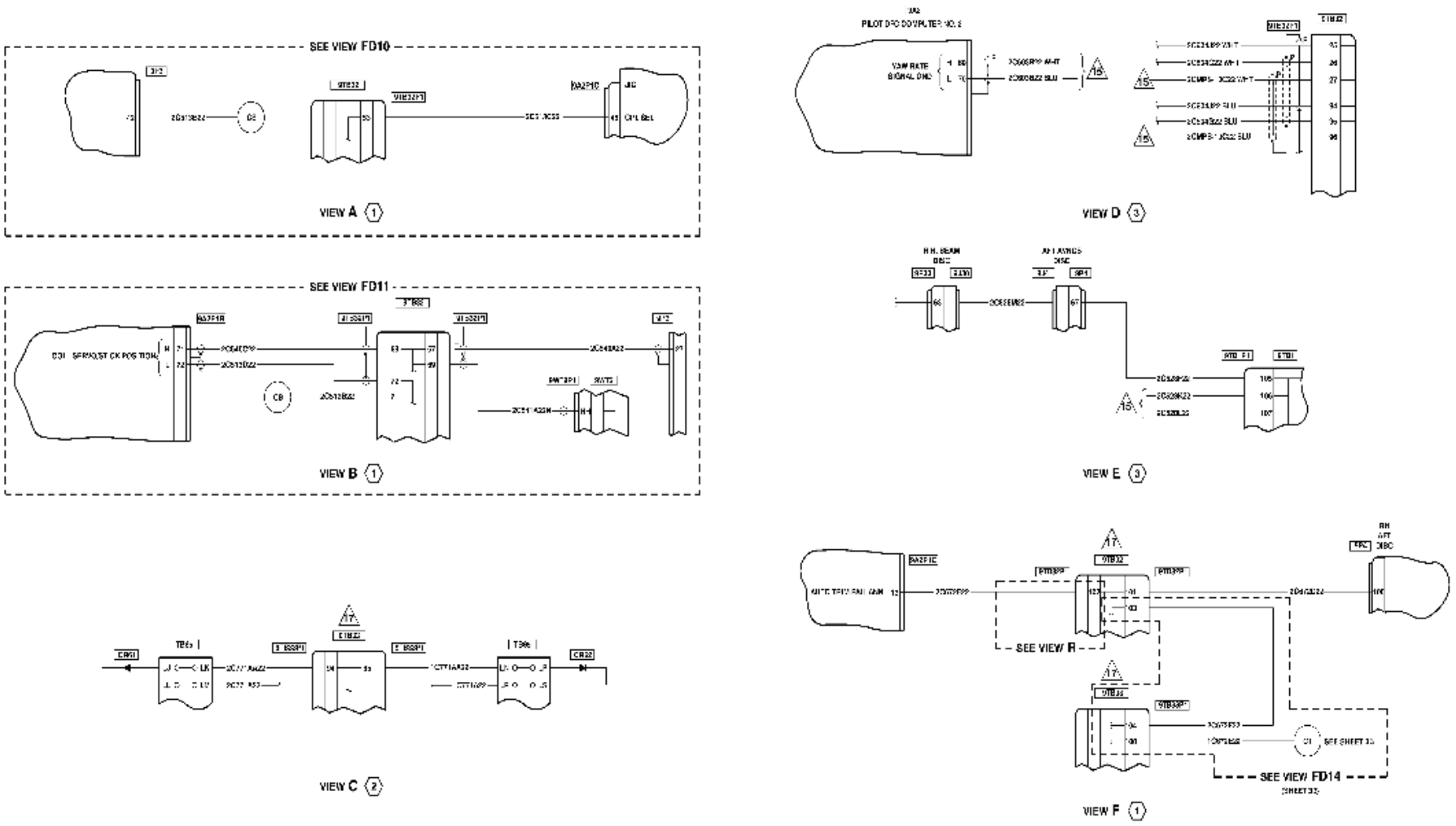
ICN-412-A-980000-B-97499-00245-A-001-01

Figure 1. Digital Automatic Flight Control Systems (Sheet 30 of 37)



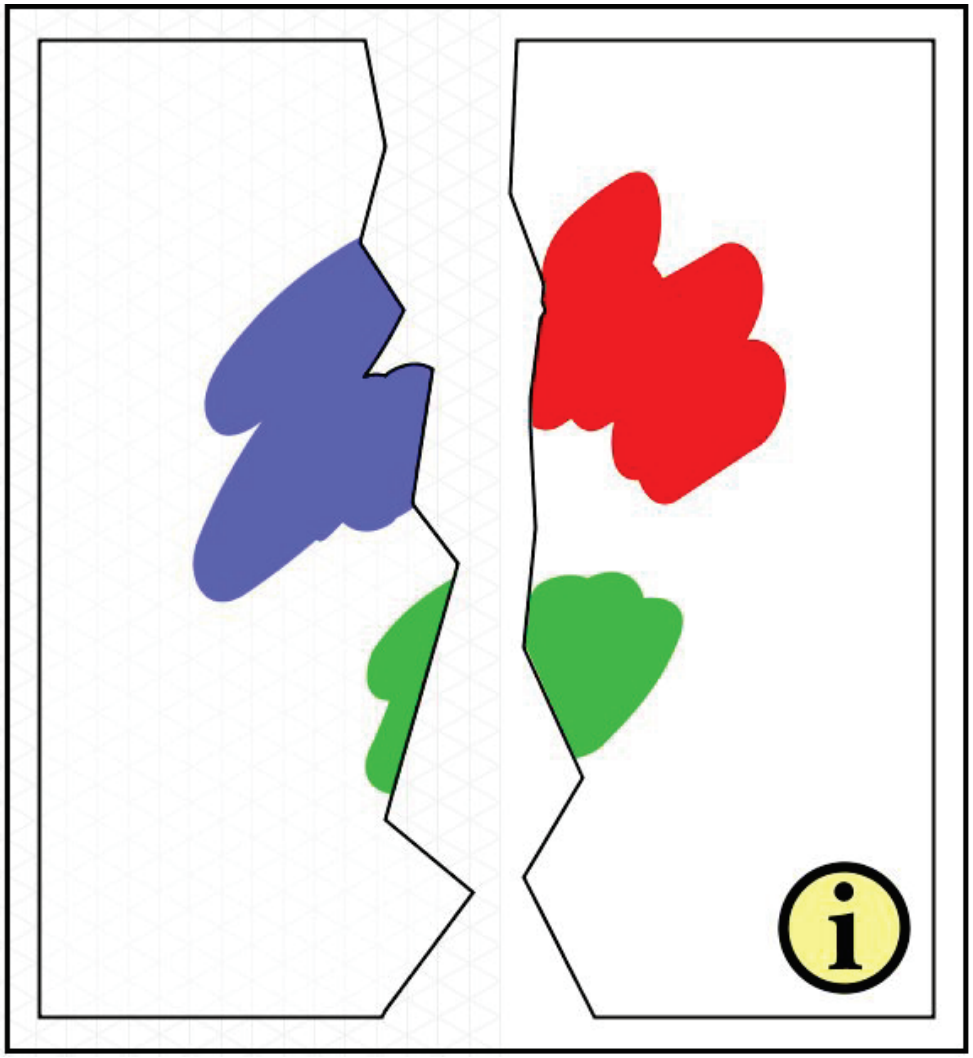
ICN-412-A-980000-B-97499-00246-A-001-01

Figure 1. Digital Automatic Flight Control Systems (Sheet 31 of 37)



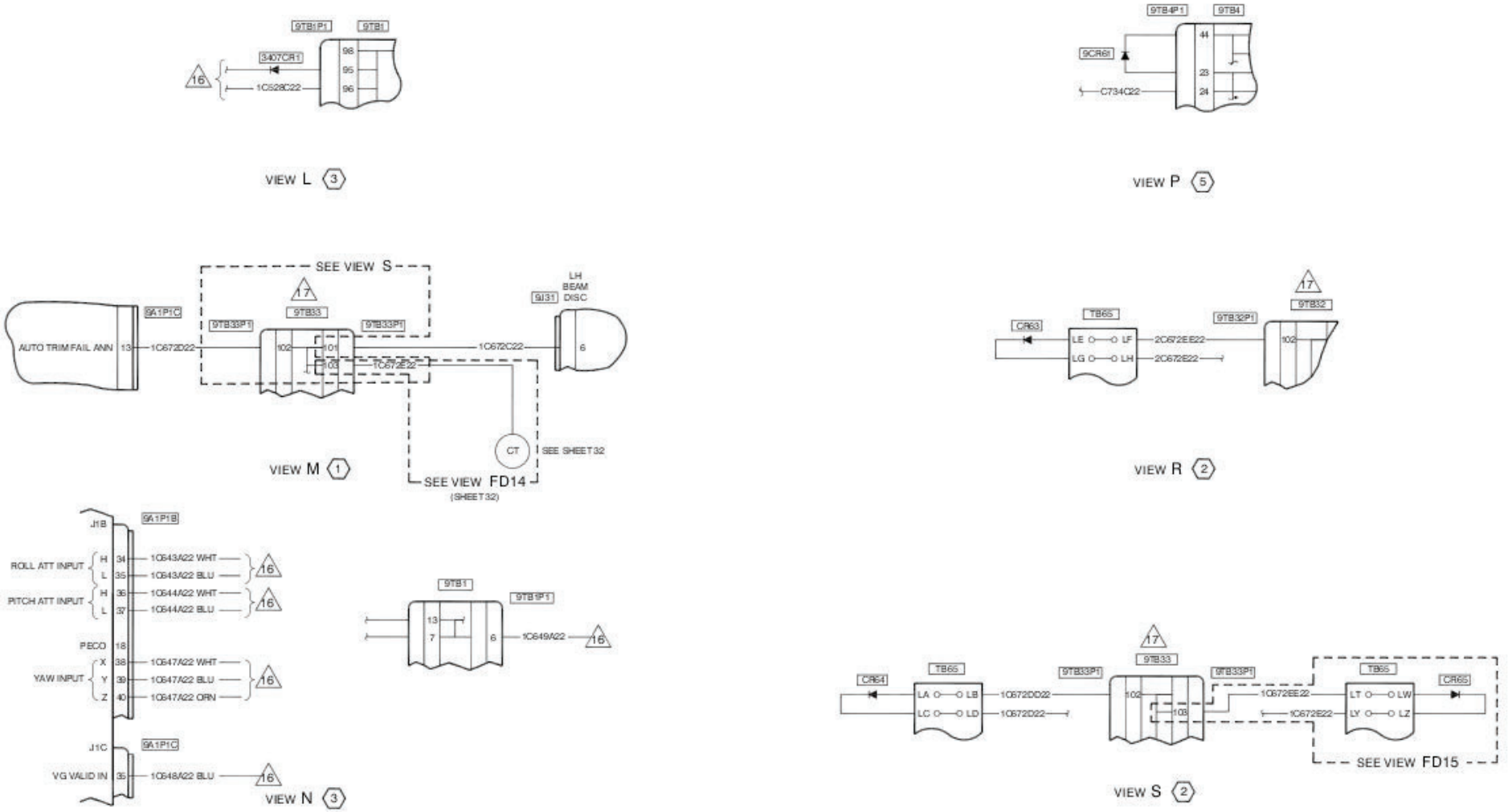
ICN-412-A-980000-B-97499-00247-A-001-01

Figure 1. Digital Automatic Flight Control Systems (Sheet 32 of 37)



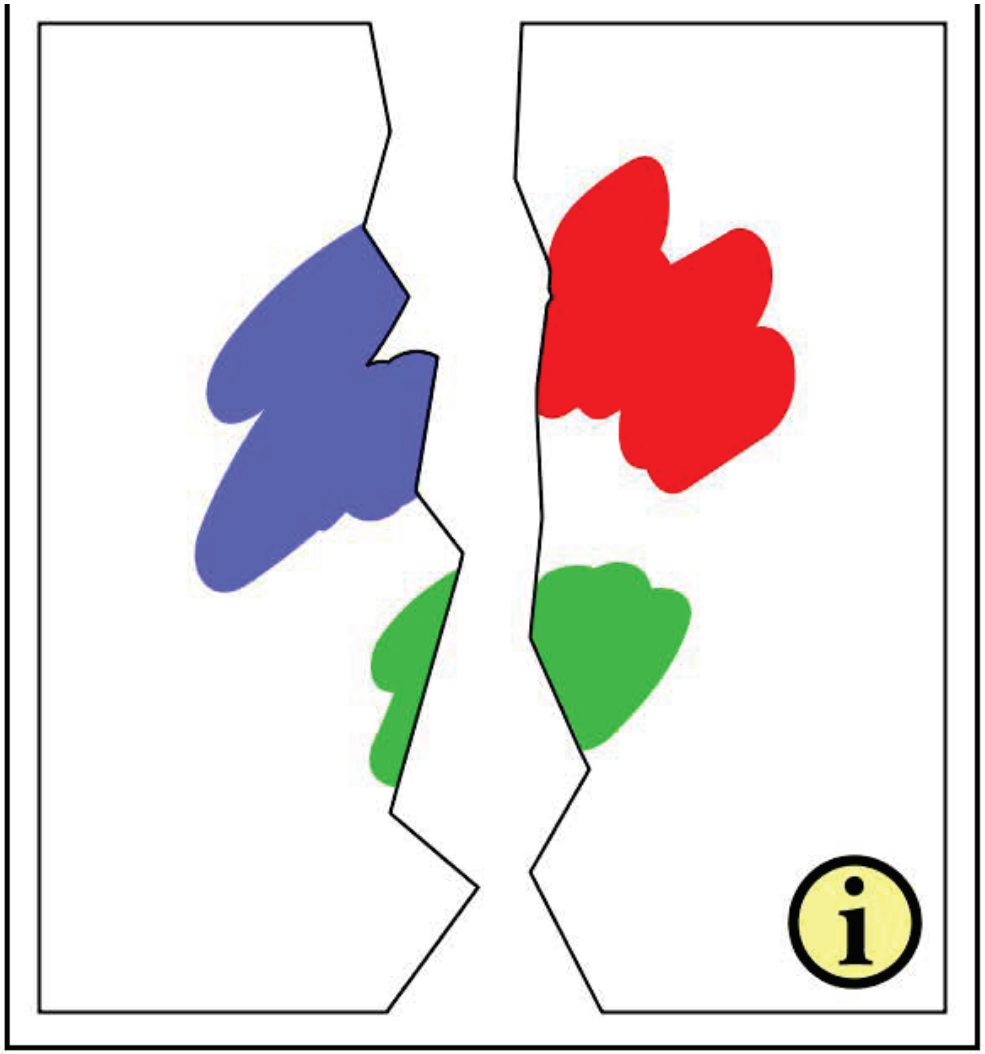
ICN-412-A-980000-B-97499-00969-A-001-01

Figure 1. Digital Automatic Flight Control Systems (Sheet 33 of 37)



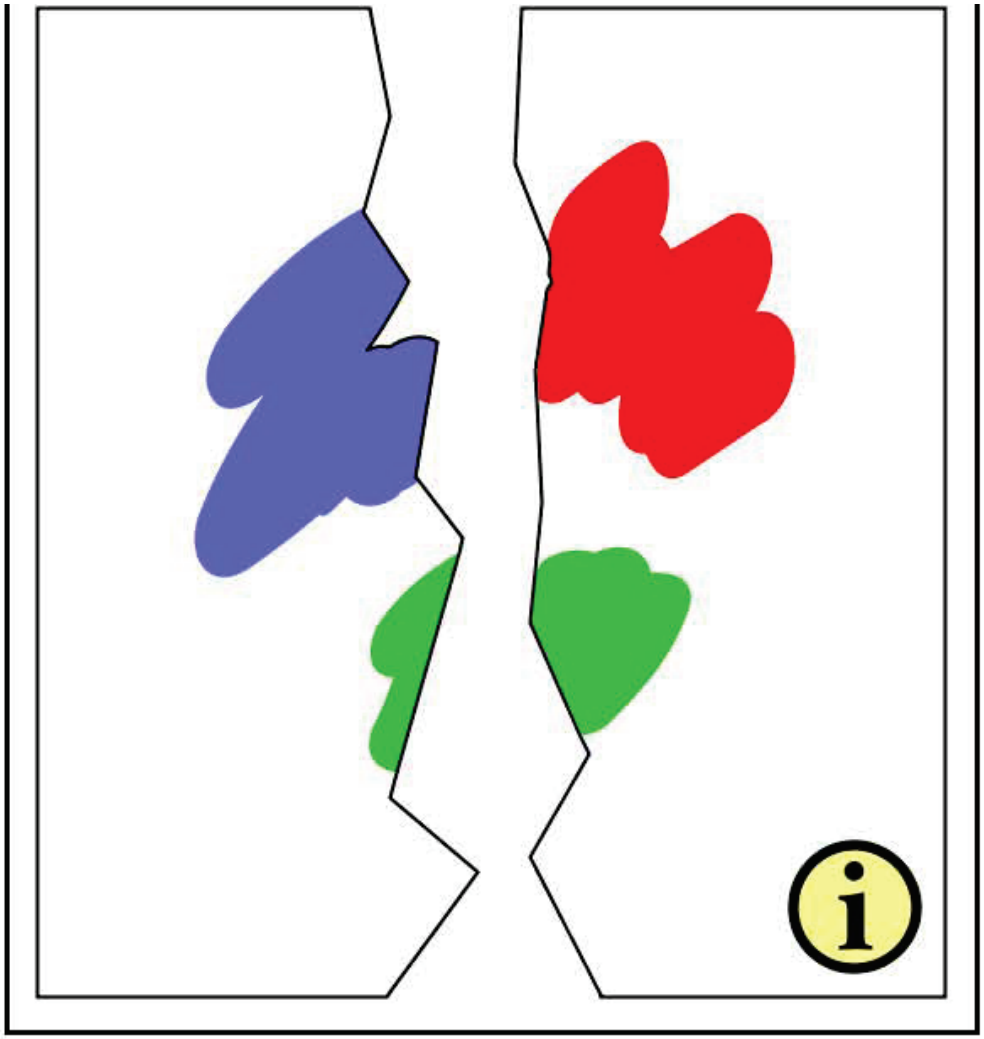
ICN-412-A-980000-B-97499-00248-A-001-01

Figure 1. Digital Automatic Flight Control Systems (Sheet 34 of 37)



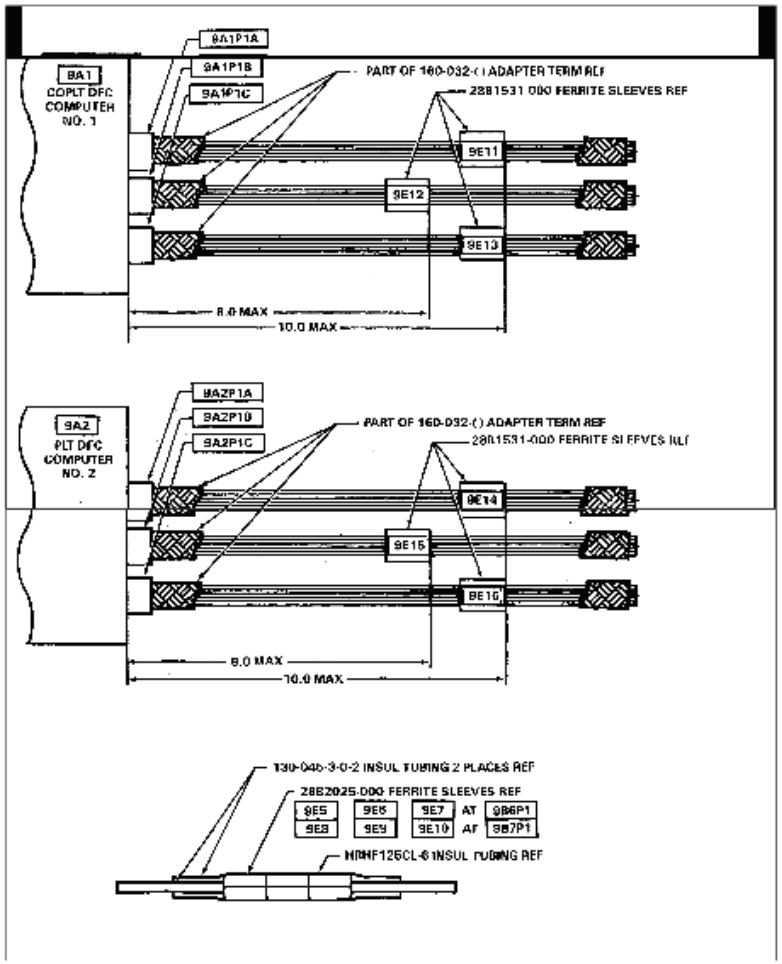
ICN-412-A-980000-B-97499-00249-A-001-01

Figure 1. Digital Automatic Flight Control Systems (Sheet 35 of 37)



ICN-412-A-980000-B-97499-00250-A-001-01

Figure 1. Digital Automatic Flight Control Systems (Sheet 36 of 37)



NOTES

1. APPLICABLE TO 3 AXIS FLIGHT DIRECTOR EQUIPPED HELICOPTERS (SEE VIEWS FD1 AND SUBSEQUENT).
2. APPLICABLE TO 4 AXIS FLIGHT DIRECTOR EQUIPPED HELICOPTERS (SEE 4 AXIS FLIGHT DIRECTOR KIT WIRING DIAGRAM).
3. APPLICABLE TO 4 AXIS FLIGHT DIRECTOR EQUIPPED HELICOPTERS WITH ELECTRONIC FLIGHT INSTRUMENTATION SYSTEM (EFIS) KIT INSTALLED (SEE EFIS KIT WIRING DIAGRAM).
4. APPLICABLE TO 4 AXIS FLIGHT DIRECTOR EQUIPPED HELICOPTERS WITH SEARCH AND RESCUE (SAR) INTERFACE KIT INSTALLED (SEE SAR INTERFACE KIT WIRING DIAGRAM).
5. SEE CAUTION AND LIGHTING WIRING DIAGRAM.
6. APPLICABLE TO HELICOPTERS WITH FLIGHT DATA RECORDER (FDR) KIT INSTALLED (SEE FLIGHT DATA RECORDER WIRING DIAGRAM (CHAPTER 97)).
7. SEE NAV NO. 1 WIRING DIAGRAM.
8. RESERVED.
9. SEE RADAR ALTIMETER 2 WIRI NG DIAGRAM (CHAPTER 97).
10. SEE FLIGHT CONTROLS AND HYDRAULICS WIRING DIAGRAM.
11. SEE ENGINE POWER PLANT AND TRANSMISSION WIRING DIAGRAM.
12. SEE WEATHER RADAR WIRING DIAGRAM (CHAPTER 97).
13. APPLICABLE TO HELICOPTERS MODIFIED WITH CARGO RELEASE KIT.
14. SEE DME NAVIGATION SYSTEM WIRING DIAGRAM (CHAPTER 97).
15. SEE AHRS 2 WIRING DIAGRAM.
16. SEE AHRS 1 WIRING DIAGRAM.
17. SEE COMPONENT REPLACEMENT.
18. APPLICABLE TO 4 AXIS FLIGHT DIRECTOR EQUIPPED HELICOPTERS ONLY S/N 36087 THRU 36171 (PRE TB 412-97- 146) AND ALL HELICOPTERS S/N 36172 AND SUBSEQUENT (POST TB 412-97-146).
18. WHEN FLIGHT DIRECTOR KIT (SEE BHT- 412-SI-41) IS INSTALLED WIRES 1C672E22, 1C672EE22 AND 2C672F22 MAY BE EITHER REMOVED ENTIRELY OR CAPPED AND STOWED.

EFFECTIVITY

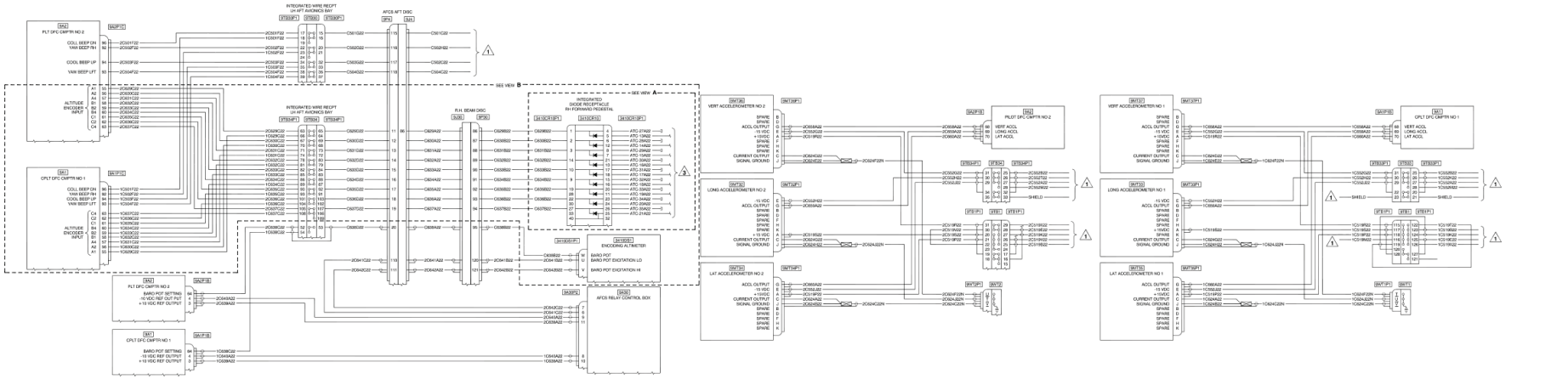
1. S/N 36101 AND SUBSEQUENT.
2. S/N 36172 AND SUBSEQUENT.
3. S/N 36248 AND SUBSEQUENT.
4. S/N 36101 TH ROUGH 36171.
5. S/N 36215 AND SUBSEQUENT (POST TB 412-07-210).

412-475-017F

ICN-412-A-980000-B-97499-00251-A-001-01

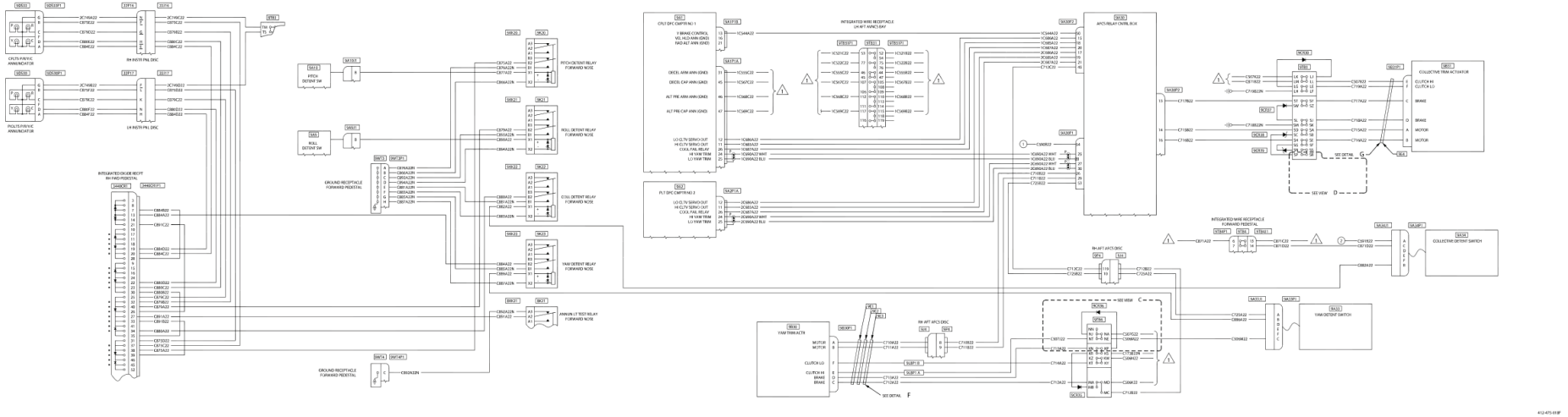
Figure 1. Digital Automatic Flight Control Systems (Sheet 37 of 37)

Intentionally blank page



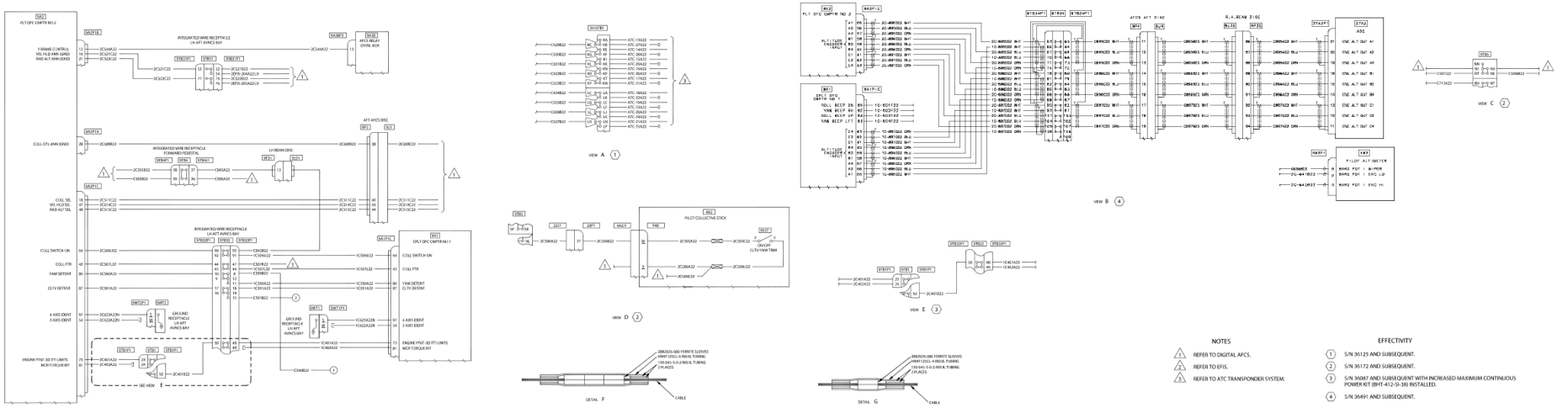
ICN-412-A-980000-B-97499-00252-A-001-01

Figure 1. 4-Axis Flight Director Kit (Sheet 1 of 3)



ICN-412-A-980000-B-97499-00253-A-001-01

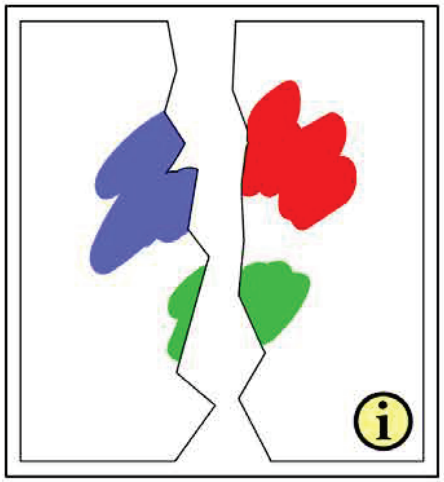
Figure 1. 4-Axis Flight Director Kit (Sheet 2 of 3)



ICN-412-A-980000-B-97499-00254-A-001-01

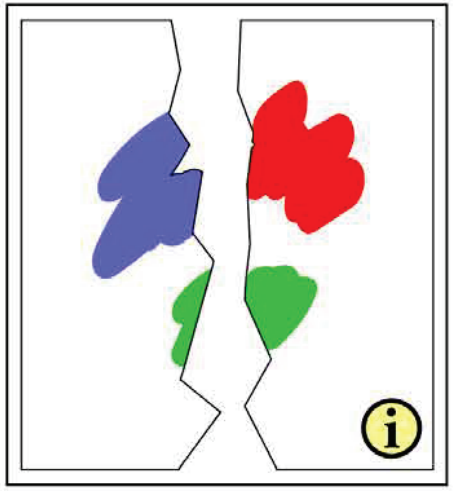
Figure 1. 4-Axis Flight Director Kit (Sheet 3 of 3)

Intentionally blank page



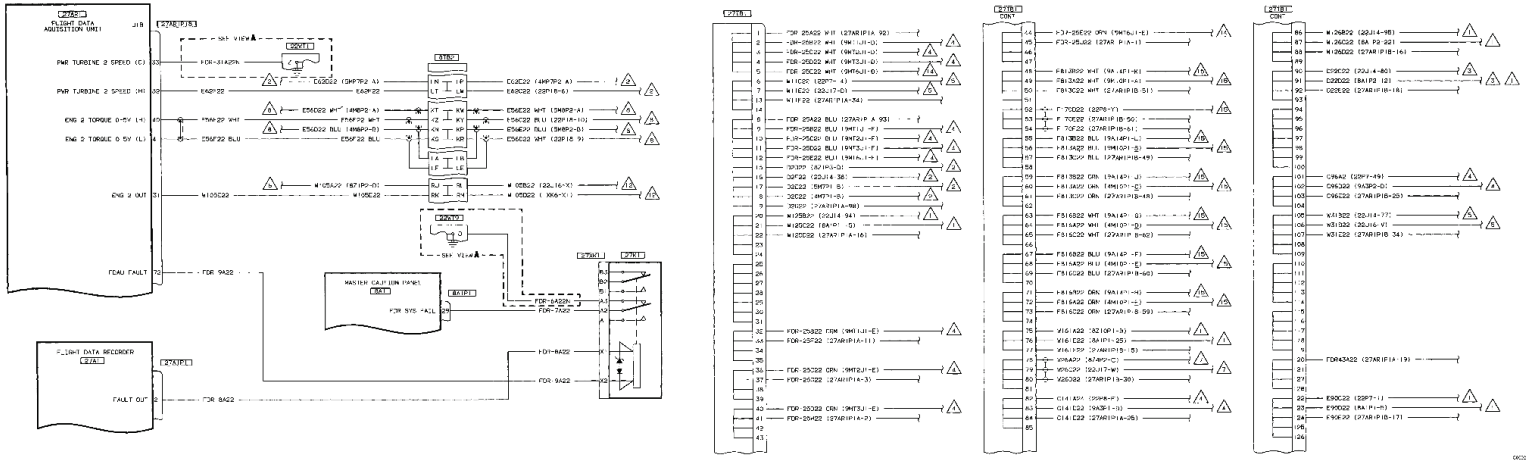
ICN-412-A-980000-B-97499-00255-A-001-01

Figure 1. Flight Data Recorder (Sheet 1 of 4)



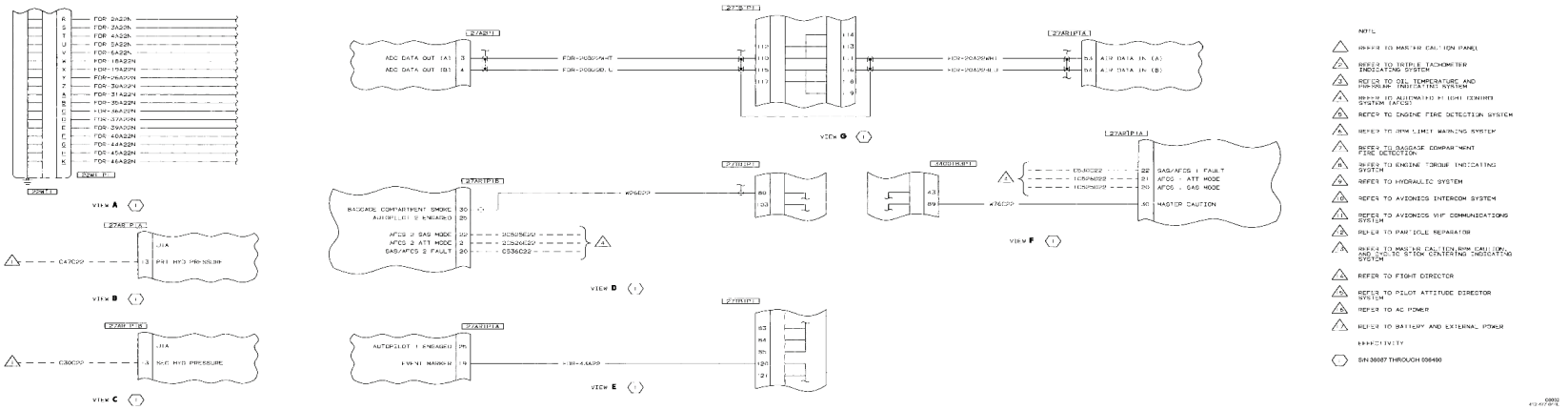
ICN-412-A-980000-B-97499-00256-A-001-01

Figure 1. Flight Data Recorder (Sheet 2 of 4)



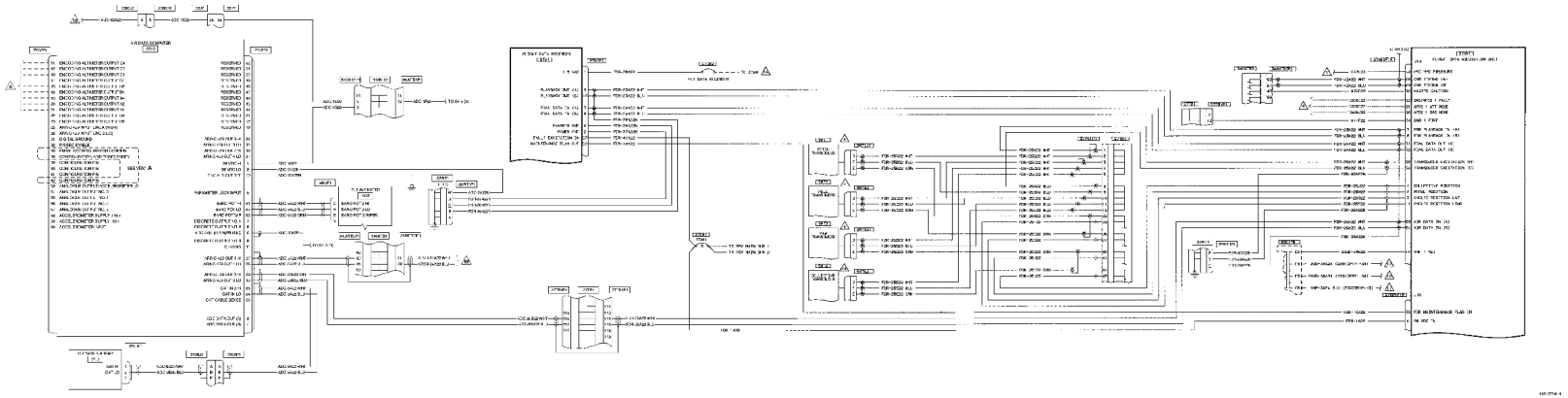
ICN-412-A-980000-B-97499-00257-A-001-01

Figure 1. Flight Data Recorder (Sheet 3 of 4)



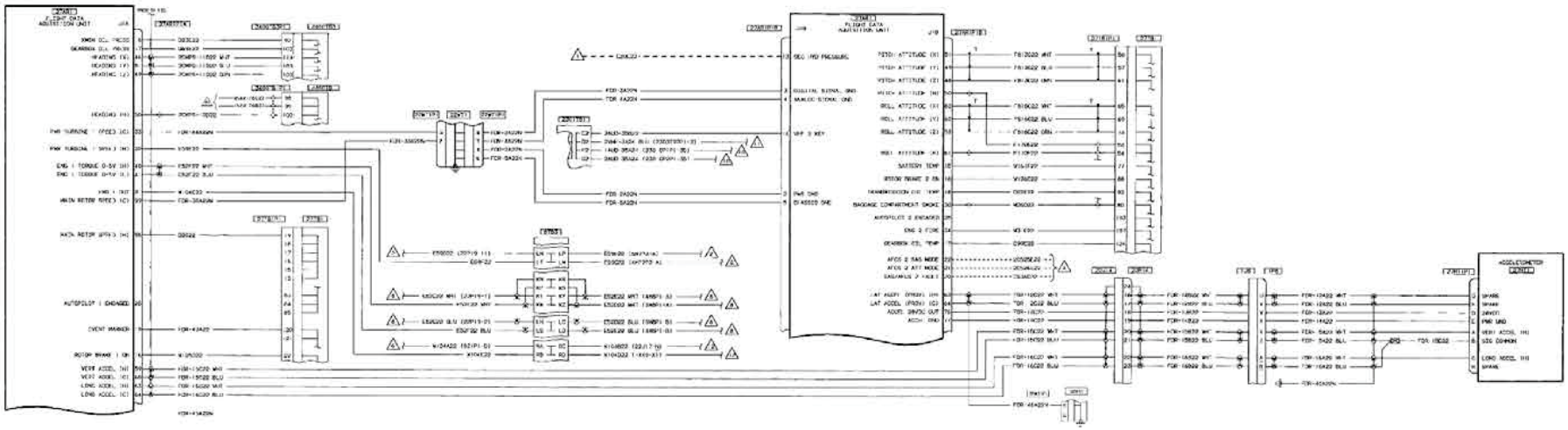
ICN-412-A-980000-B-97499-00258-A-001-01

Figure 1. Flight Data Recorder (Sheet 4 of 4)



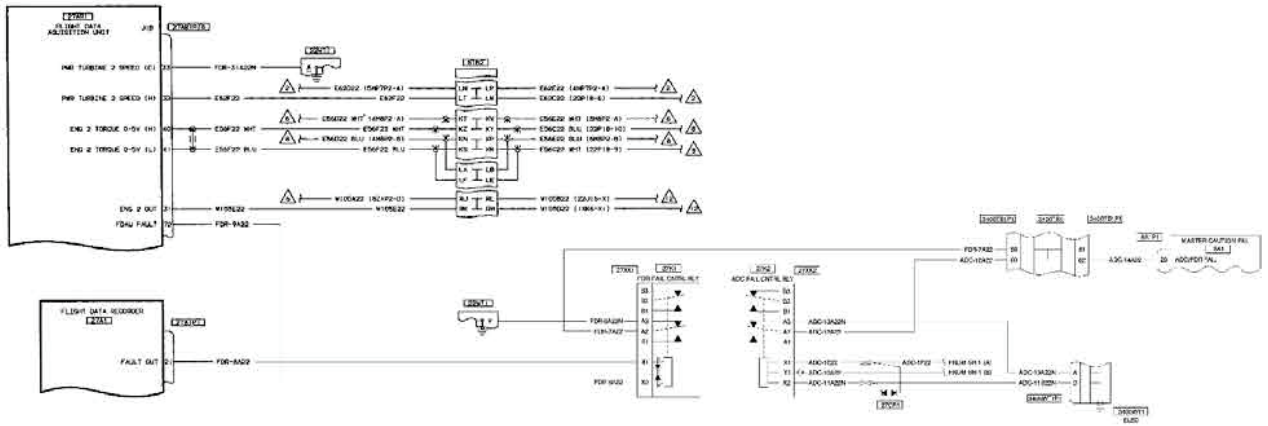
ICN-412-A-980000-B-97499-00259-A-001-01

Figure 1. Flight Data Recorder (Sheet 1 of 4)



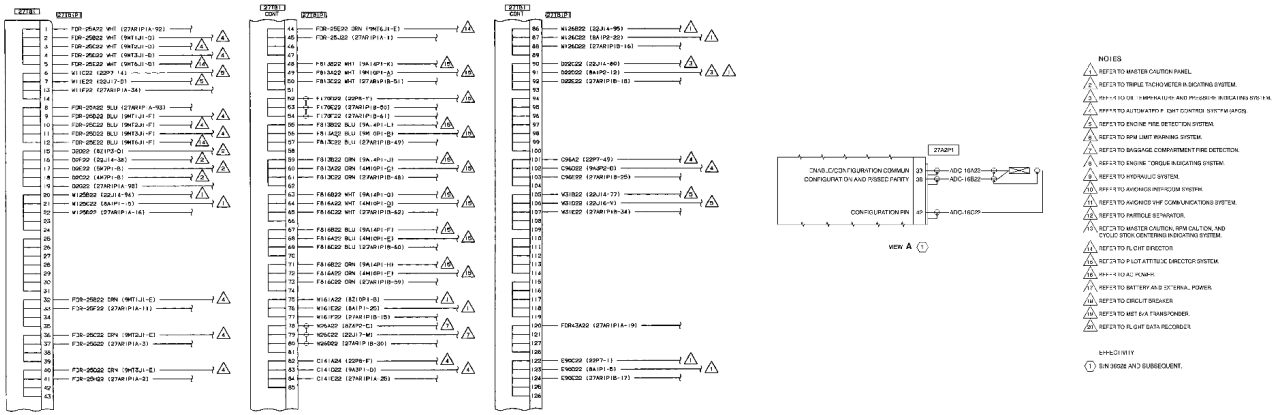
ICN-412-A-980000-B-97499-00260-A-001-01

Figure 1. Flight Data Recorder (Sheet 2 of 4)



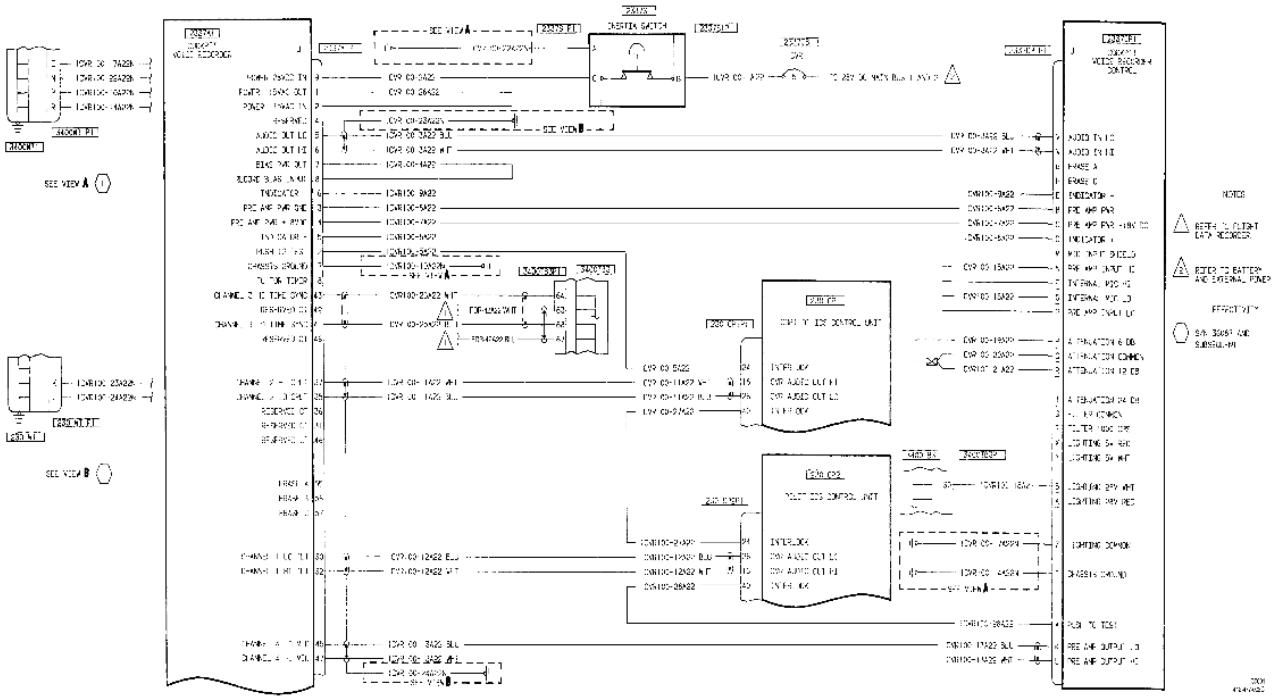
ICN-412-A-980000-B-97499-00261-A-001-01

Figure 1. Flight Data Recorder (Sheet 3 of 4)



ICN-412-A-980000-B-97499-00262-A-001-01

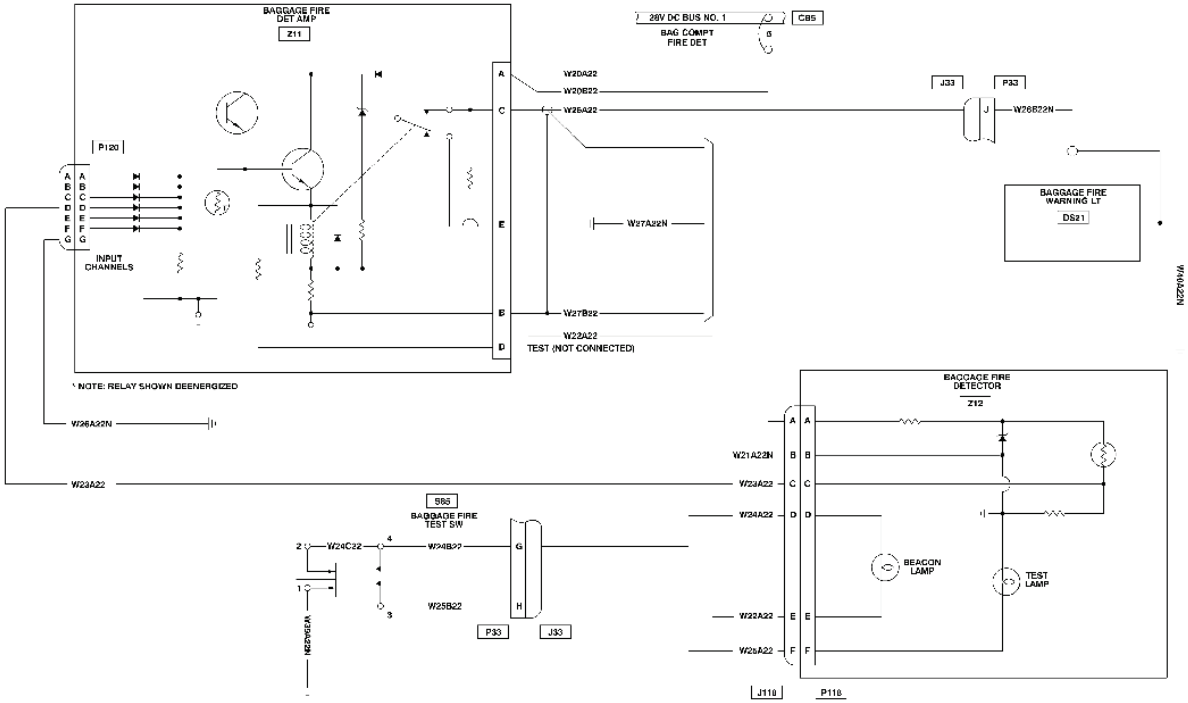
Figure 1. Flight Data Recorder (Sheet 4 of 4)



ICN-412-A-980000-B-97499-00263-A-001-01

Figure 1. Cockpit Voice Recorder (Sheet 1 of 1)

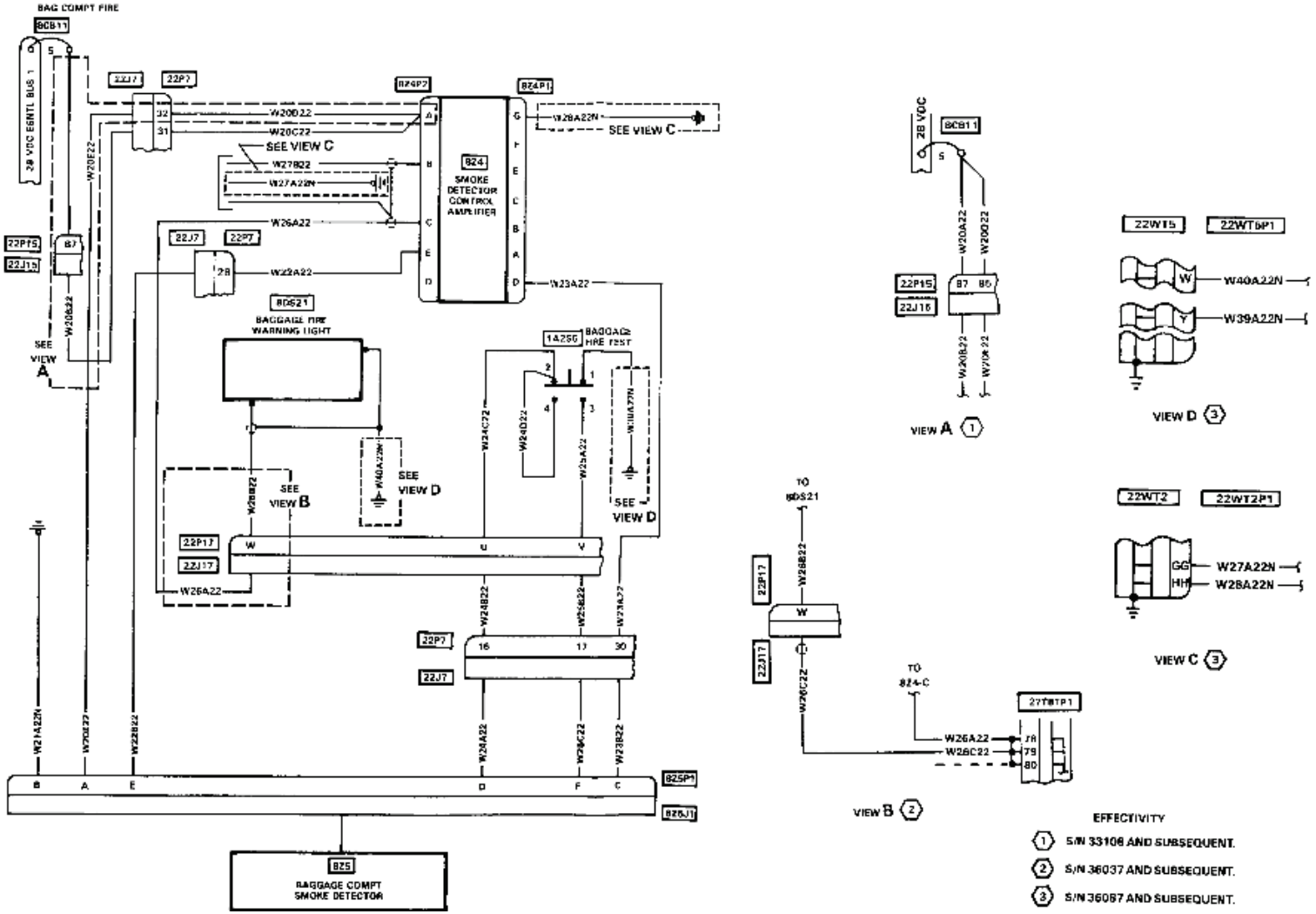
Intentionally blank page



ICN-412-A-980000-B-97499-00264-A-001-01

Figure 1. Baggage Compartment Fire Detection (Sheet 1 of 1)

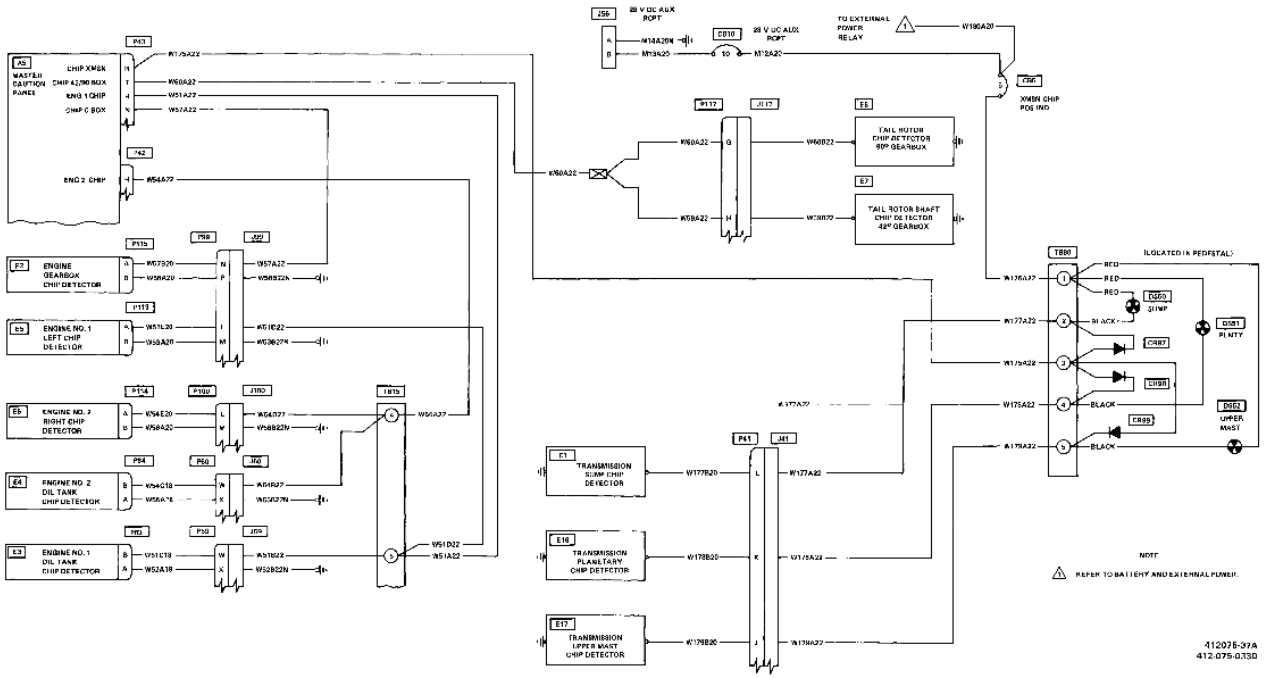
Intentionally blank page



ICN-412-A-980000-B-97499-00265-A-001-01

Figure 1. Baggage Compartment Fire Detection (Sheet 1 of 1)

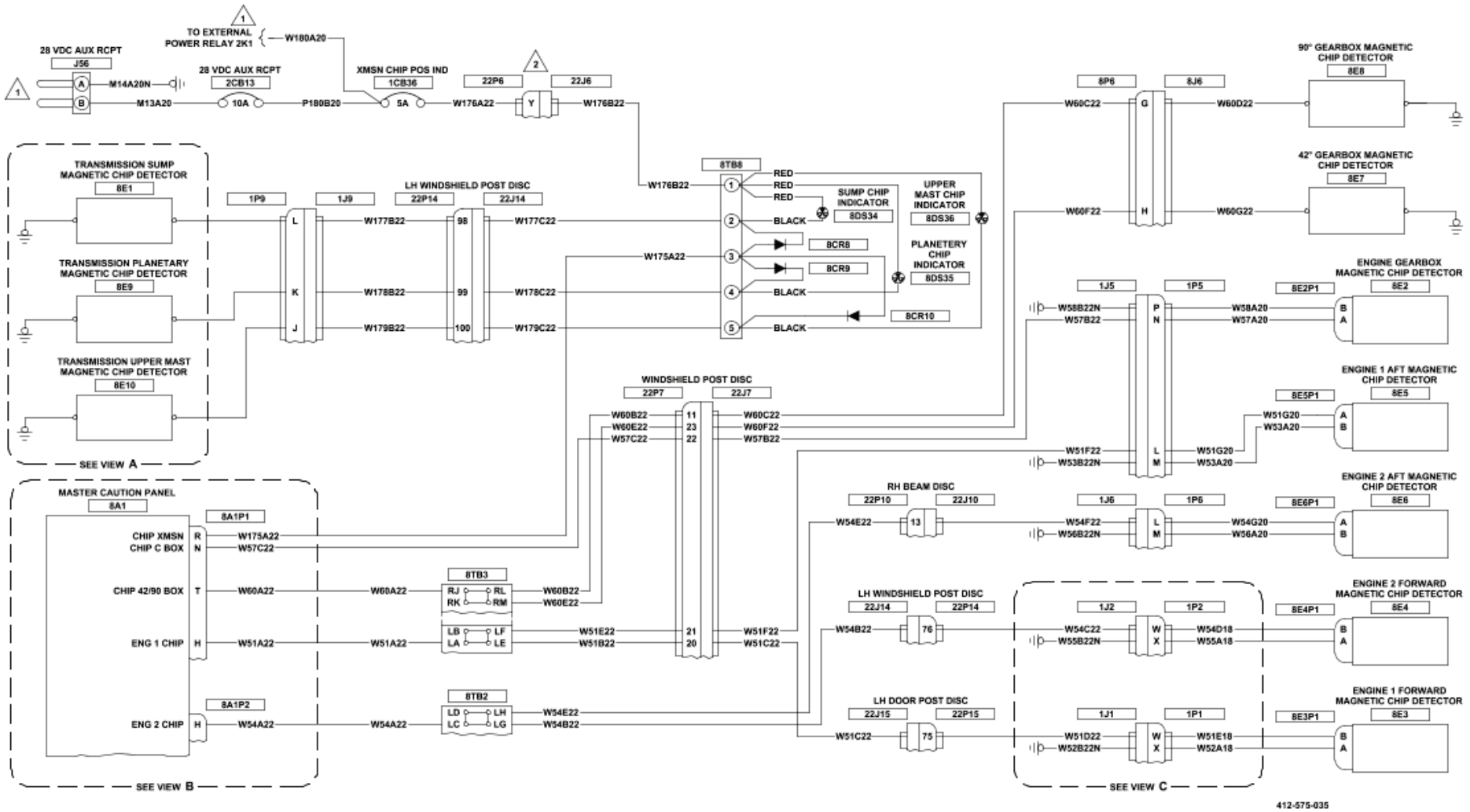
Intentionally blank page



ICN-412-A-980000-B-97499-00266-A-001-01

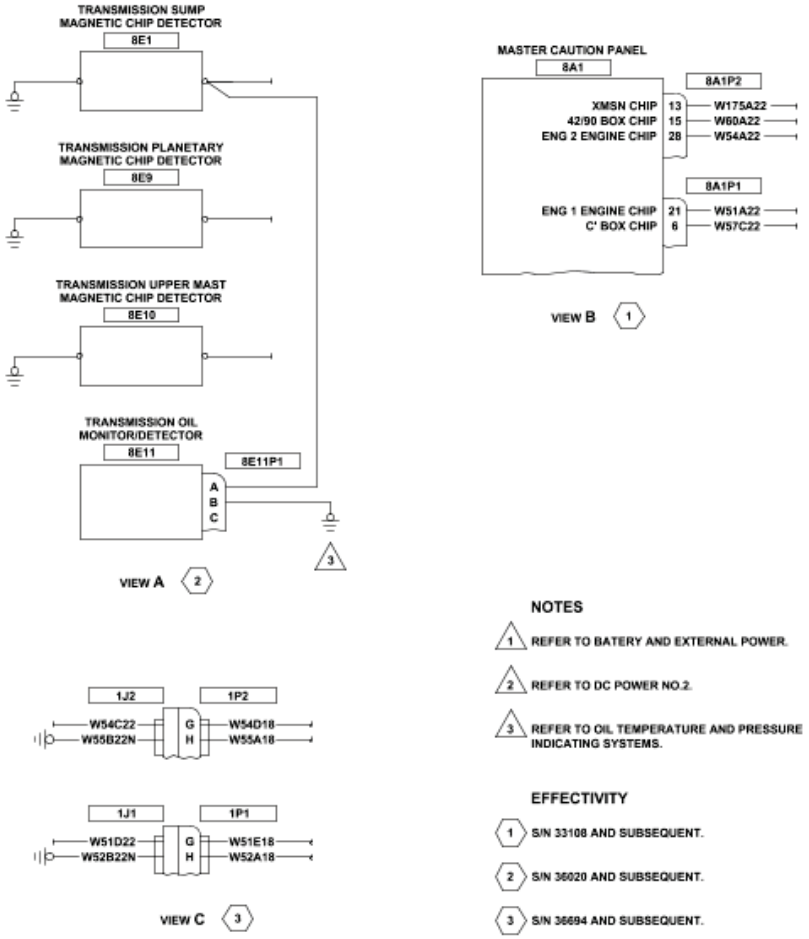
Figure 1. Chip Detectors (Sheet 1 of 1)

Intentionally blank page



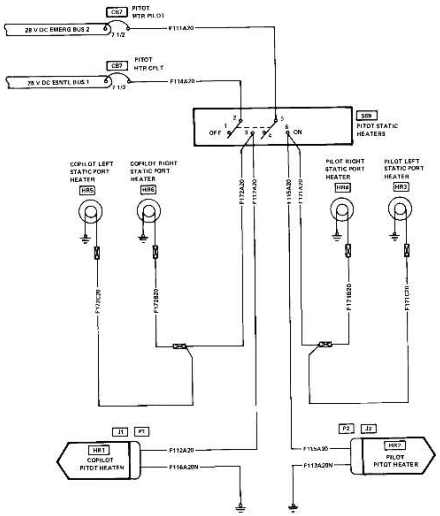
ICN-412-A-980000-B-97499-00267-A-001-01

Figure 1. Chip Detectors (Sheet 1 of 2)



ICN-412-A-980000-B-97499-00268-A-001-01

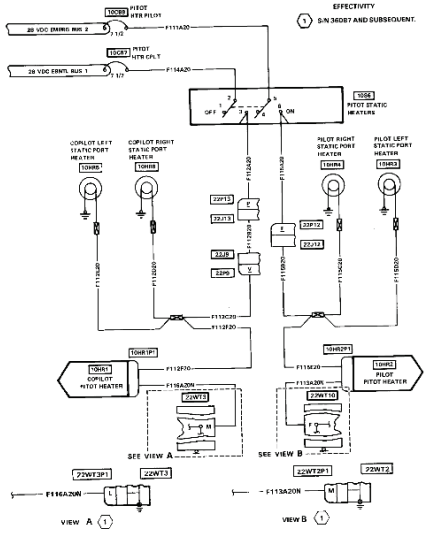
Figure 1. Chip Detectors (Sheet 2 of 2)



ICN-412-A-980000-B-97499-00269-A-001-01

Figure 1. Pitot-Static Heaters (Sheet 1 of 1)

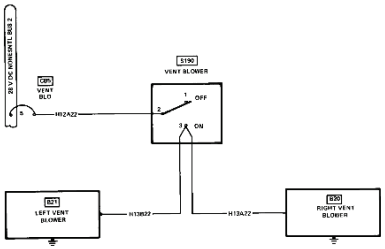
Intentionally blank page



ICN-412-A-980000-B-97499-00270-A-001-01

Figure 1. Pitot-Static Heaters (Sheet 1 of 1)

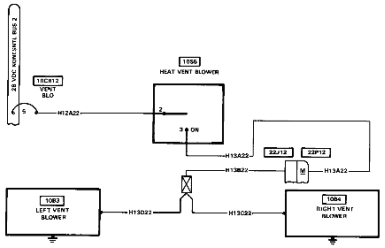
Intentionally blank page



ICN-412-A-980000-B-97499-00271-A-001-01

Figure 1. Vent Blowers (Sheet 1 of 1)

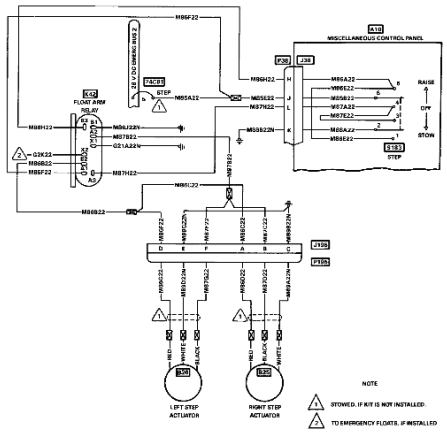
Intentionally blank page



ICN-412-A-980000-B-97499-00272-A-001-01

Figure 1. Pitot-Static Heaters (Sheet 1 of 1)

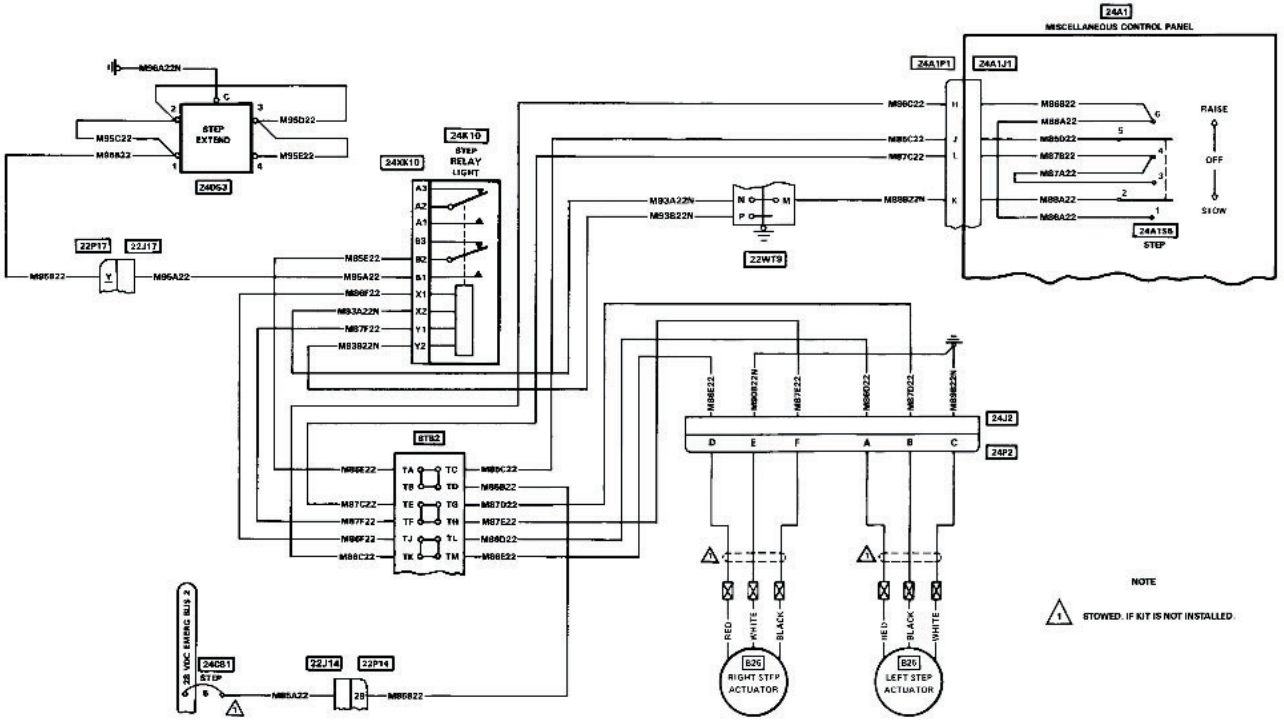
Intentionally blank page



ICN-412-A-980000-B-97499-00273-A-001-01

Figure 1. Step Actuators (Sheet 1 of 1)

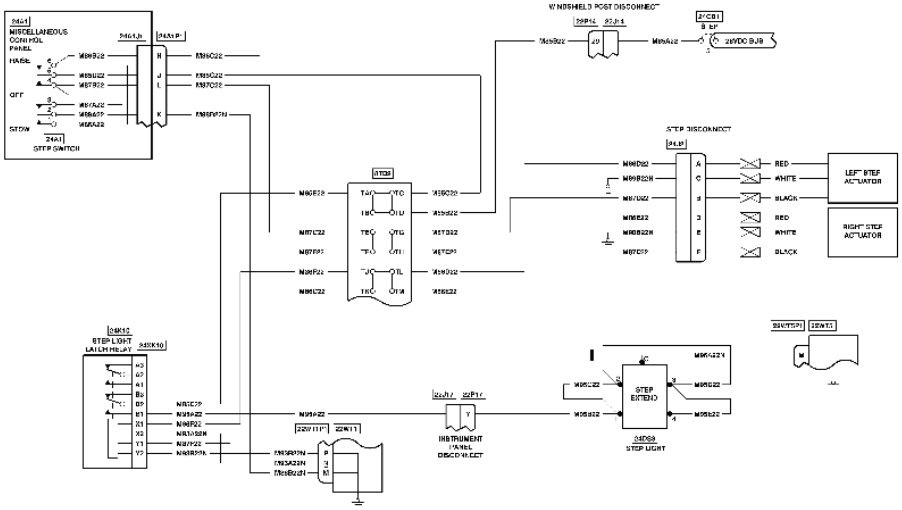
Intentionally blank page



ICN-412-A-980000-B-97499-00274-A-001-01

Figure 1. Step Actuators (Sheet 1 of 1)

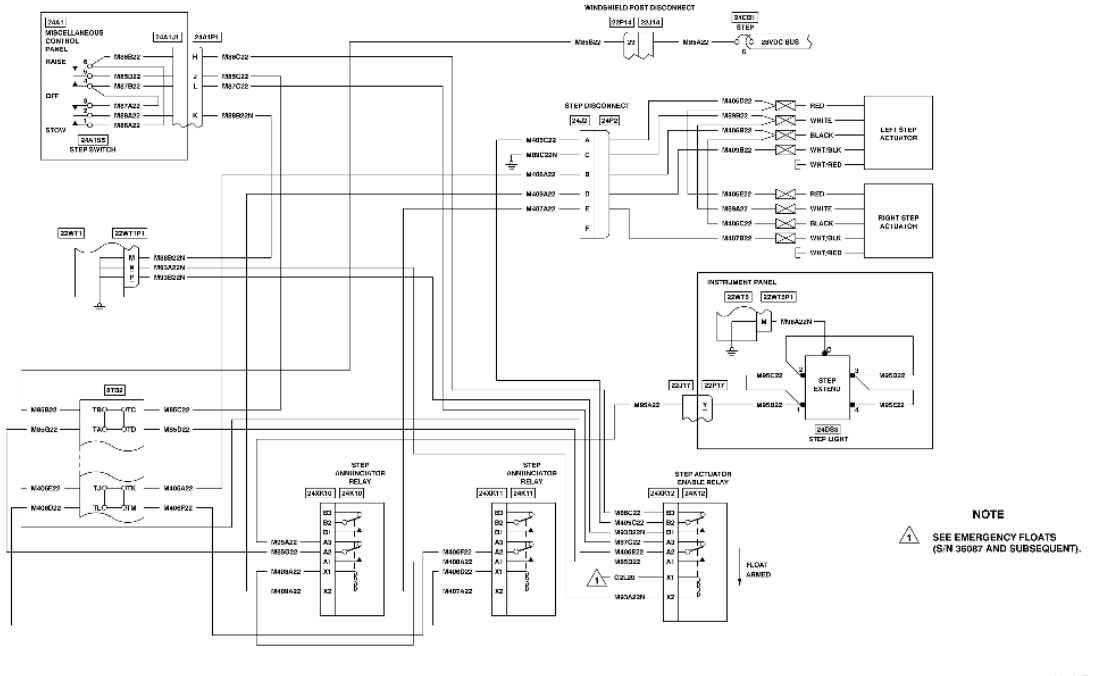
Intentionally blank page



ICN-412-A-980000-B-97499-00275-A-001-01

Figure 1. Step Actuators (Sheet 1 of 1)

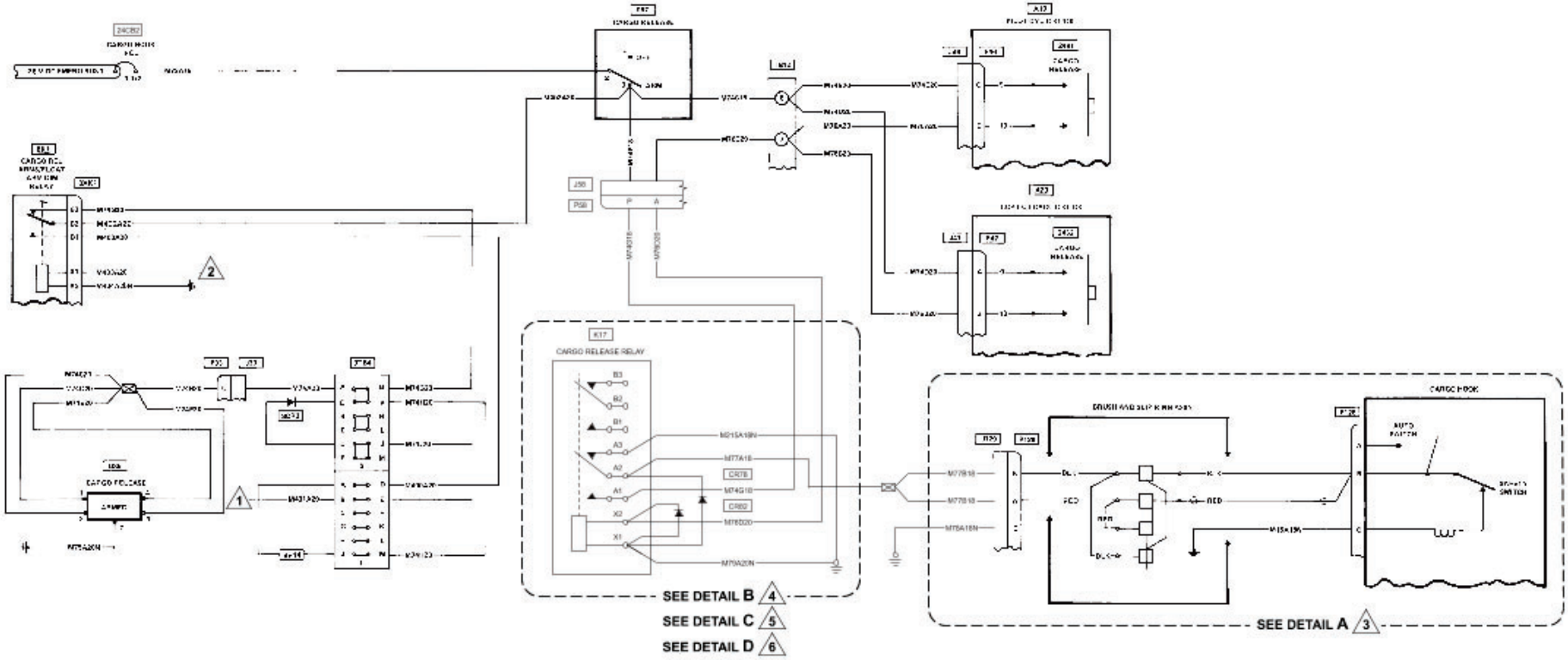
Intentionally blank page



ICN-412-A-980000-B-97499-00276-A-001-01

Figure 1. Step Actuators (Sheet 1 of 1)

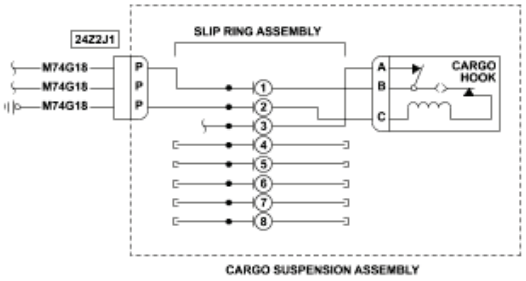
Intentionally blank page



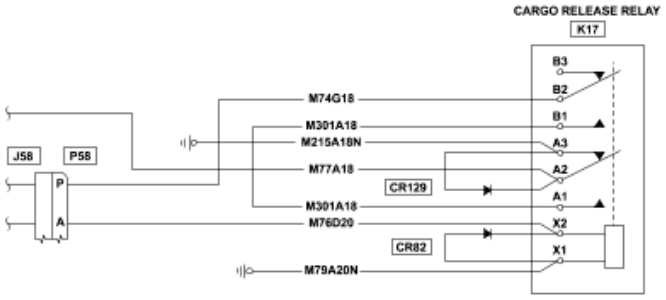
412-704-056A 412-075-036B

ICN-412-A-980000-B-97499-00277-A-001-01

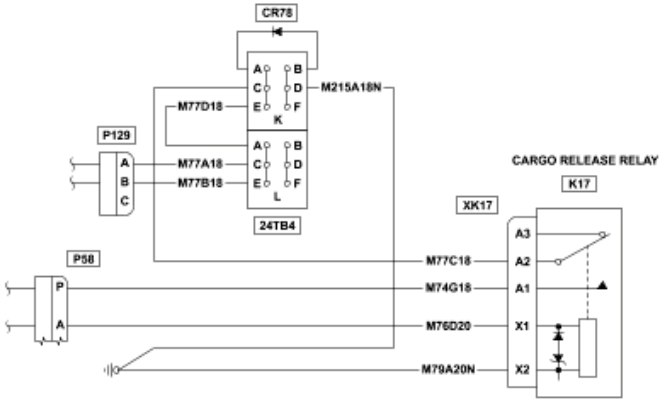
Figure 1. Cargo Hook Release (Sheet 1 of 2)



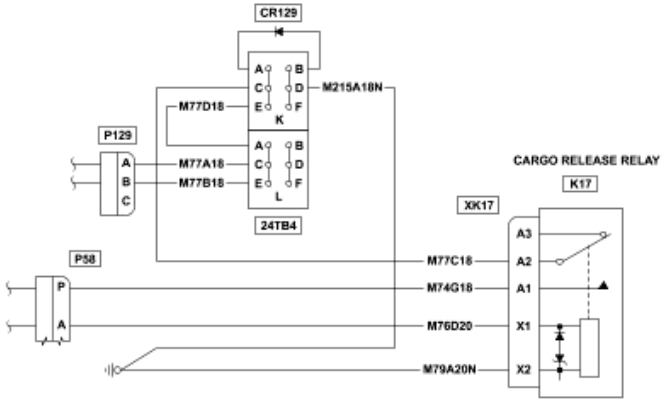
DETAIL A



DETAIL B



DETAIL C



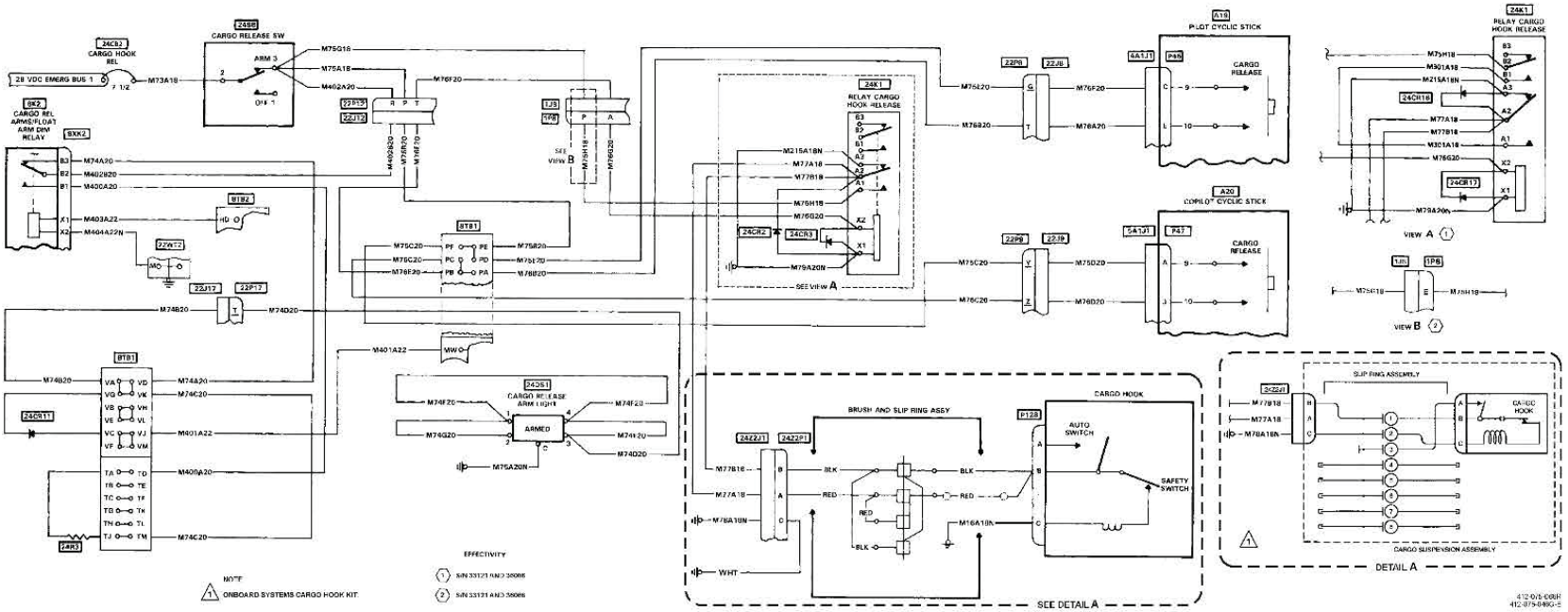
DETAIL D

NOTES

1. Refer to master caution panel.
2. Refer to engine instrument lights.
3. Onboard systems cargo hook kit.
4. Post TB 412-86-53.
5. Pre TB 412-86-53 with retrofit 412-704-056.
6. Post TB 412-86-53 with retrofit 412-704-056.

ICN-412-A-980000-B-97499-00278-A-001-01

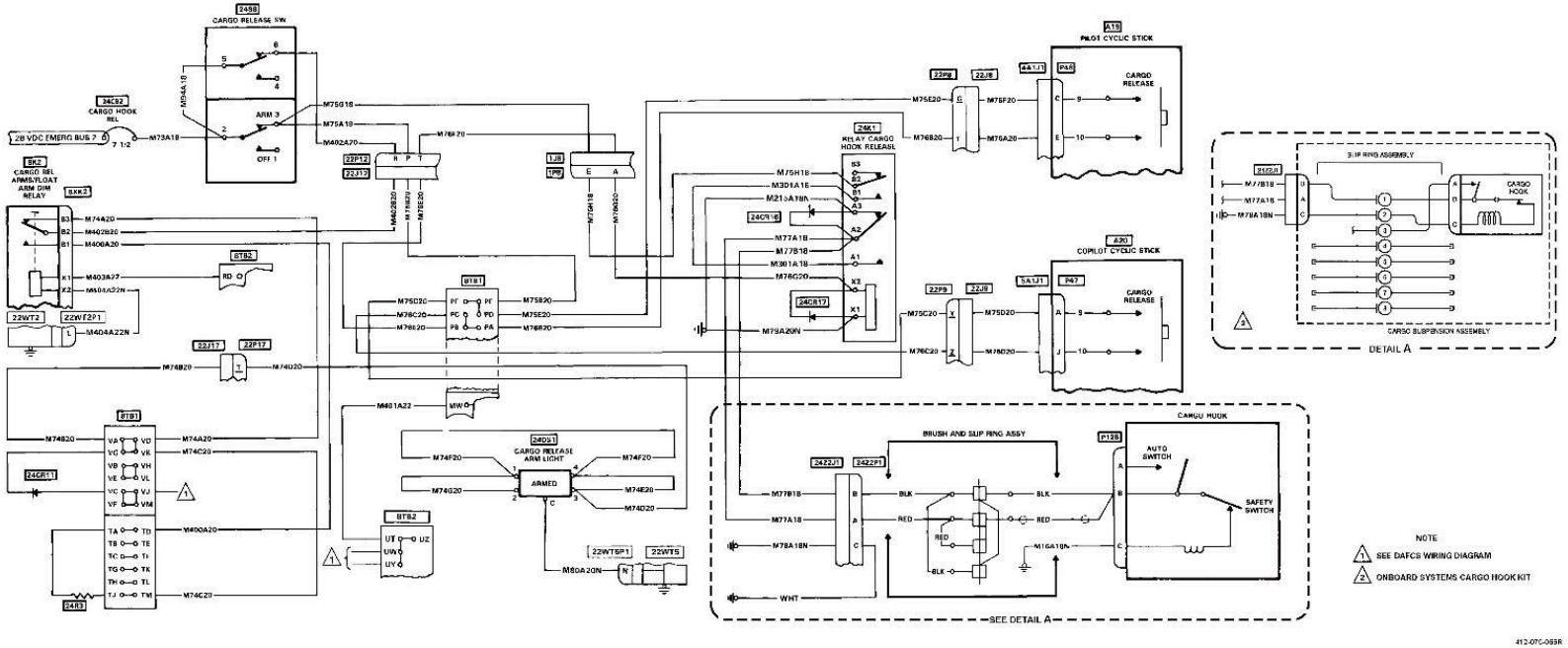
Figure 1. Cargo Hook Release (Sheet 2 of 2)



ICN-412-A-980000-B-97499-00279-A-001-01

Figure 1. Cargo Hook Release (Sheet 1 of 1)

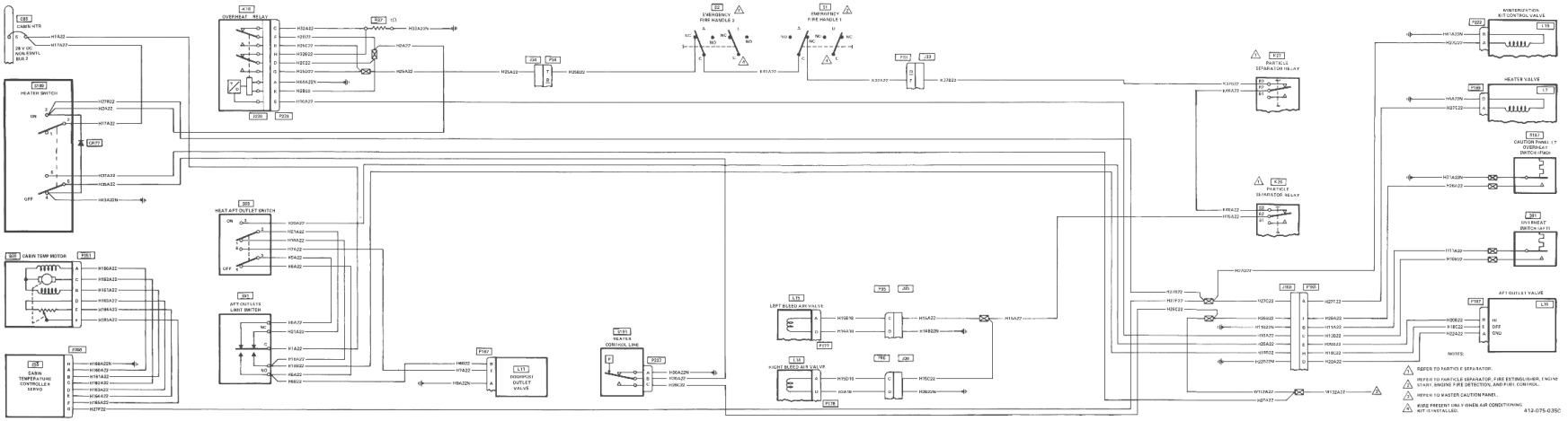
Intentionally blank page



ICN-412-A-980000-B-97499-00280-A-001-01

Figure 1. Cargo Hook Release (Sheet 1 of 1)

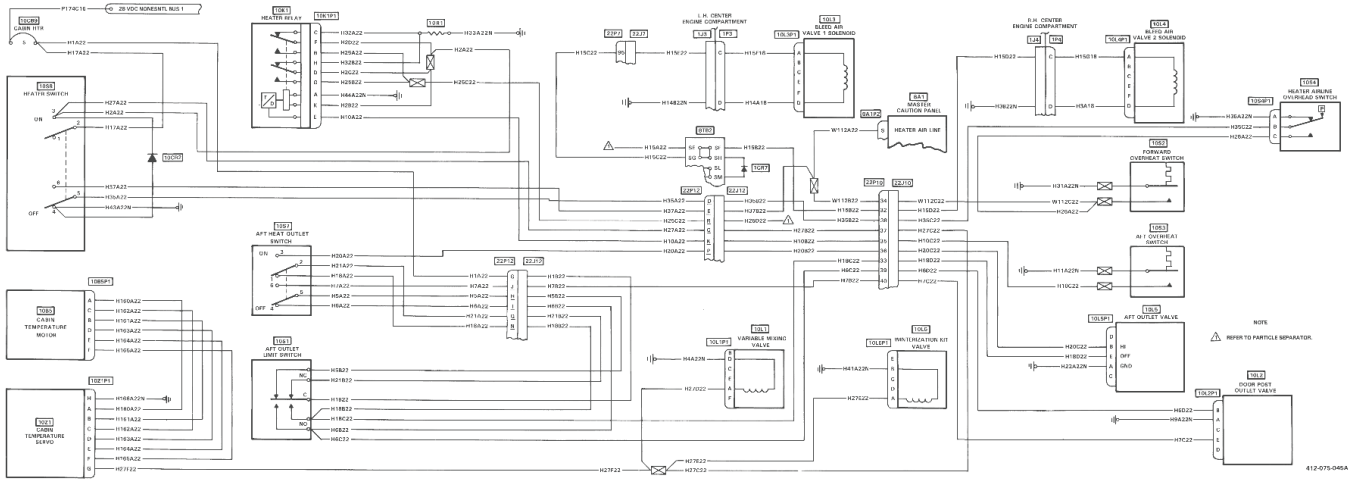
Intentionally blank page



ICN-412-A-980000-B-97499-00281-A-001-01

Figure 1. Cabin Heating (Sheet 1 of 1)

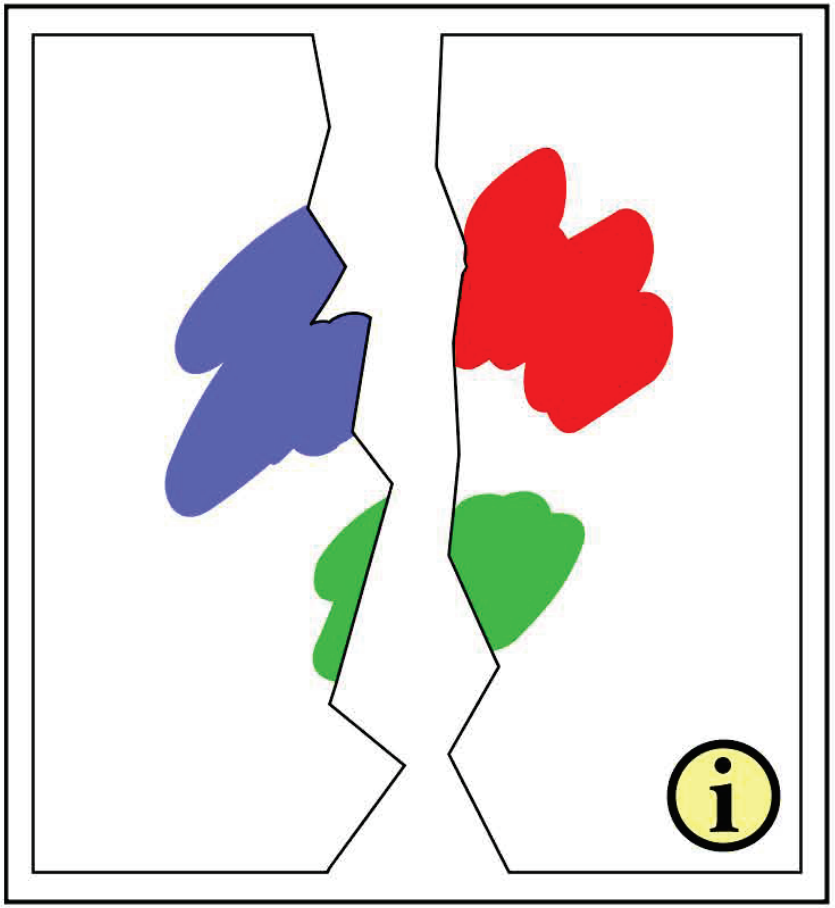
Intentionally blank page



ICN-412-A-980000-B-97499-00282-A-001-01

Figure 1. Cabin Heating (Sheet 1 of 1)

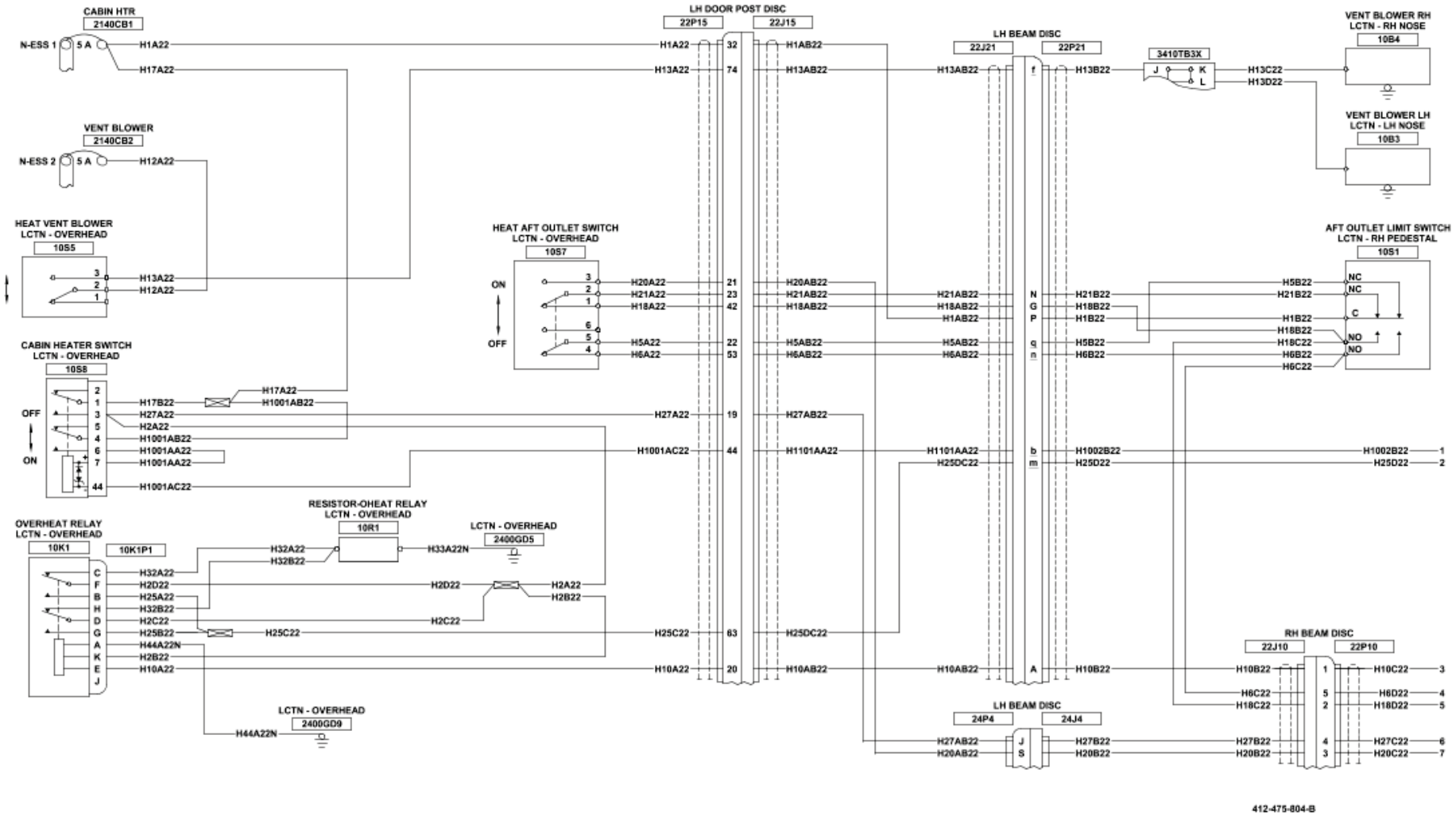
Intentionally blank page



ICN-412-A-980000-B-97499-00283-A-001-01

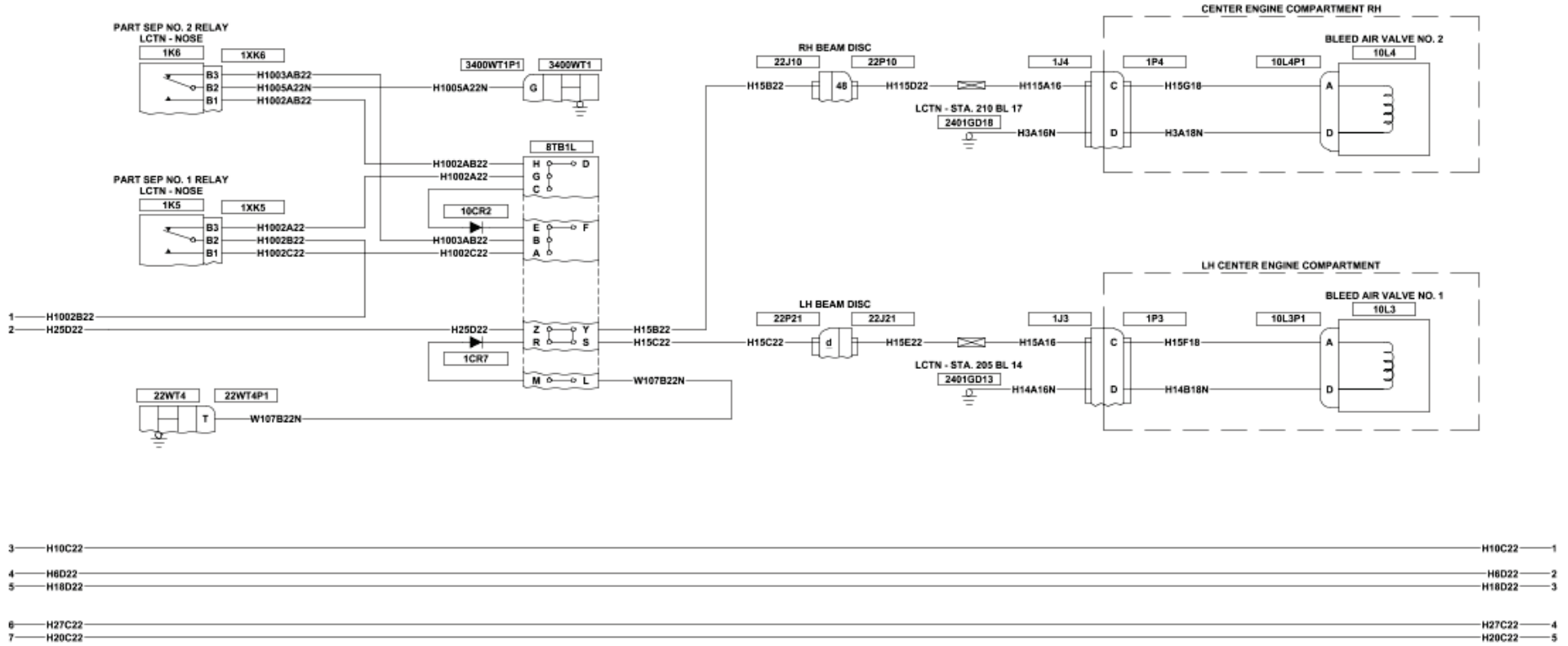
Figure 1. Cabin Heating (Sheet 1 of 1)

Intentionally blank page



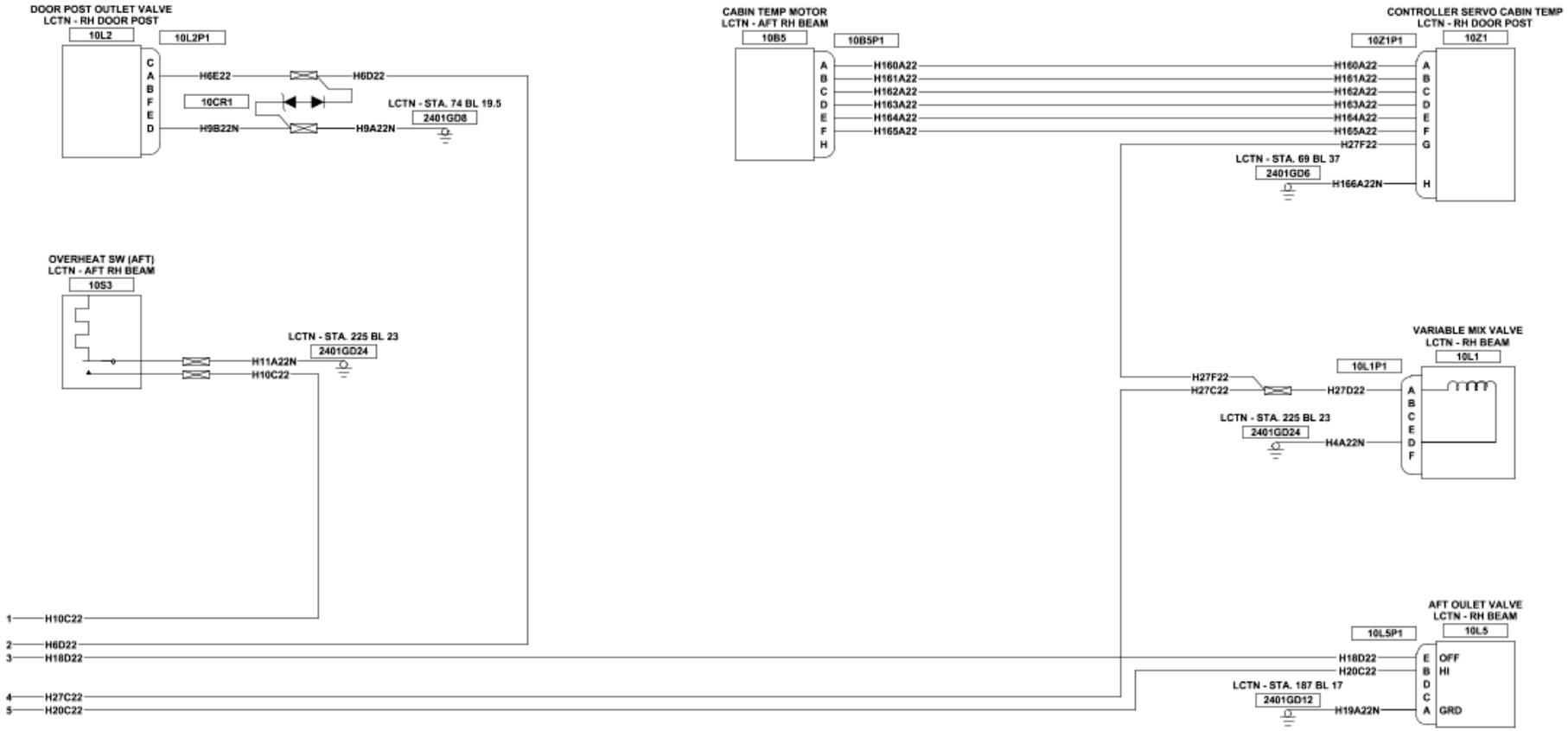
ICN-412-A-980000-B-97499-00284-A-001-01

Figure 1. Cabin Heating (Sheet 1 of 3)



ICN-412-A-980000-B-97499-00285-A-001-01

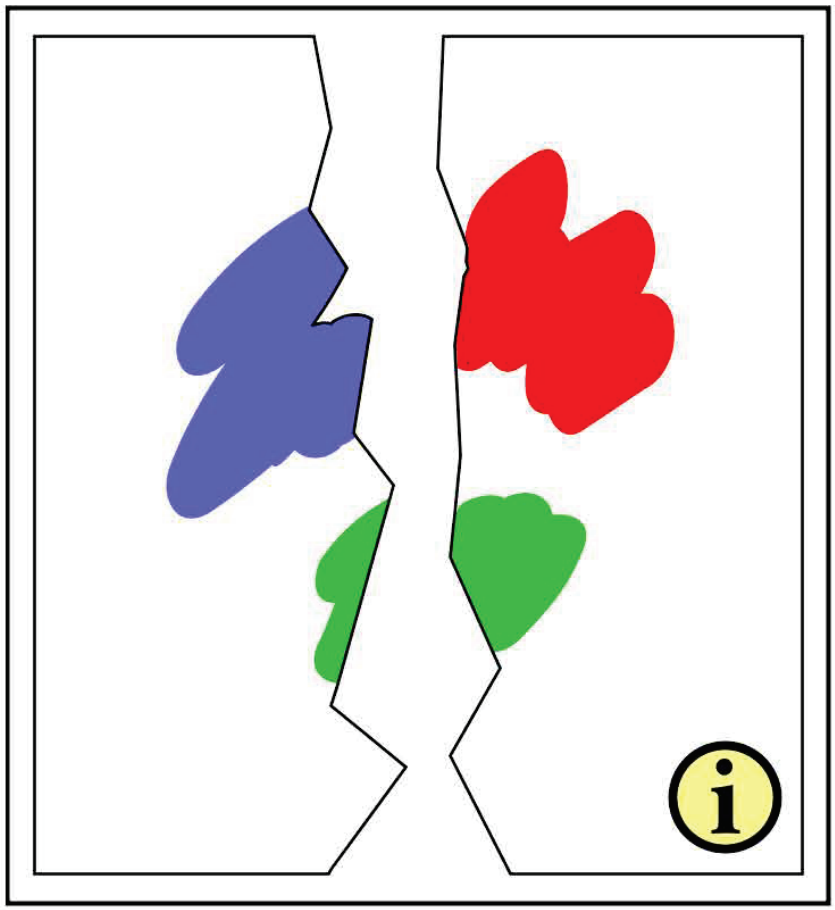
Figure 1. Cabin Heating (Sheet 2 of 3)



ICN-412-A-980000-B-97499-00286-A-001-01

Figure 1. Cabin Heating (Sheet 3 of 3)

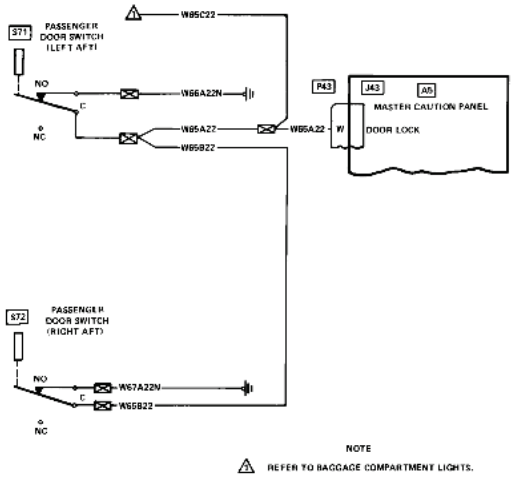
Intentionally blank page



ICN-412-A-980000-B-97499-00287-A-001-01

Figure 1. Windshield Heating (Sheet 1 of 1)

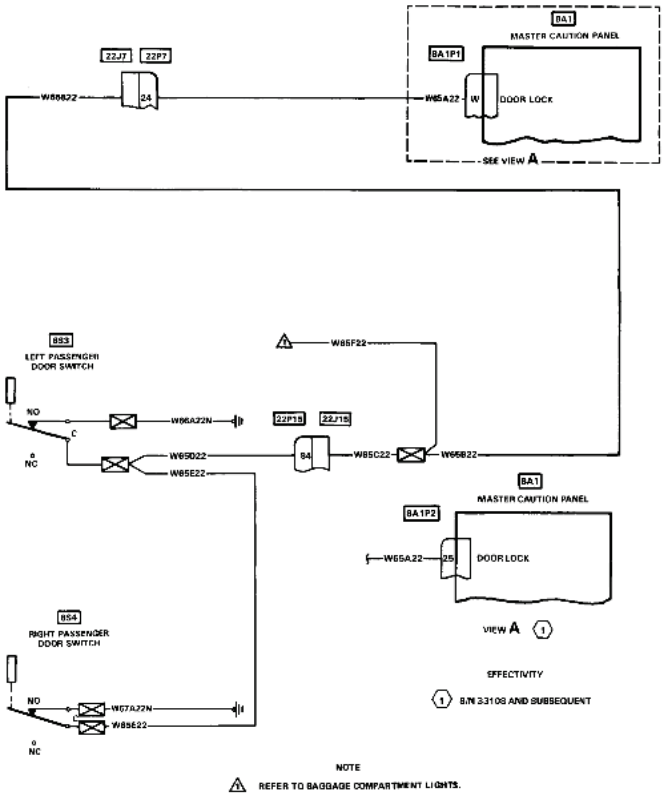
Intentionally blank page



ICN-412-A-980000-B-97499-00288-A-001-01

Figure 1. Door Switches (Sheet 1 of 1)

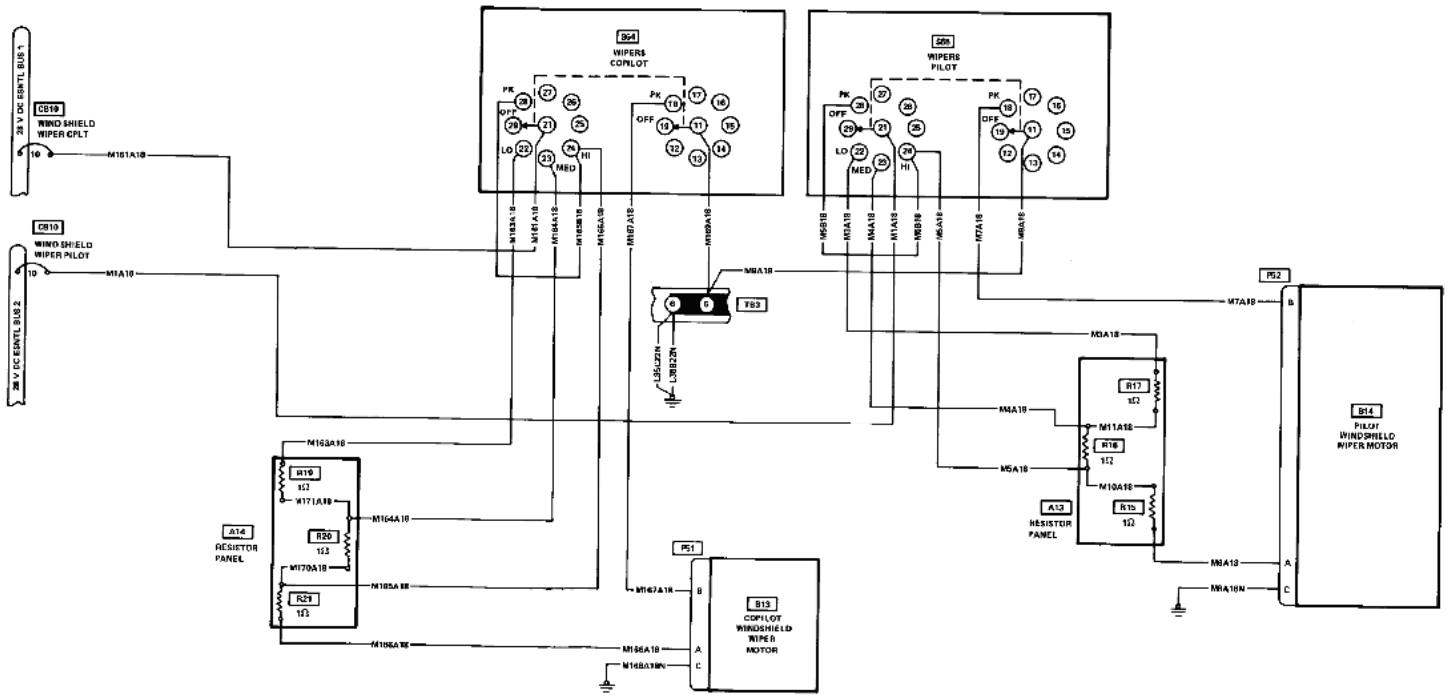
Intentionally blank page



ICN-412-A-980000-B-97499-00289-A-001-01

Figure 1. Door Switches (Sheet 1 of 1)

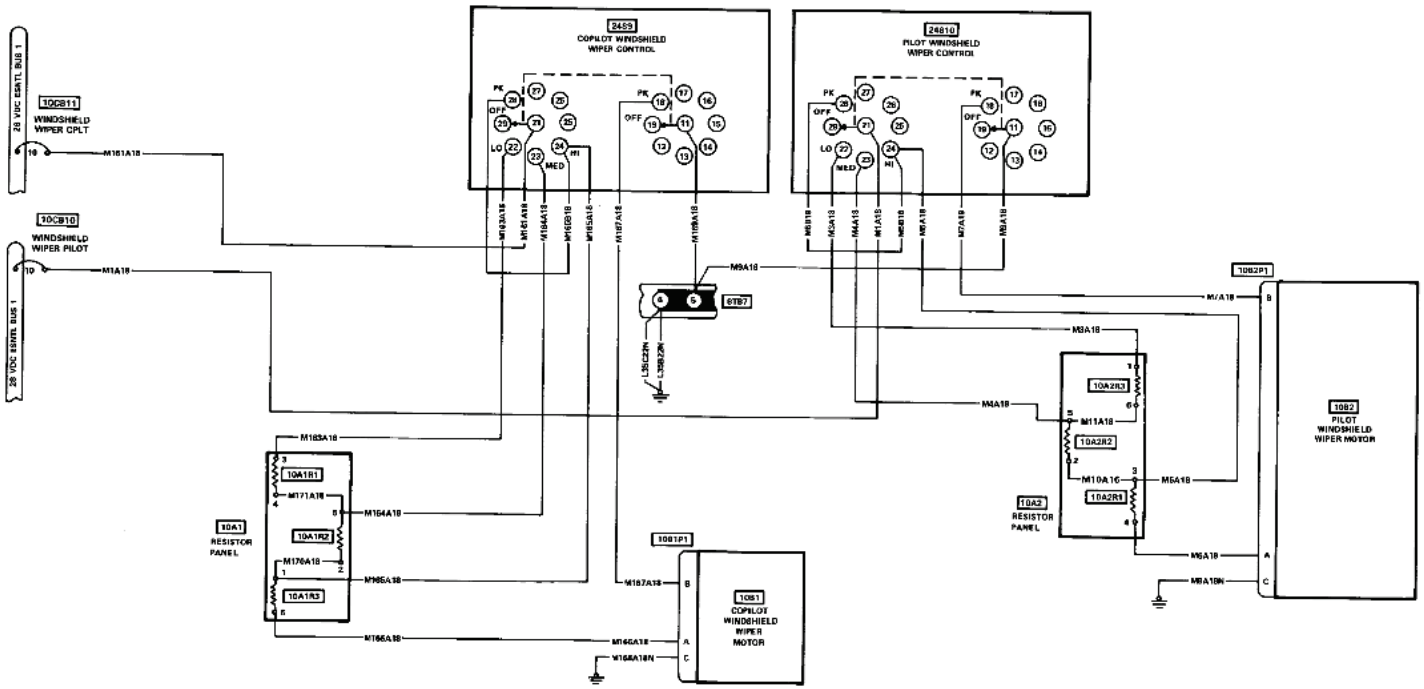
Intentionally blank page



ICN-412-A-980000-B-97499-00290-A-001-01

Figure 1. Windshield Wipers (Sheet 1 of 1)

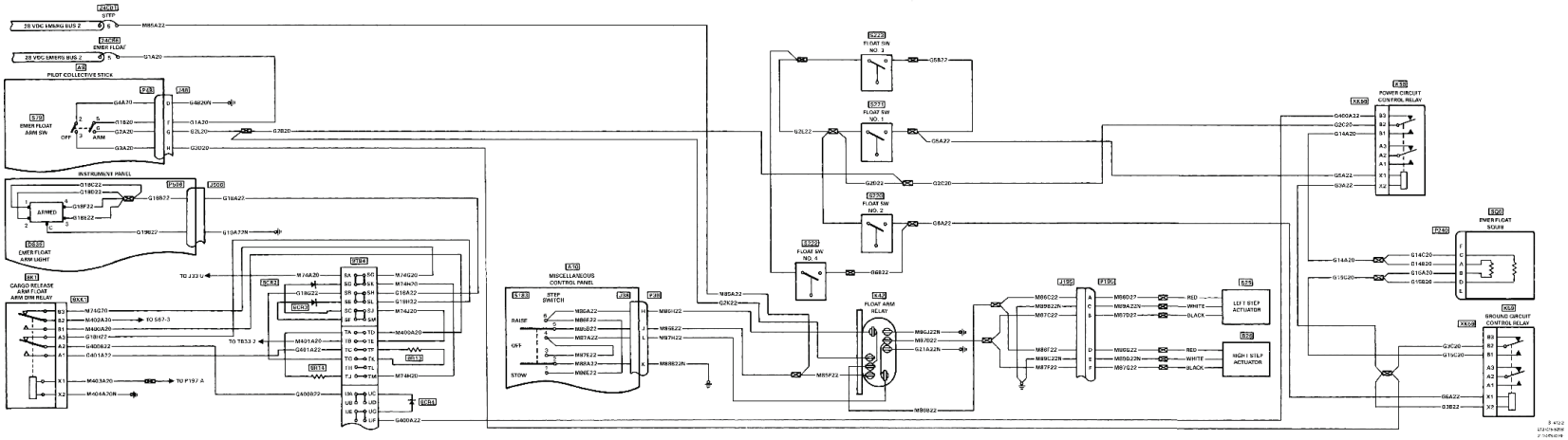
Intentionally blank page



ICN-412-A-980000-B-97499-00291-A-001-01

Figure 1. Windshield Wipers (Sheet 1 of 1)

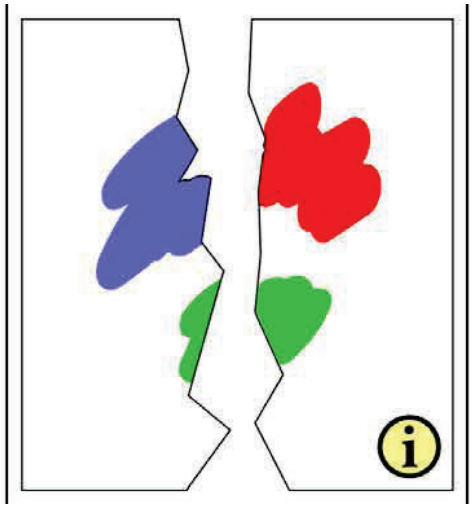
Intentionally blank page



ICN-412-A-980000-B-97499-00292-A-001-01

Figure 1. Emergency Floats (Sheet 1 of 1)

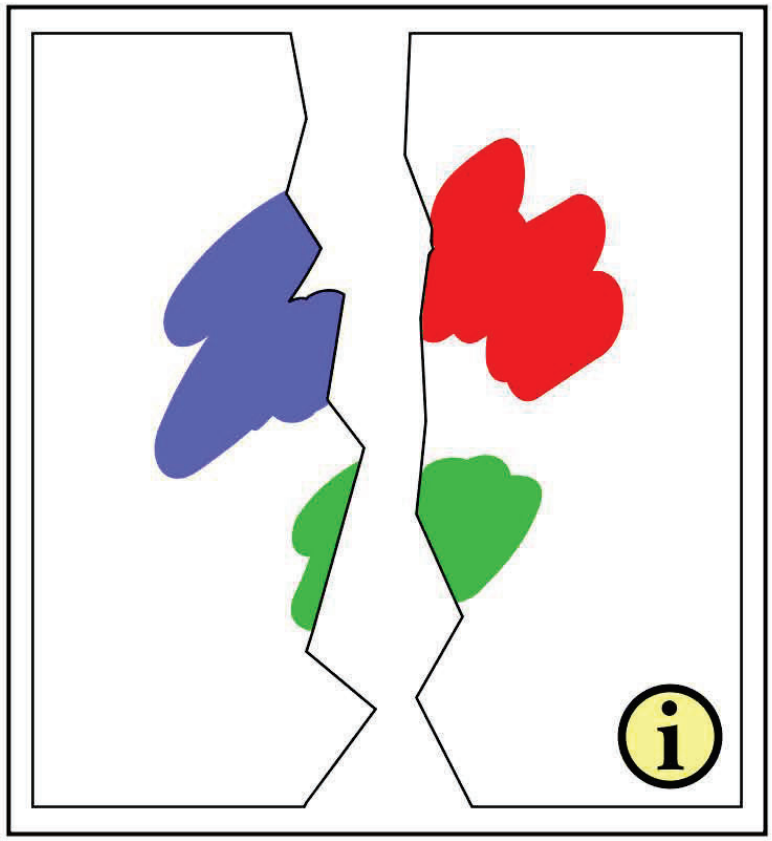
Intentionally blank page



ICN-412-A-980000-B-97499-00293-A-001-01

Figure 1. Emergency Floats (Sheet 1 of 1)

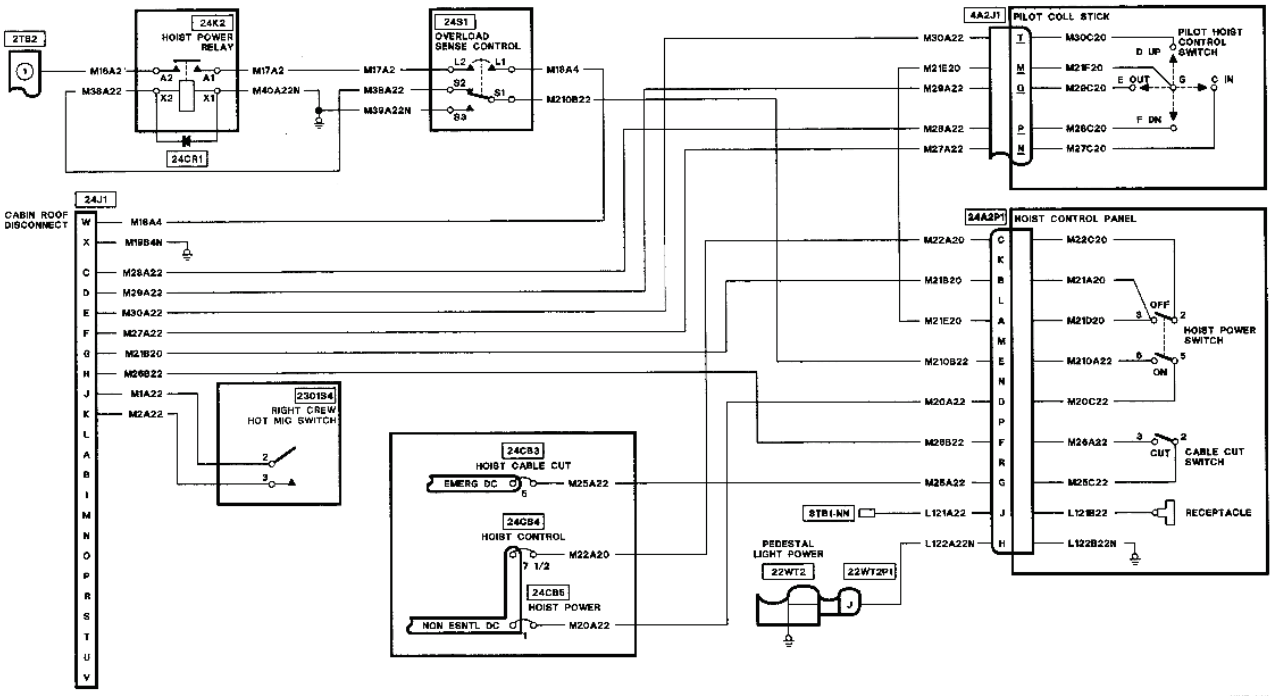
Intentionally blank page



ICN-412-A-980000-B-97499-00294-A-001-01

Figure 1. Emergency Floats (Sheet 1 of 1)

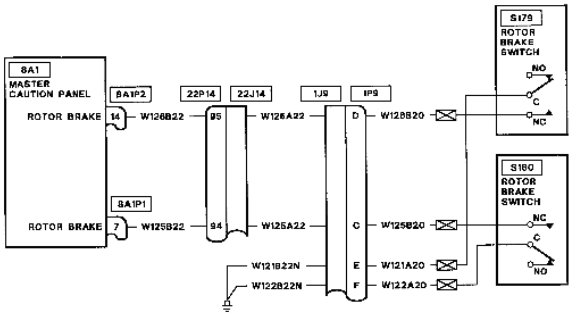
Intentionally blank page



ICN-412-A-980000-B-97499-00295-A-001-01

Figure 1. Hoist Control (Sheet 1 of 1)

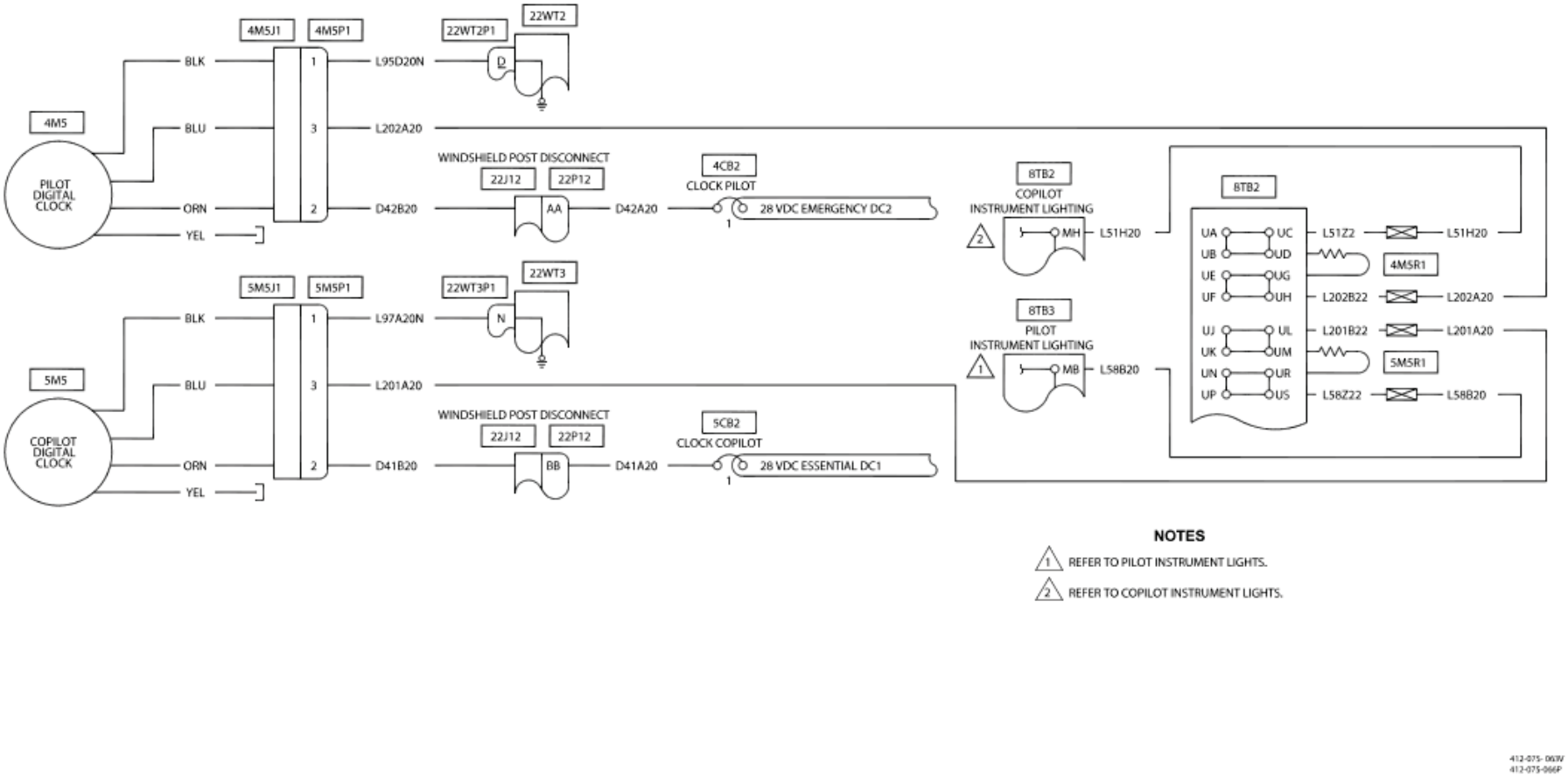
Intentionally blank page



ICN-412-A-980000-B-97499-00296-A-001-01

Figure 1. Rotor Brake (Sheet 1 of 1)

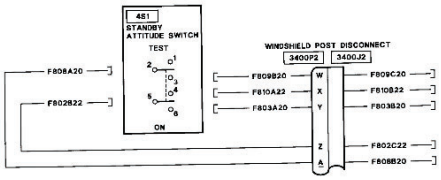
Intentionally blank page



ICN-412-A-980000-B-97499-00297-A-001-01

Figure 1. Digital Clocks (Sheet 1 of 1)

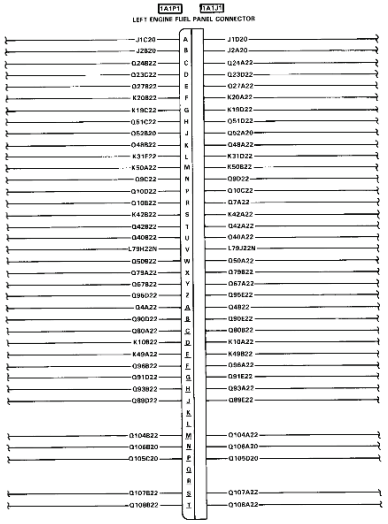
Intentionally blank page



ICN-412-A-980000-B-97499-00298-A-001-01

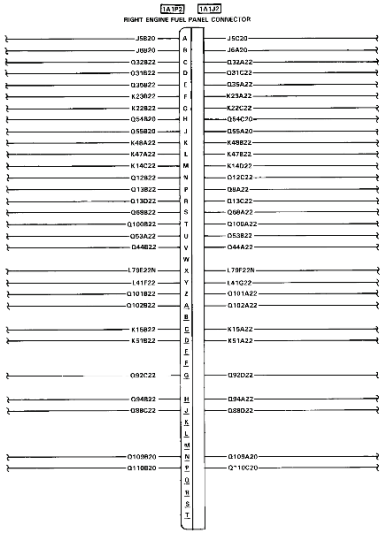
Figure 1. Standby Attitude Indicator Provisions (Sheet 1 of 1)

Intentionally blank page



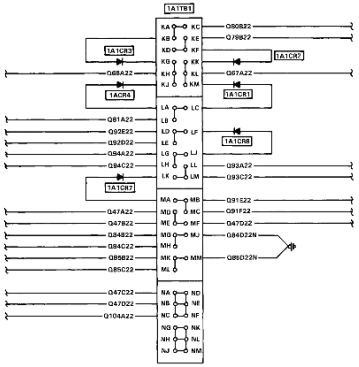
ICN-412-A-980000-B-97499-00299-A-001-01

Figure 1. Component Replacement (Sheet 1 of 87)



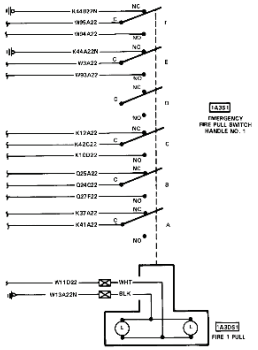
ICN-412-A-980000-B-97499-00300-A-001-01

Figure 1. Component Replacement (Sheet 2 of 87)



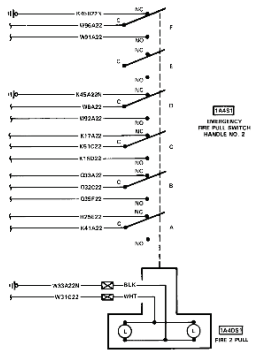
ICN-412-A-980000-B-97499-00301-A-001-01

Figure 1. Component Replacement (Sheet 3 of 87)



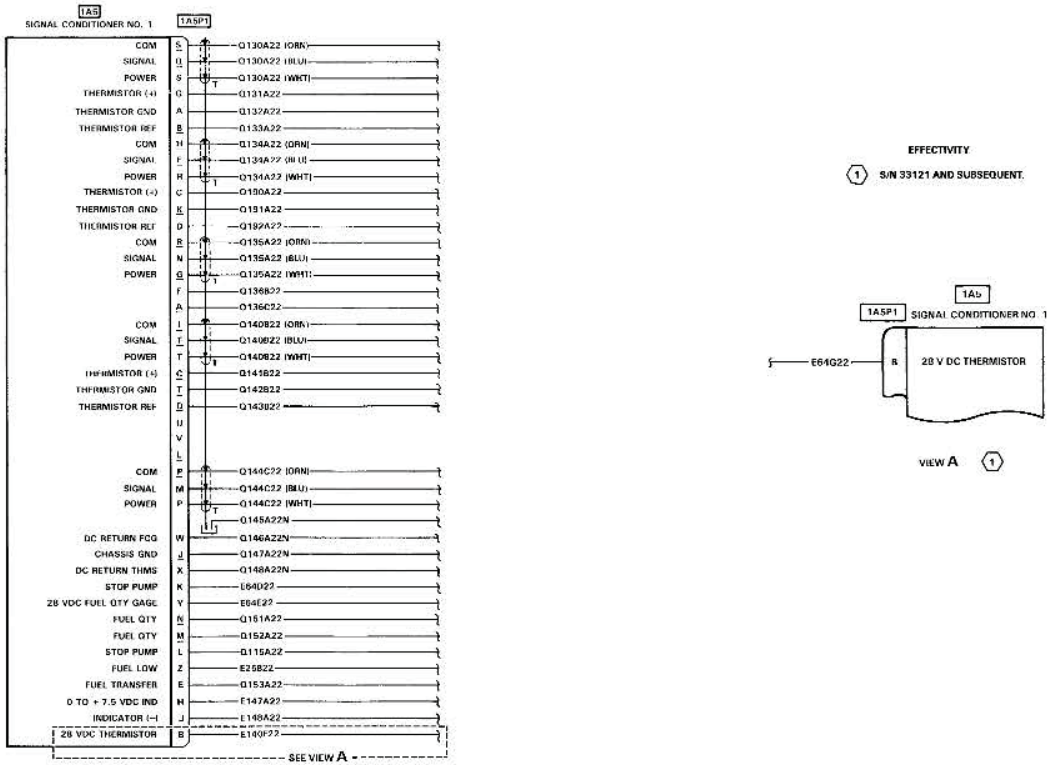
ICN-412-A-980000-B-97499-00302-A-001-01

Figure 1. Component Replacement (Sheet 4 of 87)



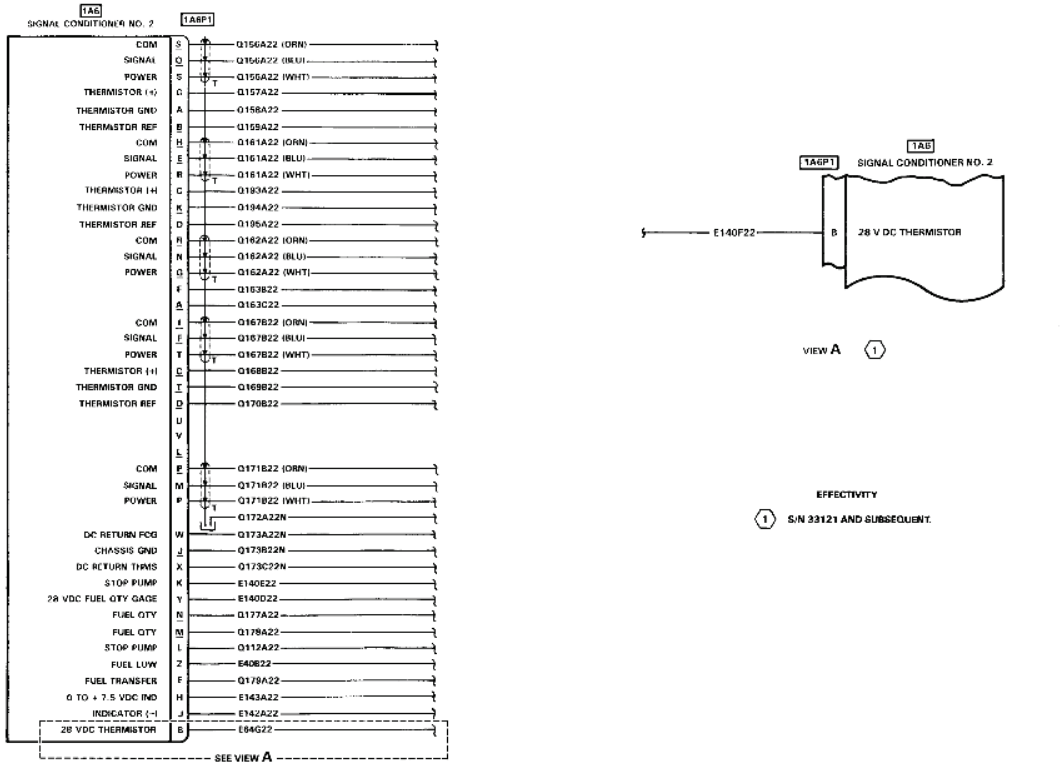
ICN-412-A-980000-B-97499-00303-A-001-01

Figure 1. Component Replacement (Sheet 5 of 87)



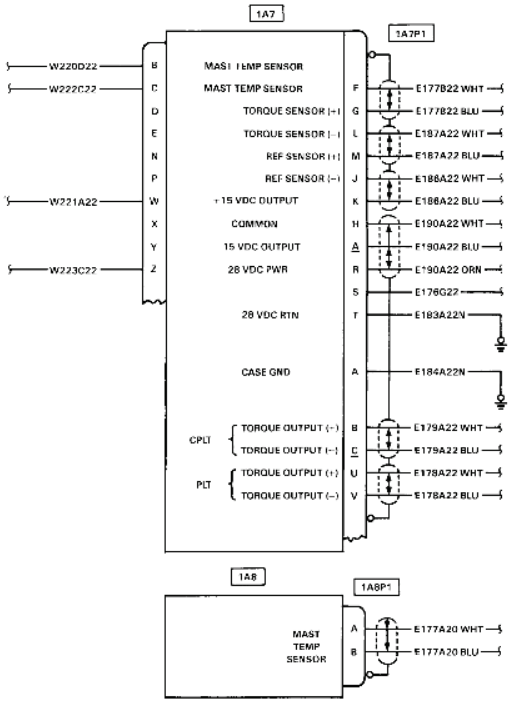
ICN-412-A-980000-B-97499-00304-A-001-01

Figure 1. Component Replacement (Sheet 6 of 87)



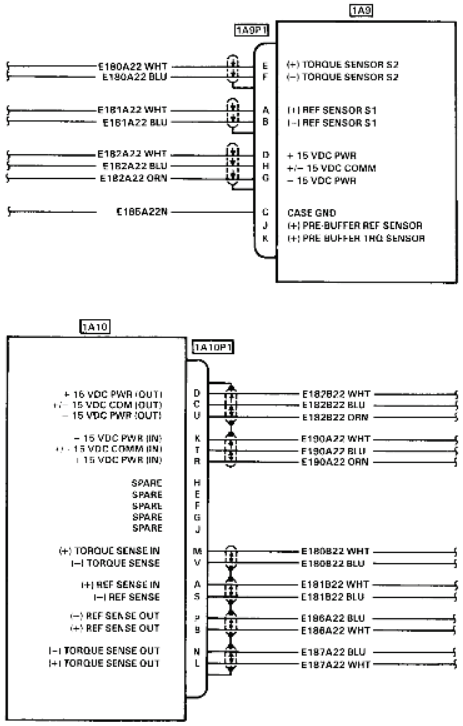
ICN-412-A-980000-B-97499-00305-A-001-01

Figure 1. Component Replacement (Sheet 7 of 87)



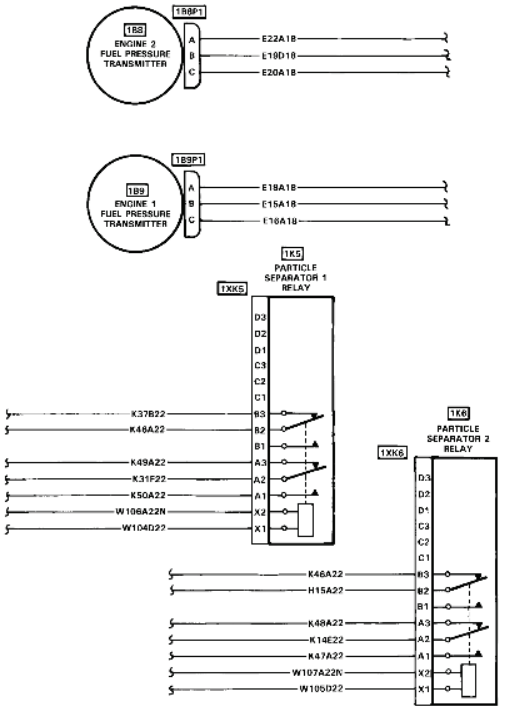
ICN-412-A-980000-B-97499-00306-A-001-01

Figure 1. Component Replacement (Sheet 8 of 87)



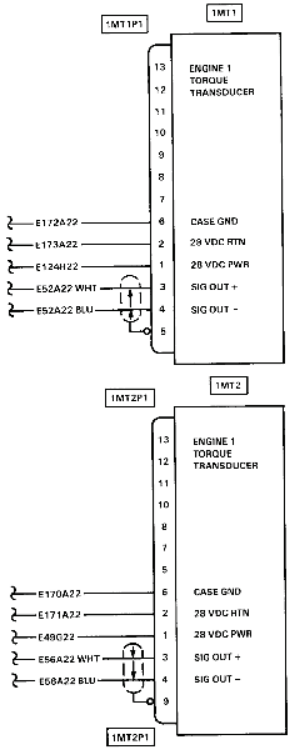
ICN-412-A-980000-B-97499-00307-A-001-01

Figure 1. Component Replacement (Sheet 9 of 87)



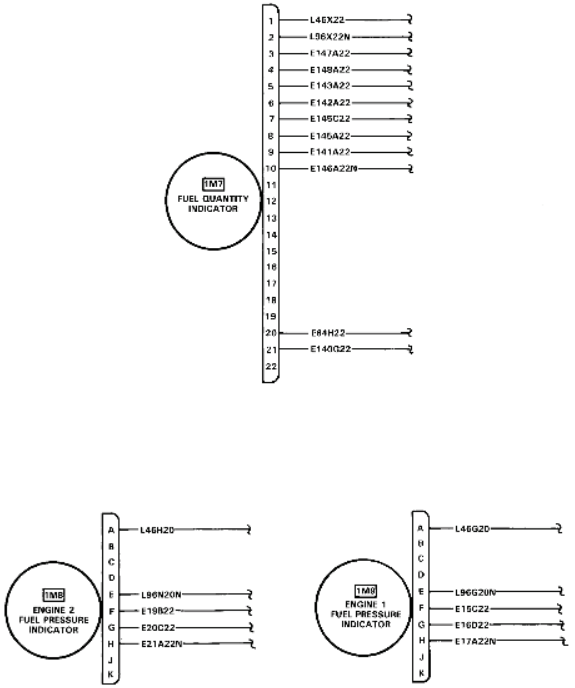
ICN-412-A-980000-B-97499-00308-A-001-01

Figure 1. Component Replacement (Sheet 10 of 87)



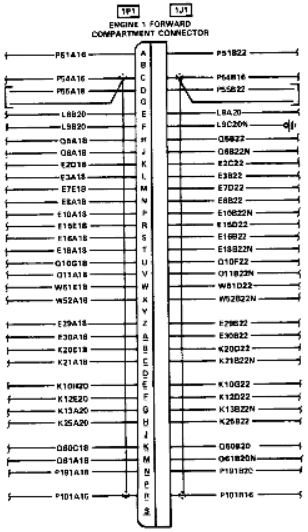
ICN-412-A-980000-B-97499-00309-A-001-01

Figure 1. Component Replacement (Sheet 11 of 87)



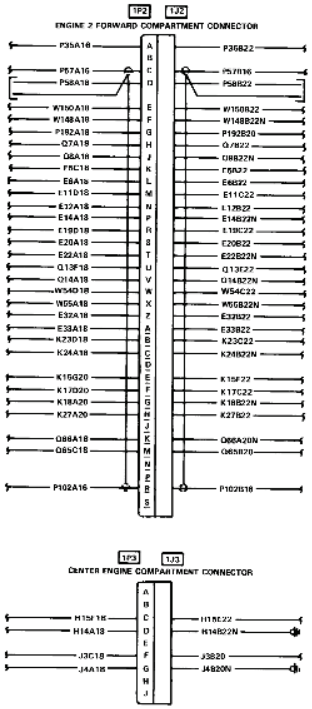
ICN-412-A-980000-B-97499-00310-A-001-01

Figure 1. Component Replacement (Sheet 12 of 87)



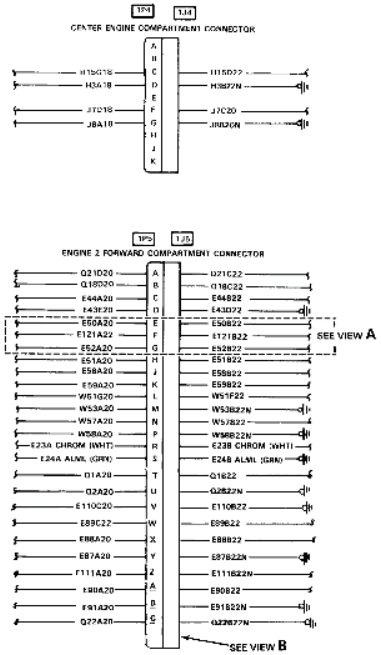
ICN-412-A-980000-B-97499-00311-A-001-01

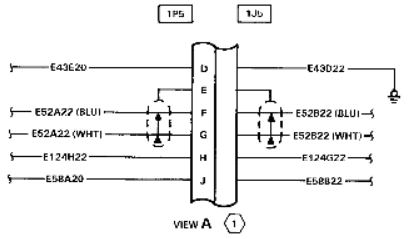
Figure 1. Component Replacement (Sheet 13 of 87)

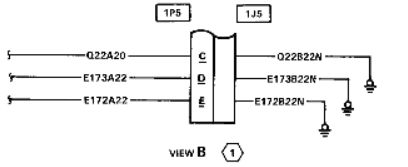


ICN-412-A-980000-B-97499-00312-A-001-01

Figure 1. Component Replacement (Sheet 14 of 87)





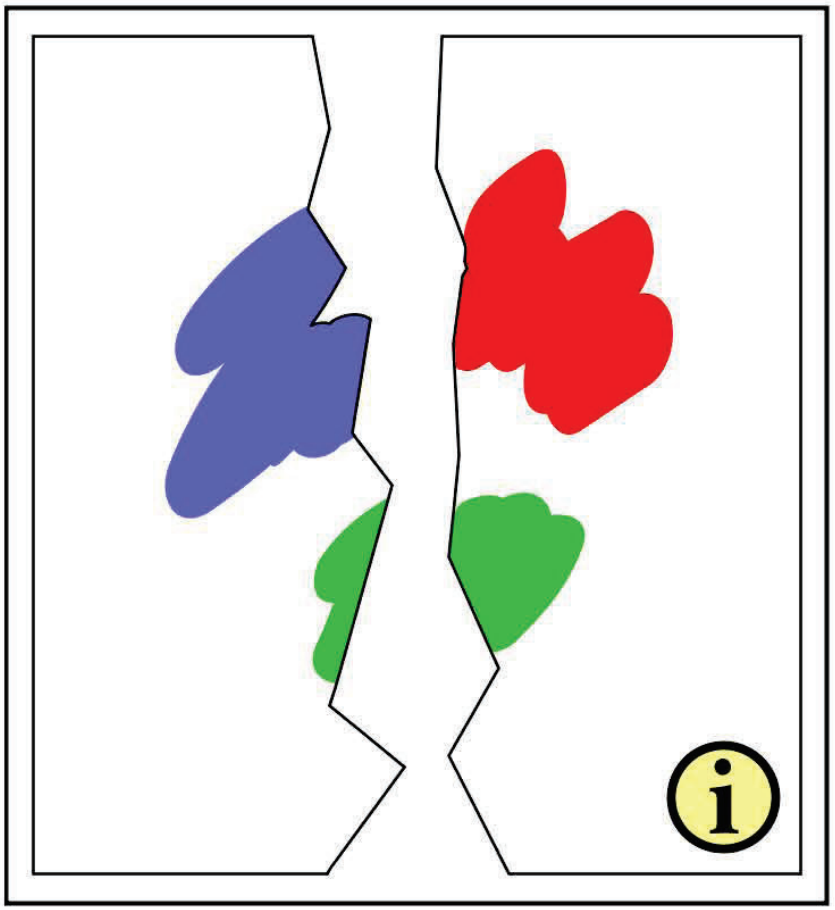


EFFECTIVITY

1 S/N 36020 AND SUBSEQUENT.

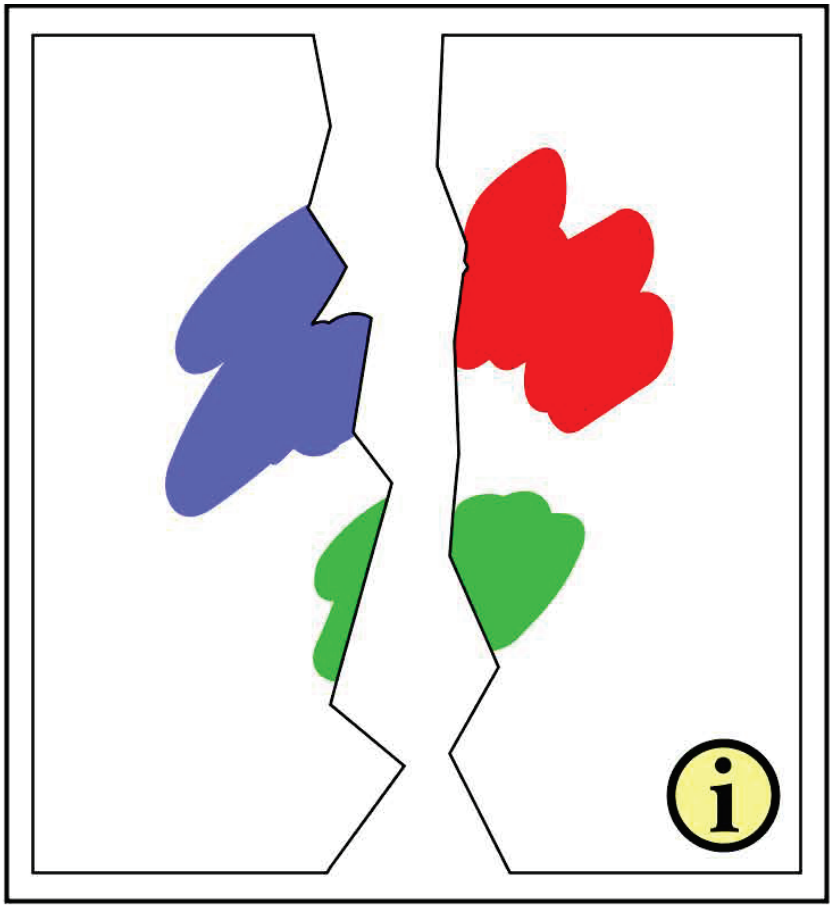
ICN-412-A-980000-B-97499-00313-A-001-01

Figure 1. Component Replacement (Sheet 15 of 87)



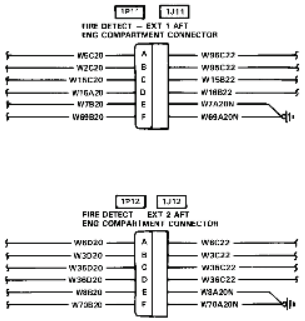
ICN-412-A-980000-B-97499-00314-A-001-01

Figure 1. Component Replacement (Sheet 16 of 87)



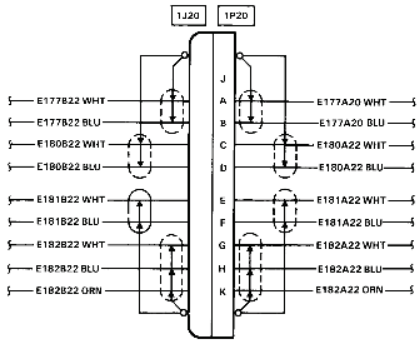
ICN-412-A-980000-B-97499-00315-A-001-01

Figure 1. Component Replacement (Sheet 17 of 87)



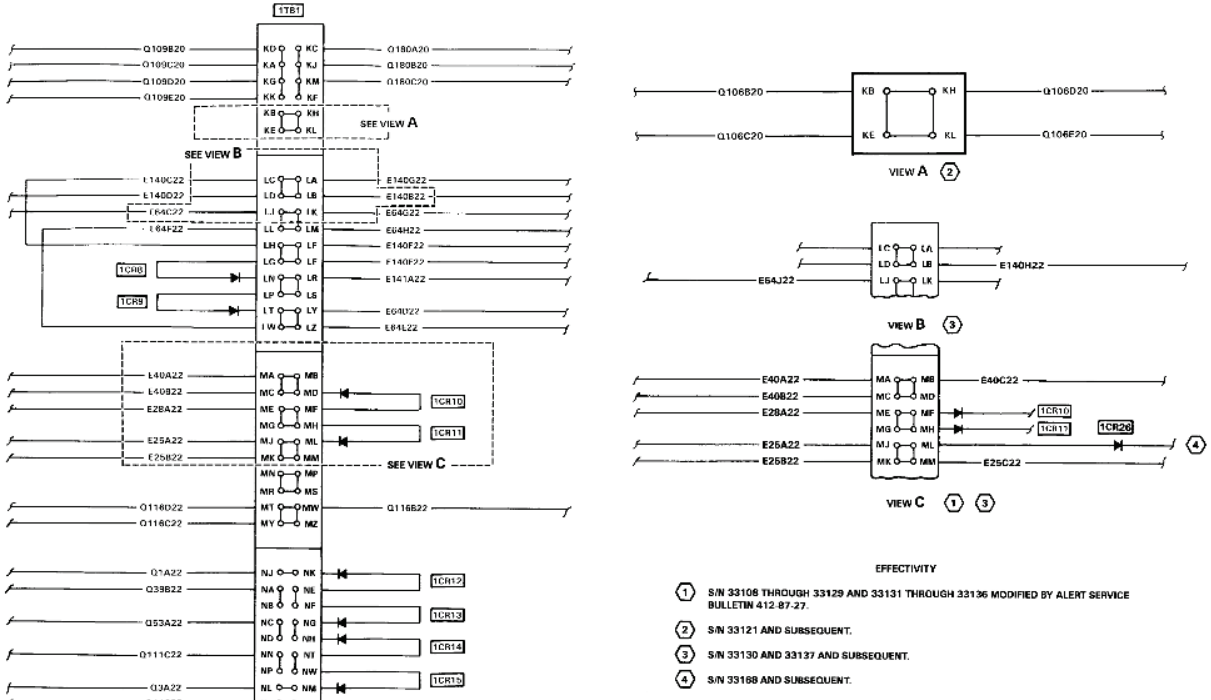
ICN-412-A-980000-B-97499-00316-A-001-01

Figure 1. Component Replacement (Sheet 18 of 87)



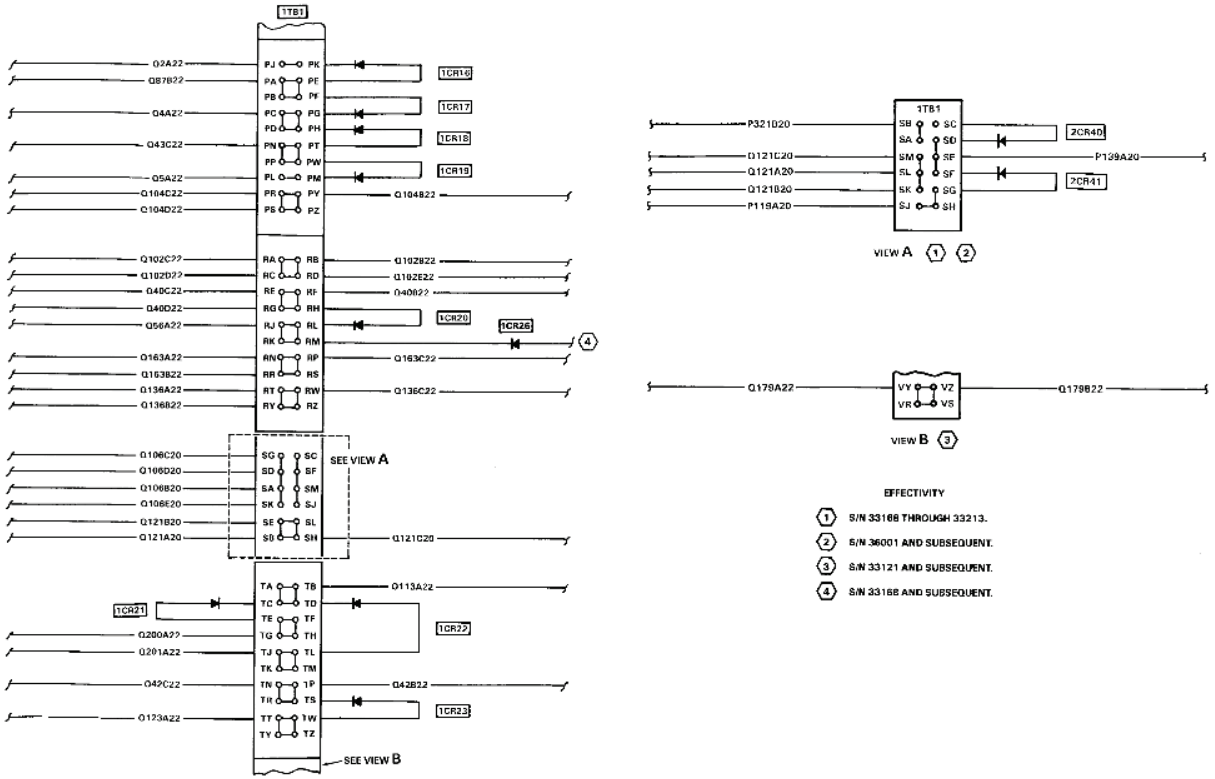
ICN-412-A-980000-B-97499-00317-A-001-01

Figure 1. Component Replacement (Sheet 19 of 87)



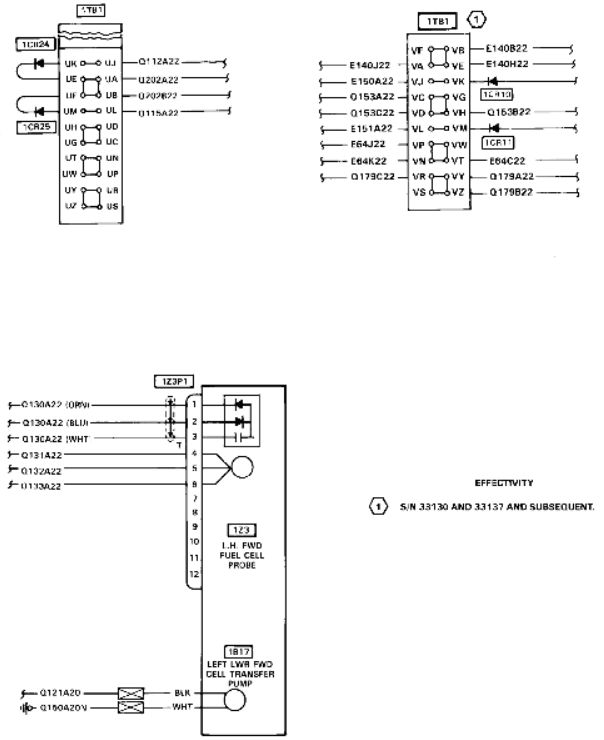
ICN-412-A-980000-B-97499-00318-A-001-01

Figure 1. Component Replacement (Sheet 20 of 87)



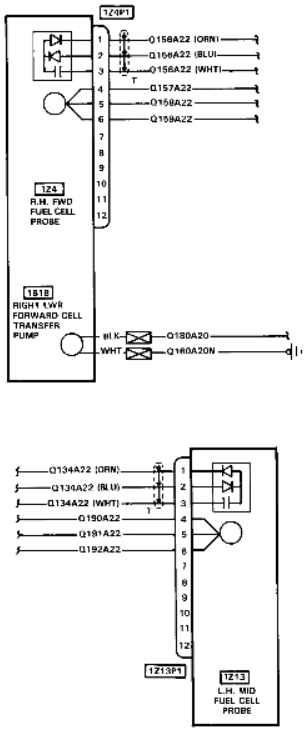
ICN-412-A-980000-B-97499-00319-A-001-01

Figure 1. Component Replacement (Sheet 21 of 87)



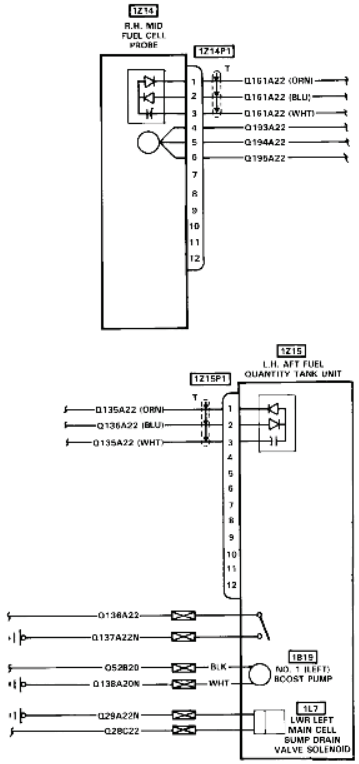
ICN-412-A-980000-B-97499-00320-A-001-01

Figure 1. Component Replacement (Sheet 22 of 87)



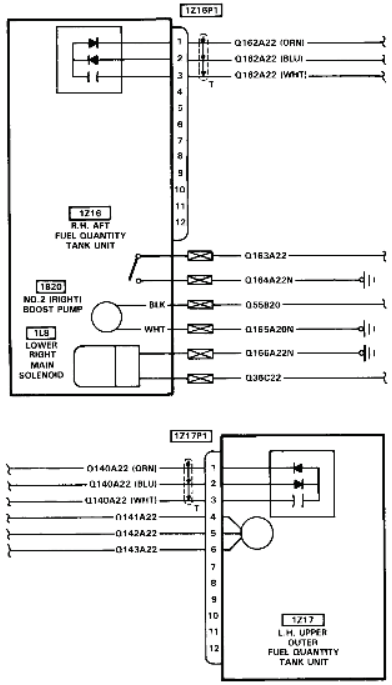
ICN-412-A-980000-B-97499-00321-A-001-01

Figure 1. Component Replacement (Sheet 23 of 87)



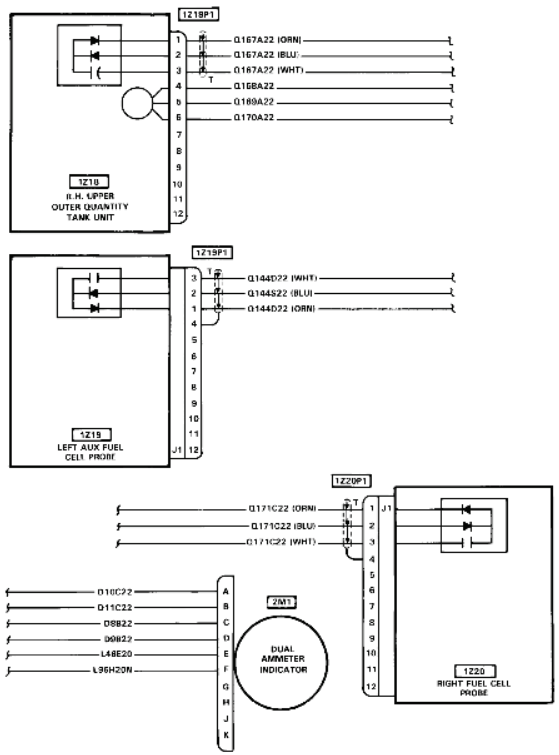
ICN-412-A-980000-B-97499-00322-A-001-01

Figure 1. Component Replacement (Sheet 24 of 87)



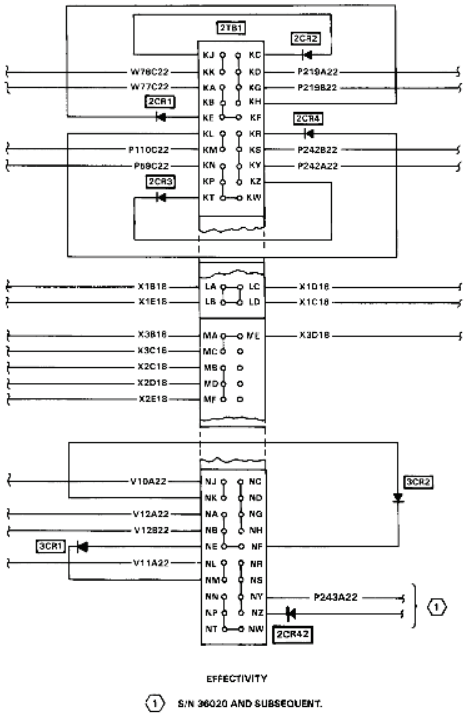
ICN-412-A-980000-B-97499-00323-A-001-01

Figure 1. Component Replacement (Sheet 25 of 87)



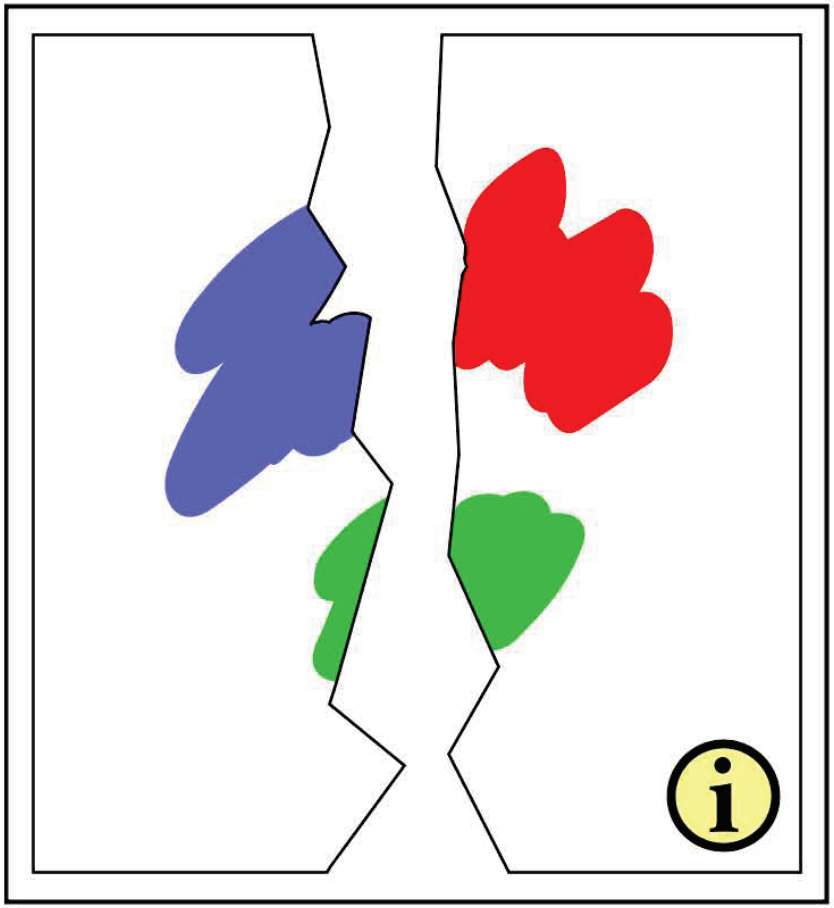
ICN-412-A-980000-B-97499-00324-A-001-01

Figure 1. Component Replacement (Sheet 26 of 87)



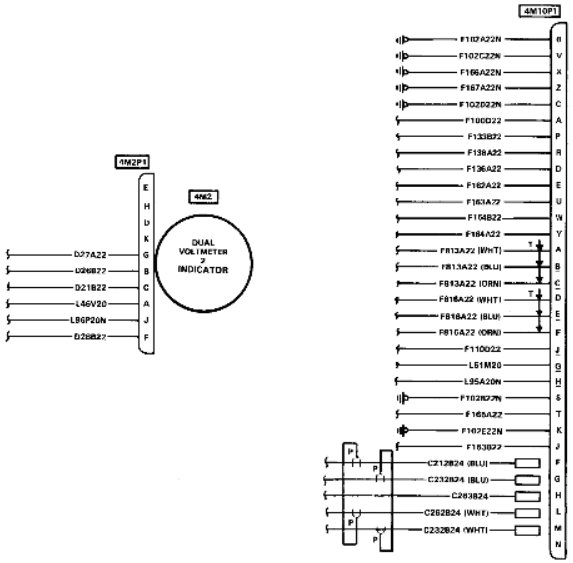
ICN-412-A-980000-B-97499-00325-A-001-01

Figure 1. Component Replacement (Sheet 27 of 87)



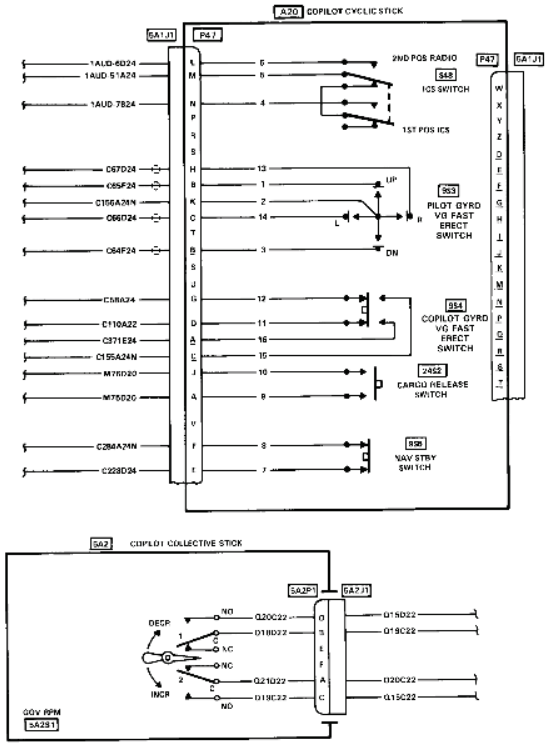
ICN-412-A-980000-B-97499-00326-A-001-01

Figure 1. Component Replacement (Sheet 28 of 87)



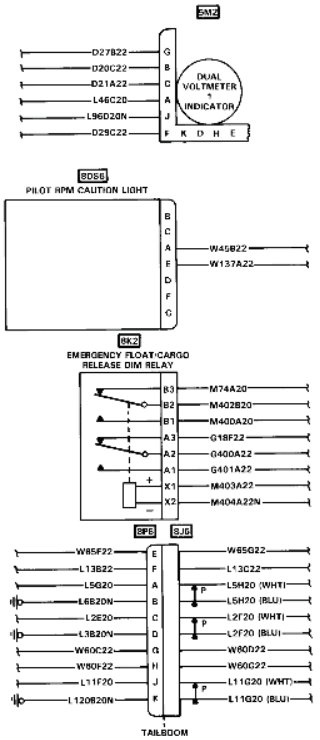
ICN-412-A-980000-B-97499-00327-A-001-01

Figure 1. Component Replacement (Sheet 29 of 87)



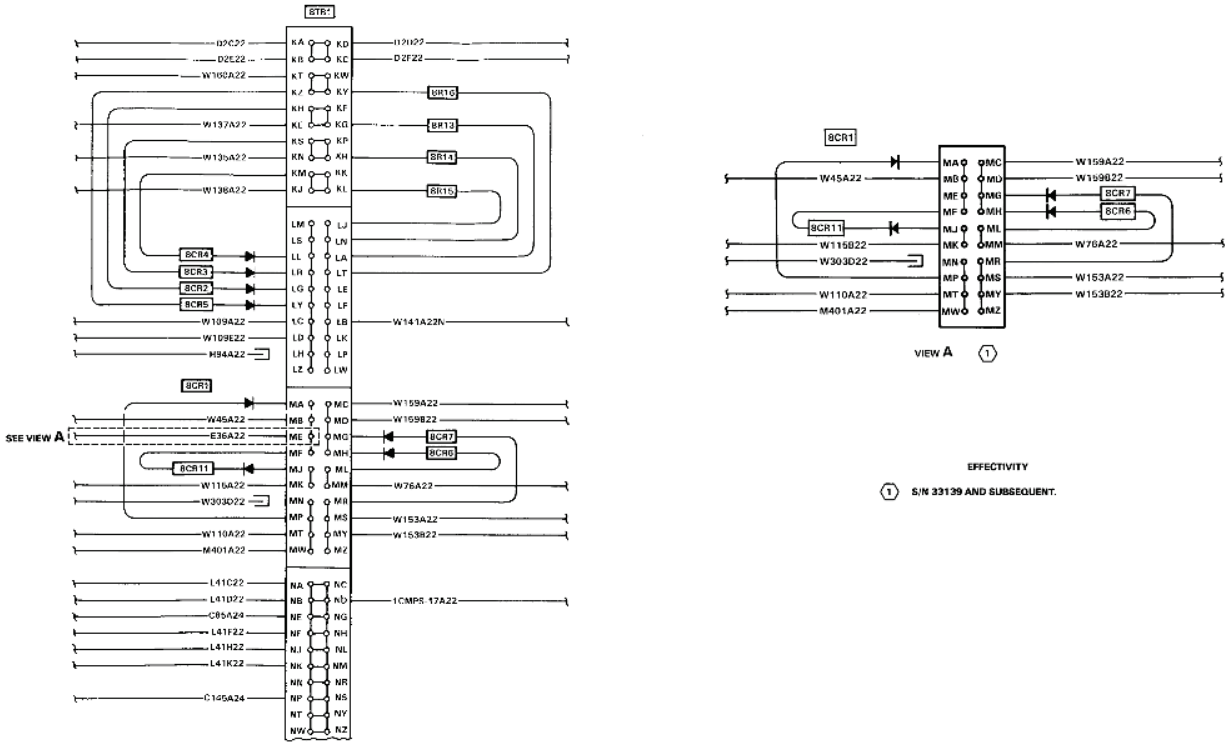
ICN-412-A-980000-B-97499-00328-A-001-01

Figure 1. Component Replacement (Sheet 30 of 87)



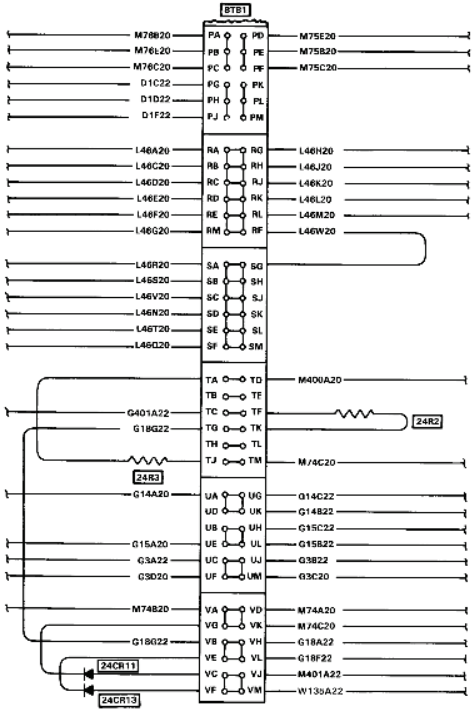
ICN-412-A-980000-B-97499-00329-A-001-01

Figure 1. Component Replacement (Sheet 31 of 87)



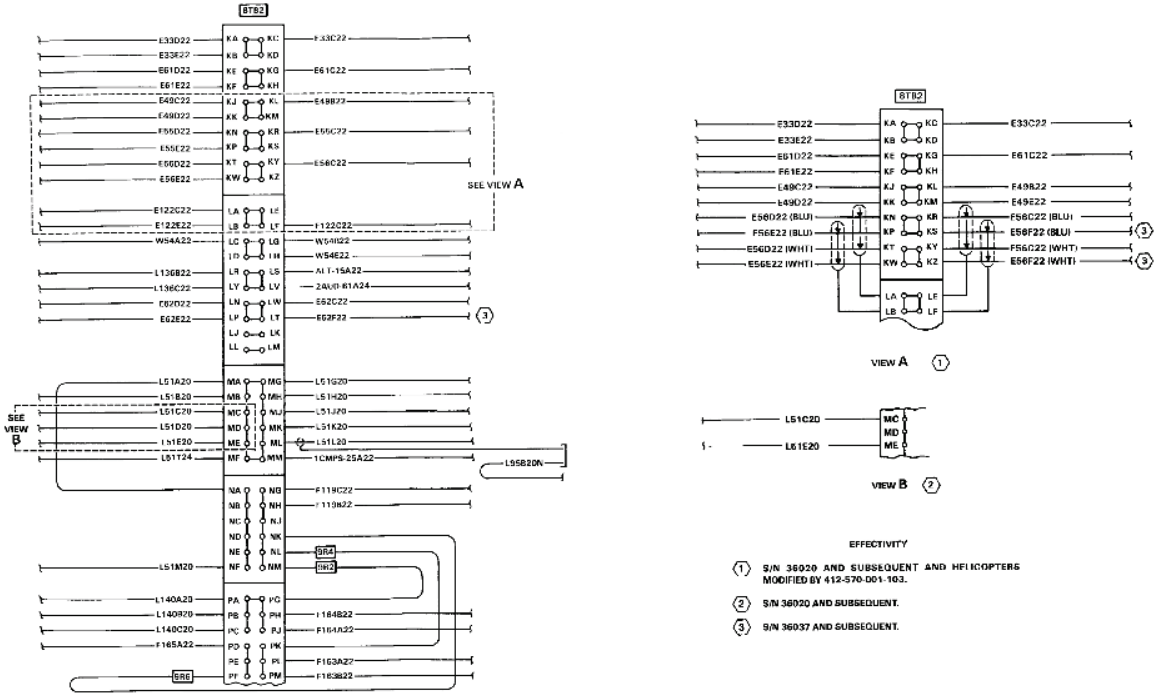
ICN-412-A-980000-B-97499-00330-A-001-01

Figure 1. Component Replacement (Sheet 32 of 87)



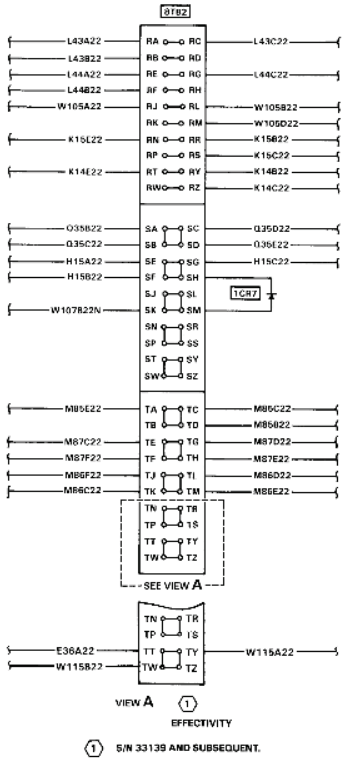
ICN-412-A-980000-B-97499-00331-A-001-01

Figure 1. Component Replacement (Sheet 33 of 87)



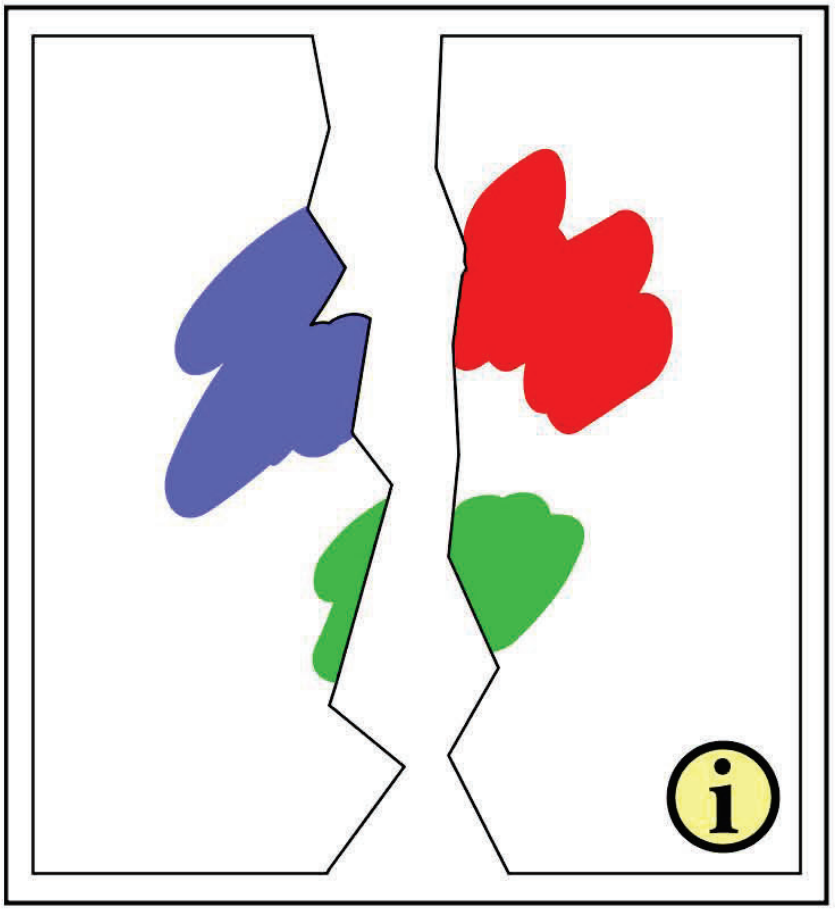
ICN-412-A-980000-B-97499-00332-A-001-01

Figure 1. Component Replacement (Sheet 34 of 87)



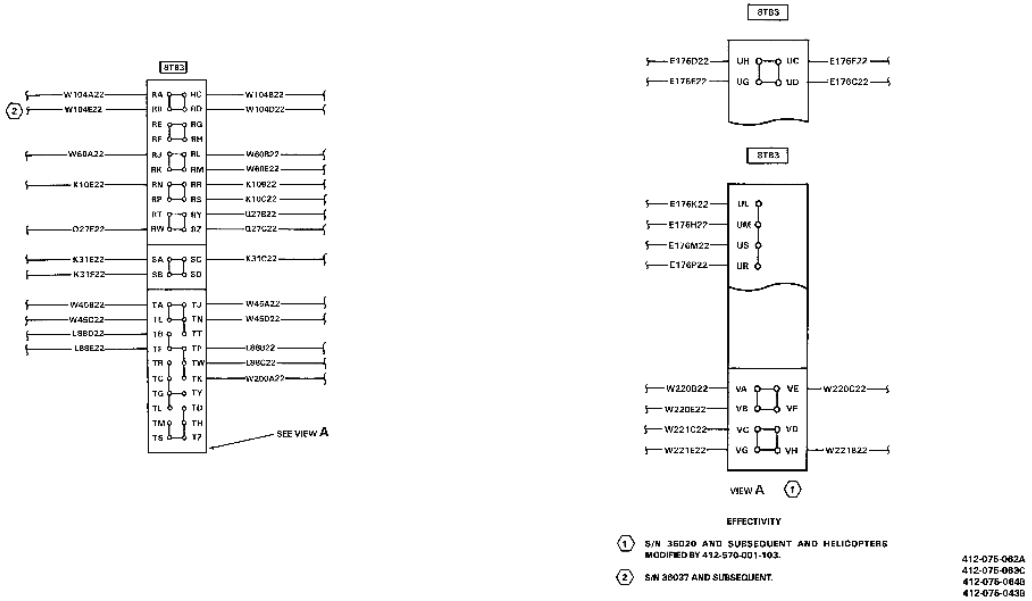
ICN-412-A-980000-B-97499-00333-A-001-01

Figure 1. Component Replacement (Sheet 35 of 87)



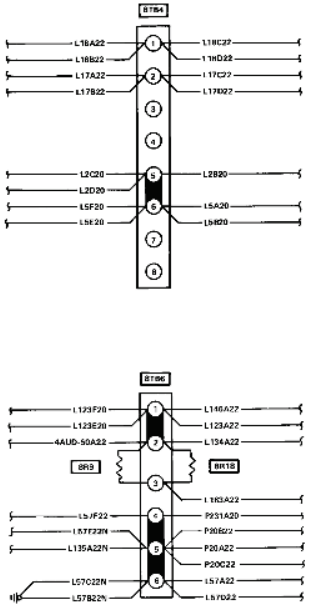
ICN-412-A-980000-B-97499-00334-A-001-01

Figure 1. Component Replacement (Sheet 36 of 87)



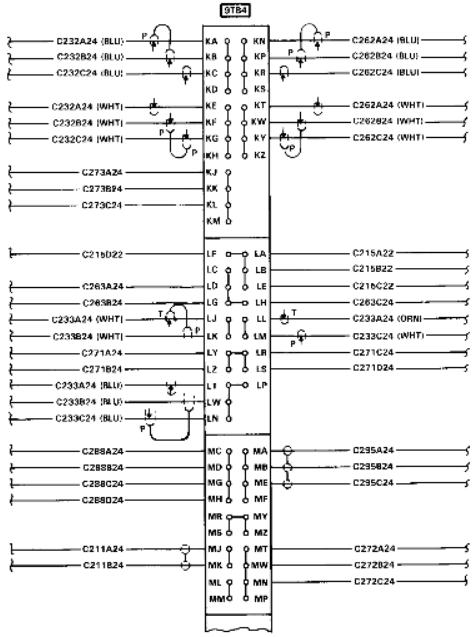
ICN-412-A-980000-B-97499-00335-A-001-01

Figure 1. Component Replacement (Sheet 37 of 87)



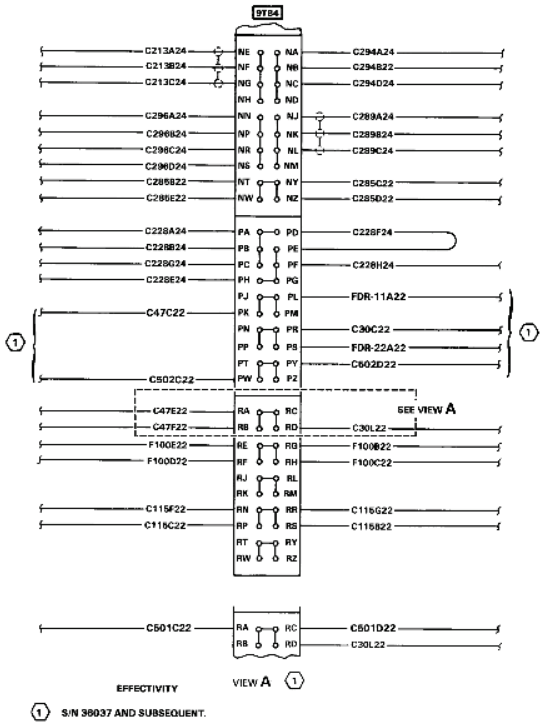
ICN-412-A-980000-B-97499-00336-A-001-01

Figure 1. Component Replacement (Sheet 38 of 87)



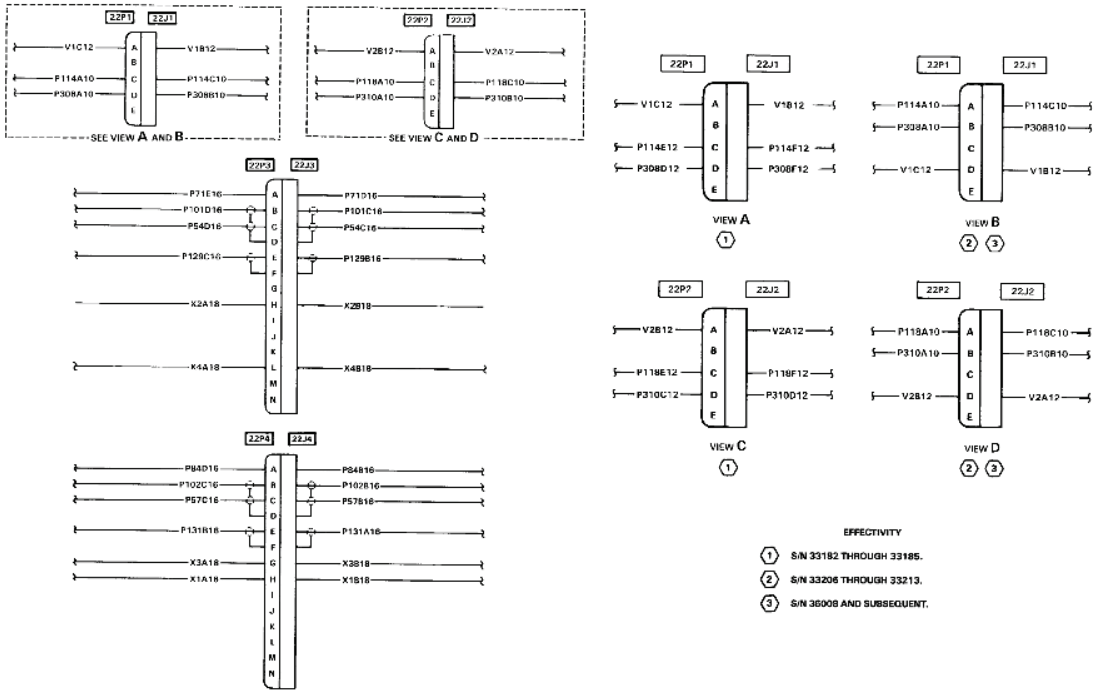
ICN-412-A-980000-B-97499-00337-A-001-01

Figure 1. Component Replacement (Sheet 39 of 87)



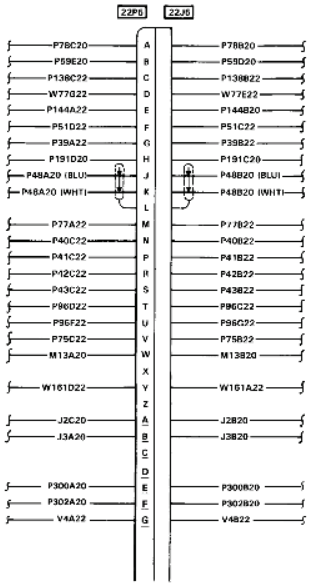
ICN-412-A-980000-B-97499-00338-A-001-01

Figure 1. Component Replacement (Sheet 40 of 87)



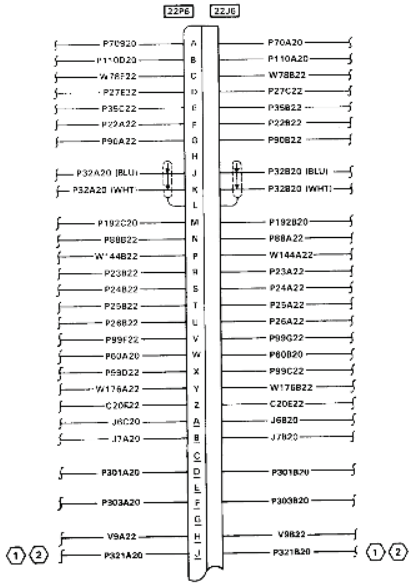
ICN-412-A-980000-B-97499-00339-A-001-01

Figure 1. Component Replacement (Sheet 41 of 87)



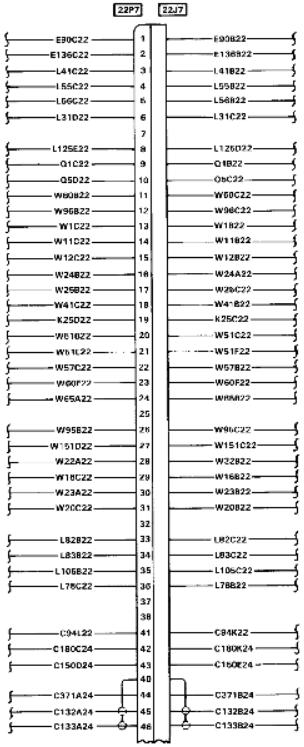
ICN-412-A-980000-B-97499-00340-A-001-01

Figure 1. Component Replacement (Sheet 42 of 87)



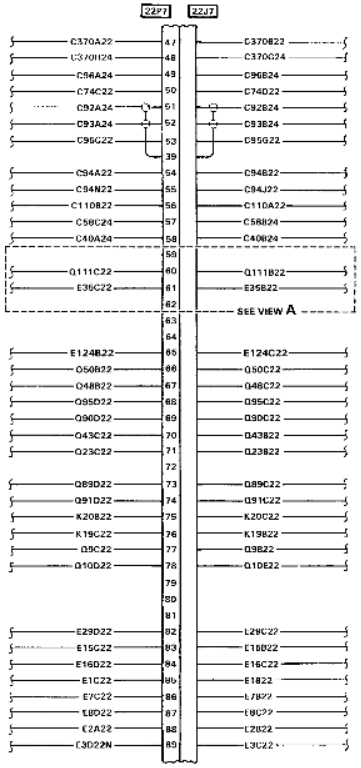
ICN-412-A-980000-B-97499-00341-A-001-01

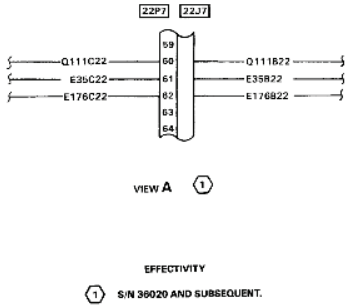
Figure 1. Component Replacement (Sheet 43 of 87)



ICN-412-A-980000-B-97499-00342-A-001-01

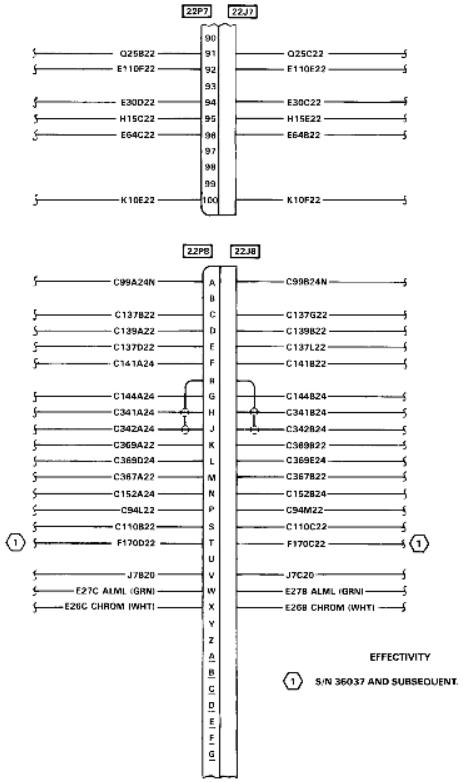
Figure 1. Component Replacement (Sheet 44 of 87)





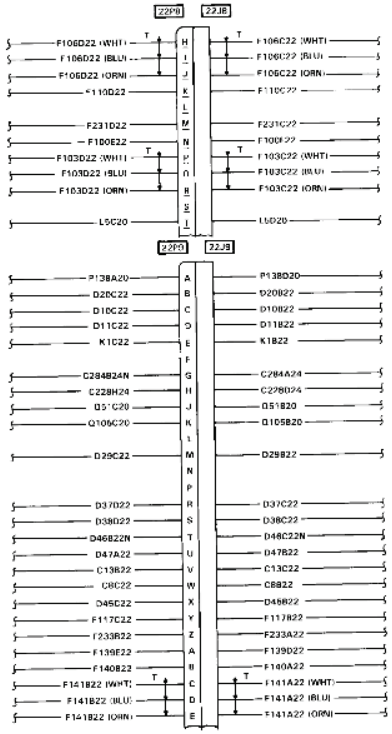
ICN-412-A-980000-B-97499-00343-A-001-01

Figure 1. Component Replacement (Sheet 45 of 87)



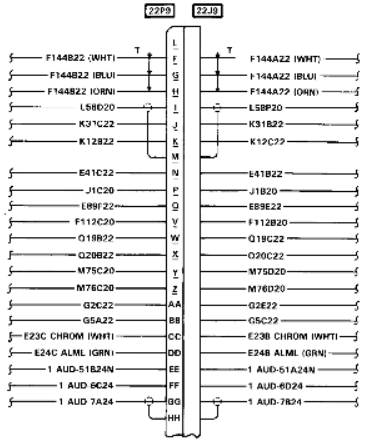
ICN-412-A-980000-B-97499-00344-A-001-01

Figure 1. Component Replacement (Sheet 46 of 87)



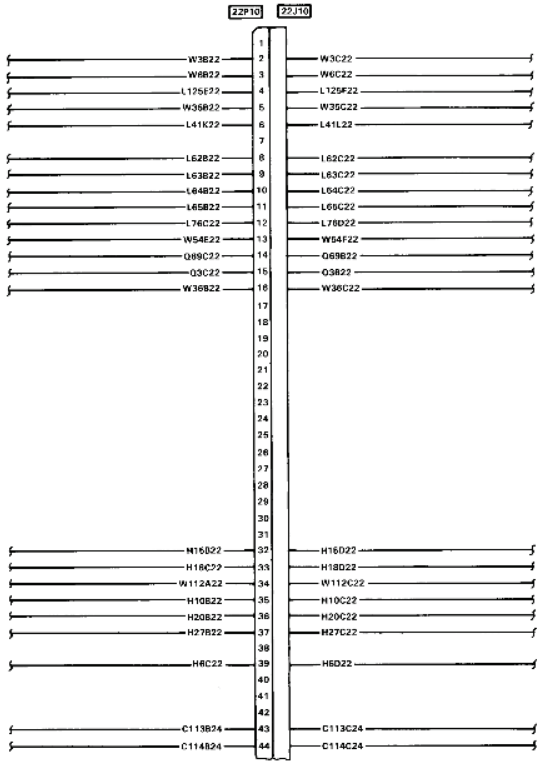
ICN-412-A-980000-B-97499-00345-A-001-01

Figure 1. Component Replacement (Sheet 47 of 87)



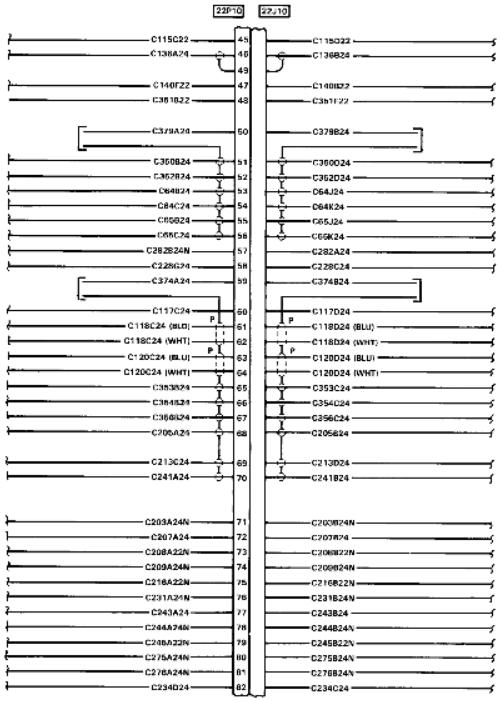
ICN-412-A-980000-B-97499-00346-A-001-01

Figure 1. Component Replacement (Sheet 48 of 87)



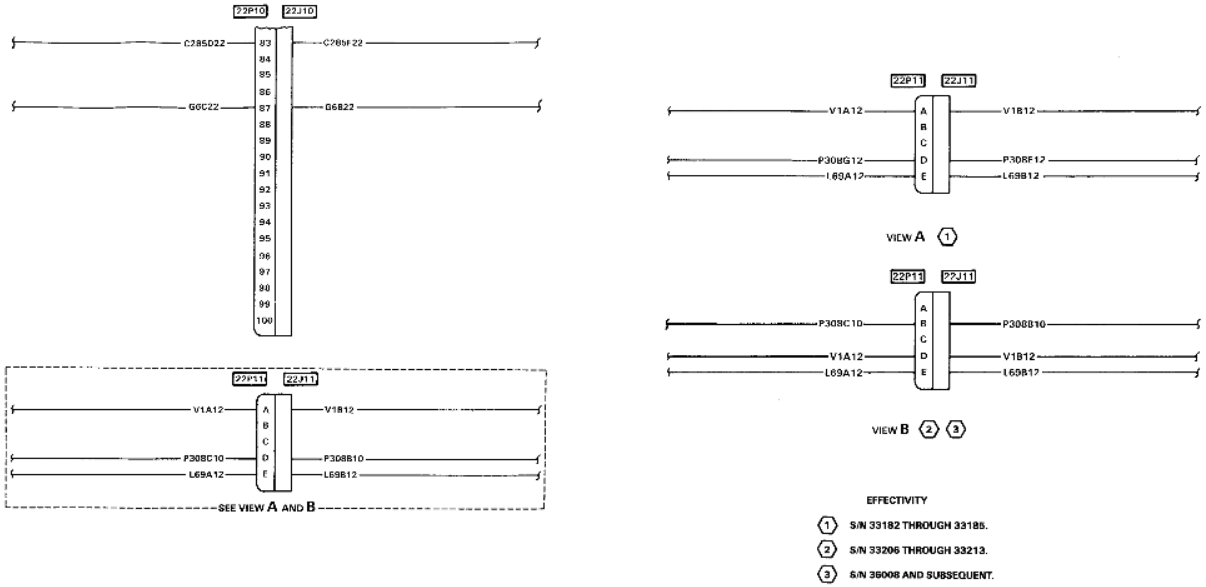
ICN-412-A-980000-B-97499-00347-A-001-01

Figure 1. Component Replacement (Sheet 49 of 87)



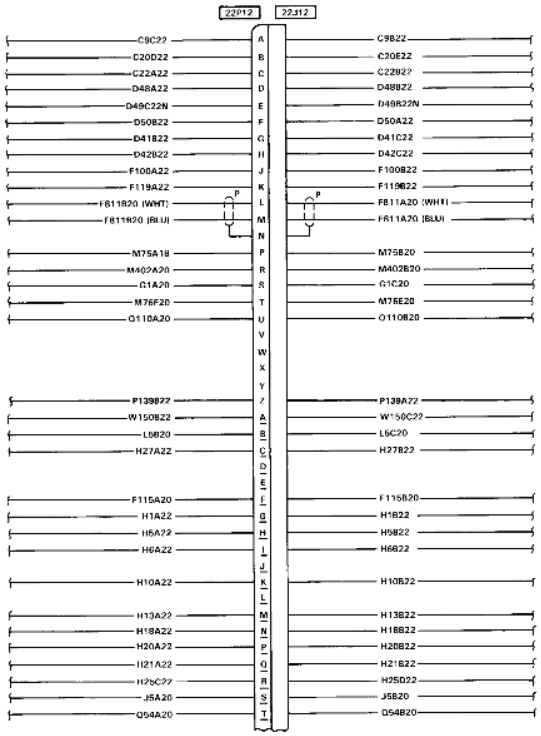
ICN-412-A-980000-B-97499-00348-A-001-01

Figure 1. Component Replacement (Sheet 50 of 87)



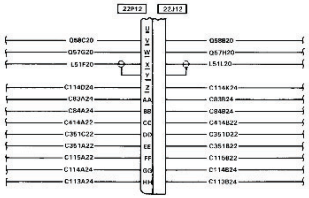
ICN-412-A-980000-B-97499-00349-A-001-01

Figure 1. Component Replacement (Sheet 51 of 87)



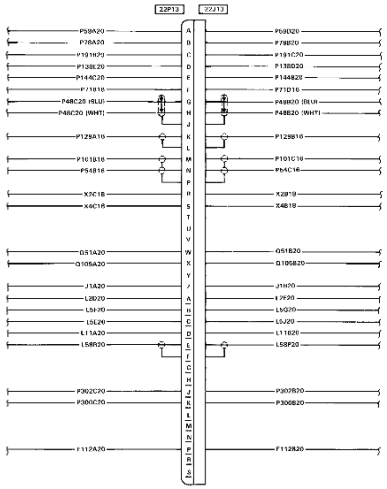
ICN-412-A-980000-B-97499-00350-A-001-01

Figure 1. Component Replacement (Sheet 52 of 87)



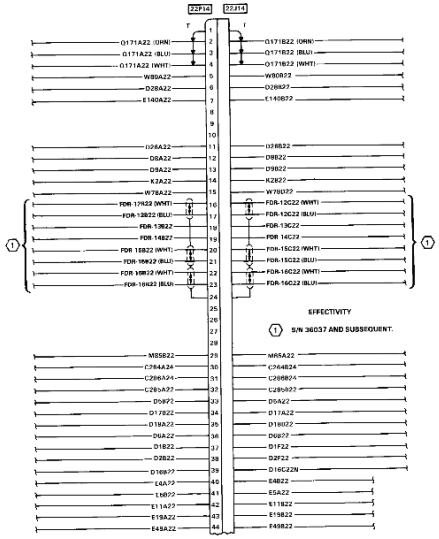
ICN-412-A-980000-B-97499-00351-A-001-01

Figure 1. Component Replacement (Sheet 53 of 87)



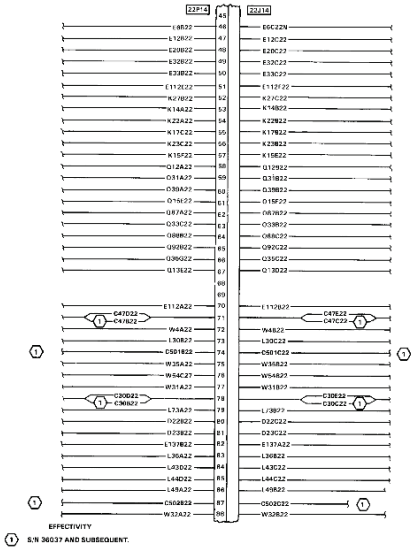
ICN-412-A-980000-B-97499-00352-A-001-01

Figure 1. Component Replacement (Sheet 54 of 87)



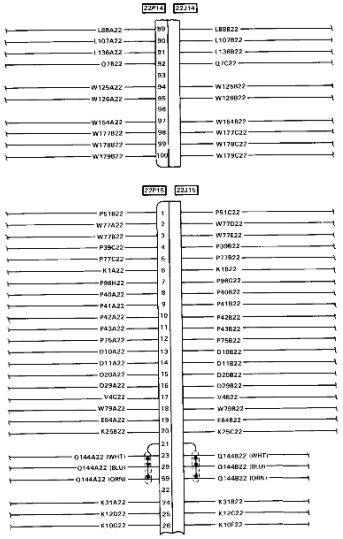
ICN-412-A-980000-B-97499-00353-A-001-01

Figure 1. Component Replacement (Sheet 55 of 87)



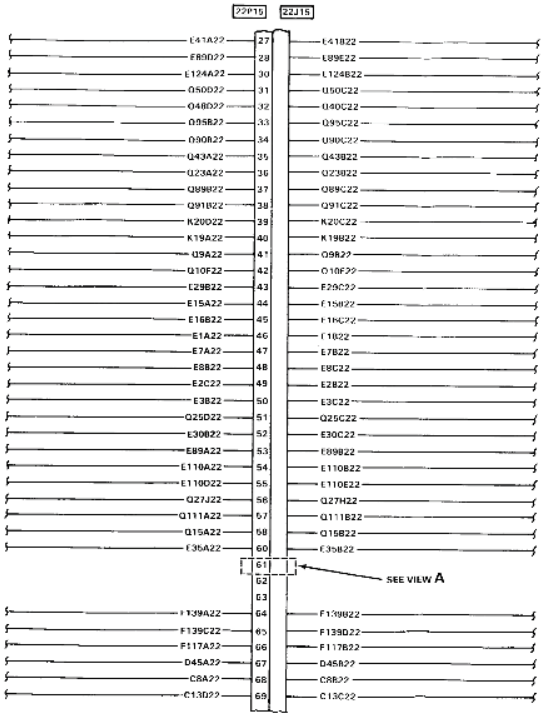
ICN-412-A-980000-B-97499-00354-A-001-01

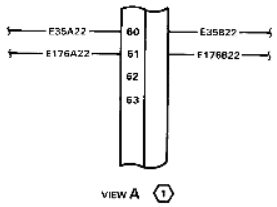
Figure 1. Component Replacement (Sheet 56 of 87)



ICN-412-A-980000-B-97499-00355-A-001-01

Figure 1. Component Replacement (Sheet 57 of 87)



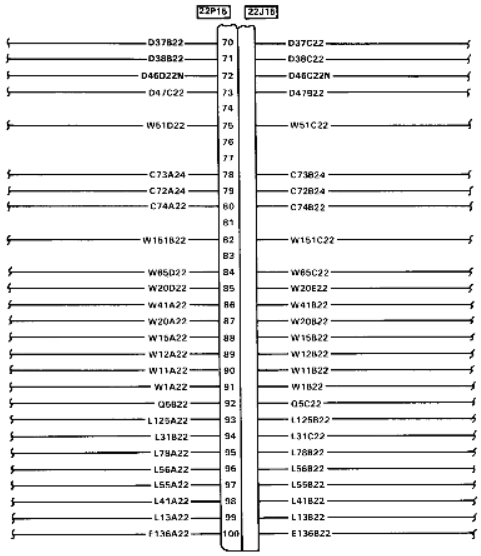


EFFECTIVITY

1 S/N 36020 AND SUBSEQUENT.

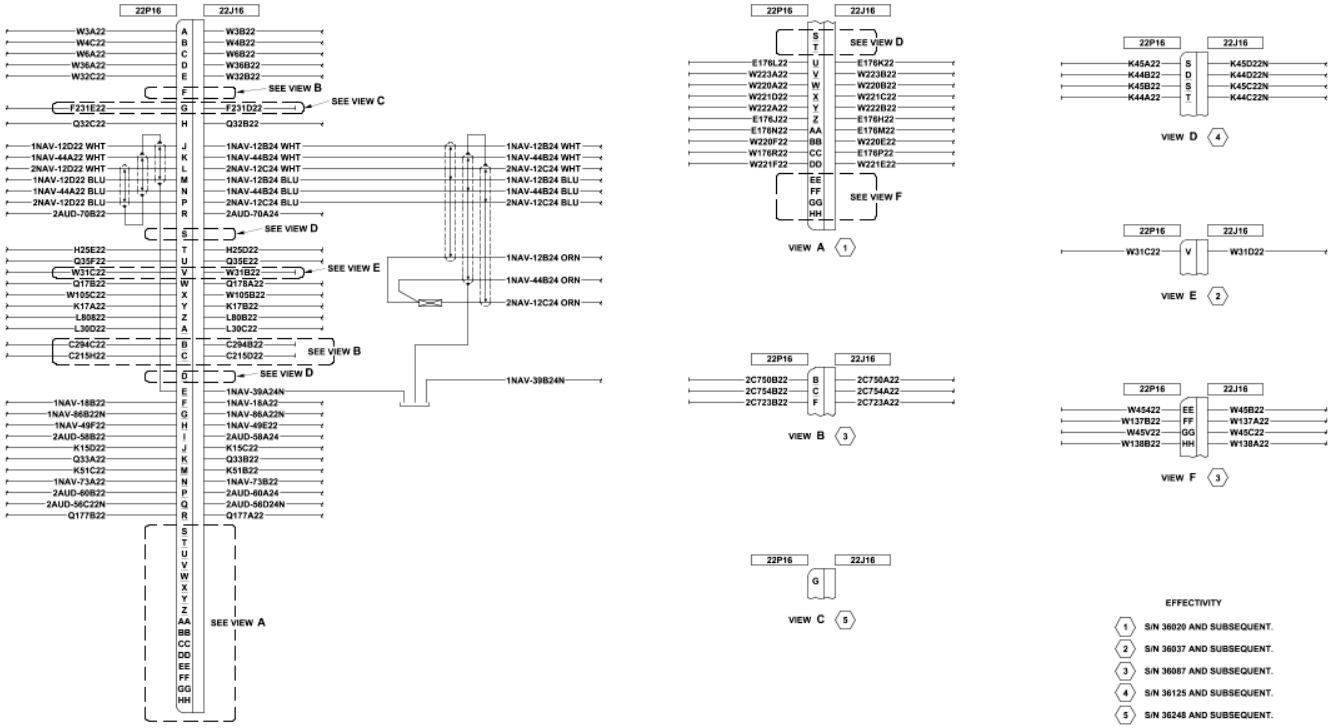
ICN-412-A-980000-B-97499-00356-A-001-01

Figure 1. Component Replacement (Sheet 58 of 87)



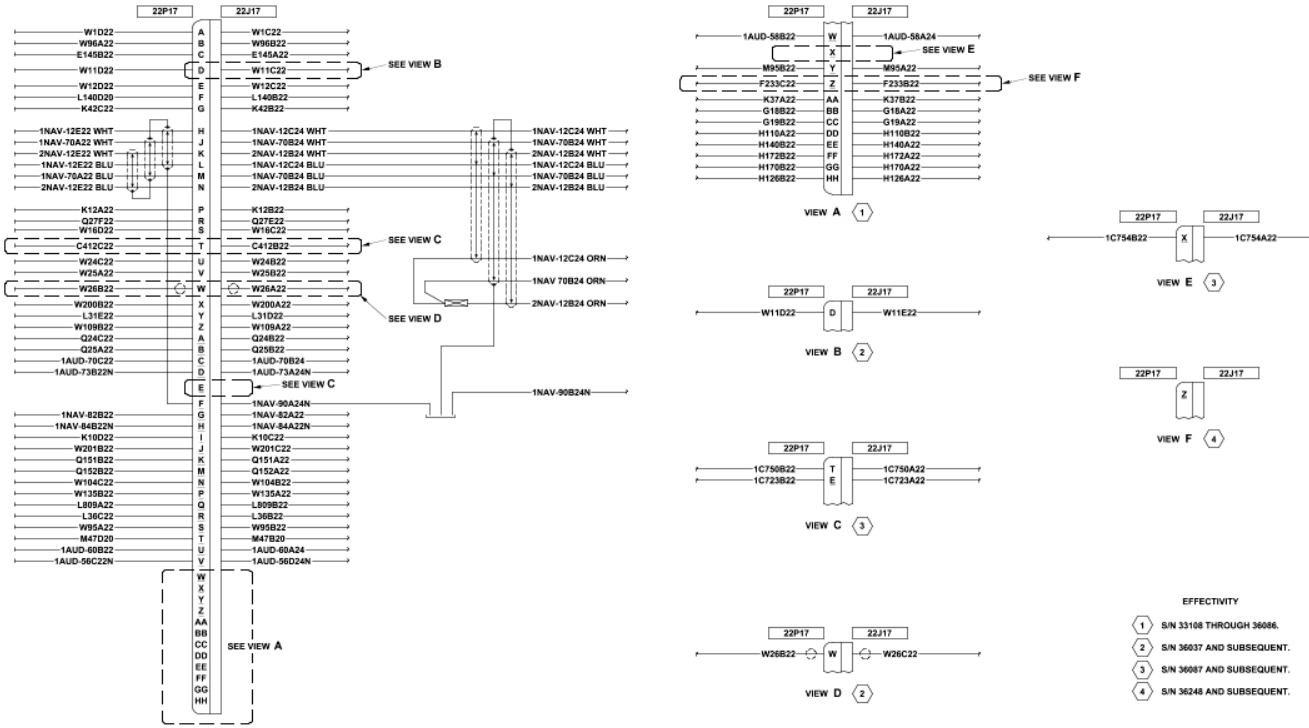
ICN-412-A-980000-B-97499-00357-A-001-01

Figure 1. Component Replacement (Sheet 59 of 87)



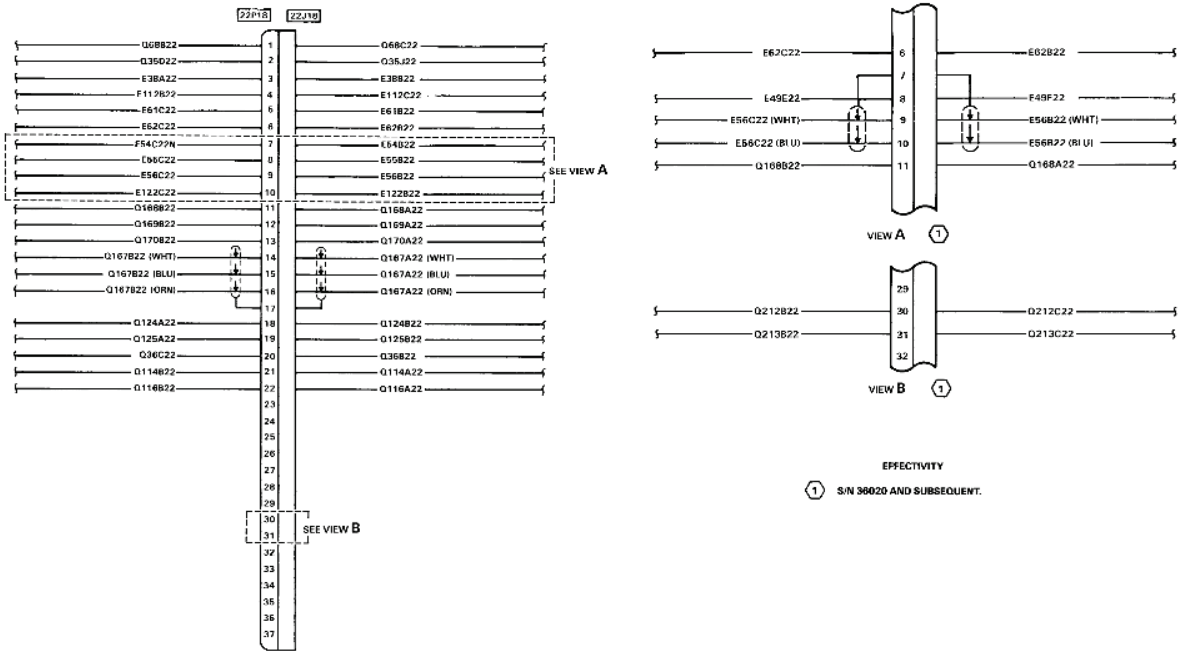
ICN-412-A-980000-B-97499-00358-A-001-01

Figure 1. Component Replacement (Sheet 60 of 87)



ICN-412-A-980000-B-97499-00359-A-001-01

Figure 1. Component Replacement (Sheet 61 of 87)

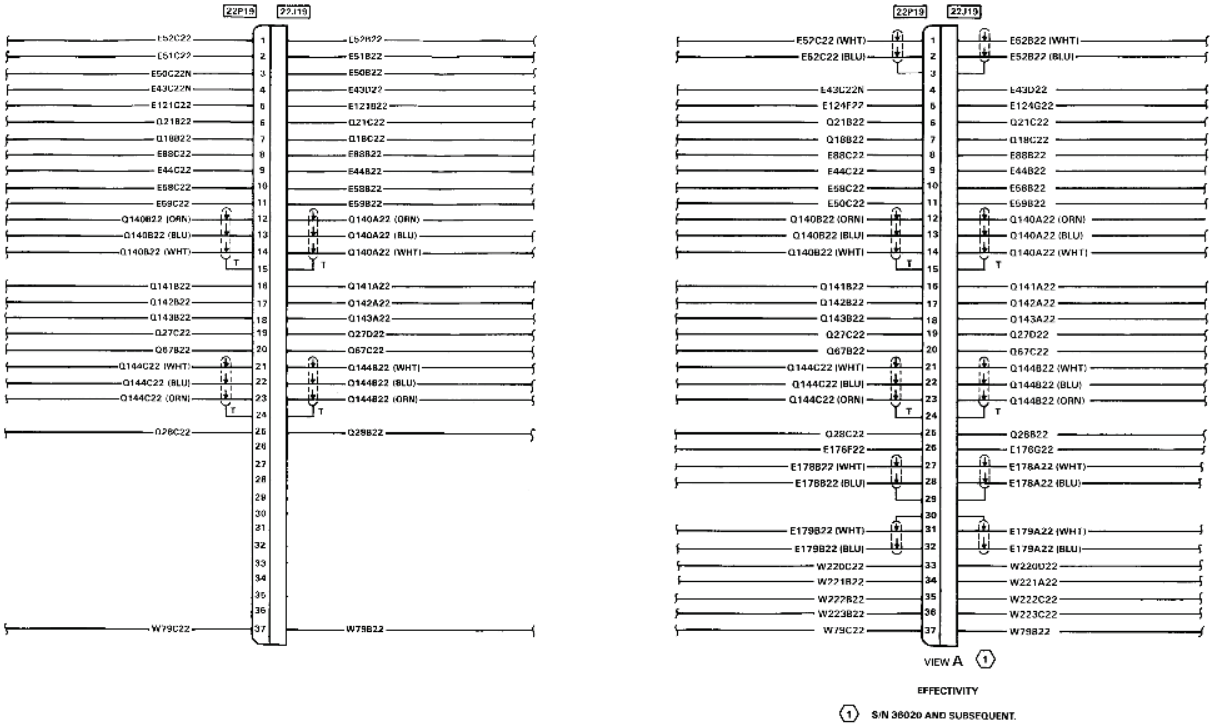


EFFECTIVITY

(1) S/N 36020 AND SUBSEQUENT.

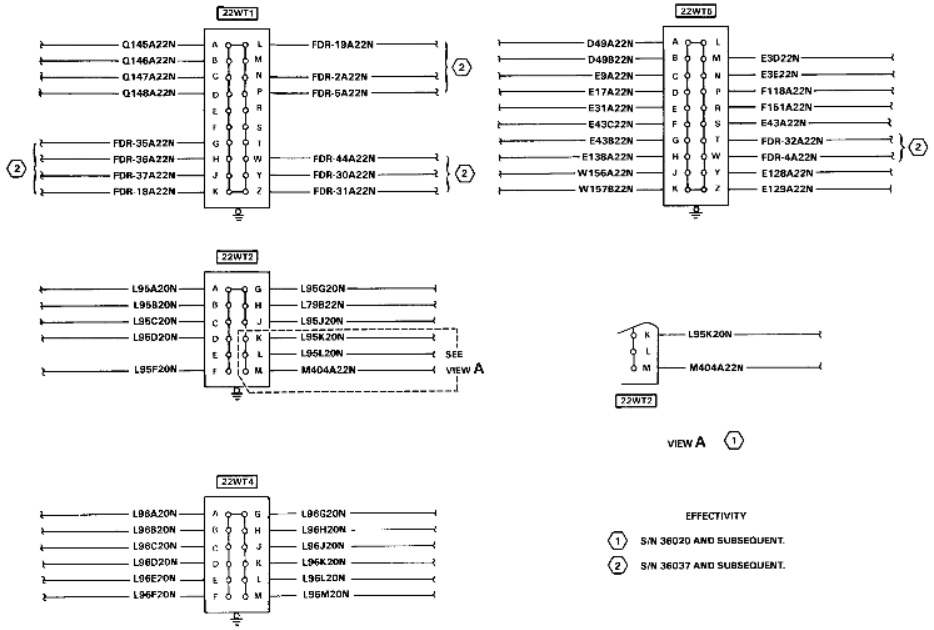
ICN-412-A-980000-B-97499-00360-A-001-01

Figure 1. Component Replacement (Sheet 62 of 87)



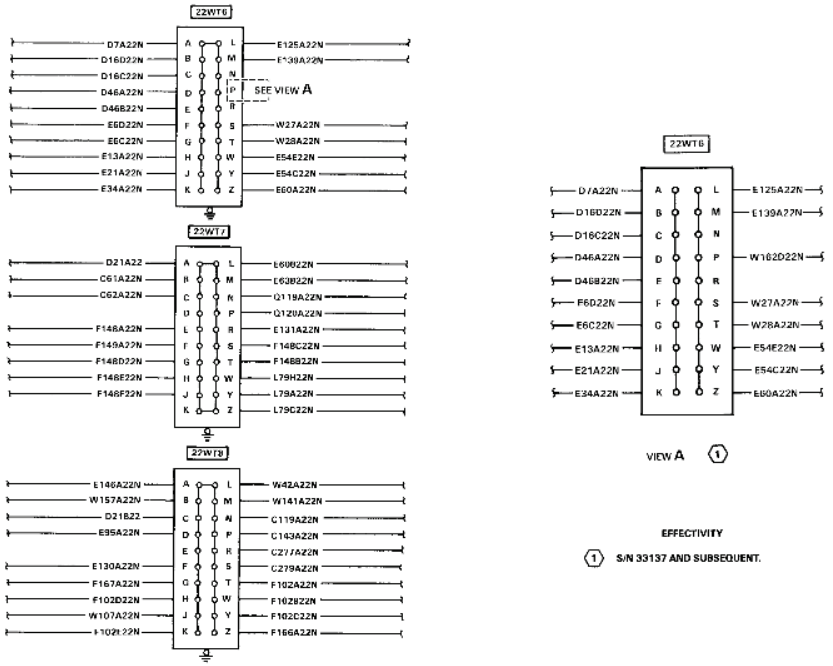
ICN-412-A-980000-B-97499-00361-A-001-01

Figure 1. Component Replacement (Sheet 63 of 87)



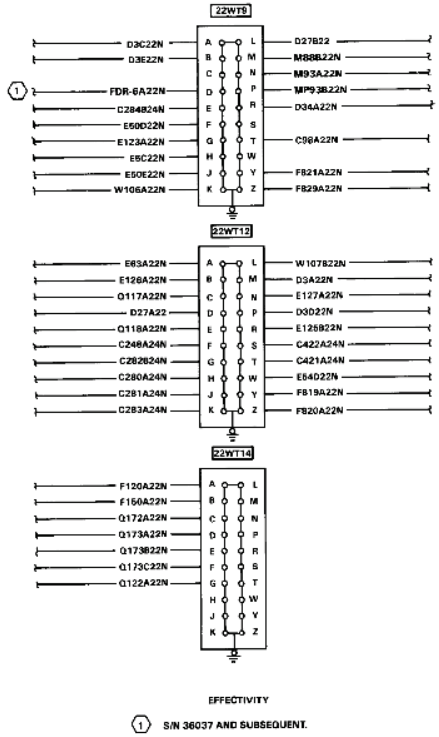
ICN-412-A-980000-B-97499-00362-A-001-01

Figure 1. Component Replacement (Sheet 64 of 87)



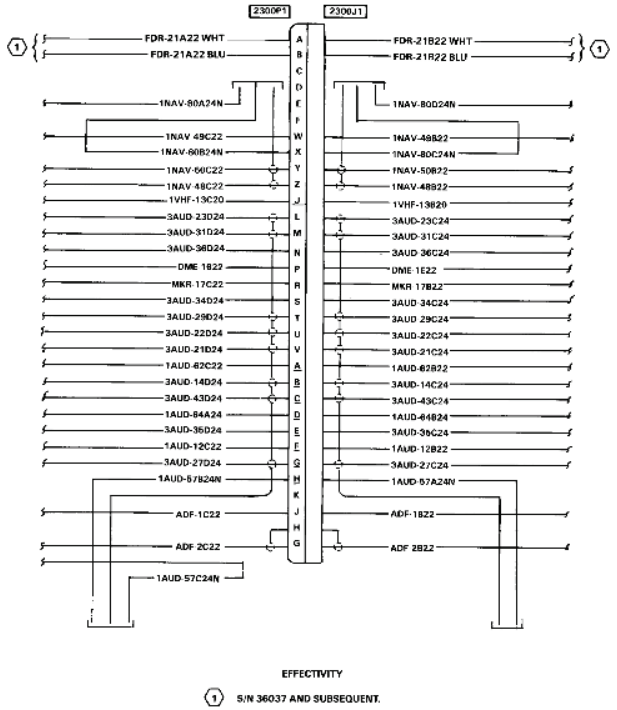
ICN-412-A-980000-B-97499-00363-A-001-01

Figure 1. Component Replacement (Sheet 65 of 87)



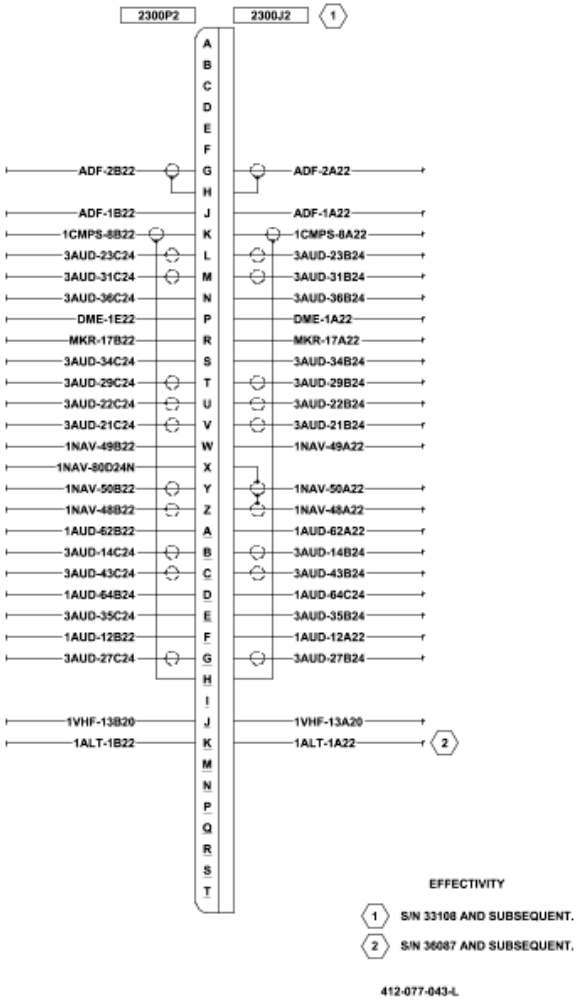
ICN-412-A-980000-B-97499-00364-A-001-01

Figure 1. Component Replacement (Sheet 66 of 87)



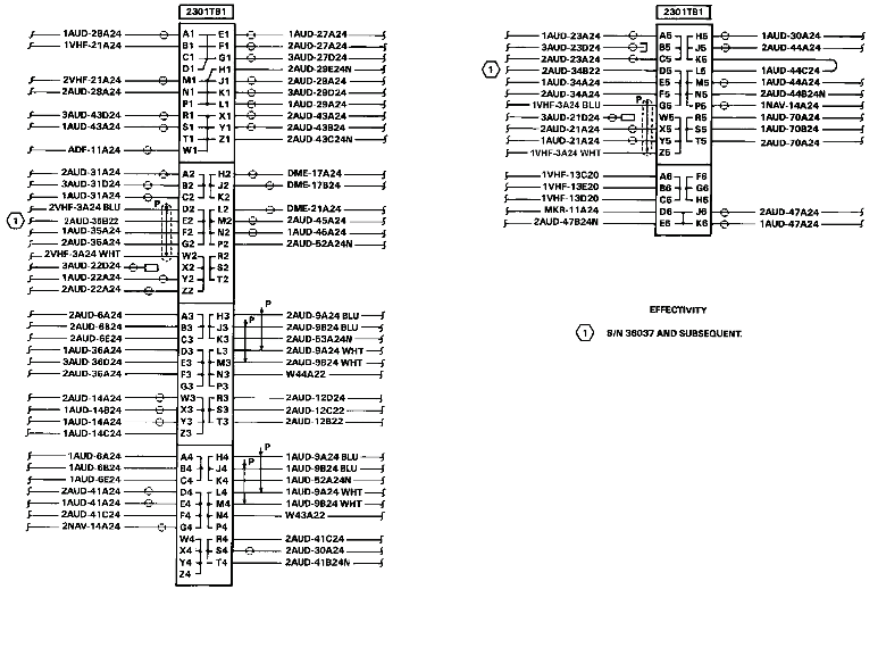
ICN-412-A-980000-B-97499-00365-A-001-01

Figure 1. Component Replacement (Sheet 67 of 87)



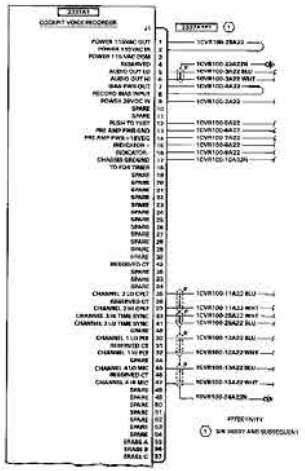
ICN-412-A-980000-B-97499-00366-A-001-01

Figure 1. Component Replacement (Sheet 68 of 87)



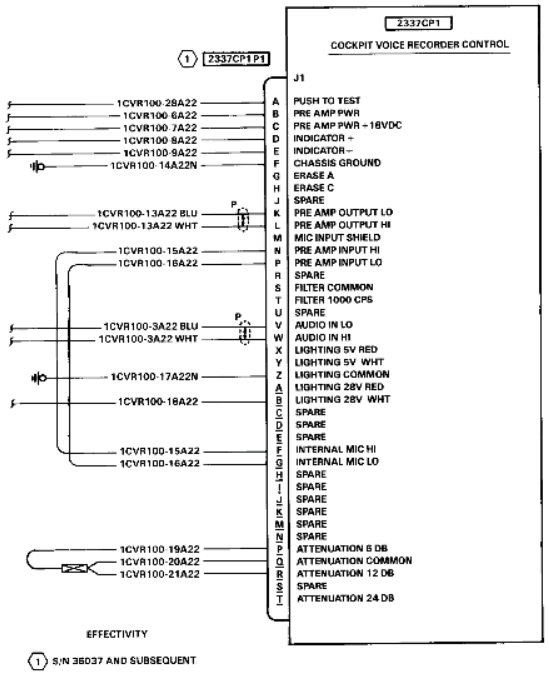
ICN-412-A-980000-B-97499-00367-A-001-01

Figure 1. Component Replacement (Sheet 69 of 87)



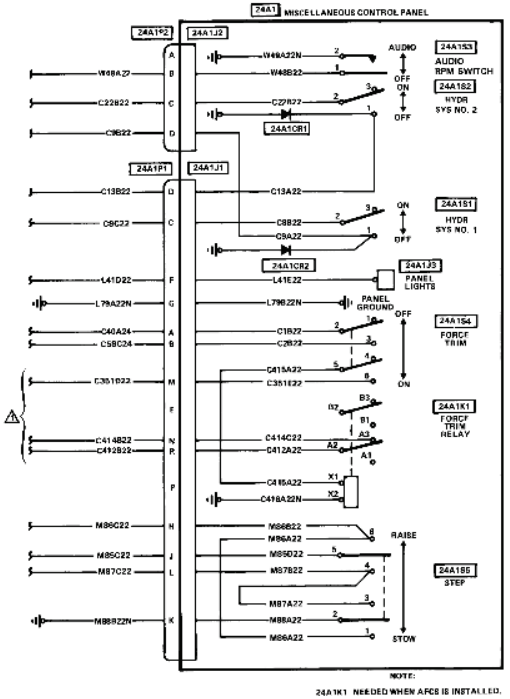
ICN-412-A-980000-B-97499-00368-A-001-01

Figure 1. Component Replacement (Sheet 70 of 87)



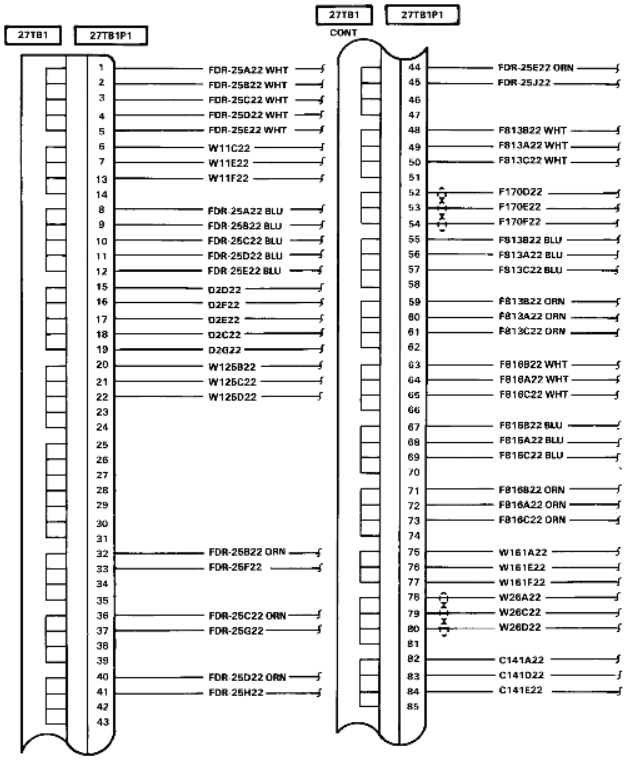
ICN-412-A-980000-B-97499-00369-A-001-01

Figure 1. Component Replacement (Sheet 71 of 87)



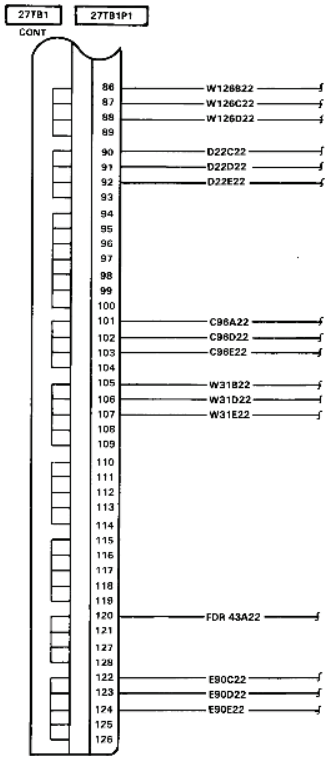
ICN-412-A-980000-B-97499-00370-A-001-01

Figure 1. Component Replacement (Sheet 72 of 87)



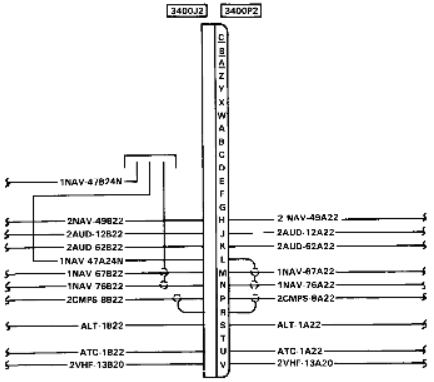
ICN-412-A-980000-B-97499-00371-A-001-01

Figure 1. Component Replacement (Sheet 73 of 87)



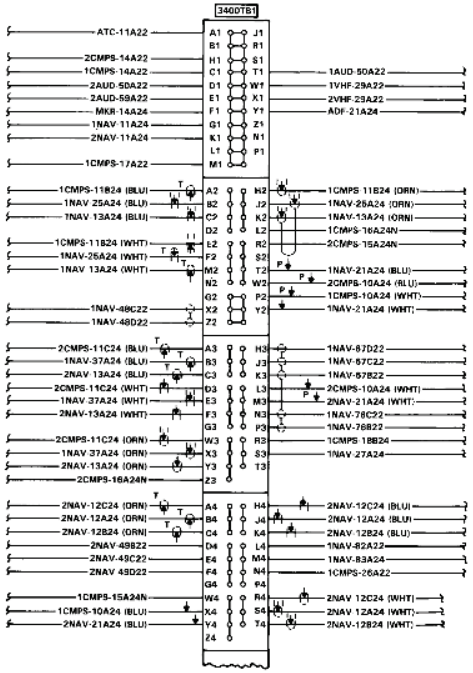
ICN-412-A-980000-B-97499-00372-A-001-01

Figure 1. Component Replacement (Sheet 74 of 87)



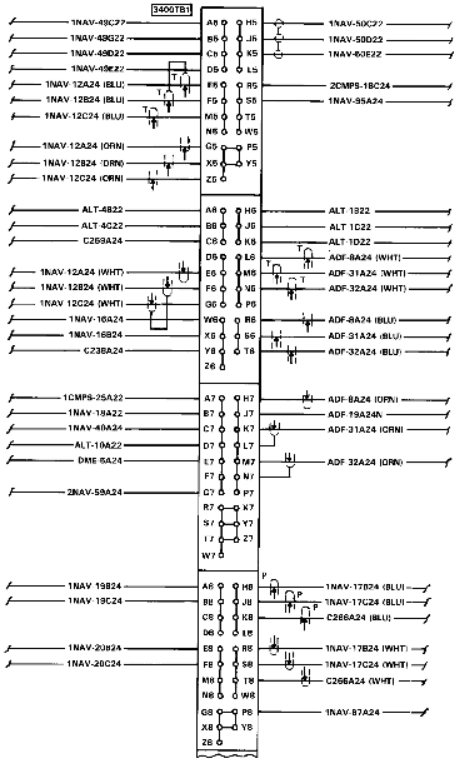
ICN-412-A-980000-B-97499-00373-A-001-01

Figure 1. Component Replacement (Sheet 75 of 87)



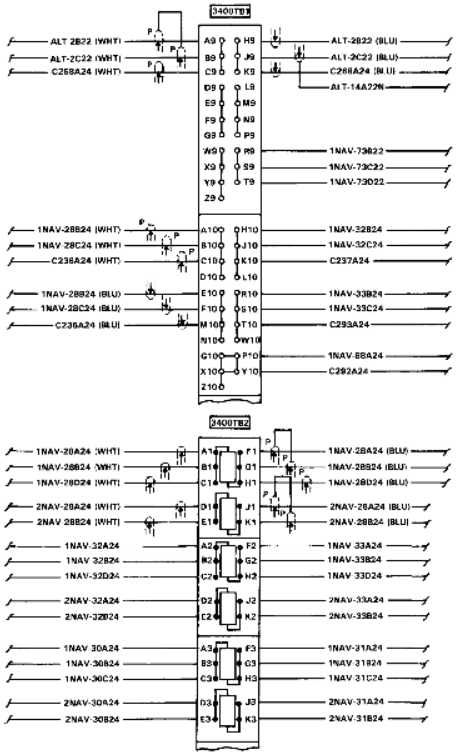
ICN-412-A-980000-B-97499-00374-A-001-01

Figure 1. Component Replacement (Sheet 76 of 87)



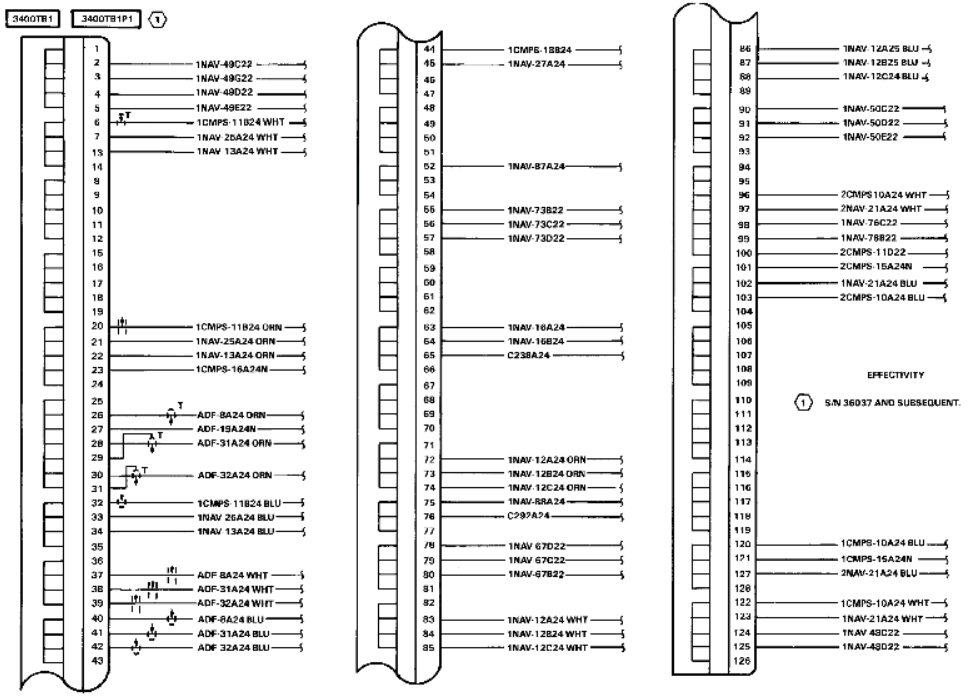
ICN-412-A-980000-B-97499-00375-A-001-01

Figure 1. Component Replacement (Sheet 77 of 87)



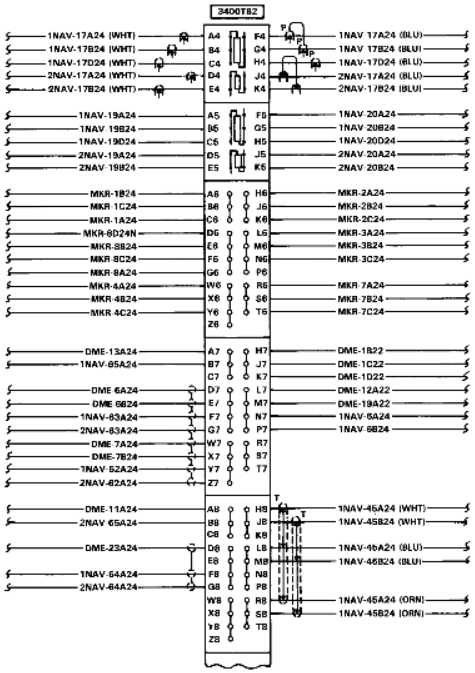
ICN-412-A-980000-B-97499-00376-A-001-01

Figure 1. Component Replacement (Sheet 78 of 87)



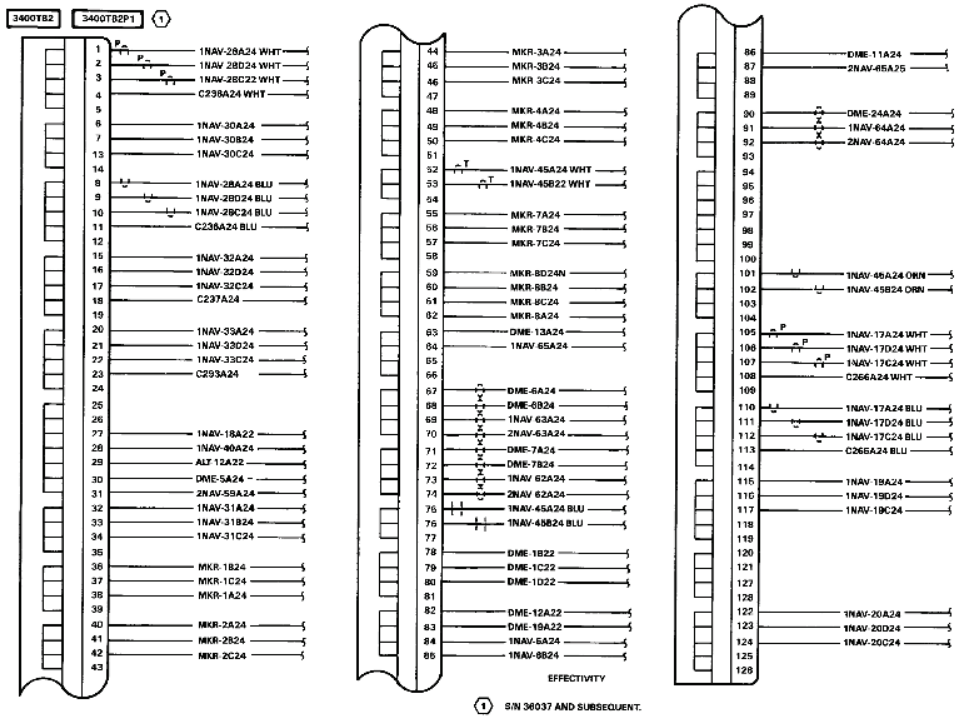
ICN-412-A-980000-B-97499-00377-A-001-01

Figure 1. Component Replacement (Sheet 79 of 87)



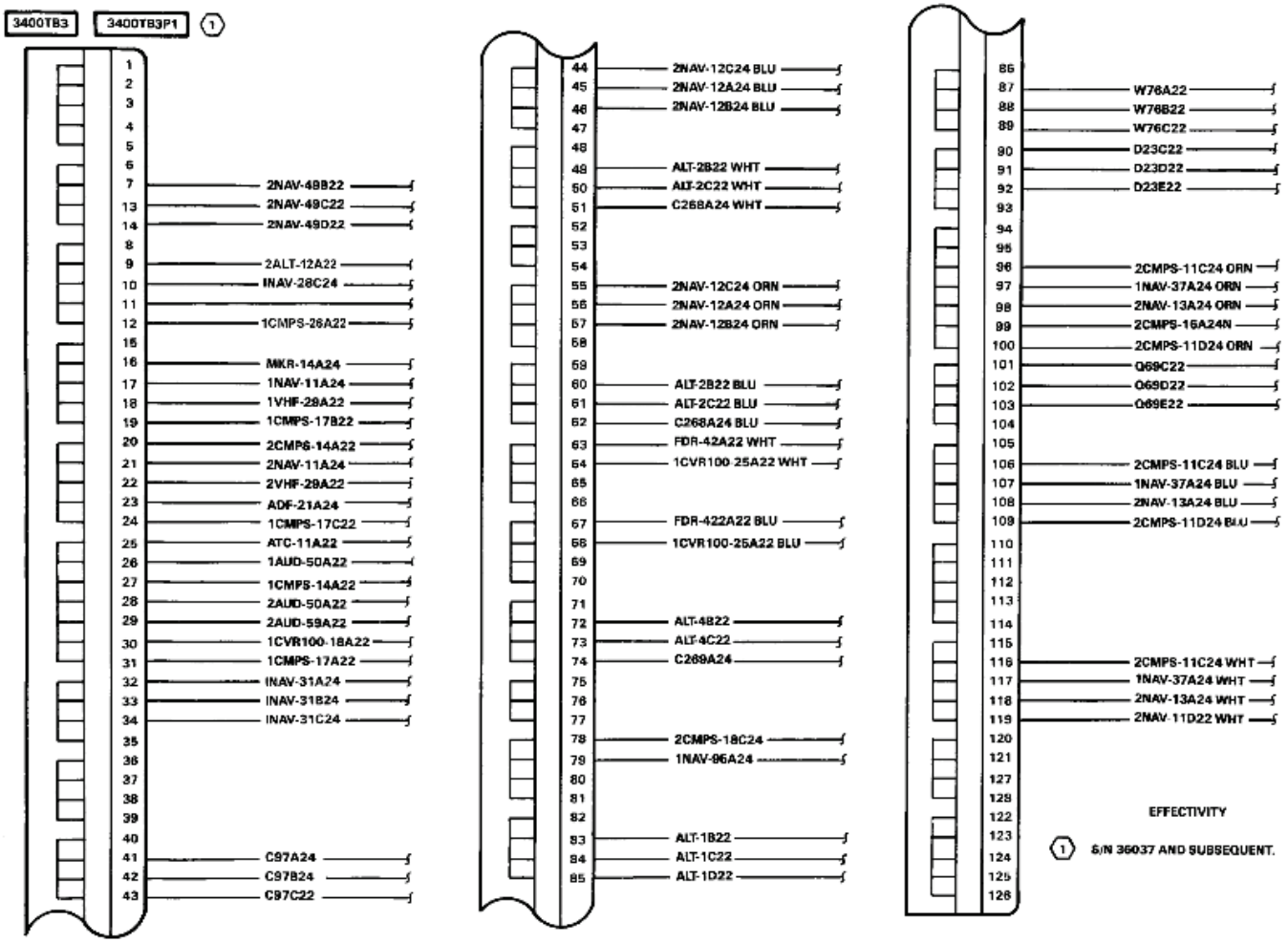
ICN-412-A-980000-B-97499-00378-A-001-01

Figure 1. Component Replacement (Sheet 80 of 87)



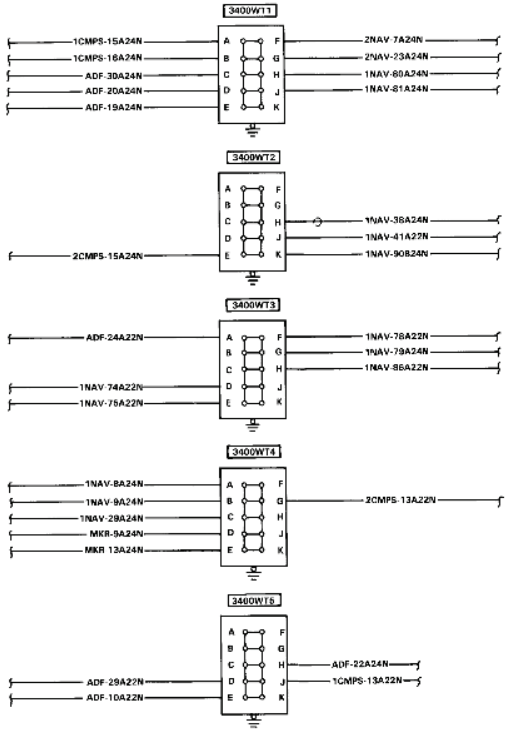
ICN-412-A-980000-B-97499-00379-A-001-01

Figure 1. Component Replacement (Sheet 81 of 87)



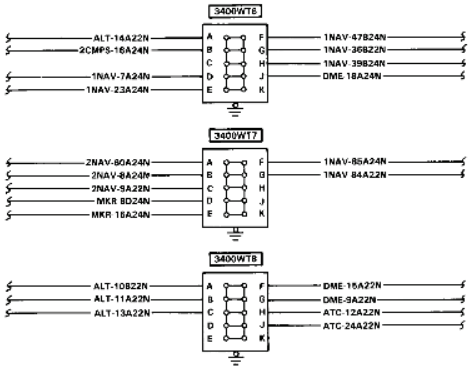
ICN-412-A-980000-B-97499-00380-A-001-01

Figure 1. Component Replacement (Sheet 82 of 87)



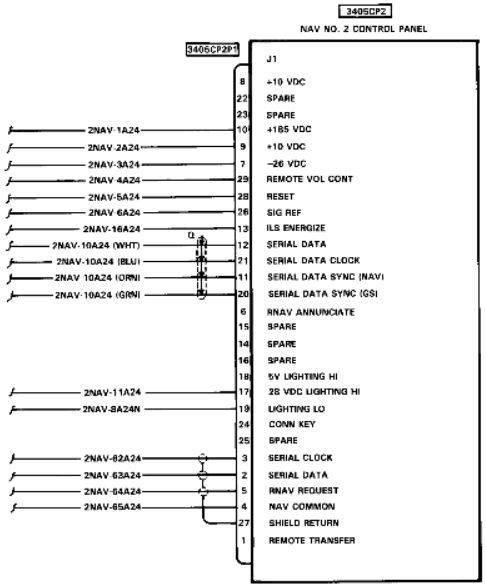
ICN-412-A-980000-B-97499-00381-A-001-01

Figure 1. Component Replacement (Sheet 83 of 87)



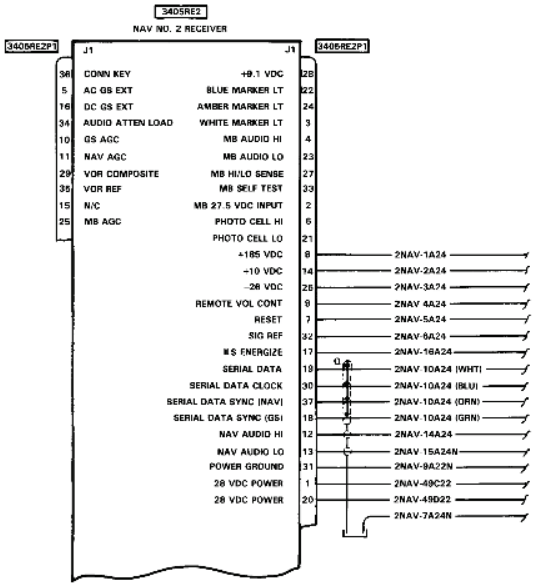
ICN-412-A-980000-B-97499-00382-A-001-01

Figure 1. Component Replacement (Sheet 84 of 87)



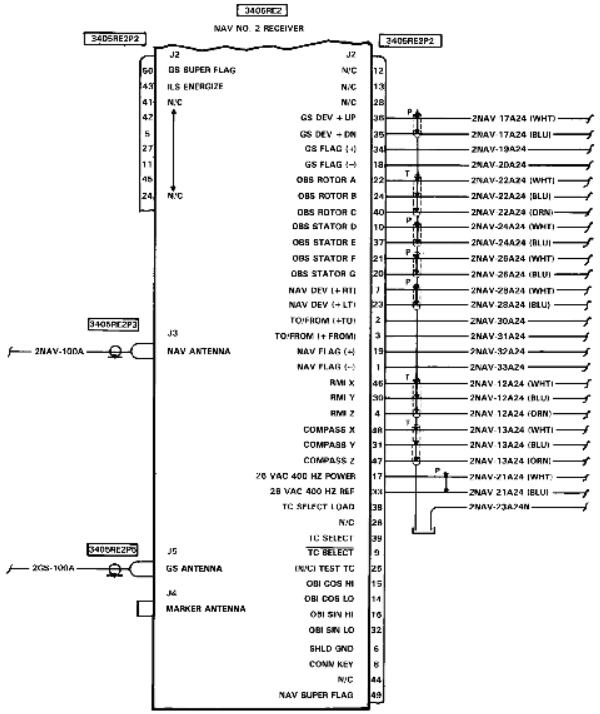
ICN-412-A-980000-B-97499-00383-A-001-01

Figure 1. Component Replacement (Sheet 85 of 87)



ICN-412-A-980000-B-97499-00384-A-001-01

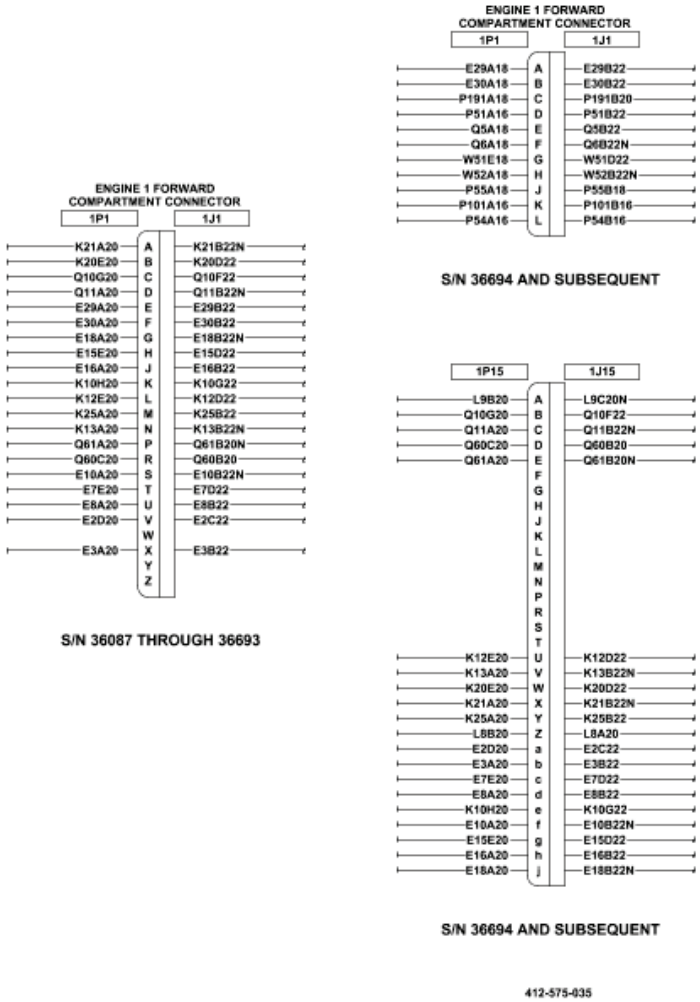
Figure 1. Component Replacement (Sheet 86 of 87)



ICN-412-A-980000-B-97499-00385-A-001-01

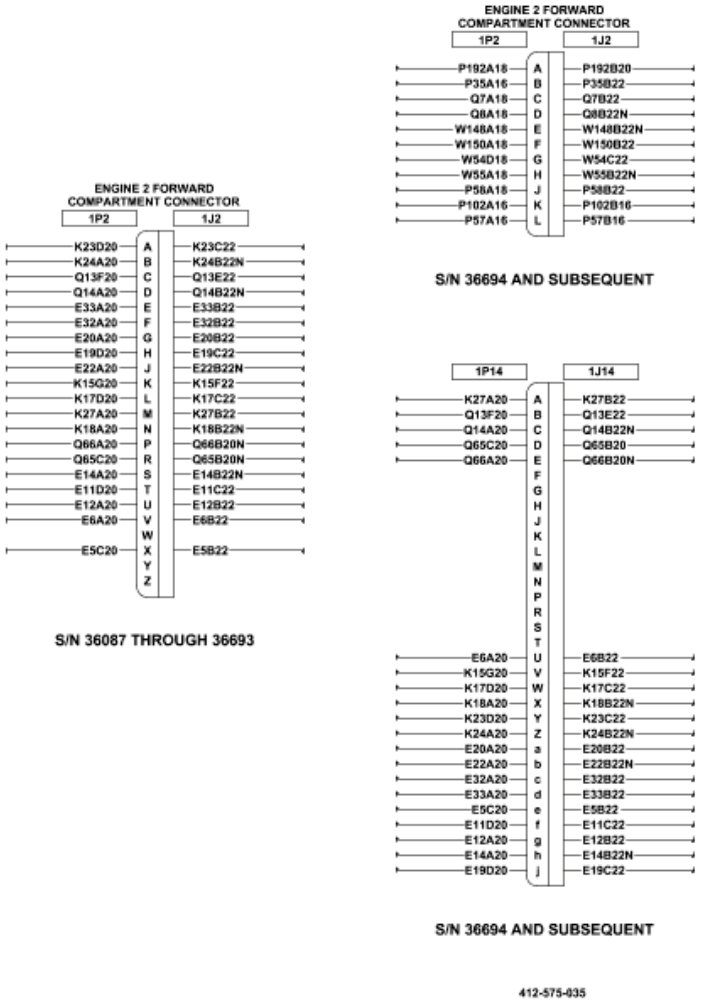
Figure 1. Component Replacement (Sheet 87 of 87)

Intentionally blank page



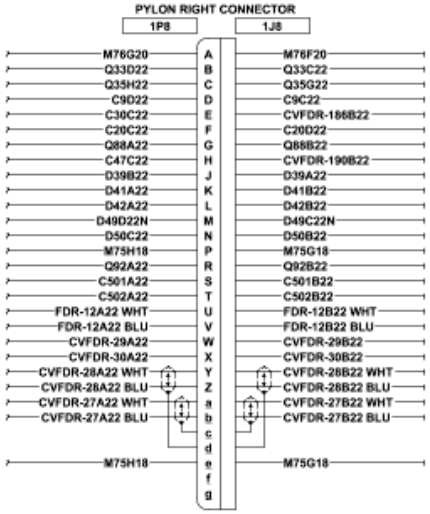
ICN-412-A-980000-B-97499-00386-A-001-01

Figure 1. Component Replacement (Sheet 1 of 104)



ICN-412-A-980000-B-97499-00387-A-001-01

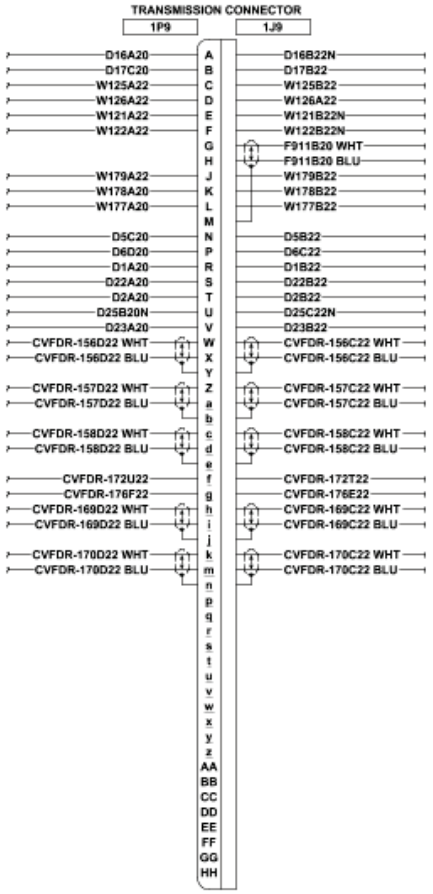
Figure 1. Component Replacement (Sheet 2 of 104)



412-075-064

ICN-412-A-980000-B-97499-00388-A-001-01

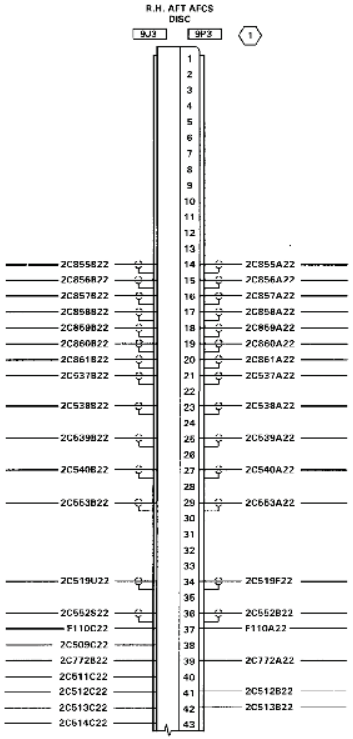
Figure 1. Component Replacement (Sheet 3 of 104)



412-075-064

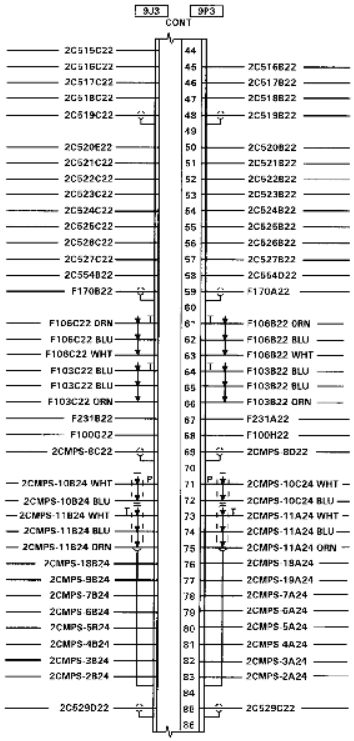
ICN-412-A-980000-B-97499-00389-A-001-01

Figure 1. Component Replacement (Sheet 4 of 104)



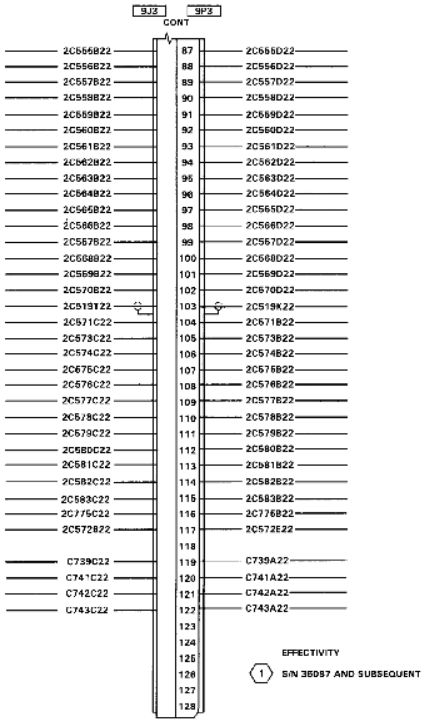
ICN-412-A-980000-B-97499-00390-A-001-01

Figure 1. Component Replacement (Sheet 5 of 104)



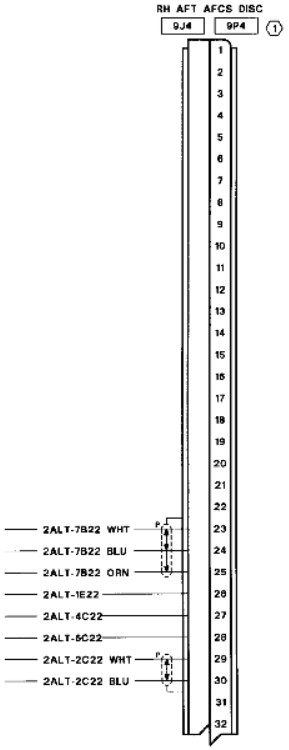
ICN-412-A-980000-B-97499-00391-A-001-01

Figure 1. Component Replacement (Sheet 6 of 104)



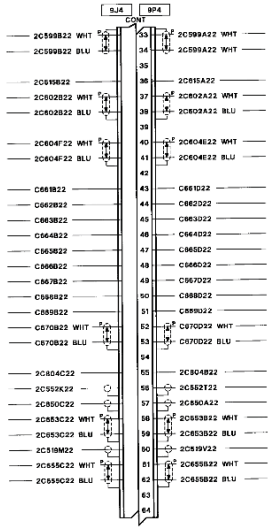
ICN-412-A-980000-B-97499-00392-A-001-01

Figure 1. Component Replacement (Sheet 7 of 104)



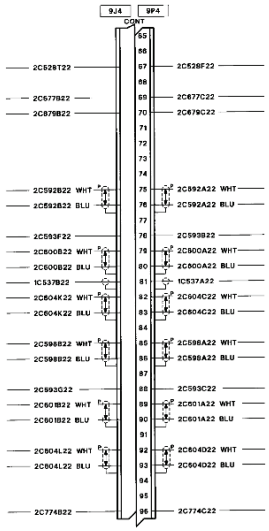
ICN-412-A-980000-B-97499-00393-A-001-01

Figure 1. Component Replacement (Sheet 8 of 104)



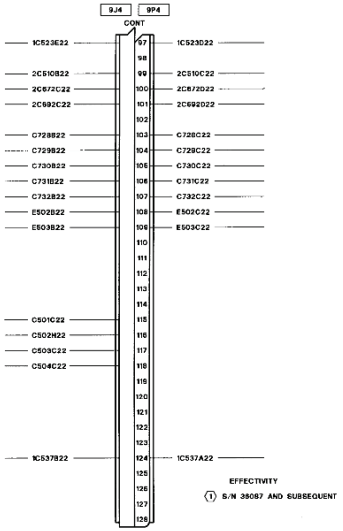
ICN-412-A-980000-B-97499-00394-A-001-01

Figure 1. Component Replacement (Sheet 9 of 104)



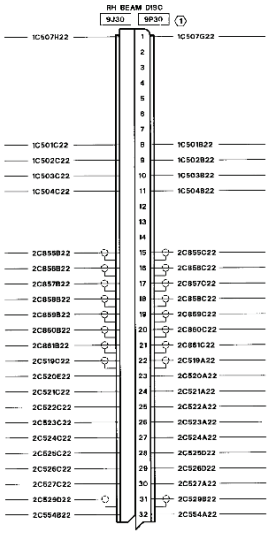
ICN-412-A-980000-B-97499-00395-A-001-01

Figure 1. Component Replacement (Sheet 10 of 104)



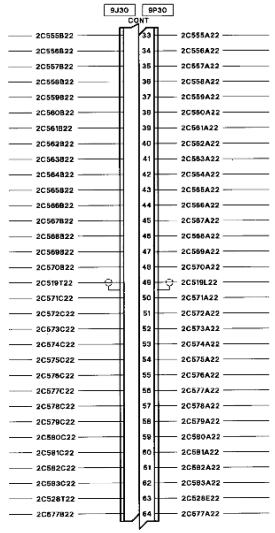
ICN-412-A-980000-B-97499-00396-A-001-01

Figure 1. Component Replacement (Sheet 11 of 104)



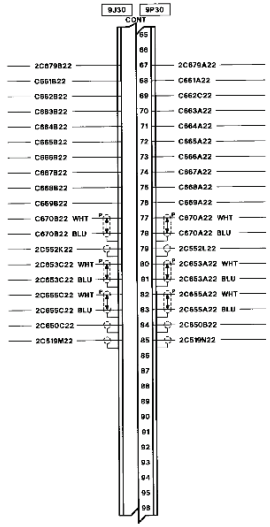
ICN-412-A-980000-B-97499-00397-A-001-01

Figure 1. Component Replacement (Sheet 12 of 104)



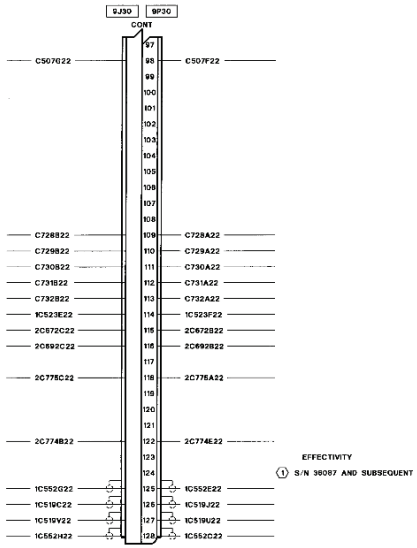
ICN-412-A-980000-B-97499-00398-A-001-01

Figure 1. Component Replacement (Sheet 13 of 104)



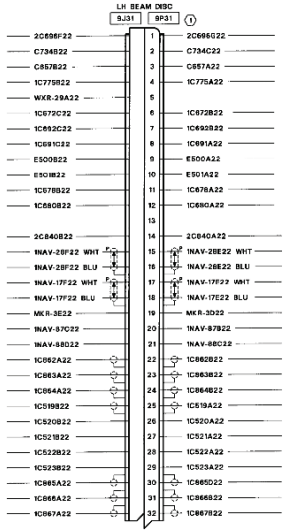
ICN-412-A-980000-B-97499-00399-A-001-01

Figure 1. Component Replacement (Sheet 14 of 104)



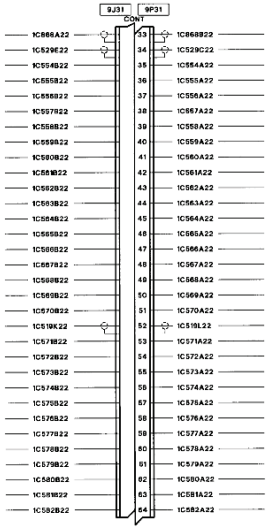
ICN-412-A-980000-B-97499-00400-A-001-01

Figure 1. Component Replacement (Sheet 15 of 104)



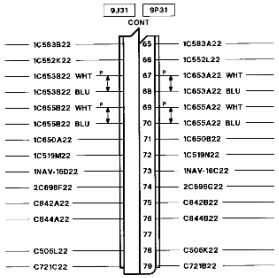
ICN-412-A-980000-B-97499-00401-A-001-01

Figure 1. Component Replacement (Sheet 16 of 104)



ICN-412-A-980000-B-97499-00402-A-001-01

Figure 1. Component Replacement (Sheet 17 of 104)

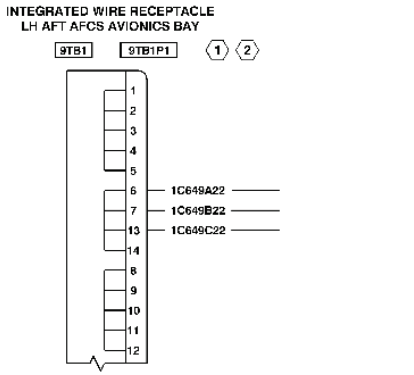


EFFECTIVITY

(1) S/N 36087 AND SUBSEQUENT

ICN-412-A-980000-B-97499-00403-A-001-01

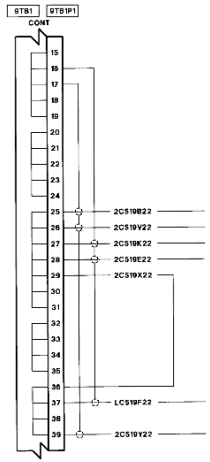
Figure 1. Component Replacement (Sheet 18 of 104)



412-475-017N

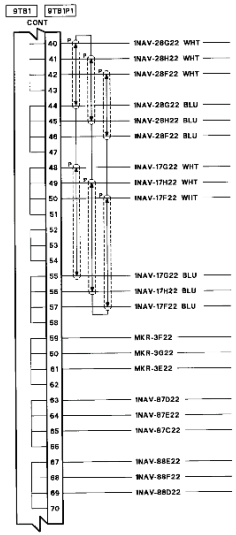
ICN-412-A-980000-B-97499-00404-A-001-01

Figure 1. Component Replacement (Sheet 19 of 104)



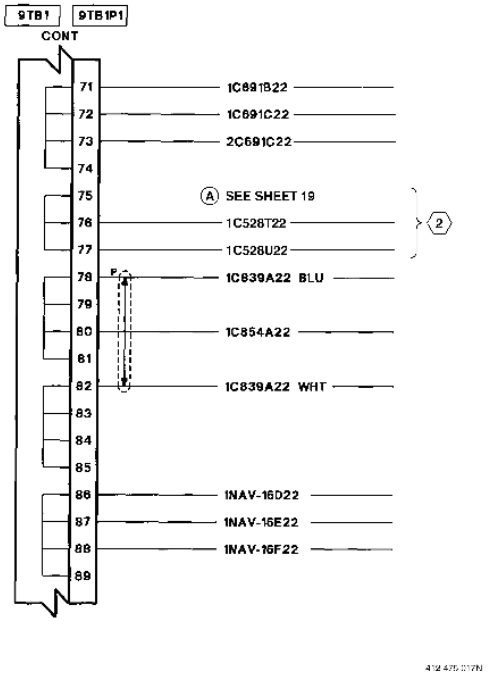
ICN-412-A-980000-B-97499-00405-A-001-01

Figure 1. Component Replacement (Sheet 20 of 104)



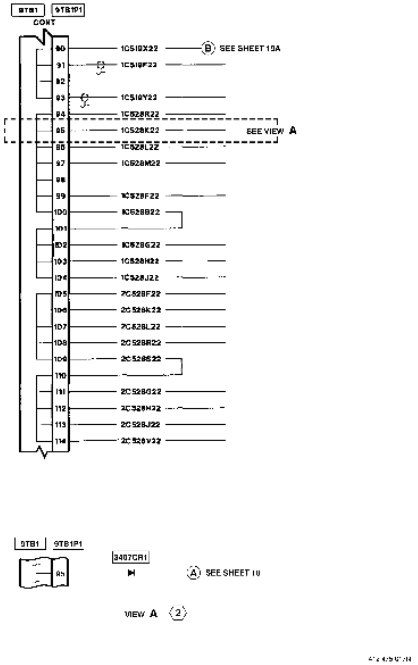
ICN-412-A-980000-B-97499-00406-A-001-01

Figure 1. Component Replacement (Sheet 21 of 104)



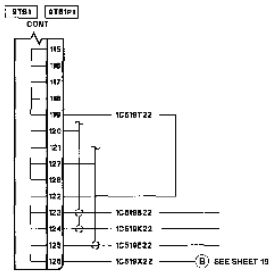
ICN-412-A-980000-B-97499-00407-A-001-01

Figure 1. Component Replacement (Sheet 22 of 104)



ICN-412-A-980000-B-97499-00408-A-001-01

Figure 1. Component Replacement (Sheet 23 of 104)



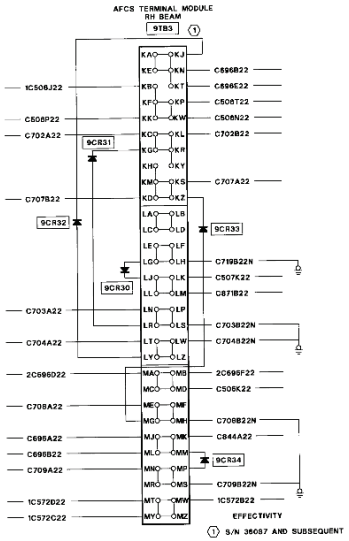
EFFECTIVITY

1 S/N 36087 AND SUBSEQUENT

2 S/N 36249 AND SUBSEQUENT

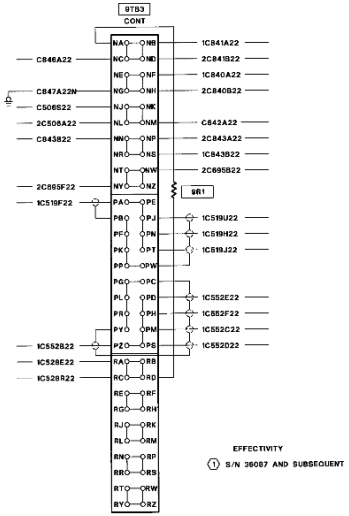
ICN-412-A-980000-B-97499-00409-A-001-01

Figure 1. Component Replacement (Sheet 24 of 104)



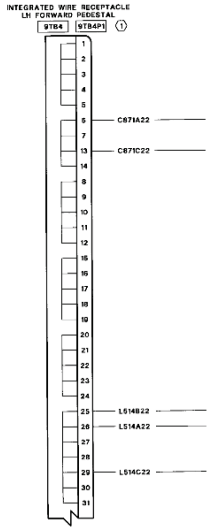
ICN-412-A-980000-B-97499-00410-A-001-01

Figure 1. Component Replacement (Sheet 25 of 104)



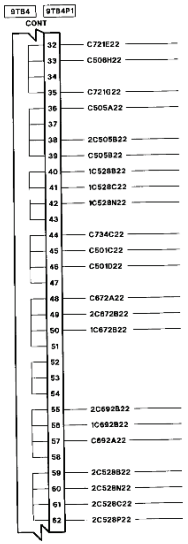
ICN-412-A-980000-B-97499-00411-A-001-01

Figure 1. Component Replacement (Sheet 26 of 104)



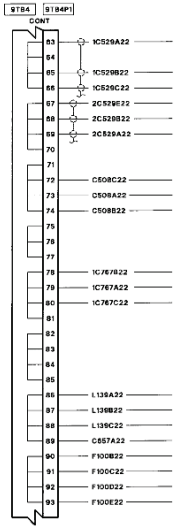
ICN-412-A-980000-B-97499-00412-A-001-01

Figure 1. Component Replacement (Sheet 27 of 104)



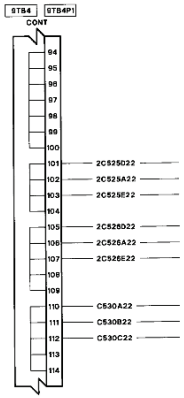
ICN-412-A-980000-B-97499-00413-A-001-01

Figure 1. Component Replacement (Sheet 28 of 104)



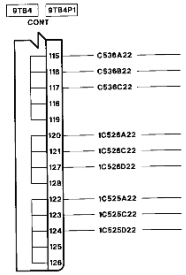
ICN-412-A-980000-B-97499-00414-A-001-01

Figure 1. Component Replacement (Sheet 29 of 104)



ICN-412-A-980000-B-97499-00415-A-001-01

Figure 1. Component Replacement (Sheet 30 of 104)

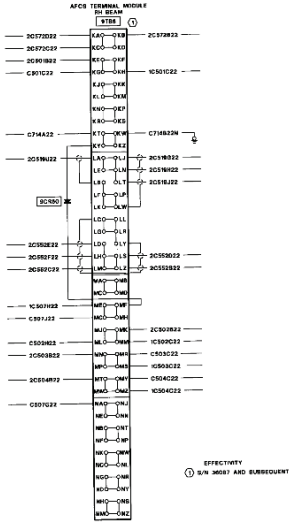


EFFECTIVITY

(1) S/N 36087 AND SUBSEQUENT

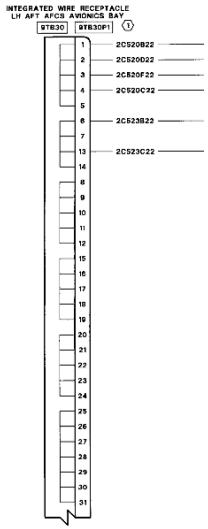
ICN-412-A-980000-B-97499-00416-A-001-01

Figure 1. Component Replacement (Sheet 31 of 104)



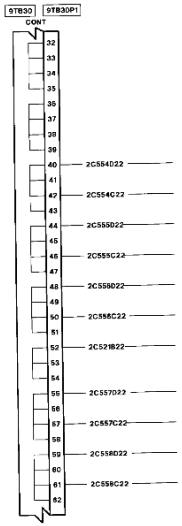
ICN-412-A-980000-B-97499-00417-A-001-01

Figure 1. Component Replacement (Sheet 32 of 104)



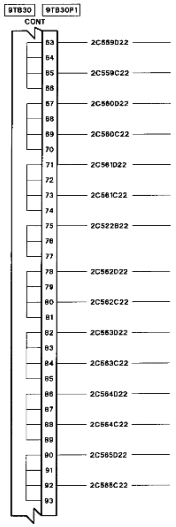
ICN-412-A-980000-B-97499-00418-A-001-01

Figure 1. Component Replacement (Sheet 33 of 104)



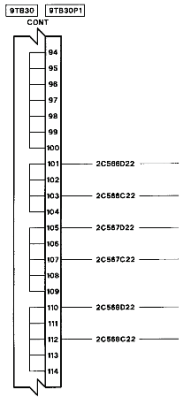
ICN-412-A-980000-B-97499-00419-A-001-01

Figure 1. Component Replacement (Sheet 34 of 104)



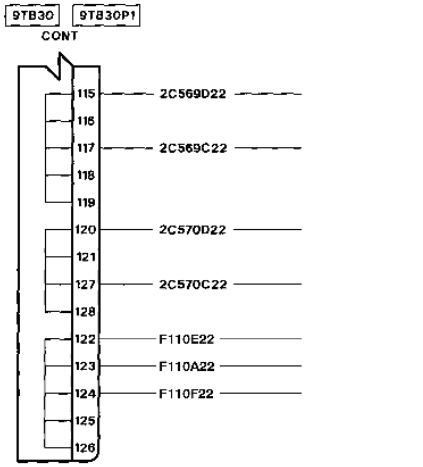
ICN-412-A-980000-B-97499-00420-A-001-01

Figure 1. Component Replacement (Sheet 35 of 104)



ICN-412-A-980000-B-97499-00421-A-001-01

Figure 1. Component Replacement (Sheet 36 of 104)



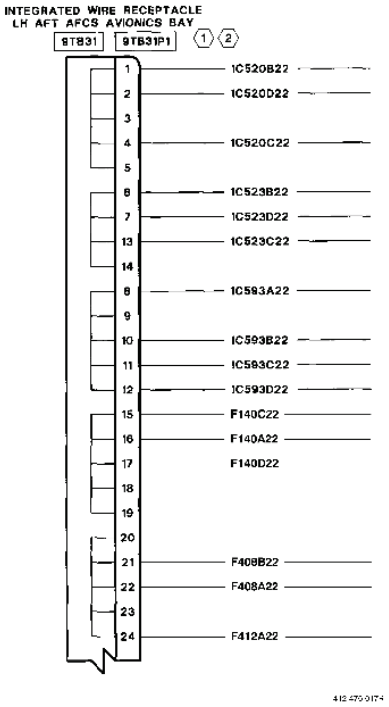
EFFECTIVITY

(1) S/N 36087 AND SUBSEQUENT

412-173-017

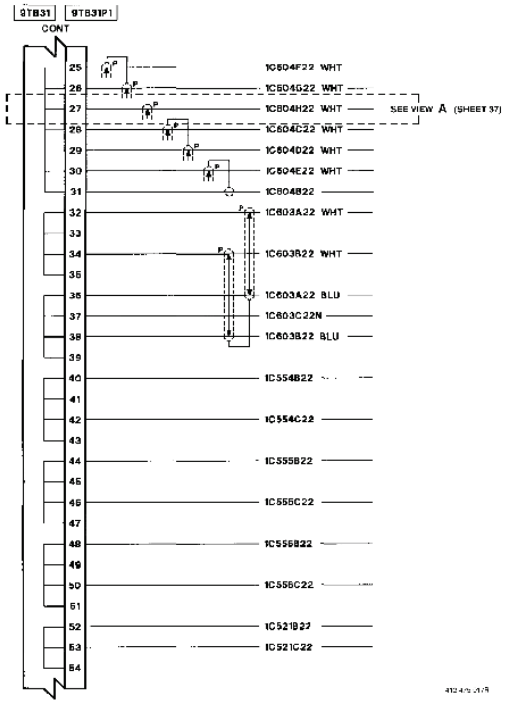
ICN-412-A-980000-B-97499-00422-A-001-01

Figure 1. Component Replacement (Sheet 37 of 104)



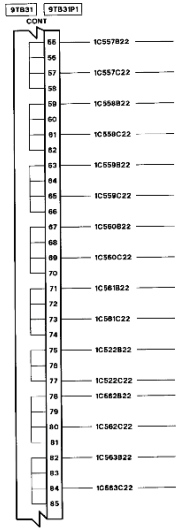
ICN-412-A-980000-B-97499-00423-A-001-01

Figure 1. Component Replacement (Sheet 38 of 104)



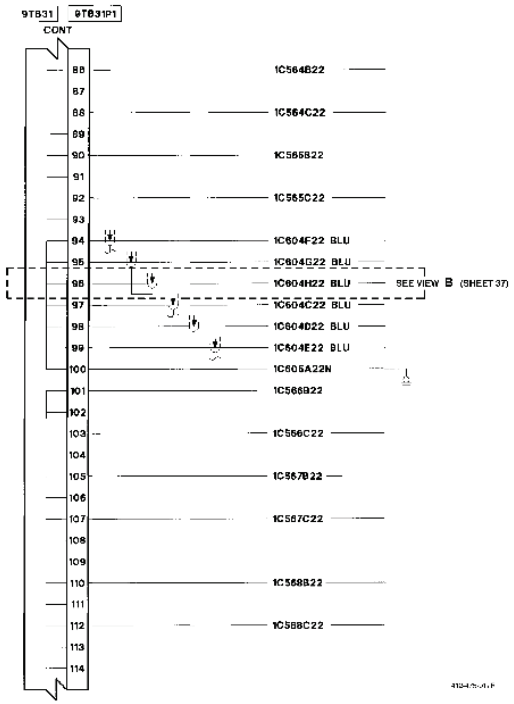
ICN-412-A-980000-B-97499-00424-A-001-01

Figure 1. Component Replacement (Sheet 39 of 104)



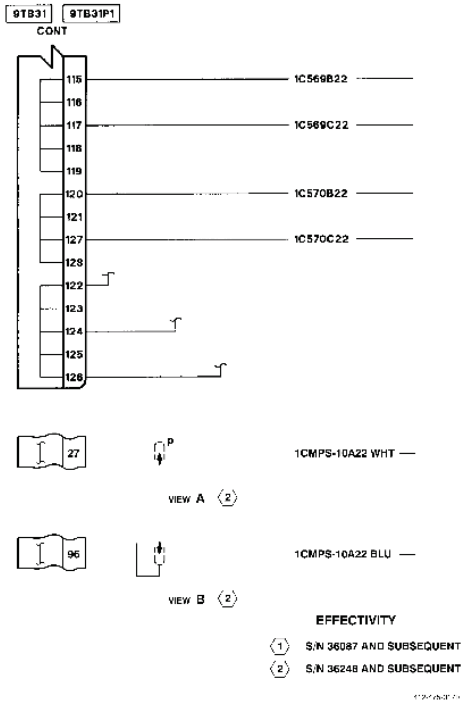
ICN-412-A-980000-B-97499-00425-A-001-01

Figure 1. Component Replacement (Sheet 40 of 104)



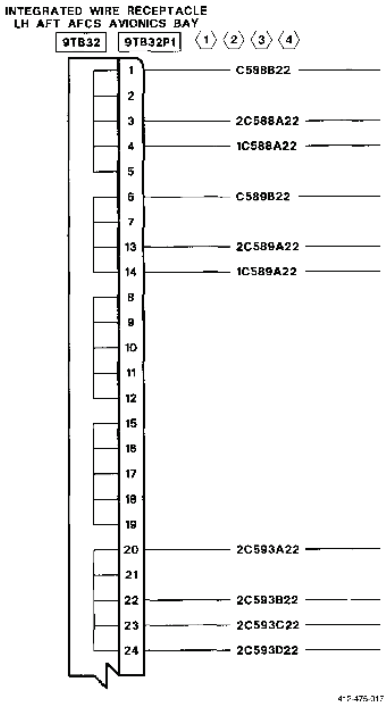
ICN-412-A-980000-B-97499-00426-A-001-01

Figure 1. Component Replacement (Sheet 41 of 104)



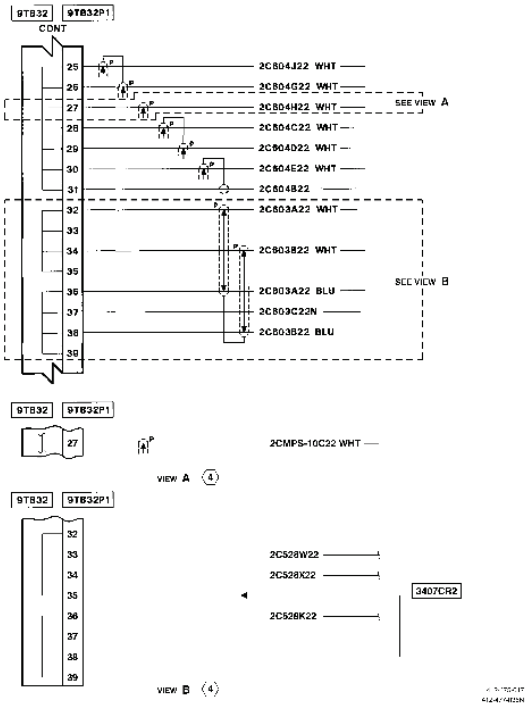
ICN-412-A-980000-B-97499-00427-A-001-01

Figure 1. Component Replacement (Sheet 42 of 104)



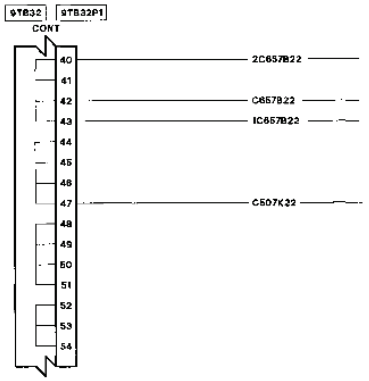
ICN-412-A-980000-B-97499-00428-A-001-01

Figure 1. Component Replacement (Sheet 43 of 104)



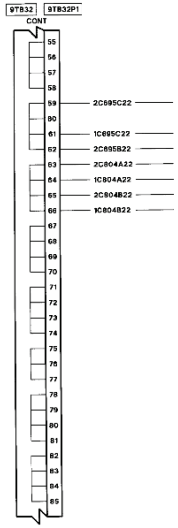
ICN-412-A-980000-B-97499-00429-A-001-01

Figure 1. Component Replacement (Sheet 44 of 104)



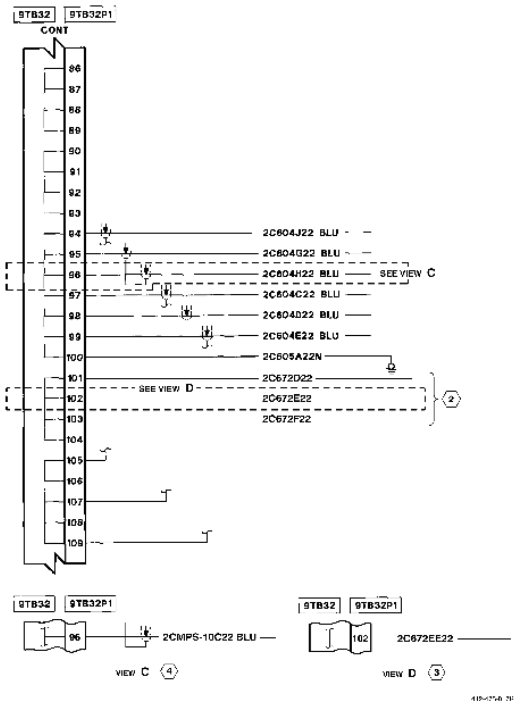
ICN-412-A-980000-B-97499-00430-A-001-01

Figure 1. Component Replacement (Sheet 45 of 104)



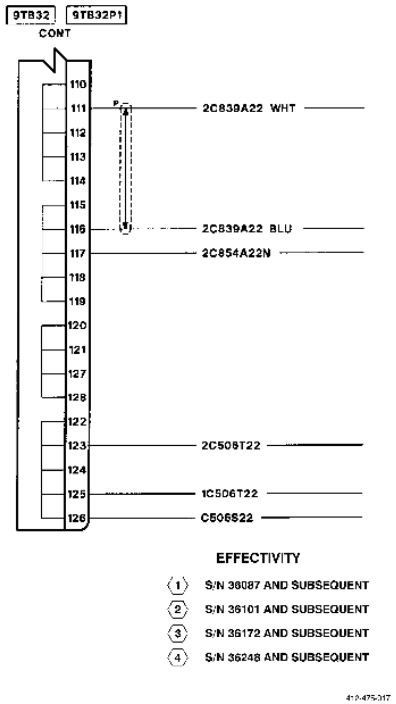
ICN-412-A-980000-B-97499-00431-A-001-01

Figure 1. Component Replacement (Sheet 46 of 104)



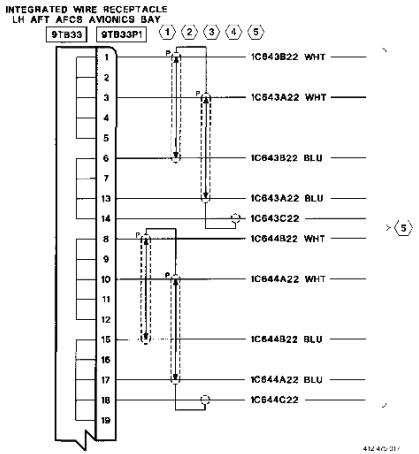
ICN-412-A-980000-B-97499-00432-A-001-01

Figure 1. Component Replacement (Sheet 47 of 104)



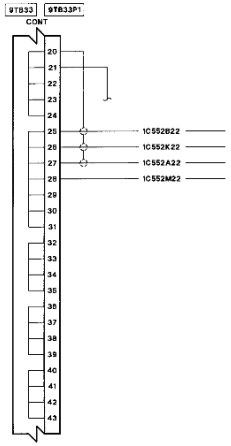
ICN-412-A-980000-B-97499-00433-A-001-01

Figure 1. Component Replacement (Sheet 48 of 104)



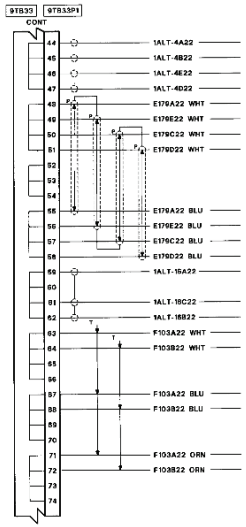
ICN-412-A-980000-B-97499-00434-A-001-01

Figure 1. Component Replacement (Sheet 49 of 104)



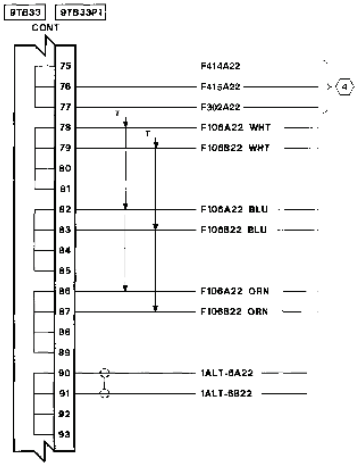
ICN-412-A-980000-B-97499-00435-A-001-01

Figure 1. Component Replacement (Sheet 50 of 104)



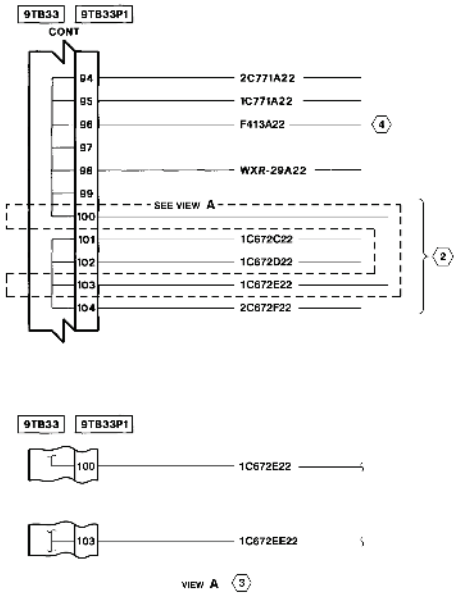
ICN-412-A-980000-B-97499-00436-A-001-01

Figure 1. Component Replacement (Sheet 51 of 104)



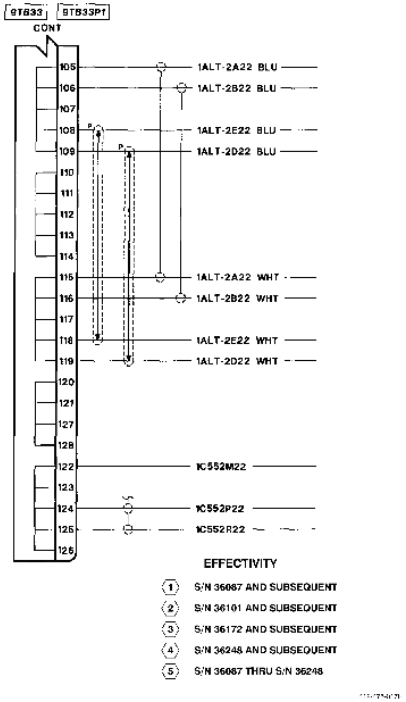
ICN-412-A-980000-B-97499-00437-A-001-01

Figure 1. Component Replacement (Sheet 52 of 104)



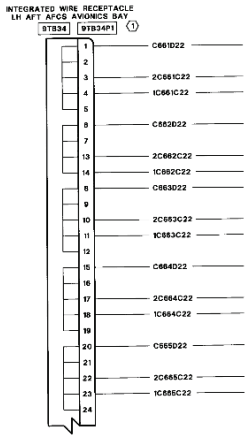
ICN-412-A-980000-B-97499-00438-A-001-01

Figure 1. Component Replacement (Sheet 53 of 104)



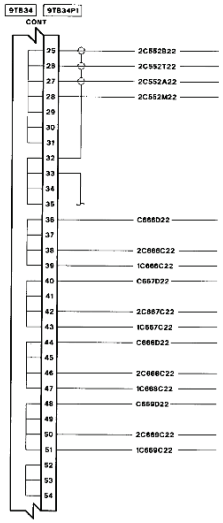
ICN-412-A-980000-B-97499-00439-A-001-01

Figure 1. Component Replacement (Sheet 54 of 104)



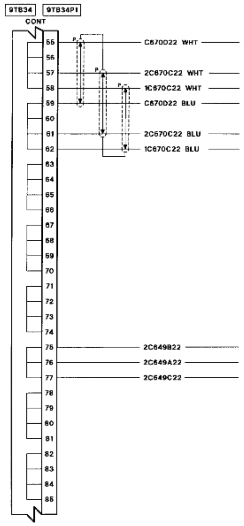
ICN-412-A-980000-B-97499-00440-A-001-01

Figure 1. Component Replacement (Sheet 55 of 104)



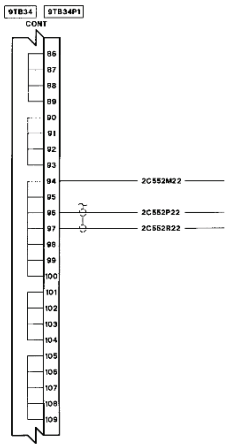
ICN-412-A-980000-B-97499-00441-A-001-01

Figure 1. Component Replacement (Sheet 56 of 104)



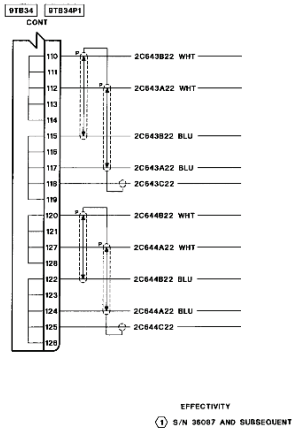
ICN-412-A-980000-B-97499-00442-A-001-01

Figure 1. Component Replacement (Sheet 57 of 104)



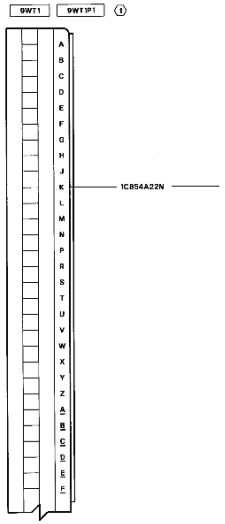
ICN-412-A-980000-B-97499-00443-A-001-01

Figure 1. Component Replacement (Sheet 58 of 104)



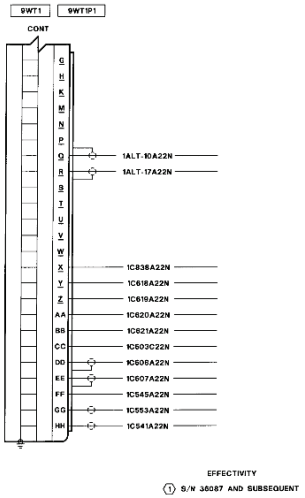
ICN-412-A-980000-B-97499-00444-A-001-01

Figure 1. Component Replacement (Sheet 59 of 104)



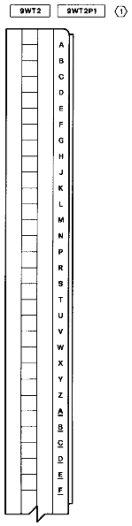
ICN-412-A-980000-B-97499-00445-A-001-01

Figure 1. Component Replacement (Sheet 60 of 104)



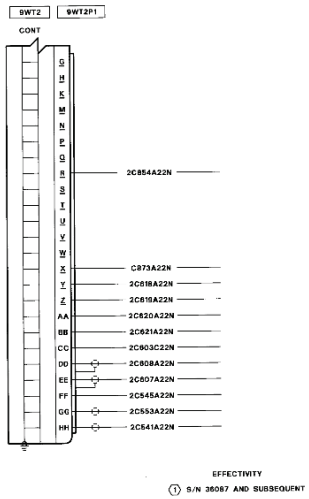
ICN-412-A-980000-B-97499-00446-A-001-01

Figure 1. Component Replacement (Sheet 61 of 104)



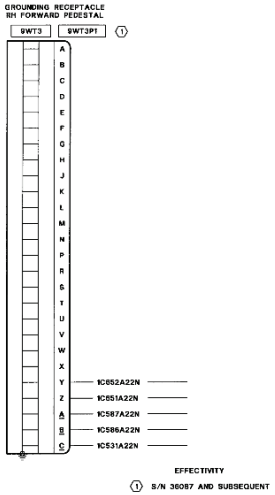
ICN-412-A-980000-B-97499-00447-A-001-01

Figure 1. Component Replacement (Sheet 62 of 104)



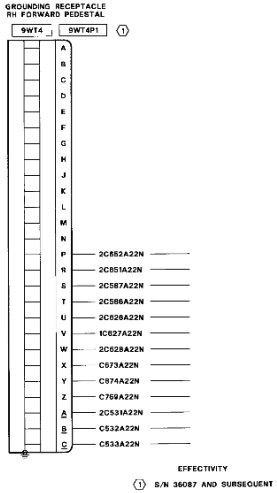
ICN-412-A-980000-B-97499-00448-A-001-01

Figure 1. Component Replacement (Sheet 63 of 104)



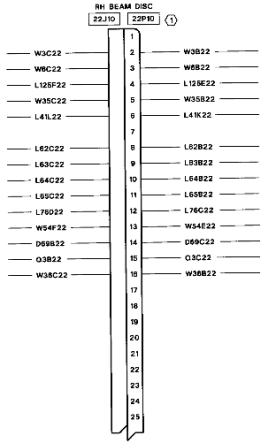
ICN-412-A-980000-B-97499-00449-A-001-01

Figure 1. Component Replacement (Sheet 64 of 104)



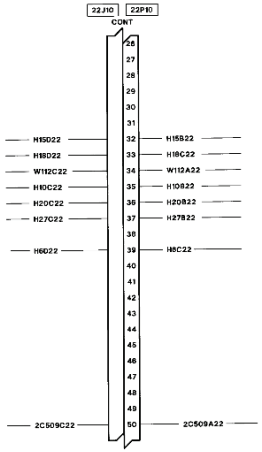
ICN-412-A-980000-B-97499-00450-A-001-01

Figure 1. Component Replacement (Sheet 65 of 104)



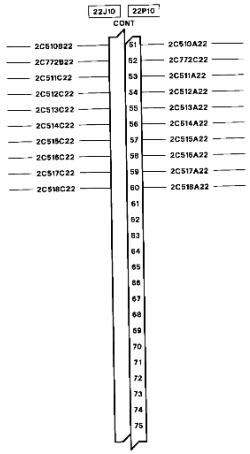
ICN-412-A-980000-B-97499-00451-A-001-01

Figure 1. Component Replacement (Sheet 66 of 104)



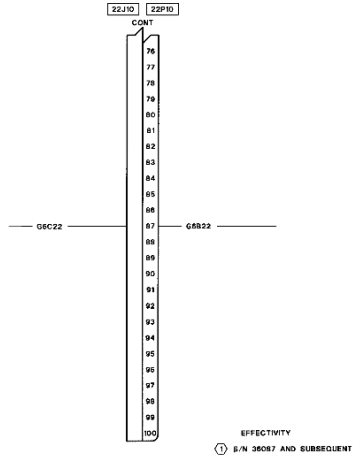
ICN-412-A-980000-B-97499-00452-A-001-01

Figure 1. Component Replacement (Sheet 67 of 104)



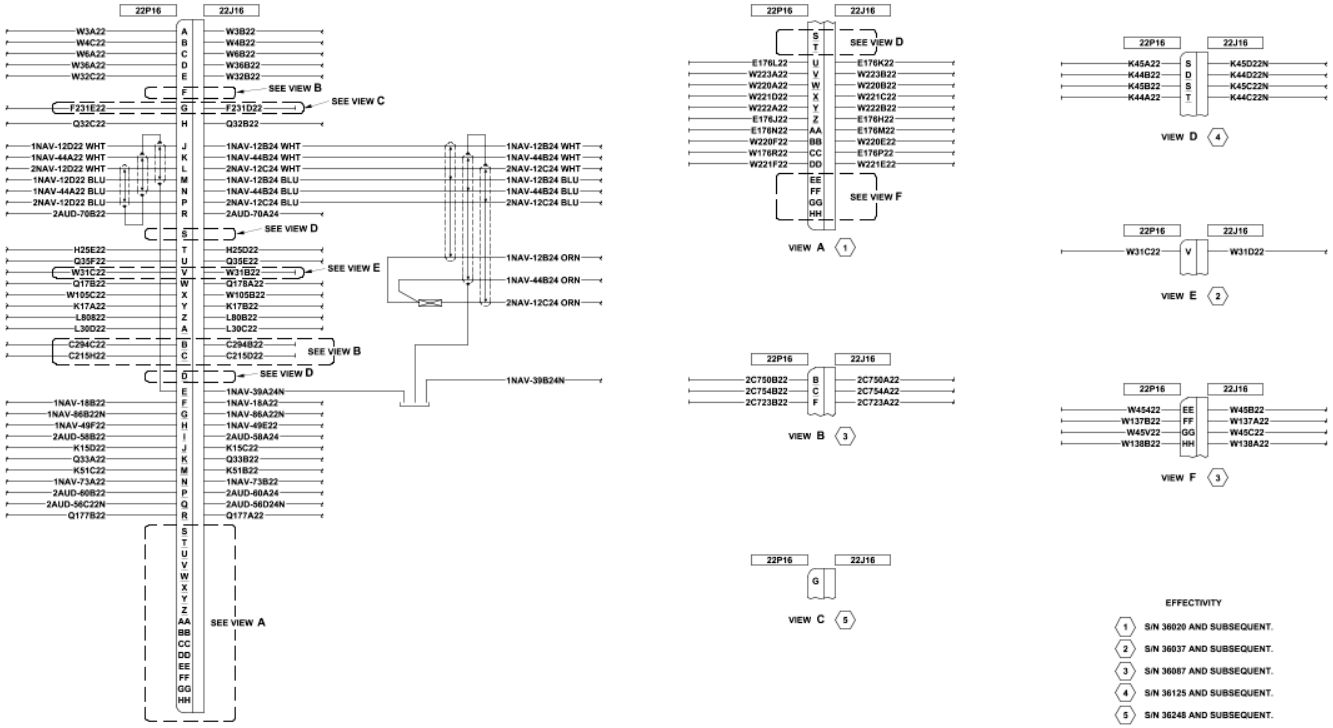
ICN-412-A-980000-B-97499-00453-A-001-01

Figure 1. Component Replacement (Sheet 68 of 104)



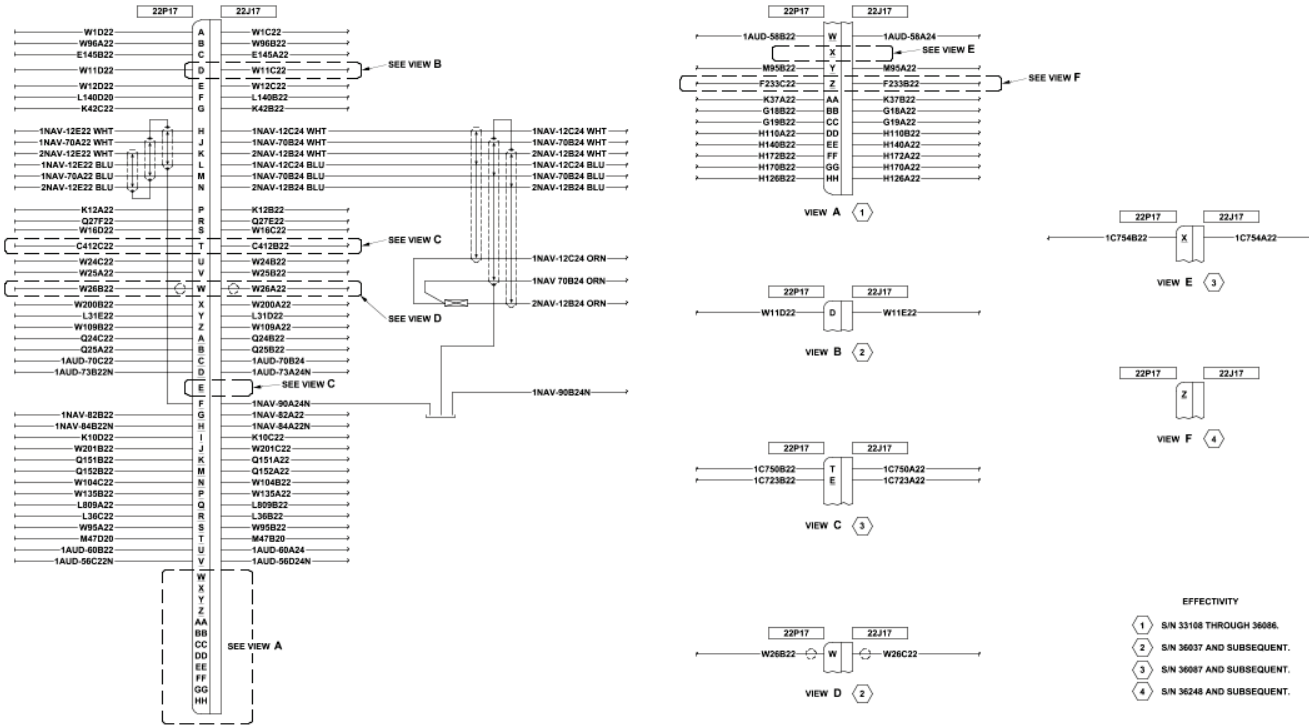
ICN-412-A-980000-B-97499-00454-A-001-01

Figure 1. Component Replacement (Sheet 69 of 104)



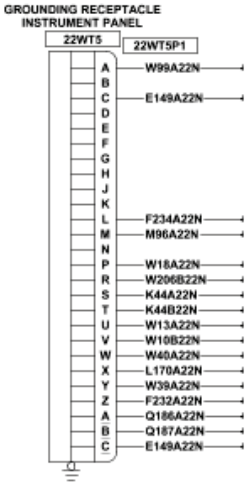
ICN-412-A-980000-B-97499-00358-A-001-01

Figure 1. Component Replacement (Sheet 70 of 104)



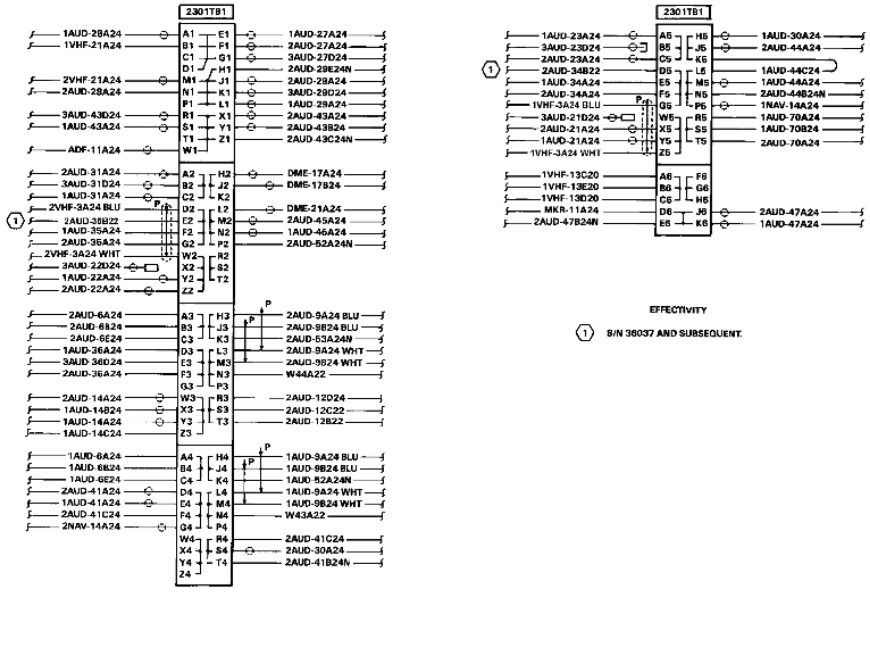
ICN-412-A-980000-B-97499-00359-A-001-01

Figure 1. Component Replacement (Sheet 71 of 104)



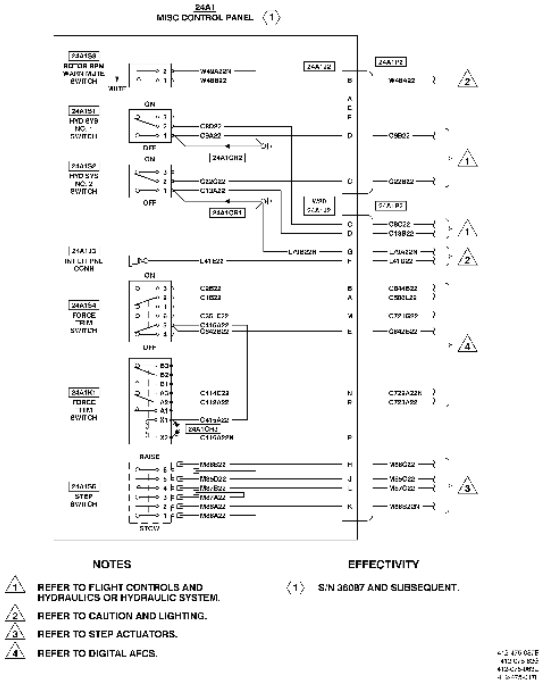
ICN-412-A-980000-B-97499-00455-A-001-01

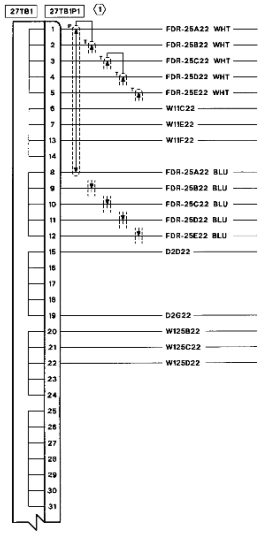
Figure 1. Component Replacement (Sheet 72 of 104)



ICN-412-A-980000-B-97499-00367-A-001-01

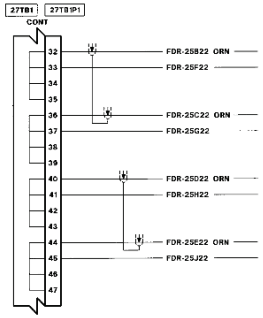
Figure 1. Component Replacement (Sheet 73 of 104)


24A1
MISC CONTROL PANEL
24A1S8
ROTOR RPM WARN/AUTIO SWITCH
24A1S51
HYD SYS NO. 1 SWITCH
24A1S56
HYD SYS NO. 2 SWITCH
24A1J3
INTERFACE CONN
24A1S54
FORCE TRIM SWITCH
24A1S6
FORCE TRIM SWITCH
24A1S55
STEP SWITCH
ON
OFF
BASE
24A1K1
24A1K2
24A1K3
24A1K4
24A1K5
24A1K6
24A1K7
24A1K8
24A1K9
24A1K10
24A1K11
24A1K12
24A1K13
24A1K14
24A1K15
24A1K16
24A1K17
24A1K18
24A1K19
24A1K20
24A1K21
24A1K22
24A1K23
24A1K24
24A1K25
24A1K26
24A1K27
24A1K28
24A1K29
24A1K30
24A1K31
24A1K32
24A1K33
24A1K34
24A1K35
24A1K36
24A1K37
24A1K38
24A1K39
24A1K40



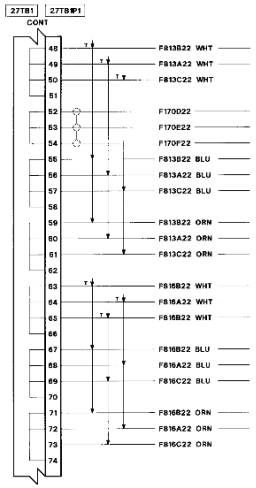
ICN-412-A-980000-B-97499-00457-A-001-01

Figure 1. Component Replacement (Sheet 75 of 104)



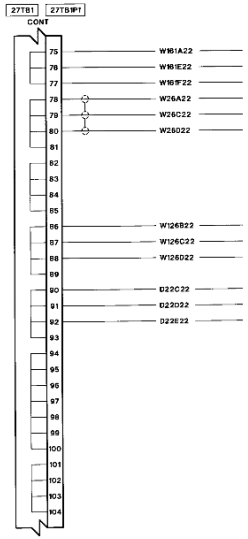
ICN-412-A-980000-B-97499-00458-A-001-01

Figure 1. Component Replacement (Sheet 76 of 104)



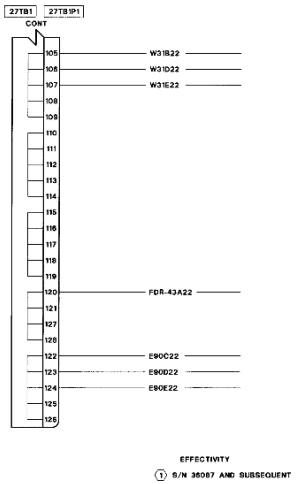
ICN-412-A-980000-B-97499-00459-A-001-01

Figure 1. Component Replacement (Sheet 77 of 104)



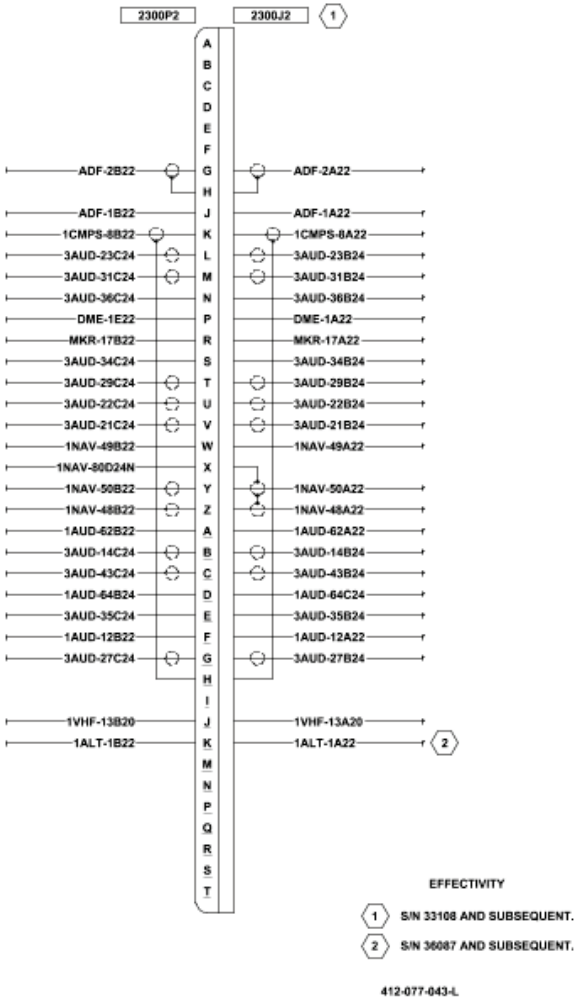
ICN-412-A-980000-B-97499-00460-A-001-01

Figure 1. Component Replacement (Sheet 78 of 104)



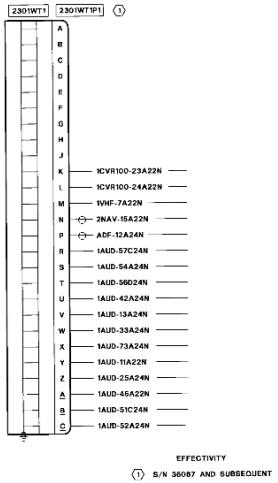
ICN-412-A-980000-B-97499-00461-A-001-01

Figure 1. Component Replacement (Sheet 79 of 104)



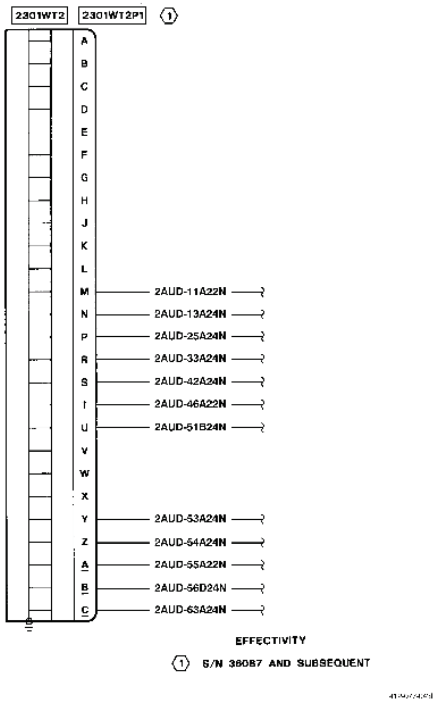
ICN-412-A-980000-B-97499-00366-A-001-01

Figure 1. Component Replacement (Sheet 80 of 104)



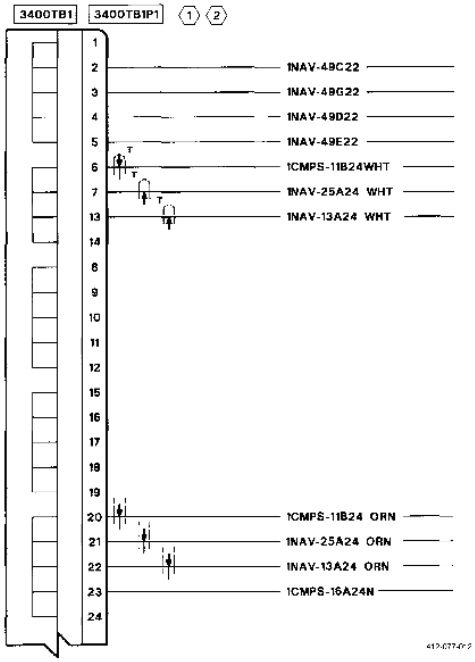
ICN-412-A-980000-B-97499-00462-A-001-01

Figure 1. Component Replacement (Sheet 81 of 104)



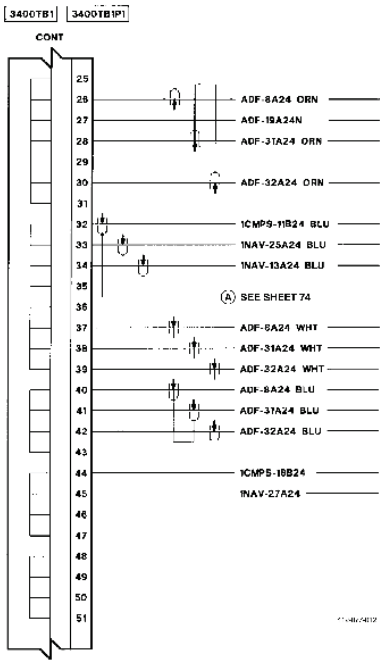
ICN-412-A-980000-B-97499-00463-A-001-01

Figure 1. Component Replacement (Sheet 82 of 104)



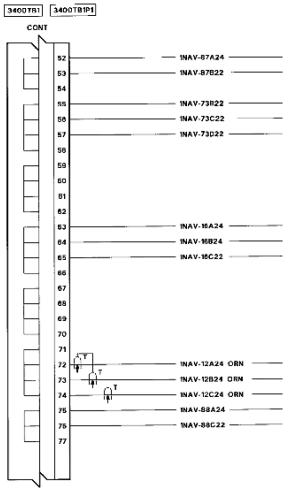
ICN-412-A-980000-B-97499-00464-A-001-01

Figure 1. Component Replacement (Sheet 83 of 104)



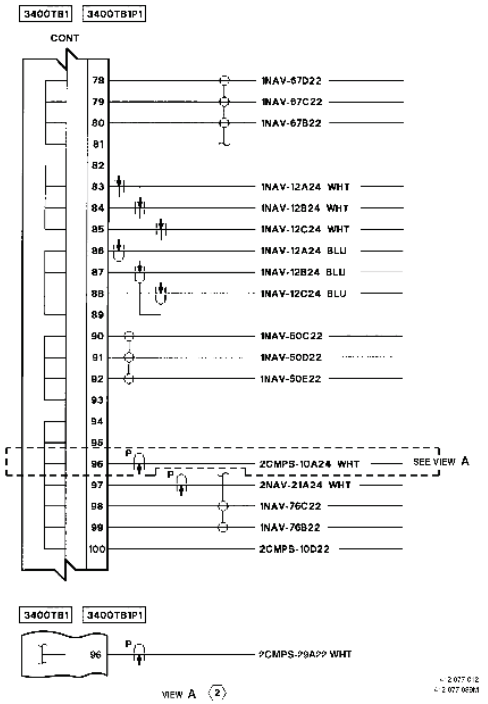
ICN-412-A-980000-B-97499-00465-A-001-01

Figure 1. Component Replacement (Sheet 84 of 104)



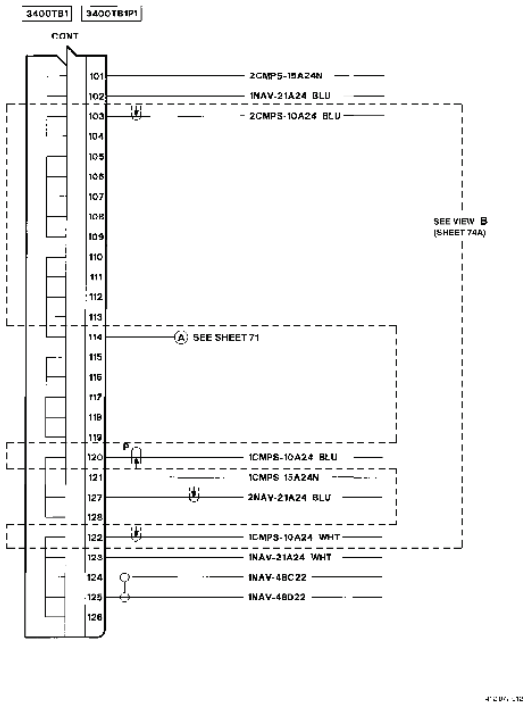
ICN-412-A-980000-B-97499-00466-A-001-01

Figure 1. Component Replacement (Sheet 85 of 104)



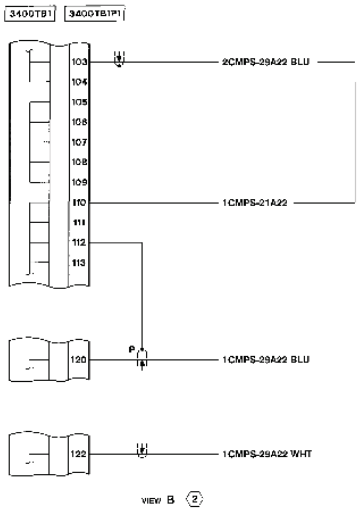
ICN-412-A-980000-B-97499-00467-A-001-01

Figure 1. Component Replacement (Sheet 86 of 104)



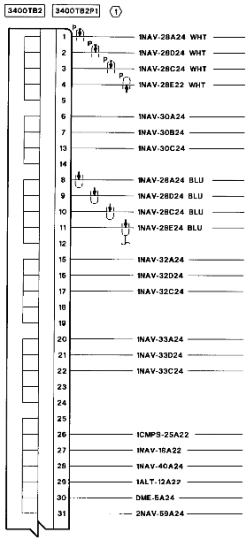
ICN-412-A-980000-B-97499-00468-A-001-01

Figure 1. Component Replacement (Sheet 87 of 104)



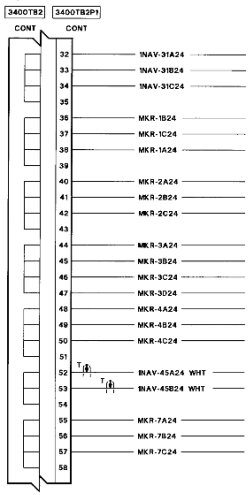
ICN-412-A-980000-B-97499-00469-A-001-01

Figure 1. Component Replacement (Sheet 88 of 104)



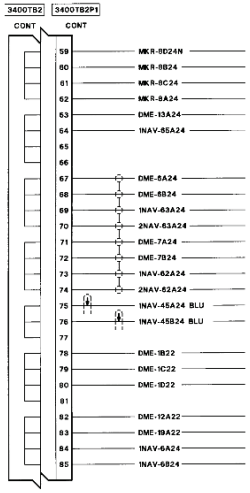
ICN-412-A-980000-B-97499-00470-A-001-01

Figure 1. Component Replacement (Sheet 89 of 104)



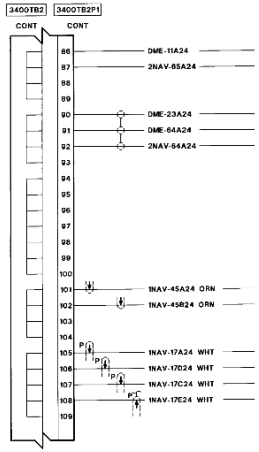
ICN-412-A-980000-B-97499-00471-A-001-01

Figure 1. Component Replacement (Sheet 90 of 104)



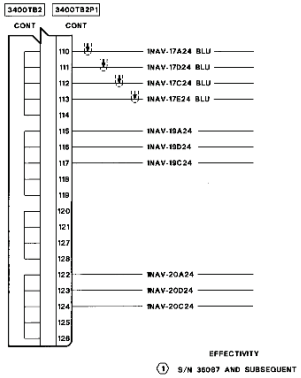
ICN-412-A-980000-B-97499-00472-A-001-01

Figure 1. Component Replacement (Sheet 91 of 104)



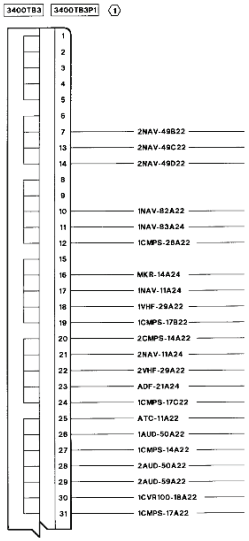
ICN-412-A-980000-B-97499-00473-A-001-01

Figure 1. Component Replacement (Sheet 92 of 104)



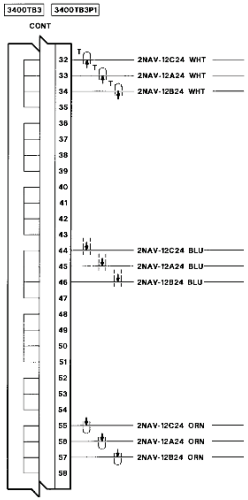
ICN-412-A-980000-B-97499-00474-A-001-01

Figure 1. Component Replacement (Sheet 93 of 104)



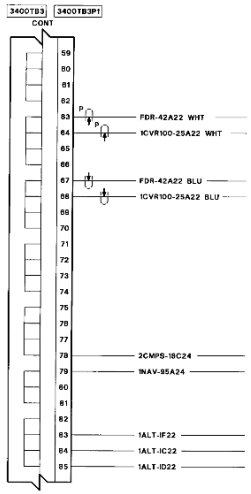
ICN-412-A-980000-B-97499-00475-A-001-01

Figure 1. Component Replacement (Sheet 94 of 104)



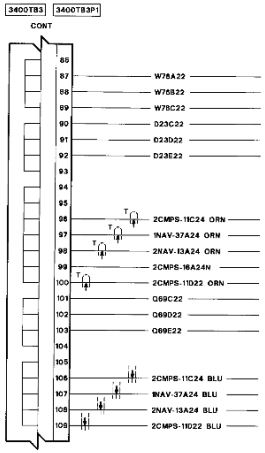
ICN-412-A-980000-B-97499-00476-A-001-01

Figure 1. Component Replacement (Sheet 95 of 104)



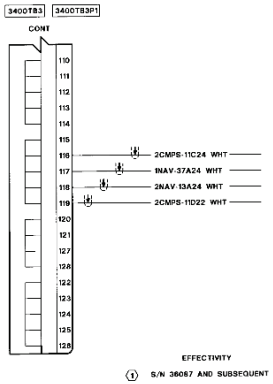
ICN-412-A-980000-B-97499-00477-A-001-01

Figure 1. Component Replacement (Sheet 96 of 104)



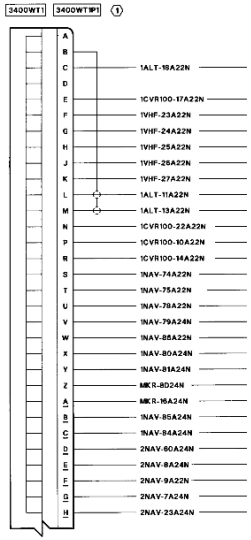
ICN-412-A-980000-B-97499-00478-A-001-01

Figure 1. Component Replacement (Sheet 97 of 104)



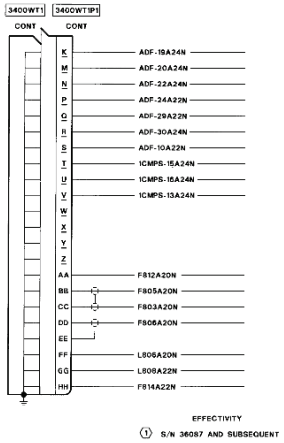
ICN-412-A-980000-B-97499-00479-A-001-01

Figure 1. Component Replacement (Sheet 98 of 104)



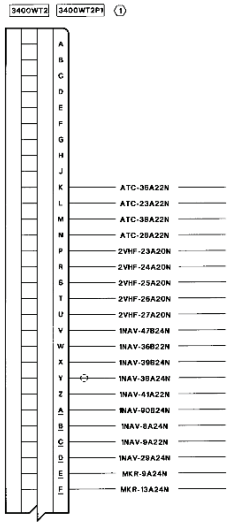
ICN-412-A-980000-B-97499-00480-A-001-01

Figure 1. Component Replacement (Sheet 99 of 104)



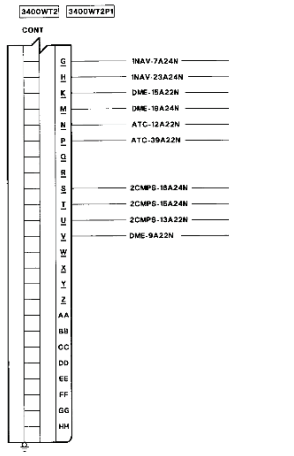
ICN-412-A-980000-B-97499-00481-A-001-01

Figure 1. Component Replacement (Sheet 100 of 104)



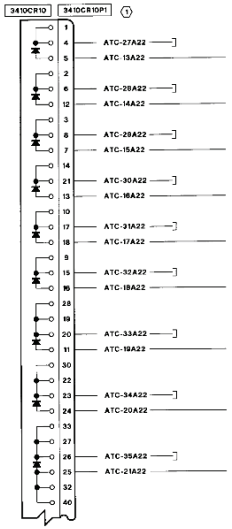
ICN-412-A-980000-B-97499-00482-A-001-01

Figure 1. Component Replacement (Sheet 101 of 104)



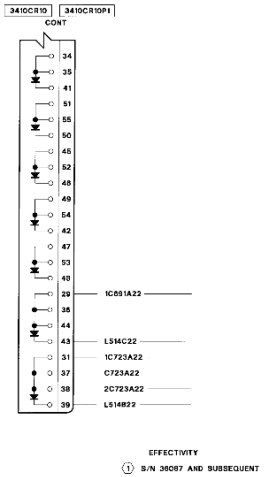
ICN-412-A-980000-B-97499-00483-A-001-01

Figure 1. Component Replacement (Sheet 102 of 104)



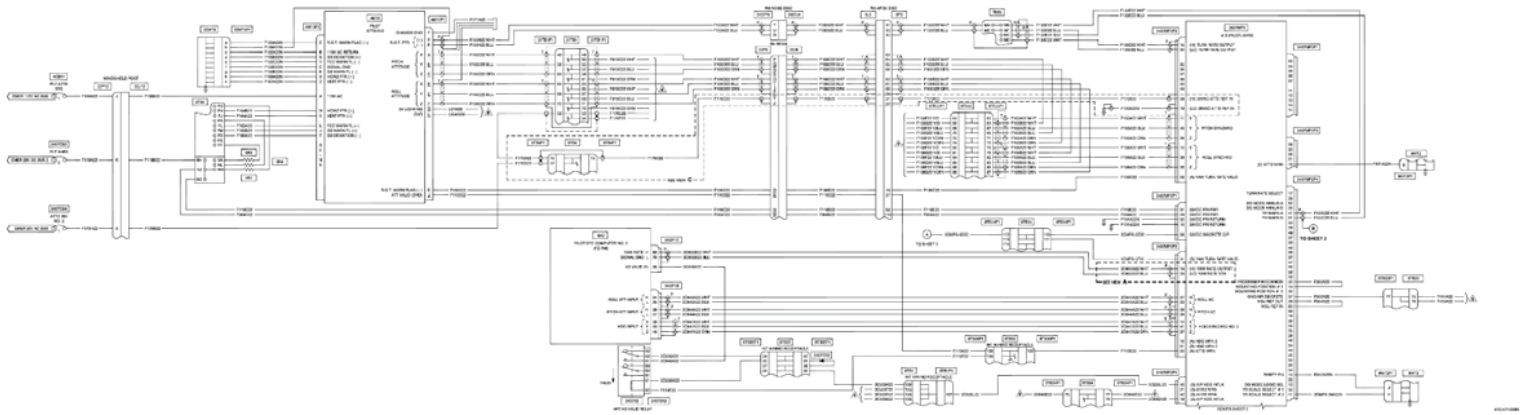
ICN-412-A-980000-B-97499-00484-A-001-01

Figure 1. Component Replacement (Sheet 103 of 104)



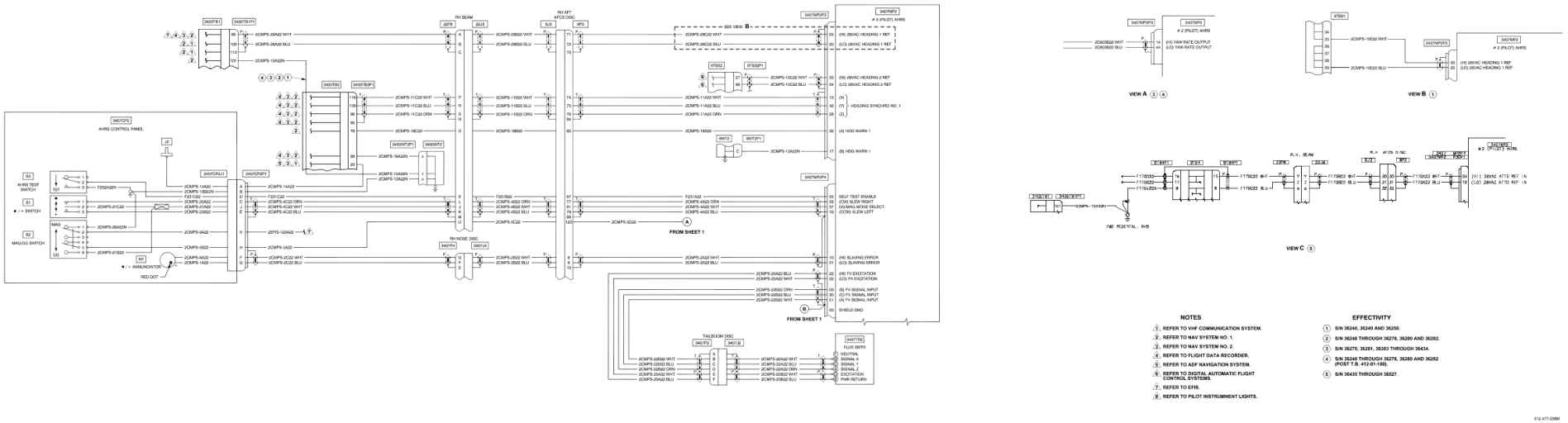
ICN-412-A-980000-B-97499-00485-A-001-01

Figure 1. Component Replacement (Sheet 104 of 104)



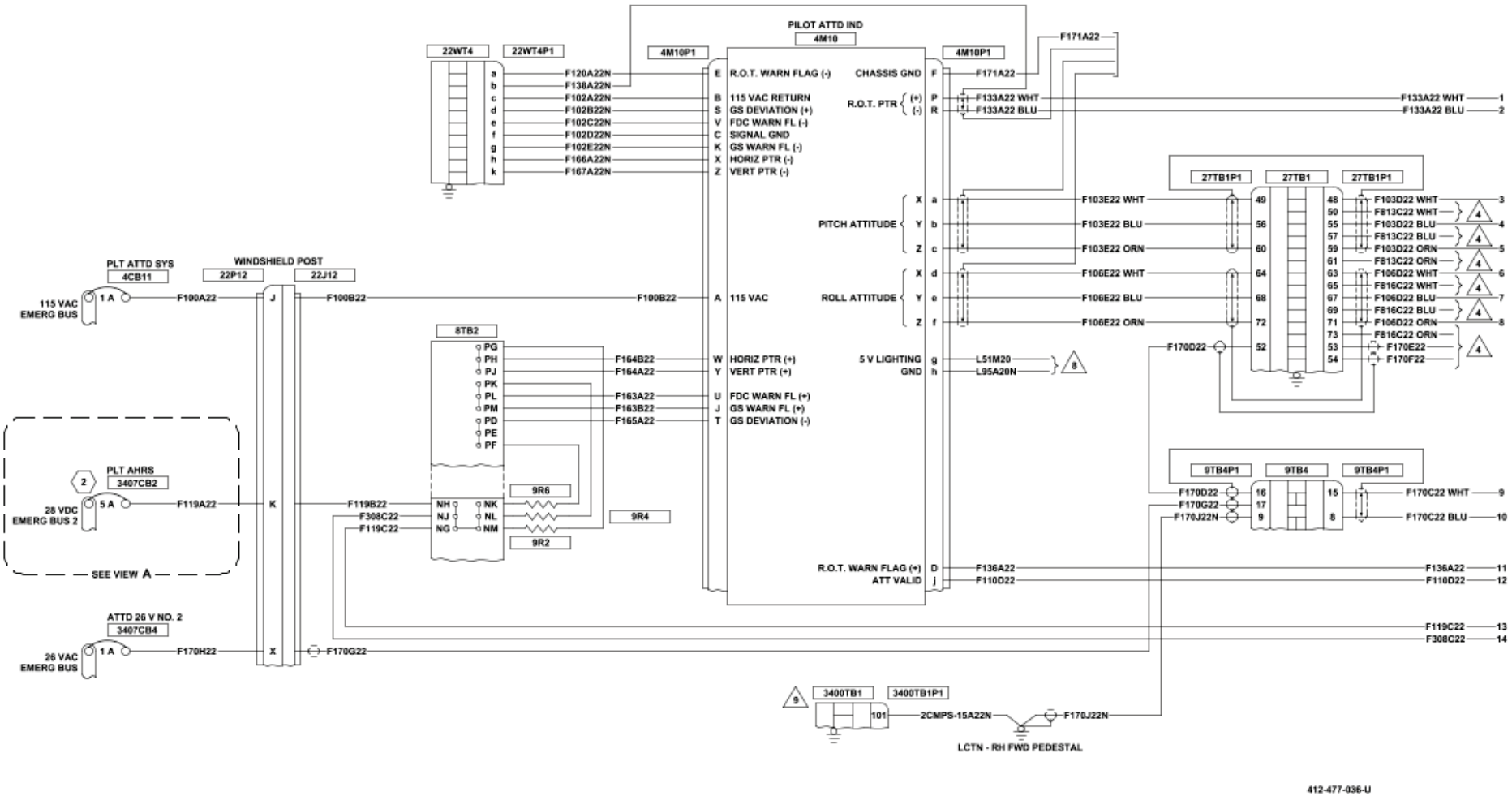
ICN-412-A-980000-B-97499-00486-A-001-01

Figure 1. Pilot AHRS 2 (LCR-92) (Sheet 1 of 2)



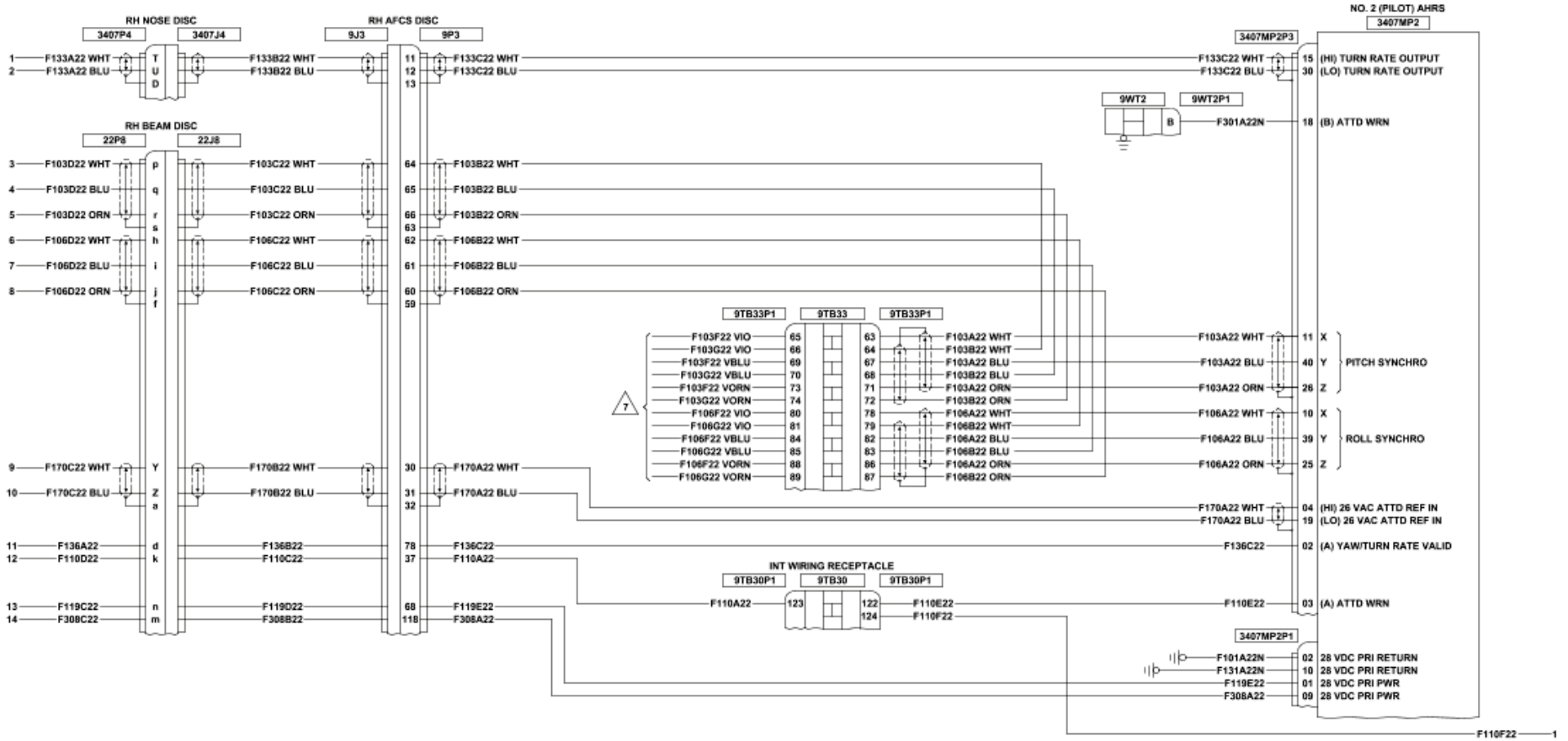
ICN-412-A-980000-B-97499-00487-A-001-01

Figure 1. Pilot AHRS 2 (LCR-92) (Sheet 2 of 2)



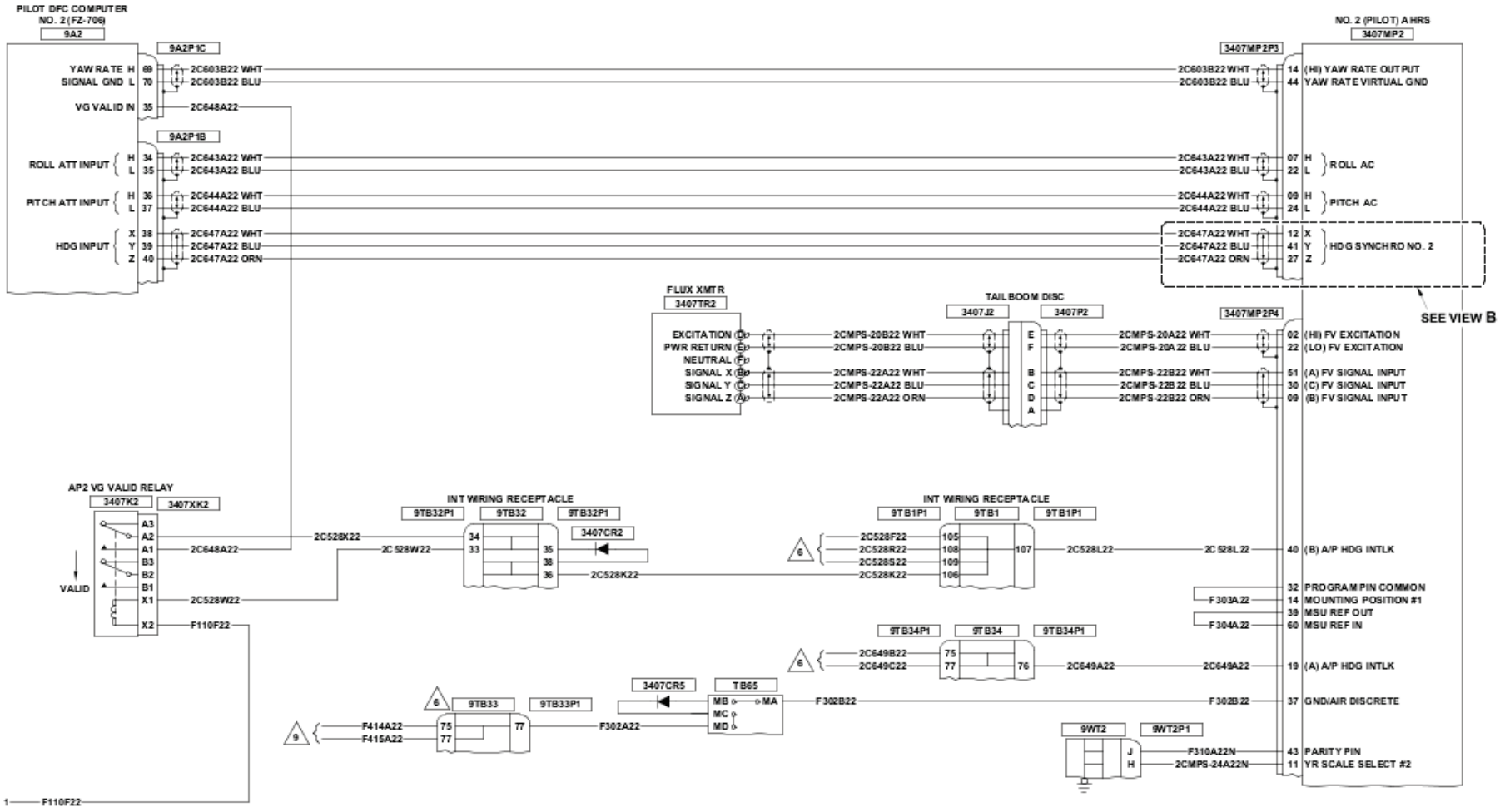
ICN-412-A-980000-B-97499-00488-A-001-01

Figure 1. Pilot AHRS 2 (LCR-100) (Sheet 1 of 5)



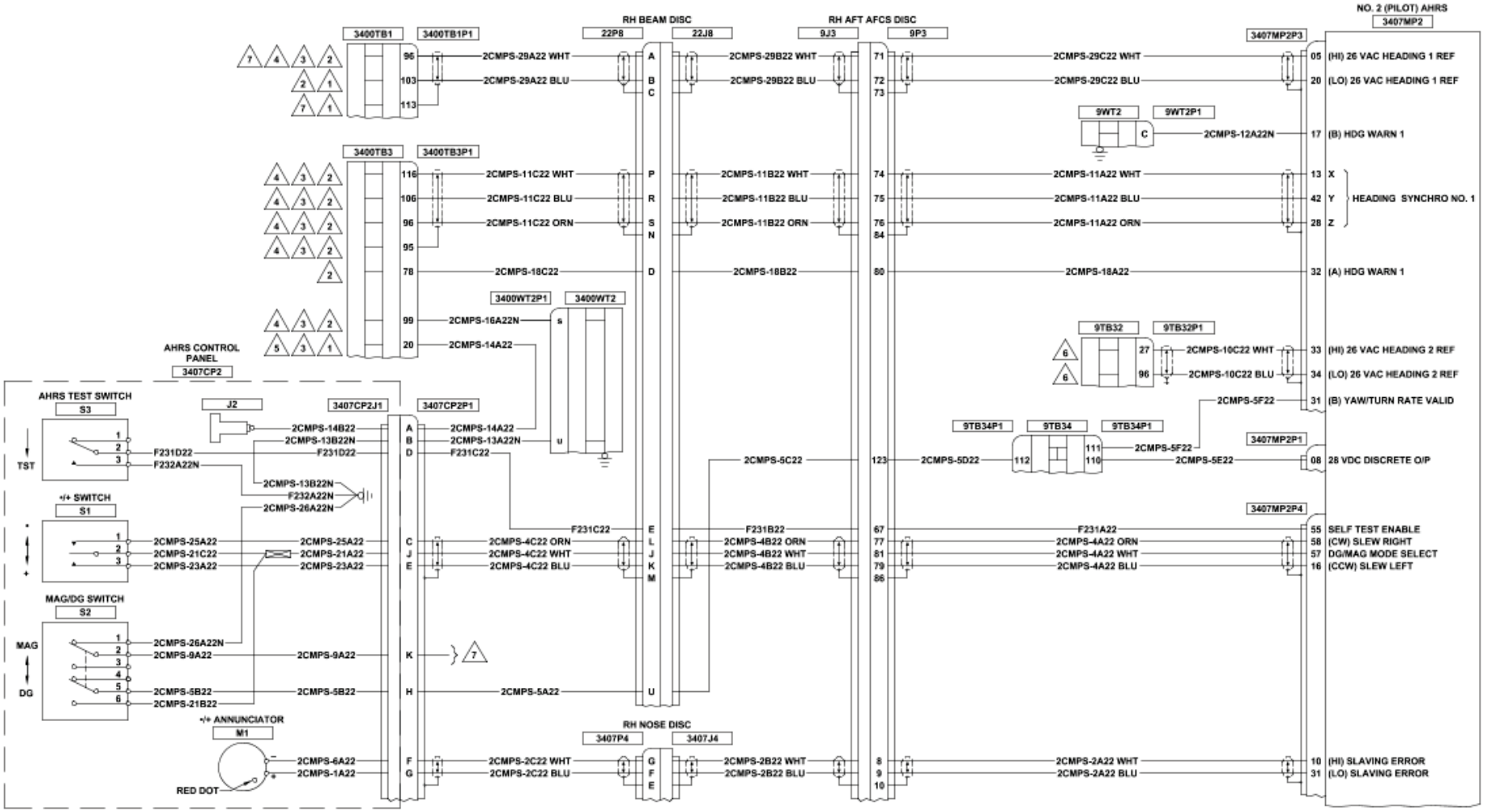
ICN-412-A-980000-B-97499-00489-A-001-01

Figure 1. Pilot AHRS 2 (LCR-100) (Sheet 2 of 5)



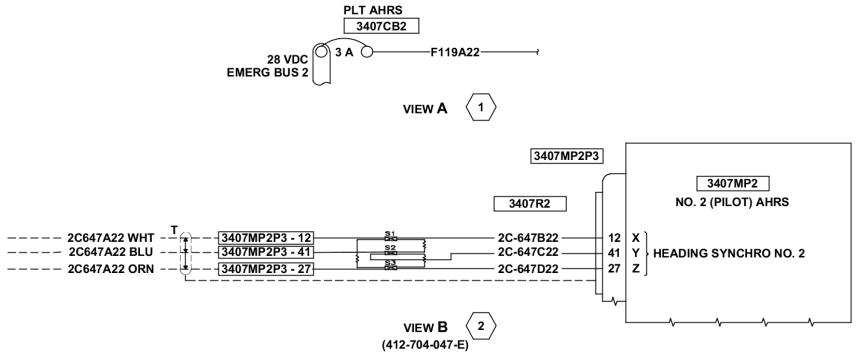
ICN-412-A-980000-B-97499-00490-A-001-01

Figure 1. Pilot AHRS 2 (LCR-100) (Sheet 3 of 5)



ICN-412-A-980000-B-97499-00491-A-001-01

Figure 1. Pilot AHRS 2 (LCR-100) (Sheet 4 of 5)



NOTES

1. REFER TO VHF COMMUNICATION SYSTEM.
2. REFER TO NAV SYSTEM NO. 1.
3. REFER TO NAV SYSTEM NO. 2.
4. REFER TO FLIGHT DATA RECORDER.
5. REFER TO ADF NAVIGATION SYSTEM.
6. REFER TO DIGITAL AUTOMATIC FLIGHT CONTROL SYSTEMS.
7. REFER TO EFIS.
8. REFER TO PILOT INSTRUMENT LIGHTS.
9. REFER TO COPILOT AHRS 1.

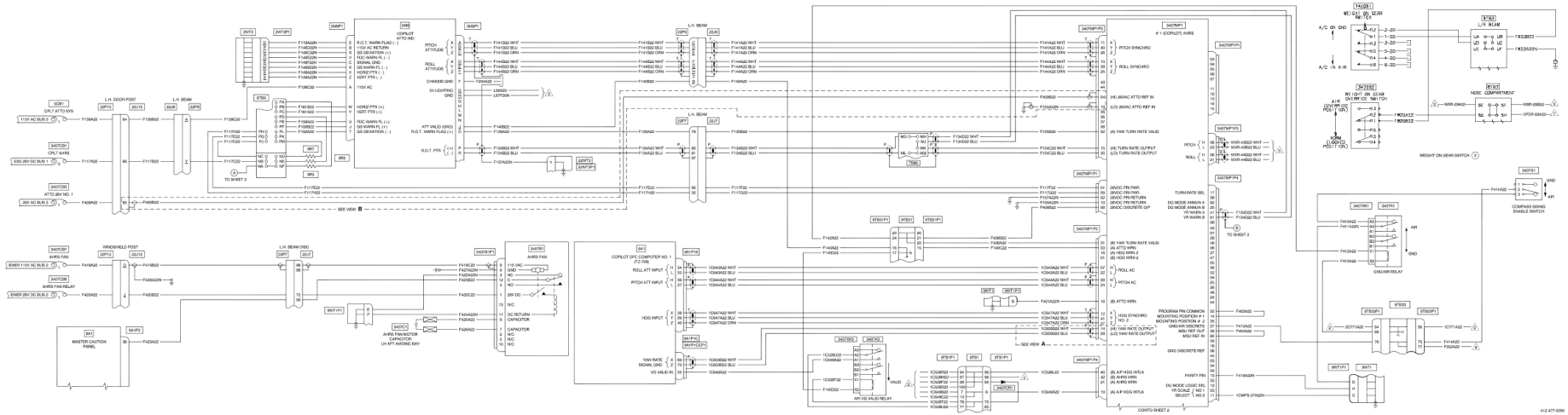
EFFECTIVITY

1. S/N 36248 THROUGH 36683
2. S/N 36684 AND SUBSEQUENT OR,
   S/N 36248 THROUGH 36683 WITH LCR100 (P/N 145130-7000 OR -7001) OR,
   S/N 36248 THROUGH 36683 WITH RETROFIT (412-704-047-121) COMPLETED OR,
   S/N 36248 THROUGH 36527 WITH RETROFIT (412-704-047-125) COMPLETED.

ICN-412-A-980000-B-97499-00492-A-001-01

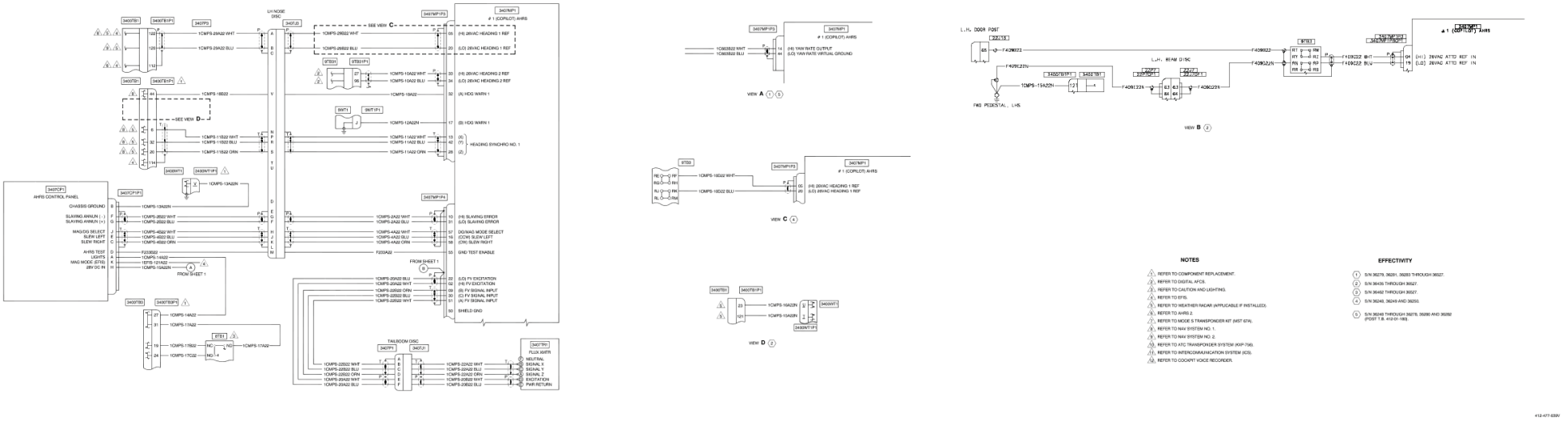
Figure 1. Pilot AHRS 2 (LCR-100) (Sheet 5 of 5)

Intentionally blank page



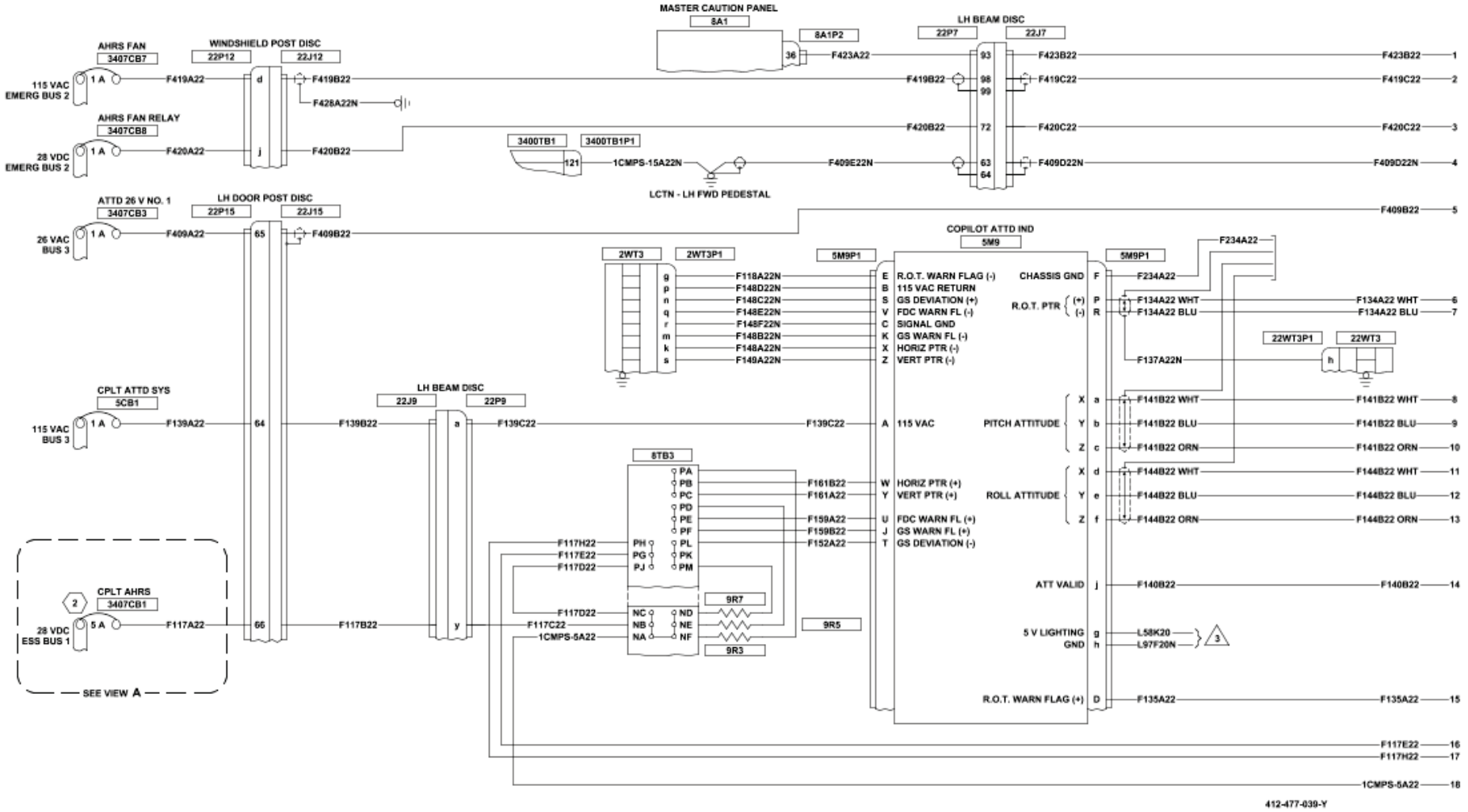
ICN-412-A-980000-B-97499-00493-A-001-01

Figure 1. Copilot AHRS 1 (LCR-92) (Sheet 1 of 2)



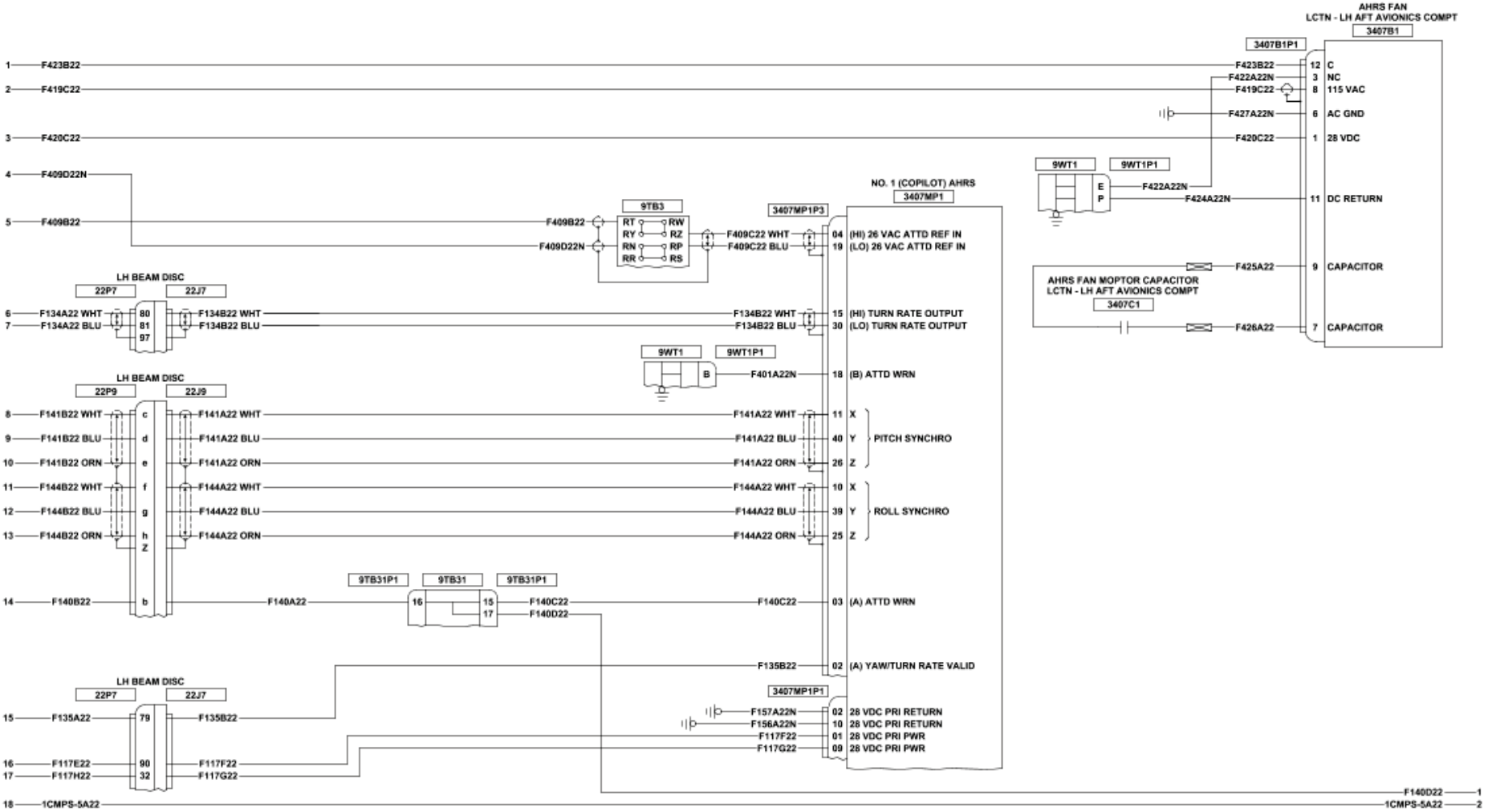
ICN-412-A-980000-B-97499-00494-A-001-01

Figure 1. Copilot AHRS 1 (LCR-92) (Sheet 2 of 2)



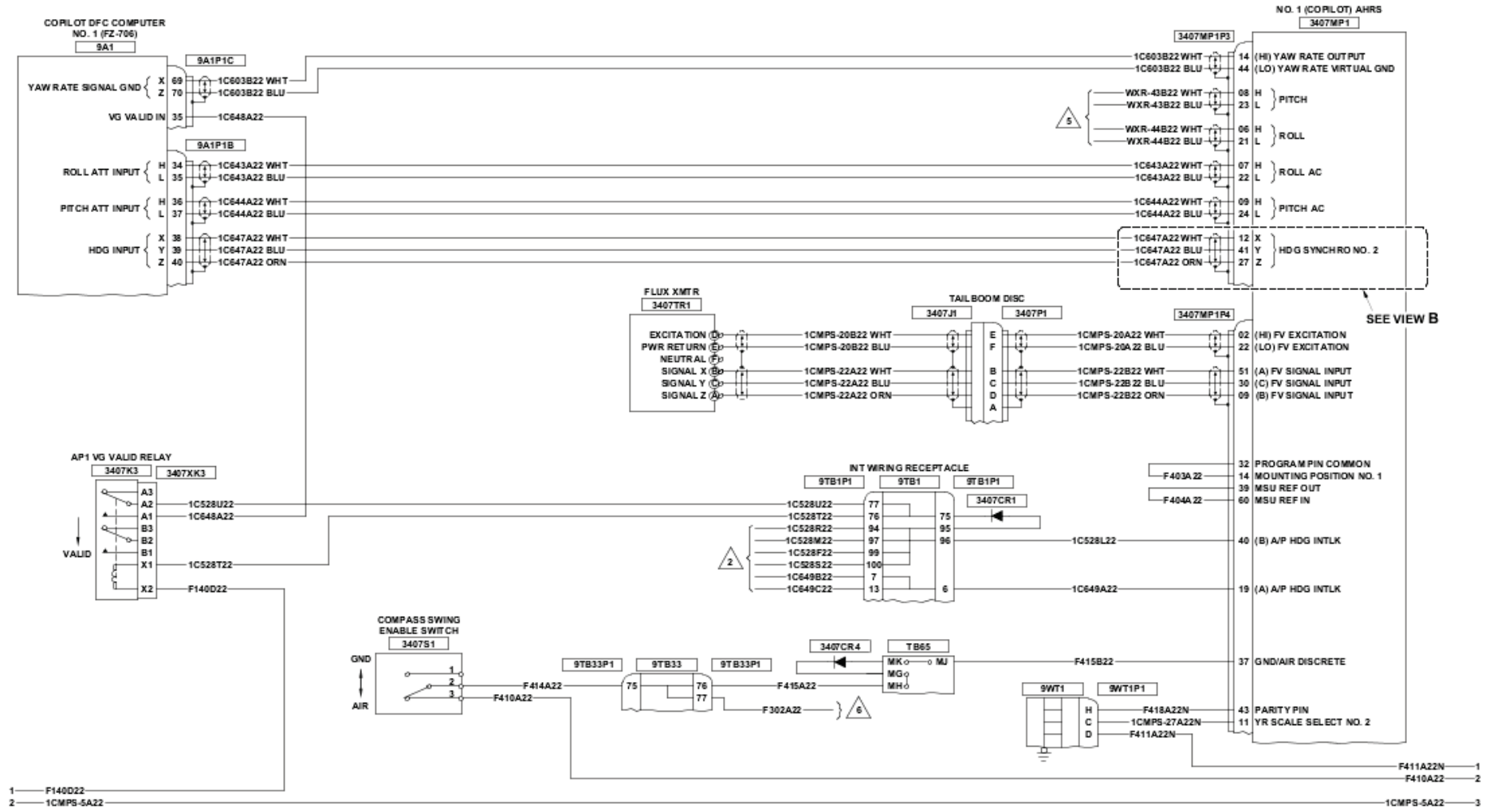
ICN-412-A-980000-B-97499-00495-A-001-01

Figure 1. Copilot AHRS 1 (LCR-100) (Sheet 1 of 5)



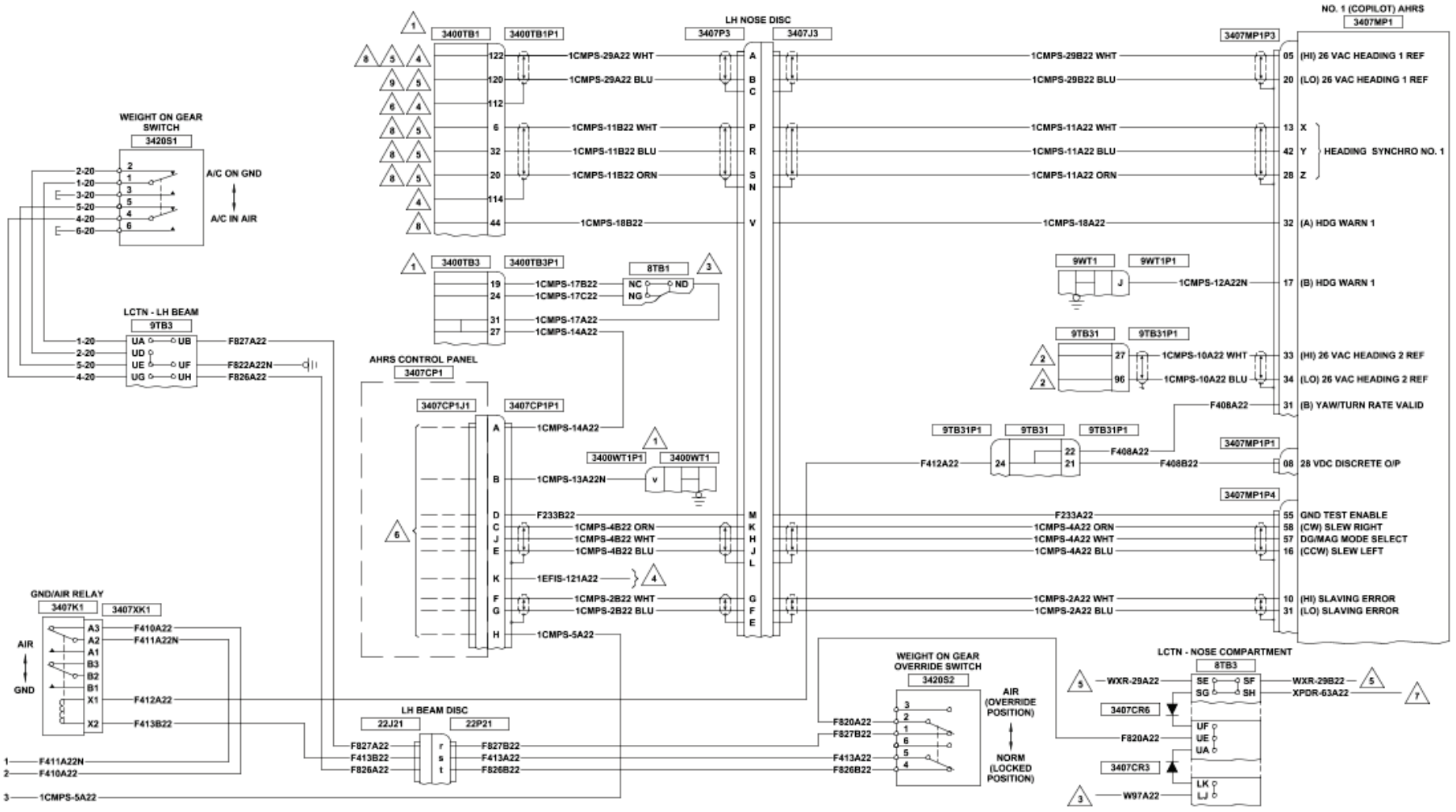
ICN-412-A-980000-B-97499-00496-A-001-01

Figure 1. Copilot AHRS 1 (LCR-100) (Sheet 2 of 5)



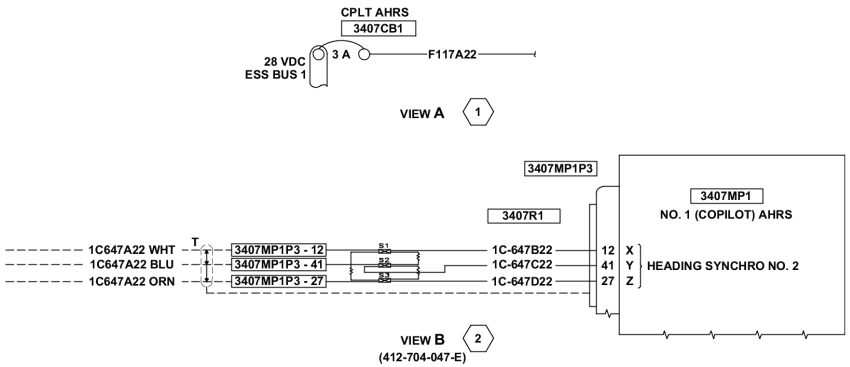
ICN-412-A-980000-B-97499-00497-A-001-01

Figure 1. Copilot AHRS 1 (LCR-100) (Sheet 3 of 5)



ICN-412-A-980000-B-97499-00498-A-001-01

Figure 1. Copilot AHRS 1 (LCR-100) (Sheet 4 of 5)



NOTES

1. REFER TO COMPONENT REPLACEMENT.
2. REFER TO DIGITAL AUTOMATIC FLIGHT CONTROL SYSTEMS.
3. REFER TO CAUTION AND LIGHTING.
4. REFER TO EFIS.
5. REFER TO WEATHER RADAR (APPLICABLE IF INSTALLED).
6. REFER TO PILOT AHRS 2.
7. REFER TO MODE S TRANSPONDER KIT (MST 67A).
8. REFER TO NAV SYSTEM NO. 1.
9. REFER TO NAV SYSTEM NO. 2.

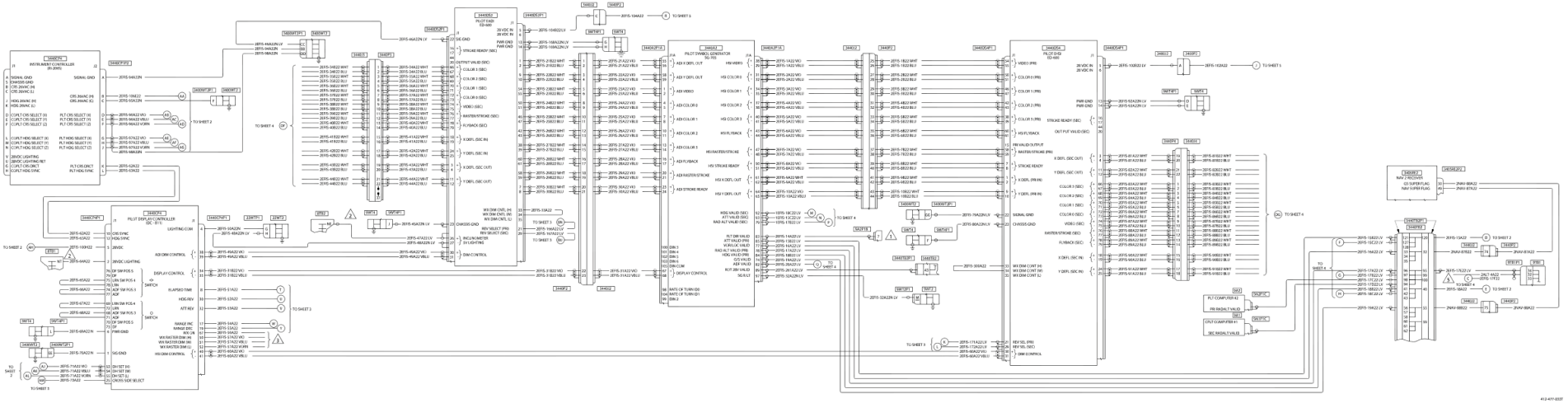
EFFECTIVITY

1. S/N 36248 THROUGH 36683
2. S/N 36684 AND SUBSEQUENT OR,
   S/N 36248 THROUGH 36683 WITH LCR100 (P/N 145130-7000 OR -7001) OR,
   S/N 36248 THROUGH 36683 WITH RETROFIT (412-704-047-121) COMPLETED
   S/N 36248 THROUGH 36527 WITH RETROFIT (412-704-047-123) COMPLETED.

ICN-412-A-980000-B-97499-00499-A-001-01

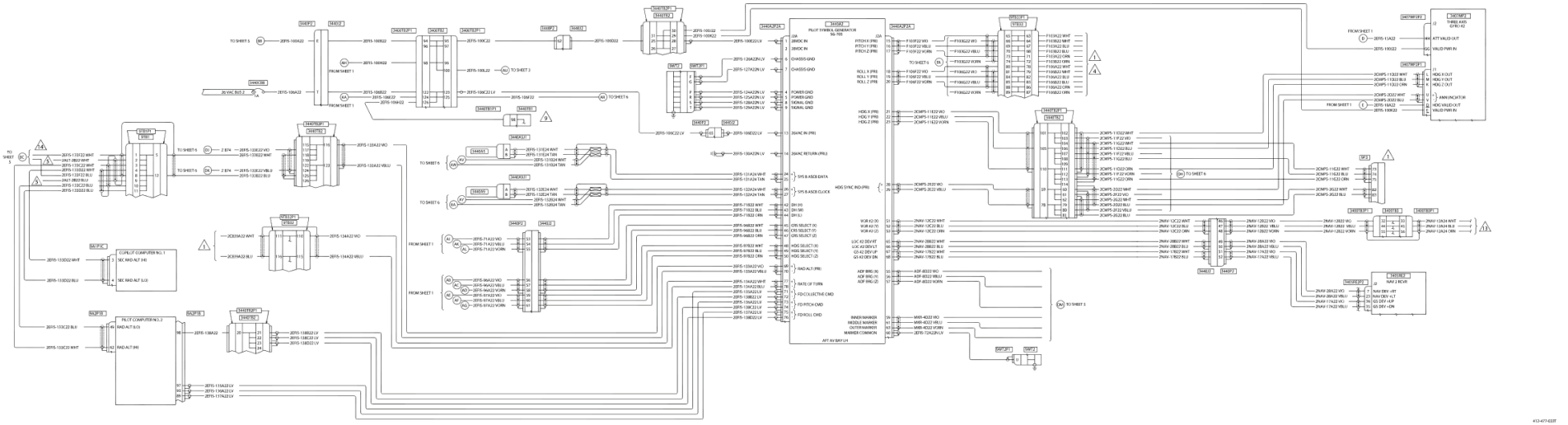
Figure 1. Copilot AHRS 1 (LCR-100) (Sheet 5 of 5)

Intentionally blank page



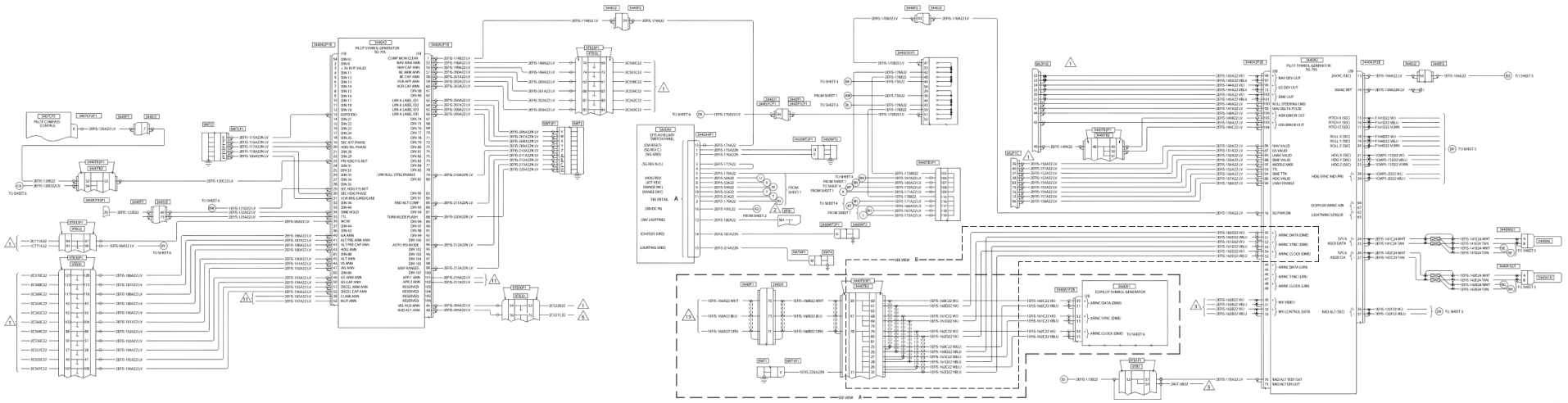
ICN-412-A-980000-B-97499-00500-A-001-01

Figure 1. Electronic Flight Instruments System (Sheet 1 of 7)



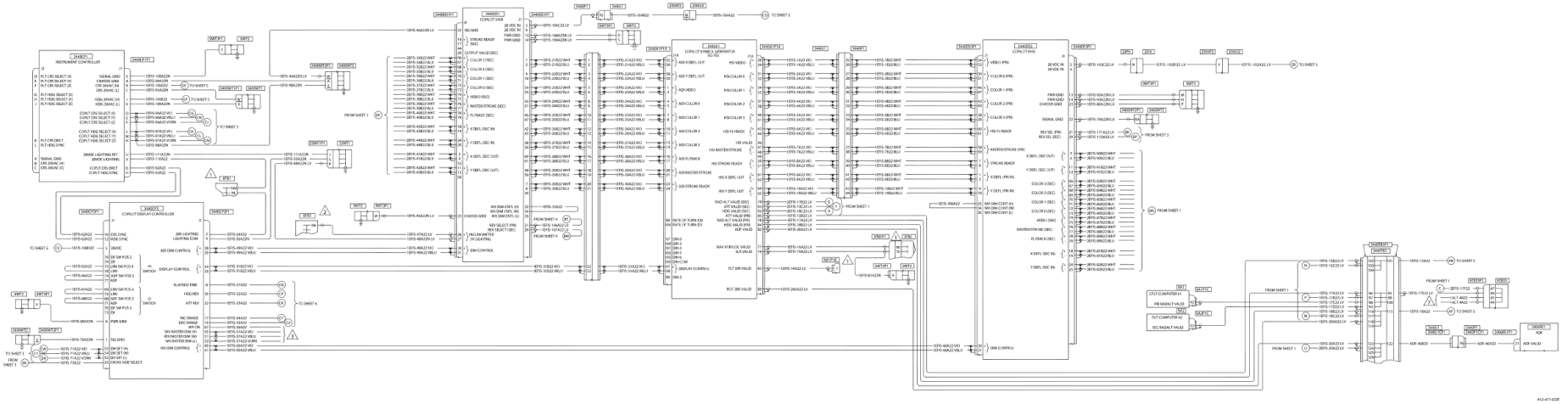
ICN-412-A-980000-B-97499-00501-A-001-01

Figure 1. Electronic Flight Instruments System (Sheet 2 of 7)



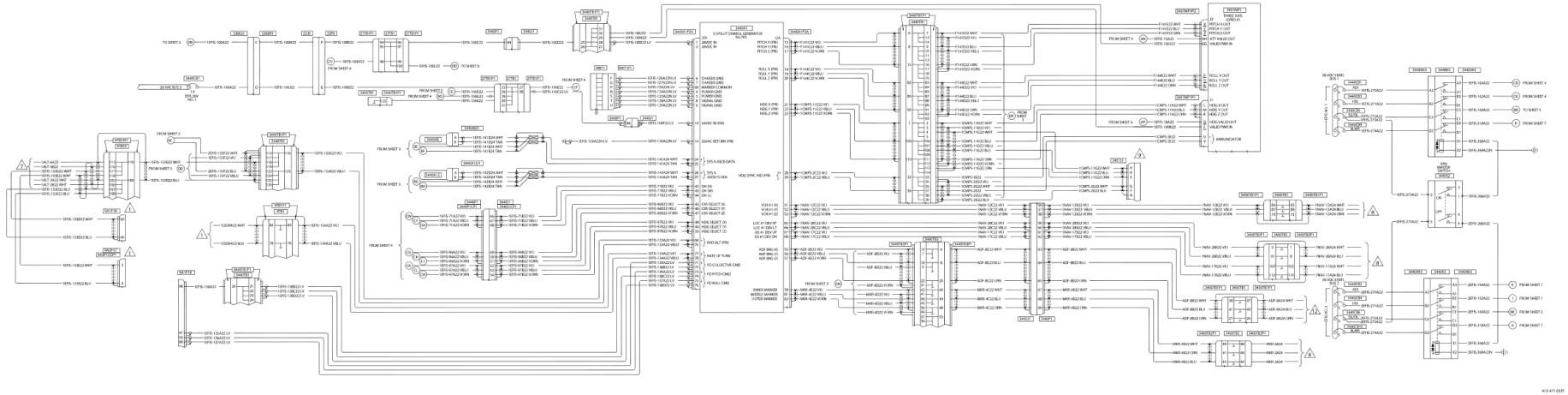
ICN-412-A-980000-B-97499-00502-A-001-01

Figure 1. Electronic Flight Instruments System (Sheet 3 of 7)



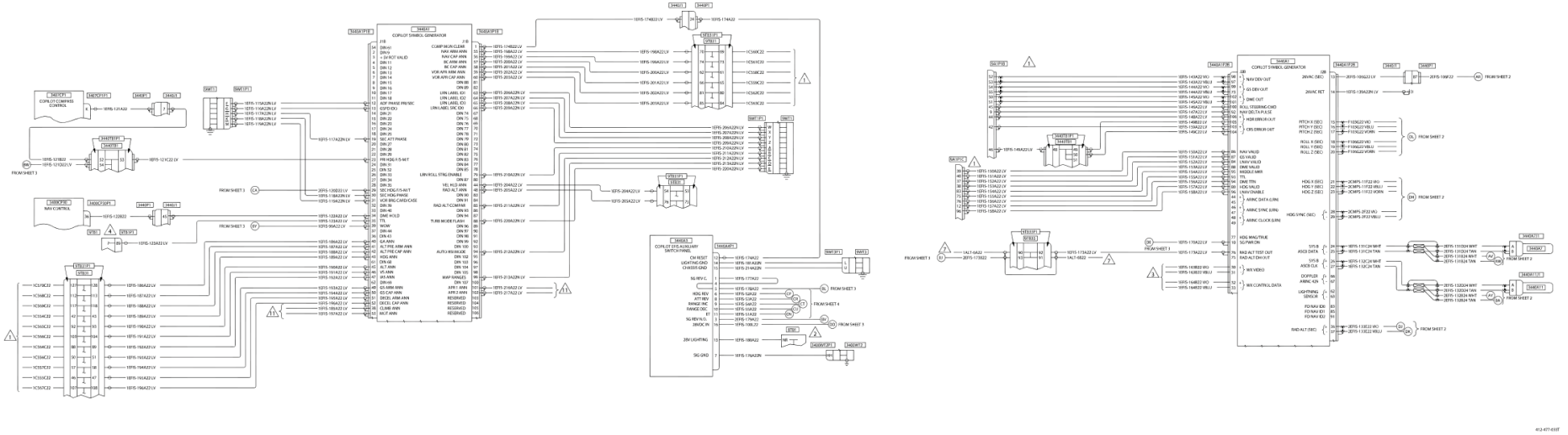
ICN-412-A-980000-B-97499-00503-A-001-01

Figure 1. Electronic Flight Instruments System (Sheet 4 of 7)



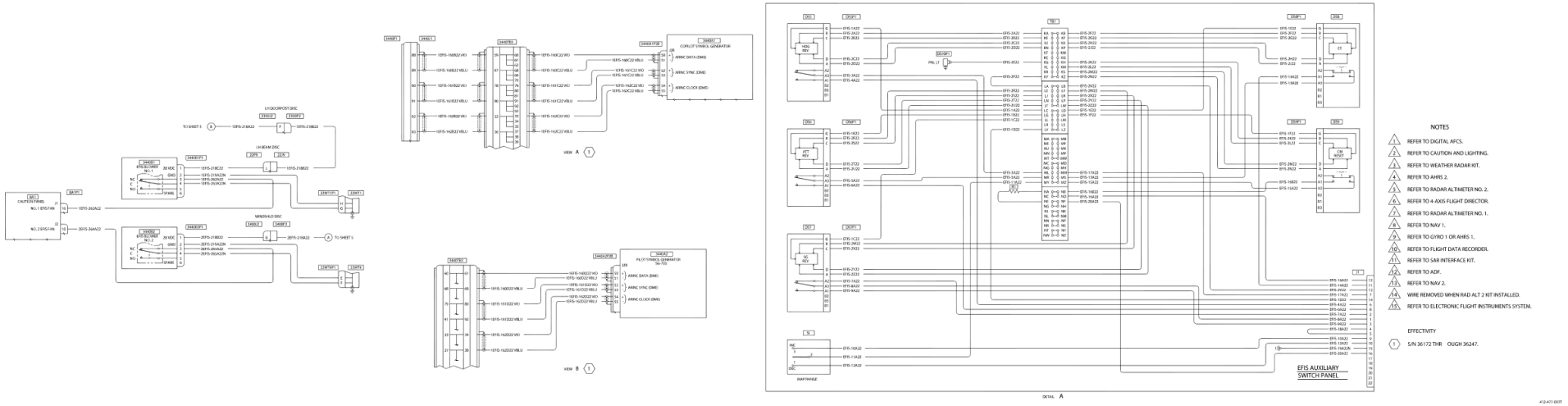
ICN-412-A-980000-B-97499-00504-A-001-01

Figure 1. Electronic Flight Instruments System (Sheet 5 of 7)



ICN-412-A-980000-B-97499-00505-A-001-01

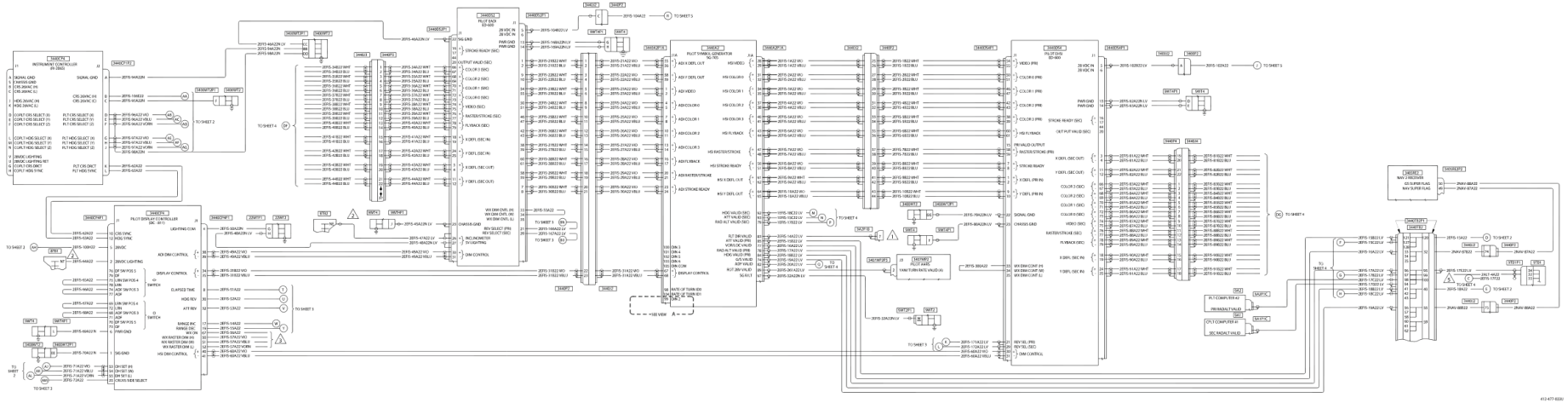
Figure 1. Electronic Flight Instruments System (Sheet 6 of 7)



ICN-412-A-980000-B-97499-00506-A-001-01

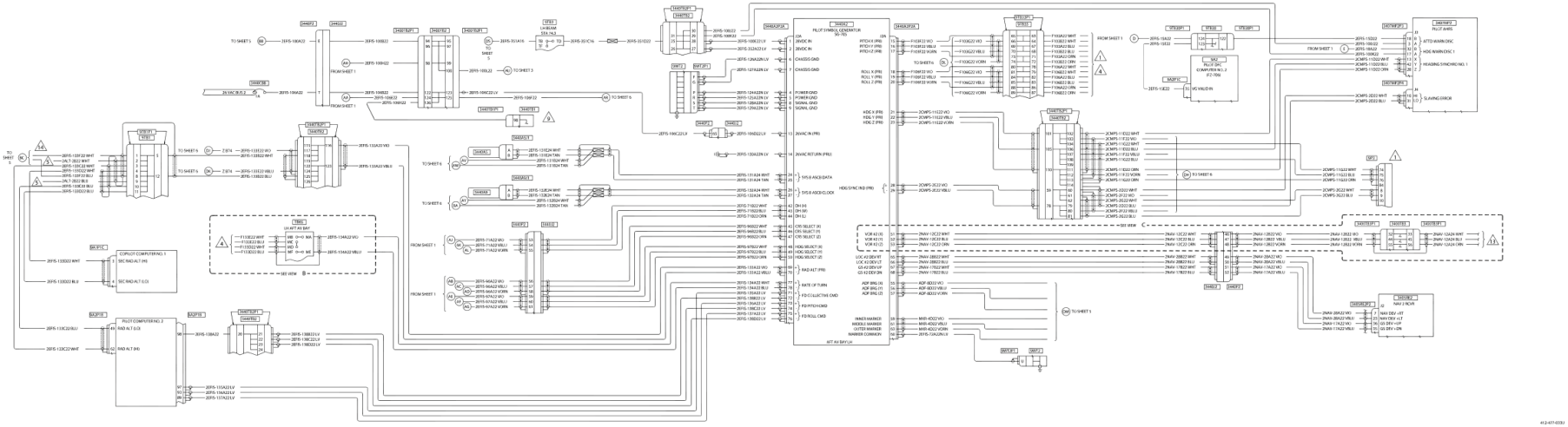
Figure 1. Electronic Flight Instruments System (Sheet 7 of 7)

Intentionally blank page



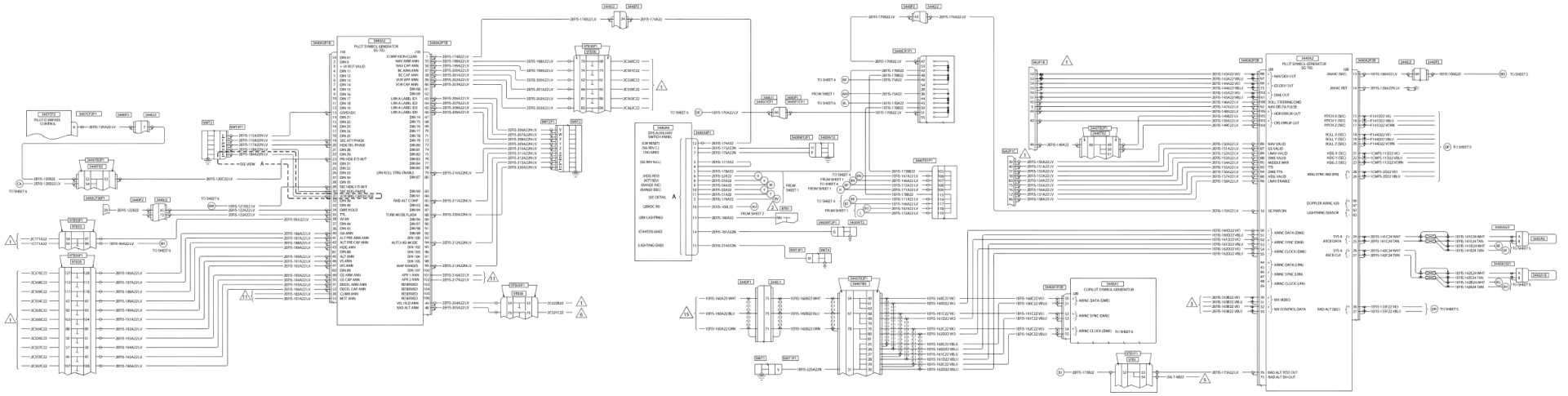
ICN-412-A-980000-B-97499-00507-A-001-01

Figure 1. Electronic Flight Instruments System (Sheet 1 of 7)



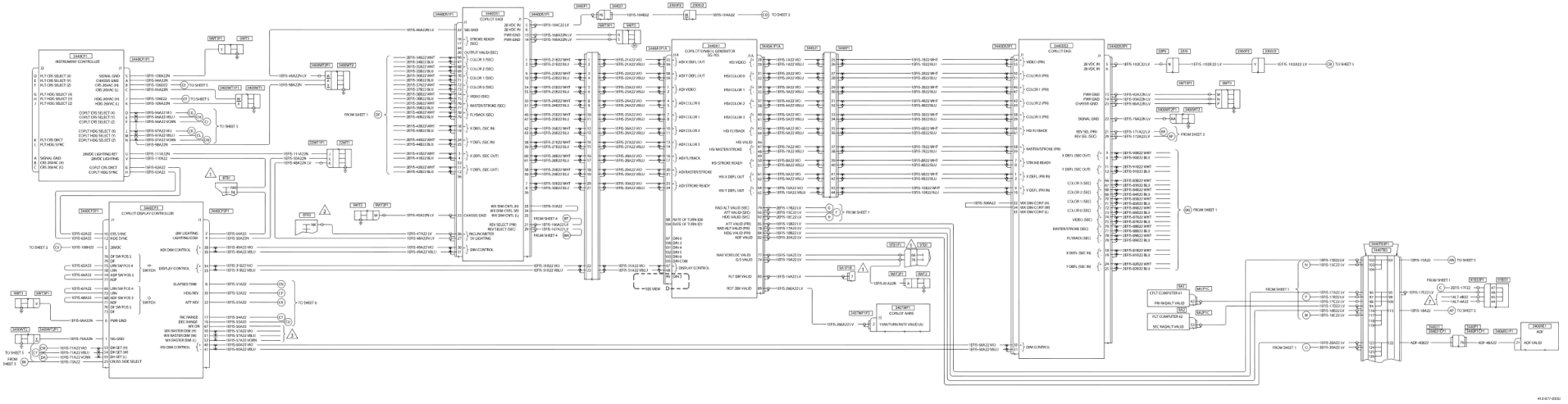
ICN-412-A-980000-B-97499-00508-A-001-01

Figure 1. Electronic Flight Instruments System (Sheet 2 of 7)



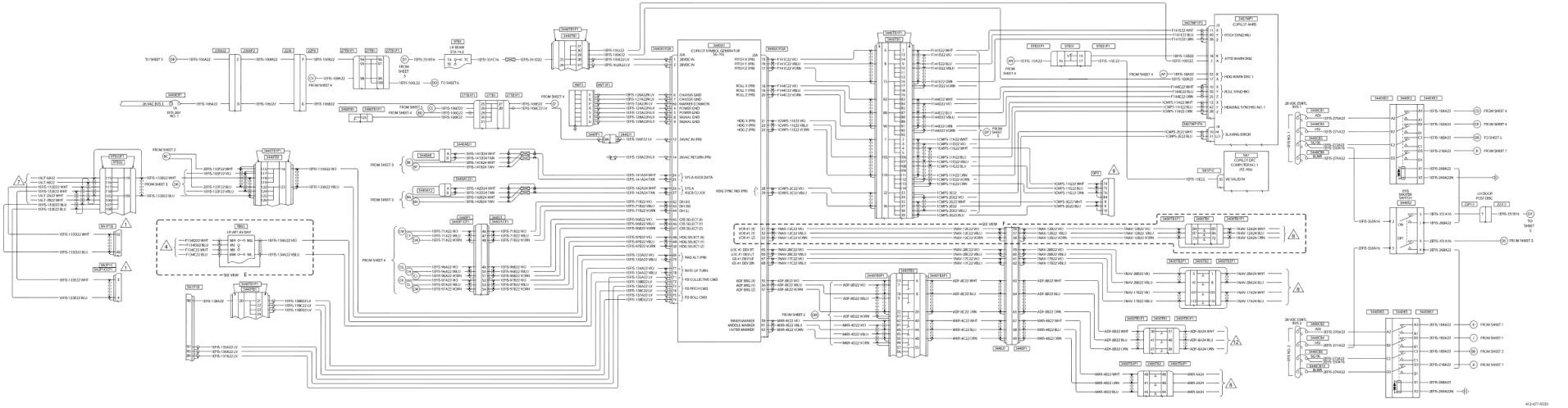
ICN-412-A-980000-B-97499-00509-A-001-01

Figure 1. Electronic Flight Instruments System (Sheet 3 of 7)



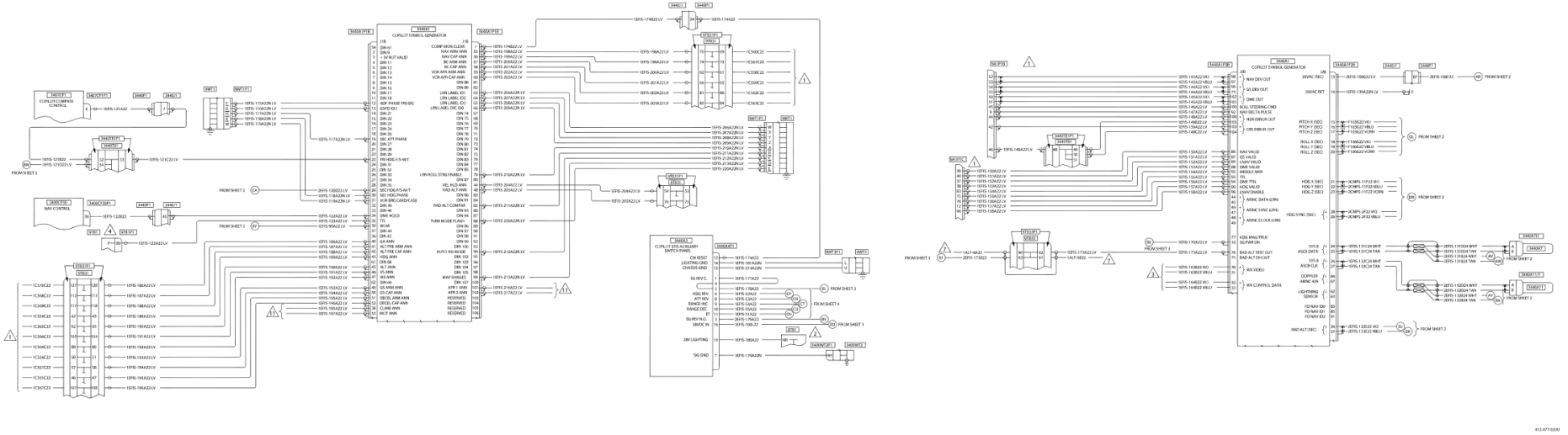
ICN-412-A-980000-B-97499-00510-A-001-01

Figure 1. Electronic Flight Instruments System (Sheet 4 of 7)



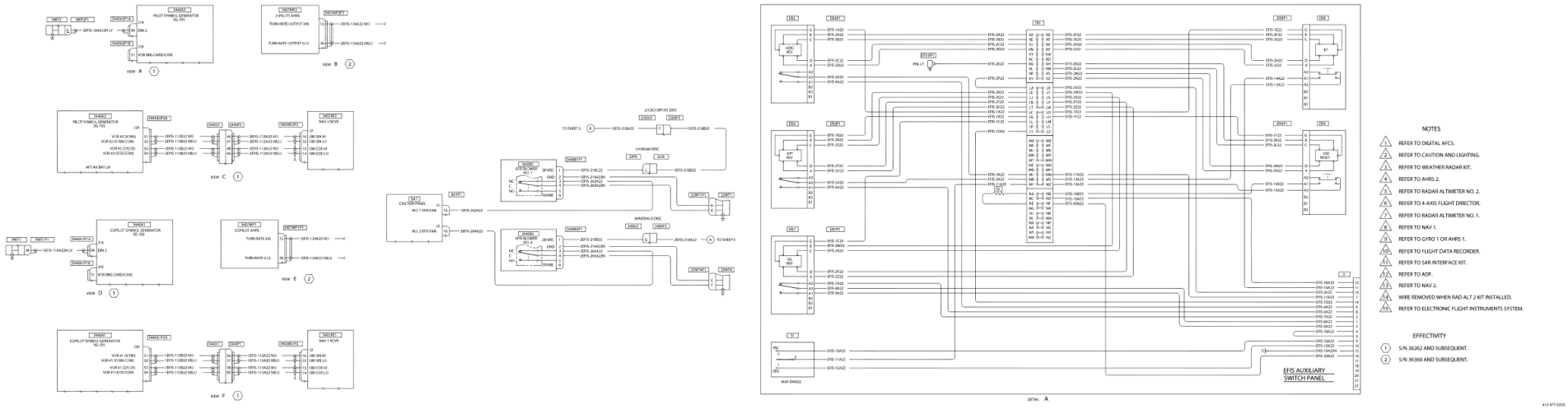
ICN-412-A-980000-B-97499-00511-A-001-01

Figure 1. Electronic Flight Instruments System (Sheet 5 of 7)



ICN-412-A-980000-B-97499-00512-A-001-01

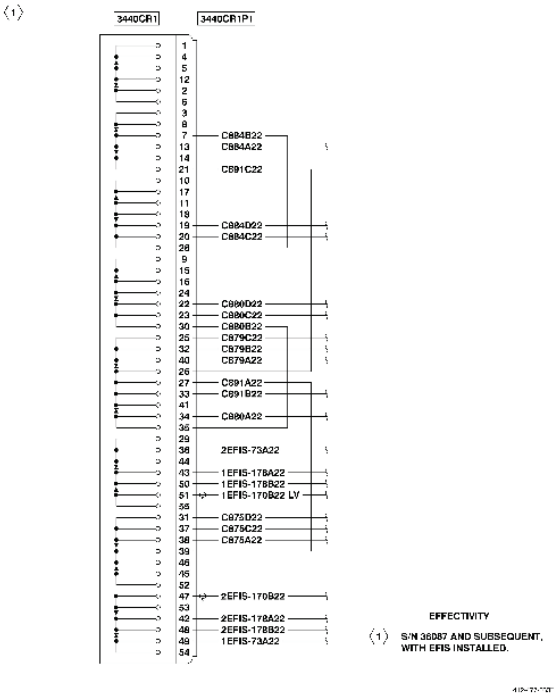
Figure 1. Electronic Flight Instruments System (Sheet 6 of 7)



ICN-412-A-980000-B-97499-00513-A-001-01

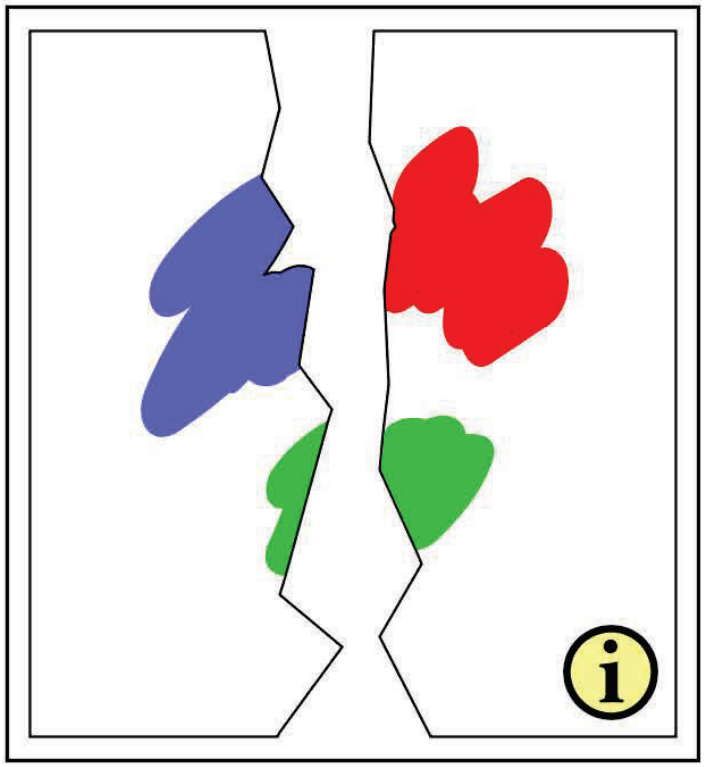
Figure 1. Electronic Flight Instruments System (Sheet 7 of 7)

Intentionally blank page



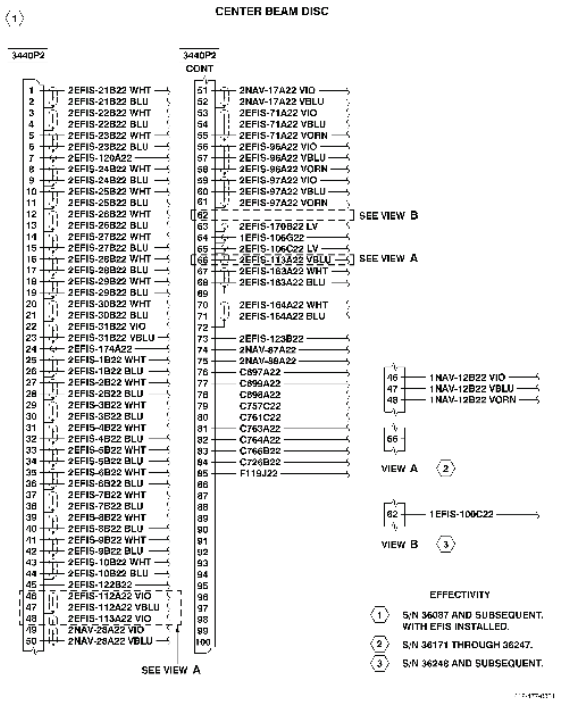
ICN-412-A-980000-B-97499-00514-A-001-01

Figure 1. Component Replacement, EFIS (Sheet 1 of 10)



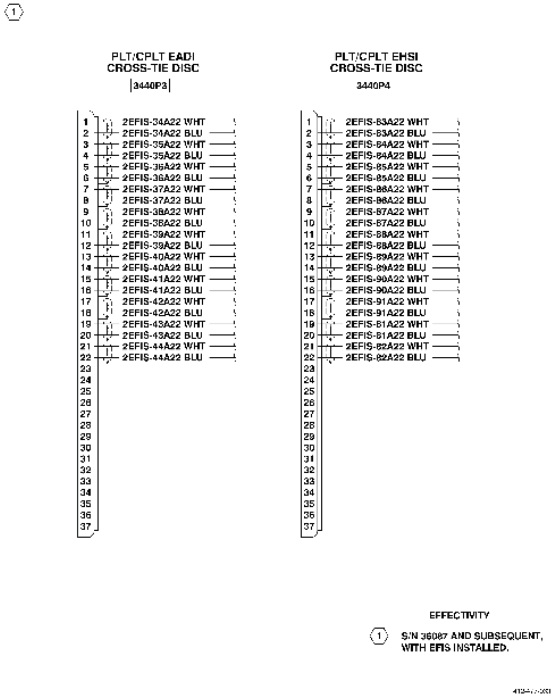
ICN-412-A-980000-B-97499-00515-A-001-01

Figure 1. Component Replacement, EFIS (Sheet 2 of 10)



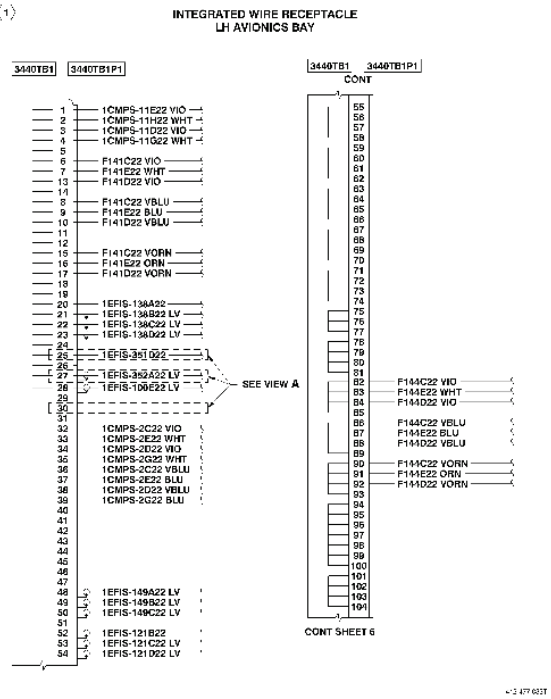
ICN-412-A-980000-B-97499-00516-A-001-01

Figure 1. Component Replacement, EFIS (Sheet 3 of 10)



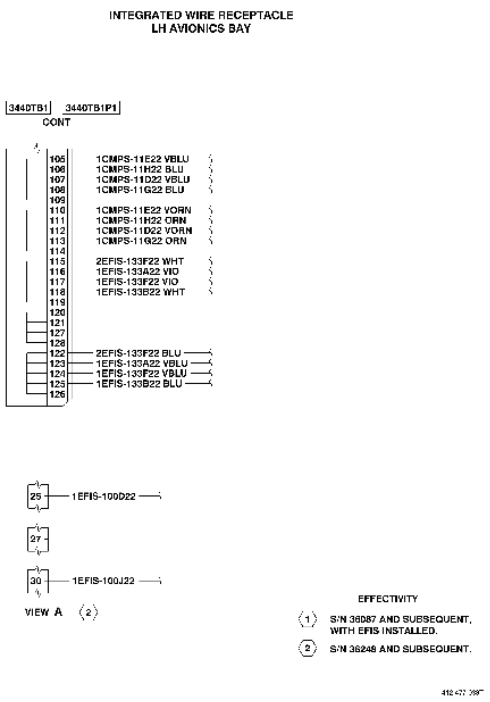
ICN-412-A-980000-B-97499-00517-A-001-01

Figure 1. Component Replacement, EFIS (Sheet 4 of 10)



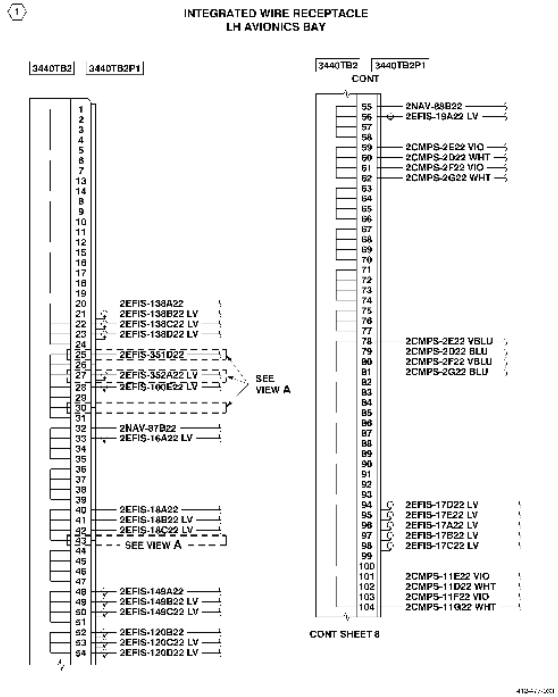
ICN-412-A-980000-B-97499-00518-A-001-01

Figure 1. Component Replacement, EFIS (Sheet 5 of 10)



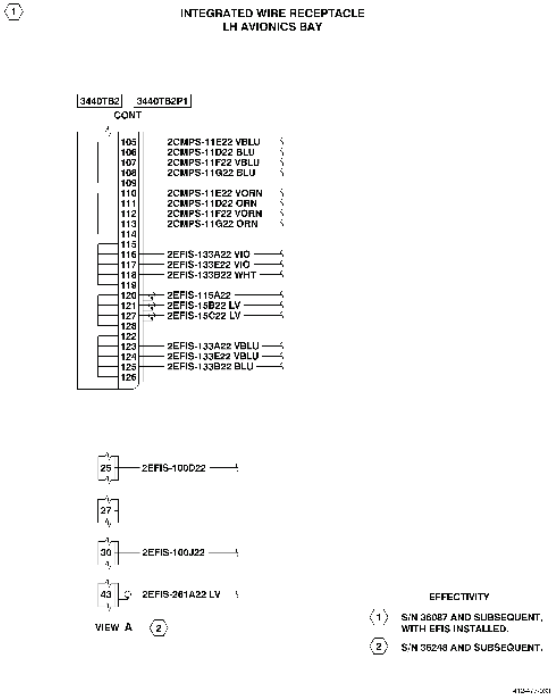
ICN-412-A-980000-B-97499-00519-A-001-01

Figure 1. Component Replacement, EFIS (Sheet 6 of 10)



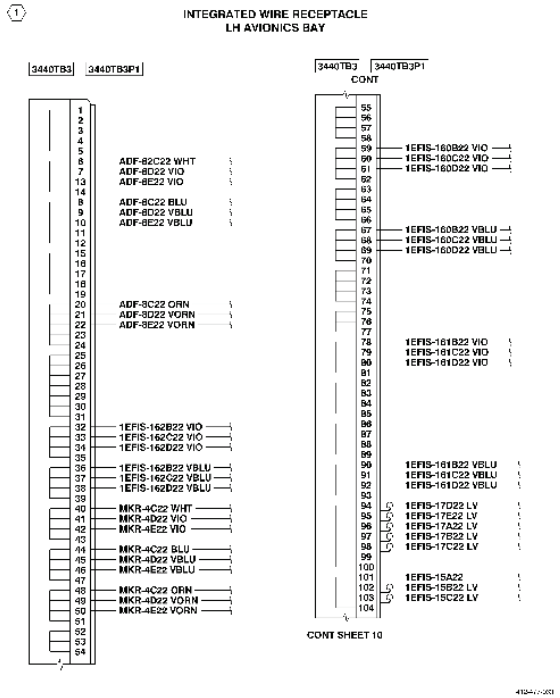
ICN-412-A-980000-B-97499-00520-A-001-01

Figure 1. Component Replacement, EFIS (Sheet 7 of 10)



ICN-412-A-980000-B-97499-00521-A-001-01

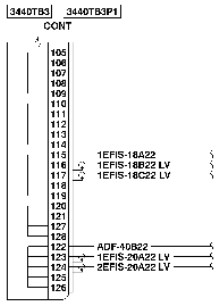
Figure 1. Component Replacement, EFIS (Sheet 8 of 10)



ICN-412-A-980000-B-97499-00522-A-001-01

Figure 1. Component Replacement, EFIS (Sheet 9 of 10)

(1) INTEGRATED WIRE RECEPTACLE
LH AVIONICS BAY



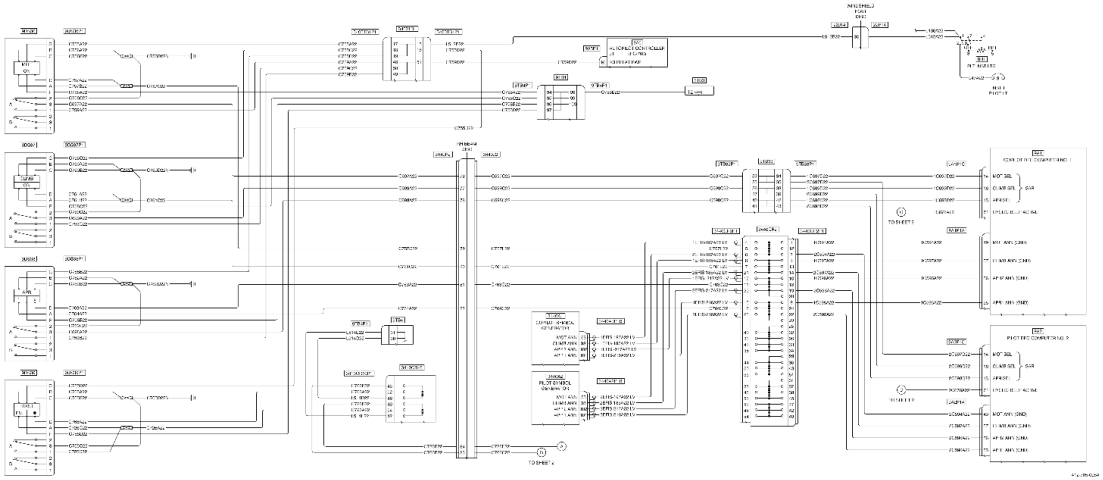
EFFECTIVITY

(1) S/N 36087 AND SUBSEQUENT, WITH EFIS INSTALLED.

412-477-399

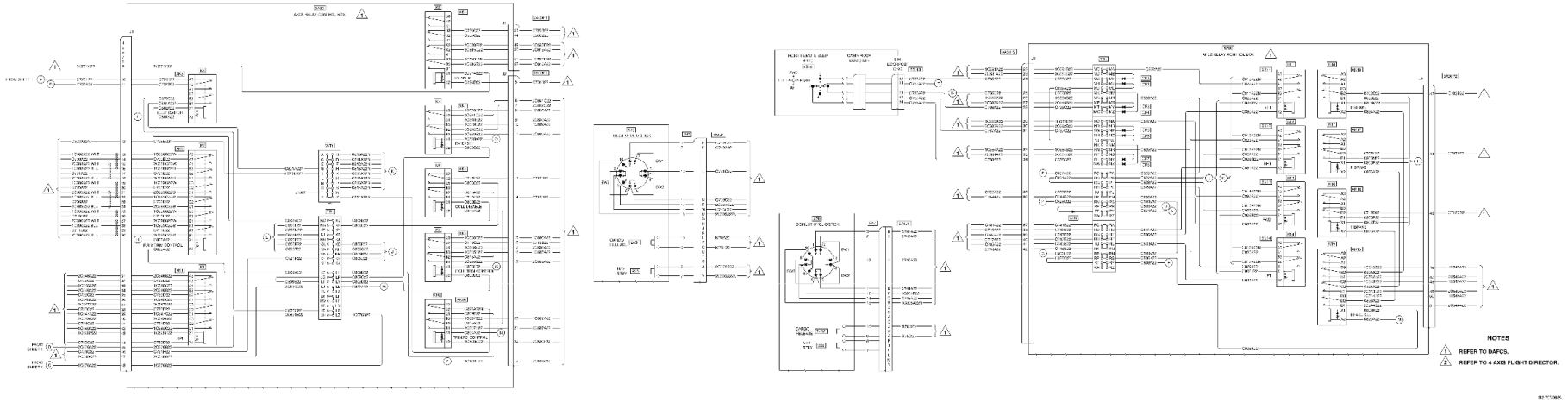
ICN-412-A-980000-B-97499-00523-A-001-01

Figure 1. Component Replacement, EFIS (Sheet 10 of 10)



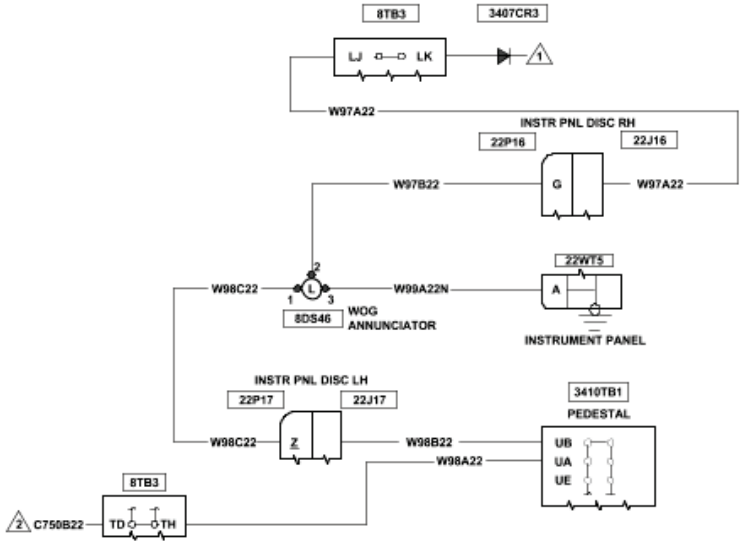
ICN-412-A-980000-B-97499-00524-A-001-01

Figure 1. SAR Interface Kit (Sheet 1 of 2)



ICN-412-A-980000-B-97499-00525-A-001-01

Figure 1. SAR Interface Kit (Sheet 2 of 2)



NOTES

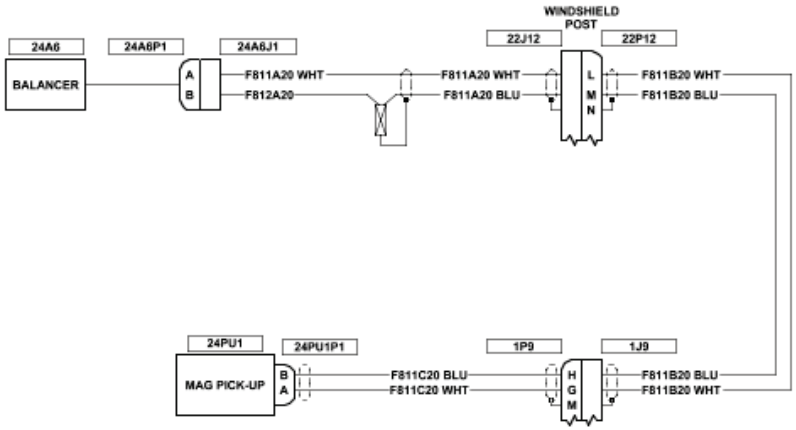
1. REFER TO COPILOT AHRS 1.
2. REFER TO DIGITAL AUTOMATIC FLIGHT CONTROL SYSTEMS.
3. FOR HELICOPTERS S/N 36248 THROUGH 36527 REFER TO BHT-412-SI-84 (412-704-039) FOR APPLICABLE WIRING DIAGRAM.

412-075-063AA

ICN-412-A-980000-B-97499-00526-A-001-01

Figure 1. Weight-on-Gear Annunciator (Sheet 1 of 1)

Intentionally blank page

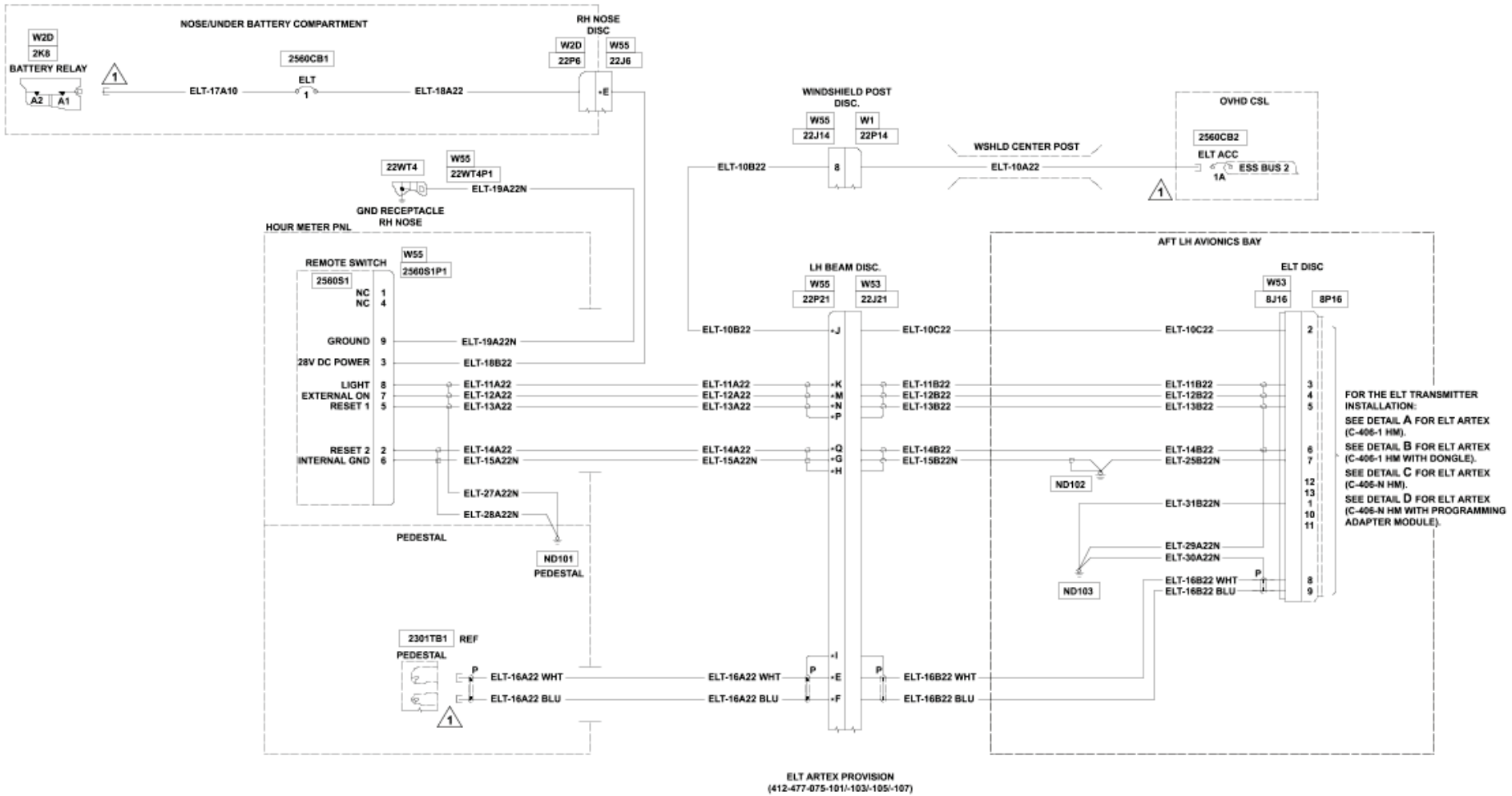


412-075-062-G 412-475-096

ICN-412-A-980000-B-97499-00527-A-001-01

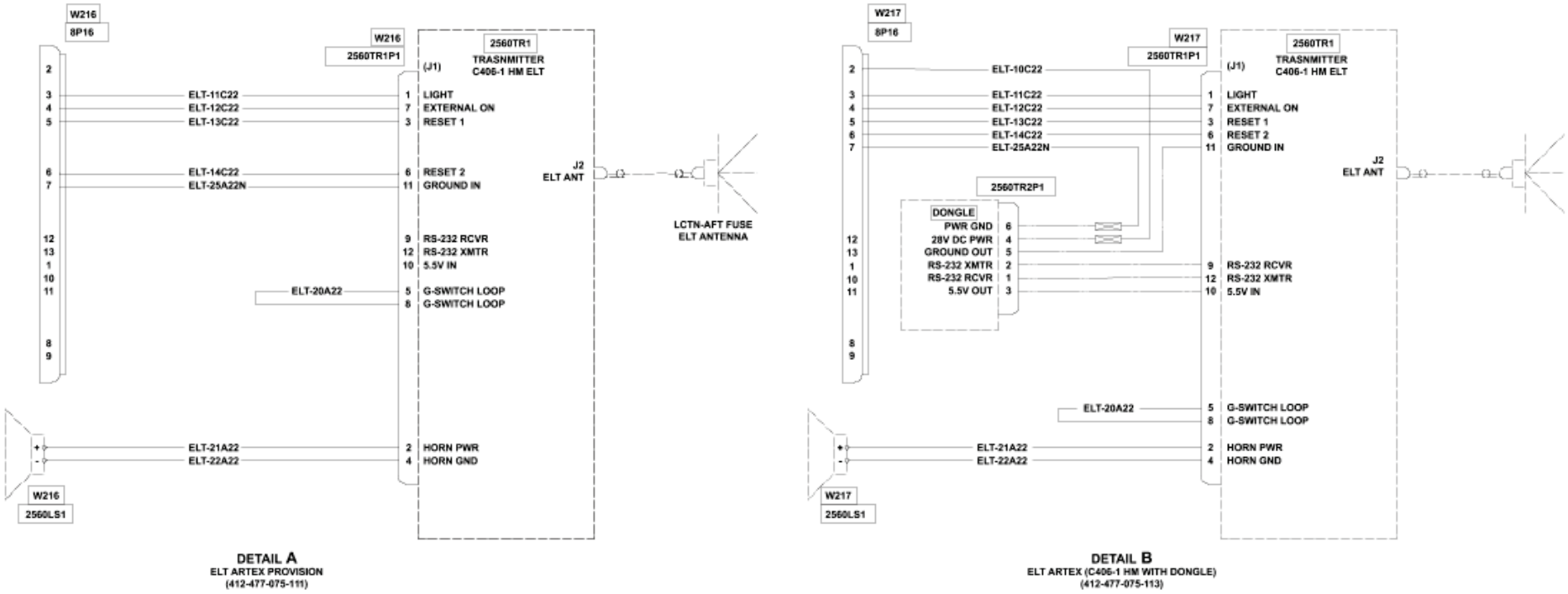
Figure 1. Rotor Track and Balance Provisions (Sheet 1 of 1)

Intentionally blank page



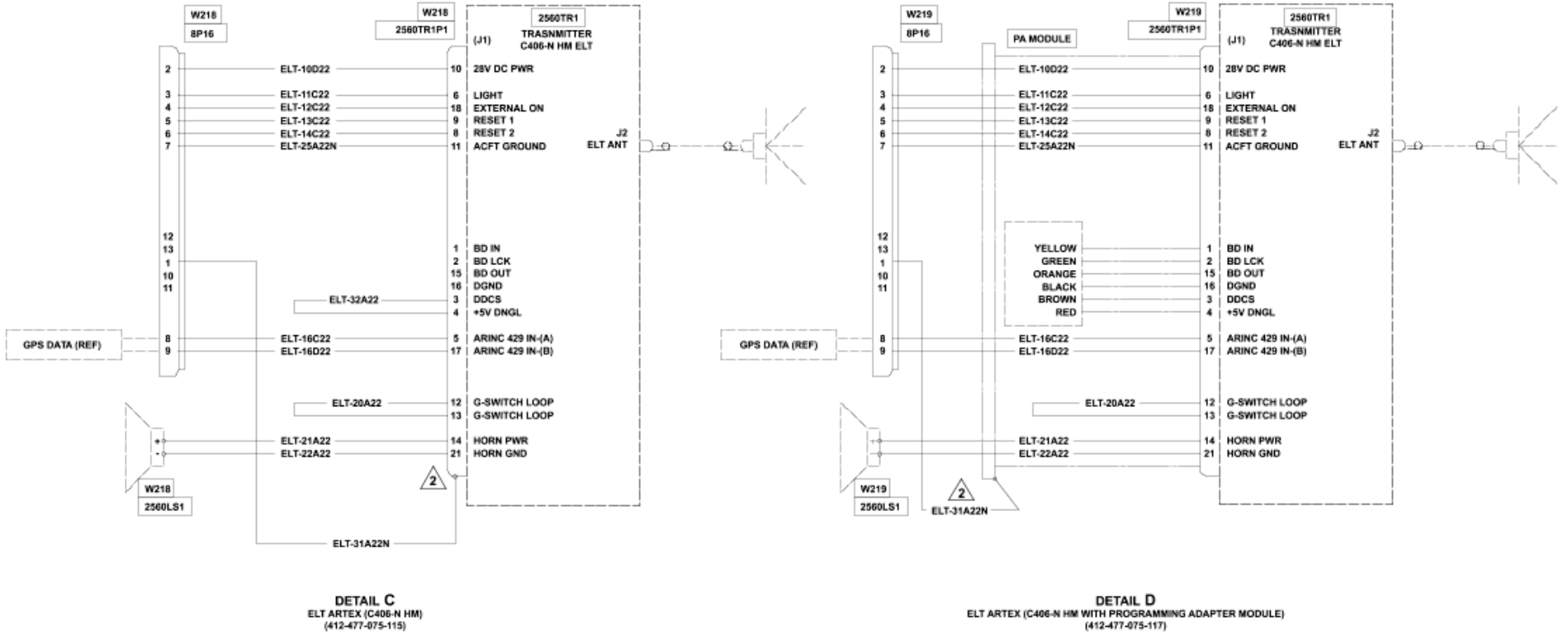
ICN-412-A-980000-B-97499-00528-A-001-01

Figure 1. Emergency Locator Transmitter (ELT ARTEX) Provision (Sheet 1 of 3)



ICN-412-A-980000-B-97499-00529-A-001-01

Figure 1. Emergency Locator Transmitter (ELT ARTEX) Provision (Sheet 2 of 3)



NOTES

1. Wires to be connected when installing the ELT transmitter.
2. Connector strain relief connected to helicopter ground.

ICN-412-A-980000-B-97499-00530-A-001-01

Figure 1. Emergency Locator Transmitter (ELT ARTEX) Provision (Sheet 3 of 3)

Intentionally blank page

CHAPTER 99
KITS

MAINTENANCE MANUAL

Intentionally blank page

List of Effective Data Modules

The listed documents are included in issue 007, dated 2025-11-18, of this publication.

C = Changed data module, N = New data module

Document title	Data module code / Seq number	Issue date	Applicable to
Maintenance Procedures - Description	DMC-412-A-00-00-00-06A-028A-A / 02623	2023-05-31	ALL
Airworthiness Limitations - Description	DMC-412-A-04-00-00-06A-288A-A / 02624	2023-05-31	ALL
Inspections and Component Overhaul Schedule - General Description	DMC-412-A-05-00-00-06A-028A-A / 02625	2023-05-31	ALL
Scheduled Inspection - Description	DMC-412-A-05-40-01-00A-042A-A / 02626	2023-05-31	ALL
Scheduled Inspection - 300 Hours/6 Calender Months	DMC-412-A-05-40-01-01A-281A-A / 02627	2023-05-31	ALL
Scheduled Inspection - 600 Hours/12 Calender Months	DMC-412-A-05-40-01-02A-281A-A / 02628	2023-05-31	ALL
Scheduled Inspection - 1200 Hours/24 Calender Months	DMC-412-A-05-40-01-03A-281A-A / 02629	2023-05-31	ALL
Maintenance Procedures - Description	DMC-412-A-00-00-00-01A-028A-A / 02630	2023-05-31	ALL
Airworthiness Limitations - Description	DMC-412-A-04-00-00-01A-288A-A / 02631	2023-05-31	ALL
Inspections and Component Overhaul Schedule - General Description	DMC-412-A-05-00-00-01A-028A-A / 02632	2023-05-31	ALL
Scheduled Inspections - Description	DMC-412-A-05-40-00-01A-042A-A / 02633	2023-05-31	ALL
Scheduled Inspections - 600 Hours/12 Calender Months	DMC-412-A-05-40-00-08A-281A-A / 02634	2023-05-31	ALL
Scheduled Inspections - 1200 Hours/24 Calender Months	DMC-412-A-05-40-00-09A-281A-A / 02635	2023-05-31	ALL
Cockpit Voice/Flight Data Recorder System - Description	DMC-412-A-97-00-01-00A-028A-A / 02636	2023-05-31	ALL
Cockpit Voice/Flight Data Recorder System - Component Description	DMC-412-A-97-00-01-00A-042A-A / 02637	2023-05-31	ALL
Parameters to Be Recorded - Description	DMC-412-A-97-00-01-00A-042B-A / 02638	2023-05-31	ALL
Cockpit Voice/Flight Data Recorder System - Troubleshooting	DMC-412-A-97-00-01-00A-422A-A / 02639	2023-05-31	ALL
Replacement of System Components - Description	DMC-412-A-97-00-01-01A-042A-A / 02640	2023-05-31	ALL
Panel Mounted Components - Removal	DMC-412-A-97-00-01-01A-520A-A / 02641	2023-05-31	ALL
Panel Mounted Components - Inspection	DMC-412-A-97-00-01-01A-280A-A / 02642	2023-05-31	ALL
Panel Mounted Components - Installation	DMC-412-A-97-00-01-01A-720A-A / 02643	2023-05-31	ALL
Shelf, Bracket or Rack Mounted Components - Removal	DMC-412-A-97-00-01-02A-520A-A / 02644	2023-05-31	ALL
Shelf, Bracket or Rack Mounted Components - Inspection	DMC-412-A-97-00-01-02A-280A-A / 02645	2023-05-31	ALL

Document title	Data module code / Seq number		Issue date	Applicable to
Shelf, Bracket or Rack Mounted Components - Installation	DMC-412-A-97-00-01-02A-720A-A / 02646		2023-05-31	ALL
Maintenance Procedures - Description	DMC-412-A-00-00-00-02A-028A-A / 02647		2023-05-31	ALL
Airworthiness Limitations - Schedule - Description	DMC-412-A-04-00-00-02A-288A-A / 02648		2023-05-31	ALL
Inspections and Component Overhaul Schedule - General Description	DMC-412-A-05-00-00-02A-028A-A / 02649		2023-05-31	ALL
Scheduled Inspections - Description	DMC-412-A-05-40-00-02A-042A-A / 02650		2023-05-31	ALL
Scheduled Inspections - 600 Hours/12 Calendar Months	DMC-412-A-05-40-00-10A-281A-A / 02651		2024-12-19	ALL
Scheduled Inspections - 1200 Hours/24 Calendar Months	DMC-412-A-05-40-00-11A-281A-A / 02652		2024-12-19	ALL
Cockpit Voice Recorder (CVR) - Description	DMC-412-A-97-00-02-00A-028A-A / 02653		2023-05-31	ALL
Cockpit Voice Recorder (CVR) - Component Description	DMC-412-A-97-00-02-00A-042A-A / 02654		2023-05-31	ALL
Recorder Download Unit (RDU) - Description	DMC-412-A-97-00-02-00A-042B-A / 02655	C	2025-11-18	ALL
CVR - Operational Test	DMC-412-A-97-00-02-00A-320A-A / 02656		2023-05-31	ALL
CVR Intelligibility - Flight Test	DMC-412-A-97-00-02-01A-320A-A / 02657		2023-05-31	ALL
ULB - Operational Test	DMC-412-A-97-00-02-02A-320A-A / 02658		2023-05-31	ALL
Cockpit Voice Recorder (CVR) - Removal	DMC-412-A-97-00-02-00A-520A-A / 02659		2023-05-31	ALL
Cockpit Voice Recorder Control unit (RCU) - Removal	DMC-412-A-97-00-02-01A-520A-A / 02660		2023-05-31	ALL
Impact Switch - Removal	DMC-412-A-97-00-02-03A-520A-A / 02661		2023-05-31	ALL
CVR Mounting Tray - Removal	DMC-412-A-97-00-02-04A-520A-A / 02662		2023-05-31	ALL
Mounting Tray - Installation	DMC-412-A-97-00-02-00A-720A-A / 02663		2023-05-31	ALL
Cockpit Voice Recorder (CVR) - Installation	DMC-412-A-97-00-02-01A-720A-A / 02664		2023-05-31	ALL
Cockpit Voice Recorder Control unit (RCU) - Installation	DMC-412-A-97-00-02-03A-720A-A / 02665		2023-05-31	ALL
Impact Switch - Installation	DMC-412-A-97-00-02-04A-720A-A / 02666		2023-05-31	ALL
CVR - Inspection	DMC-412-A-97-00-02-00A-280A-A / 02667		2023-05-31	ALL
RCU - Inspection	DMC-412-A-97-00-02-01A-280A-A / 02668		2023-05-31	ALL
Cockpit Voice Recorder (CVR) - Cleaning	DMC-412-A-97-00-02-00A-251A-A / 02669		2023-05-31	ALL
Cockpit Voice Recorder (CVR) - Repair or Replacement	DMC-412-A-97-00-02-00A-685A-A / 02670		2023-05-31	ALL

Document title	Data module code / Seq number		Issue date	Applicable to
Cockpit Voice Recorder (CVR) System - Wiring Diagram - Description	DMC-412-A-97-00-02-01A-042A-A / 02671		2023-05-31	ALL
Maintenance Procedures - Description	DMC-412-A-00-00-00-03A-028A-A / 02672		2023-05-31	ALL
Airworthiness Limitations - Schedule - Description	DMC-412-A-04-00-00-03A-288A-A / 02673		2023-05-31	ALL
Flight Data Recorder (FDR) System - General	DMC-412-A-97-00-03-00A-042A-A / 02674		2023-05-31	ALL
Flight Data Recorder (FDR) - Operational Tests	DMC-412-A-97-00-03-00A-320A-A / 02675		2023-05-31	ALL
Flight Data Recorder (FDR) System - Troubleshooting	DMC-412-A-97-00-03-00A-422A-A / 02676		2023-05-31	ALL
Flight Data Recorder (FDR) - Removal	DMC-412-A-97-00-03-01A-520A-A / 02677		2023-05-31	ALL
Flight Data Recorder (FDR) - Inspection	DMC-412-A-97-00-03-01A-280A-A / 02678		2023-05-31	ALL
Flight Data Recorder (FDR) - Cleaning	DMC-412-A-97-00-03-01A-251A-A / 02679		2023-05-31	ALL
Flight Data Recorder (FDR) - Installation	DMC-412-A-97-00-03-01A-720A-A / 02680		2023-05-31	ALL
Recorder Download Unit (RDU) - Description	DMC-412-A-97-00-03-02A-042A-A / 02681	C	2025-11-18	ALL
FDR Pre-Flight - Operational Check	DMC-412-A-97-00-03-02A-320A-A / 02682		2023-05-31	ALL
Acoustic Beacon - Test (24-month Interval) - Operational Check	DMC-412-A-97-00-03-03A-320A-A / 02683		2023-05-31	ALL
Underwater Locator Beacon (ULB) - Inspection	DMC-412-A-97-00-03-03A-280A-A / 02684		2023-05-31	ALL
Acoustic Beacon - Cleaning	DMC-412-A-97-00-03-03A-251A-A / 02685		2023-05-31	ALL
Air Data Computer (ADC) - Operational Check	DMC-412-A-97-00-03-04A-320A-A / 02686		2023-05-31	ALL
Air Data Computer (ADC) - Troubleshooting	DMC-412-A-97-00-03-04A-422A-A / 02687		2023-05-31	ALL
Air Data Computer (ADC) - Removal	DMC-412-A-97-00-03-04A-520A-A / 02688		2023-05-31	ALL
Air Data Computer (ADC) - Inspection	DMC-412-A-97-00-03-04A-280A-A / 02689		2023-05-31	ALL
Air Data Computer (ADC) - Cleaning	DMC-412-A-97-00-03-04A-251A-A / 02690		2023-05-31	ALL
Air Data Computer (ADC) - Installation	DMC-412-A-97-00-03-04A-720A-A / 02691		2023-05-31	ALL
Flight Data Acquisition Unit (FDAU) - Operational Check	DMC-412-A-97-00-03-05A-320A-A / 02692		2023-05-31	ALL
Flight Data Acquisition Unit (FDAU) - Troubleshooting	DMC-412-A-97-00-03-05A-422A-A / 02693		2023-05-31	ALL
Flight Data Acquisition Unit (FDAU) - Removal	DMC-412-A-97-00-03-05A-520A-A / 02694		2023-05-31	ALL
Flight Data Acquisition Unit (FDAU) - Inspection	DMC-412-A-97-00-03-05A-280A-A / 02695		2023-05-31	ALL

Document title	Data module code / Seq number	Issue date	Applicable to
Flight Data Acquisition Unit (FDAU) - Cleaning	DMC-412-A-97-00-03-05A-251A-A / 02696	2023-05-31	ALL
Flight Data Acquisition Unit (FDAU) - Installation	DMC-412-A-97-00-03-05A-720A-A / 02697	2023-05-31	ALL
Accelerometer - Operational Check	DMC-412-A-97-00-03-06A-320A-A / 02698	2023-05-31	ALL
Accelerometer - Troubleshooting	DMC-412-A-97-00-03-06A-422A-A / 02699	2023-05-31	ALL
Accelerometer - Removal	DMC-412-A-97-00-03-06A-520A-A / 02700	2023-05-31	ALL
Accelerometer - Inspection	DMC-412-A-97-00-03-06A-280A-A / 02701	2023-05-31	ALL
Accelerometer - Cleaning	DMC-412-A-97-00-03-06A-251A-A / 02702	2023-05-31	ALL
Accelerometer - Installation	DMC-412-A-97-00-03-06A-720A-A / 02703	2023-05-31	ALL
Flight Data Recorder (FDR) System - Repair or Replacement	DMC-412-A-97-00-03-00A-685A-A / 02704	2023-05-31	ALL
Flight Data Recorder (FDR) System - Wiring Diagram - Description	DMC-412-A-97-00-03-00A-051A-A / 02705	2023-05-31	ALL
Flight Data Recorder (FDR) - Preflight Test - Description	DMC-412-A-97-00-03-07A-042A-A / 02706	2023-05-31	ALL
Flight Data Recorder (FDR) - Preflight Test - Time Synchronization Procedure	DMC-412-A-97-00-03-07A-320A-A / 02707	2023-05-31	ALL
Flight Data Recorder (FDR) - Preflight Test - Ground Test Procedure	DMC-412-A-97-00-03-07A-136A-A / 02708	2023-05-31	ALL
Flight Data Recorder (FDR) System - Flight Test	DMC-412-A-97-00-03-07A-322A-A / 02709	2023-05-31	ALL
Flight Data Recorder System - General Description	DMC-412-A-97-00-03-00A-028A-A / 02710	2023-05-31	ALL
Flight Data Recorder System - Description	DMC-412-A-97-00-03-00A-042B-A / 02711	2023-05-31	ALL
Flight Data Recorder System - Operation	DMC-412-A-97-00-03-00A-130A-A / 02712	2023-05-31	ALL
Crash Protected Memory (A2) - Description	DMC-412-A-97-00-03-01A-042A-A / 02713	2023-05-31	ALL
Acoustic Beacon - Description	DMC-412-A-97-00-03-03A-042A-A / 02714	2023-05-31	ALL
Flight Data Recorder System - Troubleshooting	DMC-412-A-97-00-03-00A-422B-A / 02715	2023-05-31	ALL
Flight Data Recorder System - Recorder Download Unit (RDU) - Description and Operation	DMC-412-A-97-00-03-00A-090A-A / 02716	2023-05-31	ALL
Flight Data Recorder - Removal	DMC-412-A-97-00-03-00A-520A-A / 02717	2023-05-31	ALL
Flight Data Recorder - Installation	DMC-412-A-97-00-03-00A-720A-A / 02718	2023-05-31	ALL
Flight Data Recorder System - PreFlight Test - Operational Test	DMC-412-A-97-00-03-08A-320A-A / 02719	2023-05-31	ALL
Flight Data Recorder System - Beacon Test (24-Month Interval) - Operational Test	DMC-412-A-97-00-03-09A-320A-A / 02720	2023-05-31	ALL

Document title	Data module code / Seq number	Issue date	Applicable to
Flight Data Recorder System - Recorder Test - Functional Test	DMC-412-A-97-00-03-00A-340A-A / 02721	2023-05-31	ALL
Flight Data Recorder System - Inspection/Check	DMC-412-A-97-00-03-00A-280A-A / 02722	2023-05-31	ALL
Flight Data Recorder System - Cleaning	DMC-412-A-97-00-03-00A-251A-A / 02723	2023-05-31	ALL

Intentionally blank page

Maintenance Procedures - Description

Hoist HS20200-501-07 shall be maintained in accordance with Breeze-Eastern Operation and Maintenance Manual TD-99-011.

Sliding cabin door shall be maintained in accordance with the procedures defined in 412-A-52-00-00-00A-009A-A / 00774, for the sliding passenger/cargo door.

Refer to the following drawings:

- 412-709-001 - Customizing - External Hoist - Breeze
- 412-709-199 - Customizing - Sliding Cabin Door Mod R.H.S.
- 412-709-300 - Customizing - Cargo Door Roller Support
- 412-709-217 - Customizing - Breeze External Hoist Installation M412
- 412-709-319 - Customizing - Airframe Mod Hoist, M412
- 412-709-517 - Customizing - W/D & C/A Breeze External Hoist
- 412-709-704 - Customizing - Control Panel, Hoist
- 412-799-329 - Detail Parts - Hoist Installation
- 412-709-325 - Plate Mounting - Breeze Hoist
- 412-799-326 - Fitting, Breeze Hoist Detail

Intentionally blank page

Airworthiness Limitations - Description

This schedule lists the maximum airworthiness life of external hoist cutter electro-explosive device. This external hoist cable cutter device shall be removed from service before the airworthiness component life limitation is exceeded. Refer to Table 1.

Airworthiness component life limitations given to the components and assemblies are determined by experience, tests and the judgement of engineers. The lives and inspection intervals cannot be changed without the approval of the Minister of Transport.

SOME PARTS ARE INSTALLED AS ORIGINAL EQUIPMENT ON MILITARY HELICOPTERS AND MAY HAVE A LOWER AIRWORTHINESS LIFE AND/OR OVERHAUL SCHEDULE THAN WHEN USED ON A COMMERCIAL HELICOPTER. CONSEQUENTLY, PARTS THAT HAVE BEEN USED ON MILITARY HELICOPTERS SHALL NOT BE USED ON COMMERCIAL HELICOPTERS.

Neither the assignment of an airworthiness life to a component nor failure to assign an airworthiness life constitutes a warranty of any kind. The only warranty applicable to the helicopter and any component is that warranty included in the Purchase Agreement for the helicopter or the component.

This section also lists the mandatory cable cutter circuit inspection requirements. See Table 2.

Table 1. Airworthiness Limitations Schedule

PART NUMBER (1)	COMPONENT	LIFE LIMIT
KT-198	External Hoist, Cable Cutter Electro-Explosive Device	(2)

AS THE CABLE CUTTER IS CONSIDERED AN ELECTRO-EXPLOSIVE DEVICE (EED) , REMOVAL AND INSTALLATION OF CABLE SYSTEM COMPONENTS MAY HAVE TO COMPLY WITH LOCAL STANDARD AND REGULATORY PRACTICE.

NOTES:

1 Airworthiness life for part number listed applies to all successive dash numbers for that component unless otherwise specified.

2 The Maximum EED cartridge life is nine (9) years (Combination stock time and installed time). Maximum cartridge life installed is three (3) years.

Table 2. Inspection Limitations

COMPONENT	PART NUMBER	INTERVALS
Cable Cutter Circuit		Perform the tasks listed under the title CABLE CUTTER CIRCUIT in 412-A-05-40-01-01A-281A-A / 02627, every 300 hours or 6 months, whichever comes first.

Intentionally blank page

Inspections and Component Overhaul Schedule - General Description

This section contains the requirements for Scheduled Inspections for the hoist.

These inspection requirements constitute an approved inspection program designed and recommended by Bell Helicopter Textron Canada for Model 412EP helicopters equipped with Breeze External Hoist (412-709-001).

The hoist system HS20200-501-07 shall be maintained in accordance with the Breeze-Eastern Operation and Maintenance Manual TD-99-011.

The inspection intervals designed herein are the maximum allowable and should not be exceeded. When unusual local conditions, such as environmental conditions, utilization, etc., dictate, it is the responsibility of the operator to increase the scope and frequency of inspections as necessary to ensure safe operation. Relief from the requirements in this section can be requested through the local aviation authority.

NOTE

Neither the assignment of a time period for overhaul of a component nor failure to assign a time period for overhaul of a component constitutes a warranty of any kind. The only warranty applicable to the helicopter and any component is the warranty included in the Purchase Agreement for the helicopter or the component.

Intentionally blank page

Scheduled Inspection - Description

Accomplish the following Scheduled Inspections on the helicopter hoist system:

- 300 hours/6 months (412-A-05-40-01-01A-281A-A / 02627) - Accomplish each 300 hours of flight operation or 6 calendar months, whichever comes first.
- 600 hours/12 months (412-A-05-40-01-02A-281A-A / 02628) - Accomplish each 600 hours of flight operation or 12 calendar months, whichever comes first.
- 1200 hours/24 months (412-A-05-40-01-03A-281A-A / 02629) - Accomplish each 1200 hours of flight operation or 24 calendar months, whichever comes first.

Intentionally blank page

Scheduled Inspection - 300 Hours/6 Calender Months

Procedure

Table 1. 300 Hours/6 Calender Months

DATA REFERENCE	INSPECTION TASK DESCRIPTION	INITIAL MECH	INITIAL OTHER
	DATE: _____W.O. _____		
	FACILITY: _____		
	HELICOPTER S/N: _____		
	REGISTRY No.: _____		
	TOTAL TIME: _____TOTAL CYCLES: _____		
	NOTE Accomplish helicopter hoist system inspection each 300 hours flight time or 6 calendar months; whichever occurs first. Any part deemed out of tolerance or any discrepancies found by inspection should be noted and brought to the attention of Bell Helicopter Product Support Engineering.		
	GENERAL.		
	1. Review the helicopter logbook for recorded hoist system discrepancies. Correct any discrepancies.		
	2. Ensure all applicable and required inspections, Technical Bulletins (T.B.) , Alert Service Bulletins (A.S.B.) are accomplished.		
	PRELIMINARY REQUIREMENTS.		
Corrosion Control Guide, CSSD-PSE-87-001, Corrosion Control Guide.	1. Visually inspect all hoist supporting and back-up structure for corrosion, wear and damage.		
Bell Helicopter Standard Practices Manual, BHT-ALL-SPM, Standard Practices Manual; Drawing, 412-709-217, BHL/412-2041	**NOTE** If the four peripheral attaching bolts are within the torque limits, it is not necessary to verify the two hidden bolts in roof fitting BHL/412.1853-201.		
	2. Torque check all bolts shown on drawings 412-709-217 & BHL/412-2041.		
412-A-04-00-00-06A-288A-A / 02624	CABLE CUTTER CIRCUIT.		
412-709-517.	1. Ensure the ‘HOIST CABLE CUT’ circuit breaker is pulled out.		
	1.1. Disconnect the connector from the cable cutter EED on the hoist mechanism (Ref. 20A1P5).		
	1.2. Connect multimeter set to 28V DC between pins ‘C’ and ‘D’ of this connector with pin ‘C’ connected to the multimeter positive.		
	2. Push in the ‘HOIST CABLE CUT’ circuit breaker.		
	2.1. Verify that there is no voltage present between pins ‘C’ and ‘D’.		
	3. Set the multimeter to the ohms range.		
	3.1. Ensure that the resistance between pins ‘C’ and ‘D’ is less than 0.2 ohms.		
	4. Reset multimeter to the 28 V DC range and swap its leads to between Pins ‘A’ and ‘B’.		

Table 1. 300 Hours/6 Calender Months (continued)

DATA REFERENCE	INSPECTION TASK DESCRIPTION	INITIAL MECH	INITIAL OTHER
	DATE: _____W.O. _____		
	FACILITY: _____		
	HELICOPTER S/N: _____		
	REGISTRY No.: _____		
	TOTAL TIME: _____TOTAL CYCLES: _____		
	4.1. Ensure that there is no voltage present between pins ‘A’ and ‘B’ (pin ‘A’ is positive).		
	5. Change the range to read resistance.		
	5.1. Ensure that resistance between pins ‘A’ and ‘B’ is less than 0.2 ohms.		
	6. Reset multimeter to the 28 Vdc range.		
	6.1. Un-guard the ‘CABLE CUT’ switch and place it into the ‘ON’ position.		
	6.2. Ensure 28 V (within 0.5 V of ground power voltage level) exists between pins ‘A’ and ‘B’.		
	7. Change the multimeter to measure the voltage between pins ‘C’ and ‘D’.		
	7.1. Ensure 28 V (within 0.5 V of ground power voltage level) exists between pins ‘C’ and ‘D’.		
	8. Reset the ‘CABLE CUT’ switch to the ‘OFF’ position and pull out the ‘HOIST CABLE CUT’ circuit breaker.		
	8.1. Reconnect the connector (Ref. 20A1P5).		

Scheduled Inspection - 600 Hours/12 Calender Months

Procedure

Table 1. 600 Hour/12 Calendar Months

DATA REFERENCE	INSPECTION TASK DESCRIPTION	INITIAL MECH	INITIAL OTHER
	DATE: _____W.O. _____		
	FACILITY: _____		
	HELICOPTER S/N: _____		
	REGISTRY No.: _____		
	TOTAL TIME: _____TOTAL CYCLES: _____		
	NOTE Accomplish helicopter hoist system inspection each 600 hours flight time or 12 calendar months; whichever occurs first. Any part deemed out of tolerance or any discrepancies found by inspection should be noted and brought to the attention of Bell Helicopter Product Support Engineering.		
	GENERAL.		
	1. Review the helicopter logbook for recorded hoist system discrepancies. Correct any discrepancies.		
	2. Perform a complete 300 hour/6 month inspection.		
	3. Ensure all applicable and required inspections, Technical Bulletins (T.B.) , Alert Service Bulletins (A.S.B.) are accomplished.		
	STRUT ASSEMBLY BHL/412.1764-001.		
Bell Helicopter Standard Practices Manual, BHT-ALL-SPM, Standard Practices Manual; Drawing, 412-709-217, BHT/412-2041	1. Remove the two attaching bolts and inspect for corrosion and damage.		
	2. Check the self aligning bearings and strut end holes for wear and looseness.		
	3. Ensure the following bushings in attaching structure noted are within drawing limits:		
	3.1. Brace beam fitting assembly 412-709-217-113 (qty 2 holes).		
	3.2. Strut fitting - Pylon fwd face BHL/412.1800-201 (qty 1 hole).		
	3.3. Strut fitting - pylon side face ‘BHL/412.1832-202’ (qty 1 hole)		
	4. Install the two attaching bolts and torque (per 412-709-217).		
	TIE ROD ASSEMBLY BHL/412.1889-001.		
Bell Helicopter Standard Practices Manual, BHT-ALL-SPM, Standard Practices Manual; Drawing, 412-709-217, BHL/412-2041	1. Remove the two attaching bolts and inspect for corrosion and damage.		
	2. Check the self aligning bearings for wear and looseness.		
	3. Ensure the following bushings in attaching structure noted are within drawing limits:		
	3.1. Outboard tie rod fitting assembly BHL/412.1901-001 (qty 2 holes).		

Table 1. 600 Hour/12 Calendar Months (continued)

DATA REFERENCE	INSPECTION TASK DESCRIPTION	INITIAL MECH	INITIAL OTHER
	DATE: _____W.O. _____ **FACILITY: _____** **HELICOPTER S/N: _____** **REGISTRY No.: _____** **TOTAL TIME: _____TOTAL CYCLES: _____** 3.2. Tie rod attachment fitting 412-709-217-111 (qty 2 holes). 4. Install the two attaching bolts and torque (per 412-709-217).		

Scheduled Inspection - 1200 Hours/24 Calender Months

Procedure

Table 1. 1200 Hour/24 Calendar Months

DATA REFERENCE	INSPECTION TASK DESCRIPTION	INITIAL MECH	INITIAL OTHER
	DATE: _____W.O. _____		
	FACILITY: _____		
	HELICOPTER S/N: _____		
	REGISTRY No.: _____		
	TOTAL TIME: _____TOTAL CYCLES: _____		
	NOTE Accomplish helicopter hoist system inspection each 1200 hours flight time or 24 calendar months; whichever occurs first. Any part deemed out of tolerance or any discrepancies found by inspection should be noted and brought to the attention of Bell Helicopter Product Support Engineering.		
	GENERAL.		
	1. Review the helicopter logbook for recorded hoist system discrepancies. Correct any discrepancies.		
	2. Perform a complete 600 hour/12 month inspection.		
	3. Ensure all applicable and required inspections, Technical Bulletins (T.B.) , Alert Service Bulletins (A.S.B.) are accomplished.		
	WINCH ROOF BEAM FITTING BHL/412.1853-201.		
Bell Helicopter Standard Practices Manual, BHT-ALL-SPM, Standard Practices Manual; Drawing, 412-709-217	1. Remove and replace all six attaching nuts and bolts (NAS6606). Torque per 412-709-217.		
	2. Remove and reinstall the boom attaching bolt (NAS6708). Torque per 412-709-217.		
	3. Visually inspect all bolts and holes in fitting and boom for corrosion and damage. Ensure all hole diameters are within drawing limits.		
	HOIST BOOM ASSEMBLY BHL/412.1913-002.		
Drawing, 412-709-217	1. Accomplish magnetic particle inspection on the weld.		
	2. Visually inspect the hoist boom for general condition, corrosion, and damage.		

Intentionally blank page

Maintenance Procedures - Description

Maintenance is required on the L-3 Communications Cockpit Voice Flight Data Recorder (CVFDR) installation to insure the integrity of the system for recording voice data.

Maintenance is also required for the external Underwater Location Beacon (ULB) which is manufactured by the Dukane Corporation.

The following documents are required to perform this maintenance.

412-709-002	Customizing - Modular Airborne Data Recording & Acquisition System (MADRAS) , CVFDR.
412-709-210	Customizing - Provisions/Instl MADRAS (CVFDR).
165E1630-00	L3 Communications Operating Manual.
165E1903-00	L3 Communications Modular Airborne Data Recording & Acquisition System.

Intentionally blank page

Airworthiness Limitations - Description

This schedule lists the maximum airworthiness life of the ULB battery. This battery shall be removed from service before the airworthiness component life limitation is exceeded. Refer to Table 1.

Airworthiness component life limitations given to the components and assemblies are determined by experience, tests and the judgement of engineers. The lives and inspection intervals cannot be changed without the approval of the Minister of Transport.

SOME PARTS INSTALLED AS ORIGINAL EQUIPMENT ON MILITARY HELICOPTERS MAY HAVE A LOWER AIRWORTHINESS LIFE AND/OR OVERHAUL SCHEDULE THAN WHEN USED ON A COMMERCIAL HELICOPTER. CONSEQUENTLY, PARTS THAT HAVE BEEN USED ON MILITARY HELICOPTERS SHALL NOT BE USED ON COMMERCIAL HELICOPTERS.

Neither the assignment of an airworthiness life to a component nor failure to assign an airworthiness life constitutes a warranty of any kind. The only warranty applicable to the helicopter and any component is the warranty included in the Purchase Agreement for the helicopter or the component.

This section also lists the mandatory (CVFDR & ULB) inspection requirements. See Table 2.

Table 1. Airworthiness Limitations Schedule

PART NUMBER (1)	COMPONENT	LIFE LIMIT
810-2007	Underwater Location Beacon Battery	(2)

THE BATTERY USED IN THE UNDERWATER LOCATION BEACON IS A LITHIUM BATTERY. AS SUCH, REMOVAL, INSTALLATION AND DISPOSAL OF THIS BATTERY MAY HAVE TO COMPLY WITH LOCAL (HEALTH AND ENVIRONMENTAL) STANDARDS AND REGULATORY PRACTICE.

NOTES:

1 Airworthiness life for part number listed applies to all successive dash numbers for that component unless otherwise specified.

2 The Maximum battery life is six (6) years.

Table 2. Inspection Limitations

COMPONENT	PART NUMBER	INTERVALS
Microphone Integrity check		Perform the tasks listed under the title CVR FUNCTIONAL CHECKS in 412-A-05-40-00-08A-281A-A / 02634, every 600 hours or 12 months whichever comes first.
Underwater Location Beacon Function check		Perform the tasks listed under the check title ULB FUNCTIONAL CHECK in 412-A-05-40-00-09A-281A-A / 02635, every 1200 hours or 24 months whichever comes first.

Intentionally blank page

Inspections and Component Overhaul Schedule - General Description

412-A-05-00-00-00A-009A-A / 00014 contains the requirements for Scheduled Inspections for the CVFDR.

These inspection requirements constitute an approved inspection program designed and recommended by Bell Helicopter Textron Canada for Model 412EP helicopters equipped with the L-3 Communications Modular Airborne Data Recording and Acquisition System (412-709-002).

The inspection intervals designed herein are the maximum allowable and should not be exceeded. When unusual local conditions, such as environmental conditions, dictate, it is the responsibility of the operator to increase the scope and frequency of inspections as necessary to ensure safe operation. Relief from the requirements in 412-A-05-00-00-00A-009A-A / 00014 can be requested through the local aviation authority.

NOTE

Neither the assignment of a time period for overhaul of a component nor failure to assign a time period for overhaul of a component constitutes a warranty of any kind. The only warranty applicable to the helicopter and any component is the warranty included in the Purchase Agreement for the helicopter or the component.

Intentionally blank page

Scheduled Inspections - Description

Accomplish the following Scheduled Inspections on the CVFDR (412-A-05-40-00-08A-281A-A / 02634 through 412-A-05-40-00-09A-281A-A / 02635):

- 600 hours/12 months - Accomplish each 600 hours of flight operation or 12 calendar months, whichever comes first.
- 1200 hours/24 months - Accomplish each 1200 hours of flight operation or 24 calendar months, whichever comes first.

Intentionally blank page

Scheduled Inspections - 600 Hours/12 Calender Months

Procedure

Table 1. 600 Hours/12 Calender Months

DATA REFERENCE	INSPECTION TASK DESCRIPTION	INITIAL MECH	INITIAL OTHER
	DATE: _____W.O. _____		
	FACILITY: _____		
	HELICOPTER S/N: _____		
	REGISTRY NO.: _____		
	TOTAL TIME: _____TOTAL CYCLES:_____		
	NOTE Accomplish helicopter CVFDR system inspection each 600 hours flight time or 12 calendar months; whichever occurs first. Any discrepancies found by inspection should be noted and brought to the attention of Bell Helicopter Product Support Engineering.		
	GENERAL.		
	1. Review the helicopter logbook for recorded CVFDR system discrepancies. Correct any discrepancies.		
	2. Ensure all applicable and required inspections, Technical Bulletins (T.B.) , Alert Service Bulletins (A.S.B.) are accomplished.		
	PRELIMINARY REQUIREMENTS.		
	1. Ensure that L-3 Communications Portable Interface test equipment (PI) part number 17-TES0043 is available.		
	2. Ensure a rotorcraft headset is available.		
412-709-210 Customizing - Provisions/Instl MADRAS (CVFDR)	CVR FUNCTIONAL CHECKS.		
	1. Connect the PI, part number 17-TES0043 to the GSE connector on the CVFDR.		
L3 Communication Operating Manual P/N 165E1630-00	2. Configure the PI to play CH 3 only.		
	3. Announce microphone in use and time into the Pilot’s microphone and ensure the audio is heard from the PI internal speaker.		
	4. Configure the PI to play CH 2 only.		
	5. Announce microphone in use and time into the copilot’s microphone and ensure that the audio is heard from the PI internal speaker.		
	6. Configure the PI to play CH 4 only.		
	7. Announce microphone in use and time into the area microphone and ensure that the audio is heard from the PI internal speaker.		
	8. After completing these checks, configure the CVFDR such that voice recording is halted.		
	9. Using the Portable Interface, ensure that the previous recorded messages into the Pilot’s, Copilot’s and area microphone can be replayed and are on the correct channels.		

Intentionally blank page

Scheduled Inspections - 1200 Hours/24 Calender Months

Procedure

Table 1. 1200 Hours/24 Calender Months

DATA REFERENCE	INSPECTION TASK DESCRIPTION	INITIAL MECH	INITIAL OTHER
	DATE: _____W.O. _____		
	FACILITY: _____		
	HELICOPTER S/N: _____		
	REGISTRY NO.: _____		
	TOTAL TIME: _____TOTAL CYCLES:_____		
	NOTE Accomplish helicopter CVFDR system inspection each 1200 hours flight time or 24 calendar months; whichever occurs first. Any discrepancies found by inspection should be noted and brought to the attention of Bell Helicopter Product Support Engineering.		
	GENERAL.		
	1. Review the helicopter logbook for recorded CVFDR system discrepancies. Correct any discrepancies.		
	2. Perform a complete 600 hours/12 month inspection.		
	3. Ensure all applicable and required inspections, Technical Bulletins (T.B.) , Alert Service Bulletins (A.S.B.) are accomplished.		
412-709-210 Customizing - Provisions/Instl MADRAS (CVFDR)	PRELIMINARY REQUIREMENTS.		
	1. Ensure that the Dukane Ultrasonic Test Set, model 42A12, or equivalent, is available.		
L3 Communication Modular Airborne Data Recording and Acquisition system document P/N 165E1903-00	2. Ensure that operational instructions for the selected equipment are available.		
	ULB FUNCTIONAL CHECK.		
	1. Functional testing shall be performed in accordance with the requirements of Appendix D of the Installation and Operation Instruction Manual for the L-3 Communications Modular Airborne Data Recording and Acquisition System, document number 165E1903-00.		
	ULB CLEANING.		
	1. Clean the switch end of the beacon with a soft cloth and mild detergent, then dry thoroughly with a clean cloth. Clean the switch end insulator to prevent leakage currents from occurring across the switch. This will affect battery life. The water switch should be cleaned when dirt or dust becomes apparent.		

Intentionally blank page

Cockpit Voice/Flight Data Recorder System - Description

The combined Cockpit Voice and Flight Data Recorder (CVFDR) installed on the 412EP is the L-3 Communications FA2300 Series Modular Airborne Data Recording and Acquisition System (MADRAS) .

This combined unit replaces three boxes in a traditional installation: the Cockpit Voice Recorder (CVR) , the Flight Data Recorder (FDR) , and the Flight Data Acquisition Unit (FDAU) .

The Cockpit Voice/Flight Data Recorder System receives power from the +28 VDC battery bus, through the 5A MADRAS circuit breaker located in the copilot's foot well.

The system installation consists of the following components:

Equipment	Part Number
Modular Airborne Data Recording and Acquisition System (MADRAS)	2315-1200-00
CVR Control Panel	S251-1130-00
Area Microphone	S055-0067-01
Accelerometer	3001-01-101-2
Impact Switch	6895-2-6-6-6
Air Data Computer	412-375-014-101

Intentionally blank page

Cockpit Voice/Flight Data Recorder System - Component Description

The FA2300 Modular Airborne Data Recording and Acquisition System (MADRAS) is housed in a 1/2 ATR short case. It is mounted to the aircraft with a standard ARINC tray fitted with a rear-mounted connector. Installation is slightly forward of, but between, the pilot's and copilot's pedals. See drawing 412-709-210 Customizing - Provisions/Instl MADRAS for component locations. The unit has an external underwater locator beacon attached via brackets.

The FDR element of the MADRAS has a comprehensive built-in-test function. Upon detection of fault within the FDR system, the white FDR SYS FAIL annunciator on the CAUTION/WARNING/ADVISORY panel is illuminated. Detected faults are stored in nonvolatile memory for later investigation by Ground Support Equipment (GSE) .

Special to type Ground Support Equipment (GSE) is required to extract and analyze the data from the FA2300. See 165E1903-00 Installation and Operation Manual - L3 Communications Modular Airborne Data Recording & Acquisition System for detailed ground support equipment requirements.

Ground support equipment is interfaced to the FA2300 via a front mounted connector. This connector is accessed through loosening a screw and rotating a protective cover to expose the contacts.

The FDR and data acquisition subsystems have comprehensive Built-In-Test (BIT) functions. Some of these are continuously monitored during the operations of the unit. Should the BIT detect an error in operation the MADRAS sets a fail output.

This output is used to drive the FDR SYS FAIL advisory annunciator on the 412EP CAUTION/WARNING/ADVISORY panel. No pilot action is required.

The cockpit voice recorder is part of the MADRAS and has the capability to record a minimum of 30 minutes of voice data. Four audio channels are provisioned in the CVR subsystem: Channel # 2 records the copilot's hot mic audio; Channel # 3 records the pilot's hot mic audio; and Channel # 4 records the cockpit area microphone audio. Channel # 1 is not used in this installation.

Prior to each flight, the CVR must undergo a pilot-initiated self-test. A successful test is indicated by the illumination of a green indicator on the CVR control panel. Refer to BHT-412-FMS-72.4 for test details.

The CVR control panel is mounted in the pedestal panel. The front face of the unit contains a headset jack, which can be used to monitor the voice data being fed to the CVR. This monitored voice data is the sum of the audio from all the input channels of the CVR subsystem of the MADRAS. The control panel houses the preamplifier for the area microphone. Electrical power to this preamplifier is supplied directly from the MADRAS.

A test switch is provided which will permit the CVR portion of the MADRAS to perform its self-test routine. A successful self-test is indicated by the illumination of the green test indicator mounted on the control panel adjacent to the test switch.

The ERASE switch on the control panel is not utilized in this installation.

The area microphone is mounted on the aircraft centerline, underneath the glareshield. It is interfaced directly to the CVR control panel, which houses the preamplifier. From the control panel, the signal is routed to the MADRAS for recording.

To prevent the continuance of recording after an accident, an impact switch is installed in the supply line to the MADRAS. Upon detection of a preset limit of "G", the switch opens, removing all power from the MADRAS. This switch, when activated, removes power from both the FDR and CVR elements of the MADRAS. The impact switch is mounted in the nose of the rotorcraft. It is a full three-axis switch, with a switching threshold of six "G" in each of the orthogonal axes.

An air data computer is installed in the nose of the aircraft and connected to the pitot-static system. It is interfaced with the MADRAS to provide air data information.

Intentionally blank page

Parameters to Be Recorded - Description

Several flight data parameters are recorded by the MADRAS. This parameter list is organized into three distinct lists:

Mandatory Parameters as mandated by FAR135/CAR 625.

Extra 412 EP Parameters for JAA requirements.

Additional Auxiliary Parameters depending on aircraft equipment configuration.

2. Mandatory Parameters.

The following list of parameters are mandatory by CFR Part 135, Appendix C and CAR 625.33 for rotorcraft used for carrying passengers and having between 10 and 19 seats:

Relative Time	Indicated Airspeed
Altitude	Magnetic Heading
Vertical Acceleration	Longitudinal Acceleration
Pitch	Roll
Altitude Rate	Main Rotor Speed
Power Turbine Speed (both engines)	Engine Torque (both engines)
Collective Position	Pedal Position
Lateral Cyclic Position	Longitudinal Cyclic Position
Hydraulic Pressure (both systems)	Radio Transmitter Key State (all radios)
Autopilot/SAS Engaged State (all systems)	SAS Fault State

The controllable stabilator position is not recorded, as the stabilator is fixed on the 412EP.

3. Extra 412EP Parameters.

The installed system also records the following additional parameters for JAA requirements:

Engine Fire Warnings (both engines)	Engine Out Warnings (both engines)
Baggage Bay Smoke Warning	Rotor Brake Warning (both)
Event Marker	Transmission Oil Pressure Status
Transmission Oil Temperature Status	Combiner Box Oil Pressure Status
Combiner Box Oil Temperature Status	Battery Temperature Warning
Master Caution Status	Outside Air Temperature

4. Additional Auxiliary Parameters.

To make better use of the extensive capabilities of the installed system, the following list of parameters may also be recorded by the MADRAS. Not all of these parameters are available on the basic 412EP model, and some are only provided when customized installations are installed.

UTC	Present Position (lat and long)
Wind Speed and Direction	Corrected Barometric Altitude
ILS/VOR Frequencies (each Nav receiver)	LOC/GS Deviations (each Nav receiver)
VOR Bearing (each Nav receiver)	DME Frequency
DME Distance	Mast Torque
Engine Gas Producer RPM (both engines)	Engine Inter-turbine Temperature (both engines)
Heading Valid	Attitude Valid
Radio #3 Key	Radio #4 Key
Radio #5 Key	

Intentionally blank page

Cockpit Voice/Flight Data Recorder System - Troubleshooting

Refer to 165E1630-00 L3 Communications Operating Manual and 412-709-524 Customizing - W/D & CA MADRAS, when troubleshooting. Perform the following checks as necessary to isolate trouble:

Check that electrical power is supplied to the component.

Check switches, relays, and other controlling devices for operation and continuity.

Check airframe wiring, plugs and connectors for condition and continuity.

Check pitot-static lines and connections for condition and security.

Perform necessary troubleshooting of system or component to isolate malfunction.

Following troubleshooting and repair, perform self-test of system as per BHT-412-FMS-72.4 to confirm satisfactory system operation.

Intentionally blank page

Replacement of System Components - Description

The following instructions cover the replacement of CV/FDR components. Refer to drawing 412-709-210 Customizing - Provisions/Instl MADRAS for exact component locations.

NOTE

Standard repair procedures are not provided in (412-A-97-00-00-00A-028A-A / 02236). Refer to FAA publication AC 43.13-1A Aircraft Inspection and Repair Manual, which shall only apply to standard repairs provided they do not exceed the component repair limitations as published by Bell Helicopter Textron for the Model 412. Repairs shall not be accomplished if weight and balance, structural integrity, interchangeability, or operational characteristics are to be adversely affected. Contact Bell Helicopter Textron Product Support Engineering if guidance is required.

Intentionally blank page

Panel Mounted Components - Removal

Procedure

ENSURE ELECTRICAL POWER IS OFF.

1. Loosen spring-lock fasteners or mounting hardware that secures component to panel.
2. Pull component out of panel to gain access to connector(s) on rear of component.
3. Disconnect electrical connector(s) or wiring and remove component.

Intentionally blank page

Panel Mounted Components - Inspection

Procedure

1. Inspect for general condition or damage.
2. Inspect condition of connectors and related wiring.

Intentionally blank page

Panel Mounted Components - Installation

Procedure

ENSURE ELECTRICAL POWER IS OFF.

1. Connect electrical connectors or wiring to component.
2. Position component in panel.
3. Secure component to panel using spring-lock fasteners or mounting hardware.

Intentionally blank page

Shelf, Bracket or Rack Mounted Components - Removal

Procedure

ENSURE ELECTRICAL POWER IS OFF.

1. Disconnect electrical and pitot-static connectors, as applicable.
2. Remove mounting screws from shelf or bracket mounted components.
3. Loosen knurled nuts from rack and slide components out of rack.
4. Lift components from mounting.

Intentionally blank page

Shelf, Bracket or Rack Mounted Components - Inspection

Procedure

1. Inspect components for general condition or damage.
2. Inspect condition of electrical connectors and pitot-static fittings.
3. Inspect condition of mounting device and hardware.

Intentionally blank page

Shelf, Bracket or Rack Mounted Components - Installation

Procedure

ENSURE ELECTRICAL POWER IS OFF.

1. Position components onto shelf, bracket, or rack.
2. Secure components to shelf or bracket using mounting screws, or to rack by tightening knurled nuts.
3. Connect electrical and pitot-static connectors, as applicable.

NOTE

A pitot-static leak check must be carried out after any lines have been disconnected and reconnected. Refer to (412-A-95-11-05-00A-364A-A / 01807) for procedures.

Intentionally blank page

Maintenance Procedures - Description

These instructions are a supplement to the 412-A-00-00-00-00A-009A-A / 00001 when the optional Cockpit Voice Recorder kit 412-705-001-115 or 412-706-084-103/-107, and/or retrofit kit 412-704-026-101/-107 is installed.

Applicable: SN: 36301 or 36304-36333:

Cockpit Voice Recorder kit 412-705-001-115 is applicable to helicopters.

Applicable: SN: 36087-36507:

Cockpit Voice Recorder kit 412-706-084-103 is applicable to helicopters.

Applicable: SN: 36508-36637:

Cockpit Voice Recorder kit 412-706-084-107 is applicable to helicopters.

Applicable: SN: 36037-36300 or 36302 or 36303:

Retrofit kits 412-704-026-101/-107 are applicable to helicopters when Cockpit Voice Recorder kit 412-705-001-101 or 412-706-084-103 has been previously installed. Retrofit kit 412-704-026-107 supersedes kit -101.

Maintenance is required on the Cockpit Voice Recorder (CVR) system to ensure the integrity of the system for recording voice data. Refer to 412-A-00-00-00-00A-009A-A / 00001 for maintenance instructions related to basic helicopter equipment.

Maintenance is also required for the external Underwater Location Beacon (ULB) .

Intentionally blank page

Airworthiness Limitations - Schedule - Description

ALL PARTS REMOVED DUE TO REACHING THEIR LIFE LIMITS ARE DEEMED UNAIRWORTHY AND SHALL BE PERMANENTLY MARKED AS SCRAP OR PHYSICALLY DESTROYED TO THE EXTENT THAT THERE IS NO CHANCE OF REPAIR OR REINSTALLATION ON ANY HELICOPTER OR COMPONENT.

SOME PARTS ARE INSTALLED AS ORIGINAL EQUIPMENT ON BOTH MILITARY AND COMMERCIAL HELICOPTERS AND MAY HAVE A LOWER AIRWORTHINESS LIFE AND/OR OVERHAUL SCHEDULE WHEN USED ON A MILITARY HELICOPTER THAN WHEN USED ON A COMMERCIAL HELICOPTER. IN ADDITION, CIRCUMSTANCES SURROUNDING THEIR USE MAY CALL FOR OPERATION OF THE MILITARY HELICOPTER OUTSIDE THE APPROVED COMMERCIAL FLIGHT ENVELOPE. CONSEQUENTLY, PARTS THAT HAVE BEEN USED ON MILITARY HELICOPTERS MUST NOT BE USED ON COMMERCIAL HELICOPTERS.

NOTE

The operating time specified for retirement for any given part number contained in the airworthiness limitations schedule applies to all successive dash numbers for that item unless otherwise specified.

Airworthiness lives assigned to helicopter components and assemblies are based upon experience, testing and engineering judgment, and are not subject to change without FAA approval.

NOTE

Neither the assignment of a retirement life to a component nor failure to assign a retirement life constitutes a warranty of any kind. The only warranty applicable to the helicopter and any component is that warranty included in the Purchase Agreement for the helicopter or the component.

The airworthiness limitations schedule summarizes the mandatory maximum hours (life) of various components that may be used before retiring from service (Table 1). Parts not listed on the schedule have unlimited airworthiness life.

Table 1. Airworthiness Limitations Schedule

PART NUMBER (1)	COMPONENT	LIFE LIMIT
810-2007 (2)	Underwater Location Beacon Battery	Six years
C362-04270-2 (3)	Underwater Location Beacon Battery	Six years

THE BATTERY USED IN THE UNDERWATER LOCATION BEACON IS A LITHIUM BATTERY. AS SUCH, REMOVAL, INSTALLATION AND DISPOSAL OF THIS BATTERY MAY HAVE TO COMPLY WITH LOCAL (HEALTH AND ENVIRONMENTAL) STANDARDS AND REGULATORY PRACTICES.

NOTES:

1 Airworthiness life for a part number listed applies to all successive dash numbers for that component unless otherwise specified.

2 Usable in underwater locator beacon model DK100 or DK90, manufactured by Dukane Corporation.

3 Usable in underwater locator beacon model ELP-362D, manufactured by Benthos, Inc.

Intentionally blank page

Inspections and Component Overhaul Schedule - General Description

412-A-05-00-00-00A-009A-A / 00014 contains the requirements for Scheduled Inspections of the Cockpit Voice Recorder (CVR) system.

The maintenance required for the CVR is the periodic replacement of the battery in the Underwater Locator Beacon (ULB) .

The inspection intervals designed herein are the maximum allowable and should not be exceeded. When unusual local conditions, such as environmental conditions, dictate, it is the responsibility of the operator to increase the scope and frequency of inspections, as necessary, to ensure safe operation. Relief from the requirement in 412-A-05-00-00-00A-009A-A / 00014 can be requested through the local aviation authority.

NOTE

Neither the assignment of a time period for overhaul of a component nor failure to assign a time period for overhaul of a component constitutes a warranty of any kind. The only warranty applicable to the helicopter and any component is the warranty included in the Purchase Agreement for the helicopter or the component.

Intentionally blank page

Scheduled Inspections - Description

Accomplish the following Scheduled Inspections on the CVR system (412-A-05-40-00-10A-281A-A / 02651 and 412-A-05-40-00-11A-281A-A / 02652):

600 hours/12 months - Accomplish each 600 hours of flight operation or 12 calendar months, whichever comes first.

1200 hours/24 months - Accomplish each 1200 hours of flight operation or 24 calendar months, whichever comes first.

Intentionally blank page

Scheduled Inspections - 600 Hours/12 Calendar Months

Procedure

Table 1. 600 Hours/12 Calendar Months

DATA REFERENCE	INSPECTION TASK DESCRIPTION	INITIAL MECH	INITIAL OTHER
	DATE: _____ W.O.: _____ **FACILITY: _____** **HELICOPTER S/N: _____** **REGISTRY NO.: _____** **TOTAL TIME: _____TOTAL CYCLES: _____** **NOTE** Accomplish helicopter CVR system inspection each 600 hours flight time or 12 calendar months, whichever occurs first. GENERAL 1. Review the helicopter logbook for recorded CVR system discrepancies. Correct any discrepancies. 2. Ensure all applicable and required inspections, Technical Bulletins (TB) and Alert Service Bulletins (ASB) are accomplished. CVR OPERATIONAL CHECK *1. Carry out the complete Cockpit Voice Recorder (CVR) operational test procedure as per instructions in* 412-A-97-00-02-00A-320A-A / 02656. 2. Visually inspect all components for general condition or damage. Inspect connectors for signs of corrosion, damaged or missing pins, and any deformation of connector body. 3. Ensure that all components and mounting hardware are secure. Physically verify that no looseness is evident at the CVR and mounting tray installation.		

Intentionally blank page

Scheduled Inspections - 1200 Hours/24 Calendar Months

Procedure

Table 1. 1200 Hours/24 Calendar Months

DATA REFERENCE	INSPECTION TASK DESCRIPTION	INITIAL MECH	INITIAL OTHER
	DATE: _____ W.O.: _____ **FACILITY: _____** **HELICOPTER S/N: _____** **REGISTRY NO.: _____** **TOTAL TIME: _____TOTAL CYCLES: _____** **NOTE** Accomplish helicopter CVR system inspection each 1200 hours flight time or 24 calendar months, whichever occurs first. Any discrepancies found by inspection should be noted and brought to the attention of Bell Helicopter Product Support Engineering. GENERAL 1. Review the helicopter logbook for recorded CVR system discrepancies. Correct any discrepancies. 2. Perform a complete 600 hours/12 month inspection. 3. Ensure all applicable and required inspections, Technical Bulletins (TB) and Alert Service Bulletins (ASB) are accomplished.		
DELETED	ULB FUNCTIONAL CHECK *1. Carry out the complete Underwater Locator Beacon (ULB) operational test as per instructions in* 412-A-97-00-02-02A-320A-A / 02658. ULB CLEANING 1. Clean the switch end of the beacon with water, mild detergent and a soft cloth, then dry thoroughly with a clean cloth. Clean the switch end insulator to prevent leakage currents from occurring across the switch. This will affect battery life. The water switch should be cleaned when dirt or dust becomes apparent.		

Intentionally blank page

Cockpit Voice Recorder (CVR) - Description

The solid-state Cockpit Voice Recorder (CVR) Model 3257A1 is installed on the Model 412EP helicopter as an optional equipment kit (Figure 1).

The CVR receives power from both No. 1 Main DC Bus and No. 2 Main DC Bus, which supply 28 VDC to the 5A CVR circuit breaker.

The CVR installation is the same for both the 3 Axis and 4 Axis DDAFCS equipped Model 412EP helicopter.

The system installation consists of a CVR, a cockpit voice Recorder Control Unit (RCU) , an impact switch and a mounting tray.

For the component reference designator refer to Table 1.

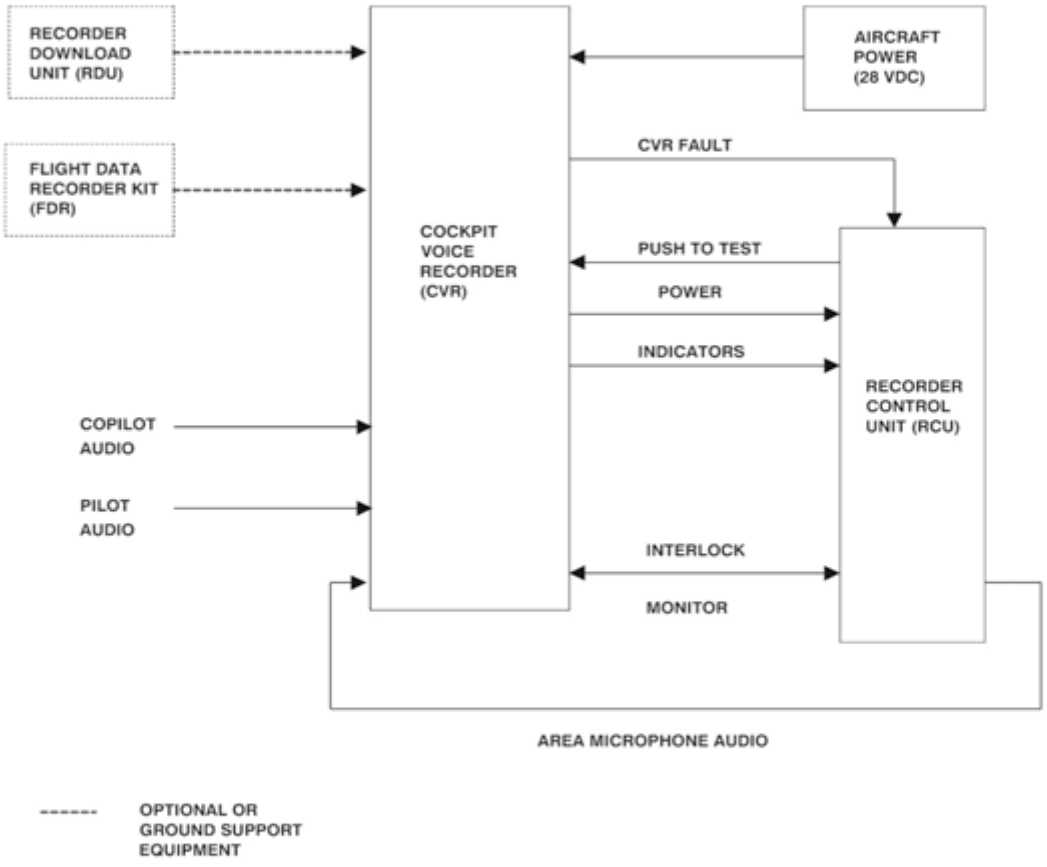
Table 1. Cockpit Voice Recorder System - Reference Table

REFERENCE DESIGNATOR	COMPONENT DESCRIPTION	LOCATION
2301CP1	Copilot ICS Control Unit (1)	Cockpit Pedestal
2301CP2	Pilot ICS Control Unit (1)	Cockpit Pedestal
2337A1	Cockpit Voice Recorder (CVR)	Nose Compartment WL 33 Shelf
2337A1P1	Connector (1)	Nose Compartment WL 33 Shelf
2337CB1	CVR Circuit Breaker (1)	Nose Compartment STA 3.00
2337CP1	Recorder Control Unit (RCU)	Cockpit Pedestal
2337CP1P1	Connector (1)	Cockpit Pedestal
2337S1	Impact Switch	Nose Compartment Underside WL 22 Shelf
27AR1	Flight Data Acquisition Unit (2)	Nose Compartment WL 22 Shelf

NOTES:

1 Basic helicopter equipment.

2 Part of flight data recorder kit (if installed).



ICN-412-A-970002-B-97499-00001-A-001-01

Figure 1. Simplified Block Diagram Cockpit Voice Recorder System (Sheet 1 of 1)

Cockpit Voice Recorder (CVR) - Component Description

The CVR (2337A1) (Figure 1) is mounted in the nose of the helicopter with a standard ARINC 404 tray. An electrical connector is located on the rear of the CVR. The unit has an external Underwater Locator Beacon (ULB) attached.

The CVR system is designed to operate whenever DC power is applied on either main DC bus. In the event of a crash, power to the CVR is shut off by means of an impact switch (2337S1) (Figure 1) installed in the nose of the helicopter. The impact switch is a three-axis switch, with a switching threshold of 6 Gs in each axis. Upon detection of the preset limit, the switch opens, removing power from the CVR.

The CVR has the capability to record at least two hours of voice data on each of the channels without combining the audio channels. This data is stored on a crash-survivable solid-state memory. Four channels of data are provisioned, three with narrow bandwidth, and one with a wide bandwidth for use by the Cockpit Area Microphone (CAM) . Two of the narrow bandwidth channels are used to record the pilot's and copilot's headphone and continuously energized microphone audio. The fourth channel is used to record the synchronization signal (timing) provided by the Flight Data Recorder (FDR) (when installed). Pilot audio is recorded on Channel 1, copilot audio on Channel 2, CAM on Channel 4 and FDR Time Correlation Signal (when used) on Channel 3. The recording automatically starts prior to engine start by switching the aircraft battery on.

The Crash Protected Memory (CPM) assembly consists of a single card within the CPM armored housing. The CPM is designed to retain, without loss or alteration, stored information without applied power for a minimum of two years.

The ULB is a self-contained locating device that aids in the underwater recovery of the CPM. It is automatically activated upon immersion in water (fresh or salt) and sends out periodic acoustic pulses for a minimum of 30 days.

The CVR Recorder Control Unit (RCU) (2337CP1) (Figure 1) is mounted in the cockpit pedestal.

The front face of the RCU contains a jack for a 600 ohm headphone, which can be used to monitor the voice data being fed to the CVR. This monitored voice data is the sum of the audio from all input channels of the CVR system. The RCU also houses the CAM and its preamplifier, which is powered by the CVR. The RCU panel lighting is supplied from the 28 VDC pedestal lighting bus and controlled by the pedestal lighting (PED LT) rheostat.

NOTE

Kits 412-705-001-115 and 412-706-084-103 and retrofit kit 412-704-026-101 have a CVR ERASE switch on the RCU. This switch is deactivated in these installations.

When the CVR is initially energized, the CVR FAIL indicator momentarily illuminates during power-up BIT. After the BIT, the CVR FAIL indicator displays the system status, as defined in Table 1. The built-in-test can also be manually initiated by pressing the CVR TEST switch.

The CVR is designed to operate with the Roanwell P/N 494-116-001-694 headsets supplied as original equipment. Other headsets with 5 ohm dynamic microphones, such as David Clark P/N H10-66 and H10-76 headsets are also acceptable.

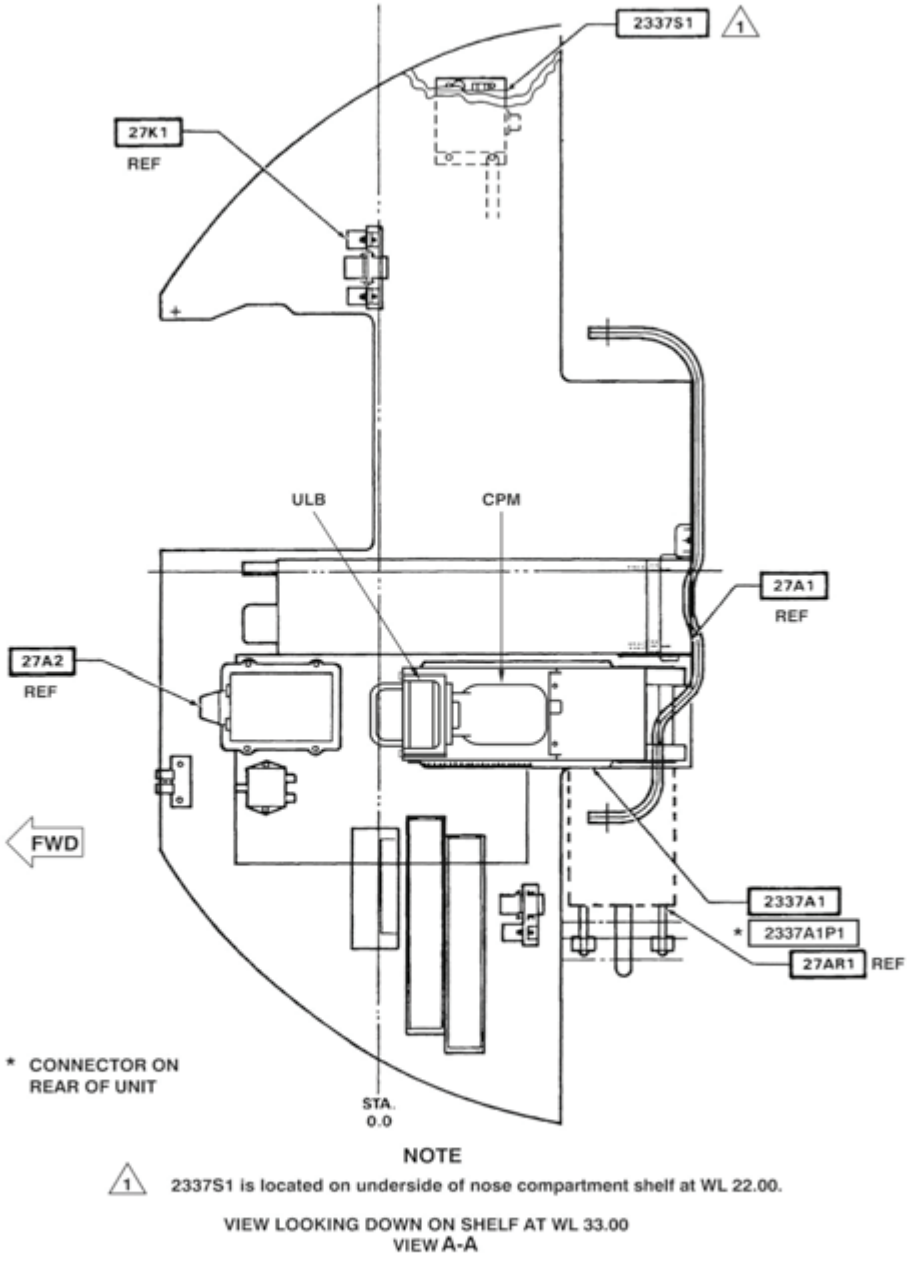
Correct CVR operation with other microphone types must be validated using the CVR intelligibility flight test in 412-A-97-00-02-01A-320A-A / 02657.

Table 1. Cockpit Voice Recorder (CVR) Indications

CVR FAIL INDICATOR	SYSTEM CONDITION
OFF	Normal Mode, CVR recording
ON	CVR recording inhibited or CVR BIT failure
ON	CVR TEST button pressed, during Power-Up or Initiated Built-In-Test (BIT)
Fast Blink (4 Hz rate)	Maintenance Mode (not recording)
Slow Blink (2 Hz rate)	Busy erasing/formatting or CPM write protect is activated
Long Blink (on for 1.8 sec., off for 0.2 sec.)	Loader Mode Active

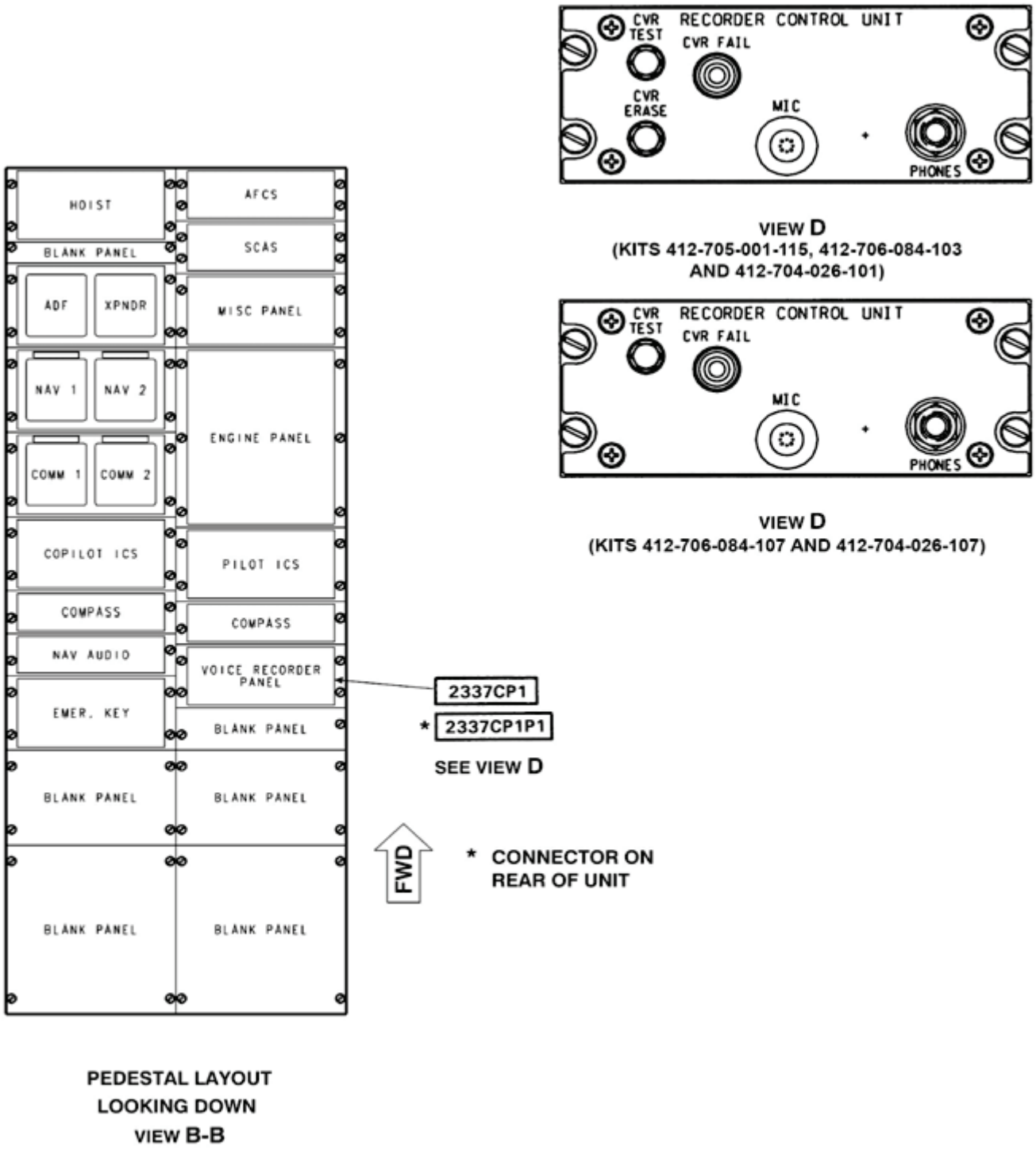
ICN-412-A-970002-B-97499-00002-A-001-01

Figure 1. Cockpit Voice Recorder (CVR) System Equipment Location (Sheet 1 of 4)



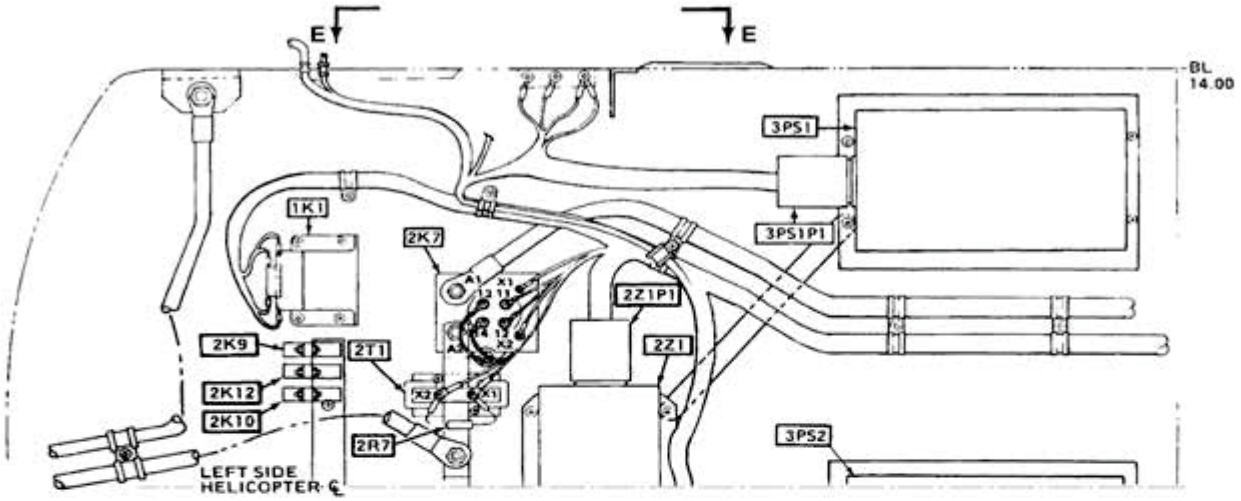
ICN-412-A-970002-B-97499-00003-A-001-01

Figure 1. Cockpit Voice Recorder (CVR) System Equipment Location (Sheet 2 of 4)

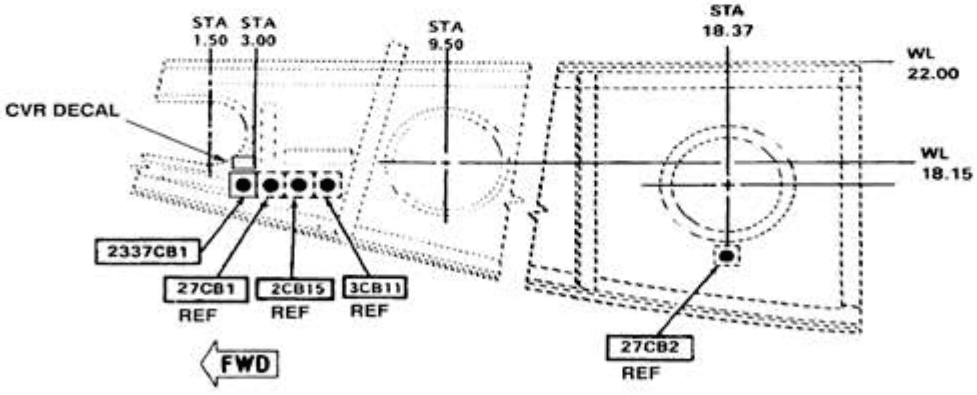


ICN-412-A-970002-B-97499-00004-A-001-01

Figure 1. Cockpit Voice Recorder (CVR) System Equipment Location (Sheet 3 of 4)



VIEW LOOKING UP AT UNDERSIDE
OF NOSE COMPARTMENT SHELF AT WL 22.00
VIEW C-C



VIEW E-E

ICN-412-A-970002-B-97499-00005-A-001-01

Figure 1. Cockpit Voice Recorder (CVR) System Equipment Location (Sheet 4 of 4)

Intentionally blank page

Recorder Download Unit (RDU) - Description

NOTE

The Recorder Download Unit is optional Ground Support Equipment and is not included as part of the CVR kit. Contact the CVR manufacturer for additional information on this equipment.

A Recorder Download Unit (RDU) is used to provide operation and maintenance data upload/download to/from the system. The RDU (Part Number PLT6254-01-01) is a PC type laptop computer with an adapter kit and a PCMCIA memory card. The adapter kit provides the communication link between the PC and CVR. The memory card provides the portable data transfer media when moving data between the RDU and Ground Support Equipment.

The RDU runs an Operational Ground Program (OGP) . The OGP communicates with the CVR through a PCMCIA interface card plugged into the laptop's PC card slot. A cable assembly interconnects the interface card to the maintenance connector on the front face of the CVR. The RDU provides the following CVR support functions:

- CVR BIT initiation and BIT data display.
- CVR memory format/initialization.
- Uploading CVR information/software.
- Downloading CVR information.

NOTE

The CVR must be removed from the helicopter for downloading.

For RDU operating instructions, refer to Extant Aerospace operation/maintenance manual. To obtain this manual and RDU information contact:

Extant Aerospace

1615 W NASA Blvd, Melbourne, FL 32901

Tel: (888) 327-7994

Mail: customerservice@extantaerospace.com

Website: www.extantaerospace.com

Intentionally blank page

CVR - Operational Test

Procedure

NOTE

The CVR is designed to operate with the Roanwell P/N 494-116-001-694 headsets supplied as original equipment. Other headsets with 5 ohm dynamic microphones, such as David Clark P/N H10-66 and H10-76 headsets are also acceptable.

Correct CVR operation with other microphone types must be validated using the CVR intelligibility flight test (412-A-97-00-02-01A-320A-A / 02657).

NOTE

A 600-ohm headphone with standard 1/4 inch headphone jack is required to monitor the CVR through the RCU.

1. Apply helicopter power to the CVR system. After power-up BIT, verify that the CVR FAIL annunciator is not ON or blinking.
2. Press and hold the CVR TEST pushbutton on the RCU while observing the CVR FAIL indicator. Verify that the indicator is illuminated while the button is pressed.
3. Release the CVR TEST pushbutton on the RCU while observing the CVR FAIL indicator. Verify that the indicator stays on for at least one second and turns off within five seconds of releasing button.
4. Open CVR circuit breaker (2337CB1) (Figure 1).
5. Disconnect the connector (2337S1P1) from the impact switch (2337S1) and verify that helicopter ground is present at pin A (Figure 2).
6. Reconnect the connector (2337S1P1) and close the CVR circuit breaker.
7. Verify that FDR and FDAU circuit breakers are opened (only if FDR kit 412-705-001-103 or 412-706-084-101 is installed).
8. Ensure the EMERG BUS 1, EMERG BUS 2, BUS 1 BATT and BUS 2 BATT circuit breakers in the nose compartment are closed.
9. Ensure the BATTERY BUS 1 and BATTERY BUS 2 switches on the overhead console are in OFF position.
10. Install pilot and copilot headsets.
11. Set BATTERY BUS 1 switch to ON position.
12. Verify that CVR FAIL indicator is not blinking or ON continuously. Depress and hold the CVR TEST pushbutton on the RCU. The CVR FAIL indicator shall illuminate while the button is pressed. Release the CVR TEST pushbutton on the RCU and verify that the indicator stays ON for at least one second and turns OFF within five seconds after releasing the CVR TEST button.
13. Plug the 600-ohm monitoring headphone into the phone jack on the RCU and verify that one high pitch tone followed by three low pitch tones are audible when the CVR TEST button is pressed.
14. Set BATTERY BUS 1 switch to OFF and BATTERY BUS 2 switch to ON.
15. Repeat Step 12. and Step 13..
16. Set BATTERY BUS 1 switch to ON.
17. Cover the area microphone located on the RCU. Place ICS Volume Rotary switch on pilot's and copilot's ICS control panel in normal operating position.
18. Speak into the pilot headset microphone, without keying the ICS. Verify that the audio can be heard on the headphone connected to the RCU. The announcement should be clear and intelligible.
19. Repeat Step 18. using the copilot headset.
20. Speak into the pilot headset microphone, with ICS keyed. Verify that the audio can be heard on the headphone connected to the RCU. The announcement should be clear and intelligible.

21. Repeat Step 20. using the copilot headset.

22. Disconnect the pilot and copilot headsets and uncover the area microphone. Speak towards the area microphone on the RCU. Verify that the audio can be heard on the headphone connected to the RCU. The announcement should be clear and intelligible.

NOTE

Perform Step 23. only if the FDR kit 412-705-001-103 or 412-706-084-101 is installed.

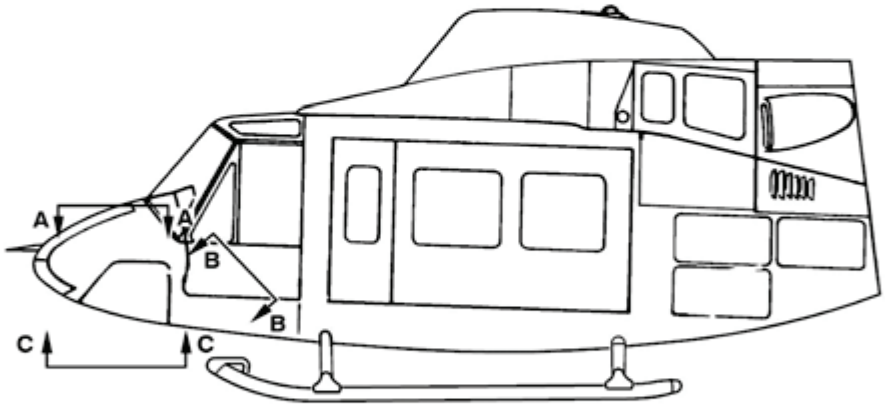
23. Close the FDR and FDAU circuit breakers. Verify that the FDR Time Correlation Signal can be heard on the headset connected to the RCU. This is a Frequency Shift Keying modulated signal transmitted once per second and identifiable as a 4000 Hz (approximate) tone.

24. Vary the pedestal lighting (PED LT) rheostat from OFF to maximum brightness (BRT) and verify if the lighting levels of the RCU front panel vary uniformly with the rheostat control.

25. Remove the 600-ohm headphone from RCU.

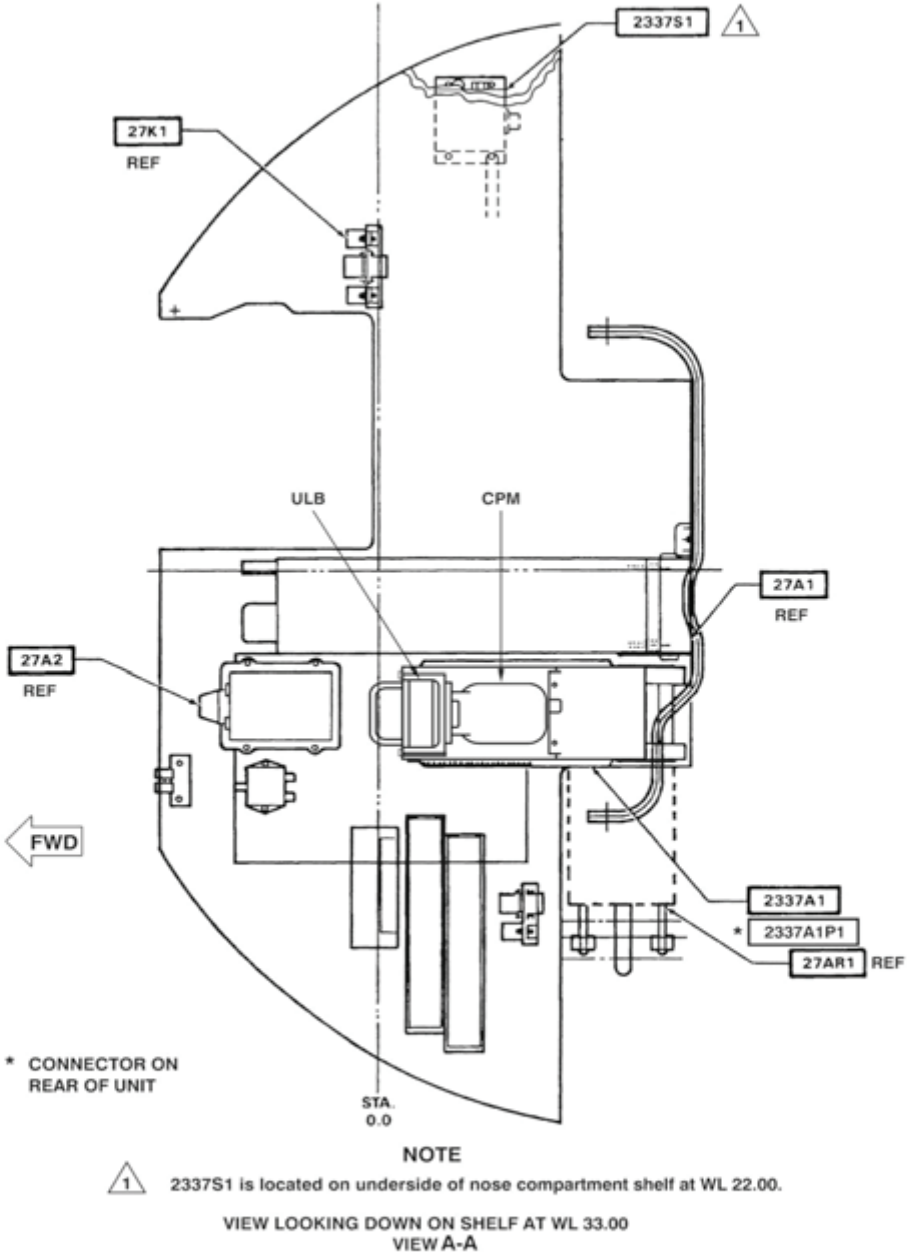
Table 1. CVR - Troubleshooting

INDICATION	ACTION
CVR FAIL indicator blinking.	1. Verify connections from RCU to CVR. 2. Check helicopter wiring between CVR pins 7 and 8. 3. Upload Control Program (for long blink condition).
CVR FAIL “ON” > 5 sec. after CVR power applied.	Replace the CVR. Replace the control unit (RCU).
CVR FAIL is not illuminated while CVR TEST button pressed.	1. Verify CVR is present and powered. 2. Check helicopter wiring interconnection between CVR pin 13 and RCU pin B. Also verify interconnection between CVR pin 14 and RCU pin C. This verifies RCU power input connection to CVR source. 3. Check CVR pin 23 with RCU pin i. 4. Replace the RCU. 5. Replace CVR.
CVR FAIL does not remain on for at least one second after CVR TEST button released.	1. Check helicopter wiring between CVR pin 12 and RCU pin A. 2. Check helicopter wiring between CVR pin 23 and RCU pin i. 3. Replace CVR. 4. Replace RCU.
CVR FAIL “ON” > 5 sec. after CVR TEST button released.	1. Replace CVR.
No audio is audible at monitor headphone.	1. Check Intercom System (ICS) is operational. 2. Check helicopter wiring continuity between ICS control unit and CVR (Figure 2). 3. Verify continuity between CVR pin 5 and RCU pin V and between CVR pin 6 and RCU pin W.
Area microphone audio is not audible at monitor headphone (but other channels are).	1. Verify continuity between CVR pins 13, 14, 45 and 47 RCU pin K and RCU pins B, C, K and L. At RCU verify continuity: N with f and P with g. 2. Replace RCU. 3. Replace CVR.
Some areas of RCU front panel are not illuminated.	1. Replace RCU.
Entire indicator panel is not illuminated.	1. Verify panel illumination voltage (28V) present at RCU pin b and ground present at pin Z. 2. Replace RCU.



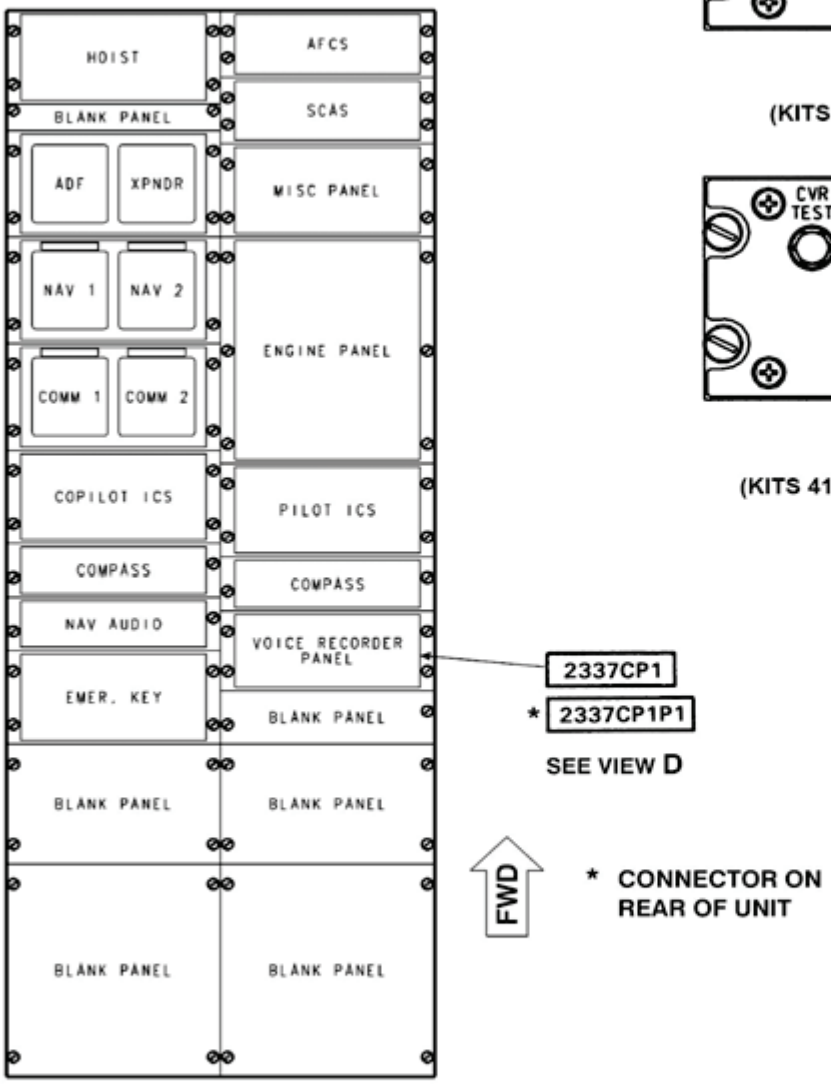
ICN-412-A-970002-B-97499-00002-A-001-01

Figure 1. Cockpit Voice Recorder (CVR) System Equipment Location (Sheet 1 of 4)

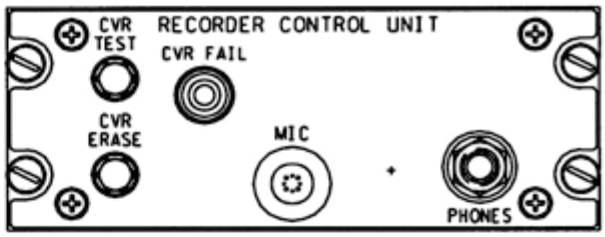


ICN-412-A-970002-B-97499-00003-A-001-01

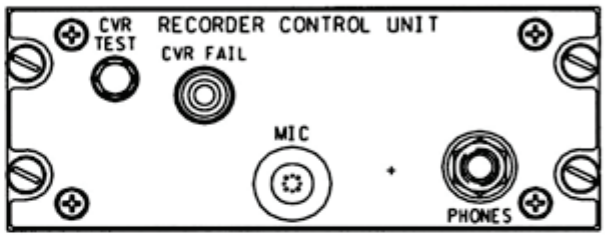
Figure 1. Cockpit Voice Recorder (CVR) System Equipment Location (Sheet 2 of 4)



PEDESTAL LAYOUT
LOOKING DOWN
VIEW B-B



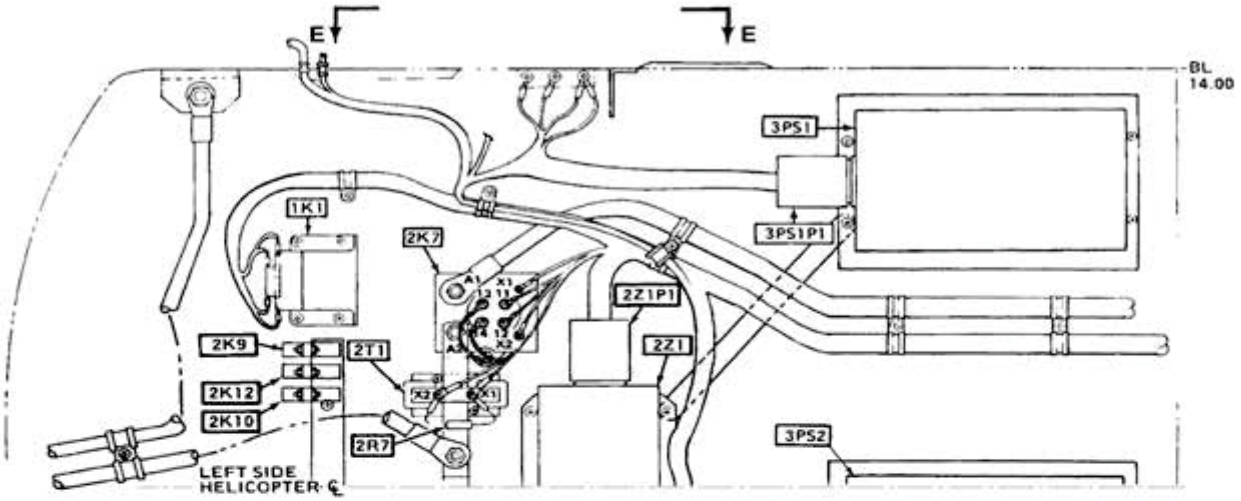
VIEW D
(KITS 412-705-001-115, 412-706-084-103
AND 412-704-026-101)



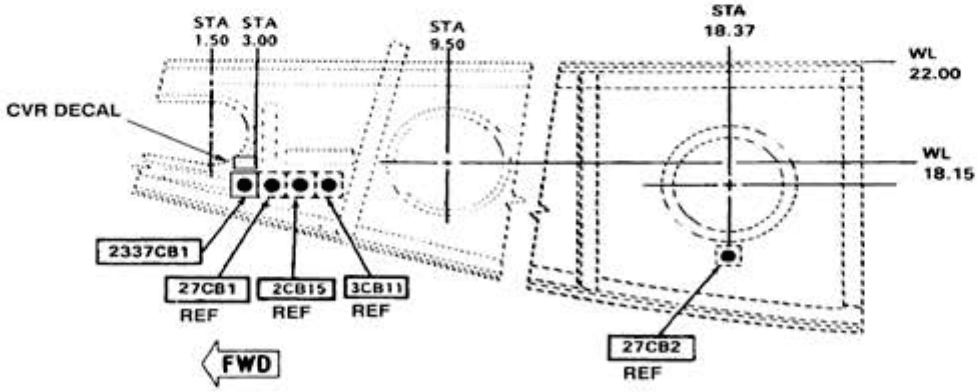
VIEW D
(KITS 412-706-084-107 AND 412-704-026-107)

ICN-412-A-970002-B-97499-00004-A-001-01

Figure 1. Cockpit Voice Recorder (CVR) System Equipment Location (Sheet 3 of 4)



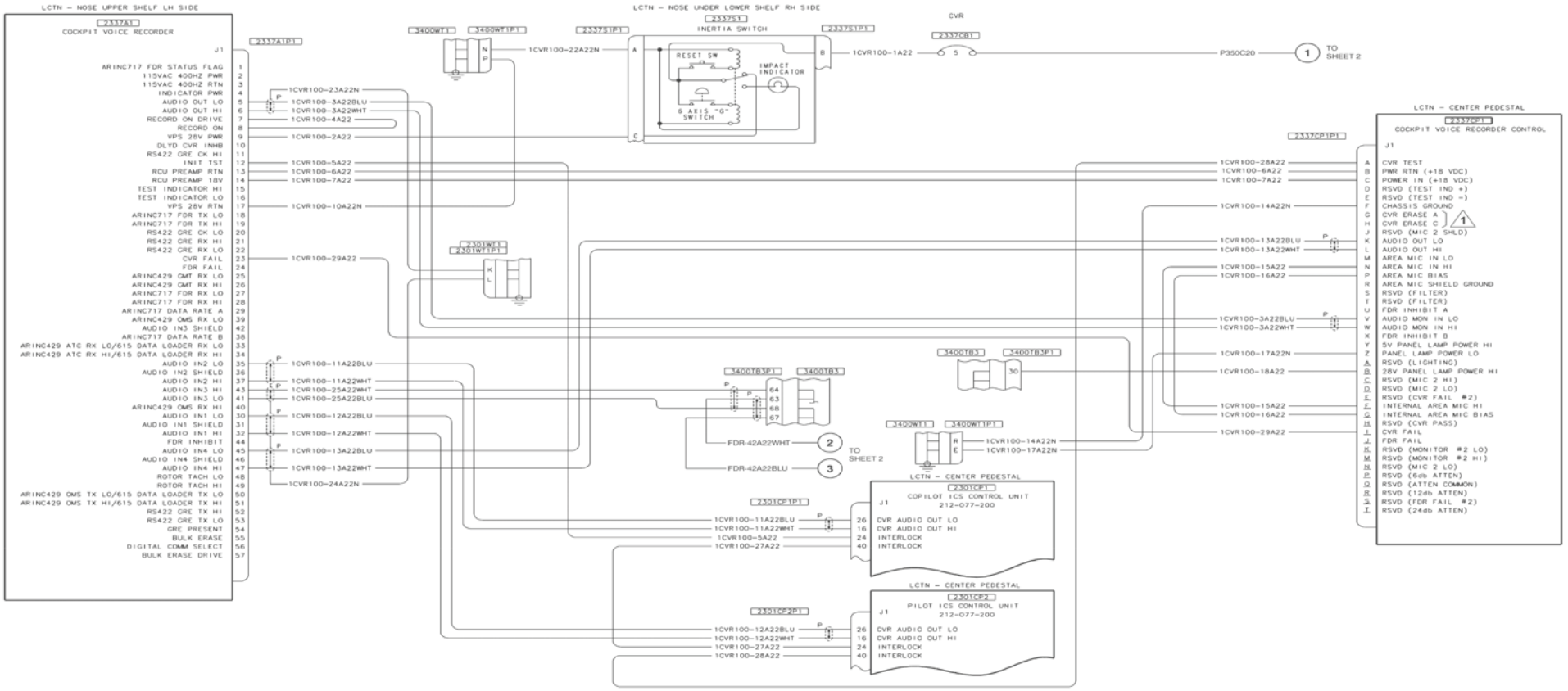
VIEW LOOKING UP AT UNDERSIDE
OF NOSE COMPARTMENT SHELF AT WL 22.00
VIEW C-C



VIEW E-E

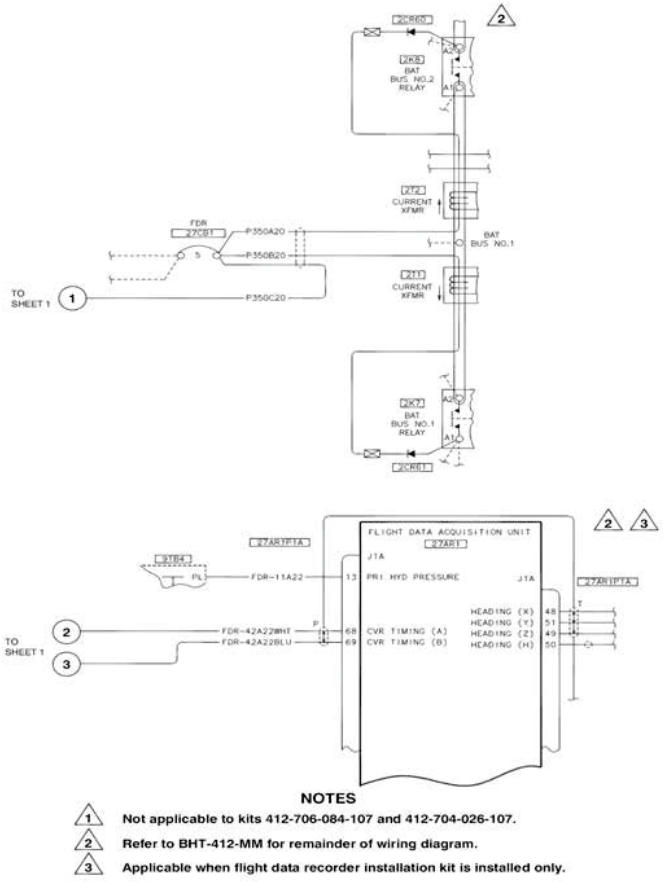
ICN-412-A-970002-B-97499-00005-A-001-01

Figure 1. Cockpit Voice Recorder (CVR) System Equipment Location (Sheet 4 of 4)



ICN-412-A-970002-B-97499-00006-A-001-01

Figure 2. Model 3257C1 Cockpit Voice Recorder (CVR) System Wiring Diagram (Sheet 1 of 2)



NOTES

1. Not applicable to kits 412-706-084-107 and 412-704-026-107.
2. Refer to BHT-412-MM for remainder of wiring diagram.
3. Applicable when flight data recorder installation kit is installed only.

ICN-412-A-970002-B-97499-00007-A-001-01

Figure 2. Model 3257C1 Cockpit Voice Recorder (CVR) System Wiring Diagram (Sheet 2 of 2)

CVR Intelligibility - Flight Test

Preliminary requirements

Support equipment

Nomenclature	Identification No.	Qty	Remarks
Recorder Downloader Unit (RDU)	- PLT6254-01-01	1	
28 VDC Power Supply	- Locally Available	1	
Connector (2337A1P1)	- M81659/29A2-0026	1	
Headset	- 494-116-001-694	1	or Equivalent

Procedure

NOTE

The CVR is designed to operate with the Roanwell headset (494-116-001-694) supplied as original equipment. Other headsets with 5 ohm dynamic microphones, such as David Clark P/N H10-66 and H10-76 headsets are also acceptable.

Correct CVR operation with other microphone types must be validated using the flight test procedure in Table 1.

1. This test verifies the operational intelligibility of the Cockpit Voice Recorder (CVR) . Do this test in one session.

In order to download the data after the test, a RDU is required (412-A-97-00-02-00A-042B-A / 02655). It is also necessary to use a 28 vdc power supply (Locally Available) to power the CVR after it is removed from the helicopter. Locally manufacture a cable assembly with connector (2337a1p1) (M81659/29A2-0026) to supply 28 VDC to the CVR (Figure 1).

1.1. Apply helicopter power to the CVR system. After power-up BIT, verify that the CVR FAIL annunciator is not ON or blinking.

1.2. Press and hold the CVR TEST pushbutton on the RCU while observing the CVR FAIL indicator. Verify that the indicator is illuminated while the button is pressed.

1.3. Release the CVR TEST pushbutton on the RCU while observing the CVR FAIL indicator. Verify that the indicator stays on for at least one second and turns off within 5 seconds of releasing button.

NOTE

Make sure the area microphone on the RCU front panel is uncovered and clear of obstructions.

During flight, the area microphone records engine, rotor, and structural noises and is not intended to record intelligible voice. All crew communication is recorded by hot microphones.

1.4. Do the steps, in order, as listed in Table 1.

1.5. After engine shutdown, download and confirm an acceptable level of intelligibility of each individual track.

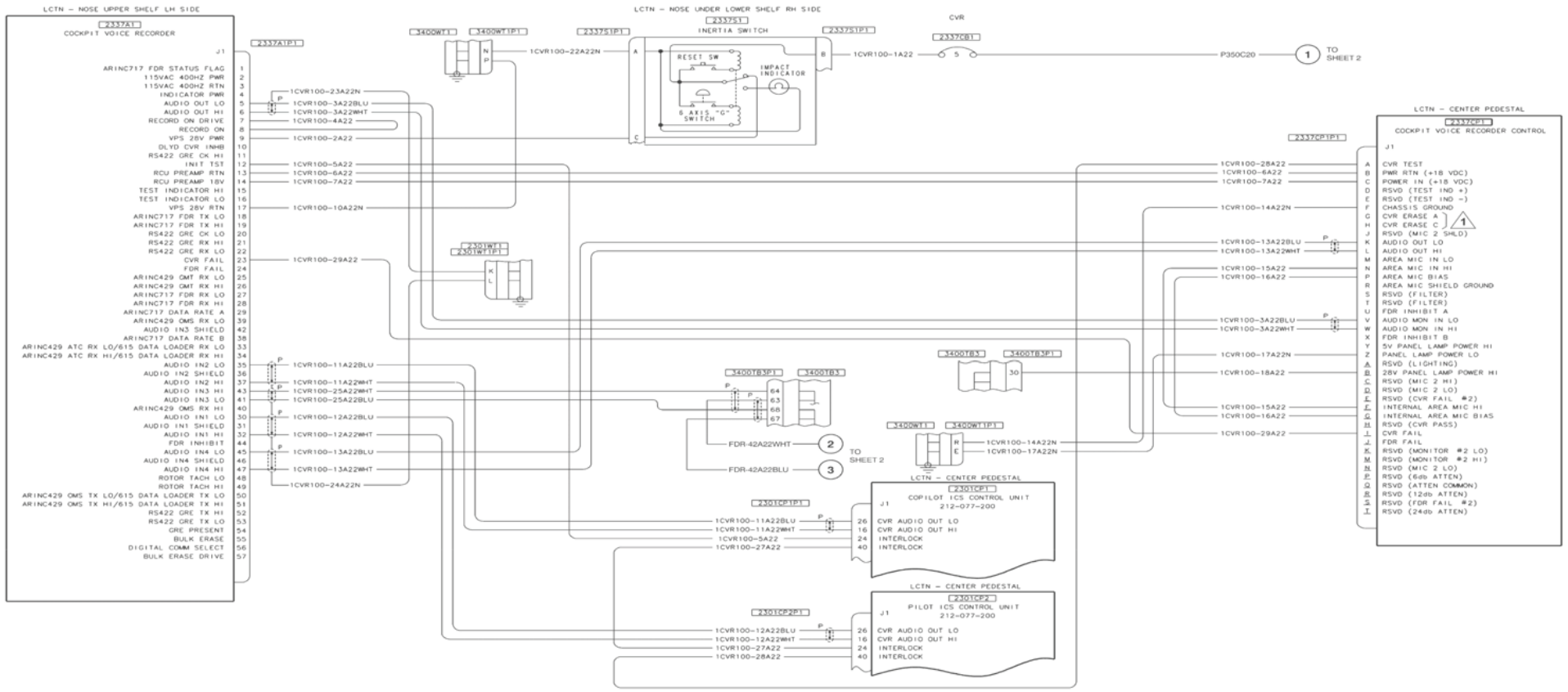
1.6. Record CVR session onto a file. Identify the file as CVR_XXXXX_YYMMDD, where XXXXX is the helicopter serial number and YYMMDD is the date of the intelligibility test.

Table 1. CVR Intelligibility - Flight Test

CREW	MICROPHONE STATUS	ANNOUNCEMENT
		PRESTART
Pilot	Unkeyed	Pilot position unkeyed, date, time, helicopter S/N
Copilot	Unkeyed	Copilot position unkeyed, date, time, helicopter S/N
Pilot	Unkeyed	Pilot position, area microphone, date, time, helicopter S/N
Copilot	Unkeyed	Copilot position, area microphone, date, time, helicopter S/N
		ENGINE START

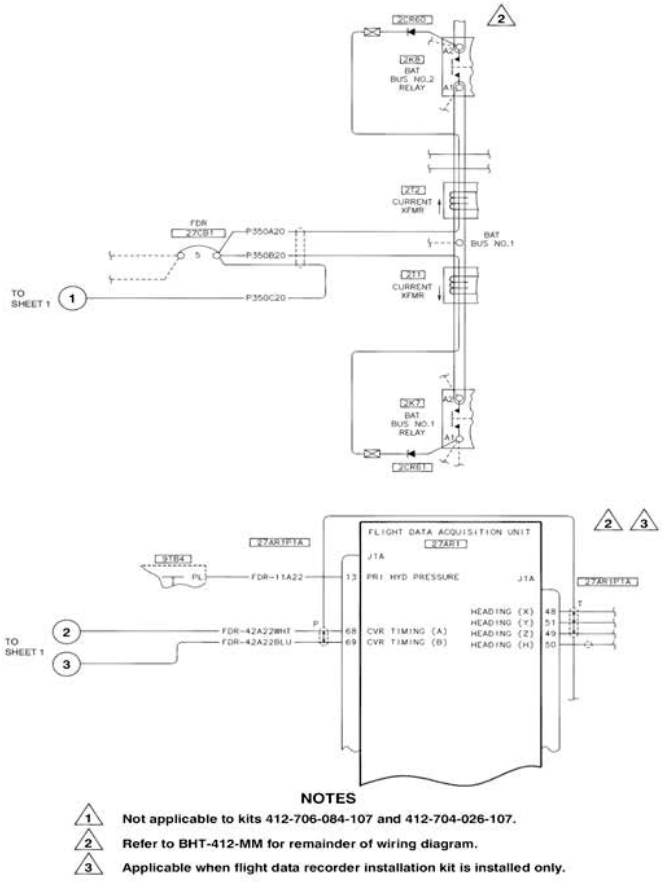
Table 1. CVR Intelligibility - Flight Test (continued)

CREW	MICROPHONE STATUS	ANNOUNCEMENT
Pilot	Keyed	Engine No., starter engaged, ITT °C start peak, and starter disengaged
Pilot	Keyed	Engine No., starter engaged, ITT °C start peak, and starter disengaged
		TAKEOFF
Pilot	Keyed	Pilot position keyed commencing takeoff
Copilot	Unkeyed	Copilot position unkeyed, airspeed knots, rate of climb ft/min
		TEST AREA
NOTE: On each audio ICS panel adjust audio volume to mid range for NAV1, NAV2, ADF, DME, COM1 and COM2. Mute all except required COM. Select required signal and un-mute long enough to record audio.		
Pilot	Keyed	Pilot position NAV1...., NAV2...., ADF...., DME....
Copilot	Keyed	Copilot position NAV1...., NAV2...., ADF...., DME....
Pilot	Keyed	Pilot position No.1 COM check and No. 2 COM check
Copilot	Keyed	Copilot position No.1 COM check and No. 2 COM check
Pilot	Keyed	Pilot position No.1 Cabin PA check
Copilot	Keyed	Copilot position No.1 Cabin PA check
		AT V_{NE}
NOTE: Open windows and vents. Turn on vent blowers. After test, close windows and vents. Set blowers OFF.		
Pilot	Keyed	Pilot position windows, vents open, blowers ON, airspeed knots
Copilot	Keyed	Copilot position windows, vents open, blowers ON, airspeed knots
		SPEED REDUCTION AND AUTOROTATION
Pilot	Keyed	Speed reduction, cruise airspeed knots
Pilot	Keyed	Entering autorotation descent. Activated low rotor RPM audio. Power recovery.
		1 MINUTE HOVER
Pilot	Keyed	1 minute hover
Copilot	Keyed	1 minute hover
		LANDING
NOTE: Monitor approach, landing, and conversations with tower. After shutdown, remove power from CVR.		
Pilot	Keyed	Pilot position conclusion of CVR test, date, time, helicopter S/N
Copilot	Keyed	Copilot position conclusion of CVR test, date, time, helicopter S/N



ICN-412-A-970002-B-97499-00006-A-001-01

Figure 1. Model 3257C1 Cockpit Voice Recorder (CVR) System Wiring Diagram (Sheet 1 of 2)



NOTES

1. Not applicable to kits 412-706-084-107 and 412-704-026-107.
2. Refer to BHT-412-MM for remainder of wiring diagram.
3. Applicable when flight data recorder installation kit is installed only.

ICN-412-A-970002-B-97499-00007-A-001-01

Figure 1. Model 3257C1 Cockpit Voice Recorder (CVR) System Wiring Diagram (Sheet 2 of 2)

ULB - Operational Test

Preliminary requirements

Support equipment

Nomenclature	Identification No.	Qty	Remarks
Acoustic Beacon Test Set	- Benthos ATS-260	1	
-	- Dukane PL-3	1	

Procedure

1. Verify that the battery expiration date listed on the acoustic beacon housing is not expired.

2. Attach the acoustic test set cable clip to the beacon.

3. Hold down the button on the test set handle labeled PUSH TO TEST.

4. Apply the test set probe to the water activated switch.

5. Check the battery condition by observing the green and the red indicators on the test set. Indicators will read as follows:

5.1. Green Indicator - The battery is good.

5.2. Red Indicator - The battery shall be replaced.

6. Check the beacon operating condition by observing the amber indicator and listening for an audible tone from the test set. Indicator and tone information will be as follows:

6.1. Amber Indicator Flashing and Tone Audible - The beacon is operating properly.

7. Release PUSH TO TEST button and remove test clip from beacon.

Intentionally blank page

Cockpit Voice Recorder (CVR) - Removal

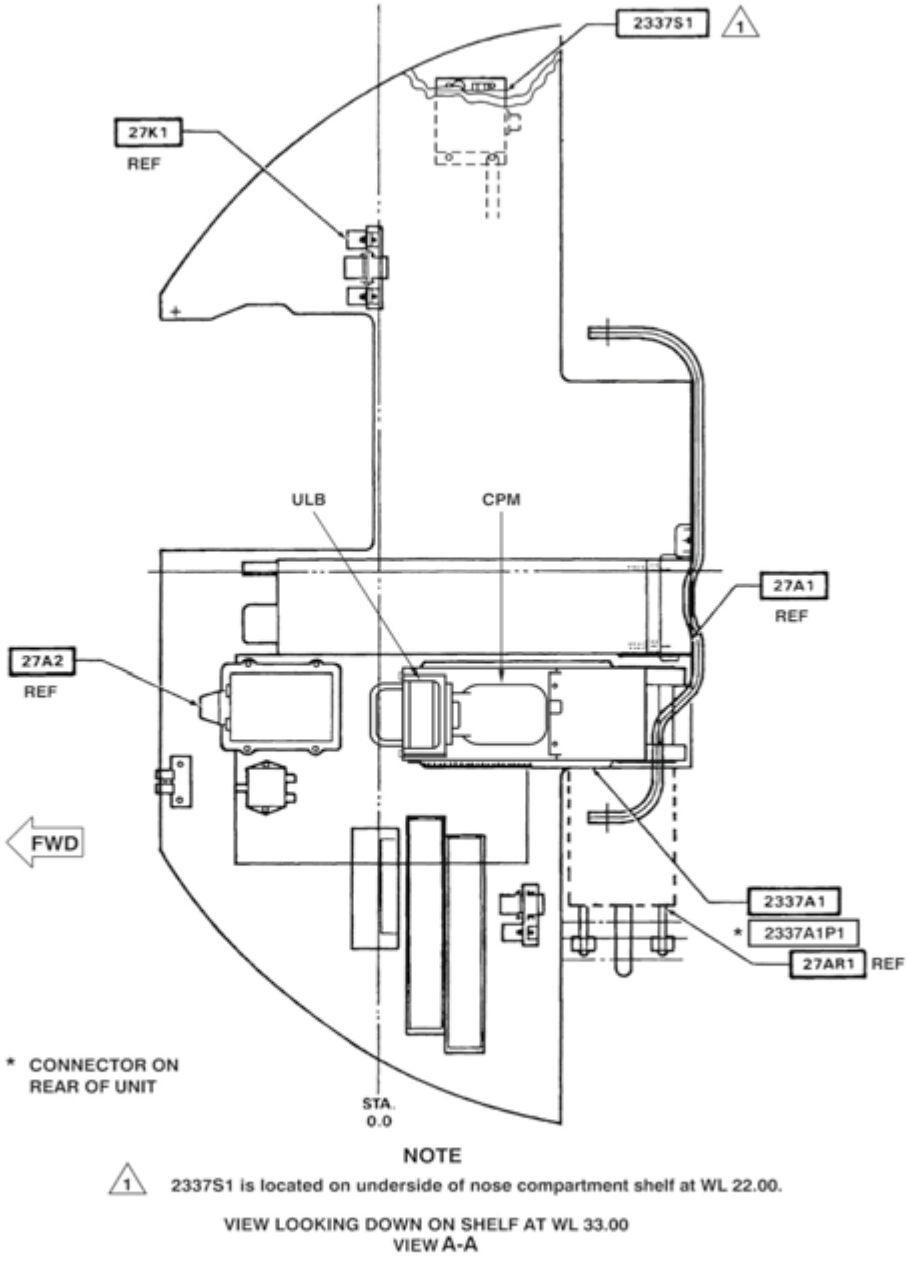
Procedure

ENSURE ELECTRICAL POWER IS OFF.

1. Ensure CVR system circuit breaker (2337CB1) is open (Figure 1).
2. Loosen the two fasteners on front of CVR (2337A1) (Figure 1) far enough to free the two hook supports.
3. Remove Cockpit Voice Recorder (2337A1) by pulling straight out of tray.

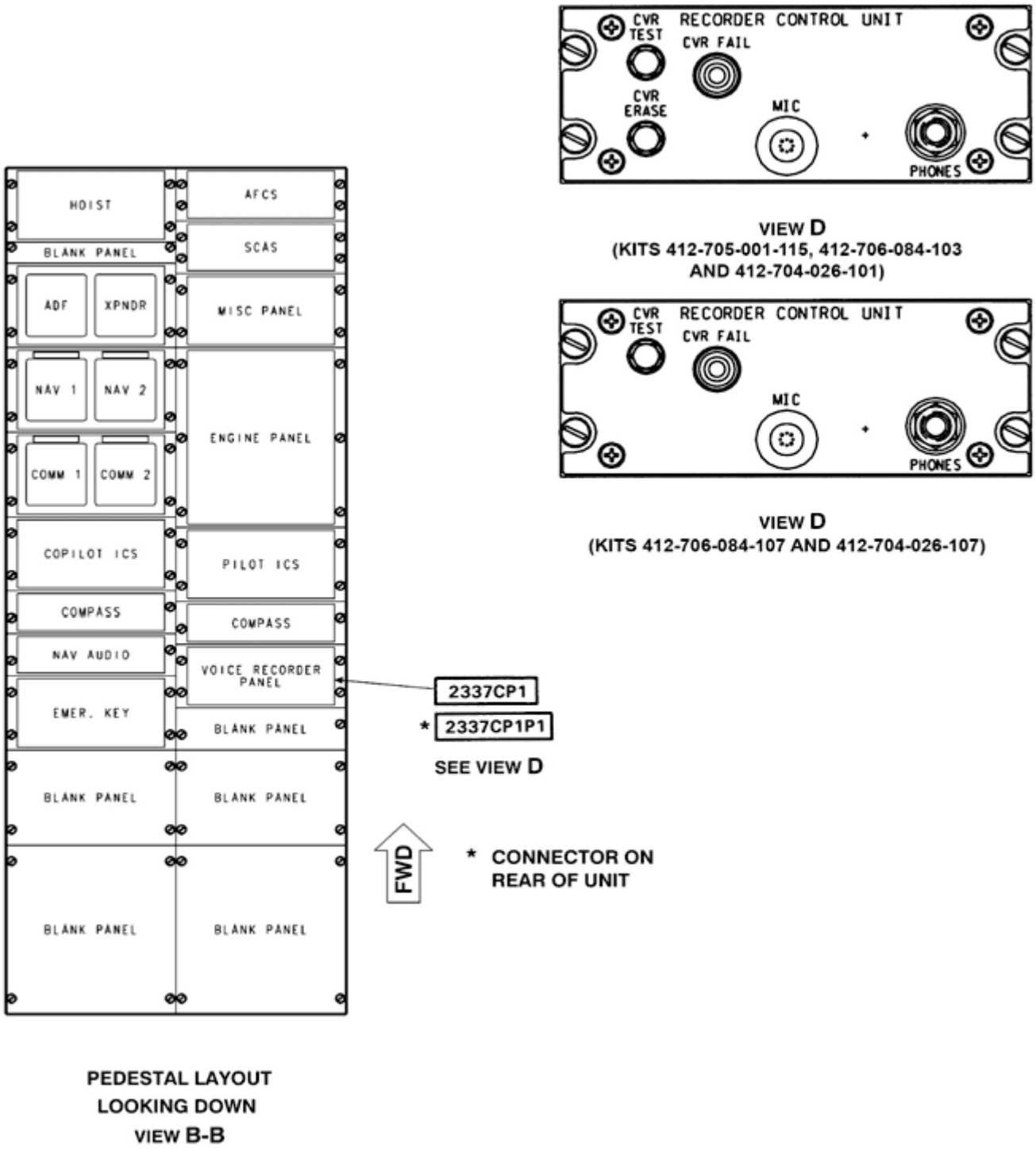
ICN-412-A-970002-B-97499-00002-A-001-01

Figure 1. Cockpit Voice Recorder (CVR) System Equipment Location (Sheet 1 of 4)



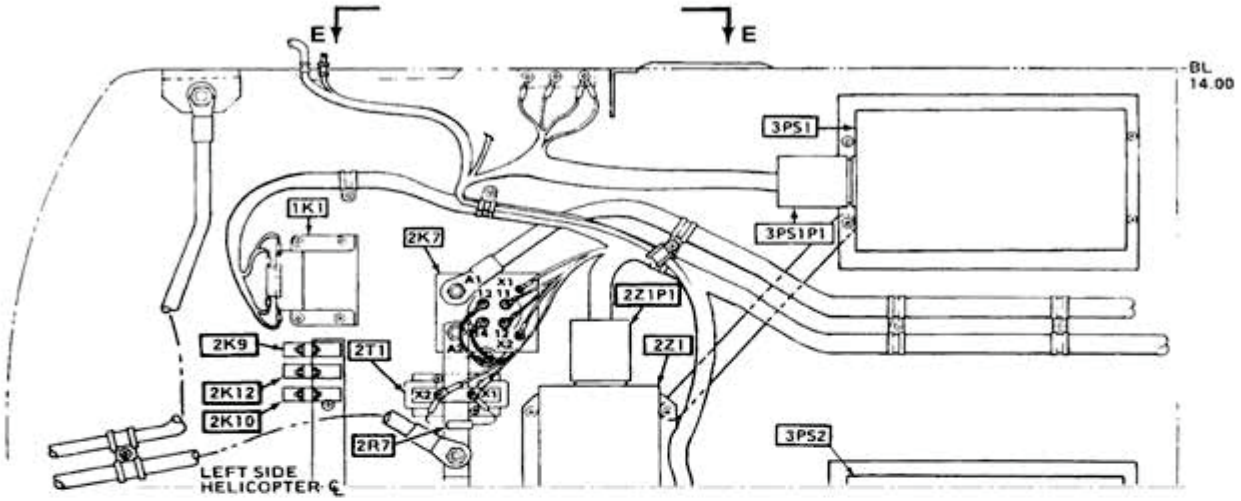
ICN-412-A-970002-B-97499-00003-A-001-01

Figure 1. Cockpit Voice Recorder (CVR) System Equipment Location (Sheet 2 of 4)

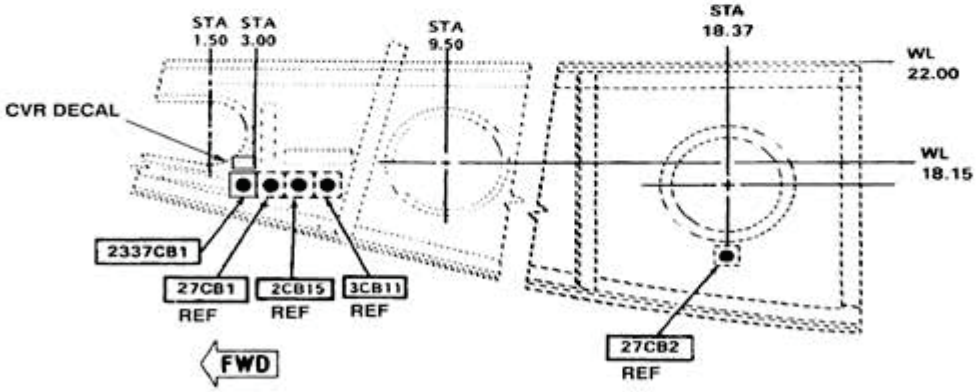


ICN-412-A-970002-B-97499-00004-A-001-01

Figure 1. Cockpit Voice Recorder (CVR) System Equipment Location (Sheet 3 of 4)



VIEW LOOKING UP AT UNDERSIDE
OF NOSE COMPARTMENT SHELF AT WL 22.00
VIEW C-C



VIEW E-E

ICN-412-A-970002-B-97499-00005-A-001-01

Figure 1. Cockpit Voice Recorder (CVR) System Equipment Location (Sheet 4 of 4)

Intentionally blank page

Cockpit Voice Recorder Control unit (RCU) - Removal

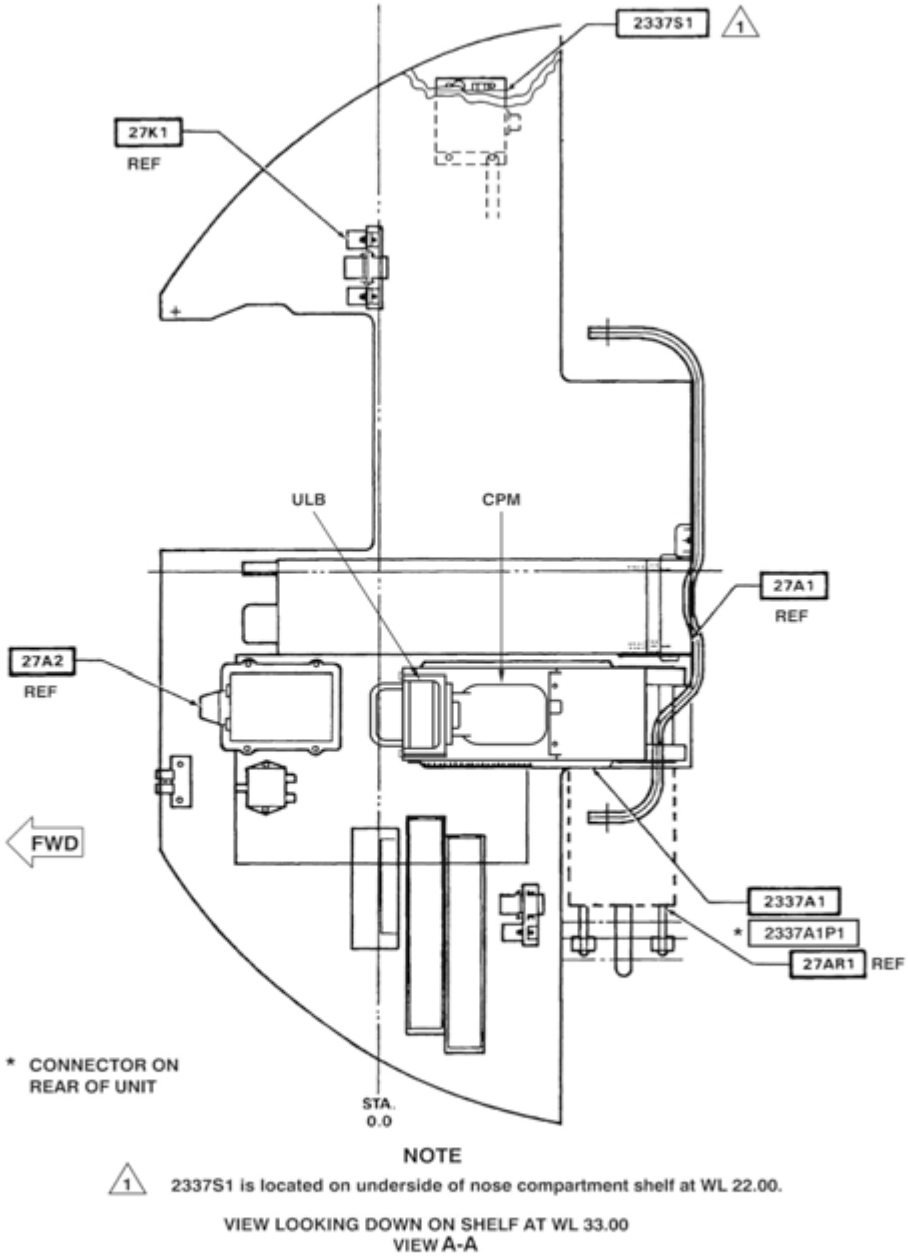
Procedure

ENSURE ELECTRICAL POWER IS OFF.

1. Ensure CVR system circuit breaker (2337CB1) is open.
2. Loosen spring-lock fasteners that secure Recorder Control Unit (RCU) to pedestal.
3. Lift RCU (2337CP1) (Figure 1) far enough out of pedestal to permit access to electrical connector (2337CP1P1) at back of unit.
4. Disconnect electrical connector (2337CP1P1) at rear of unit. Lift clear.

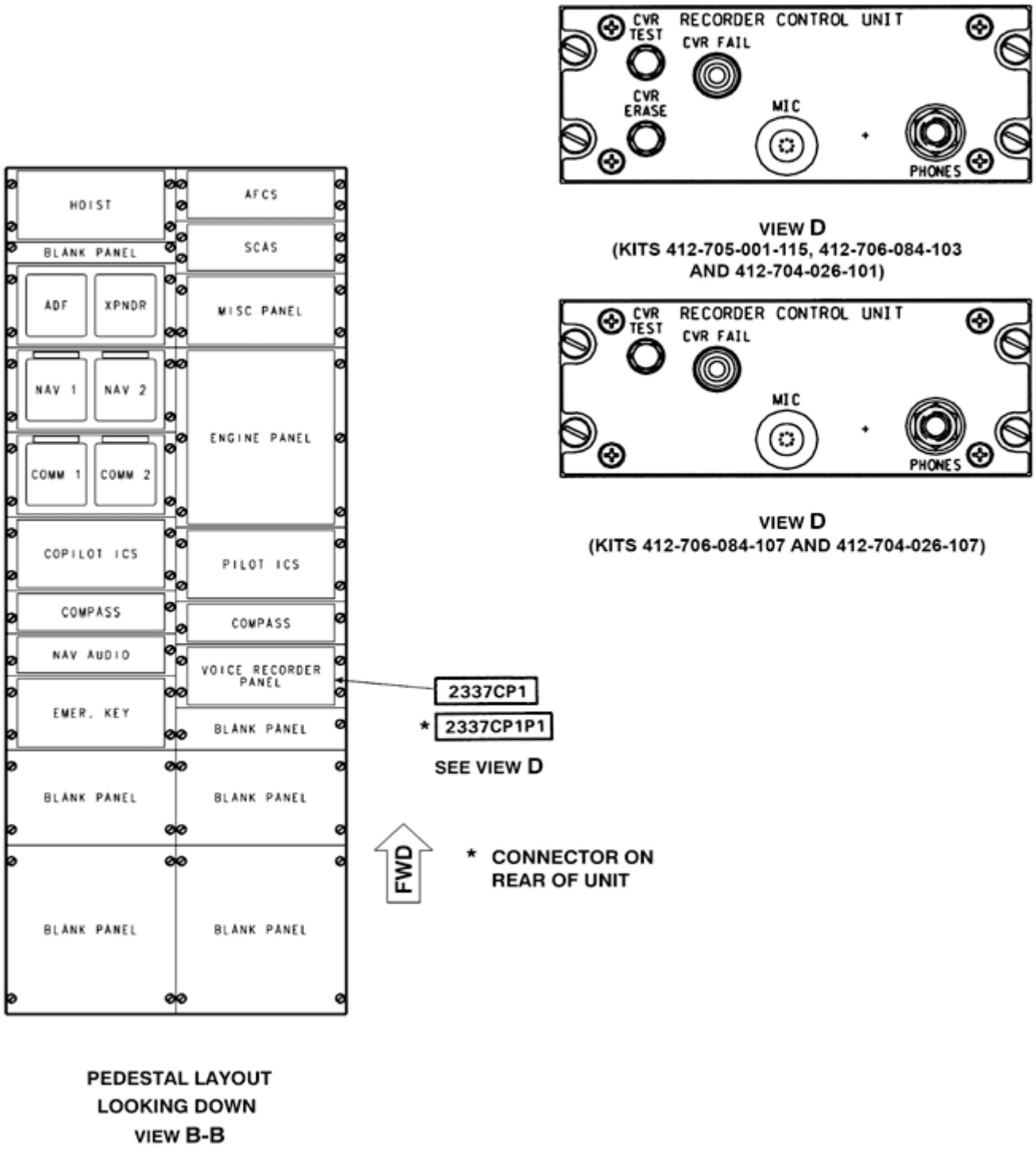
ICN-412-A-970002-B-97499-00002-A-001-01

Figure 1. Cockpit Voice Recorder (CVR) System Equipment Location (Sheet 1 of 4)



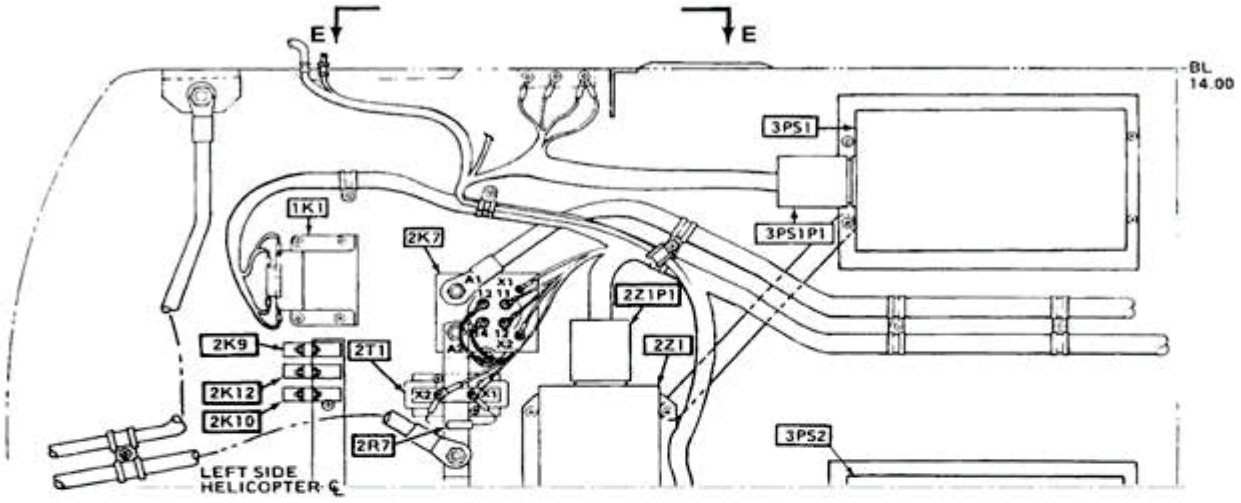
ICN-412-A-970002-B-97499-00003-A-001-01

Figure 1. Cockpit Voice Recorder (CVR) System Equipment Location (Sheet 2 of 4)

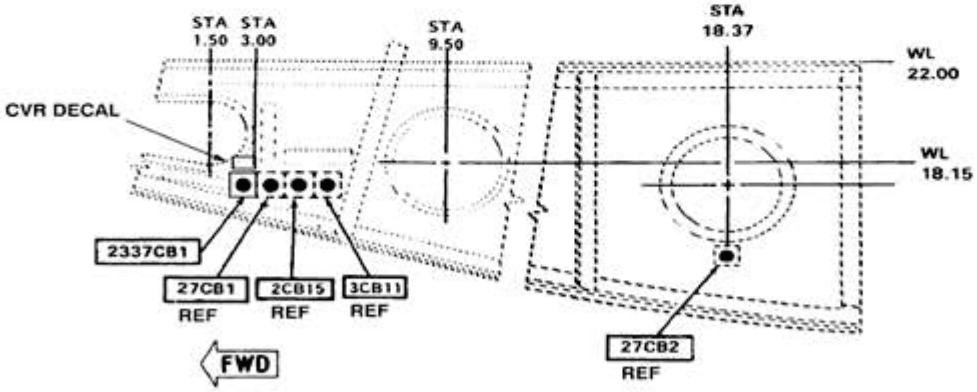


ICN-412-A-970002-B-97499-00004-A-001-01

Figure 1. Cockpit Voice Recorder (CVR) System Equipment Location (Sheet 3 of 4)



VIEW LOOKING UP AT UNDERSIDE
OF NOSE COMPARTMENT SHELF AT WL 22.00
VIEW C-C



VIEW E-E

ICN-412-A-970002-B-97499-00005-A-001-01

Figure 1. Cockpit Voice Recorder (CVR) System Equipment Location (Sheet 4 of 4)

Intentionally blank page

Impact Switch - Removal

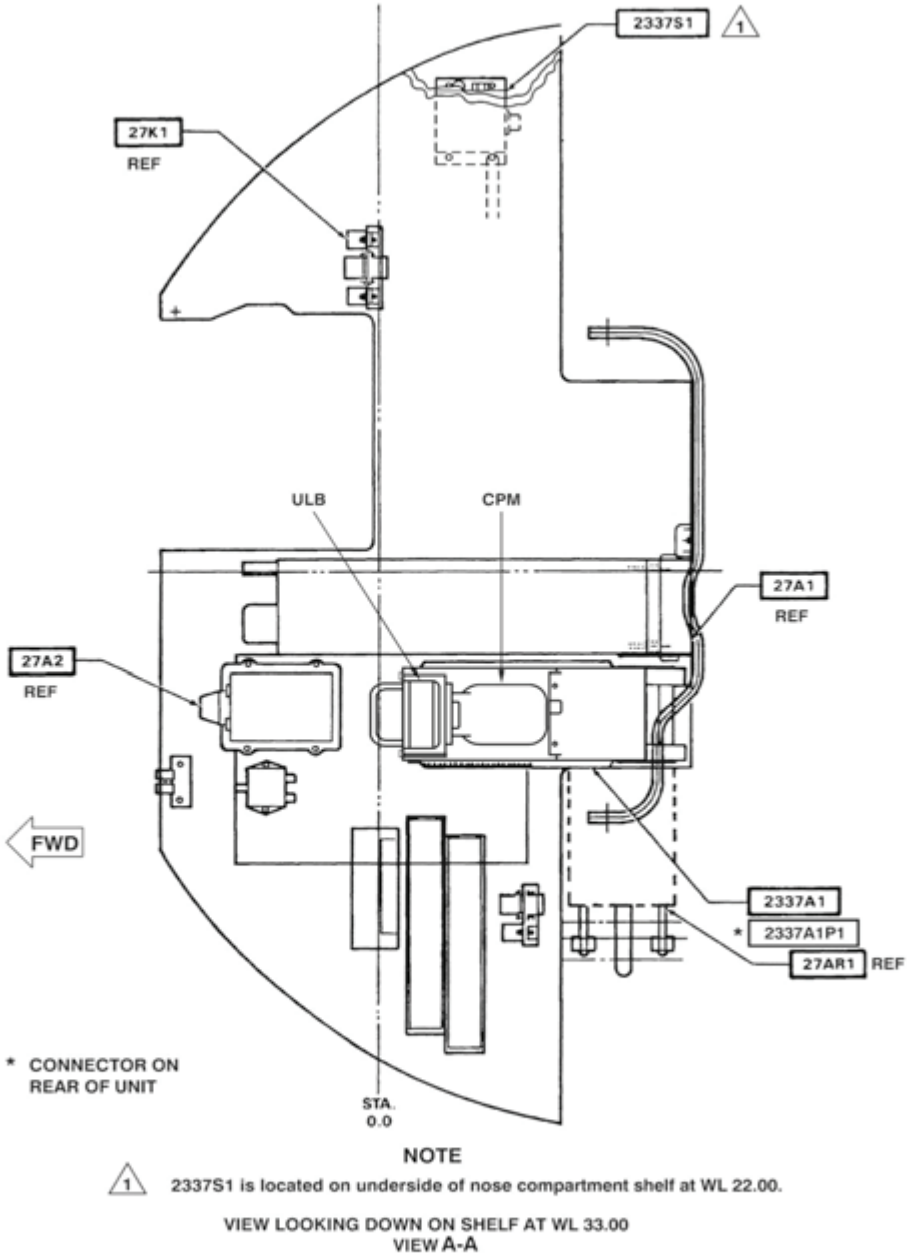
Procedure

ENSURE ELECTRICAL POWER IS OFF.

1. Remove and retain screw that secures bonding strap to the impact switch.
2. Remove and retain screws that secure the impact switch (2337S1) (Figure 1) to mounting location.
3. Disconnect electrical connector at back of unit.
4. Remove impact switch.

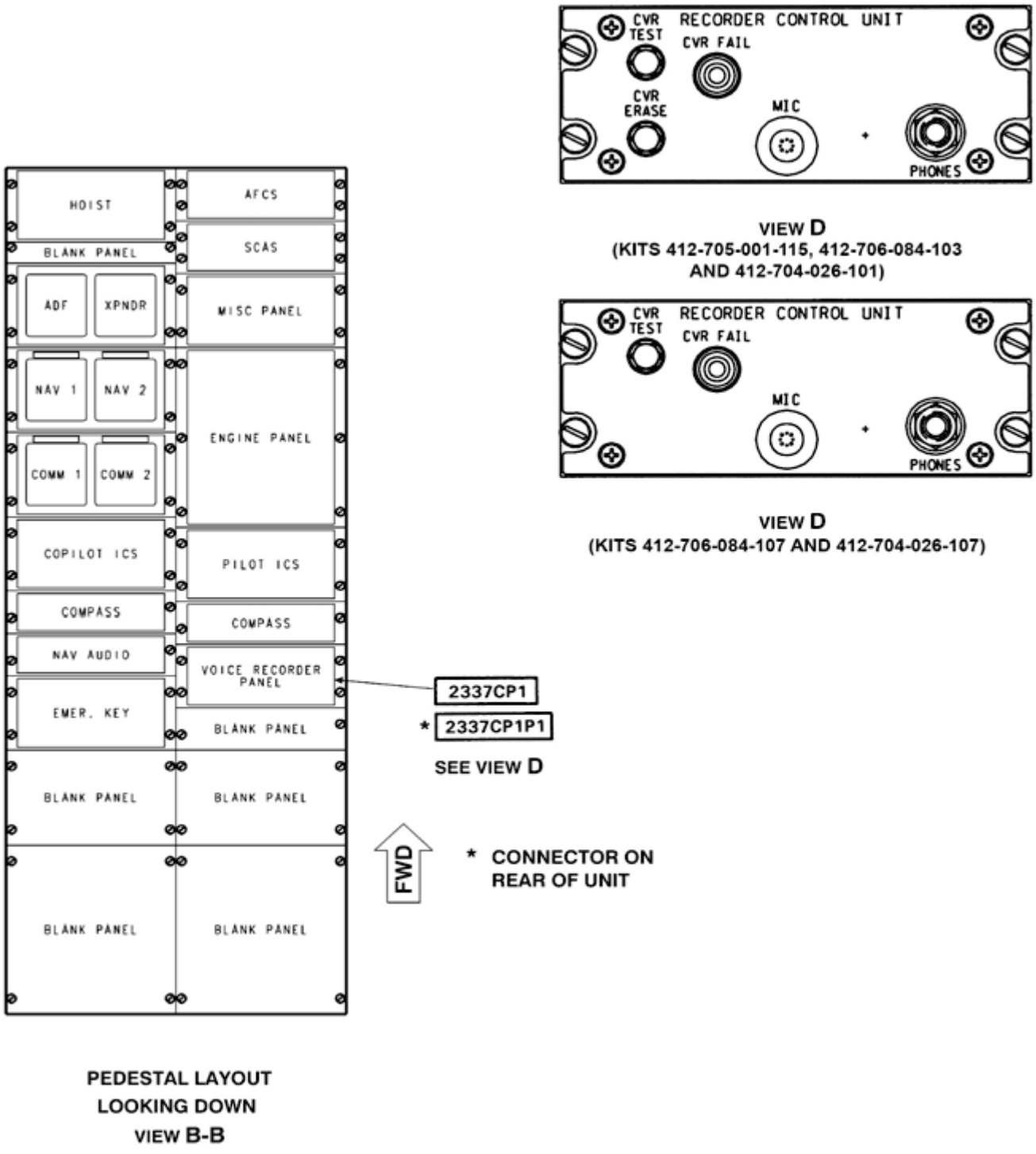
ICN-412-A-970002-B-97499-00002-A-001-01

Figure 1. Cockpit Voice Recorder (CVR) System Equipment Location (Sheet 1 of 4)



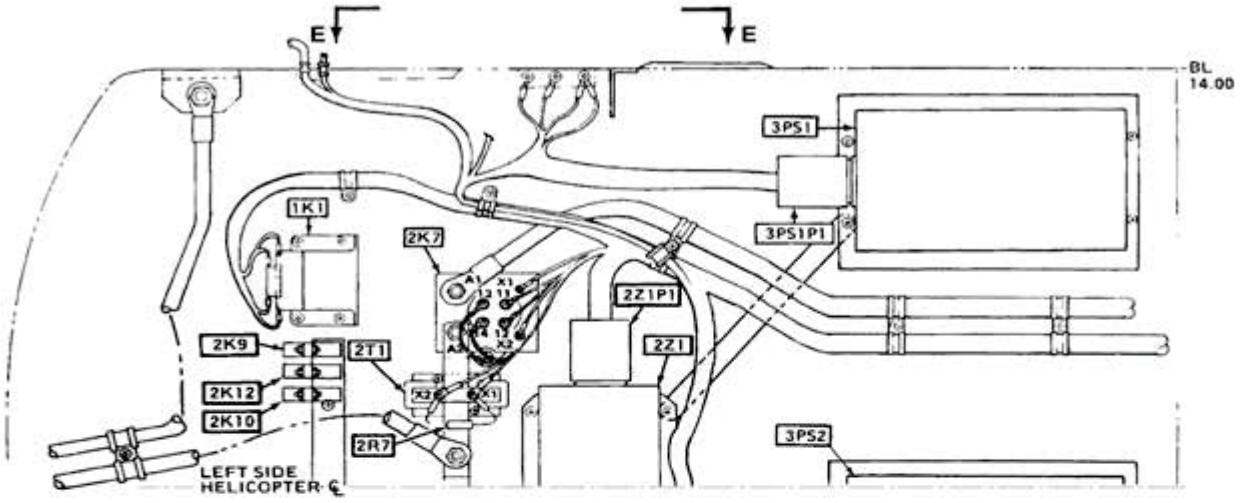
ICN-412-A-970002-B-97499-00003-A-001-01

Figure 1. Cockpit Voice Recorder (CVR) System Equipment Location (Sheet 2 of 4)

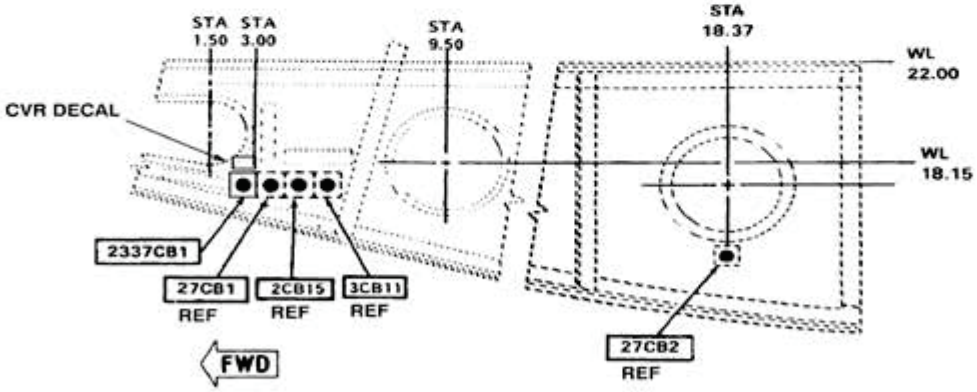


ICN-412-A-970002-B-97499-00004-A-001-01

Figure 1. Cockpit Voice Recorder (CVR) System Equipment Location (Sheet 3 of 4)



VIEW LOOKING UP AT UNDERSIDE
OF NOSE COMPARTMENT SHELF AT WL 22.00
VIEW C-C



VIEW E-E

ICN-412-A-970002-B-97499-00005-A-001-01

Figure 1. Cockpit Voice Recorder (CVR) System Equipment Location (Sheet 4 of 4)

Intentionally blank page

CVR Mounting Tray - Removal

Procedure

ENSURE ELECTRICAL POWER IS OFF.

1. Remove and retain screws that secure mounting tray and spacer, retain all hardware for reinstallation.

Intentionally blank page

Mounting Tray - Installation

Procedure

ENSURE ELECTRICAL POWER IS OFF.

1. Position mounting tray and spacer, secure with mounting screws.
2. Verify bonding for mounting tray is less than 2.5 milliohms.

Intentionally blank page

Cockpit Voice Recorder (CVR) - Installation

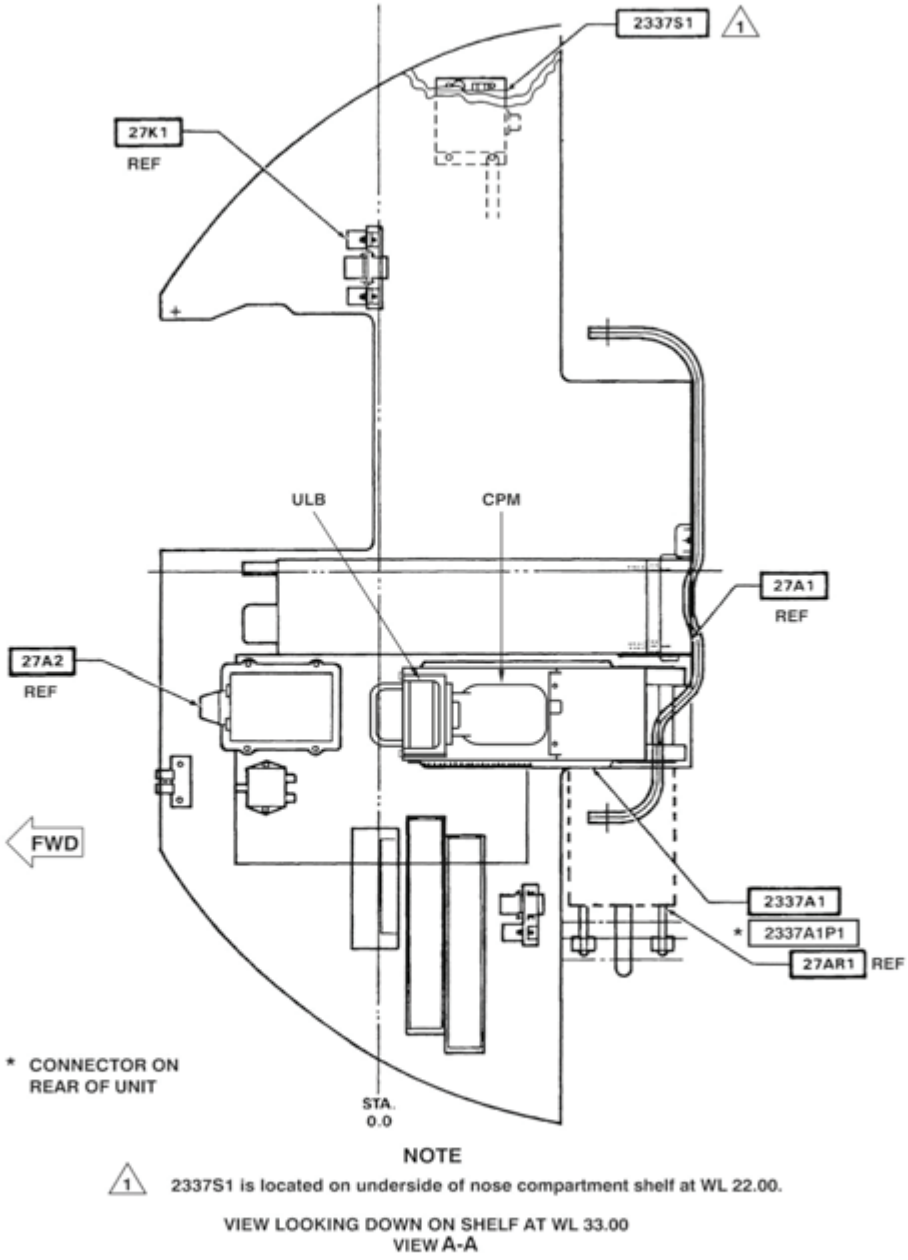
Procedure

ENSURE ELECTRICAL POWER IS OFF.

1. Ensure CVR system circuit breaker (2337CB1) (Figure 1) is open.
2. Position Cockpit Voice Recorder (2337A1) on mounting tray and push carefully into place.
3. Place the two tray-locking mechanisms over CVR crow feet secure by turning locking mechanism handtight.
4. Verify that the CVR is securely mounted to the mounting tray and no looseness is evident.

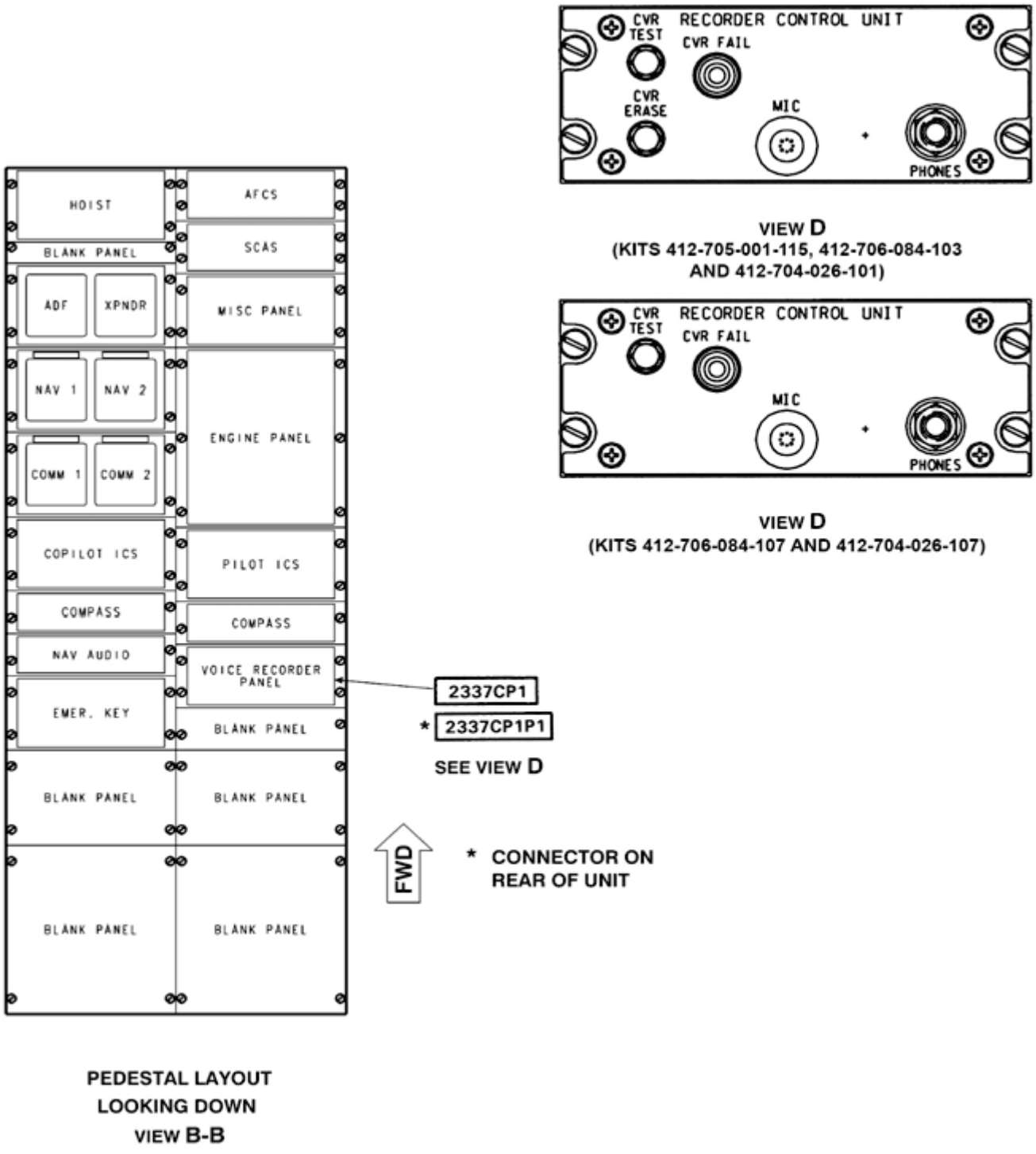
ICN-412-A-970002-B-97499-00002-A-001-01

Figure 1. Cockpit Voice Recorder (CVR) System Equipment Location (Sheet 1 of 4)



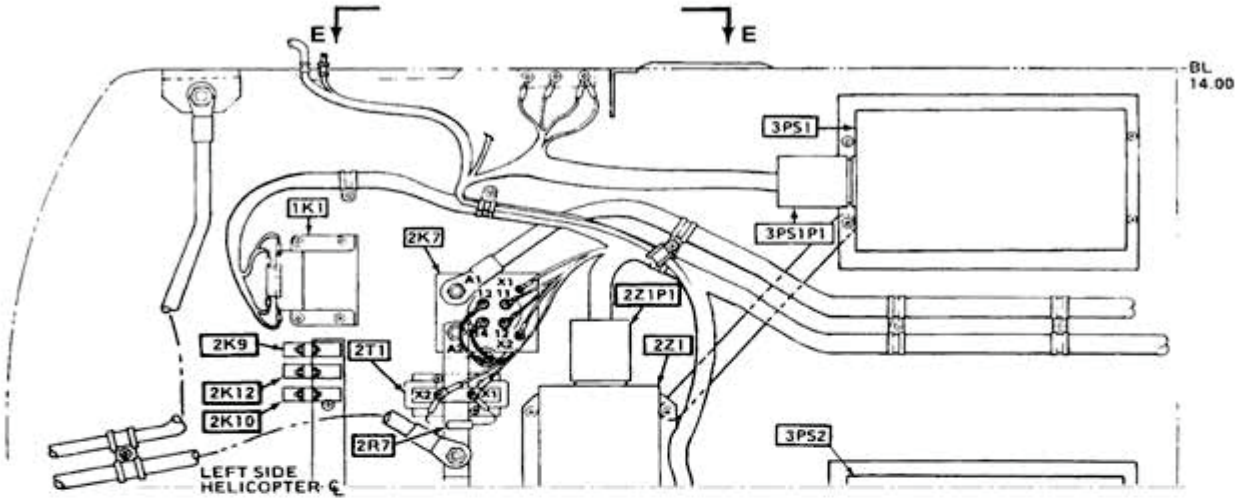
ICN-412-A-970002-B-97499-00003-A-001-01

Figure 1. Cockpit Voice Recorder (CVR) System Equipment Location (Sheet 2 of 4)

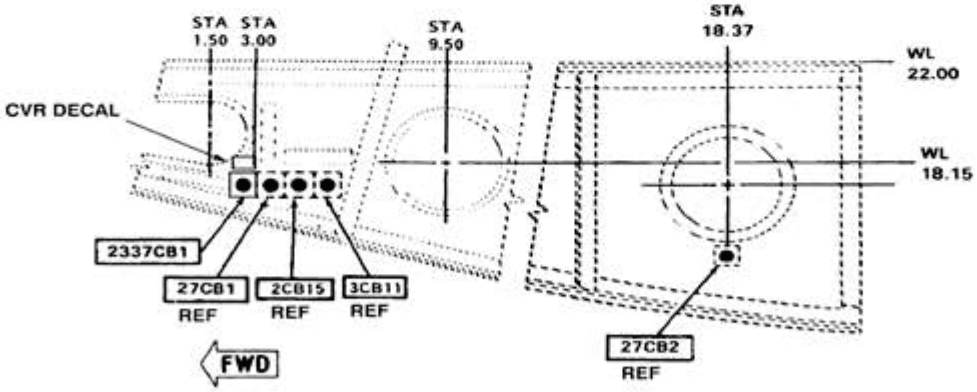


ICN-412-A-970002-B-97499-00004-A-001-01

Figure 1. Cockpit Voice Recorder (CVR) System Equipment Location (Sheet 3 of 4)



VIEW LOOKING UP AT UNDERSIDE
OF NOSE COMPARTMENT SHELF AT WL 22.00
VIEW C-C



VIEW E-E

ICN-412-A-970002-B-97499-00005-A-001-01

Figure 1. Cockpit Voice Recorder (CVR) System Equipment Location (Sheet 4 of 4)

Intentionally blank page

Cockpit Voice Recorder Control unit (RCU) - Installation

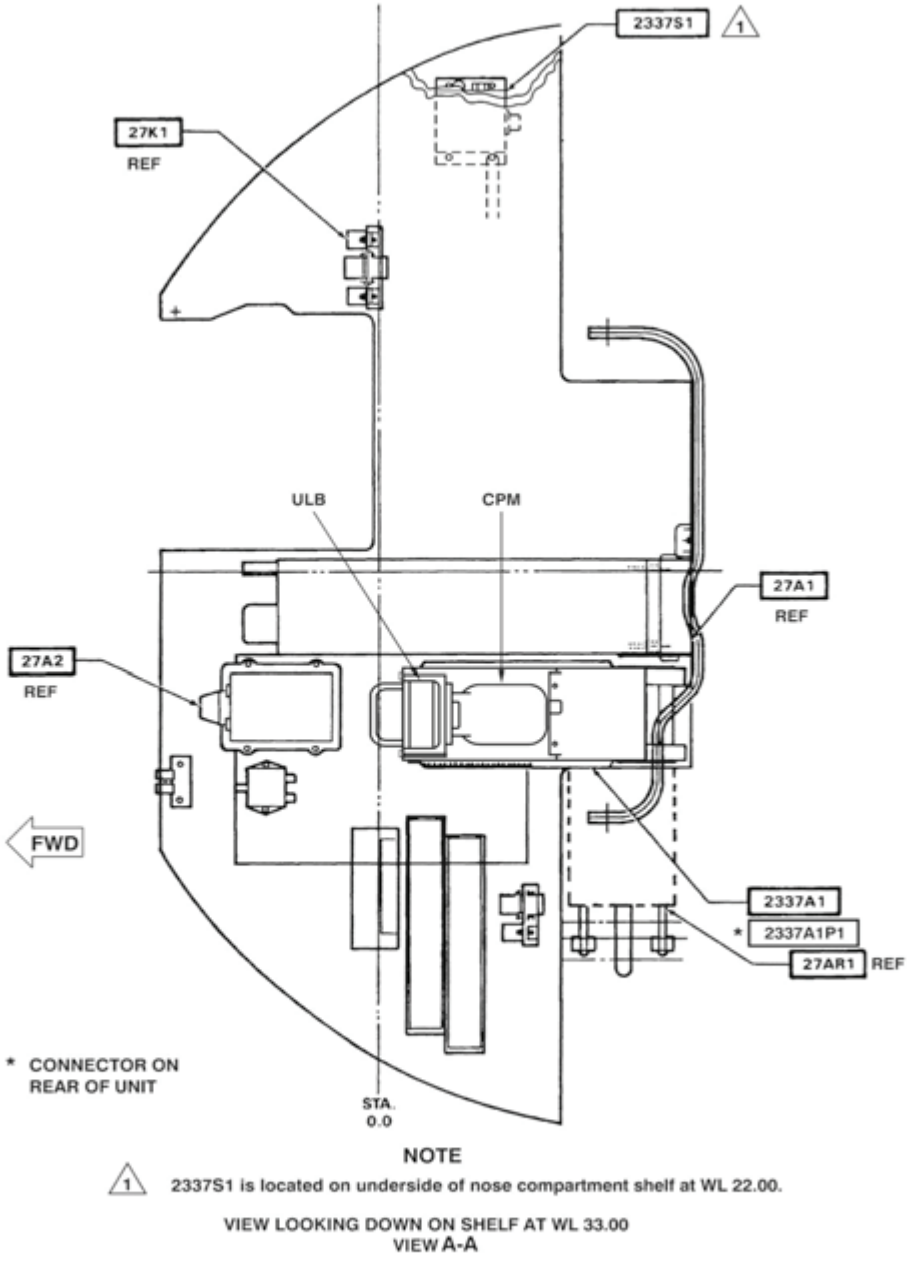
Procedure

ENSURE ELECTRICAL POWER IS OFF.

1. Ensure that CVR system circuit breaker (2337CB1) (Figure 1) is open.
2. Connect electrical connector (2337CP1P1) to back of control panel.
3. Position Recorder Control Unit (2337CP1) in pedestal and secure with spring lock fasteners.

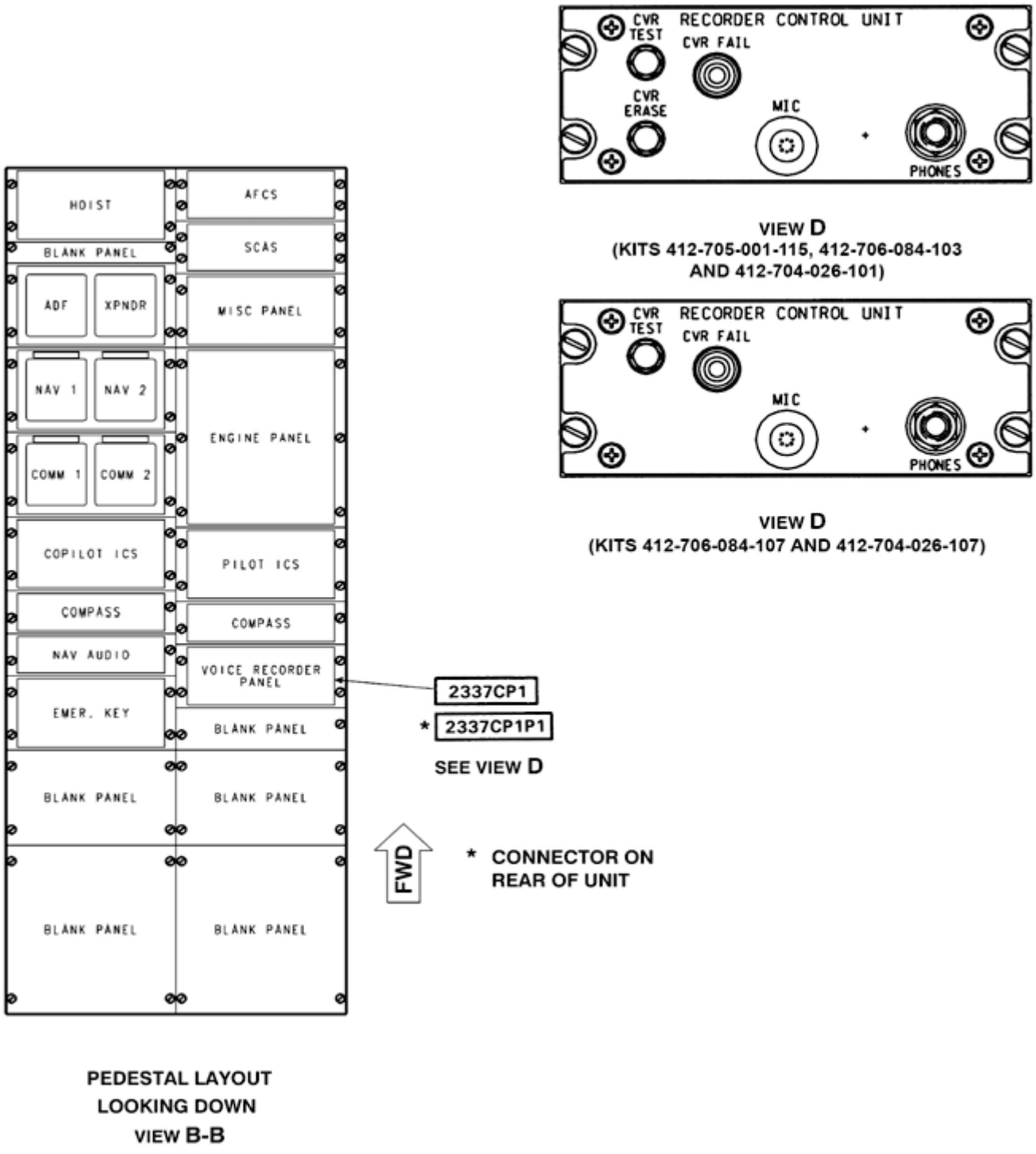
ICN-412-A-970002-B-97499-00002-A-001-01

Figure 1. Cockpit Voice Recorder (CVR) System Equipment Location (Sheet 1 of 4)



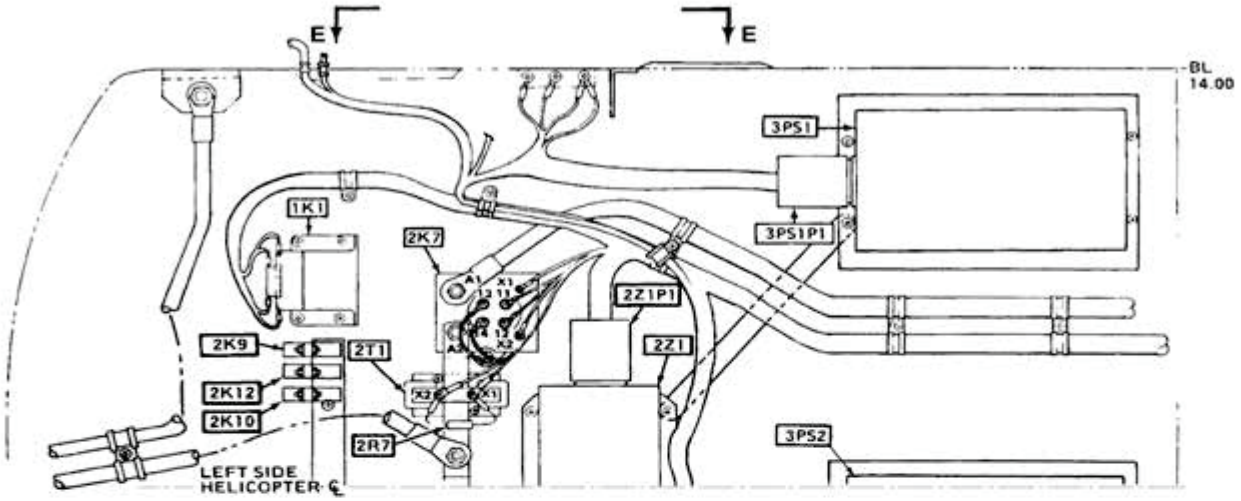
ICN-412-A-970002-B-97499-00003-A-001-01

Figure 1. Cockpit Voice Recorder (CVR) System Equipment Location (Sheet 2 of 4)

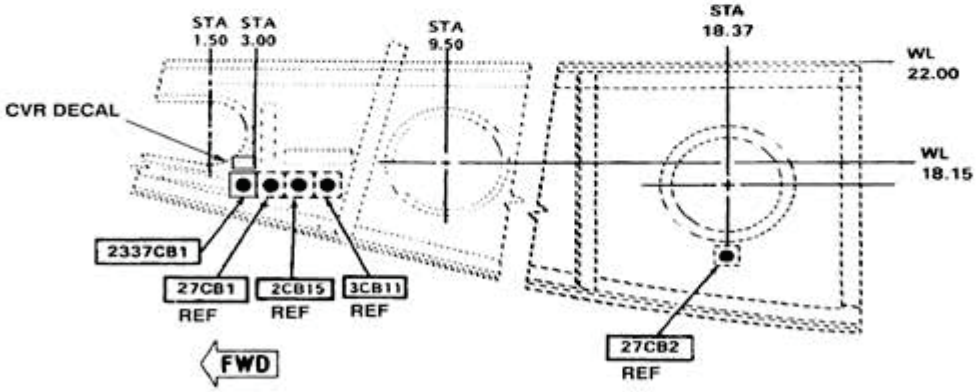


ICN-412-A-970002-B-97499-00004-A-001-01

Figure 1. Cockpit Voice Recorder (CVR) System Equipment Location (Sheet 3 of 4)



VIEW LOOKING UP AT UNDERSIDE
OF NOSE COMPARTMENT SHELF AT WL 22.00
VIEW C-C



VIEW E-E

ICN-412-A-970002-B-97499-00005-A-001-01

Figure 1. Cockpit Voice Recorder (CVR) System Equipment Location (Sheet 4 of 4)

Intentionally blank page

Impact Switch - Installation

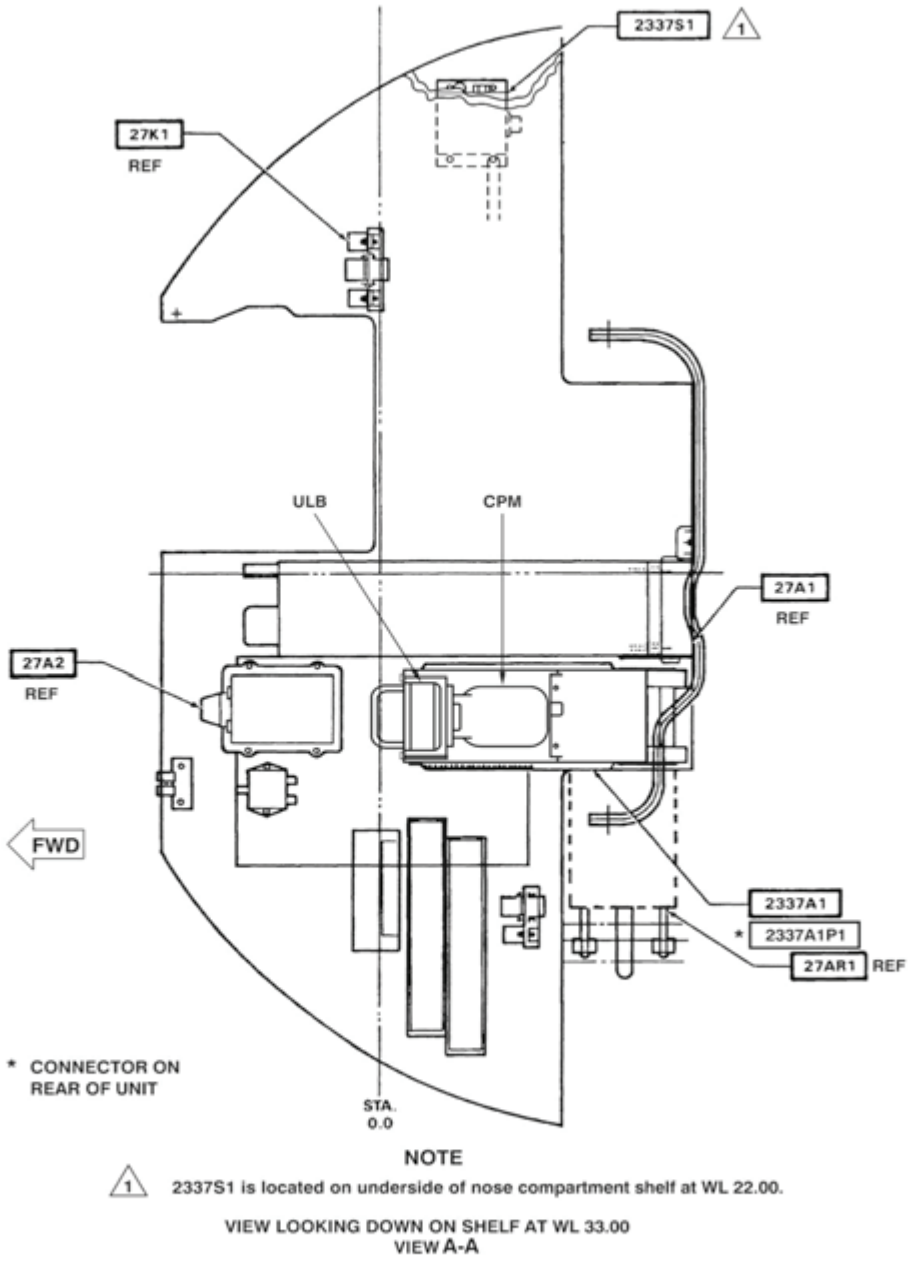
Procedure

ENSURE ELECTRICAL POWER IS OFF.

1. Position impact switch (2337S1) (Figure 1) in place and secure with mounting screws.
2. Connect electrical connector to back of the impact switch.
3. Install bonding strap to impact switch and measure the resistance of electrical bond. Verify bonding is less than 10 milliohms.

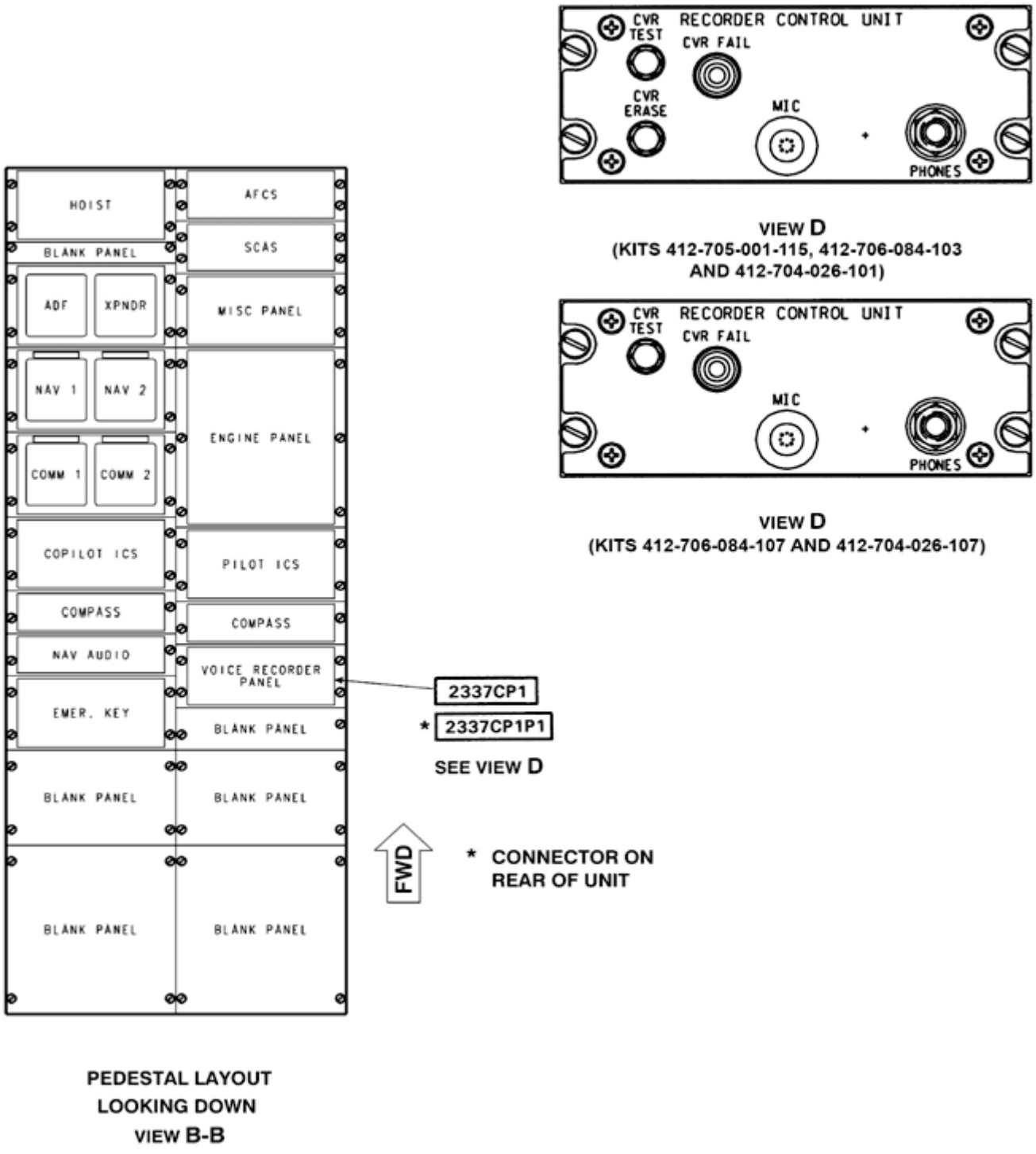
ICN-412-A-970002-B-97499-00002-A-001-01

Figure 1. Cockpit Voice Recorder (CVR) System Equipment Location (Sheet 1 of 4)



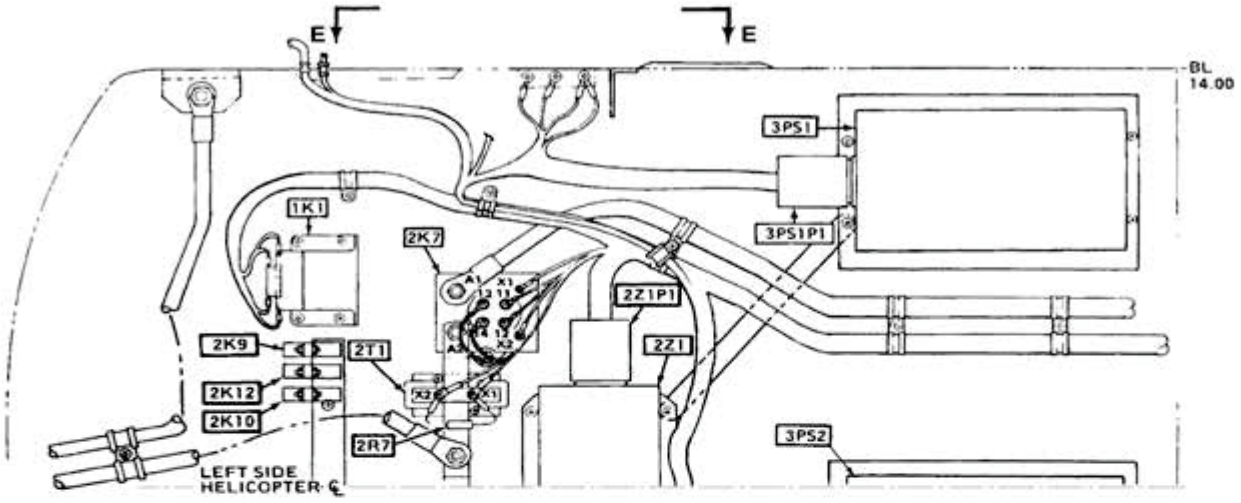
ICN-412-A-970002-B-97499-00003-A-001-01

Figure 1. Cockpit Voice Recorder (CVR) System Equipment Location (Sheet 2 of 4)

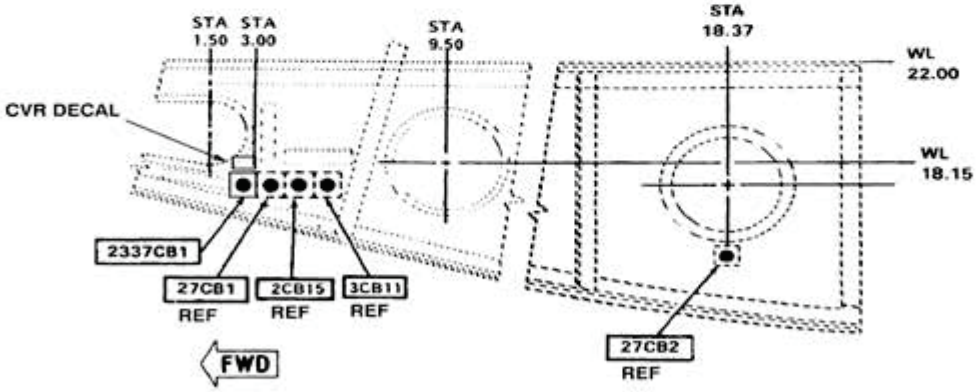


ICN-412-A-970002-B-97499-00004-A-001-01

Figure 1. Cockpit Voice Recorder (CVR) System Equipment Location (Sheet 3 of 4)



VIEW LOOKING UP AT UNDERSIDE
OF NOSE COMPARTMENT SHELF AT WL 22.00
VIEW C-C



VIEW E-E

ICN-412-A-970002-B-97499-00005-A-001-01

Figure 1. Cockpit Voice Recorder (CVR) System Equipment Location (Sheet 4 of 4)

Intentionally blank page

CVR - Inspection

Procedure

1. Visually inspect components for general condition or damage.
2. Visually inspect electrical connector for damaged/bent pins, corrosion or cracks.
3. Visually inspect condition of all mounting hardware for corrosion or wear.
4. Verify that the CVR is securely mounted to the mounting tray and no looseness is evident.

Intentionally blank page

RCU - Inspection

Procedure

1. Visually inspect RCU exterior surfaces for damage or missing attaching screws and turnlock fasteners.
2. Visually inspect front panel nomenclatures and markings for legibility.
3. Visually inspect indicator for cracks, damaged lenses, etc.
4. Visually inspect area microphone for dust or dirt accumulation or other visible contamination or damage.
5. Visually inspect condition of all mounting hardware for corrosion or wear.
6. Visually inspect switches for damage and security.

Intentionally blank page

Cockpit Voice Recorder (CVR) - Cleaning

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Isopropyl alcohol	C-385	AR	

Procedure

isopropyl alcohol (C-385) IS FLAMMABLE AND TOXIC. KEEP AWAY FROM HEAT, SPARKS, AND OPEN FLAME. AVOID PROLONGED OR REPEATED BREATHING OF VAPOR. USE ONLY IN A WELL-VENTILATED AREA.

EXERCISE CARE WHEN CLEANING THE INDICATING PANEL TO PREVENT SCRATCHING THE SURFACE COATING. SCRATCHES IN THE SURFACE COATING MAY NOT BE EVIDENT UNTIL THE DISPLAY IS OPERATED.

1. Remove moisture and loose dirt with a soft clean cloth.
2. Remove dust and dirt from panel control with a soft clean cloth. To remove fingerprints or difficult dirt, use a cloth moistened with water; if necessary, a mild detergent may be used to remove grease or other substances not effectively cleaned with water only.
3. Thoroughly dry surfaces.
4. Clean connector contacts using a soft-bristled brush moistened with isopropyl alcohol (C-385).
5. Clean the switch end of the beacon with water, mild detergent and a soft cloth, then dry thoroughly with a clean cloth. Clean the switch end insulator to prevent leakage currents from occurring across the switch. This will affect battery life. The water switch should be cleaned when dirt or dust becomes apparent.

Intentionally blank page

Cockpit Voice Recorder (CVR) - Repair or Replacement

Procedure

1. Replace defective or inoperative units.
2. Further repairs, other than removal and replacement, must be conducted by an authorized repair facility.

Intentionally blank page

Cockpit Voice Recorder (CVR) System - Wiring Diagram - Description

The Cockpit Voice Recorder (CVR) system wiring diagram is provided to assist maintenance personnel in understanding the circuit and components installed in the helicopter, and in troubleshooting and tracing inoperative and malfunctioning circuits. Standard symbols are used where possible.

2. Wire Identification

All wires on the systems wiring diagram are identified by codes as they are marked in the helicopter. The code or wire number indicates the circuit function, wire number, wire segment number, and wire conductor size. Refer to (412-A-98-00-00-00A-042A-A / 02456) for wire code definitions.

Intentionally blank page

Maintenance Procedures - Description

Applicable: SN: 36087-Subsequent:

These instructions are a supplement to the 412-A-00-00-00-00A-009A-A / 00001 when the Smiths Flight Data Recorder (FDR) installation kit (412-706-084-101) is installed. Flight Data Recorder kit 412-706-084-101 is applicable.

Refer to the 412-A-00-00-00-00A-009A-A / 00001 for maintenance instructions related to basic helicopter equipment.

Maintenance is required on the FDR system to ensure the integrity of the system for recording data.

Maintenance is also required for the Underwater Locator Beacon (ULB) .

Intentionally blank page

Airworthiness Limitations - Schedule - Description

ALL PARTS REMOVED DUE TO REACHING THEIR LIFE LIMITS ARE DEEMED UNAIRWORTHY AND SHALL BE PERMANENTLY MARKED AS SCRAP OR PHYSICALLY DESTROYED TO THE EXTENT THAT THERE IS NO CHANCE OF REPAIR OR REINSTALLATION ON ANY HELICOPTER COMPONENT.

SOME PARTS ARE INSTALLED AS ORIGINAL EQUIPMENT ON BOTH MILITARY AND COMMERCIAL HELICOPTERS AND MAY HAVE LOWER AIRWORTHINESS LIFE AND/OR OVERHAUL SCHEDULE WHEN USED ON A MILITARY HELICOPTER THAN WHEN USED ON A COMMERCIAL HELICOPTER. IN ADDITION, CIRCUMSTANCES SURROUNDING THEIR USE MAY CALL FOR OPERATION OF THE MILITARY HELICOPTER OUTSIDE THE APPROVED COMMERCIAL FLIGHT ENVELOPE. CONSEQUENTLY, PARTS THAT HAVE BEEN USED ON MILITARY HELICOPTERS MUST NOT BE USED ON COMMERCIAL HELICOPTERS.

The airworthiness limitations schedule summarizes the mandatory maximum hours (life) of various components that may be used before retiring from service (Table 1). Parts not listed on the schedule have unlimited airworthiness life.

NOTE

The operating time specified for retirement for any given part number contained in the airworthiness limitations schedule applies to all successive dash numbers for that item unless otherwise specified.

Airworthiness lives assigned to helicopter components and assemblies are based upon experience, testing and engineering judgment, and are not subject to change without FAA approval.

NOTE

Neither the assignment of a retirement life to a component nor failure to assign a retirement life constitutes a warranty of any kind. The only warranty applicable to the helicopter and any component is that warranty included in the Purchase Agreement for the helicopter or the component.

Table 1. Airworthiness Limitations Schedule

COMPONENT	PART NUMBER (1)	LIFE LIMIT
Underwater Locator Beacon Battery	810-2007/K (2)	6 years
Underwater Locator Beacon Battery	C362-04270-2 (3)	6 years
THE BATTERY USED IN THE UNDERWATER LOCATOR BEACON IS A LITHIUM BATTERY. AS SUCH, REMOVAL, INSTALLATION AND DISPOSAL OF THIS BATTERY MAY HAVE TO COMPLY WITH LOCAL (HEALTH AND ENVIRONMENTAL) STANDARDS AND REGULATORY PRACTICES.		

NOTES:

1 Airworthiness life for a part number listed applies to all successive dash numbers for that component unless otherwise specified.

2 Usable in underwater locator beacon model DK100/90, manufactured by Dukane Corporation.

3 Usable in underwater locator beacon model ELP-3620, manufactured by Benthos, Inc.

Intentionally blank page

Flight Data Recorder (FDR) System - General

Preliminary requirements

Support equipment

Nomenclature	Identification No.	Qty	Remarks
Flight Data Recorder (FDR) BHT-412-SI-73	- 177045-01-01	1	(Smith's) (Kit 412-706-084)
Accelerometer	- 3001-01-101-2	1	(Magnetek)
Flight Data Acquisition Unit (FDAU)	- 412-375-013-101	1	
Air Data Computer (ADC) BHT-412-SI-73	- 412-375-014-101	1	(Kit 412-706-084)
Air Data Computer (ADC) BHT-412-SI-86	- D51610-2202	1	(Penny & Giles) (Kit 412-706-108)

Procedure

1. The solid-state FDR Model 3257D1 is installed on the Model 412EP helicopter as an optional equipment kit (Figure 1). The FDR (27A1) receives 115 VAC power from the FDR circuit breaker (27CB2).

2. For kit 412-706-084-101, the Flight Data Acquisition Unit (FDAU) (27AR1) and the Air Data Computer (ADC) (27A2) receive power from main DC Bus 1 and main DC Bus 2 to the FDAU circuit breakers (27CB1).

3. For kit 412-706-084-105, the FDAU (27AR1) receives power from main DC Bus 1 and main DC Bus 2 to the FDAU circuit breaker (27CB1). For kit 412-706-108-101/103, the ADC (27A2) receives power from essential DC Bus 1 to the ADC circuit breaker (27CB11).

4. The system installation consists of an FDR equipped with an Underwater Locator Beacon (ULB) , an accelerometer, an Air Data Computer (ADC) , a Flight Data Acquisition Unit (FDAU) , a relay, and two mounting trays. Kit 412-706-108-101/103 adds a second relay for the ADC fail monitoring.

5. For the component reference designator, refer to Table 1.

6. The FDR (27A1) (Figure 2) is mounted in the nose of the helicopter with a standard ARINC tray fitted with a rear-mounted electrical connector. The unit has an external ULB attached to it. The ULB is a highly reliable, rugged acoustic pinger designed to assist in the recovery of and investigation into helicopters that have gone down over water.

7. The accelerometer (27MT1) (Figure 2) is installed at STA 129.0. The accelerometer is a dual-axis instrument that transmits longitudinal and vertical acceleration to the FDAU.

8. The ADC (27A2) (Figure 2) is installed in the nose of the helicopter. The ADC converts pitot/static pressures into electrical signals and transfers them to the FDAU.

9. The FDAU (27AR1) (Figure 2) is installed in the nose of the helicopter. The FDAU collects data and transmits it through an external terminal (ARINC 717 and/or RS-422) to the FDR, which records the data in crash protected memory. At least the last 25 hours of flight data is retained, even at recording rates of up to 512 words per second.

10. The FDR incorporates a maintenance connector (J8) on the front panel that is used for testing the unit, on-aircraft maintenance, or downloading the recorded data. The front connector includes power, control discretes, and high-speed RS-422 and ethernet channels. The ARINC 717 playback signal is available on the front connector to assist in troubleshooting.

11. The FDR system includes ground equipment for on-aircraft maintenance, return-to-service testing in the shop, and flight data playback/analysis. The ground equipment consists of the flight line computer (Recorder Download Unit (RDU)), the return-to-service test set (Flight Recorder Analysis Test Station (FRATS)), and the associated software.

12. FDR FAIL (FDR_FAIL recorder output discrete) is turned on at power-up. It is cleared when all Power-up BIT (PUB) tests have passed. Then, if no failures are detected, a series of conditions are checked to see if a blink condition applies. There are three blink rates used by the FDR system: a long blink, 4Hz, and 2Hz. The system conditions indicated by FDR FAIL are identified in Table 2.

13. The FDR system is designed to record:

- Flight control positions

- Pitch and roll attitudes
- Heading
- Accelerations
- Airspeed
- Vertical speed
- Altitude
- Outside air temperature
- Engine 1 and 2 torque
- Engine 1 and 2 rotor speeds
- Individual discrete warnings

14. The data is collected by the FDAU and is stored in the FDR. The ADC converts pitot/static pressures into electrical signals for storage. The system will be functioning whenever power is applied to main DC Bus and to AC Bus No. 3.

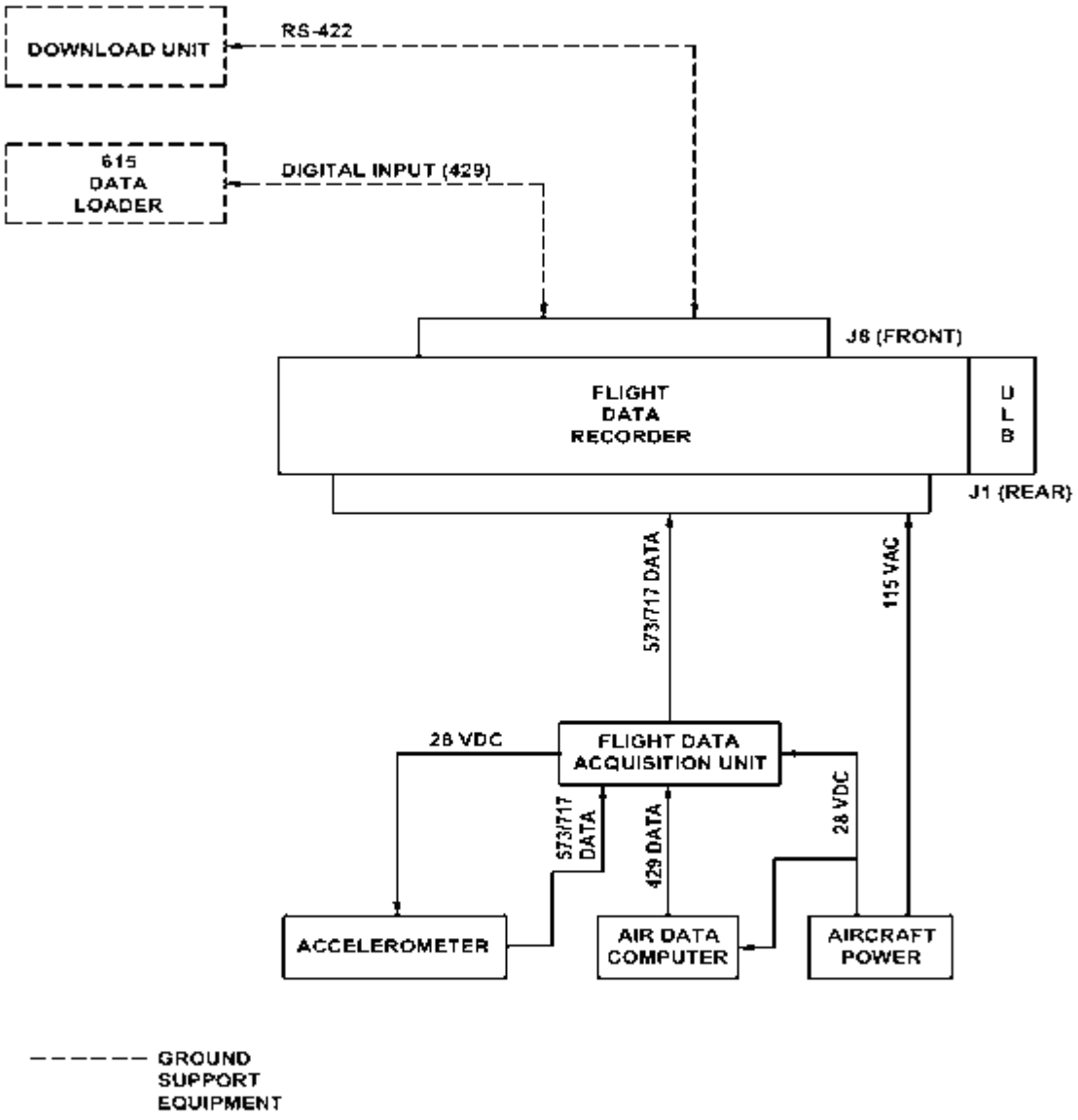
15. The system also sends out an audio synchronization pulse to the CVR. The FDR System Block Diagram (Figure 3) shows the data to be recorded by the FDR.

Table 1. Flight Data Recorder (FDR) System Reference Table

REFERENCE DESIGNATOR	COMPONENT DESCRIPTION	LOCATION
27A1	Flight Data Recorder (FDR)	Nose Compartment, WL 33 Shelf
27A2	Air Data Computer (ADC)	Nose Compartment, WL 33 Shelf
27AR1	Flight Data Acquisition Unit (FDAU)	Nose Compartment, WL 22 Shelf
27K1	Relay	Nose Compartment, WL 33 Shelf
27MT1	Accelerometer	Pylon Area, STA 129.0

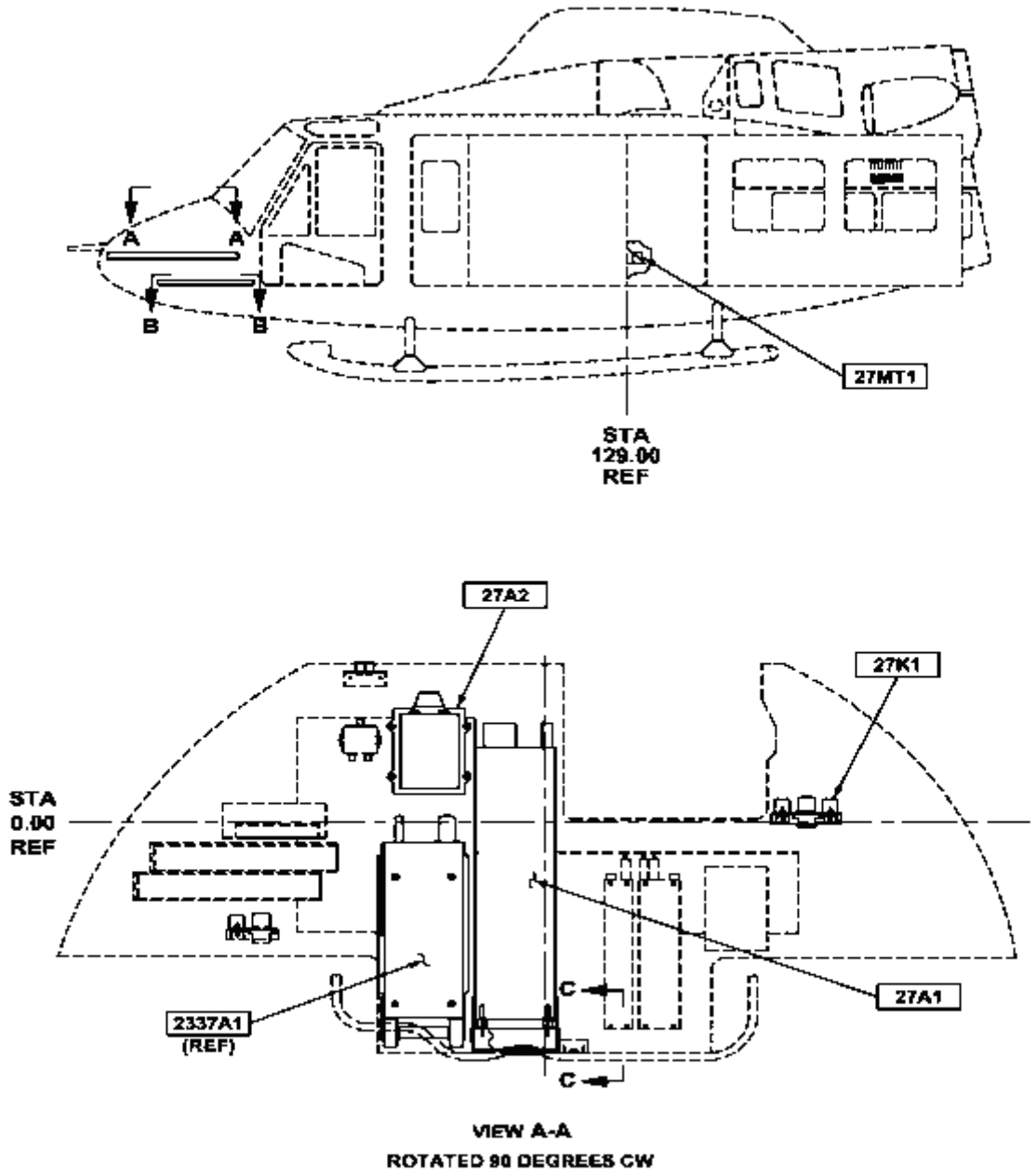
Table 2. Flight Data Recorder (FDR) Indication

FDR FAIL	SYSTEM CONDITION
OFF	Normal Mode, FDR recording
ON	FDR recording inhibited, during Power-up or Initiated Built-in Test, or FDR BIT failure
Fast Blink (4 Hz rate)	Maintenance Mode (not recording)
Slow Blink (2 Hz rate)	Busy erasing/formatting or CPM write protect is activated
Long Blink (on for 1.8 sec., off for 0.2 sec.)	Loader Mode Active



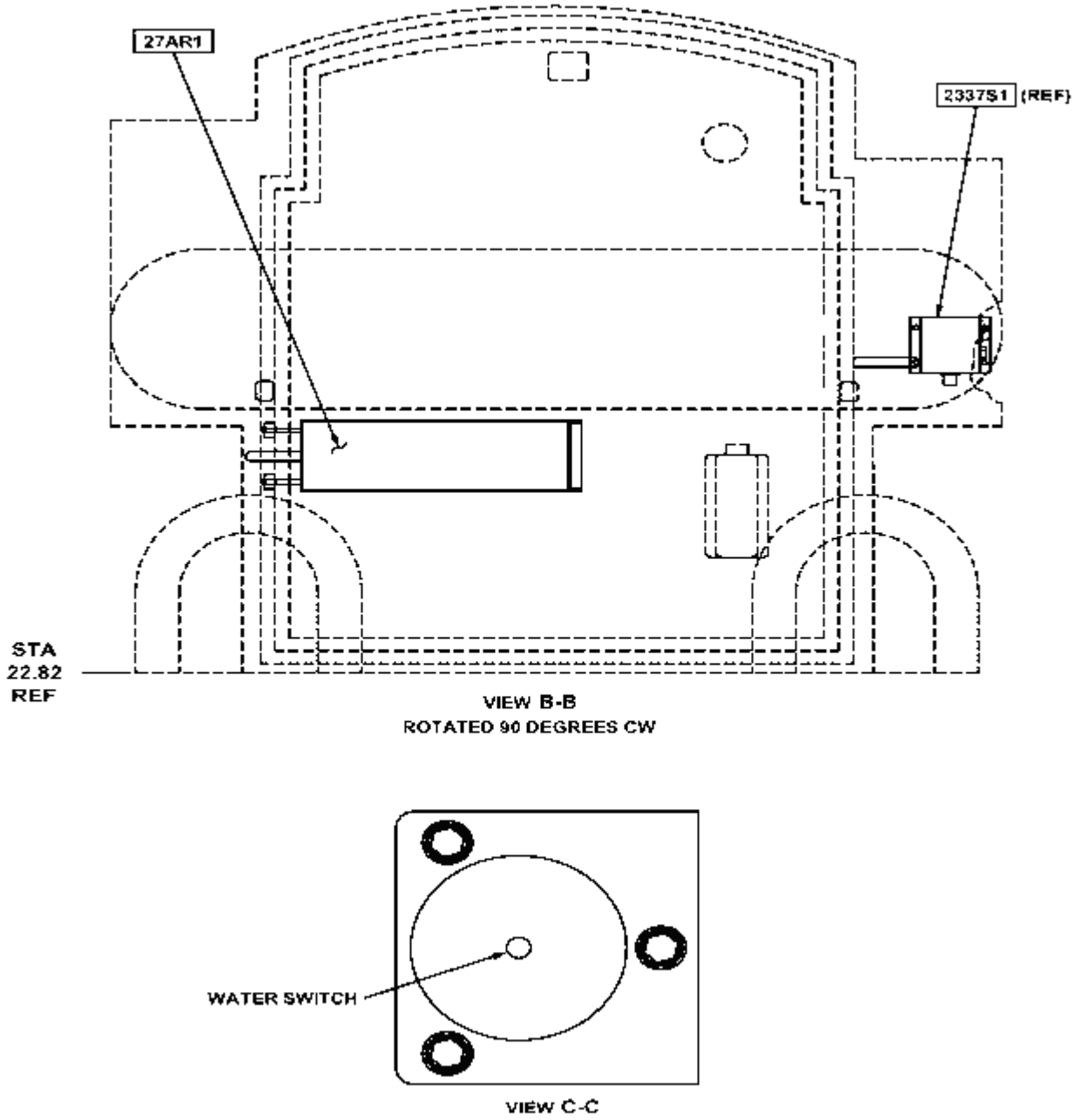
ICN-412-A-970003-B-97499-00001-A-001-01

Figure 1. Simplified Block Diagram Flight Data Recorder System (Sheet 1 of 1)



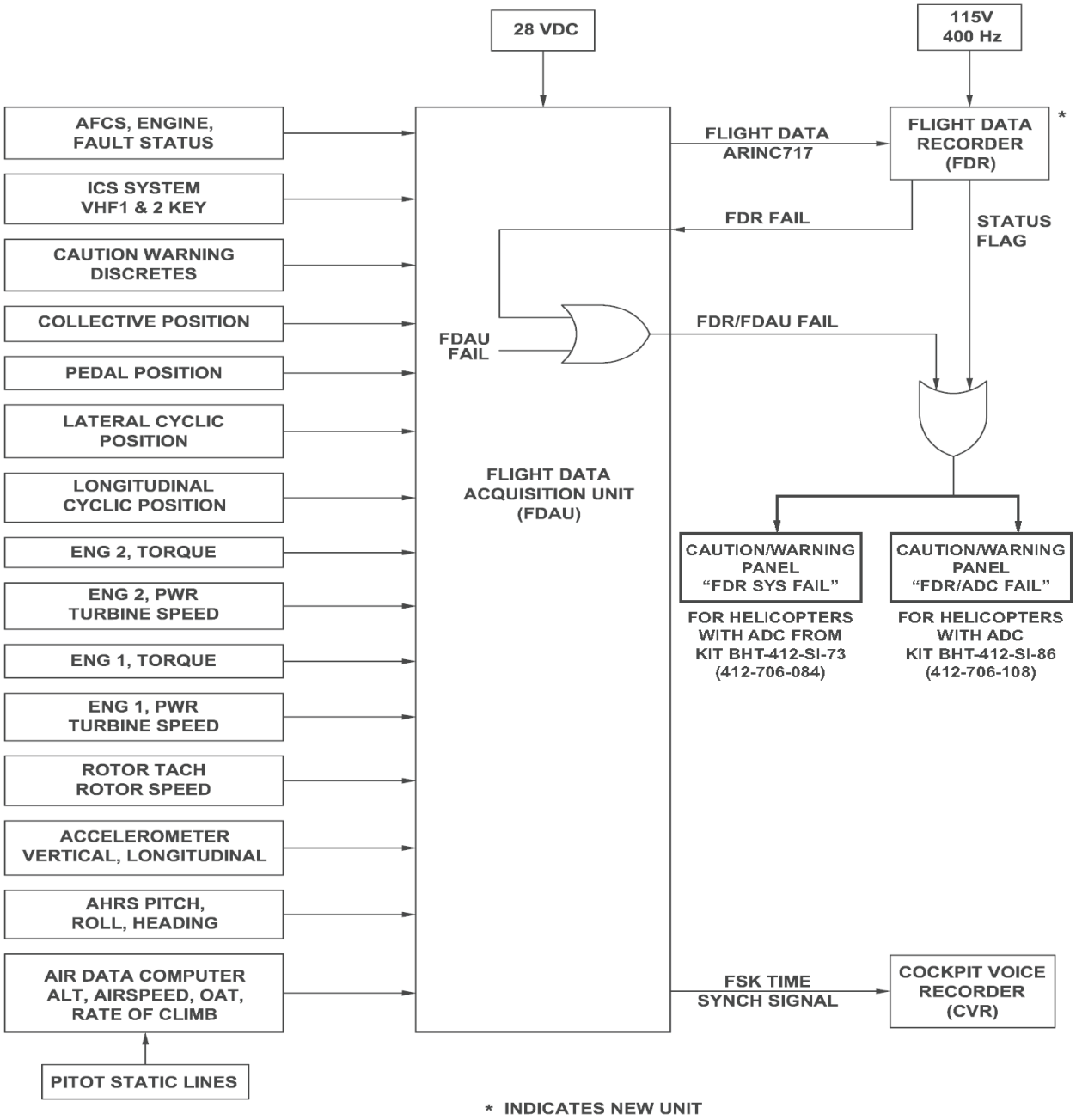
ICN-412-A-970003-B-97499-00002-A-001-01

Figure 2. Flight Data Recorder (FDR) System Equipment Location (Sheet 1 of 2)



ICN-412-A-970003-B-97499-00003-A-001-01

Figure 2. Flight Data Recorder (FDR) System Equipment Location (Sheet 2 of 2)



ICN-412-A-970003-B-97499-00004-A-001-01

Figure 3. FDR System Block Diagram (Sheet 1 of 1)

Flight Data Recorder (FDR) - Operational Tests

Procedure

1. The operational tests of the FDR consist of a preflight test (412-A-97-00-03-07A-042A-A / 02706) and a flight test (412-A-97-00-03-07A-322A-A / 02709).

Intentionally blank page

Flight Data Recorder (FDR) System - Troubleshooting

To troubleshoot the FDR system, refer to 412-A-97-00-03-00A-422B-A / 02715.

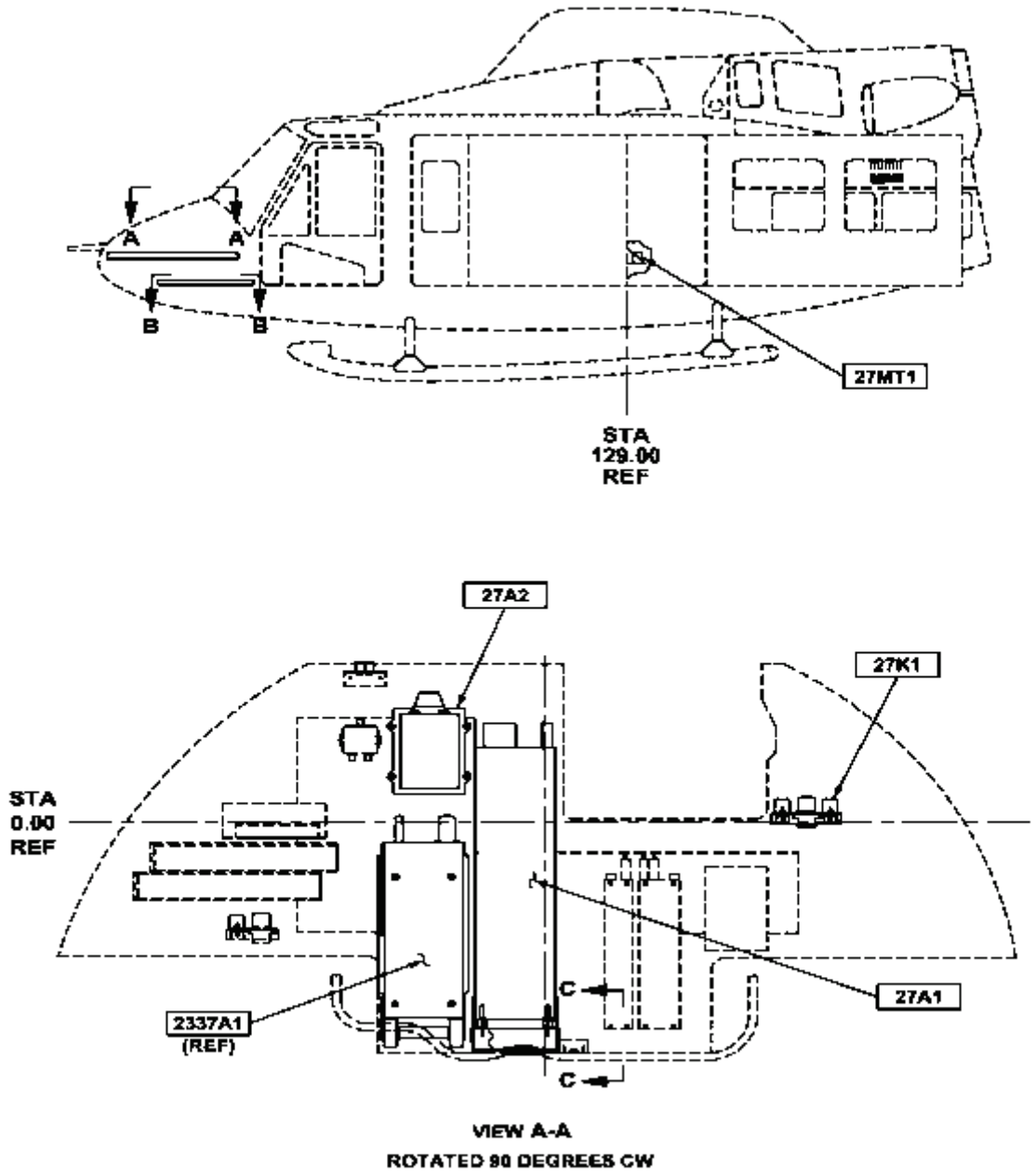
Intentionally blank page

Flight Data Recorder (FDR) - Removal

Procedure

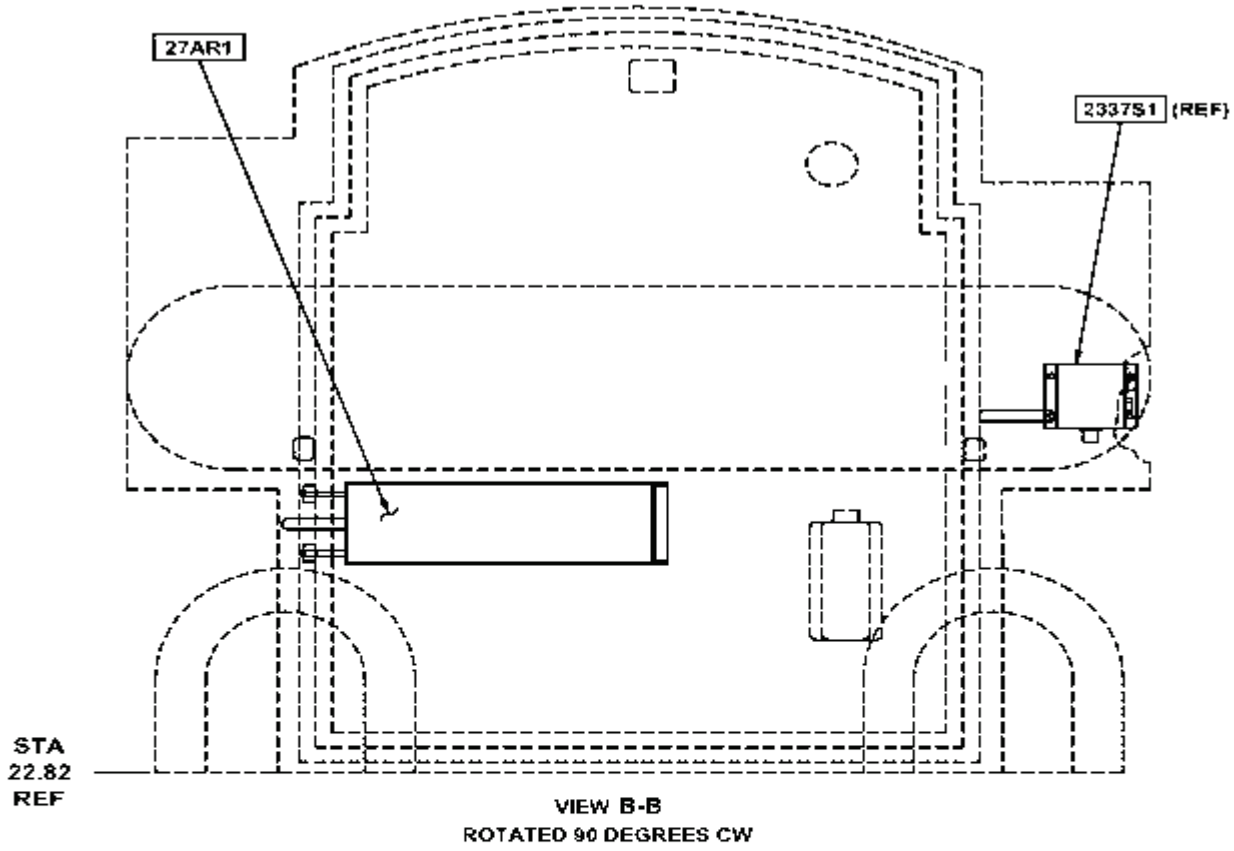
ENSURE ELECTRICAL POWER IS OFF.

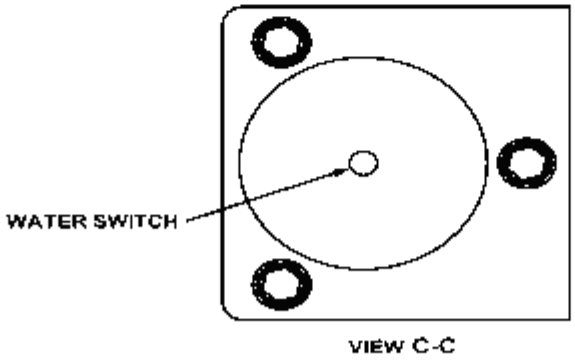
1. Ensure FDR circuit breaker (27CB2) is open.
2. Loosen the two fasteners on front of FDR (27A1) (Figure 1) far enough to free the two hook supports.
3. Remove FDR (27A1) by pulling straight out of tray.



ICN-412-A-970003-B-97499-00002-A-001-01

Figure 1. Flight Data Recorder (FDR) System Equipment Location (Sheet 1 of 2)





ICN-412-A-970003-B-97499-00003-A-001-01

Figure 1. Flight Data Recorder (FDR) System Equipment Location (Sheet 2 of 2)

Intentionally blank page

Flight Data Recorder (FDR) - Inspection

Procedure

1. Visually inspect components for general condition or damage.
2. Visually inspect electrical connector for damaged/bent pins, corrosion, or cracks.
3. Visually inspect condition of all mounting hardware for corrosion or wear.
4. Inspect stenciled markings for legibility.

Intentionally blank page

Flight Data Recorder (FDR) - Cleaning

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Isopropyl alcohol	C-385	AR	

Procedure

1. Clean chassis surfaces with paper towel or soft clean cloth moistened in mild detergent, then dry thoroughly.
2. Remove detergent residue with paper towel or soft clean cloth.
3. Remove stubborn dirt/grease from chassis using a lint-free cloth or a soft-bristle brush moistened with isopropyl alcohol (C-385).
4. Thoroughly dry surfaces.
5. Clean connector contacts using a soft-bristled brush moistened with isopropyl alcohol (C-385).
6. Thoroughly dry surfaces.

Intentionally blank page

Flight Data Recorder (FDR) - Installation

Procedure

1. Ensure FDR circuit breaker (27CB2) is open.
2. Position FDR (27A1) on mounting tray and push carefully into place.
3. Place the two tray-locking mechanism over FDR crow feet. Secure by turning locking mechanism hand tight.
4. Verify that the FDR is securely mounted to the mounting tray and no looseness is evident.
5. Close FDR circuit breaker (27CB2).
6. Verify that the FDR SYS FAIL light is extinguished.

Intentionally blank page

Recorder Download Unit (RDU) - Description

NOTE

The Recorder Download Unit (RDU) is optional ground support equipment and is not included as part of the FDR kit. Contact the FDR manufacturer for additional information on this equipment.

An RDU is used to provide operation and maintenance data upload/download to/from the system. The RDU (PLT6254-01-01) is a PC type laptop computer with an adapter kit and a PCMCIA memory card. The adapter kit provides the communication link between the PC and the recorder. The memory card provides the portable data transfer media when moving data between the RDU and ground support equipment.

The RDU runs an IDARS Operational Ground Program (IOGP) . The IOGP communicates with the recorder through a PCMCIA interface card plugged into the laptop's PC card slot. A cable assembly interconnects the interface card to the maintenance connector on the front face of the recorder. The RDU provides the following recorder support functions:

- Flight data download, file format conversion, and display
- Recorder BIT initiation and BIT data display
- Recorder memory format/initialize
- Uploading recorder software

The RDU consists of the following:

PART NUMBER	NOMENCLATURE
Pentium III, 300 Mhz or higher, PCMCIA, 20Gb, Win 2000 or higher	Portable Computer
177090-07	FDR PCMCIA Interface Kit
553170	Integrated Ground Software (IGS) tool for display of helicopter specific parameter information
553340	IDARS Operational Ground Program (IOGP) software tool for retrieval/eraser of FDR data
554588-02-01	ASCDB File

The PCMCIA interface is used with the IOGP software for downloading of stored data or erasing the data stored on the FDR.

In download mode, the data stored on the FDR is retrieved and saved by the IOGP as a Recorded Data File (RDF) in a format necessary for post download analysis using the IGS.

The IGS software tool is used to decompress the retrieved RDF and present it in engineering units that can be easily interpreted by the user for post flight analysis.

For RDU operating instructions, refer to Extant Aerospace operation/maintenance manual for FDR model No. 3257D1. To obtain this manual and RDU information, contact:

Extant Aerospace

1615 W NASA Blvd, Melbourne, FL 32901

Tel: (888) 327-7994

Mail: customerservice@extantaerospace.com

Website: www.extantaerospace.com

Intentionally blank page

FDR Pre-Flight - Operational Check

Procedure

1. The test described in Table 1 provides flight crew check of the recorder. Minimum test interval is once per day or at crew change.

NOTE

Recorder Display Unit (RDU) should not be connected to the recorder during this procedure. RDU connection will cause FDR FAIL to flash indicating maintenance mode.

Table 1. Pre-Flight Test Failure Isolation Chart

ACTION	VERIFY
Apply helicopter AC and DC power to the recorder system.	Verify that the FDR FAIL annunciator (FDR_FAIL output of the recorder) is off.

Intentionally blank page

Acoustic Beacon - Test (24-month Interval) - Operational Check

Procedure

1. The recorder may be equipped with a Dukane or a Benthos acoustic locator beacon. The following paragraphs provide the locator beacon test procedure using each manufacturer's test set.

1.1. Benthos Beacon

1.1.1. This test confirms operation of the locator beacon using the Benthos ATS-260 test set.

1.1.1.1. Verify that the battery expiration date listed on the acoustic locator beacon housing is not expired.

1.1.1.2. Attach the acoustic test set cable clip to the locator beacon.

1.1.1.3. Hold down the button on the test set handle labeled PUSH TO TEST.

1.1.1.4. Apply the test set probe to the water activated switch.

1.1.1.5. Check the battery condition by observing the green and the red indicators on the test set. Indicators will read as follows:

- Green indicator - The battery is good.
- Red indicator - The battery should be replaced.

1.1.1.6. Check the beacon operating condition by observing the amber indicator and listening for a tone from the test set. Indicator and tone information will be as follows:

- Amber indicator flashing and tone audible - The beacon is operating properly.

1.1.1.7. Release PUSH TO TEST button and remove test clip from beacon.

1.2. Dukane Beacon - Operational Check

1.2.1. This test confirms operation of a Dukane locator beacon using the Dukane PL-3 pinglite test set.

NOTE

A rapid or accelerating pulse rate usually indicates a faulty circuit. If normal operation does not occur, return beacon to manufacturer for service.

1.2.1.1. Apply spring end of test set to water switch; ensure that center spring touches water switch pin.

1.2.1.2. Listen for three to four audible pings and observe that PL-3 LED flashes.

1.3. Dukane Beacon - Battery Test

1.3.1. Use a high impedance voltmeter (input impedance of 10M ohms) to perform the following procedure:

1.3.1.1. Make sure the case and water switch are clean and dry prior to testing. If in doubt, clean as prescribed in 412-A-97-00-03-03A-251A-A / 02685.

1.3.1.2. Place the negative lead of a high impedance voltmeter on the water switch pin and the positive lead to the beacon case or the mounting kit, if already installed.

1.3.1.3. Measure the battery voltage. The beacon is operable at a minimum acceptable voltage of 2.97 volts. If the battery voltage is below the minimum, beacon battery requires replacement.

Intentionally blank page

Underwater Locator Beacon (ULB) - Inspection

Procedure

NOTE

Beacon battery must be replaced by the date stamped on case (6-year interval).

1. Verify battery expiration date (date stamped on beacon case).

Intentionally blank page

Acoustic Beacon - Cleaning

Procedure

1. Clean the end cap with mild detergent, then dry thoroughly.
2. Dry beacon thoroughly with a soft clean cloth.

Intentionally blank page

Air Data Computer (ADC) - Operational Check

Procedure

1. Perform a ground test procedure of the ADC (412-A-97-00-03-07A-136A-A / 02708).

Intentionally blank page

Air Data Computer (ADC) - Troubleshooting

Procedure

1. To troubleshoot the ADC, please contact Product Support Engineering.

Intentionally blank page

Air Data Computer (ADC) - Removal

Procedure

ENSURE ELECTRICAL POWER IS OFF.

Applicable: BHT-412-SI-86: installed:

1. Ensure FDAU circuit breaker (27CB1) is open. For helicopters with the ADC, ensure the ADC circuit breaker (27CB11) is open.
2. Remove pitot and static lines.
3. Remove and retain screws and washers that secure ADC and retain all hardware for reinstallation.
4. Disconnect 27A2P1 connector.

Intentionally blank page

Air Data Computer (ADC) - Inspection

Procedure

1. Visually inspect ADC for damage to case.
2. Visually inspect electrical connector for damaged/bent pins, corrosion, or cracks.
3. Visually inspect condition of pitot and static line connectors for damaged threads or cracking.
4. Visually inspect condition of all mounting hardware for corrosion or wear.

Intentionally blank page

Air Data Computer (ADC) - Cleaning

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Isopropyl alcohol	C-385	AR	

Procedure

1. Clean chassis surfaces with paper towel or soft clean cloth moistened in mild detergent, then dry thoroughly.
2. Remove detergent residue with paper towel or soft clean cloth.
3. Remove stubborn dirt/grease from chassis using a lint-free cloth or a soft-bristle brush moistened with isopropyl alcohol (C-385).
4. Remove debris and clean static line ports/connectors using a lint-free cloth moistened with isopropyl alcohol (C-385).
5. Thoroughly dry surfaces.
6. Clean connector contacts using a soft-bristled brush moistened with isopropyl alcohol (C-385).
7. Thoroughly dry surfaces.

Intentionally blank page

Air Data Computer (ADC) - Installation

Procedure

Applicable: BHT-412-SI-86: Installed:

1. Ensure FDAU circuit breaker (27CB1) is open. For helicopters with the ADC, ensure the ADC circuit breaker (27CB11) is open.

2. Connect 27A2P1 connector.

3. Position ADC (27A2) on mounting tray and secure with four screws and washers.

TORQUE ALL PLASTIC FITTINGS TO 15 ±5 INCH-POUNDS (1.695 ±0.56 NM).

4. Install pitot and static lines.

NOTE

Ensure all lines and fittings to be installed are free of water traps; use drain caps (50-012-02N400) to meet these requirements.

5. Close FDAU circuit breaker (27CB1) and, if applicable, the ADC Circuit breaker (27CB11).

6. Perform a pitot-static leak test (412-A-95-11-05-00A-364A-A / 01807).

Intentionally blank page

Flight Data Acquisition Unit (FDAU) - Operational Check

Procedure

1. Perform a ground test procedure of the FDAU (412-A-97-00-03-07A-136A-A / 02708).

Intentionally blank page

Flight Data Acquisition Unit (FDAU) - Troubleshooting

Procedure

1. To troubleshoot the FDAU, please contact Product Support Engineering.

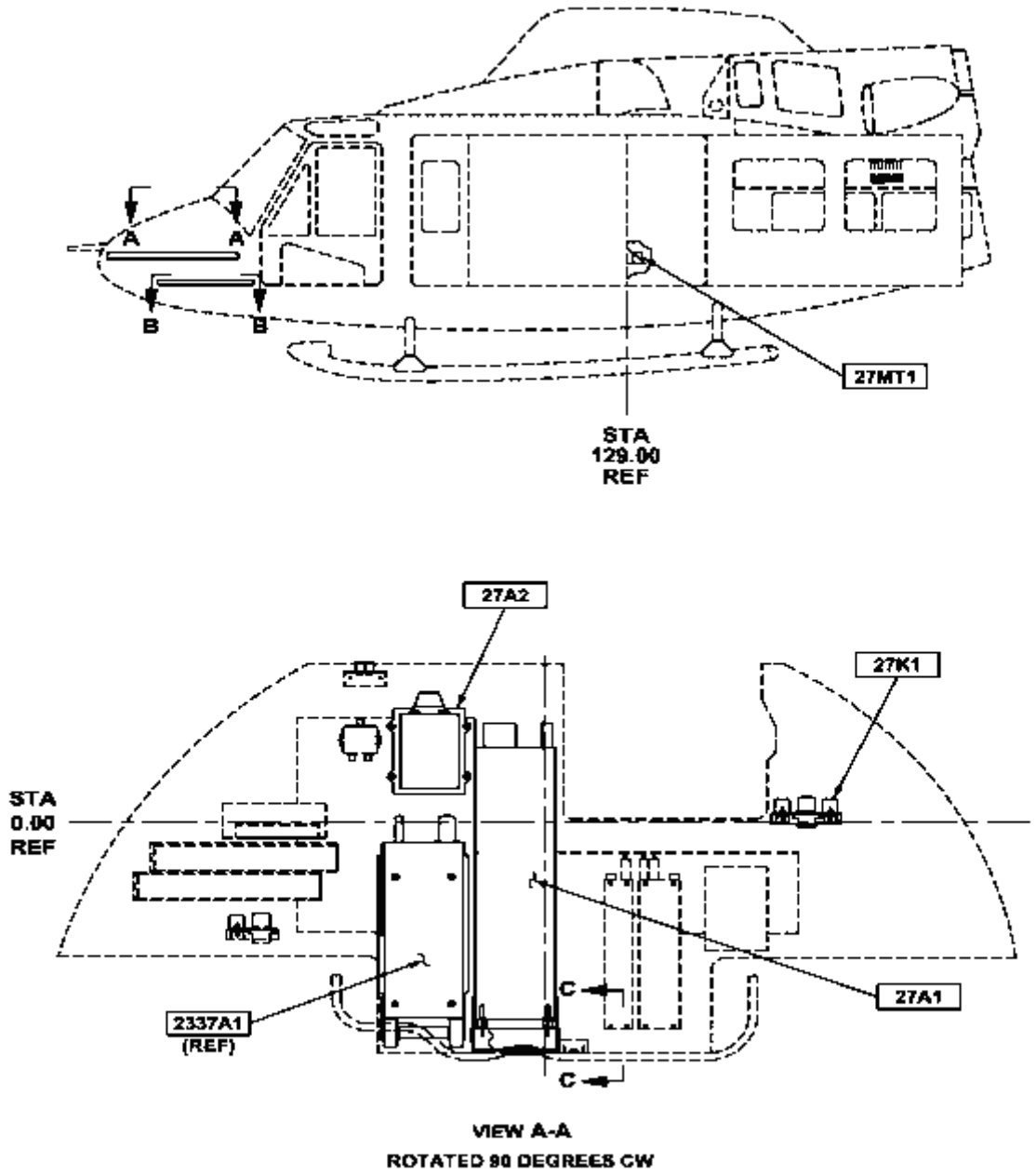
Intentionally blank page

Flight Data Acquisition Unit (FDAU) - Removal

Procedure

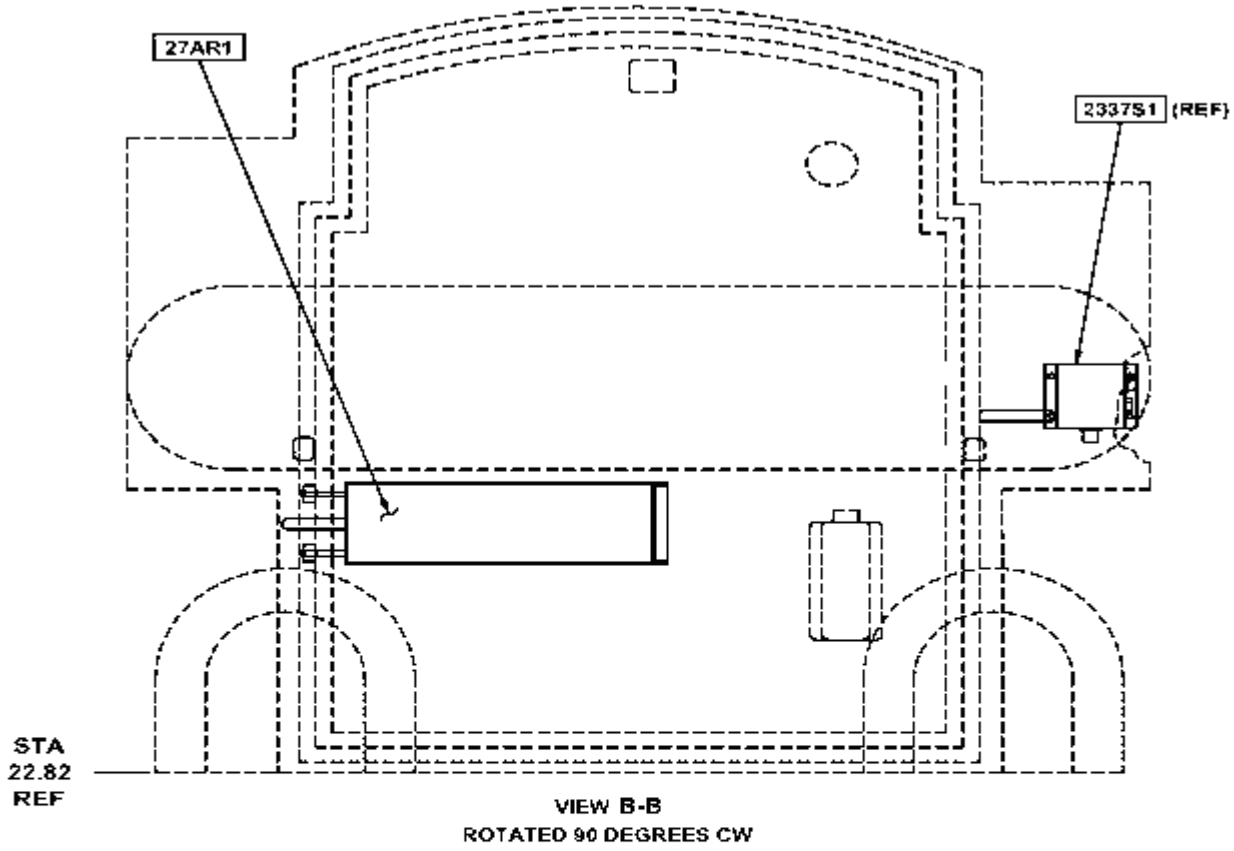
ENSURE ELECTRICAL POWER IS OFF.

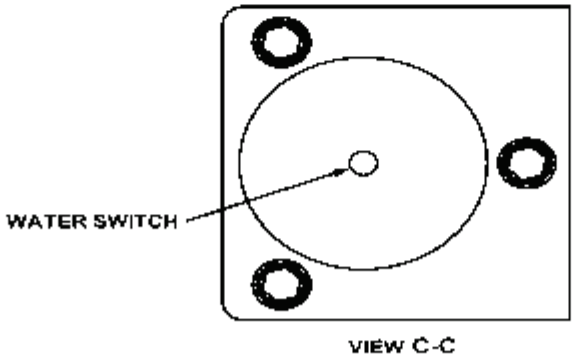
1. Ensure FDAU circuit breaker (27CB1) is open.
2. Loosen the two fasteners on front of FDAU (27AR1) (Figure 1) far enough to free the two hook supports.
3. Remove FDAU by pulling straight out of tray.



ICN-412-A-970003-B-97499-00002-A-001-01

Figure 1. Flight Data Recorder (FDR) System Equipment Location (Sheet 1 of 2)





ICN-412-A-970003-B-97499-00003-A-001-01

Figure 1. Flight Data Recorder (FDR) System Equipment Location (Sheet 2 of 2)

Intentionally blank page

Flight Data Acquisition Unit (FDAU) - Inspection

Procedure

1. Visually inspect components for general condition or damage.
2. Visually inspect electrical connector for damaged/bent pins, corrosion, or cracks.
3. Visually inspect condition of all mounting hardware for corrosion or wear.

Intentionally blank page

Flight Data Acquisition Unit (FDAU) - Cleaning

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Isopropyl alcohol	C-385	AR	

Procedure

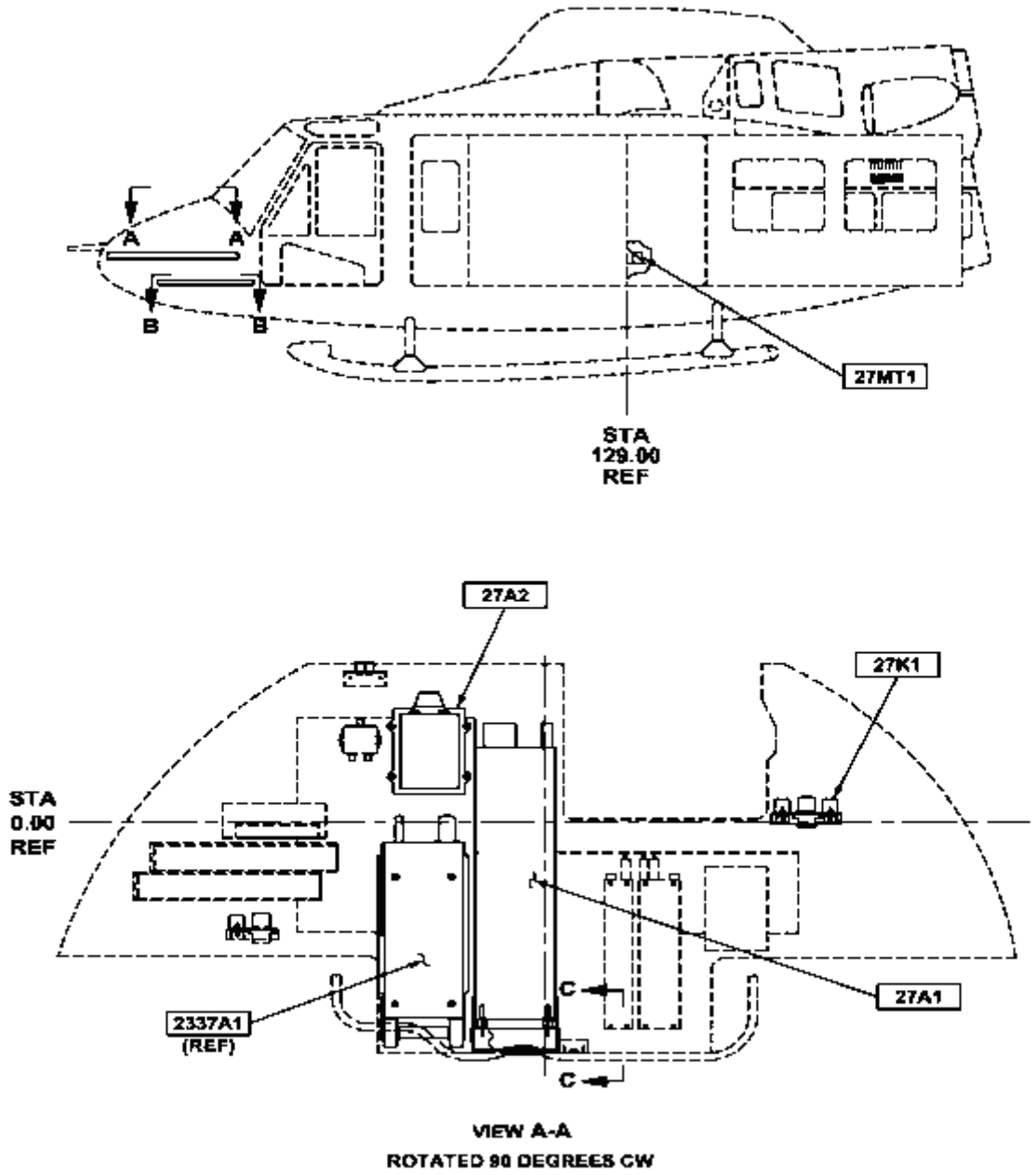
1. Clean chassis surfaces with paper towel or soft clean cloth moistened in mild detergent, then dry thoroughly.
2. Remove detergent residue with paper towel or soft clean cloth.
3. Remove stubborn dirt/grease from chassis using a lint-free cloth or a soft-bristle brush moistened with isopropyl alcohol (C-385).
4. Thoroughly dry surfaces.
5. Clean connector contacts using a soft-bristled brush moistened with isopropyl alcohol (C-385).
6. Thoroughly dry surfaces.

Intentionally blank page

Flight Data Acquisition Unit (FDAU) - Installation

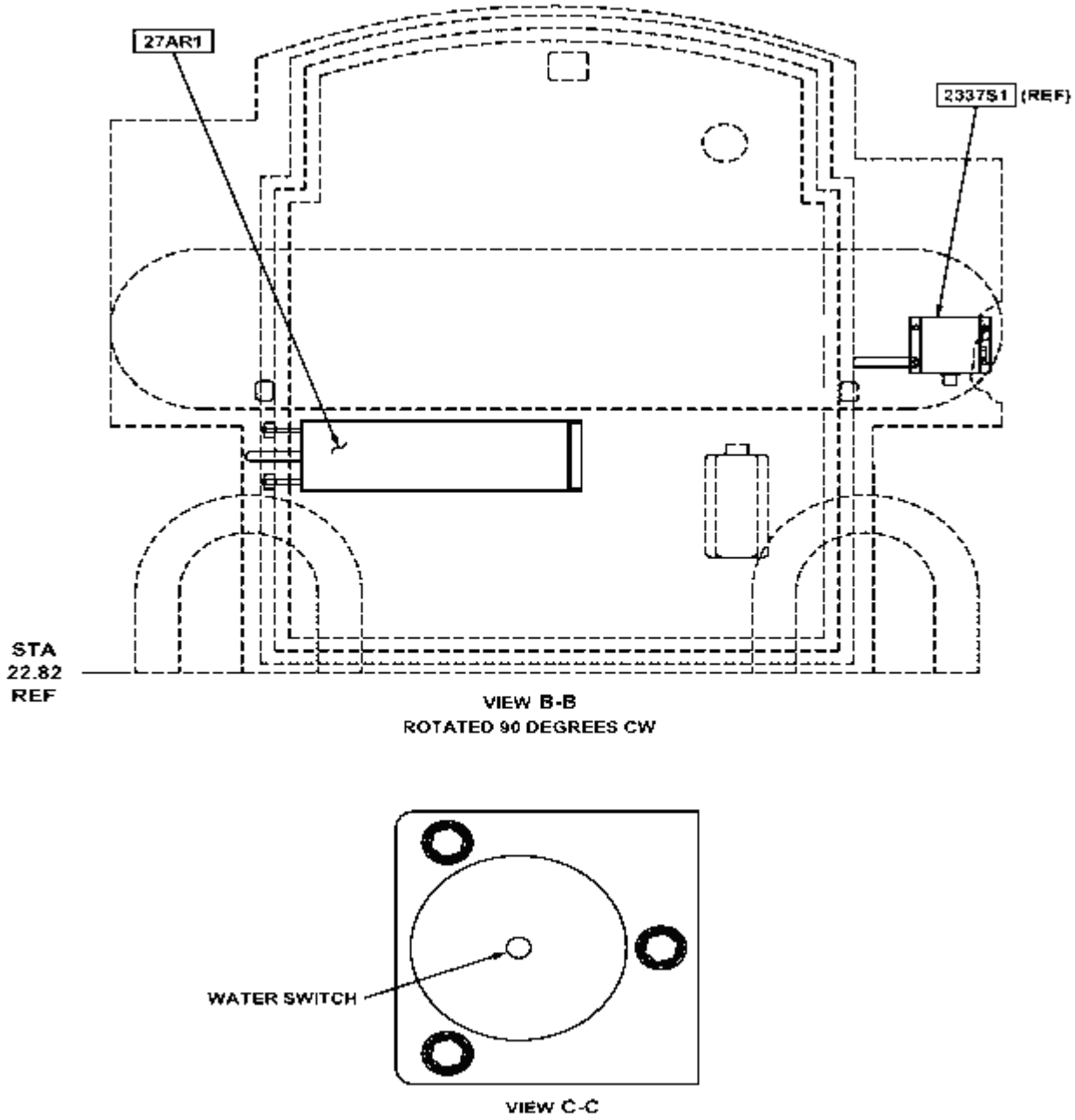
Procedure

1. Ensure FDAU circuit breaker (27CB1) is open.
2. Position FDAU (27AR1) (Figure 1) on mounting tray and push it carefully into place.
3. Place the two tray-locking mechanisms over FDAU crow feet and secure by turning locking mechanism until hand tight.
4. Close FDAU circuit breaker (27CB1).
5. Perform a ground test procedure of the FDR (412-A-97-00-03-07A-136A-A / 02708).



ICN-412-A-970003-B-97499-00002-A-001-01

Figure 1. Flight Data Recorder (FDR) System Equipment Location (Sheet 1 of 2)



ICN-412-A-970003-B-97499-00003-A-001-01

Figure 1. Flight Data Recorder (FDR) System Equipment Location (Sheet 2 of 2)

Intentionally blank page

Accelerometer - Operational Check

Procedure

1. Perform a ground test procedure of the accelerometer (412-A-97-00-03-07A-136A-A / 02708).

Intentionally blank page

Accelerometer - Troubleshooting

Procedure

1. Remove the accelerometer (27MT1) (412-A-97-00-03-06A-520A-A / 02700), except leave it connected to 27MT1P1.
2. Remove FDAU (27AR1) (412-A-97-00-03-05A-520A-A / 02694).
3. Put a jumper wire between connector 27AR1P1B-6 and 27AR1P1B-76, and connect a ground wire to 27AR1P1B-77.
4. Close FDAU circuit breaker (27CB1).
5. Refer to Test Table (Dual-axial Accelerometer) (Figure 1) and measure output voltage with accelerometer (27MT1) in axial null, +90° and -90° positions for each applicable axis.
6. If the output voltages are not within the specified ranges, the accelerometer should be returned to the factory for repair or overhaul.

AXIS	POSITION		OUTPUT
VERTICAL	0° AXIAL NULL		MEASURE BETWEEN 27AR1P1A-59 (H) AND 27AR1P1A-60 (C) 1.800 VDC ±25 mV
	+90° +1G		MEASURE BETWEEN 27AR1P1A-59 (H) AND 27AR1P1A-60 (C) 2.333 VDC ±25 mV
	-90° -1G		MEASURE BETWEEN 27AR1P1A-59 (H) AND 27AR1P1A-60 (C) 1.267 VDC ±25 mV
	0° AXIAL NULL		MEASURE BETWEEN 27AR1P1B-63 (H) AND 27AR1P1B-64 (C) 2.600 VDC ±25 mV
LATERAL	+90° +1G		MEASURE BETWEEN 27AR1P1B-63 (H) AND 27AR1P1B-64 (C) 5.000 VDC ±25 mV
	-90° -1G		MEASURE BETWEEN 27AR1P1B-63 (H) AND 27AR1P1B-64 (C) 0.2000 VDC ±25 mV
LONGITUDINAL	0° AXIAL NULL		MEASURE BETWEEN 27AR1P1A-63 (H) AND 27AR1P1A-64 (C) 2.600 VDC ±25 mV
	+90° +1G		MEASURE BETWEEN 27AR1P1A-63 (H) AND 27AR1P1A-64 (C) 5.000 VDC ±25 mV
	-90° -1G		MEASURE BETWEEN 27AR1P1A-63 (H) AND 27AR1P1A-64 (C) 0.2000 VDC ±25 mV

ICN-412-A-970003-B-97499-00005-A-001-01

Figure 1. Test Table (Dual-axial Accelerometer) (Sheet 1 of 1)

Accelerometer - Removal

Procedure

ENSURE ELECTRICAL POWER IS OFF.

1. Ensure FDAU circuit breaker (27CB1) is open.
2. At STA 129.0, locate accelerometer (27MT1).
3. Disconnect connector (27MT1P1).
4. Remove and retain screws and washers that secure accelerometer (27MT1) and retain all hardware for reinstallation.
5. Remove accelerometer (27MT1).

Intentionally blank page

Accelerometer - Inspection

Procedure

1. Visually inspect accelerometer for damage to case.
2. Visually inspect for oil damage.
3. Visually inspect electrical connector for damaged/bent pins, corrosion, or cracks.
4. Visually inspect condition of all mounting hardware for corrosion or wear.

Intentionally blank page

Accelerometer - Cleaning

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Isopropyl alcohol	C-385	AR	
Cleaning compound	C-533	AR	

Procedure

1. Remove stubborn dirt/grease from accelerometer chassis using a soft-bristle brush moistened in cleaning compound (C-533).
2. Dry clean with paper towel or soft clean cloth.
3. Thoroughly dry surfaces.
4. Clean connector contacts using a soft-bristled brush moistened with isopropyl alcohol (C-385).
5. Thoroughly dry surfaces.
6. At STA 129.0, position accelerometer (27MT1) in its mounting position and secure with two screws and washers.

Intentionally blank page

Accelerometer - Installation

Procedure

1. Ensure FDAU circuit breaker (27CB1) is open.
2. Connect connector (27MT1P1).
3. Close FDAU circuit breaker (27CB1).

Intentionally blank page

Flight Data Recorder (FDR) System - Repair or Replacement

Procedure

1. Replace defective or inoperative units.

NOTE

ULB battery can be locally replaced. Contact vendor for appropriate procedures.

2. Further repairs, other than removal and replacement, must be conducted by an authorized repair facility.

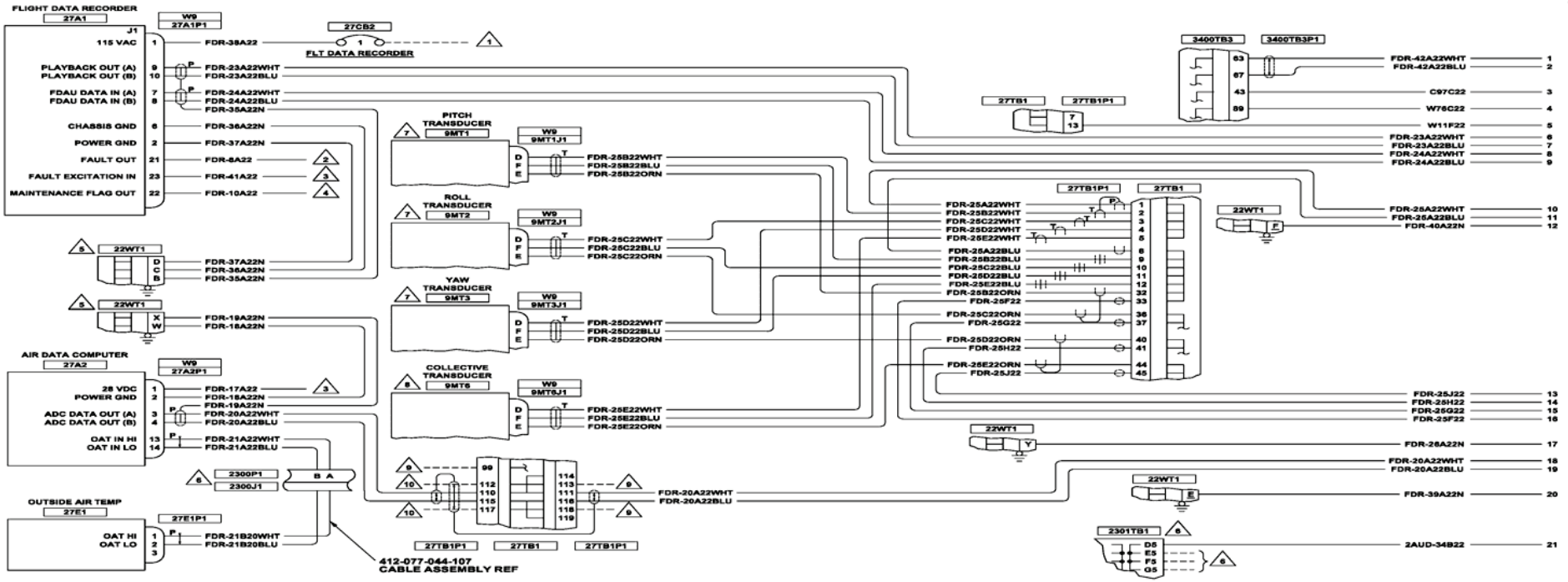
Intentionally blank page

The FDR system wiring diagram (Figure 1 and Figure 2) is provided to assist maintenance personnel in understanding the circuit and components installed in the helicopter, and in troubleshooting and tracing inoperative and malfunctioning circuits. Standard symbols are used where possible.

2. Wire Identification

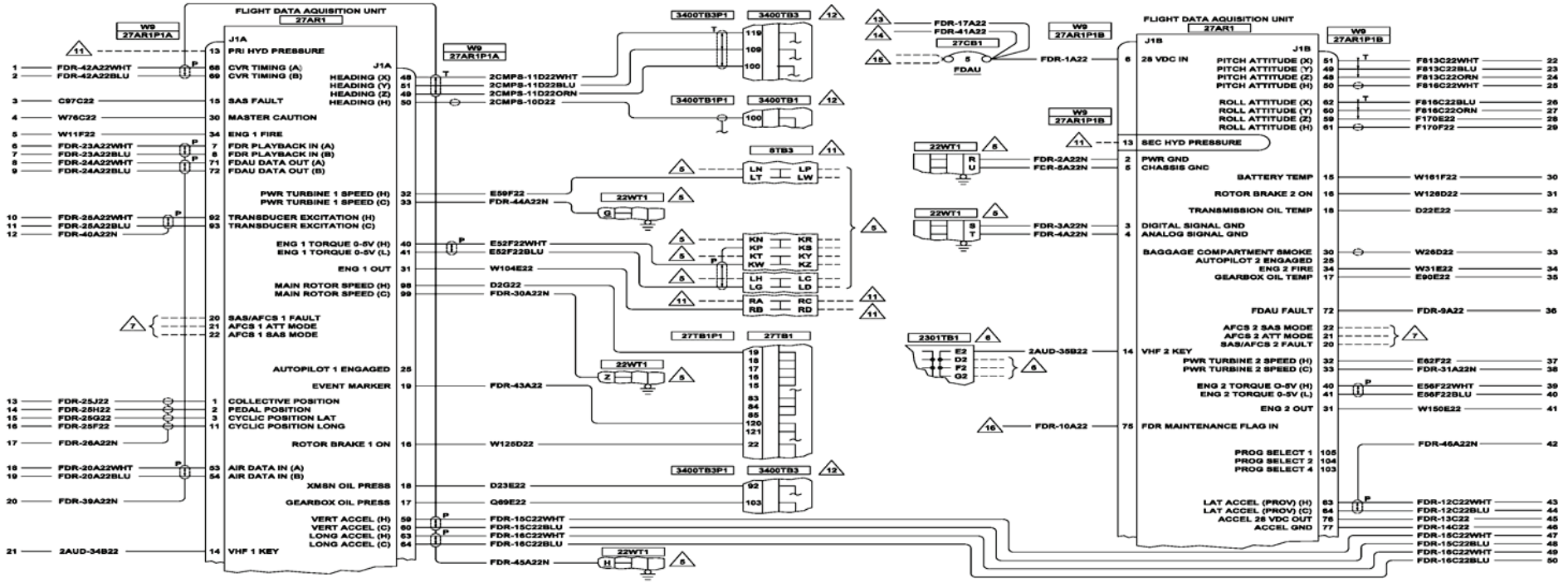
All wires on the system wiring diagram (Figure 1 and Figure 2) are identified by codes as they are marked in the helicopter. The code or wire number indicates the circuit function, wire number, wire segment number, and wire conductor size. Refer to (412-A-98-00-00-00A-042A-A / 02456) for wire code definitions.

Intentionally blank page



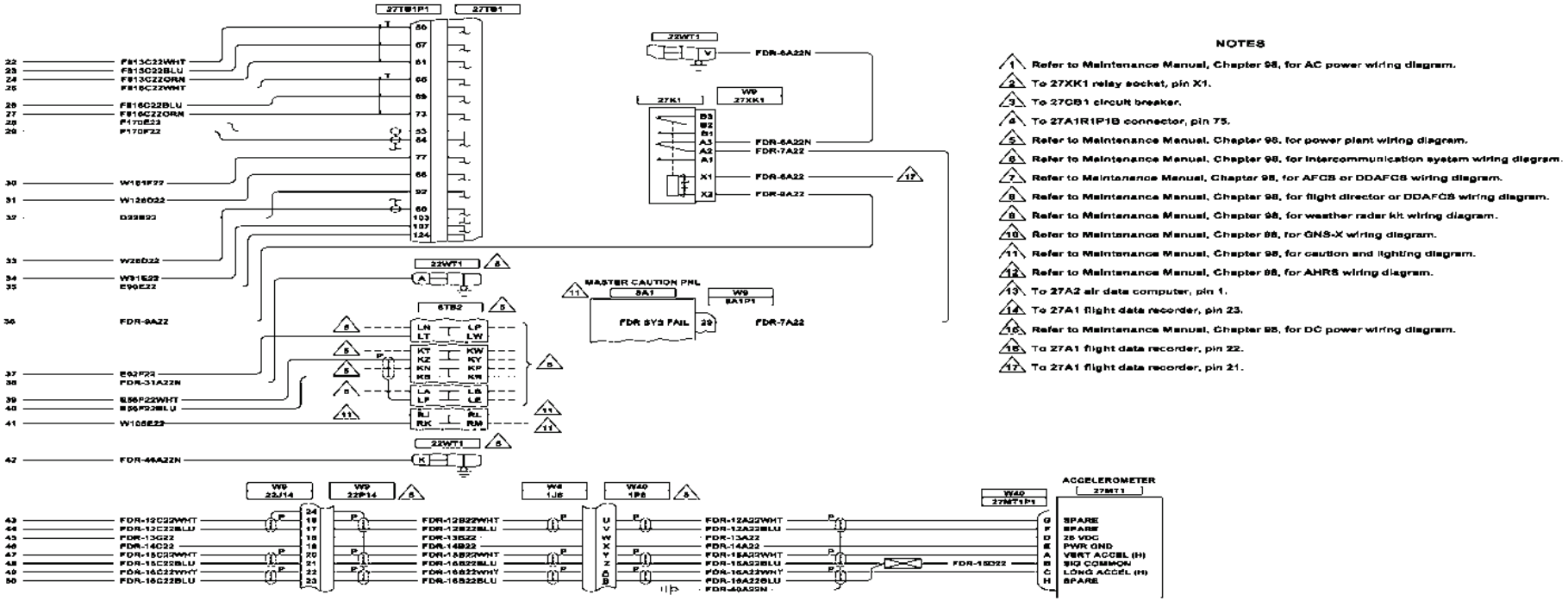
ICN-412-A-970003-B-97499-00006-A-001-01

Figure 1. Wiring Diagram - Flight Data Recorder (BHT-412-SI-73) (Sheet 1 of 3)



ICN-412-A-970003-B-97499-00007-A-001-01

Figure 1. Wiring Diagram - Flight Data Recorder (BHT-412-SI-73) (Sheet 2 of 3)



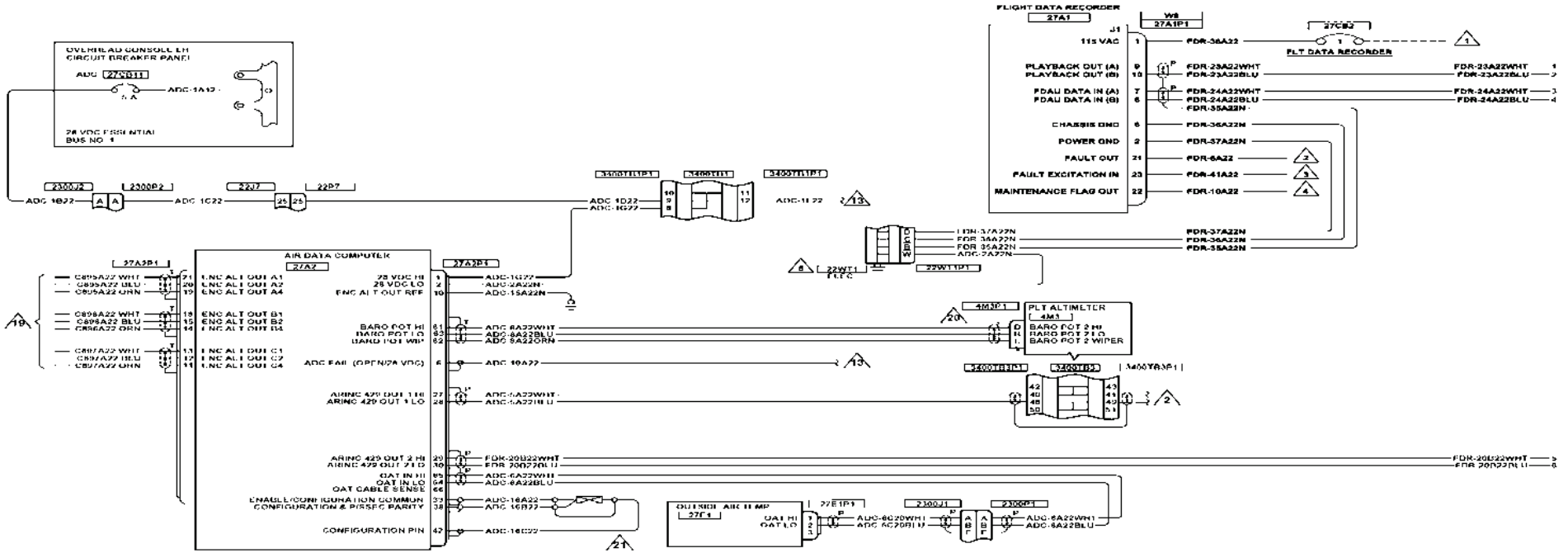
NOTES

1. Refer to Maintenance Manual, Chapter 98, for AC power wiring diagram.
2. To 27XK1 relay socket, pin X1.
3. To 27CB1 circuit breaker.
4. To 27A1R1P1B connector, pin 75.
5. Refer to Maintenance Manual, Chapter 98, for power plant wiring diagram.
6. Refer to Maintenance Manual, Chapter 98, for intercommunication system wiring diagram.
7. Refer to Maintenance Manual, Chapter 98, for AFCS or DDAFCS wiring diagram.
8. Refer to Maintenance Manual, Chapter 98, for flight director or DDAFCS wiring diagram.
9. Refer to Maintenance Manual, Chapter 98, for weather radar kit wiring diagram.
10. Refer to Maintenance Manual, Chapter 98, for GNS-X wiring diagram.
11. Refer to Maintenance Manual, Chapter 98, for caution and lighting diagram.
12. Refer to Maintenance Manual, Chapter 98, for AHRS wiring diagram.
13. To 27A2 air data computer, pin 1.
14. To 27A1 flight data recorder, pin 23.
15. Refer to Maintenance Manual, Chapter 98, for DC power wiring diagram.
16. To 27A1 flight data recorder, pin 22.
17. To 27A1 flight data recorder, pin 21.

ICN-412-A-970003-B-97499-00008-A-001-01

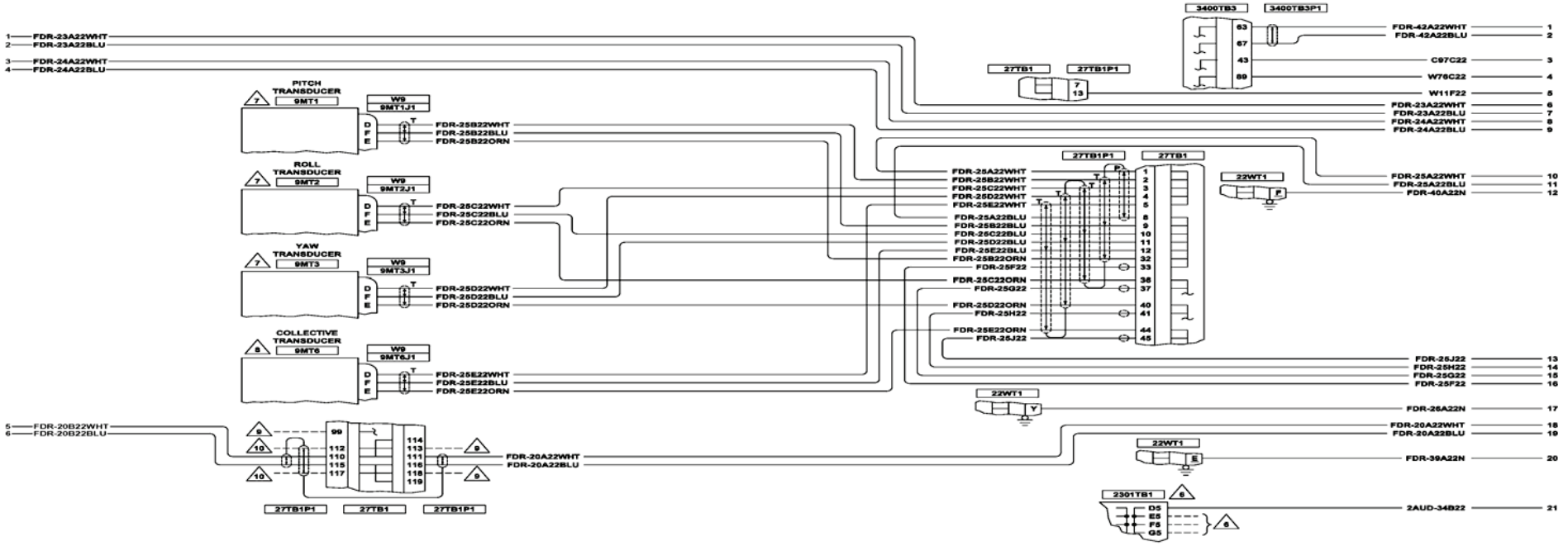
Figure 1. Wiring Diagram - Flight Data Recorder (BHT-412-SI-73) (Sheet 3 of 3)

Intentionally blank page



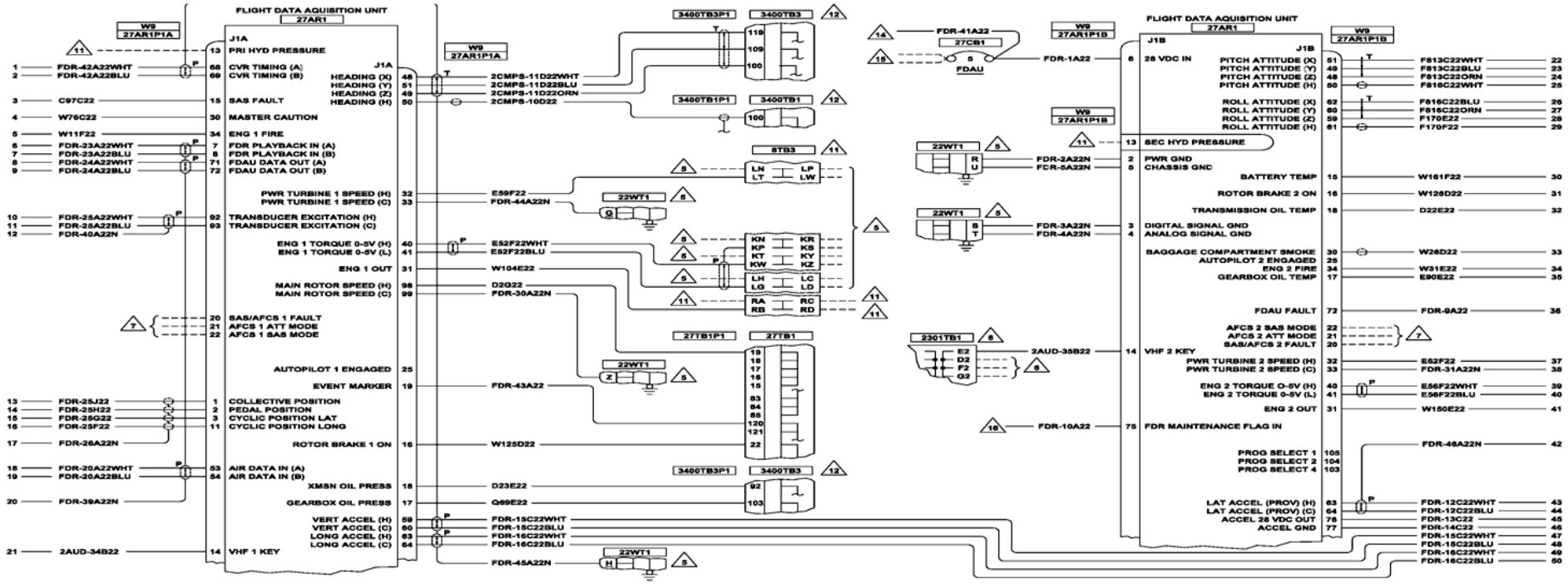
ICN-412-A-970003-B-97499-00009-A-001-01

Figure 2. Flight Data Recorder (BHT-412-SI-73) with ADC (BHT-412-SI-86) - Wiring Diagram (Sheet 1 of 4)



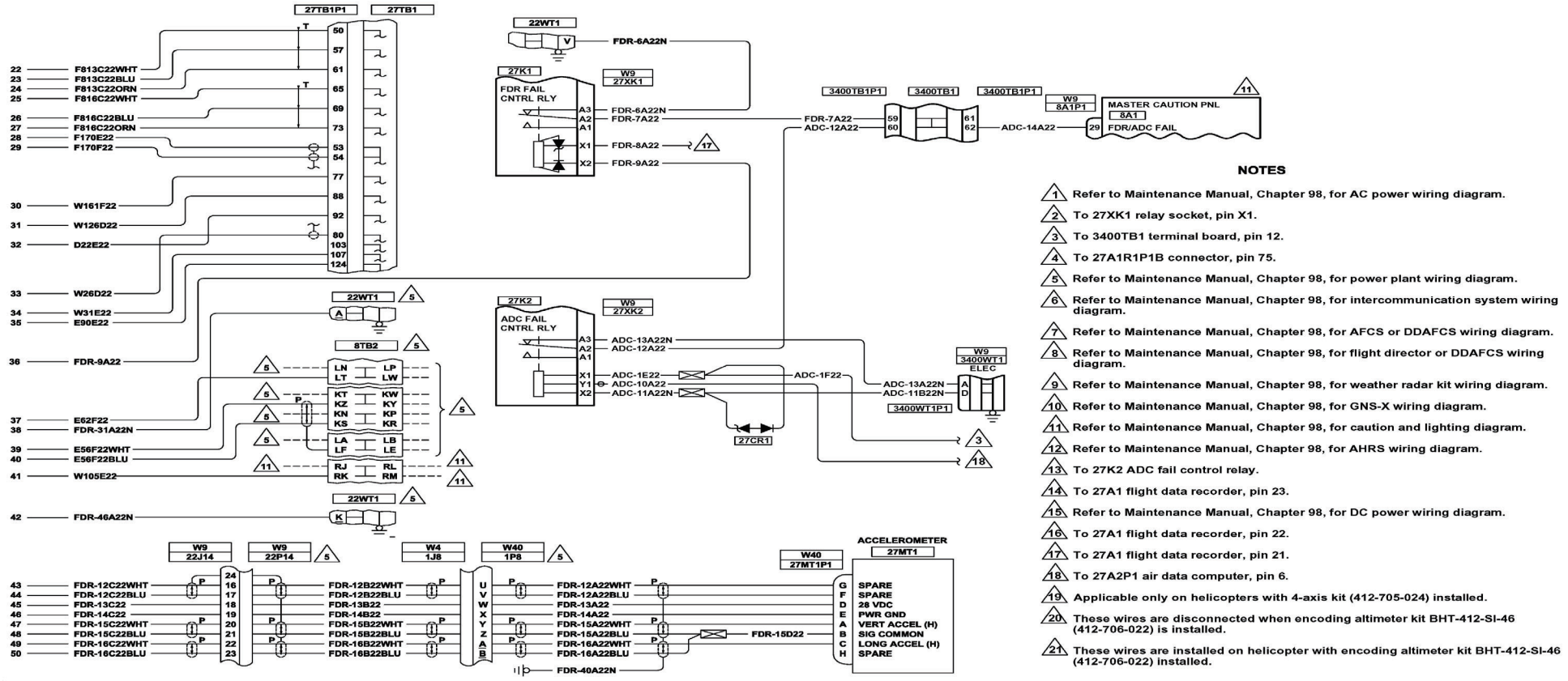
ICN-412-A-970003-B-97499-00010-A-001-01

Figure 2. Flight Data Recorder (BHT-412-SI-73) with ADC (BHT-412-SI-86) - Wiring Diagram (Sheet 2 of 4)



ICN-412-A-970003-B-97499-00011-A-001-01

Figure 2. Flight Data Recorder (BHT-412-SI-73) with ADC (BHT-412-SI-86) - Wiring Diagram (Sheet 3 of 4)



NOTES

1. Refer to Maintenance Manual, Chapter 98, for AC power wiring diagram.
2. To 27XK1 relay socket, pin X1.
3. To 3400TB1 terminal board, pin 12.
4. To 27A1R1P1B connector, pin 75.
5. Refer to Maintenance Manual, Chapter 98, for power plant wiring diagram.
6. Refer to Maintenance Manual, Chapter 98, for intercommunication system wiring diagram.
7. Refer to Maintenance Manual, Chapter 98, for AFCS or DDAFCS wiring diagram.
8. Refer to Maintenance Manual, Chapter 98, for flight director or DDAFCS wiring diagram.
9. Refer to Maintenance Manual, Chapter 98, for weather radar kit wiring diagram.
10. Refer to Maintenance Manual, Chapter 98, for GNS-X wiring diagram.
11. Refer to Maintenance Manual, Chapter 98, for caution and lighting diagram.
12. Refer to Maintenance Manual, Chapter 98, for AHRS wiring diagram.
13. To 27K2 ADC fail control relay.
14. To 27A1 flight data recorder, pin 23.
15. Refer to Maintenance Manual, Chapter 98, for DC power wiring diagram.
16. To 27A1 flight data recorder, pin 22.
17. To 27A1 flight data recorder, pin 21.
18. To 27A2P1 air data computer, pin 6.
19. Applicable only on helicopters with 4-axis kit (412-705-024) installed.
20. These wires are disconnected when encoding altimeter kit BHT-412-SI-46 (412-706-022) is installed.
21. These wires are installed on helicopter with encoding altimeter kit BHT-412-SI-46 (412-706-022) installed.

ICN-412-A-970003-B-97499-00012-A-001-01

Figure 2. Flight Data Recorder (BHT-412-SI-73) with ADC (BHT-412-SI-86) - Wiring Diagram (Sheet 4 of 4)

Flight Data Recorder (FDR) - Preflight Test - Description

The FDR system shall be preflight tested in accordance with the time synchronization procedure to provide reference data verifying the installation.

Intentionally blank page

Flight Data Recorder (FDR) - Preflight Test - Time Synchronization Procedure

Procedure

NOTE

Ensure that the FDAU Relative Time, the cockpit digital clock (or stop watch), FDR, and CVR elapsed time are correlated to the same time as accurately as possible. This is essential for the accuracy of the post download correlation of the FDR recorded data with those recorded by test engineers.

NOTE

FDAU Relative Time is the internally generated relative time counter or frame counter. The FDAU combines this data with other parameters in order to facilitate the Flight Data ARINC 717 data stream output. The time counter resets every 16,380 seconds (273 minutes). As such, download FDR and CVR recordings after each 4-hour block of testing, and reset the FDR memory.

1. To accomplish this, apply power on the helicopter or pull the FDAU, FDR, and CVR circuit breakers. Start the FDAU, FDR, and CVR recording cycle by setting the respective circuit breakers back in at same moment as the cockpit digital clock elapsed time counter set.

Intentionally blank page

Flight Data Recorder (FDR) - Preflight Test - Ground Test Procedure

Procedure

1. For ground test procedure, refer to the table that follows.

The following equipment, or equivalent, is required during this procedure:

- Pitot/Static tester
- Ground Power Unit (GPU)
- Hydraulic test set

FLIGHT DATA RECORDER (KIT 412-706-084-101/105)
GROUND TEST PROCEDURES

MODEL: 412

HELICOPTER NO.: _____

PILOT/EXAMINER: _____
DATE: _____

1.1. Visual Inspection

( )

Inspect the equipment installation to detect faulty mechanical mountings or electrical connections. Remove the connector cover from the front of the FDR and connect the test cable.

1.2. FDR Status Indication

1.2.1. Perform the following steps and note results:

• FDR FAIL Mode

( )

With FDR circuit breaker pulled out and the ADC circuit breaker in, ensure that the FDR SYS FAIL or FDR/ADC FAIL light is illuminated on the Caution Panel.

• FDR Normal Mode

( )

With FDR circuit breaker in and the ADC circuit breaker in, ensure that the FDR SYS FAIL or FDR/ADC FAIL light is extinguished on the caution panel after a short power-up BIT test.

• FDR Busy Erase/Formatting

( )

Connect J8 connector to RDU and verify that FDR SYS FAIL or FDR/ADC FAIL light is illuminated and that FDR FAIL light Slow Blink (2 Hz rate) is observed on the FDAU.

( )

Clear memory with the RDU (format memory).

1.3. Air Data Parameters (Using Pitot/Static Tester Applied to the Copilot's System and the AUX PITOT STATIC Switches Set to NORM)

( )

Time synchronization
Power Cycle: 1
Elapsed Time: 0

( )

1.3.1. Set the copilot’s altimeter to 29.92 in Hg.

( )

1.3.2. Connect the pitot static test set to the copilot pitot tube and static.
Elapsed Time

( )

1.3.3. Check altitude as follows:
Power Cycle: _____
_____

( )

1.3.3.1. Simulate the altitudes below and verify that the RDU readout is within acceptable limits. Record the values.

APPLIED ALTITUDE COPILOT’S ALTIMETER RDU READING PRES_ALT (FEET) (Feet) (Feet)
Min
Max
Elapsed Time

( )

-500
_____
-600
_____

-400

_____ ( )

0

_____

-100

_____

100

_____ ( )

500

_____

400

_____

600

_____ ( )

5000

_____

4850

_____

5150

_____ ( )

10000

_____

9850

_____

10150

_____

Elapsed Time ( )

1.3.4. Check altitude rate as follows:
Power Cycle: _____

_____ ( )

1.3.4.1. Simulate the altitude rates below and verify that the RDU readout is within acceptable limits. Record the values.

APPLIED ALTITUDE RATE
COPILOT'S VERTICAL SPEED
RDU READING ALT_RATE (FEET/MINUTE)
(Feet/minute)
(Feet/minute)
Min
Max

Elapsed Time ( )

0

_____

-250

_____

250

_____ ( )

500

_____

250

_____

750

_____ ( )

2000

_____

1750

_____

2250

_____

( )

-500
-750
-250

( )

-2000
-2250
-1750

Elapsed Time

( )

1.3.5. Check airspeed as follows:
Power Cycle: _____

( )

1.3.5.1. Simulate the airspeeds below and verify that the RDU readout is within acceptable limits. Record the values.

APPLIED AIRSPEED
COPILOT'S INDICATED AIRSPEED
RDU READING AIRSPEED (KNOTS)

2. Flight Data Download Procedure

2.1. Connect RDU to J8 connector on front panel of FDR.

2.2. Double-click IOGP icon or select IOGP from Start menu on RDU laptop computer.

2.3. After IOGP has started, select "Download!" from the top menu of the IOGP.

2.4. Modify/Enter Aircraft and Recorder Serial Number data as necessary in the following substeps. If data is already correct, select "Continue Download". If data is not correct, proceed as follows:

NOTE

This operation only affects the downloaded file data and does not modify the documentary data stored in recorder memory. Refer to the Aircraft Documentary Data Upload/Display procedure (412-A-97-00-03-00A-130A-A / 02712) to change the recorder's stored documentary data.

2.4.1. Select "Modify Enterable Data".

2.4.2. Enter Aircraft Serial Number and/or Recorder Serial Number.

2.4.3. Select "Download Data".

2.4.4. Select "Continue Download".

2.5. From the "Download Recorded Data" window, select (by clicking on) each file type to be downloaded. A check mark will appear next to each file type selected. The recorder file types may include:

2.5.1. A717 ARINC 717 Flight Data

2.5.2. R422 RS422 Flight Data

2.5.3. BitL Built-in Test (BIT) Log

NOTE

If desired, individual sessions may be downloaded by selecting "Download By: Session". This will launch a "session(s) to download" window that enables individual session selection. Building the session menu may take several minutes.

2.6. Click "Download".

2.7. From the "Download File" window, select desired drive, directory, and enter a valid filename. The file extension must remain.rdf.

2.8. Click "Save" and wait for the download operation to complete.

2.9. Disconnect J8 connector.

Flight Data Recorder (FDR) System - Flight Test

Procedure

1. Flight testing of the FDR system shall be in accordance with the following procedures to provide reference data verifying the installation.

For the purposes of this test, both a ground run and a flight test will be performed.

The RDU shall be used to collect ground and in-flight data and cross check the data against the flight crew instruments.

1.1. Test Setup

1.1.1. Prior to start of testing, perform the following:

1.1.1.1. Confirm that the FDR recorded data from the ground tests (412-A-97-00-03-07A-136A-A / 02708) have been downloaded and are acceptable.

FLIGHT DATA RECORDER (KIT 412-706-084-101/105)

FLIGHT TEST PROCEDURES - GROUND RUN

MODEL: 412 **HELICOPTER NO.:** _____

PILOT/EXAMINER: _____ **DATE:** _____

1.2. Ground Run Procedure

1.2.1. Prior to Engine Startup

( ) 1.2.1.1. Perform a time synchronization (412-A-97-00-03-07A-320A-A / 02707).

( ) Time synchronization — Power Cycle: 1 — Elapsed Time: 0

( ) 1.2.1.2. Set the copilots altimeter to 29.92 in Hg.

( ) 1.2.1.3. Verify each of the following on the RDU flight test report page and that it corresponds with the helicopter indications.

	HELICOPTER INDICATION	PARAMETER	RDU READING		Elapsed Time
( )	ENG 1 OUT (Light ON)	ENG 1 OUT	ENG1_OUT (1=OUT)	_____	_____
( )	ENG 2 OUT (Light ON)	ENG 2 OUT	ENG2_OUT (1=OUT)	_____	_____
( )	XMSN OIL PRESS (Light ON)	XMSN Oil P	XMSN_OP_LOW (0=LOW)	_____	_____
( )	XMSN OIL TEMP (Light OFF)	XMSN Oil T	XMSN_OT_HIGH (0=HIGH)	_____	_____
( )	CBOX OIL PRESS (Light ON)	GBox Oil P	GBOX_OP_LOW (0=LOW)	_____	_____
( )	CBOX OIL TEMP (Light OFF)	GBox Oil T	GBOX_OT_HIGH (0=HIGH)	_____	_____
( )	Master Caution (Light ON)	MSTR CAUT	MASTER_CAU (1=ON)	_____	_____
( )	No. 1 HYDRAULIC (Light ON)	PRIM HYD	LOW_HYD_PRI (1=LOW)	_____	_____
( )	No. 2 HYDRAULIC (Light ON)	SEC HYD	LOW_HYD_SEC (1=LOW)	_____	_____
( )	BATT HOT (Light OFF)	BATT TEMP	BATT_HOT (0=HIGH)	_____	_____

FLIGHT DATA RECORDER (KIT 412-706-084-101/105)

FLIGHT TEST PROCEDURES - GROUND RUN

MODEL: 412 **HELICOPTER NO.: _____**

PILOT/EXAMINER: _____ **DATE: _____**

( ) 1.2.1.4. Record values below and verify that the RDU figures are within the stated limits of the Indicator reading.

HELICOPTER INDICATION	PARAMETER	RDU READING		Elapsed Time
0% (Triple Torque Ind)	_____T-RQ1	TURB1Q (%) >0% <5%	_____	_____
0% (Triple Torque Ind)	_____T-RQ2	TURB2Q (%) >0% <5%	_____	_____

1.2.2. Ground Run

( ) 1.2.2.1. After both engines are started and stabilized to idle verify the following RDU indications and that they correspond to the helicopter indications as shown.

HELICOPTER INDICATION	PARAMETER	RDU READING		Elapsed Time
ENG 1 OUT (Light OFF)	ENG 1 OUT	ENG1_OUT (1=OUT)	_____	_____
ENG 2 OUT (Light OFF)	ENG 2 OUT	ENG2_OUT (1=OUT)	_____	_____
XMSN OIL PRESS (Light OFF)	XMSN Oil P	XMSN_OP_LOW (0=LOW)	_____	_____
XMSN OIL TEMP (Light OFF)	XMSN Oil T	XMSN_OT_HIGH (0=HIGH)	_____	_____
CBOX OIL PRESS (Light OFF)	GBox Oil P	GBOX_OP_LOW (0=LOW)	_____	_____
CBOX OIL TEMP (Light OFF)	GBox Oil T	GBOX_OT_HIGH (0=HIGH)	_____	_____
Master Caution (Light OFF)	MSTR CAUT	MASTER_CAU (1=ON)	_____	_____
No. 1 HYDRAULIC (Light OFF)	PRIM HYD	LOW_HYD_PRI (1=LOW)	_____	_____
No. 2 HYDRAULIC (Light OFF)	SEC HYD	LOW_HYD_SEC (1=LOW)	_____	_____
BATT HOT (Light OFF)	BATT TEMP	BATT_HOT (0=HIGH)	_____	_____

( ) 1.2.2.2. Isolate Hydraulic System #2 by turning OFF the #1 hydraulic system and Event Mark. Verify following:

HELICOPTER INDICATION	PARAMETER	RDU READING		Elapsed Time
No. 1 HYDRAULIC (Light ON)	PRIM HYD	LOW_HYD_PRI (1=LOW)	_____	_____
No. 2 HYDRAULIC (Light OFF)	SEC HYD	LOW_HYD_SEC (1=LOW)	_____	_____

( ) 1.2.2.3. Turn Hydraulic System #1 back ON and Event Mark, and turn OFF hydraulic system #2. Verify the following:

HELICOPTER INDICATION	PARAMETER	RDU READING	Elapsed Time

FLIGHT DATA RECORDER (KIT 412-706-084-101/105)

FLIGHT TEST PROCEDURES - GROUND RUN

MODEL: 412 **HELICOPTER NO.:** _____

PILOT/EXAMINER: _____ **DATE:** _____

( )	No. 1 HYDRAULIC (Light OFF)	PRIM HYD	LOW_HYD_PRI (1=LOW)	_____	_____
( )	No. 2 HYDRAULIC (Light ON)	SEC HYD	LOW_HYD_SEC (1=LOW)	_____	_____
( )	1.2.2.4. Record values below and verify that the RDU figures are within the stated limits of the Indicator reading (engines at idle).				
	HELICOPTER INDICATION	**PARAMETER**	**RDU READING**		Elapsed Time
( )	Triple Tach Ind	_____ RPM1	TURB1_SPEED (±5%)	_____	_____
( )	Triple Tach Ind	_____ RPM2	TURB2_SPEED (±5%)	_____	_____
( )	Triple Tach Ind	_____R-PMR	ROTOR_SPEED (±5%)	_____	_____
( )	Triple Torque Ind	_____T-RQ1	TURB1Q (±5%)	_____	_____
( )	Triple Torque Ind	_____T-RQ2	TURB2Q (±5%)	_____	_____

FLIGHT DATA RECORDER (KIT 412-706-084-101/105)

FLIGHT TEST PROCEDURES - FLIGHT TESTS

MODEL: 412 **HELICOPTER NO.:** _____

PILOT/EXAMINER: _____ **DATE:** _____

1.3. Flight Test Procedure

NOTE

All flight maneuvers are subject to the pilots discretion. All altitudes are to be compared to the copilot's altimeter. Prior to take-off, align the pilots compass.

For each test point stabilize in a steady state condition for approximately 5 seconds prior to taking cockpit instruments readings.

( ) 1.3.1. Take off and note the following:.

HELICOPTER INDICATION	**INSTRUMENTS**	**RDU READING**		Elapsed Time
Take-off Heading	_____ ___	MAG_HEADING (±5°)	____ __	_____

( ) 1.3.2. Hover for 30 seconds. Record the following and verify that the RDU readings are within the status limits of the flight instruments:

HELICOPTER INDICATION	**INSTRUMENTS**	**RDU READING**		Elapsed Time
Altitude	_____ ___	PRES_ALT (±100 FT)	____ __	_____

FLIGHT DATA RECORDER (KIT 412-706-084-101/105)

FLIGHT TEST PROCEDURES - FLIGHT TESTS

MODEL: 412 **HELICOPTER NO.: _____**

PILOT/EXAMINER: _____ **DATE: _____**

( )	Airspeed	_____	AIRSPEED (±10 Knots)	____	_____
( )	Rotor RPM	_____	ROTOR_SPEED (±5%)	____	_____
( )	RPM ENG 1	_____	TURB1_SPEED (±5%)	____	_____
( )	RPM ENG 2	_____	TURB2_SPEED (±5%)	____	_____
( )	TRQ ENG 1	_____	TURB1Q (±5%)	____	_____
( )	TRQ ENG 2	_____	TURB2Q (±5%)	____	_____

( ) 1.3.3. Fly Straight and level at 80 knots for approximately 1 minute. Record the following and verify that the RDU readings are within the status limits of the flight instruments:

	HELICOPTER INDICATION	INSTRUMENTS	RDU READING		Elapsed Time
( )	Altitude	_____	PRES_ALT (±100 FT)	____	_____
( )	Airspeed	_____	AIRSPEED (±10 Knots)	____	_____

( ) 1.3.4. Climb at 1000 fpm. Record the following and verify that the RDU readings are within ±5% of the flight instruments:

	HELICOPTER INDICATION	INSTRUMENTS	RDU READING		Elapsed Time
( )	Rate of Climb	_____	ALT_RATE (±250 FPM)	____	_____
( )	TRQ ENG 1	_____	TURB1Q (±5%)	____	_____
( )	TRQ ENG 2	_____	TURB2Q (±5%)	____	_____
( )	Pitch Angle	_____	PITCH_ATT (±2°)	____	_____

( ) 1.3.5. Accelerate to V_h. Record the following and verify that the RDU readings are within the status limits of the flight instruments:

	HELICOPTER INDICATION	INSTRUMENTS	RDU READING		Elapsed Time
( )	Altitude	_____	PRES_ALT (±100 FT)	____	_____
( )	Airspeed	_____	AIRSPEED (±10 Knots)	____	_____
( )	Rotor RPM	_____	ROTOR_SPEED (±5%)	____	_____

FLIGHT DATA RECORDER (KIT 412-706-084-101/105)

FLIGHT TEST PROCEDURES - FLIGHT TESTS

MODEL: 412 **HELICOPTER NO.: _____**

PILOT/EXAMINER: _____ **DATE: _____**

( )	TRQ ENG 1	_____ ___	TURB1Q (±5%)	____ __		_____
( )	TRQ ENG 2	_____ ___	TURB2Q (±5%)	____ __		_____
( )	OAT	_____ ___	AIR_TEMP (±2°C)	____ __		_____
( )	1.3.6. Descend at 1000 fpm. Record the following and verify that the RDU readings are within ±5% of the flight instruments:					
	HELICOPTER INDICATION	**INSTRU-MENTS**	**RDU READING**			Elapsed Time
( )	Rate of Descent	_____ ___	ALT_RATE (±250 FPM)	____ __		_____
( )	Airspeed	_____ ___	PRES_ALT (±10 Knots)	____ __		_____
( )	1.3.7. Initiate a steady state turn at 30° of bank, constant speed and altitude, for a full 360°. Repeat the turn in the other direction. Record the following and verify that the RDU readings are within the status limits of the flight instruments:					
	HELICOPTER INDICATION	**INSTRU-MENTS**	**RDU READING**			Elapsed Time
( )	Roll Angle (cw)	_____ ___	ROLL_ATT (±2°)	____ __		_____
( )	Roll Angle (ccw)	_____ ___	ROLL_ATT (±2°)	____ __		_____
( )	1.3.8. Nose up at a pitch angle of 20°. Record the following and verify that the RDU readings are within the status limits of the flight instruments:					
	HELICOPTER INDICATION	**INSTRU-MENTS**	**RDU READING**			Elapsed Time
( )	Pitch Angle	_____ ___	PITCH_ATT (±2°)	____ __		_____
( )						
( )	1.3.9. Nose down at a pitch angle of 10°. Record the following and verify that the RDU readings are within the status limits of the flight instruments:					
	HELICOPTER INDICATION	**INSTRU-MENTS**	**RDU READING**			Elapsed Time
( )	Pitch Angle	_____ ___	PITCH_ATT (±2°)	____ __		_____
( )	1.3.10. Return to the airfield land and note the following:					
	HELICOPTER INDICATION	**INSTRU-MENTS**	**RDU READING**			Elapsed Time
( )	Approach Heading	_____ ___	MAG_HEAD-ING (±5%)	____ __		_____
( )	1.3.11. Shut down engine and apply external power to helicopter.					
( )	1.3.12. Install RDU to FDR and retrieve the flight data into a file on the hard disk of computer.					

FLIGHT DATA RECORDER (KIT 412-706-084-101/105)

FLIGHT TEST PROCEDURES - FLIGHT TESTS

MODEL: 412 **HELICOPTER NO.: _____**

PILOT/EXAMINER: _____ **DATE: _____**

( )	1.3.13. Name the file as follows: ft360XXYY/MM/DD	
( )	1.3.14. Enter description as follows:	
( )	Flight Test: S/N 360XXYY/MM/DD Pilot:_____ Engr:_____	
( )	1.3.15. Disconnect the RDU from the FDR. Remove the test cable and re-connect the connector on the front face of the FDR.	
( )	1.3.16. Review the file and check that parameters are within the stated limits.	
	1.3.17. Post Flight Test Analysis **NOTE** The portable computer will automatically save the aircraft file “ft360XXYY/MM/DD” under drive C:\IOGP.	
( )	1.3.17.1. Review the data collected and verify that all parameters are within the stated limits.	
( )	1.3.17.2. Make two copies of the flight test data from the internal memory of the RDU to floppy discs. One is for the end customer and one is for quality assurance. **NOTE** ASCB file is normally found in C:\IGS\ASCDBS.	

Flight Data Recorder System - General Description

1. Purpose

This manual has been prepared for use as an operation/maintenance manual to provide source data for helicopter maintenance manuals. It describes the operation and level 1 maintenance activities in support of the 3257D1 Flight Data Recorder.

2. Content

This manual contains all the normal level 1 operation/maintenance data in support of the 3257D1 Flight Data Recorder. In performing these activities, a Recorder Download Unit (RDU) is used to provide operation and maintenance data upload/download to/from the recorder. This is a laptop computer with a serial cable that connects to the recorder. This manual also documents the function and operation of the RDU.

3. References

Refer to the following publications for detailed description of the Recorder Display Unit (RDU) software applications:

- RM 1000 Reference Manual for Windows Data Review Tool.
- RM 1005 Reference Manual for PC Common DECOMP.
- RM 1020 Operational Ground Program for the Integrated Data Acquisition and Recording System Reference Manual.

Recorder level two maintenance is documented in:

- Flight Data Recorder Component Maintenance Manual (Smiths Aerospace Manual TDM 2120 (31-30-03)).

4. Abbreviations

The following is a list of abbreviations used throughout this manual and those generated during Recorder Download Unit (RDU) operation:

ABBREVIATION	MEANING
ARINC	Aeronautical Radio, Inc.
ASC	Aircraft Specific Configuration
ASCDB	Aircraft Specific Configuration Database
ATC	Air Traffic Control
ATP	Acceptance Test Program
BIT	Built-In Test
BitL	BIT Log
CP	Control Program
CPM	Crash Protected Memory
CPU	Central Processing Unit
DB	Decibel
DECOMP	Decompression
DLYD	Delayed
EEPROM	Electrically Erasable Programmable Read Only Memory
FDAU	Flight Data Acquisition Unit
FDR	Flight Data Recorder
FPGA	Field Programmable Gate Array
FRATS	Flight Recorder Analysis Test Station
GMT	Greenwich Mean Time
GND	Ground
GRE	Ground Readout Equipment
HZ	Hertz

ABBREVIATION	MEANING
IAW	In Accordance With
I/O	Input/Output
IOGP	IDARS Operational Ground Program
LBS	Pounds
LP	Loader Program
LRU	Line Replaceable Unit
MON	Monitor
MRT	Multi-channel Receiver/Transmitter
mS	Millisecond
mW	Milliwatt
NVM	Non-Volatile Memory
PUB	Power-Up BIT
OGP	Operational Ground Program
OMS	On-board Maintenance System
OUT	Output
PC	Personal Computer
PCMCIA	Personal Computer Memory Card International Association
PWR	Power
RDF	Recorder Data Format
RDU	Recorder Download Unit
RES	Reserved
RIPS	Recorder Independent Power Supply
RSVD	Reserved
RTN	Return
RX	Receive
TDM	Technical Data Manual
TTL	Transistor Logic
TX	Transmit
VAC	Volts Alternating Current
VDC	Volts Direct Current
VPU	Voice Processing Unit
VRMS	Volts Root Mean Square
WinDRT™	Windows Data Review Tool

Flight Data Recorder System - Description

1. Identification

The model 3257D1 is a 1/2 ATR Long Flight Data Recorder (FDR) that includes built-in crash protected data storage and an acoustic locating beacon.

MODEL	FUNCTION
3257D1	1/2 ATR Long Flight Data Recorder (FDR)

2. Description

The Flight Data Recorder (3257D1) (Figure 1) is a solid-state data recorder with self-contained armored memory. The recorder receives flight data through an external terminal (ARINC 717 and/or RS-422) and records the data in crash protected memory. The recorder complies with form factor, mounting, connector, and electrical characteristics of ARINC 747 and will function properly if interchanged with an ARINC 573 FDR. At least the last 25 hours of flight data is retained, even at recording rates up to 512 words per second.

Leading particulars for the recorder are provided in Table 1.

Refer to Figure 2 for a block diagram of the system interconnection with optional system component connections identified.

The recorder incorporates a maintenance connector (J8) on the front panel that is used for testing the unit, on-aircraft maintenance, or downloading the recorded data. The front connector includes power, control discretes, and high speed RS-422 and ethernet channels. The recorder can be powered from the front connector using an optional battery pack to permit downloading the data without applying helicopter power. The RS-422 channel is also available at the rear connector for wiring to an optional remote download connector on the helicopter. The ARINC 717 playback signal is available on the front connector to assist in troubleshooting.

The recorder system includes ground equipment for on-aircraft maintenance, return to service testing in the shop, and flight data playback/analysis. The ground equipment consists of the flight line computer (Recorder Download Unit (RDU)), the return to service test set (Flight Recorder Analysis Station (FRATS)), and the associated software.

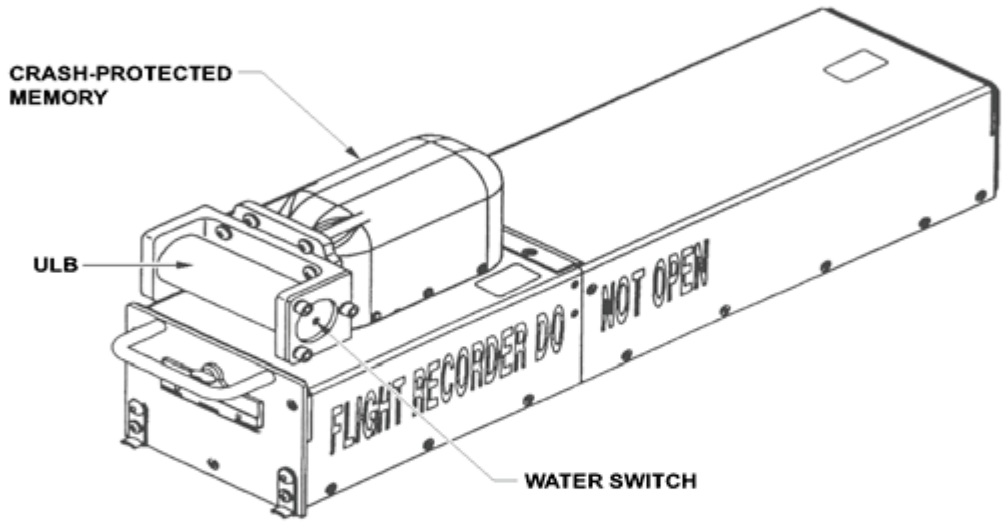
Typical interconnection of a 3257D1 based system is illustrated in Figure 3.

Table 1. Leading Particulars

PHYSICAL	DESCRIPTION
SIZE	6.94 in. High x 4.88 in. Wide x 19.25 in. Deep (17.62 cm High X 12.39 cm Wide X 48.89 cm Deep) NOTE: Depth excludes handle and hooks.
WEIGHT	10 pounds (4.54 Kg) (MAX)
MOUNTING	M85731/1-11C Hold-down Hooks (2) ARINC 404A 1/2 ATR Long
COOLING	Free Convection and Radiation
CONNECTORS: • J1 • MAINTENANCE	 CANNON DPX2MA-57P00-34-2201 (Mates with DPX2MA-57S00-33-XX01 or equivalent) DB-25
OPERATIONAL	
INPUT POWER • POWER	115 VAC/400HZ/1-PHASE or 28 VDC < 12 WATTS
INTERFACES • DISCRETES	

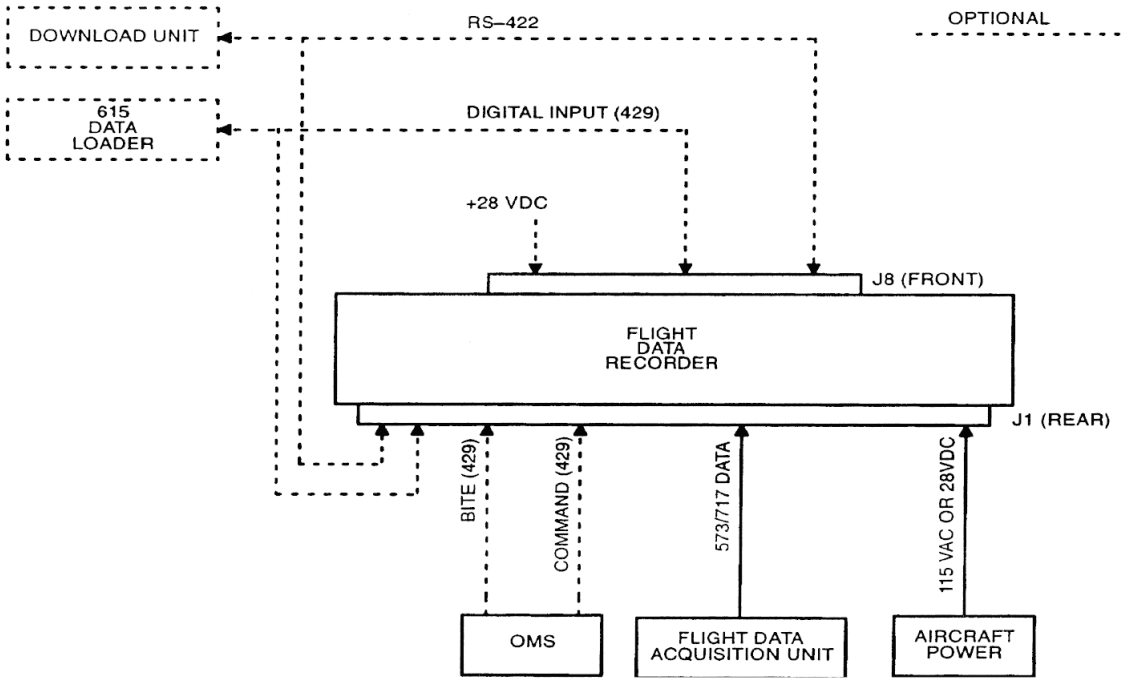
Table 1. Leading Particulars (continued)

PHYSICAL	DESCRIPTION
- INPUTS	4 TTL (OPEN/GROUND TYPE)
- OUTPUTS	1 OPEN/GND
	1 INDICATOR_PWR/GND
COMMUNICATIONS	
- RS-422	1 Asynchronous to 1.5M BAUD
	1 Asynchronous or Isochronous to 1.5M
- ARINC 717	1 HARVARD Bi-phase Interface
- ARINC 429	2 INPUTS, 1 OUTPUT
- ETHERNET	1 CHANNEL, 10 BaseT



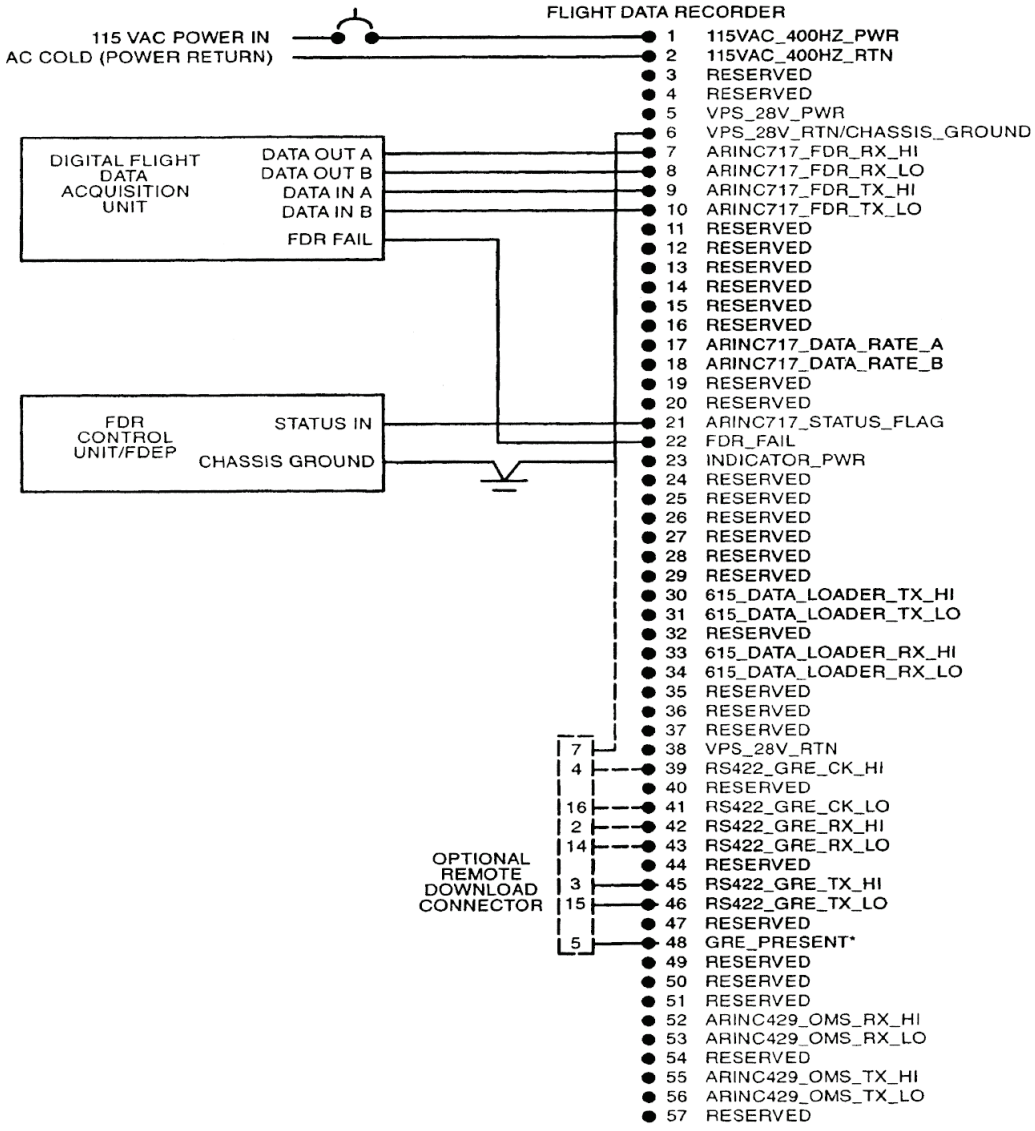
ICN-412-A-970003-B-97499-00013-A-001-01

Figure 1. Recorder System Identification View (Sheet 1 of 1)



ICN-412-A-970003-B-97499-00014-A-001-01

Figure 2. Recorder System Block Diagram (Sheet 1 of 1)



ICN-412-A-970003-B-97499-00015-A-001-01

Figure 3. 3257D1 System Interconnect Diagram (Sheet 1 of 1)

Intentionally blank page

Flight Data Recorder System - Operation

Procedure

1. Operation

This section provides a description of FDR status annunciation and procedures for normal operation, data download, and maintenance activities. These include:

- Record Flight Data in Crash Protected Memory,
- Connecting Recorder Download Unit (RDU) to the Recorder,
- Flight Data Download,
- Flight Data Conversion,
- Displaying Flight Data,
- Aircraft Documentary Data Upload/Display,
- Memory Format/Initialize,
- Uploading Recorder Software,
- Verify Recorder Software.

The following paragraphs provide a description of each recorder control input or discrete output with the recorder installed on the helicopter as a complete FDR system.

1.1. FDR Status Indication

FDR FAIL (FDR_FAIL recorder output discrete) is turned on at power-up. It is cleared when all Power-Up BIT (PUB) tests have passed. Then, if no failures are detected, a series of conditions are checked to see if a blink condition applies. There are three blink rates used by the recorder system, long blink, 4 Hz, and 2 Hz. The system conditions indicated by FDR FAIL are as identified in Table 1.

1.2. Operating/Maintenance Procedures

1.2.1. Record Flight Data in Crash Protected Memory

Under normal circumstances, power is applied to the recorder when the helicopter is powered and recording takes place automatically.

1.2.2. Connecting RDU to Recorder

In order to perform certain maintenance tasks, such as downloading recorded data and displaying BIT information, the RDU must be connected to the recorder. Use the following procedure to connect the RDU to the recorder front panel maintenance connector (J8) or remote GRE connector:

NOTE

RDU may be connected to the recorder with the RDU and/or recorder system power already applied. Therefore, order of the following steps is not important.

1.2.2.1. Connect the RDU interface cable to the recorder front panel maintenance connector (located behind access door).

1.2.2.2. Close circuit breakers supplying power to the recorder/helicopter if not already done.

1.2.2.3. Power-up the RDU.

1.2.3. Flight Data Download

1.2.3.1. Connect RDU to the recorder front panel maintenance connector (J8) or remote GRE connector (Step 1.2.2.).

1.2.3.2. Double-click IOGP icon or select IOGP from Start menu.

1.2.3.3. After IOGP has started, select the following from the top menu of the IOGP: Download!

1.2.3.4. Modify/Enter Aircraft and Recorder Serial Number data as necessary in the following substeps. If data is already correct, select “Continue Download” and proceed to Step 1.2.3.5..

NOTE

This operation only affects the downloaded file data and does not modify the documentary data stored in recorder memory. Refer to the Aircraft Documentary Data Upload/Display procedure to change the recorder's stored documentary data.

1.2.3.4.1. Select “Modify Enterable Data”.

1.2.3.4.2. Enter Aircraft Serial Number and/or Recorder Serial Number.

1.2.3.4.3. Select “Download Data”.

1.2.3.4.4. Select “Continue Download”.

1.2.3.5. From the “Download Recorded Data” window, select (by clicking on) each file type to be downloaded. A check mark will appear next to each file type selected. The recorder file types may include:

A717	ARINC 717 Flight data
R422	RS422 flight data
BitL	Built-In Test (BIT) Log

NOTE

If desired, individual sessions may be downloaded by selecting Download By: Session. This will launch a “sessions(s) to download” window that enables individual session selection. Building the session menu may take several minutes.

1.2.3.6. Click “Download”.

1.2.3.7. From the “Download File” window, select desired drive, directory, and enter a valid filename. The file extension must remain .rdf.

1.2.3.8. Click “Save” and wait for the download operation to complete.

1.2.4. Flight Data Conversion

The following procedure provides data conversion of the .rdf file created in the Flight Data Download procedure. This enables viewing of the flight data using the Windows Data Review Tool (WinDRT™) , Smiths Aerospace part number 549651). The data is converted using the Smiths Aerospace PC DECOMP program (549342). Detailed operation of the DECOMP tool is provided in the PC Common DECOMP Reference Manual (RM1005).

1.2.4.1. Double-click PC DECOMP icon or select PC DECOMP from START menu.

1.2.4.2. Select ADD (PC Decomp toolbar). The “Select download files to process” window will appear.

1.2.4.3. Select the directory/filename of the file to be converted (this will be the path/filename used during data download).

1.2.4.4. Proceed to Step 1.2.4.5. if the ASCDB column identifies the appropriate application specific configuration database file. Otherwise, click on CHG ASC in PC Decomp toolbar. Select path/filename of the appropriate ASCDB file.

NOTE

c:\PC Decomp\ASCDBs is a folder created during PC Decomp application installation. This folder is a convenient location for ASCDB storage.

1.2.4.5. Click on GO from the PC Decomp toolbar. Conversion will be initiated and the Status window will indicate “Complete” when the converted data file (.ddf) has been created.

NOTE

Data types may include flight data, maintenance data, etc. Errors encountered during conversion are logged in the “<filename>.LOG” file. PC Decomp File Preferences allows the user to provide the log file name or select automatic filename generation using the current date. Refer to RM1005 for any errors encountered during file conversion.

1.2.5. Displaying Flight Data

The Windows Data Review Tool (WinDRT™) provides display of converted data files (DDF). Specific data may be selected by the user and displayed in tabular columns of data or as graphical plots. Detailed operation of the WinDRT™ tool is provided in Windows Data Review Tool (WinDRT™) Reference Manual (RM1000).

1.2.5.1. Double-click WinDRT™ icon or select WinDRT™ from the START menu.

1.2.5.2. Select Open DDF from the File Menu (or use Open button on toolbar).

1.2.5.3. Enter the path/filename of the data to be viewed (.ddf file generated by PC Decomp).

1.2.5.4. Click OK to open the selected file.

Once the DDF has been opened, the first block of data contained in the DDF will be displayed. Additional toolbars will become available once a file has been opened. The name of the DDF will be added to the File menu's list of most recently opened files.

1.2.6. Aircraft Documentary Data Upload/Display

Helicopter information (Aircraft Serial Number, Recorder Serial Number, etc.) may be viewed and/or updated using the following procedure:

NOTE

The following procedure provides modification/entry of enterable parameters. Maintenance - > Identify provides parameter viewing without the data modification option.

1.2.6.1. Connect RDU to the recorder front panel maintenance connector (J8) or remote GRE connector (Step 1.2.2.).

1.2.6.2. Double-click the IOGP icon or select IOGP from the Start menu.

1.2.6.3. Select Maintenance -> Aircraft Information from the top menu.

1.2.6.4. Click on “Modify Enterable Data”.

1.2.6.5. Enter appropriate information and click “Upload Data”.

1.2.7. Memory Format/Initialize

Formatting the memories is necessary only when the recorder is moved from one aircraft to another or the operator wishes to clear the flight data from the CPM. Various header information is written into the flash memory, partitions are created and data zeroed.

Three memory formatting operations are available in the IOGP Maintenance - Format Memories menu:

- Format Crash Protected Memory,
- Clear BIT Information,
- Installation Format.

The Format Crash Protected Memory and Installation Format commands erase flight data from the CPM. The same erase action is initiated in either command and takes several seconds for completion. The Installation Format command also includes an erase verification process and therefore requires the additional time to read all erased memory. Installation Format can be selected when security directives require verification of the erase process.

The Clear BIT Information command clears the Built-In Test history and log files.

FORMATTING THE MEMORIES CAUSES ANY DATA PREVIOUSLY RECORDED IN CRASH-PROTECTED MEMORY TO BE DISCARDED. DOWNLOAD THE DATA BEFORE FORMATTING THE MEMORY.

NOTE

Format Crash Protected Memory and Clear BIT Information are automatically performed during a control program upload. It is not necessary to perform these commands as an added step to the upload.

1.2.7.1. Connect RDU to the recorder (Step 1.2.2.).

1.2.7.2. Double-click IOGP icon or select IOGP from Start menu.

1.2.7.3. Select the following from the top menu of the IOGP: Maintenance -> Format Memories.

1.2.7.4. In the “Format/clear memory” window, click on the desired format action(s). A check mark will appear beside each action selected. The options are:

- Format Crash Protected Memory - Erases and formats crash protected memory (less than 10 seconds).
- Clear BIT Information - Clears the fault history and BIT log.

- Installation Format - Erases and formats crash protected memory. Includes erase process verification (reads entire memory content).

1.2.7.5. Click “Format/Clear”.

UPLOADING NEW SOFTWARE CAUSES ANY DATA PREVIOUSLY RECORDED IN CRASH-PROTECTED MEMORY TO BE DISCARDED. REGULATIONS MAY REQUIRE DOWNLOAD OF THE FLIGHT DATA BEFORE BEGINNING THE UPLOAD PROCEDURE.

1.2.8. Uploading Recorder Software

All appropriate software is loaded into the recorder prior to initial shipment and following any warranty repair actions. If a field update is provided, the new software may be loaded as follows:

1.2.8.1. Ensure appropriate executable software files (.sr extensions) are stored on one of the RDU drives.

1.2.8.2. Connect RDU to recorder (Step 1.2.2.).

1.2.8.3. Double-click IOGP icon or select IOGP from the START menu.

1.2.8.4. Select Upload -> Load from the IOGP menu.

1.2.8.5. On the “Warning” window, click “Upload”.

1.2.8.6. Select the directory and filename of the software file. Click “Select”.

NOTE

Ensure that power is not interrupted to the recorder during the upload process. If this happens, the upload operation must be restarted from the beginning.

1.2.8.7. Optional: Upload helicopter information by performing the “Upload/Display Aircraft Information” procedure (Step 1.2.6.).

NOTE

If new definition files (.def) are provided with a software update, they must be copied into the IOGP folder of the RDU. In addition, ensure any new application specific configuration database (.asc) files are copied into the ASCDBs folder of the RDU.

1.2.9. Verify Recorder Software

The following procedure verifies the recorder control program against a copy stored in the RDU.

1.2.9.1. Ensure executable software file (.sr extension) is stored on a RDU drive.

1.2.9.2. Connect RDU to the recorder (Step 1.2.2.).

1.2.9.3. Double-click IOGP icon or select IOGP from the START menu.

1.2.9.4. Select Upload -> Verify from the IOGP menu.

1.2.9.5. Select directory/filename of the file. Click “Select”. The verify process proceeds automatically. Errors are annunciated by a “verify error” statement.

2. Theory of Operation

The recorder contains the internal circuitry necessary to provide a flight data recorder as defined by ARINC 757, refer to Figure 1. It receives flight data via helicopter data busses and stores this data in solid state memory devices contained within the armored housing. The FDR includes processors, memory, and power supplies, plus interfacing and control circuitry and Input/Output (I/O) ports for information transfer/retrieval to/from other systems and equipment.

Refer to Table 2 for pin definition of connector (J1).

The recorder consists of a Voice Processing Unit (VPU) CCA, Crash-Protected Memory (CPM) , and an acoustic locating beacon.

The VPU CCA contains all the circuitry necessary to interface with:

- ARINC 717 Harvard Bi-phase serial bus for receiving/recording flight data,
- ARINC 429 bus for providing BIT Status and responding to OMS commands,
- RS-422 bus for receiving/recording flight data,
- Ethernet bus for the purpose of downloading recorded information,
- 615 Data Loader.

Refer to Table 3 for signal definition of each connector pin of the recorder maintenance connector (J8) on the front panel of the LRU.

2.1. Voice Processing Unit and Power Supply CCA (A1)

2.1.1. Power

The recorder operates on 115 VAC 400 Hz single phase primary or 28 VDC alternative power. The recorder power supply is located on the A1 circuit card assembly and provides ±5, ±12, and +3.3 VDC outputs used by the recorder.

2.1.2. Discretes

2.1.2.1. Input Discretes

The recorder provides four Transistor Logic (TTL) input discretes at the Line Replaceable Unit (LRU) connector. These are Open/Ground discretes with 1K ohms to Vcc with 1 mA sink current. They are intended for recorder configuration/control and not for switching by external equipment. This input type will accept sustained voltage to 5.5 volts (with respect to helicopter chassis ground) without damage.

2.1.2.1.1. ARINC 717 Data Rate Configuration Inputs (TTL Open/Ground)

Two Aircraft Configuration input discretes are provided to select the ARINC 717 data rate, as described in Step 2.1.4..

DATA_RATE_A	DATA_RATE_B	RATE (WORDS/SEC)
OPEN	OPEN	64
OPEN	GND	128
GND	OPEN	256
GND	GND	512

2.1.2.2. GRE Present (TTL Open/Ground)

GRE_PRESENT* input discrete serves to halt all recording and place the recorder in a mode ready for downloading or other ground maintenance.

2.1.2.3. Loader Command (TTL Open/Ground) (Front connector only)

The LOADER_CMD* discrete is used during Acceptance Test Procedure (ATP) and unit maintenance to enable modes not needed in operational use. It should not be wired in the helicopter.

2.1.2.4. Output Discretes

The VPU provides two discrete output interfaces at the LRU connector:

FDR_FAIL and ARINC717_FDR_STATUS_FLAG. The FDR_FAIL discrete is Open/Ground type where a logic 1 is an open and a logic 0 is <1.3 V when sinking a maximum of 100mA.

The ARINC717_FDR_STATUS_FLAG is connected to a ground or the voltage present on the INDICATOR_PWR input.

2.1.2.4.1. FDR_FAIL

The FDR_FAIL output discrete is asserted (opened) when power is off, recorder reset, or during the initiated BIT. It is cleared when all tests have passed. Then, if no failures are detected, a series of conditions are checked to see if the blink conditions apply. There are three blink rates used by the recorder. They are long blink, 4 Hz, and 2 Hz. The long blink is defined as on for 1.8 seconds and off for 0.2 second. The blink rates identify the following system states:

FDR FAIL	SYSTEM CONDITION
OFF	Normal Mode, FDR recording
ON	During Power-Up Built-In Test (BIT) , or recorder failure

FDR FAIL	SYSTEM CONDITION
Fast Blink (4 Hz)	Maintenance Mode (not recording)
Slow Blink (2 Hz)	Busy erasing/formatting or CPM write protect is activated
Long Blink (on 1.8, off 0.2 sec)	Loader Mode Active

The logic sense of FDR_FAIL is defined such that a fail indication is given by an open. In this way, removing the recorder is sensed by other systems as a fail indication.

2.1.2.4.2. Status Flag

The ARINC717_FDR_STATUS_FLAG output discrete electrically implements the ARINC 747 Status Flag interface to the control panel. The output is used when the recorder is used with an ARINC 717 compliant FDR control panel. The discrete is grounded if any of the following conditions occur:

- ARINC 717 data stream not present.
- ARINC 717 bit rate not specified by the configuration discretes.
- ARINC 717 sync words not in proper sequence and spacing.
- Recorder not in recording mode.
- Hardware failures preventing recording.

The status flag is implemented with a relay that connects ARINC717_FDR_STATUS_FLAG to ground or the voltage supplied on the INDICATOR_PWR input.

The voltage sent to Status Flag is provided by the Indicator Power input, and is used to indicate the operational state of the FDR system. Status Flag defaults to the Ground state on power-up, indicating a system failure has been detected. Status Flag indicates the health of the FDR System, rather than just the FDR LRU. This includes monitoring the presence, bit rate, and validity of the data stream received from the FDAU.

2.1.3. RS-422 Serial Interfaces

The recorder provides two independent external RS-422 serial interfaces.

2.1.3.1. Ground Readout Equipment (GRE) RS-422 Serial Interface

The recorder provides an electrically isolated RS-422 serial interface for the purpose of communications between the recorder and the Ground Readout Equipment (RDU or FRATS). The serial interface has separate differential transmit and receive channels and meets all RS-422A electrical requirements. This serial interface can be operated in either an isochronous mode or an asynchronous mode. Differential clock lines are provided for operating in the isochronous mode. Baud rates up to 1.5M baud are supported by this serial interface.

This RS-422 serial interface is for reading out previously recorded data from the CPM, initiating and interrogating self test, and loading new operational software.

Some applications may use the GRE RS-422 interface as an additional flight data recording input. In these applications, GRE cannot be connected without disconnecting the recorder from the flight data input.

2.1.3.2. Debug RS-422 Serial Interface

The recorder provides an electrically isolated RS-422 serial interface for the purpose of Debug communication. The serial interface has separate differential transmit and receive channels and meets all RS-422A electrical requirements. This serial interface can be operated in the asynchronous mode only. Baud rates up to 1.5M baud are supported by this serial interface.

2.1.4. ARINC 717

The recorder provides one serial data interface in accordance with (IAW) ARINC 717, (Characteristic for Digital FDRs). The incoming data stream on the ARINC 717 bus is sampled and converted to a digital bit stream internal to the recorder for the purpose of storing the data in the CPM.

The default data rate is 64 (12-bit) words per second, and the maximum data rate is 512 (12-bit) words per second (6,144 bits per second). The data rate is configured by the DATA_RATE input discretes.

The ARINC717_FDR_RX (FDR DATA IN) input channel is wrapped to the ARINC717_FDR_TX (FDR DATA OUT) output channel unless any of the following conditions occur:

- Recorder hardware failure that prevents recording.
- Recorder not in recording mode.

- ARINC 717 input data stream not present.

2.1.5. ARINC 429

The recorder provides two ARINC 429 channels. They are dedicated to On-board Maintenance System (OMS) and 615 data loader interfaces.

2.1.5.1. On-board Maintenance System (OMS) Channel

The ARINC429_OMS_RX (OMS IN) input and ARINC429_OMS_TX (OMS OUT) outputs of the recorder provide BIT Status (once per second) and respond to commands from the OMS.

2.1.5.2. 615 Data Loader Channel

The 615_DATA_LOADER_RX and TX channels support ARINC 615 data loaders. The 615 transmit channel is shared with the OMS transmit channel internal to the recorder.

Table 1. FDR FAIL Summary

FDR FAIL	SYSTEM CONDITION
OFF	Normal Mode, FDR recording
ON	FDR recording inhibited, during Power-Up or Initiated Built-In Test, or FDR BIT failure
Fast Blink (4 Hz rate)	Maintenance Mode (not recording)
Slow Blink (2 Hz rate)	Busy erasing/formatting or CPM write protect is activated
Long Blink (on for 1.8 sec, off for 0.2 sec)	Loader Mode Active

Table 2. 3257D1 Connector (J1) Pin Definition

PIN	I/O	SIGNAL NAME	NOTES
1	IN	115VAC_400HZ_PWR	AC power input, high, 400Hz, 115VAC
2	IN	115VAC_400HZ_RTN	AC power input, low, 400Hz, 115VAC
3	-	RESERVED	
4	-	RESERVED	
5	IN	VPS_28V_PWR	DC power input, high, 28VDC
6	IN	VPS_28V_RTN/CHASSIS_GROUND	DC power input, low, 28VDC
7	IN	ARINC717_FDR_RX_HI	ARINC-717 serial channel input, high
8	IN	ARINC717_FDR_RX_LO	ARINC-717 serial channel input, low
9	OUT	ARINC717_FDR_TX_HI	(FDR DATA OUT A) ARINC717 serial channel out
10	OUT	ARINC717_FDR_TX_LO	(FDR DATA OUT B) ARINC717 serial channel out
11	-	RESERVED	
12	-	RESERVED	
13	-	RESERVED	
14	-	RESERVED	
15	-	RESERVED	
16	-	RESERVED	
17	IN	ARINC717_DATA_RATE_A	(DATA RATE A) Open/Gnd input discrete
18	IN	ARINC717_DATA_RATE_B	(DATA RATE B) Open/Gnd input discrete
19	-	RESERVED	
20	-	RESERVED	
21	OUT	ARINC717_FDR_STATUS_FLAG	INDICATOR_PWR/Gnd output, Gnd = Fail
22	OUT	FDR_FAIL	Open/Gnd - Open = fail

Table 2. 3257D1 Connector (J1) Pin Definition (continued)

PIN	I/O	SIGNAL NAME	NOTES
23	IN	INDICATOR_PWR	This voltage is output on ARINC717_FDR_STATUS_FLAG when FDR OK
24	-	RESERVED	
25	-	RESERVED	
26	-	RESERVED	
27	-	RESERVED	
28	-	RESERVED	
29	-	RESERVED	
30	IN	615_DATA_LOADER_TX_HI	ARINC 429 serial channel input, high
31	IN	615_DATA_LOADER_TX_LO	ARINC 429 serial channel input, low
32	-	RESERVED	
33	IN	615_DATA_LOADER_RX_HI	ARINC 429 serial channel input, high
34	IN	615_DATA_LOADER_RX_LO	ARINC 429 serial channel input, low
35	-	RESERVED	
36	-	RESERVED	
37	-	RESERVED	
38	IN	VPS_28V_RTN	28VDC power input (return)
39	IN	RS422_GRE_CK_HI	Ground Readout Equipment Serial Clock
40	-	RESERVED	
41	IN	RS422_GRE_CK_LO	Ground Readout Equipment Clock (LO)
42	IN	RS422_GRE_RX_HI	Ground Readout Eq. Serial Channel Input, High
43	IN	RS422_GRE_RX_LO	Ground Readout Eq. Serial Channel Input, Low
44	-	RESERVED	
45	OUT	RS422_GRE_TX_HI	Ground Readout Eq. Serial Channel Output, High
46	OUT	RS422_GRE_TX_LO	Ground Readout Eq. Serial Channel Output, Low
47	-	RESERVED	
48	IN	GRE_PRESENT*	Open/Gnd - Ground indicates GRE attached
49	-	RESERVED	
50	-	RESERVED	
51	-	RESERVED	
52	IN	ARINC429_OMS_RX_HI	ARINC-429 Serial Channel Input, High
53	IN	ARINC429_OMS_RX_LO	ARINC-429 Serial Channel Input, Low
54	-	RESERVED	
55	OUT	ARINC429_OMS_TX_HI	ARINC-429 Serial Channel Output, High
56	OUT	ARINC429_OMS_TX_LO	ARINC-429 Serial Channel Output, Low
57	-	RESERVED	

Table 3. Recorder Maintenance Connector (J8) Pin Definition

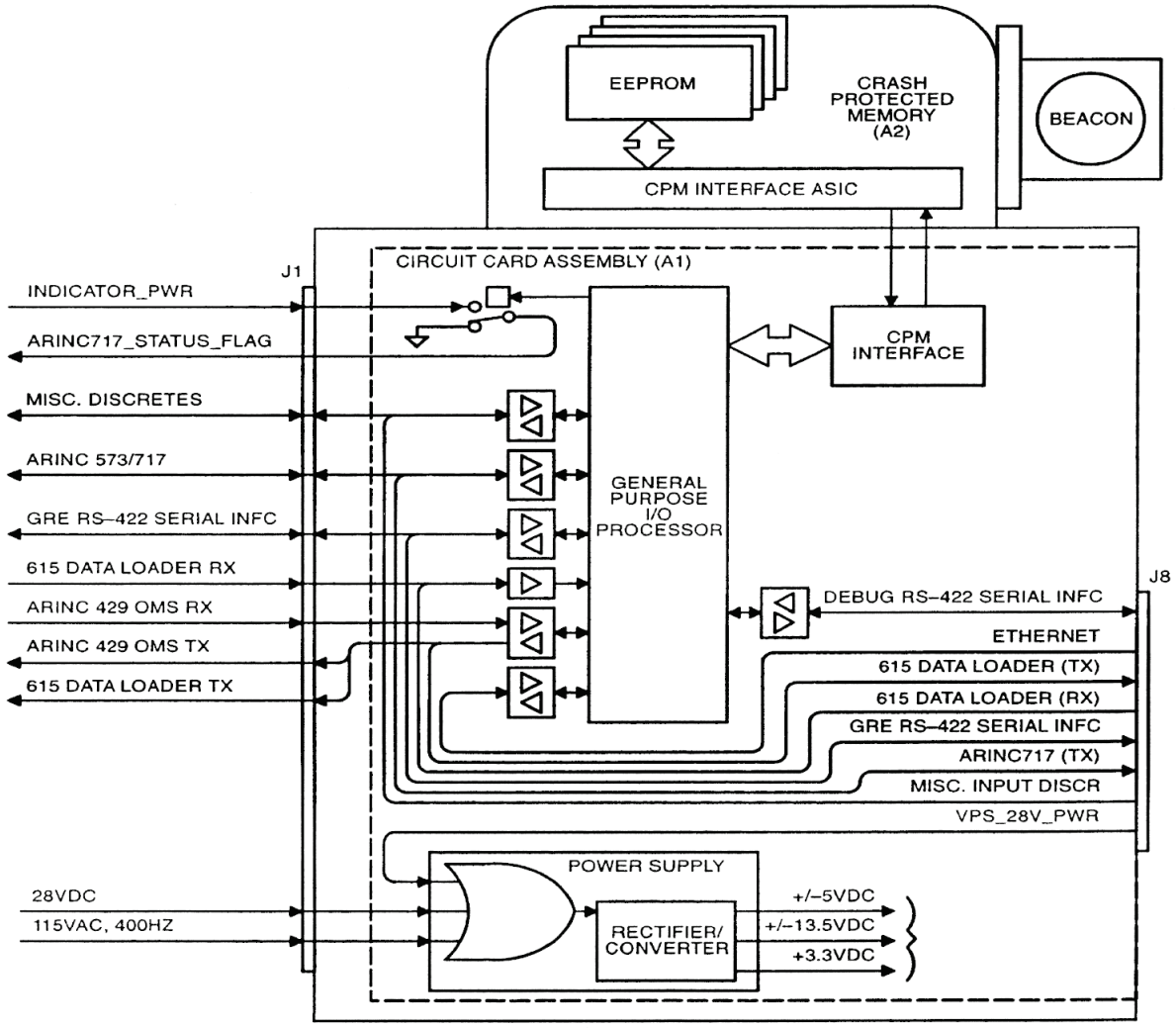
PIN	I/O	SIGNAL NAME	NOTES
1	IN	LAB_RST*	Reserved
2	IN	RS422_GRE_RX_HI	Ground Readout Equipment RS422, receive
3	OUT	RS422_GRE_TX_HI	Ground Readout Equipment RS422, transmit

Table 3. Recorder Maintenance Connector (J8) Pin Definition (continued)

PIN	I/O	SIGNAL NAME	NOTES
4	IN	RS422_GRE_CLK_HI	Ground Readout Equipment RS422, clock
5	IN	GRE_PRESENT*	Gnd - Ground Readout Equipment present
6	OUT	ARINC717_FDR_TX_HI	FOR DATA OUT A
7	IN	VPS_28V_RTN	Battery Operation Input (Diode isolated)
8	IN	RS422_DEBUG_RX_HI	Asynchronous, ≤ 1.5M
9	OUT	RS422_DEBUG_TX_HI	Asynchronous, ≤ 1.5M
10	IN	615_DATA_LOADER_RX_H1	ARINC 615 Data Loader
11	OUT	615_DATA_LOADER_TX_HI	ARINC 615 Data Loader
12	IN	ENET_TPRX_HI	Ethernet, Receive, High
13	OUT	ENET_TPTX_HI	Ethernet, Transmit, High
14	IN	RS422_GRE_RX_LO	Ground Readout Equipment RS422, receive
15	OUT	RS422_GRE_TX_LO	Ground Readout Equipment RS422, transmit
16	IN	RS422_GRE_CLK_LO	Ground Readout Equipment RS422, clock
17	IN	LOADER_CMD*	Ground forces unit to stay in Loader Program
18	OUT	ARINC717_FDR_TX_LO	FDR DATA OUT B
19	IN	VPS_28V_PWR	Battery Operation Input (Diode isolated)
20	IN	RS422_DEBUG_RX_LO	Asynchronous, ≤ 1.5M
21	OUT	RS422_DEBUG_TX_LO	Asynchronous, ≤ 1.5M
22	IN	615_DATA_LOADER_RX_LO	ARINC 615 Data Loader
23	OUT	615_DATA_LOADER_TX_LO	ARINC 615 Data Loader
24	IN	ENET_TPRX_LO	Ethernet, Receive, Low
25	OUT	ENET_TPTX_LO	Ethernet, Transmit, Low

Table 4. Input Discrete Summary

DISCRETE	LOCATION	TYPE
ARINC717_DATA_RATE_A	Rear	TTL Open/Ground
ARINC717_DATA_RATE_B	Rear	TTL Open/Ground
GRE_PRESENT*	Front/Rear	TTL Open/Ground
LOADER_CMD*	Front	TTL Open/Ground



ICN-412-A-970003-B-97499-00016-A-001-01

Figure 1. Recorder Block Diagram (Sheet 1 of 1)

Crash Protected Memory (A2) - Description

The Crash Protected Memory (CPM) provides a 25 hour flight data storage capacity via Flash Electrically Erasable Programmable Read Only Memory (EEPROM) . The CPM is mounted on top of the recorder chassis. The CPM assembly consists of a single circuit card with an armored housing. The memory is designed to retain stored information without applied power for a minimum of 2 years.

Intentionally blank page

Acoustic Beacon - Description

The acoustic beacon is a self-contained locating device which assists in underwater recovery. It is automatically activated upon water immersion and then sends out periodic acoustic pulses for a minimum of 30 days. The beacon is mounted on top of the recorder, providing optimal acoustic energy dispersion. The beacon is contained in a cylindrical, watertight aluminum case capable of operation during deep water immersion to 20,000 feet.

The beacon is powered by a battery with a useful shelf life of 6 years. When submersed, the beacon provides an acoustic pulse of 37.5 kHz, with pulse length > 9 mS and a repetition rate not less than 0.9 pulses per second.

Intentionally blank page

Flight Data Recorder System - Troubleshooting

1. General

The following fault isolation procedures guide troubleshooting based on the results of operational tests performed in the Test section of this manual. Refer to the Test section and perform the Preflight Operational Test prior to beginning the fault isolation provided in this section.

Refer to the BIT Data Display section for troubleshooting intermittent failures that are not identified while performing the Preflight Operational Test. This allows display of the recorder BIT history to identify failures that occurred during past operation.

2. Fault Isolation

Failure isolation begins with observation of the FDR FAIL annunciator during the FDR Preflight Operational Test. Failing conditions are listed with their associated preflight test number in Table 1. Use the maintenance action column to guide the failure isolation process.

Failures identified during the Recorder Functional Test procedure are isolated in a like manner using the isolation steps identified in Table 2.

3. BIT Data Display

If a fault within the recorder system occurs, Built-In Test (BIT) results are logged into BIT files which may be reviewed to determine the nature and source of the fault indication. This can be especially useful in reviewing fault history to identify an “intermittent” failure.

Connect RDU to the recorder (412-A-97-00-03-00A-130A-A / 02712).

Double-click the IOGP icon or select IOGP from the Start menu.

Select the following from the top menu of the IOGP: Maintenance -> BIT functions -> Display results

The IOGP provides a display of the BIT data including history, results (current), and summary. Figure 1 shows a typical display (application specific) of BIT history.

Each BIT test history counter content is provided in hexadecimal (Hex Value column) and decimal (right side of window). During flight, a history counter is incremented when the specific test transitions from healthy (passing) to failing. Successive failures of the same test will not increment the history counter unless the test passes and then later returns to a failing condition.

Figure 2 illustrates Results and Summary word portions of a typical BIT display. It should be noted that this display is application specific. The example illustrated in Figure 2 indicates that OMS inputs are not supplying data.

The last section of the BIT display is BIT Summary Words. System operation and fault conditions are summarized in this section. Bit 12 of summary word 1 is reserved and does not indicate a system failure. RDU connection disables FDR recording and sets bit 13.

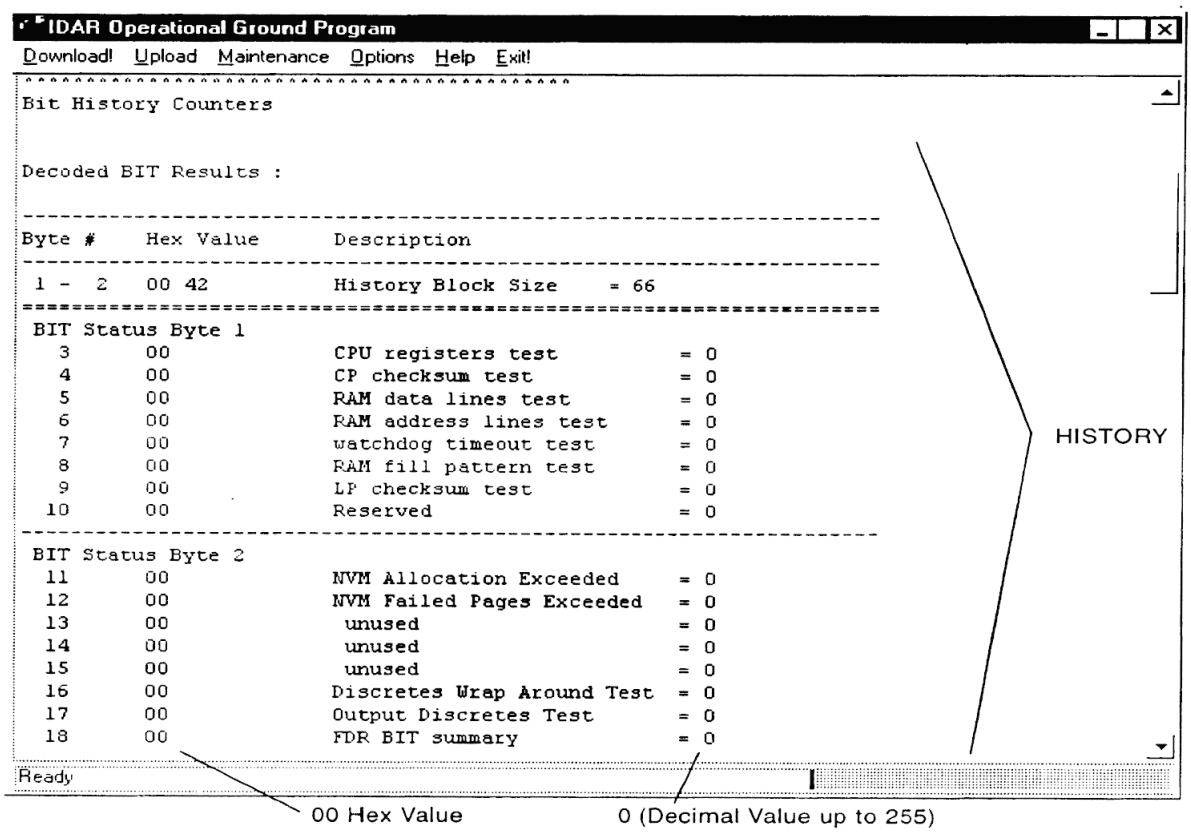
The recorder also produces a bit log file that can be downloaded in the same manner as flight data. The bit log file is updated at startup, each time a bit flag logic state toggles, or at least once per hour. The bit log file is accessible using the “Download!” function of the IOGP. The file appears as “BitL” in the Download Recorded Data window. Following file conversion using PC Decomp, the file can be reviewed in the same manner as flight data using the WinDRT™ application.

Table 1. Preflight Test Failure Isolation Chart

STEP	CONDITION	MAINTENANCE ACTION
1.	FDR FAIL output discrete blinking	a. Verify RDU is not connected to the recorder. b. Upload Control Program (If Long Blink condition as described in Description and Operation).
	FDR FAIL output discrete on > 5 sec after recorder power applied	a. Replace recorder.
2.	FDR Status Flag On	a. Verify operation of the FDAU. b. Replace recorder.

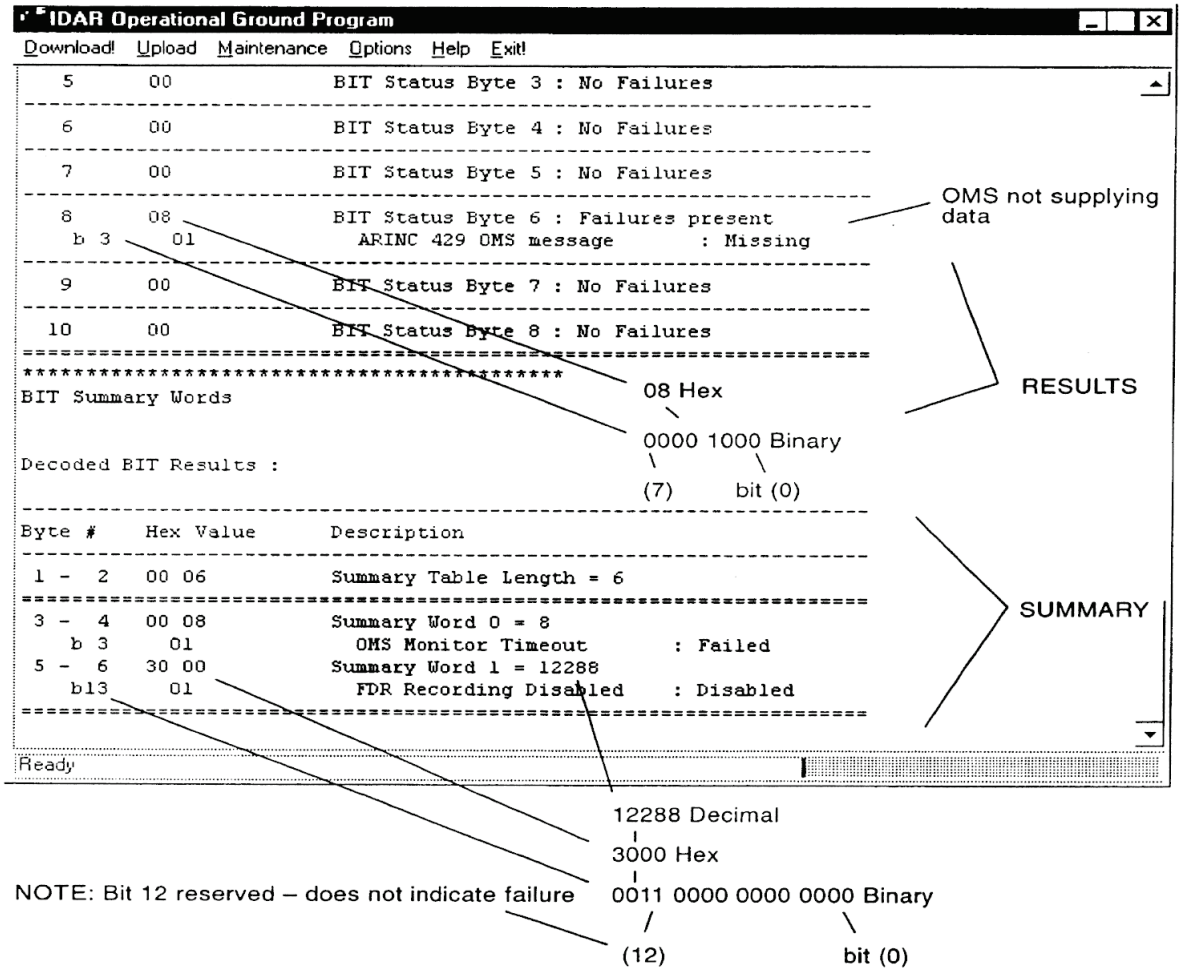
Table 2. Functional Test Failure Isolation Chart

STEP	CONDITION	MAINTENANCE ACTION
1.	FDR Preflight Test Fails	Refer to Table 1.
2.	Beacon battery date expiration, operational test indicates battery replacement, or operational check failure.	Replace beacon or recorder w/beacon.
3.	FDR download data corrupted or other data recording failures.	a. Verify FDAU operation. b. Replace recorder. c. Check helicopter wiring for the ARINC 717 data and wrapback signals.



ICN-412-A-970003-B-97499-00017-A-001-01

Figure 1. Typical BIT Results Display (History Section) (Sheet 1 of 1)



ICN-412-A-970003-B-97499-00018-A-001-01

Figure 2. Typical BIT Results Display (Results and Summary Sections) (Sheet 1 of 1)

Flight Data Recorder System - Recorder Download Unit (RDU) - Description and Operation

1. Identification

The Recorder Download Unit (RDU) (PLT6254-01-01) is a PC type laptop computer with an adapter kit and a PCMCIA memory card. The adapter kit provides the communication link between the PC and the recorder. The memory card provides the portable data transfer media when moving data between the RDU and Ground Support Equipment. Equipment consists of:

- Laptop computer (PC),
- Adapter Kit (Interface card and cable assembly),
- Memory Card (PCMCIA).

2. Description

The RDU is supplied with a laptop computer that runs an Operational Ground Program (OGP) , refer to Figure 1. The OGP communicates with the recorder through a PCMCIA interface card plugged into the laptop's PC card slot. A cable assembly interconnects the interface card to the maintenance connector on the front face of the recorder. The RDU provides the following recorder support functions:

- Flight data download, file format conversion, and display,
- Recorder BIT initiation and BIT data display,
- Recorder memory format/initialize,
- Uploading recorder software.

A utility called PC DECOMP is used to convert the raw flight data into a file format usable by data reviewing software. The data is viewed using a utility called the Windows Data Review Tool (WinDRT™) . Specific data may be selected and displayed in tabular columns of data or as graphical plots.

The PLT6254-01 (RDU) hardware is listed in Table 1.

Refer to Table 2 for a list of test station software.

3. Hardware Installation

Power-up the laptop and perform any preliminary setup prescribed by the computer manufacturer.

ENSURE INTERFACE CARD ASSEMBLY IS INSERTED WITH ORIENTATION CONSISTENT WITH MANUFACTURERS OPERATING INSTRUCTIONS. FORCING THE CARD INTO THE SLOT WITH IMPROPER ORIENTATION (UPSIDE DOWN) MAY RESULT IN DAMAGE TO THE PC AND/OR INTERFACE CARD.

Insert the PCMCIA interface card into a PC card slot of the laptop observing orientation as prescribed in the laptop operating instructions/reference manual. Mechanical resistance during card insertion (approximately 3/8" from full insertion) may indicate that the card is upside down. If the computer recognizes the card, proceed to Paragraph 3.4. If, when inserting the card, Windows presents a dialog window about new hardware installation, proceed to the next step.

Proceed through the windows dialog and insert the floppy disk from the PCMCIA kit when prompted. It may be necessary to browse the floppy drive and select the data folder to enable driver recognition. When the installation is complete, an icon will appear in the Task Bar Tray.

Remove the PCMCIA interface card from the PC card slot using the card ejection button. If an “Unexpected PC Card Removal” window opens, select the “Don't display this warning again.” option.

4. Software Installation

4.1. Operational Ground Program

NOTE

The PCMCIA interface card must be installed in the PC during this process to allow RS-422 communications device selection.

The Operational Ground Program (IOGP) is installed using the procedure in the Reference Manual for the Operational Ground Program for the Integrated Data Acquisition and Recording System (RM 1020). Follow the instructions for installation.

During IOGP installation, the “Enable Clock/Calendar Functions?” should be answered “No”.

If prompted “Warning! No RS-422 communications device has been selected as the default. Please select one from the RS-422 configuration panel”. Select “MPAP-200SI PCMCIA” for the Device and activate “Use Automatic Settings” within the Communications Options window. Typically, the IOGP can automatically determine the I/O address of the desired device. If the address determined automatically does not seem to be correct, the address can be entered manually. Refer to RM1020 for details.

Following IOGP program installation, copy the applicable Control Program (.sr) and Definition file(s) (.def) into the IOGP folder. These files are supplied with the recorder software load.

4.2. PC DECOMP

Install PC DECOMP using the procedures provided in the PC Common DECOMP Reference Manual RM1005. When prompted for installation location, enter “c:\PCDECOMP”.

Following PC DECOMP installation, copy the applicable Application Specific Configuration Data Base(s) (supplied with recorder software load) to the ASCDBs folder.

Optional: The PC DECOMP default file source folder can be changed to any desired location. Since the default output file location of the IOGP is the IOGP folder, the user may choose to set this location for the default source folder of PC DECOMP.

Right click the Decomp icon (on the Windows desktop or Start menu).

Select “Properties”.

Change the “Start in:” path to the desired default file source folder i.e. C:\iogp. Click “Apply” to initiate the path change.

4.3. Windows Data Review Tool

Install Windows Data Review Tool (WinDRT™) using the procedures provided in the Windows Data Review Tool Reference Manual RM1000.

Optional: The WinDRT™ default file source folder can be changed to any desired location. It may be convenient to setup the default source folder to match the output file location used by PC Decomp. Review the “Preferences” selection in PC Decomp to determine the output file location.

Right click the WinDRT™ icon (on the Windows desktop or Start menu).

Select Properties

Change the “Start in:” path to the desired default file source folder i.e. C:\iogp. Click “Apply” to initiate the path change.

5. RDU Software Description

5.1. Operational Ground Program

The IOGP is used for downloading data from and uploading data to the recorder. The IOGP provides the following features:

- Download recorded flight data,
- Download documentary data (such as serial numbers),
- Upload aircraft information/software,
- Initiate Built-in Test (BIT) and download BIT results,
- Format and initialize data recorder memory.

5.2. Data Conversion Program (PC DECOMP)

DECOMP converts the downloaded raw aircraft flight data (.RDF) into a usable format (.DDF) for the WinDRT™ program. Figure 2 and the following paragraphs give a brief description of the various files used with DECOMP. Refer to the PC Common DECOMP Reference Manual (RM1005) for additional information.

5.2.1. Aircraft Specific Configuration Database

The Aircraft Specific Configuration Database (ASCDB) is a file which provides information that the DECOMP software uses to convert aircraft specific flight data. This file is tailored to the specific aircraft operational flight program configuration. The ASCDB defines to DECOMP the format of the raw data created. For each parameter, this includes the bit length and the position within the recording data frame. For conversion to engineering units, it includes the parameter's scale factor and optionally, an equation for performing simple arithmetic operations. For reports generation, the ASCDB includes labelling and display formatting information as well as data set definitions for grouping parameters into ordered sets.

5.2.2. Cross Reference File

The Cross Reference File (.XRF) tells the DECOMP software which ASCDB it should use to process the data in the RDF file being converted.

5.2.3. Recorder Data Format

An .RDF file is raw flight data downloaded from the recorder. The filename for the RDF file is provided by the user at download time.

5.2.4. Decompressed Data Format

The Decompressed Data File (.DDF) is the primary output from DECOMP. A DDF is a binary file with variable length records. DDFs are named to match the name of the RDF they are made from.

5.3. WINDOWS DATA REVIEW TOOL (WINDRT™)

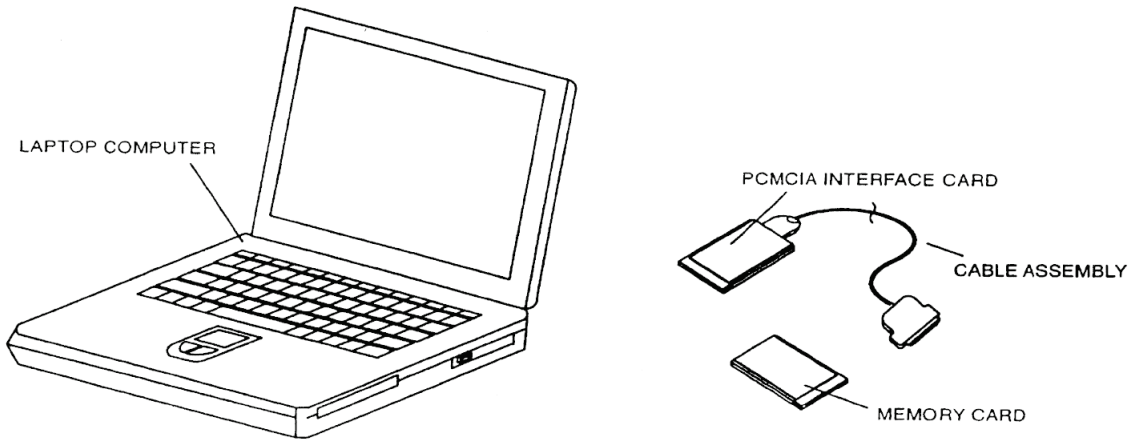
The WinDRT™ allows the user to view aircraft data and recorder BIT information downloaded from the recorder. The aircraft data from the recorder must first receive format conversion using PC DECOMP. The aircraft data can be displayed in tabular or graphical format. Graphical displays can show plots of up to eight parameters against time. A graphical display can be synchronized with a tabular display so that both windows are showing the same portion of the data. Refer to the Windows Data Review Tool Reference Manual RM1000 for WinDRT™ operation instructions.

Table 1. RDU Hardware

NOMENCLATURE	PART NO.	FUNCTION
Laptop Computer, PC - 366 MHz P2, 8.1 GB Hard Drive Minimum: 166Mhz Pentium CPU 13.3" XGA Display 32 MB RAM 1 GB Hard Drive 3.5" Floppy Disk Drive 2 - Type II PCMCIA Receptacles 87 - key keyboard and Touchpad Software: Windows 95 Version 4.00.95 B	Panasonic CF-47G4CJUMM or equivalent.	Aircraft level display of built-in test data.
Interface Assembly Kit, Personal Computer Card, RS-422	173518-07	Interconnects PC to recorder maintenance connector.
Interface Card Assembly, PCMCIA Type II (p/o interface assy kit)	174682-01	
Cable Assembly, Special Purpose, Electrical (p/o interface assy kit)	174600-08	
Memory Card, 128 MB (Minimum)	SDP3BI-128-101	Download/Upload data transfer.

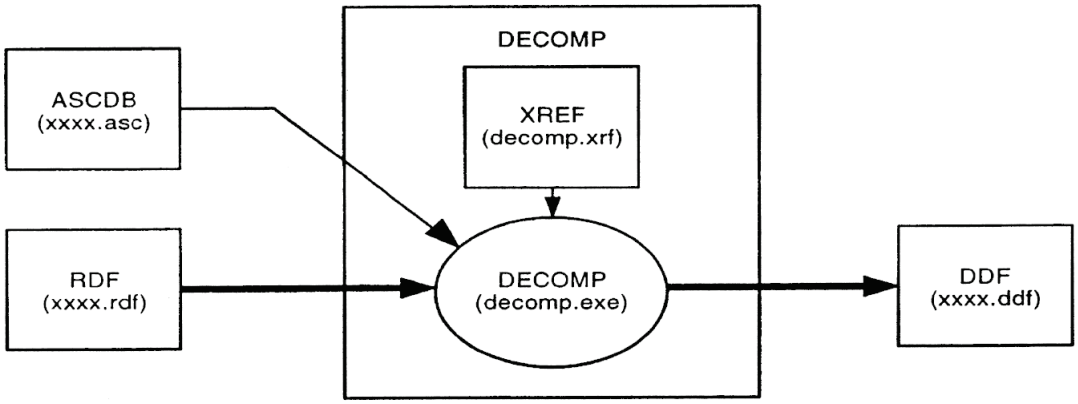
Table 2. RDU Software

DESCRIPTION	PART NO.	USED FOR
Operational Ground Program (IOGP)	552240-XX-XX	Recorder communication
Windows Data Review Tool (WinDRT™))	549651-XX-XX	Data file display/Review
PC DECOMP	549342-XX-XX	Data file Format Conversion
Aircraft Specific Configuration Database	Xxxxxx	Defines Data Format
MS-Windows 95 or 98	-	PC Operating Platform



ICN-412-A-970003-B-97499-00019-A-001-01

Figure 1. RDU Identification View (Sheet 1 of 1)



ICN-412-A-970003-B-97499-00020-A-001-01

Figure 2. DECOMP Block Diagram (Sheet 1 of 1)

Flight Data Recorder - Removal

Procedure

1. Ensure system circuit breakers are pulled.
2. Loosen the two fasteners on front of the recorder far enough to free the two hook supports.
3. Remove recorder from the ARINC 404 tray by pulling straight out of rack.

Intentionally blank page

Flight Data Recorder - Installation

Procedure

1. Ensure system circuit breakers are pulled.
2. Install recorder in ARINC 404 tray and push carefully into place, making sure that rear connectors are seated properly.
3. Place the two tray-locking mechanisms over recorder crow feet and tighten snugly.

Intentionally blank page

Flight Data Recorder System - PreFlight Test - Operational Test Procedure

1. Introduction

This section provides the operational/functional tests used in support of Level 1 maintenance of the flight data recorder. Topics covered in this section include:

- System preflight operational test,
- Underwater locator beacon functional test (24-month interval),
- System functional test (as needed).

2. Testing Requirements

The recorder is tested at standard atmospheric conditions and normal room temperature.

3. Required Test Equipment

The test equipment required for testing is listed in Table 1.

4. FDR Preflight Operational Test

This test, included in Table 2, provides flight crew check of the recorder. Minimum test interval is once per day or at crew change.

NOTE

Recorder Display Unit (RDU) should not be connected to the recorder during this procedure. RDU connection will cause FDR FAIL to flash indicating maintenance mode.

Table 1. Required Test Equipment

NOMENCLATURE	(1) MANUFACTURER MODEL/PART NO.	USED FOR
(2) Acoustic Beacon Test Set	Benthos ATS-260	Benthos beacon functional test
(2) Acoustic Beacon Test Set	Dukane PL-3	Dukane beacon functional test
(2) Digital Multimeter	Fluke 177	Battery test of Dukane beacon

NOTES:

1 Equivalent substitutes may be used for listed items.

2 FDR may be supplied with Dukane or Benthos beacon.

Table 2. Preflight Test Failure Isolation Chart

STEP	ACTION	VERIFY
1.	Apply helicopter AC and DC power to the recorder system	Verify that the FDR FAIL annunciator (FDR_FAIL output of the recorder) is off.
2.	-	Verify that FDR STATUS FLAG (ARINC717_FDR_STATUS FLAG output of recorder) is off (FDAU must be powered-up).

Intentionally blank page

Flight Data Recorder System - Beacon Test (24-Month Interval) - Operational Test

Procedure

1. The recorder may be equipped with Dukane or a Benthos acoustic beacon. The following paragraphs provide the beacon test procedure using each manufacturer's test set.

1.1. Benthos Beacon

This test confirms operation of the locator beacon using the Benthos ATS-260 test set.

1.1.1. Verify the battery expiration date listed on the acoustic beacon housing is not expired.

1.1.2. Attach the acoustic test set cable clip to the beacon.

1.1.3. Hold down the button on the test set handle labeled PUSH TO TEST.

1.1.4. Apply the test set probe to the water activated switch.

1.1.5. Check the battery condition by observing the green and the red indicators on the test set. Indicators will read as follows:

- Green Indicator - The battery is good.
- Red Indicator - The battery should be replaced.

1.1.6. Check the beacon operating condition by observing the amber indicator and listening for a tone from the test set. Indicator and tone information will be as follows:

- Amber Indicator Flashing and Tone Audible - The beacon is operating properly.

1.1.7. Release PUSH TO TEST button and remove test clip from beacon.

1.2. Dukane Beacon

This test confirms operation of a Dukane locator beacon using the Dukane PL-3 pinglite test set.

1.2.1. Operational Test

NOTE

A rapid or accelerating pulse rate usually indicates a faulty circuit. If normal operation does not occur, return beacon to manufacturer for service.

1.2.1.1. Apply spring end of test set to water switch, ensure that center spring touches water switch pin.

1.2.1.2. Listen for three to four audible pings and PL-3 LED flashes.

1.2.2. Battery Test

Use a high impedance voltmeter (Input impedance of 10M ohms) to perform the following procedure:

1.2.2.1. Make sure the case and water switch are clean and dry prior to testing. If in doubt, clean as prescribed in CLEANING section.

1.2.2.2. Place the negative lead of a high impedance voltmeter on the water switch pin and the positive lead to the beacon case or the mounting kit, if already installed.

1.2.2.3. Measure the battery voltage. The beacon is operable at a minimum acceptable voltage of 2.97 Volts. If the battery voltage is below the minimum, beacon battery requires replacement.

Intentionally blank page

Flight Data Recorder System - Recorder Test - Functional Test

Procedure

1. This functional test provides the following operational checks in addition to those verified by preflight test:

- Flight data record validation.
- Acoustic Beacon.

2. The test should be performed anytime that the recorder system function is in question.

2.1. Perform the recorder preflight Test procedure.

2.2. Perform the Beacon Test procedure.

2.3. Perform download of the FDR data (412-A-97-00-03-00A-130A-A / 02712). Process the data and verify that no dropouts or corruptions were reported. Verify that recorded values for key parameters represent a valid flight profile matching the particulars of the flight that was recorded.

Intentionally blank page

Flight Data Recorder System - Inspection/Check Procedure

1. Acoustic Beacon

Verify beacon battery expiration date (date stamped on beacon case). Beacon battery must be replaced by the date stamped on case (6-year interval).

2. Recorder Chassis and Covers

2.1. Check recorder chassis assembly and covers for deformation, dents, punctures, badly worn surfaces, cracks, damaged or missing fasteners and hardware, or damaged handle.

2.2. Inspect for corrosion and scratches, that require work in finishing department.

2.3. Inspect stenciled markings for legibility.

2.4. Inspect for loose or missing labels and reflective tape and check that labels are securely attached and legibility of lettering.

3. Recorder Connectors

3.1. Inspect connector body for damage or deformation.

3.2. Inspect for cracked or broken insulator and for contacts that are missing, corroded, broken, or deformed and out of alignment.

Intentionally blank page

Flight Data Recorder System - Cleaning

Procedure

1. Cleaning Agents

Table 1 lists cleaning agents required to clean the recorder.

NOTE

Equivalent substitutes may be used for items listed.

2. Procedures

Clean the recorder when necessary.

2.1. Acoustic Beacon

The beacon end cap with the water switch should remain free and clear of dirt, grease and dust. The end cap should be cleaned with detergent (Table 1) and distilled water. After cleaning, dry beacon thoroughly with a clean cloth.

2.2. Recorder Chassis

2.2.1. Clean chassis surfaces using paper towel or soft cloth moistened with detergent (Table 1) and distilled water.

2.2.2. Remove detergent residue using paper towel or soft cloth moistened with distilled water.

2.2.3. Thoroughly dry surfaces.

ISOPROPYL ALCOHOL IS FLAMMABLE AND TOXIC. KEEP AWAY FROM HEAT, SPARKS, AND OPEN FLAME. AVOID PROLONGED OR REPEATED BREATHING OF VAPOR. USE ONLY IN A WELL-VENTILATED AREA.

2.2.4. Remove stubborn dirt/grease from chassis using a lint-free cloth or soft bristle brush moistened with isopropyl alcohol (Table 1). Thoroughly dry surfaces.

2.2.5. Clean connector as follows:

2.2.5.1. Remove dust and loose dirt using a soft-bristled brush.

ISOPROPYL ALCOHOL IS FLAMMABLE AND TOXIC. KEEP AWAY FROM HEAT, SPARKS, AND OPEN FLAME. AVOID PROLONGED OR REPEATED BREATHING OF VAPOR. USE ONLY IN A WELL-VENTILATED AREA.

2.2.5.2. Clean connector contacts using a soft-bristled brush moistened with isopropyl alcohol (Table 1). Thoroughly dry surfaces.

Table 1. Required Cleaning Agents

MATERIAL	MANUFACTURER'S NAME AND ADDRESS	APPLICATION
Liqui-Nox Liquid Detergent 21837-005	Alconox Inc. 215 Park Ave, South New York, NY 10003 (CAGE 17534)	General cleaning.
Isopropyl Alcohol TT-E-735	- - -	Used to clean stubborn dirt or grease.

Intentionally blank page

Maintenance Procedures - General Description

This manual provides shipboard operations instructions for Bell 412 series model helicopter manufactured by Bell Helicopter Inc. and Bell Helicopter Textron, a Division of Textron Canada LTD. This includes characteristics of 412 series helicopter, operating conditions and interface and additional services required on shipboard. This manual shall be used in conjunction with 412/412EP Flight Manual and 412-A-00-00-00-00A-009A-A / 00001.

For components unique to this application, see drawing 412-799-129.

Intentionally blank page

Performance, Weight and Dimensions - Description

Principal performance, weight, and dimensions of helicopter are given in Table 1 and Figure 1.

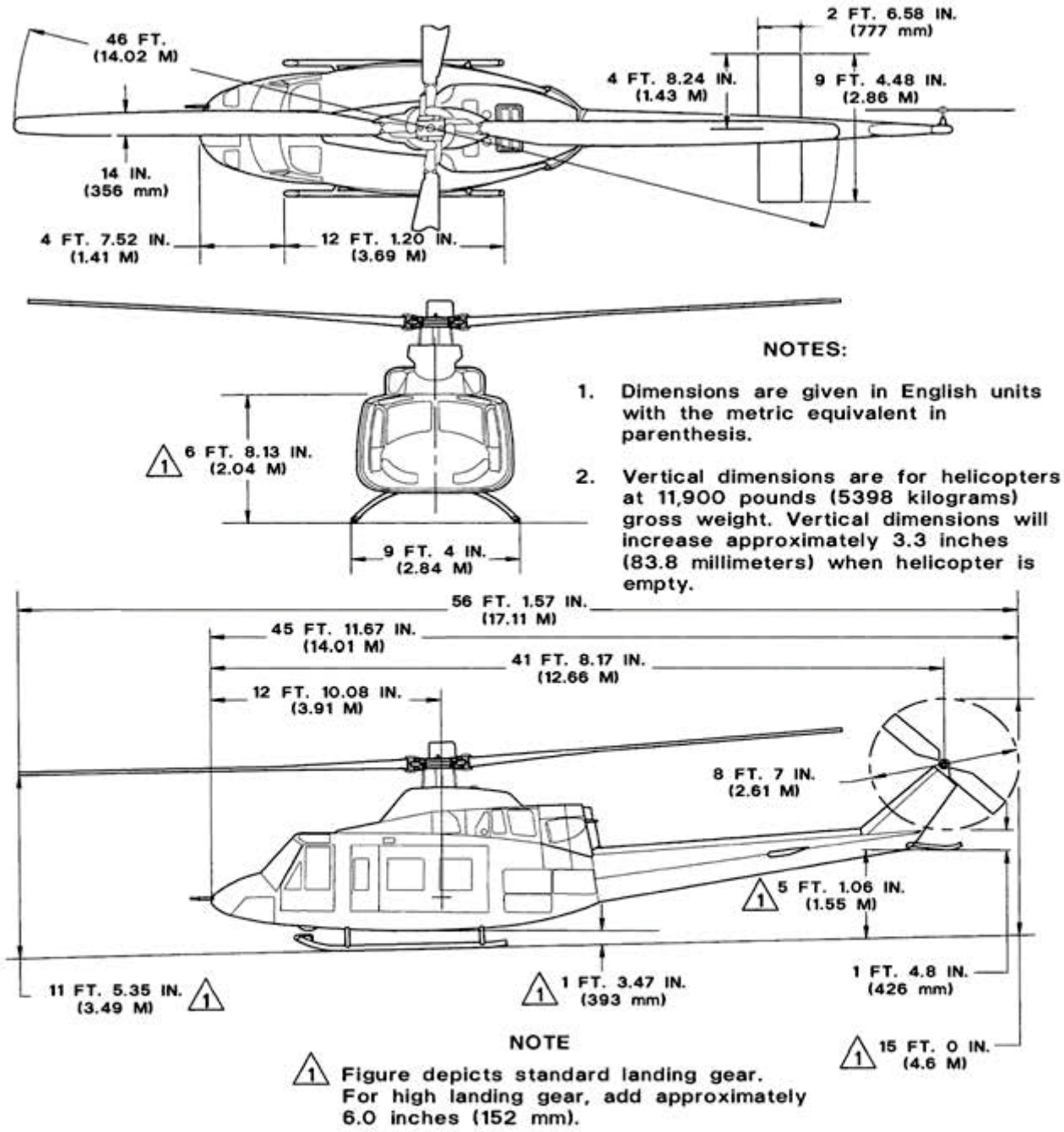
Table 1. Performance, Weights and Dimensions

Performance		
(Standard day at 11,900 lbs gross weight except where indicated)	English	Metric
Maximum Speed, Sea Level (Vne)	140 kts	259 km/hr
Maximum Cruise Speed, Sea Level	122 kts	226 km/hr
Best Rate of Climb, Takeoff Power, Sea Level	1780 ft/min	9 m/sec
Hover Ceiling In-Ground Effect (T.O.P.)	10200 ft	3110 m
Hover Ceiling Out-of-Ground Effect (T.O.P.)	5200 ft	1585 m
Twin Engine Service Ceiling	16500 ft	5029 m
Single Engine Service Ceiling (30-Minutes Power)	7600 ft	2316 m
Range of Long Range Cruise Speed (127 knots at 5,000 feet)		
With 30-minutes reserve	338 NMI	626 km
With no reserve	402 NMI	745 km
Fuel Consumption at 127 knots at 5,000 feet	712 lb/hr	323 kg/hr
Weights		
Maximum Take-Off Gross Weight	11900 lbs	5409 kg
Weight Empty (standard (utility) configuration)	6799 lbs	3084 kg
Useful Load	5101 lbs	2314 kg
Dimensions		
Fuselage Length	45 ft 11 in.	17.19 m
Fuselage Width	8 ft 4 in.	2.5 m
Length Over-all (including rotors)	56 ft 2 in.	17.1 m
Height Over-all (to tip of tail rotor) at 11,900 lbs. gross weight	15 ft 0 in.	4.6 m
Width (including horizontal stabilizer	9 ft 5 in.	2.9 m
Main Rotor Diameter (blade tip circle)	46 ft 0 in.	14.0 m
Tail Rotor Diameter (blade tip circle)	8 ft 7 in.	2.6 m
Landing Gear	9 ft 4 in.*	2.84 m
Passenger Cabin Length	7 ft 8 in.	2.3 m
Passenger Cabin Width	8 ft 0 in.	2.4 m
Passenger Cabin Height	4 ft 1 in.	1.2 m

***NOTE:**

CENTER TO CENTER OF STANDARD LANDING GEAR SKID TUBES IS 100 IN. (UNLOADED).

CENTER TO CENTER OF HIGH LANDING GEAR SKID TUBES IS 110 IN. (UNLOADED).



ICN-412-A-000000-B-97499-00056-A-001-01

Figure 1. Airframe Principal Dimensions (Sheet 1 of 1)

Shipboard Deck Operation Capabilities - Description

Deck operation capabilities for landing, blade folding, stability, towing, and tiedown have been determined using a database of FFG-7 frigate ship motion parameters and 412 limit load capabilities. Operating capabilities are expressed in terms of ship deck pitch and roll angles. Pitch and roll angles are a matched pair of numbers resulting from a ship operating in a particular sea state.

Intentionally blank page

Summary of Deck Operating Capabilities - Description

Table 1 defines shipboard capabilities for Deck Landing, Blade Fold, Towing, and Tiedown.

Deck landing capability is determined considering helicopter sink speed, relative wind speed and direction, accelerations due to ship deck pitch and roll and helicopter landing gear limit load capability. Pilot reaction to deck motion is also considered in determining this capability.

Blade folding operation capability is determined using accelerations due to ship deck pitch and roll, limit loads of blade attachment, and relative wind speed.

Towing capability is determined based on helicopter stability limits without tiedown cables. Helicopter stability capabilities are discussed in 412-A-00-00-01-05A-042A-A / 02729.

Tiedown capabilities are determined for Normal, Moderate, and Heavy weather conditions. Capabilities at these conditions are based on accelerations due to ship deck pitch and roll angles and limit load capabilities of landing gear, mooring lugs, and mooring cables.

Table 1. Shipboard Operations Capability

11,900 lb Max G. W.					
OPERATION	**SEA STATE**	**ROLL (deg)**	**PITCH (deg)**	**PERIOD (sec)**	**WIND ± 30° TO HEAD**
Deck Landing	3	10°	*	-	35 Knots @ 10° 30 Knots @ ±30°
Deck Landing	3	10°	*	-	65 Knots @ 10° 35 Knots crosswind
Deck Towing	3	13°	6.5°	-	35 Knot wind from any direction
Main rotor blades folding	3	15°	15°	-	60 Knot wind from any direction
Main rotor blades folded on supports	3	15°	15°	-	60 Knot wind from any direction
Moored blades folded one cable on each tiedown fitting	3	7°	2°	9	45 Knot wind from any direction (Normal Weather)
Moored blades folded three cables on each tiedown fitting	4	13°	8°	Roll 9 sec Pitch 6 sec. 3m Heave in 6 sec.	45 Knot wind from any direction (Moderate Weather)
In Hanger blades folded	7	45°	8°	Roll 13 sec Pitch 6 sec. 3m Heave in 6 sec.	0 Knot in hanger (Heavy Weather)

NOTE: 40° is limit of available published data as per Report SPD-738-01 shown in (1) Table 2. 45° is extrapolated from this report.

* See flight manual.

Table 2. Sea States

SHIPBOARD MOTION PARAMETERS FOR FGG-7 AT 10 KNOTS (1) UPPER SEA STATE 1							
Relative Wave heading Angle	**Load Factors in Local horizontal Axis System (2)**		**Ship Pitch and Roll (2)**		**Load Factors in Ship Deck Axis System**		
	Ny	**Nz**	**pitch**	**roll**	**nx(s)**	**ny(s)**	**nz(s)**
0	0.017	1.046	0.671	1.471	-0.012	0.044	1.045
15	0.019	1.047	0.671	1.586	-0.012	0.048	1.046

Table 2. Sea States (continued)

SHIPBOARD MOTION PARAMETERS FOR FGG-7 AT 10 KNOTS (1)							
UPPER SEA STATE 1							
Relative Wave heading Angle	Load Factors in Local horizontal Axis System (2)		Ship Pitch and Roll (2)		Load Factors in Ship Deck Axis System		
	Ny	Nz	pitch	roll	nx(s)	ny(s)	nz(s)
30	0.022	1.048	0.657	1.914	-0.012	0.057	1.047
45	0.025	1.049	0.643	2.393	-0.012	0.069	1.047
60	0.028	1.049	0.614	2.907	-0.011	0.081	1.047
75	0.029	1.048	0.586	3.343	-0.011	0.090	1.044
90	0.029	1.044	0.550	3.614	-0.010	0.095	1.040
105	0.027	1.039	0.514	3.700	-0.009	0.094	1.035
120	0.024	1.033	0.486	3.585	-0.009	0.089	1.029
135	0.020	1.026	0.471	3.278	-0.008	0.079	1.023
150	0.016	1.019	0.464	2.828	-0.008	0.066	1.017
165	0.012	1.014	0.464	2.364	-0.008	0.054	1.012
180	0.011	1.012	0.464	2.157	-0.008	0.049	1.011
UPPER SEA STATE 2							
Relative Wave heading Angle	Load Factors in Local horizontal Axis System (2)		Ship Pitch and Roll (2)		Load Factors in Ship Deck Axis System		
	Ny	Nz	pitch	roll	nx(s)	ny(s)	nz(s)
0	0.037	1.100	1.444	3.165	-0.028	0.098	1.096
15	0.040	1.101	1.444	3.411	-0.028	0.106	1.096
30	0.047	1.104	1.413	4.117	-0.027	0.126	1.097
45	0.054	1.106	1.383	5.147	-0.027	0.153	1.097
60	0.060	1.106	1.321	5.253	-0.026	0.180	1.093
75	0.063	1.103	1.260	7.190	-0.024	0.200	1.086
90	0.062	1.095	1.183	7.774	-0.023	0.210	1.077
105	0.059	1.084	1.106	7.958	-0.021	0.208	1.065
120	0.052	1.070	1.045	7.713	-0.020	0.195	1.053
135	0.044	1.055	1.014	7.052	-0.019	0.173	1.042
150	0.035	1.041	0.999	6.084	-0.018	0.145	1.031
165	0.027	1.029	0.999	5.085	-0.018	0.118	1.023
180	0.023	1.026	0.999	4.640	-0.018	0.106	1.020
UPPER SEA STATE 3							
Relative Wave heading Angle	Load Factors in Local horizontal Axis System (2)		Ship Pitch and Roll (2)		Load Factors in Ship Deck Axis System		
	Ny	Nz	pitch	roll	nx(s)	ny(s)	nz(s)
0	0.051	1.137	1.979	4.337	-0.039	0.137	1.129
15	0.055	1.139	1.979	4.674	-0.039	0.148	1.130
30	0.064	1.143	1.937	5.643	-0.039	0.176	1.130
45	0.075	1.146	1.895	7.053	-0.038	0.215	1.127
60	0.082	1.146	1.811	8.569	-0.036	0.252	1.120
75	0.086	1.141	1.727	9.854	-0.034	0.280	1.109
90	0.085	1.131	1.621	10.650	-0.032	0.293	1.095

Table 2. Sea States (continued)

SHIPBOARD MOTION PARAMETERS FOR FGG-7 AT 10 KNOTS (1)							
UPPER SEA STATE 1							
Relative Wave heading Angle	Load Factors in Local horizontal Axis System (2)		Ship Pitch and Roll (2)		Load Factors in Ship Deck Axis System		
	Ny	Nz	pitch	roll	nx(s)	ny(s)	nz(s)
105	0.080	1.115	1.516	10.910	-0.030	0.290	1.079
120	0.072	1.095	1.432	10.570	-0.027	0.271	1.064
135	0.080	1.075	1.390	9.664	-0.026	0.239	1.050
150	0.047	1.056	1.369	8.338	-0.025	0.200	1.037
165	0.037	1.040	1.369	6.969	-0.025	0.163	1.028
180	0.032	1.035	1.369	6.359	-0.025	0.146	1.024
UPPER SEA STATE 4							
Relative Wave heading Angle	Load Factors in Local horizontal Axis System (2)		Ship Pitch and Roll (2)		Load Factors in Ship Deck Axis System		
	Ny	Nz	pitch	roll	nx(s)	ny(s)	nz(s)
0	0.066	1.177	2.570	5.632	-0.053	0.181	1.164
15	0.071	1.180	2.570	6.070	-0.053	0.196	1.165
30	0.083	1.185	2.515	7.328	-0.052	0.234	1.164
45	0.097	1.189	2.461	9.160	-0.052	0.285	1.158
60	0.107	1.189	2.351	11.130	-0.049	0.334	1.145
75	0.112	1.183	2.242	12.800	-0.046	0.371	1.128
90	0.111	1.170	2.105	13.840	-0.043	0.387	1.108
105	0.104	1.150	1.969	14.160	-0.039	0.382	1.088
120	0.093	1.125	1.859	13.730	-0.037	0.357	1.070
135	0.078	1.098	1.805	12.550	-0.035	0.314	1.054
150	0.062	1.072	1.777	10.830	-0.033	0.261	1.041
165	0.048	1.052	1.777	9.050	-0.033	0.213	1.031
180	0.042	1.045	1.777	8.257	-0.032	0.191	1.028
UPPER SEA STATE 5							
Relative Wave heading Angle	Load Factors in Local horizontal Axis System (2)		Ship Pitch and Roll (2)		Load Factors in Ship Deck Axis System		
	Ny	Nz	pitch	roll	nx(s)	ny(s)	nz(s)
0	0.117	1.315	4.560	9.993	-0.105	0.343	1.270
15	0.127	1.319	4.560	10.770	-0.105	0.370	1.268
30	0.148	1.328	4.463	13.000	-0.103	0.442	1.257
45	0.172	1.336	4.366	16.250	-0.102	0.538	1.231
60	0.190	1.336	4.172	19.740	-0.097	0.629	1.190
75	0.198	1.324	3.978	22.700	-0.092	0.693	1.142
90	0.197	1.301	3.735	24.550	-0.085	0.718	1.099
105	0.185	1.265	3.493	25.130	-0.077	0.704	1.065
120	0.165	1.222	3.299	24.350	-0.070	0.653	1.043
135	0.138	1.174	3.202	22.270	-0.066	0.572	1.032
150	0.109	1.128	3.153	19.210	-0.062	0.474	1.028
165	0.084	1.093	3.153	16.060	-0.060	0.383	1.026

Table 2. Sea States (continued)

SHIPBOARD MOTION PARAMETERS FOR FGG-7 AT 10 KNOTS (1)							
UPPER SEA STATE 1							
Relative Wave heading Angle	**Load Factors in Local horizontal Axis System (2)**		**Ship Pitch and Roll (2)**		**Load Factors in Ship Deck Axis System**		
	Ny	**Nz**	**pitch**	**roll**	**nx(s)**	**ny(s)**	**nz(s)**
180	0.074	1.080	3.153	14.650	-0.059	0.344	1.024
UPPER SEA STATE 6							
Relative Wave heading Angle	Load Factors in Local horizontal Axis System (2)		Ship Pitch and Roll (2)		Load Factors in Ship Deck Axis System		
	Ny	Nz	pitch	roll	nx(s)	ny(s)	nz(s)
0	0.187	1.502	7.273	15.940	-0.190	0.589	1.381
15	0.202	1.509	7.273	17.180	-0.191	0.635	1.371
30	0.236	1.524	7.119	20.740	-0.189	0.756	1.331
45	0.274	1.536	6.964	25.920	-0.186	0.913	1.252
60	0.303	1.535	6.654	31.490	-0.178	1.055	1.142
75	0.317	1.517	6.345	36.210	-0.168	1.146	1.029
90	0.314	1.480	5.958	39.150	-0.154	1.173	0.943
105	0.296	1.423	5.560	40.000	-0.138	1.137	0.895
120	0.263	1.354	5.262	38.840	-0.124	1.050	0.885
135	0.220	1.277	5.107	35.520	-0.114	0.918	0.908
150	0.174	1.204	5.029	30.640	-0.106	0.761	0.943
165	0.135	1.149	5.029	25.810	-0.101	0.616	0.974
180	0.118	1.127	5.029	23.370	-0.099	0.553	0.984
UPPER SEA STATE 7							
Relative Wave heading Angle	Load Factors in Local horizontal Axis System (2)		Ship Pitch and Roll (2)		Load Factors in Ship Deck Axis System		
	Ny	Nz	pitch	roll	nx(s)	ny(s)	nz(s)
0	0.363	1.600	14.100	30.910	-0.390	1.109	1.145
15	0.392	1.605	14.100	33.310	-0.391	1.182	1.086
30	0.458	1.614	13.730	40.000	-0.383	1.359	0.907
45	0.531	1.617	10.750	40.000	-0.302	1.428	0.876
60	0.587	1.604	8.452	40.000	-0.236	1.470	0.838
75	0.614	1.590	7.009	40.000	-0.194	1.485	0.814
90	0.609	1.513	6.087	40.000	-0.160	1.434	0.761
105	0.573	1.438	5.560	40.000	-0.139	1.359	0.728
120	0.510	1.350	5.418	40.000	-0.127	1.254	0.702
135	0.426	1.256	5.752	40.000	-0.126	1.130	0.683
150	0.338	1.170	6.566	40.000	-0.134	1.006	0.673
165	0.261	1.106	7.855	40.000	-0.151	0.904	0.671
180	0.228	1.081	8.609	40.000	-0.162	0.862	0.672
UPPER SEA STATE 8							
Relative Wave heading Angle	Load Factors in Local horizontal Axis System (2)		Ship Pitch and Roll (2)		Load Factors in Ship Deck Axis System		
	Ny	Nz	pitch	roll	nx(s)	ny(s)	nz(s)

Table 2. Sea States (continued)

SHIPBOARD MOTION PARAMETERS FOR FGG-7 AT 10 KNOTS (1) UPPER SEA STATE 1							
Relative Wave heading Angle	Load Factors in Local horizontal Axis System (2)		Ship Pitch and Roll (2)		Load Factors in Ship Deck Axis System		
	Ny	Nz	pitch	roll	nx(s)	ny(s)	nz(s)
0	0.554	2.487	18.250	40.000	-0.779	1.943	1.453
15	0.598	2.507	16.940	40.000	-0.730	2.000	1.453
30	0.699	2.551	13.730	40.000	-0.605	2.128	1.449
45	0.811	2.588	10.750	40.000	-0.483	2.256	1.426
60	0.896	2.585	8.452	40.000	-0.380	2.330	1.383
75	0.937	2.530	7.009	40.000	-0.309	2.332	1.322
90	0.930	2.420	6.087	40.000	-0.257	2.259	1.246
105	0.875	2.253	5.560	40.000	-0.218	2.112	1.155
120	0.779	2.047	5.410	40.000	-0.193	1.907	1.060
135	0.651	1.820	5.752	40.000	-0.182	1.663	0.969
150	0.515	1.605	6.566	40.000	-0.184	1.419	0.890
165	0.399	1.440	7.855	40.000	-0.197	1.223	0.836
180	0.348	1.376	8.609	40.000	-0.206	1.141	0.818

1 Ship Performance Department; Report No. SPD-738-01; "A Non-Aviation Ship Motion Data Base for the DD 963, CG 26, FF 1052, FFG 7, and the FF 1040 Ship Classes"; dated December 1979.

2 The ny and nz load factors and the ship pitch and roll angles represent the highest expected value in 1000 successive wave encounters.

Intentionally blank page

Landing Gear and Deck Interface Reaction Loads - Description

Interface loads and reactions between helicopter and ship deck are given in Table 1.

Static Gear Loads are for 11,900 lbs GW helicopter at extremes of CG envelope with a vertical load factor of 1.

Limit Gear Loads for Landing and Mooring is a summary of maximum design loads which include ship deck and helicopter accelerations, as well as landing gear dynamic response.

Table 1. Landing Gear/Deck Interface Reaction Loads

11,900 lbs Max. G. W.			
		Main Landing Gear Aft CG Limit	**Main Landing Gear Fwd CG Limit**
Landing Gear dimensions			
Cross tube centerline to centerline	in.	91.38	91.38
*Skid tube centerline to centerline	in.	110.0	110.0
Static Gear Loads (1g)			
Static Gear Loads (1g) Max Vertical Load	lb	11900	11900
*Fwd cross tube deflection	in.	3.47	5.12
*Aft cross tube deflection	in.	6.50	5.64
Limit Gear Loads	lb	19955 total	19865 total
Landing	lb	7706 aft cross tube	6639 aft cross tube
Vv = 6.56 ft/sec	lb	2271 fwd cross tube	3293 fwd cross tube
Limit Gear Loads Mooring:			
normal (SS3)	lb	5685	4988
moderate (SS4)	lb	6011	5293
heavy (SS7)	lb	9567	8861
Ultimate Deck Reaction	lb	14350	13292
ultimate = limit x 1.5			

* Dimensions are approximate.

Intentionally blank page

Helicopter Shipboard Stability - Description

Shipboard helicopter stability is determined using resultant load from combining helicopter weight, ship deck accelerations, relative wind speed, and direction. Helicopter becomes unstable when this combined load vector is outside of crosshatched area shown in Figure 1.

Helicopter Configurations. Critical configurations are minimum and maximum gross weights at most forward center of gravity shown in Table 1.

Table 2 presents Roll-over Stability limits for critical combinations of sea state, wind speed, and wind direction. (Example: With rotor stopped, at Sea State 5 with wind at 90°, helicopter is stable at wind speeds up to 58 knots.) Stability limits are presented for rotors stopped and rotors turning.

Table 1. Gross Weight

Gross Weight (lbs)	Center of Gravity	
	STA (in.)	WL (in.)
11900	133.5 - 143.0	49.9 - 50.4
9500	130.0 - 146.0	49.9 - 50.4

Table 2. Helicopter Shipboard Stability

	Sea State			
*Wind Direction	2	3	4	5
Rotor Stopped***				
6799 lb	Wind Speed at Aircraft Stability Limit (Knots)			
30	*	*	*	*
60	*	*	*	66
90	102	94	85	58
120	*	*	*	86
150	*	*	*	*
11900 lb				
30	*	*	*	*
60	*	*	*	*
90	*	*	*	*
120	*	*	*	*
150	*	*	*	*
Rotor turning				
6799 lb				
30	**	**	**	**
60	**	**	**	**
90	**	**	**	**
120	**	**	**	**
150	**	**	**	**
11900 lb				
30	**	**	**	**
60	**	**	**	**
90	**	**	**	**
120	**	**	**	**
150	**	**	**	**

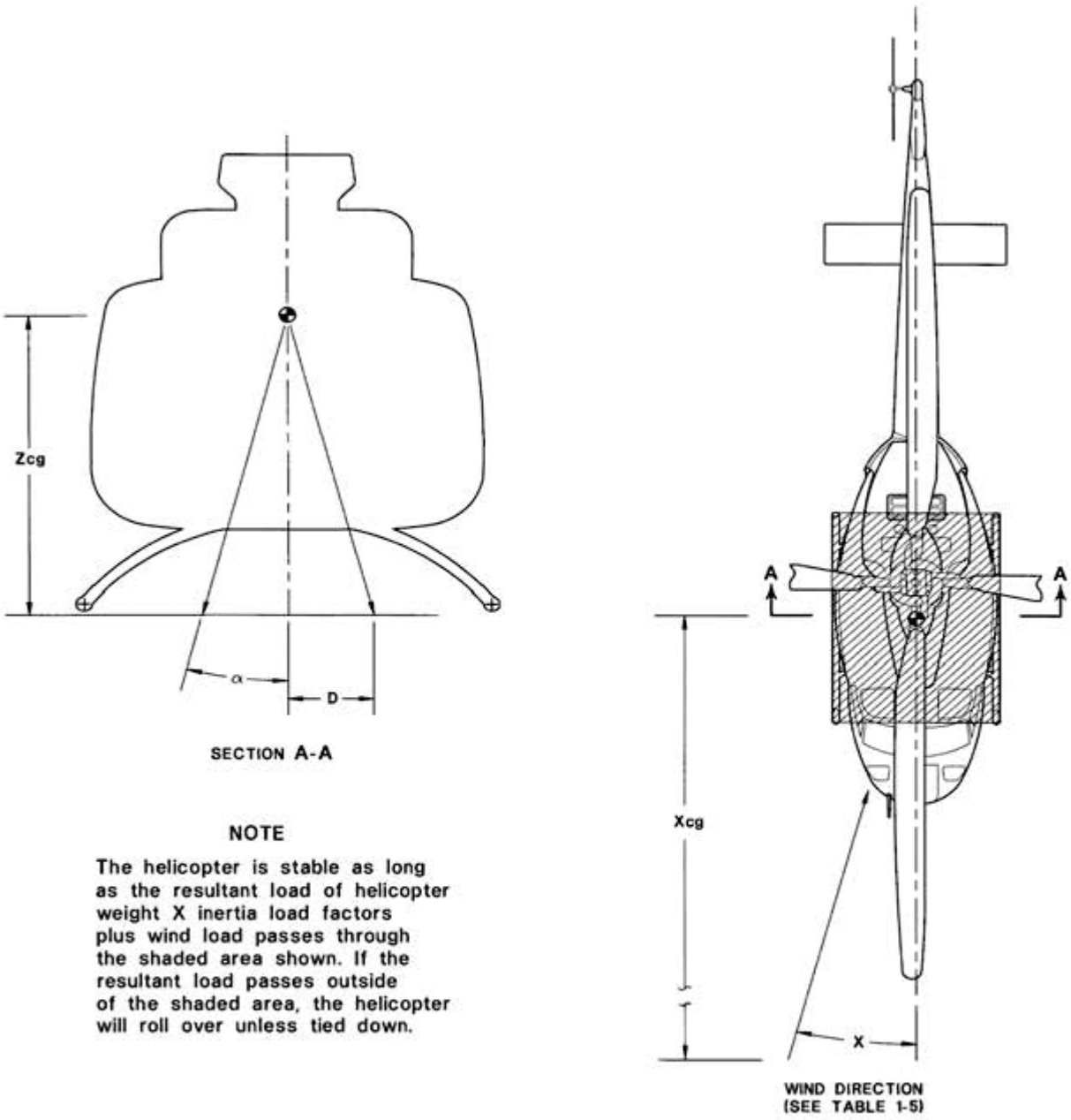
Table 2. Helicopter Shipboard Stability (continued)

	Sea State			
***Wind Direction**	**2**	**3**	**4**	**5**

*The helicopter will not turn over in this sea state when subjected to a wind velocity up to 100 knots, although deck conditions may allow helicopter to slide before rollover. The landing gear skid tubes may need to be blocked against side motion to prevent skidding.

**See flight manual for wind restrictions.

***Blade restraint is limited to 60 knot wind.



ICN-412-A-000001-B-97499-00001-A-001-01

Figure 1. Helicopter Stability (Sheet 1 of 1)

Intentionally blank page

Ship Motion and Dynamic Characteristics - Description

Table 1 presents dynamic characteristics for FFG-7 Frigate operating at 10 knots and at 45° relative wave heading with critical modal wave period for associated sea state. This condition is considered to be a conservative representation of overall range of ship speed, relative wave heading, and wave modal period. Table 2 presents a more comprehensive account of statistical ship motion data. Figure 1 shows relationship between vertical acceleration and deck pitching motion. Figure 2 shows relationship between lateral acceleration and deck rolling motion.

Table 1. Dynamic Characteristics

Dynamic Characteristics of FFG-7 Frigate (1)							
	Sea State						
	2	3	4	5	6	7	8
Roll Angle	8.0	10.9	14.2	25.1	40.0	40.0	40.0
Pitch Angle	1.4	2.0	2.6	4.6	7.3	14.1	18.3
Shipboard Load Factor (Ship Local Axis System)							
Lateral	±0.21	±0.29	±0.39	±0.72	±1.17	±1.49	±2.33
Vertical	1±0.11	1±0.15	1±0.19	1±0.34	1±0.54	1±0.62	1±1.59

NOTE:

1 These are the maximum values occuring separately in each Sea State. (Refer to Table 2 for a more detailed view of the shipboard motion parameters.)

Table 2. Sea States

SHIPBOARD MOTION PARAMETERS FOR FGG-7 AT 10 KNOTS (1)							
UPPER SEA STATE 1							
Relative Wave heading Angle	Load Factors in Local horizontal Axis System (2)		Ship Pitch and Roll (2)		Load Factors in Ship Deck Axis System		
	Ny	Nz	pitch	roll	nx(s)	ny(s)	nz(s)
0	0.017	1.046	0.671	1.471	-0.012	0.044	1.045
15	0.019	1.047	0.671	1.586	-0.012	0.048	1.046
30	0.022	1.048	0.657	1.914	-0.012	0.057	1.047
45	0.025	1.049	0.643	2.393	-0.012	0.069	1.047
60	0.028	1.049	0.614	2.907	-0.011	0.081	1.047
75	0.029	1.048	0.586	3.343	-0.011	0.090	1.044
90	0.029	1.044	0.550	3.614	-0.010	0.095	1.040
105	0.027	1.039	0.514	3.700	-0.009	0.094	1.035
120	0.024	1.033	0.486	3.585	-0.009	0.089	1.029
135	0.020	1.026	0.471	3.278	-0.008	0.079	1.023
150	0.016	1.019	0.464	2.828	-0.008	0.066	1.017
165	0.012	1.014	0.464	2.364	-0.008	0.054	1.012
180	0.011	1.012	0.464	2.157	-0.008	0.049	1.011
UPPER SEA STATE 2							
Relative Wave heading Angle	Load Factors in Local horizontal Axis System (2)		Ship Pitch and Roll (2)		Load Factors in Ship Deck Axis System		
	Ny	Nz	pitch	roll	nx(s)	ny(s)	nz(s)
0	0.037	1.100	1.444	3.165	-0.028	0.098	1.096
15	0.040	1.101	1.444	3.411	-0.028	0.106	1.096
30	0.047	1.104	1.413	4.117	-0.027	0.126	1.097

Table 2. Sea States (continued)

SHIPBOARD MOTION PARAMETERS FOR FGG-7 AT 10 KNOTS (1)							
UPPER SEA STATE 1							
Relative Wave heading Angle	Load Factors in Local horizontal Axis System (2)		Ship Pitch and Roll (2)		Load Factors in Ship Deck Axis System		
	Ny	Nz	pitch	roll	nx(s)	ny(s)	nz(s)
45	0.054	1.106	1.383	5.147	-0.027	0.153	1.097
60	0.060	1.106	1.321	5.253	-0.026	0.180	1.093
75	0.063	1.103	1.260	7.190	-0.024	0.200	1.086
90	0.062	1.095	1.183	7.774	-0.023	0.210	1.077
105	0.059	1.084	1.106	7.958	-0.021	0.208	1.065
120	0.052	1.070	1.045	7.713	-0.020	0.195	1.053
135	0.044	1.055	1.014	7.052	-0.019	0.173	1.042
150	0.035	1.041	0.999	6.084	-0.018	0.145	1.031
165	0.027	1.029	0.999	5.085	-0.018	0.118	1.023
180	0.023	1.026	0.999	4.640	-0.018	0.106	1.020
UPPER SEA STATE 3							
Relative Wave heading Angle	Load Factors in Local horizontal Axis System (2)		Ship Pitch and Roll (2)		Load Factors in Ship Deck Axis System		
	Ny	Nz	pitch	roll	nx(s)	ny(s)	nz(s)
0	0.051	1.137	1.979	4.337	-0.039	0.137	1.129
15	0.055	1.139	1.979	4.674	-0.039	0.148	1.130
30	0.064	1.143	1.937	5.643	-0.039	0.176	1.130
45	0.075	1.146	1.895	7.053	-0.038	0.215	1.127
60	0.082	1.146	1.811	8.569	-0.036	0.252	1.120
75	0.086	1.141	1.727	9.854	-0.034	0.280	1.109
90	0.085	1.131	1.621	10.650	-0.032	0.293	1.095
105	0.080	1.115	1.516	10.910	-0.030	0.290	1.079
120	0.072	1.095	1.432	10.570	-0.027	0.271	1.064
135	0.080	1.075	1.390	9.664	-0.026	0.239	1.050
150	0.047	1.056	1.369	8.338	-0.025	0.200	1.037
165	0.037	1.040	1.369	6.969	-0.025	0.163	1.028
180	0.032	1.035	1.369	6.359	-0.025	0.146	1.024
UPPER SEA STATE 4							
Relative Wave heading Angle	Load Factors in Local horizontal Axis System (2)		Ship Pitch and Roll (2)		Load Factors in Ship Deck Axis System		
	Ny	Nz	pitch	roll	nx(s)	ny(s)	nz(s)
0	0.066	1.177	2.570	5.632	-0.053	0.181	1.164
15	0.071	1.180	2.570	6.070	-0.053	0.196	1.165
30	0.083	1.185	2.515	7.328	-0.052	0.234	1.164
45	0.097	1.189	2.461	9.160	-0.052	0.285	1.158
60	0.107	1.189	2.351	11.130	-0.049	0.334	1.145
75	0.112	1.183	2.242	12.800	-0.046	0.371	1.128
90	0.111	1.170	2.105	13.840	-0.043	0.387	1.108
105	0.104	1.150	1.969	14.160	-0.039	0.382	1.088

Table 2. Sea States (continued)

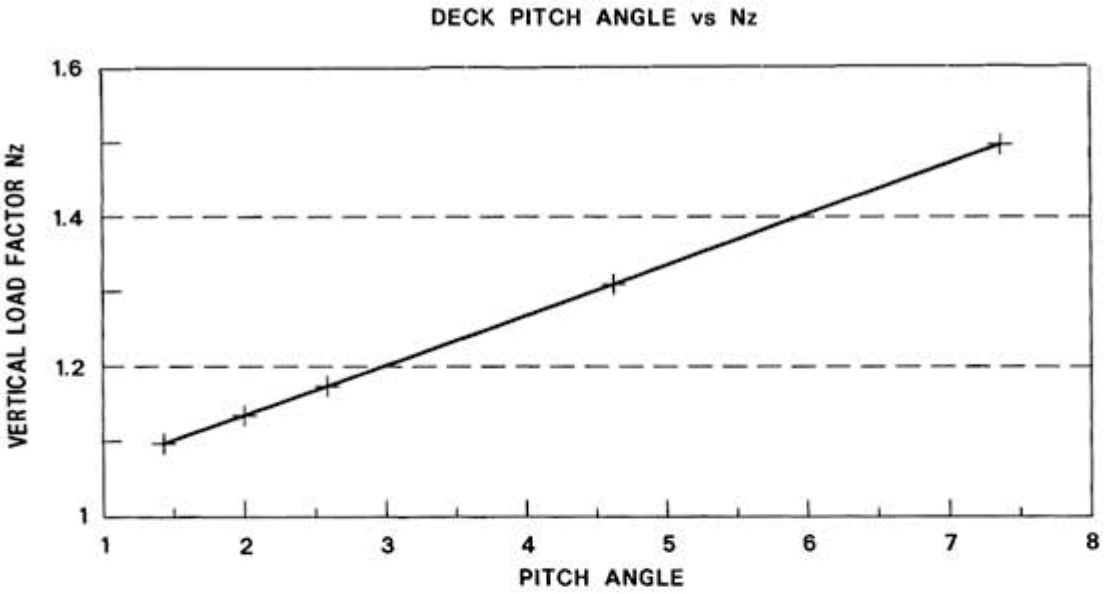
SHIPBOARD MOTION PARAMETERS FOR FGG-7 AT 10 KNOTS (1)							
UPPER SEA STATE 1							
Relative Wave heading Angle	Load Factors in Local horizontal Axis System (2)		Ship Pitch and Roll (2)		Load Factors in Ship Deck Axis System		
	Ny	Nz	pitch	roll	nx(s)	ny(s)	nz(s)
120	0.093	1.125	1.859	13.730	-0.037	0.357	1.070
135	0.078	1.098	1.805	12.550	-0.035	0.314	1.054
150	0.062	1.072	1.777	10.830	-0.033	0.261	1.041
165	0.048	1.052	1.777	9.050	-0.033	0.213	1.031
180	0.042	1.045	1.777	8.257	-0.032	0.191	1.028
UPPER SEA STATE 5							
Relative Wave heading Angle	Load Factors in Local horizontal Axis System (2)		Ship Pitch and Roll (2)		Load Factors in Ship Deck Axis System		
	Ny	Nz	pitch	roll	nx(s)	ny(s)	nz(s)
0	0.117	1.315	4.560	9.993	-0.105	0.343	1.270
15	0.127	1.319	4.560	10.770	-0.105	0.370	1.268
30	0.148	1.328	4.463	13.000	-0.103	0.442	1.257
45	0.172	1.336	4.366	16.250	-0.102	0.538	1.231
60	0.190	1.336	4.172	19.740	-0.097	0.629	1.190
75	0.198	1.324	3.978	22.700	-0.092	0.693	1.142
90	0.197	1.301	3.735	24.550	-0.085	0.718	1.099
105	0.185	1.265	3.493	25.130	-0.077	0.704	1.065
120	0.165	1.222	3.299	24.350	-0.070	0.653	1.043
135	0.138	1.174	3.202	22.270	-0.066	0.572	1.032
150	0.109	1.128	3.153	19.210	-0.062	0.474	1.028
165	0.084	1.093	3.153	16.060	-0.060	0.383	1.026
180	0.074	1.080	3.153	14.650	-0.059	0.344	1.024
UPPER SEA STATE 6							
Relative Wave heading Angle	Load Factors in Local horizontal Axis System (2)		Ship Pitch and Roll (2)		Load Factors in Ship Deck Axis System		
	Ny	Nz	pitch	roll	nx(s)	ny(s)	nz(s)
0	0.187	1.502	7.273	15.940	-0.190	0.589	1.381
15	0.202	1.509	7.273	17.180	-0.191	0.635	1.371
30	0.236	1.524	7.119	20.740	-0.189	0.756	1.331
45	0.274	1.536	6.964	25.920	-0.186	0.913	1.252
60	0.303	1.535	6.654	31.490	-0.178	1.055	1.142
75	0.317	1.517	6.345	36.210	-0.168	1.146	1.029
90	0.314	1.480	5.958	39.150	-0.154	1.173	0.943
105	0.296	1.423	5.560	40.000	-0.138	1.137	0.895
120	0.263	1.354	5.262	38.840	-0.124	1.050	0.885
135	0.220	1.277	5.107	35.520	-0.114	0.918	0.908
150	0.174	1.204	5.029	30.640	-0.106	0.761	0.943
165	0.135	1.149	5.029	25.810	-0.101	0.616	0.974
180	0.118	1.127	5.029	23.370	-0.099	0.553	0.984

Table 2. Sea States (continued)

SHIPBOARD MOTION PARAMETERS FOR FGG-7 AT 10 KNOTS (1)							
UPPER SEA STATE 1							
Relative Wave heading Angle	**Load Factors in Local horizontal Axis System (2)**		**Ship Pitch and Roll (2)**		**Load Factors in Ship Deck Axis System**		
	Ny	**Nz**	**pitch**	**roll**	**nx(s)**	**ny(s)**	**nz(s)**
UPPER SEA STATE 7							
Relative Wave heading Angle	Load Factors in Local horizontal Axis System (2)		Ship Pitch and Roll (2)		Load Factors in Ship Deck Axis System		
	Ny	Nz	pitch	roll	nx(s)	ny(s)	nz(s)
0	0.363	1.600	14.100	30.910	-0.390	1.109	1.145
15	0.392	1.605	14.100	33.310	-0.391	1.182	1.086
30	0.458	1.614	13.730	40.000	-0.383	1.359	0.907
45	0.531	1.617	10.750	40.000	-0.302	1.428	0.876
60	0.587	1.604	8.452	40.000	-0.236	1.470	0.838
75	0.614	1.590	7.009	40.000	-0.194	1.485	0.814
90	0.609	1.513	6.087	40.000	-0.160	1.434	0.761
105	0.573	1.438	5.560	40.000	-0.139	1.359	0.728
120	0.510	1.350	5.418	40.000	-0.127	1.254	0.702
135	0.426	1.256	5.752	40.000	-0.126	1.130	0.683
150	0.338	1.170	6.566	40.000	-0.134	1.006	0.673
165	0.261	1.106	7.855	40.000	-0.151	0.904	0.671
180	0.228	1.081	8.609	40.000	-0.162	0.862	0.672
UPPER SEA STATE 8							
Relative Wave heading Angle	Load Factors in Local horizontal Axis System (2)		Ship Pitch and Roll (2)		Load Factors in Ship Deck Axis System		
	Ny	Nz	pitch	roll	nx(s)	ny(s)	nz(s)
0	0.554	2.487	18.250	40.000	-0.779	1.943	1.453
15	0.598	2.507	16.940	40.000	-0.730	2.000	1.453
30	0.699	2.551	13.730	40.000	-0.605	2.128	1.449
45	0.811	2.588	10.750	40.000	-0.483	2.256	1.426
60	0.896	2.585	8.452	40.000	-0.380	2.330	1.383
75	0.937	2.530	7.009	40.000	-0.309	2.332	1.322
90	0.930	2.420	6.087	40.000	-0.257	2.259	1.246
105	0.875	2.253	5.560	40.000	-0.218	2.112	1.155
120	0.779	2.047	5.410	40.000	-0.193	1.907	1.060
135	0.651	1.820	5.752	40.000	-0.182	1.663	0.969
150	0.515	1.605	6.566	40.000	-0.184	1.419	0.890
165	0.399	1.440	7.855	40.000	-0.197	1.223	0.836
180	0.348	1.376	8.609	40.000	-0.206	1.141	0.818

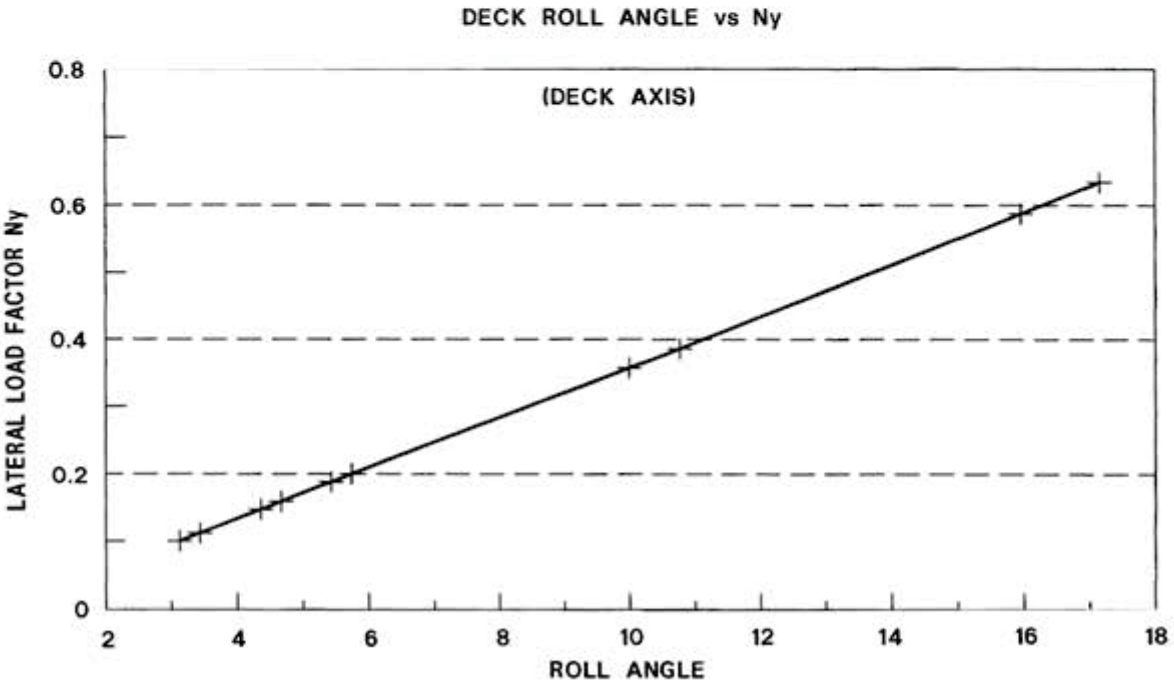
1 Ship Performance Department;·Report No. SPD-738-01; "A Non-Aviation Ship Motion Data Base for the DD 963, CG 26, FF 1052, FFG 7, and the FF 1040 Ship Classes"; dated December 1979.

2 The ny and nz load factors and the ship pitch and roll angles represent the highest expected value in 1000 successive wave encounters.



ICN-412-A-000001-B-97499-00002-A-001-01

Figure 1. Shipboard Deck Pitch Angle vs Nz (Sheet 1 of 1)



ICN-412-A-000001-B-97499-00003-A-001-01

Figure 2. Deck Roll Angle vs Ny (Sheet 1 of 1)

Requirements - General Description

This chapter depicts shipboard lashing and mooring line configurations at sea states and handling requirements.

Lashing is partial horizontal restraint of helicopter which may be accomplished with rotors turning or stopped.

Mooring is full restraint with rotors stopped.

Intentionally blank page

Lashing - Maintenance Procedure

Procedure

TIEDOWN LINES SHALL NOT BE ATTACHED TO ANY FUSELAGE HARDPOINT FITTINGS WHEN THE ROTOR IS TURNING.

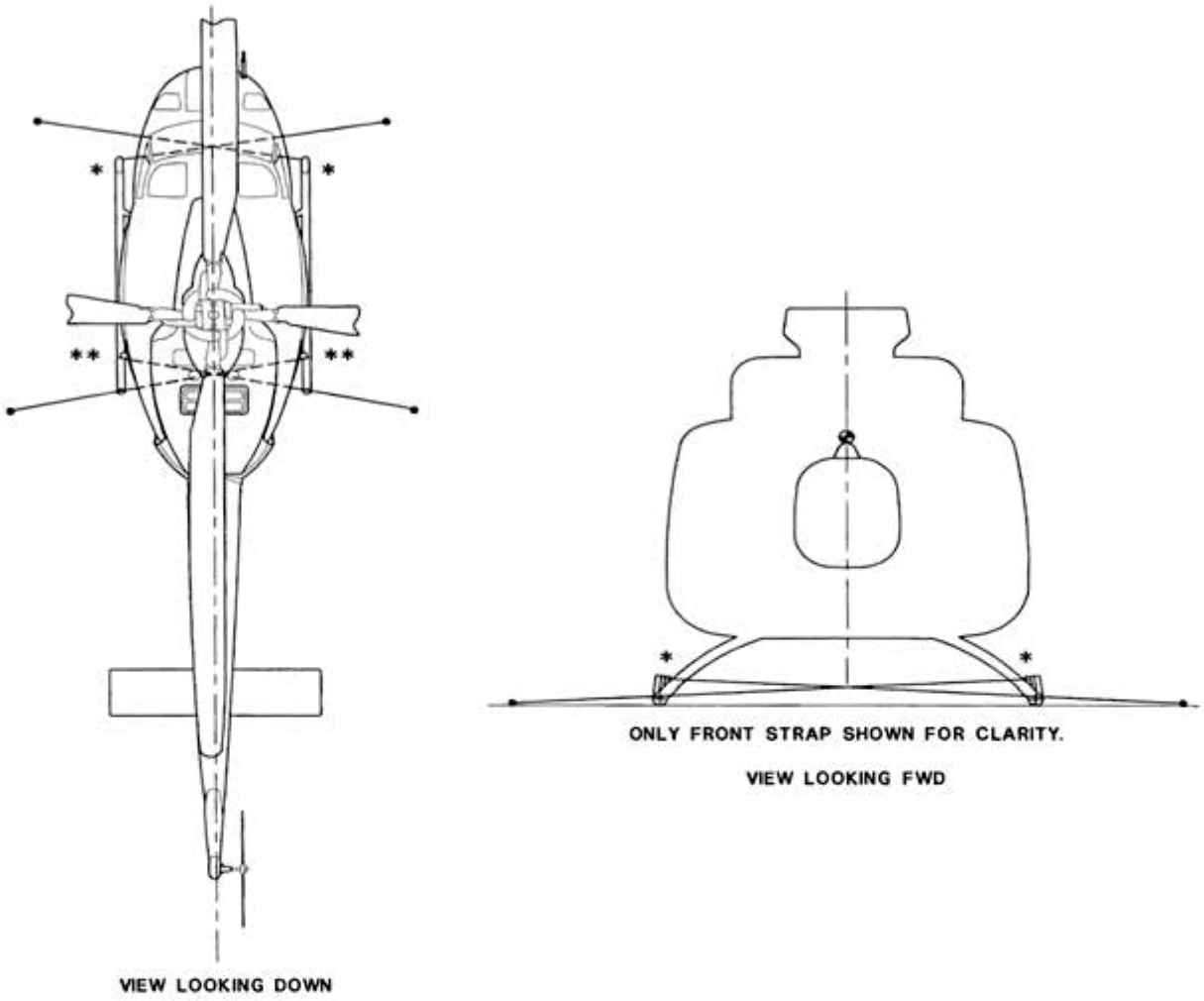
1. If lashing upon landing with rotors still turning, the following conditions must not be exceeded Figure 1:

1.1. Lashing lines must be hand tightened only, with a minimum of 1 inch slack.

1.2. Roll: 10°

1.3. Pitch: 10°

1.4. Wind: 60 knots head on, or 35 knots cross wind.



NOTES

* Tow skid ring loops

** Aft cross-tube saddles or aft tow rings if installed.

• Nearest available deck tiedown

Lashing lines must be hand tightened only with a minimum of 1.0 inch (25.4 mm) slack.

ICN-412-A-021000-B-97499-00001-A-001-01

Figure 1. Lashing (Rotor Turning) (Sheet 1 of 1)

Mooring - Normal Weather - Maintenance Procedure

Procedure

WARNING

TIEDOWN LINES SHALL NOT BE ATTACHED TO ANY FUSELAGE HARDPOINT FITTINGS WHEN THE ROTOR IS TURNING.

1. Under normal weather conditions sea state 3 or less with helicopter on deck, the following conditions must not be exceeded Figure 1:

1.1. Mooring lines must be hand tightened only, with a minimum of 1 inch slack.

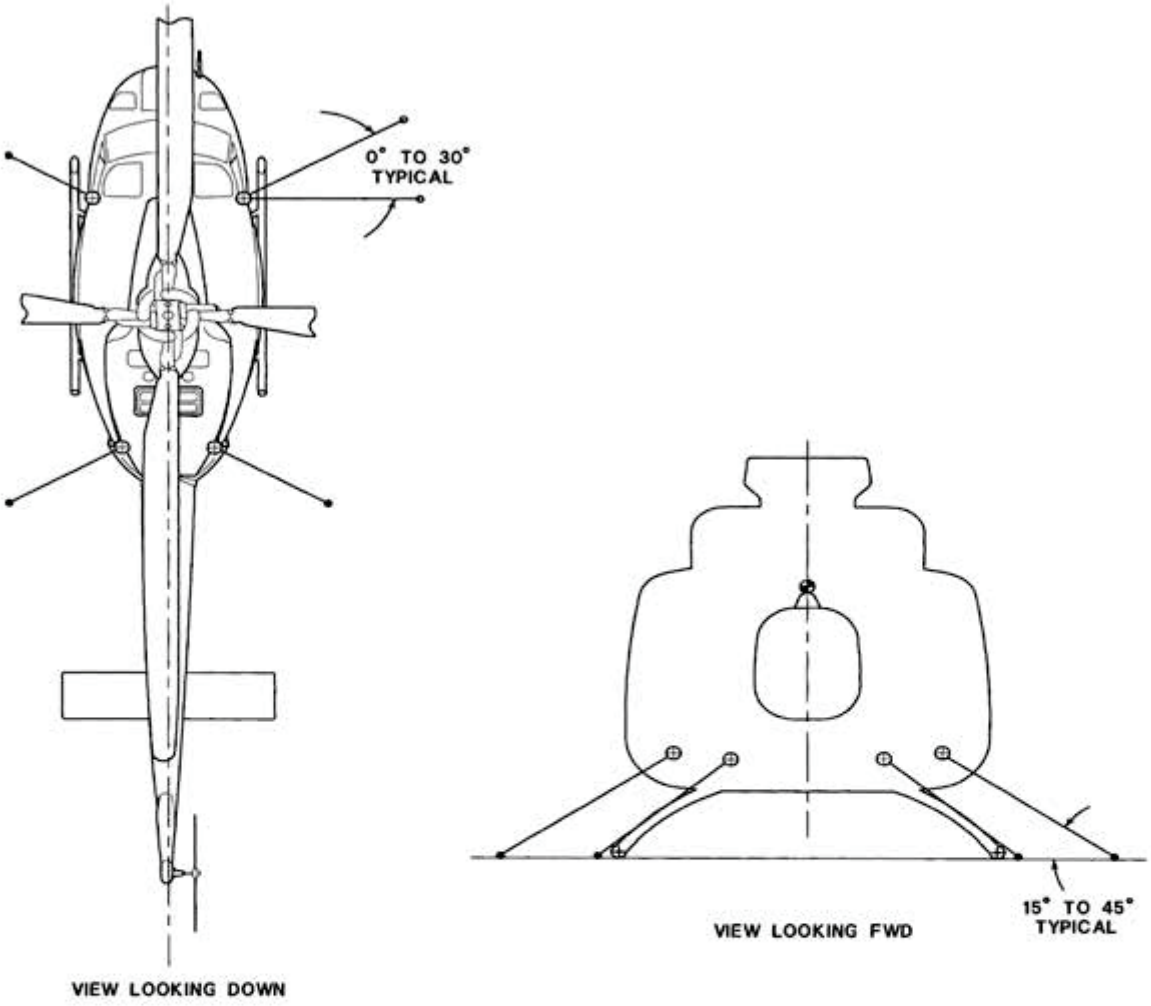
1.2. Roll: 7°

1.3. Pitch: 2°

1.4. Wind: 60 knots any direction.

2. Install main rotor folding kit and forward cross-tube strap (between forward tow rings) at wind speeds above 45 knots.

3. Recommend use of high gross weight towing strap on aft cross-tube.



NOTES

Install main rotor blade folding kit and forward crosstube strap at wind speeds above 45 knots (See figure 2-5).

⊖ Fuselage tiedown rings.

ICN-412-A-021000-B-97499-00002-A-001-01

Figure 1. Mooring-Normal Weather (Sea State 3 or Less) on Deck (Rotor Stopped) (Sheet 1 of 1)

Mooring - Moderate Weather on Deck - Maintenance Procedure

Procedure

TIEDOWN LINES SHALL NOT BE ATTACHED TO ANY FUSELAGE HARDPOINT FITTINGS WHEN THE ROTOR IS TURNING.

1. In moderate weather sea states 4 or less, the following conditions must not be exceeded Figure 1:

1.1. Install mooring lines.

1.2. Roll: 13°

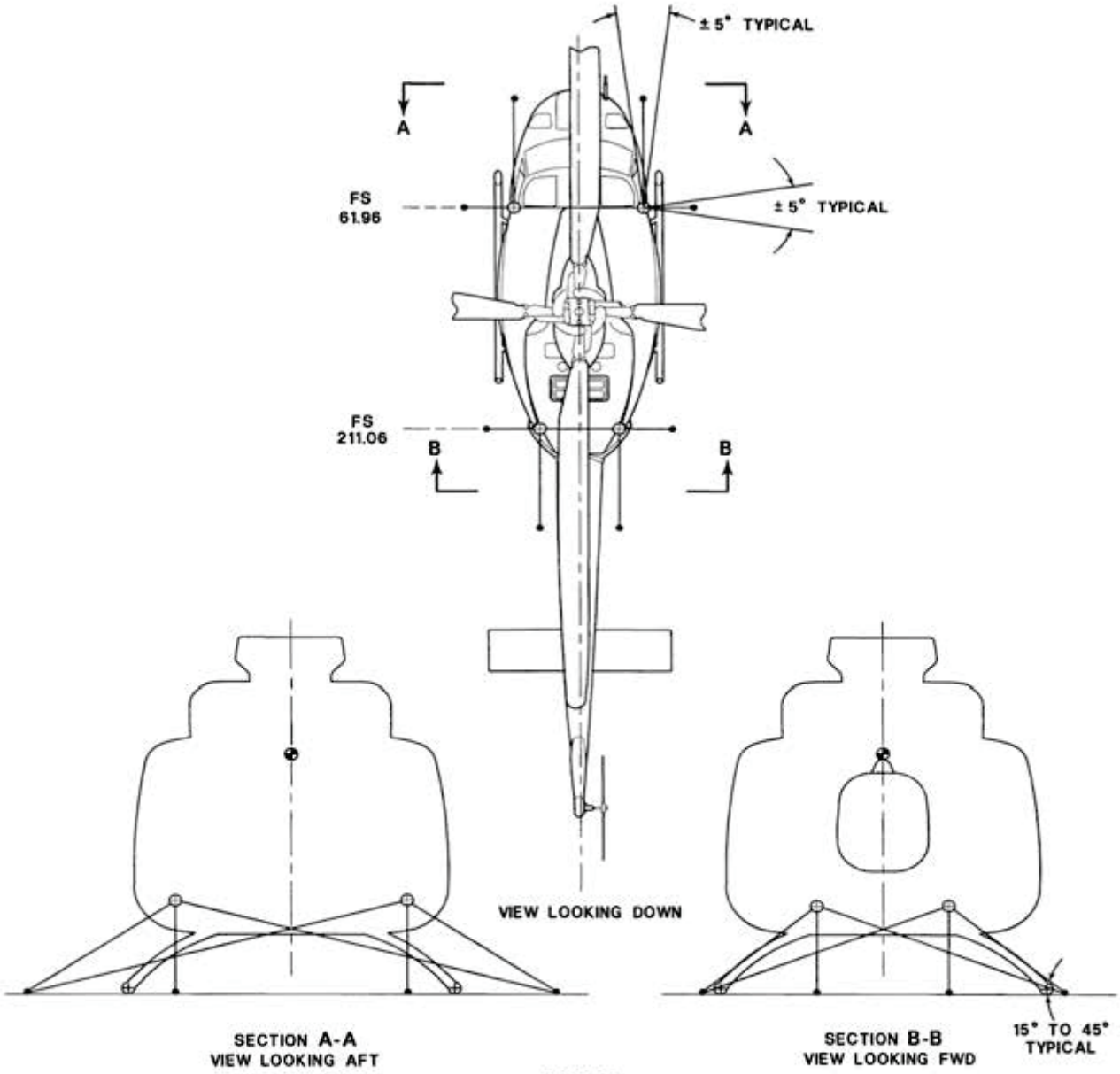
1.3. Pitch: 8°

1.4. Wind: 60 knots any direction.

2. Install main rotor blade folding kit and forward cross-tube strap (between forward tow rings) at wind speeds above 45 knots.

3. Use of high gross weight towing strap on aft cross-tube is required.

4. Use of forward cross-tube strap is required for roll angles or over 10°.



NOTES

Install main rotor blade folding kit and forward crosstube strap at wind speeds above 45 knots. (See figure 2-5).

⊕ Fuselage tiedown rings.

ICN-412-A-021000-B-97499-00003-A-001-01

Figure 1. Mooring-Moderate Weather (Sea State 4 or Less) on Deck (Sheet 1 of 1)

Mooring - Heavy Weather in Hanger - Maintenance Procedure

Procedure

1. In heavy weather at sea state greater than 4, the following conditions must not be exceeded Figure 1:

1.1. Roll: 45°

1.2. Pitch: 8°

1.3. Wind: N/A (in hangar).

2. Install main rotor blade folding kit prior to moving in hangar.

NOTE

It is recommended that jacks be secured to floor to prevent movement.

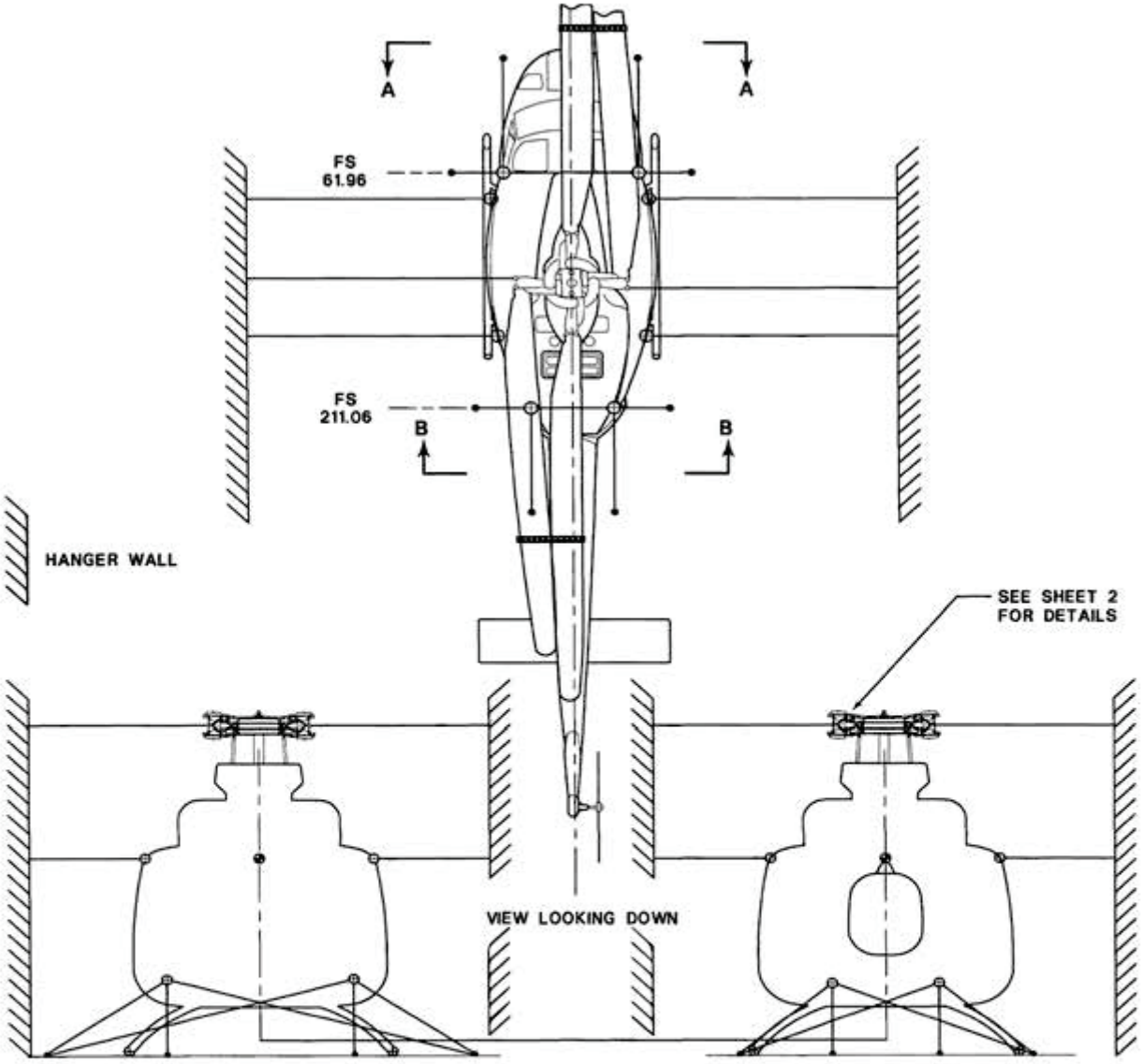
3. Helicopter shall be stabilized with jacks at four (4) jack points as follows.

3.1. Refer to 412-A-00-00-00-00A-009A-A / 00001 for jacking procedures.

3.2. Jack until ram contacts jack fittings.

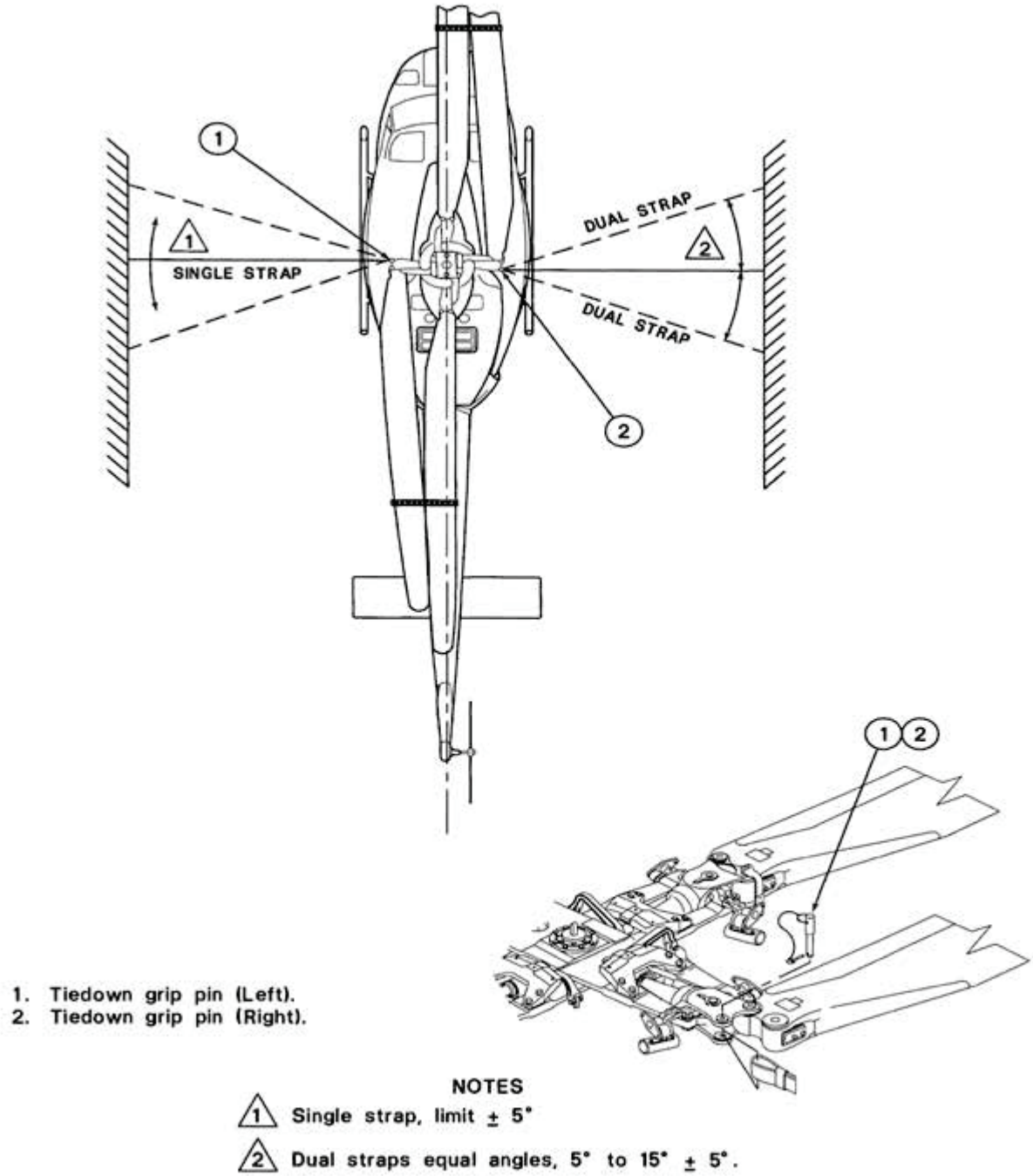
3.3. Stabilize helicopter by jacking approximately 0.5 inch (12.7 mm) leaving some weight bearing on skids.

3.4. Secure helicopter with mooring lines as shown in Figure 1.



ICN-412-A-021000-B-97499-00004-A-001-01

Figure 1. Mooring-Heavy Weather (Sea State Greater Than 4) in Hangar (Sheet 1 of 2)



ICN-412-A-021000-B-97499-00005-A-001-01

Figure 1. Mooring-Heavy Weather (Sea State Greater Than 4) in Hangar (Sheet 2 of 2)

Intentionally blank page

Secure Mast - Maintenance Procedure

Procedure

1. During heavy weather with the main rotor blades folded and helicopter moored in hanger it is necessary to reduce lateral motion of mast and rotor head with respect to the airframe (Figure 1, Sheet 2).

1.1. Secure mast as follows.

1.1.1. If installed, remove main rotor grip pins (1) from left and (2) from right side rotor hub. Pins may be temporarily installed in the rotor blade grip.

1.1.2. Insert 3 inch (76.2 mm) minimum width nylon webbing, (minimum breaking strength of 10,000 lbs) with a loop of at least 6 inch (162.4 mm) diameter formed in the end, into gap in the rotor hub.

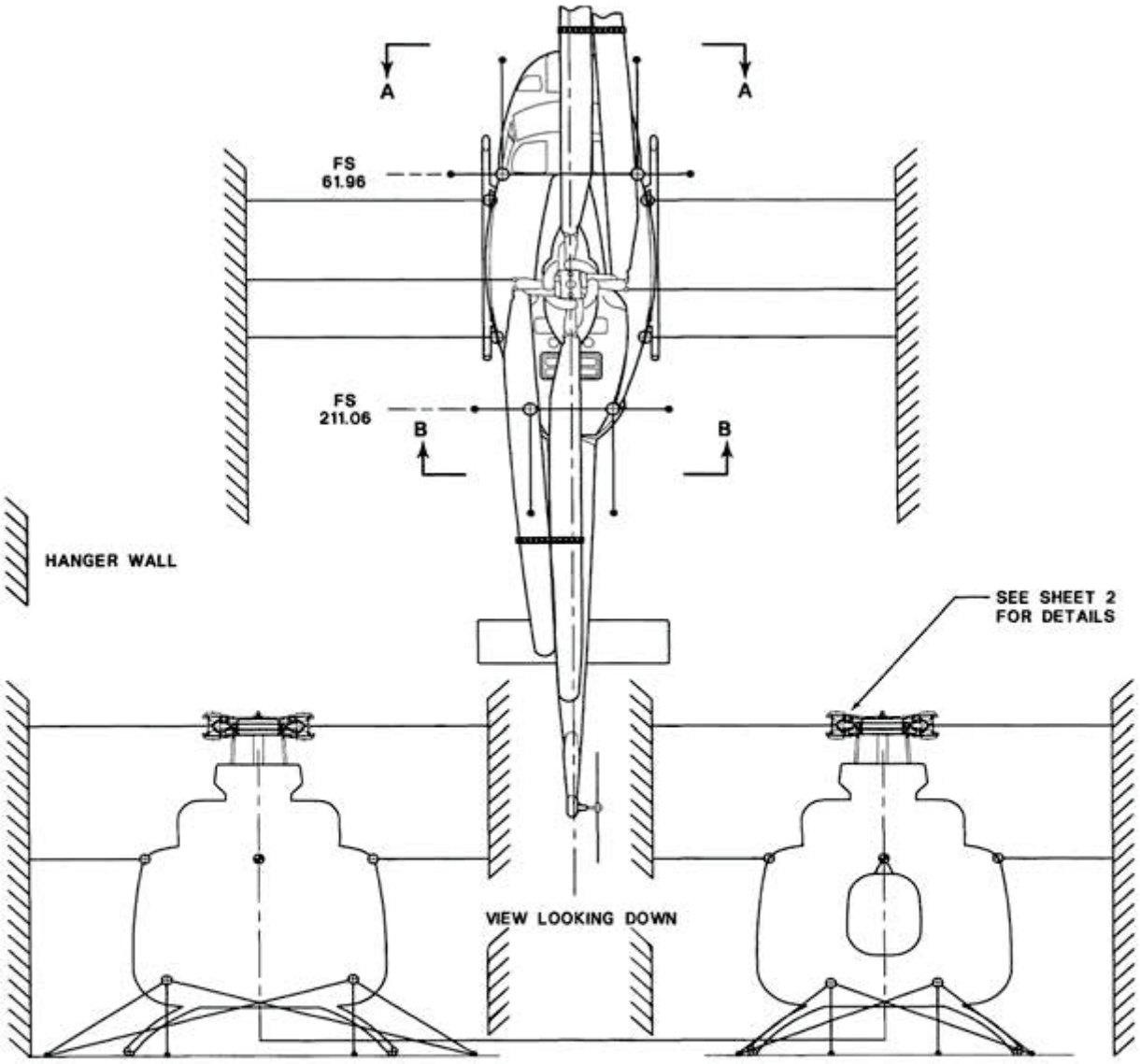
1.1.3. Insert special shipboard tiedown grip pins (1 and 2) into empty main rotor hub grip pin holes trapping web loop.

DO NOT USE FLIGHT GRIP PIN FOR MOORING PURPOSES, USE ONLY THE SPECIAL MOORING PIN.

1.1.4. Attach outboard ends of the webbed mooring line to hanger wall. Webbing must be positioned horizontally and directly outboard within (5° angles) from rotor as not to apply rotational forces on rotor.

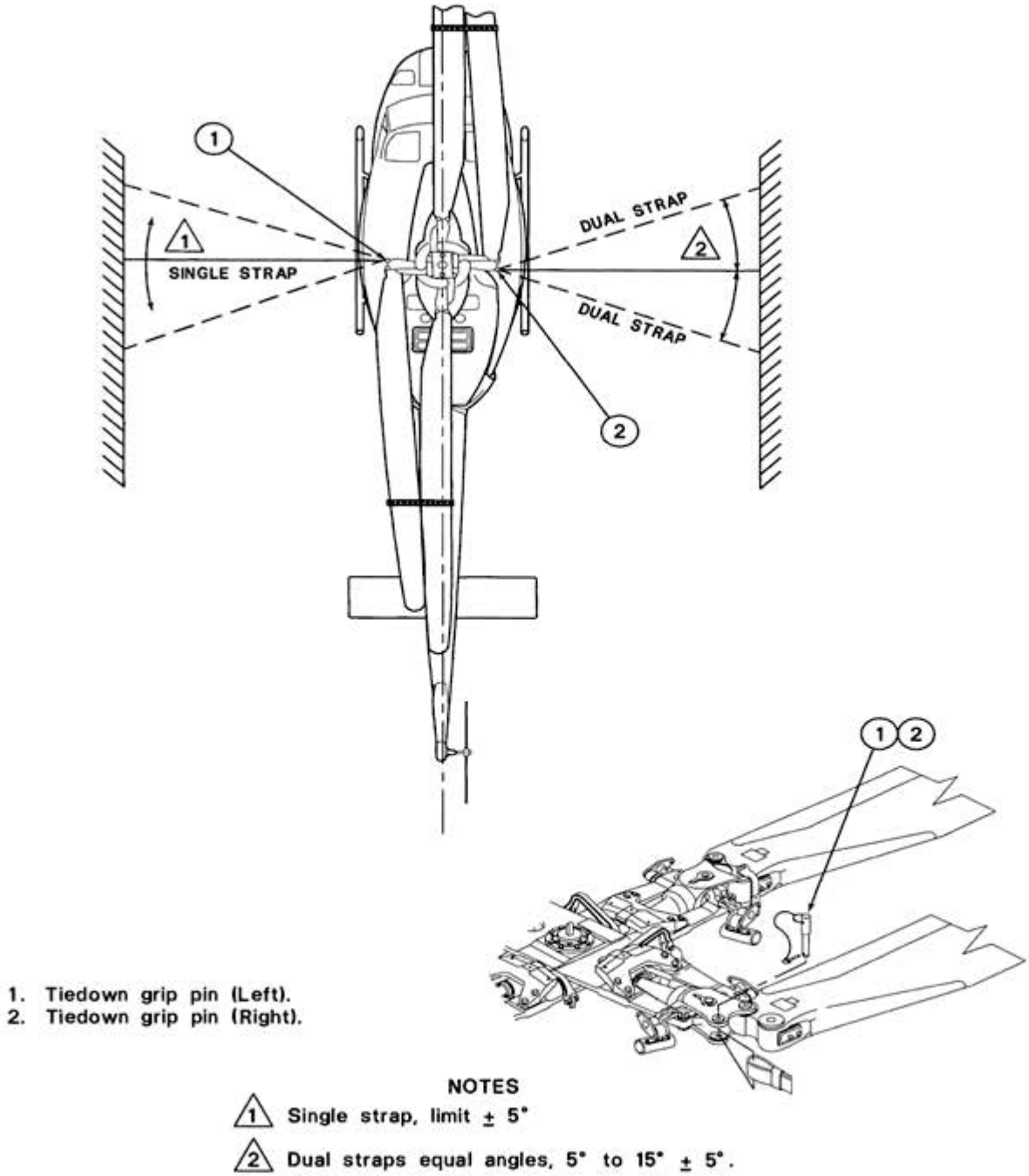
1.1.5. If limit of 5° angles are not practical, run two lines at equal angles 5° to 15° ± 5° to allow for variable positioning of helicopter.

1.1.6. Apply equal tension left and right mooring lines to prevent lateral movement of the mast and hub.



ICN-412-A-021000-B-97499-00004-A-001-01

Figure 1. Mooring-Heavy Weather (Sea State Greater Than 4) in Hangar (Sheet 1 of 2)



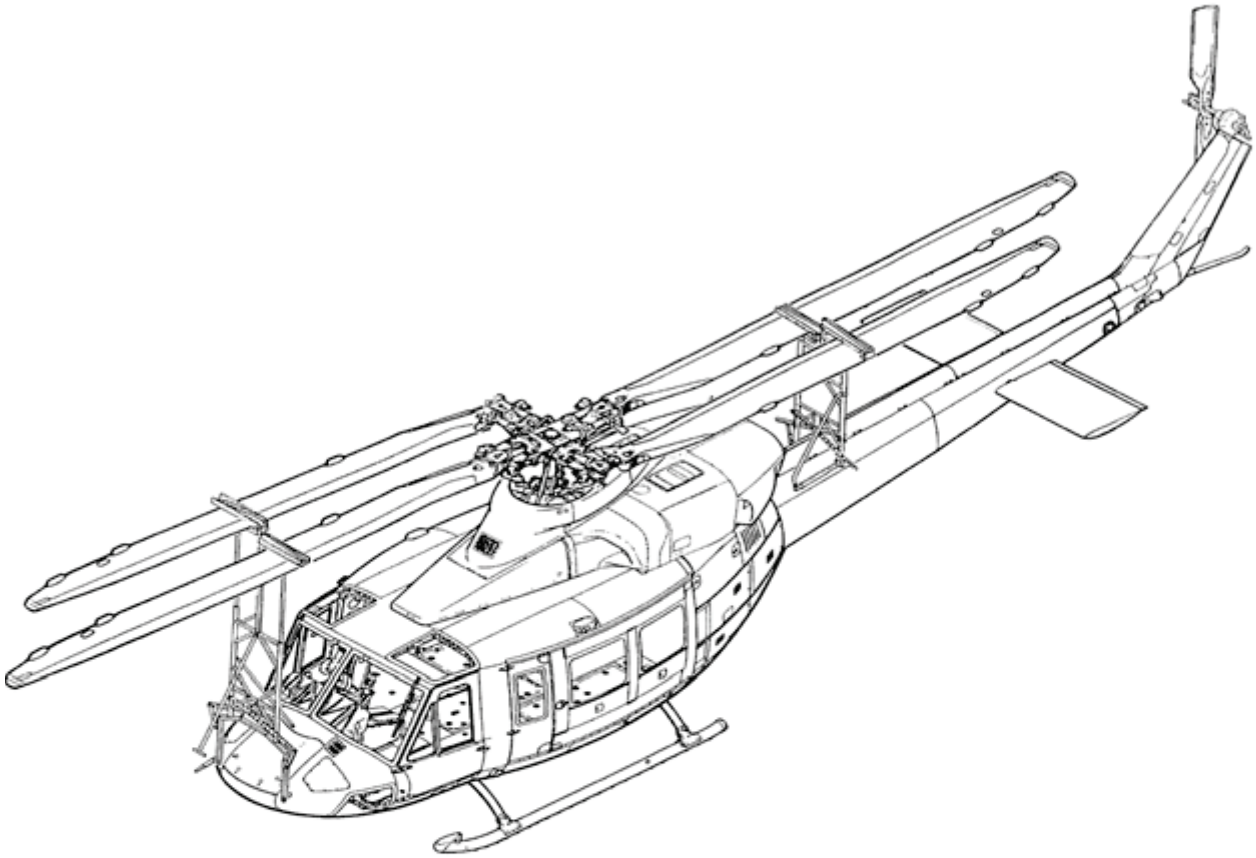
ICN-412-A-021000-B-97499-00005-A-001-01

Figure 1. Mooring-Heavy Weather (Sea State Greater Than 4) in Hangar (Sheet 2 of 2)

Intentionally blank page

Main Rotor Blade Folding Kit - Description

Fold main rotor blades into main rotor blade folding kit (Figure 1).



ICN-412-A-021000-B-97499-00006-A-001-01

Figure 1. Main Rotor Blade Folding Kit (Sheet 1 of 1)

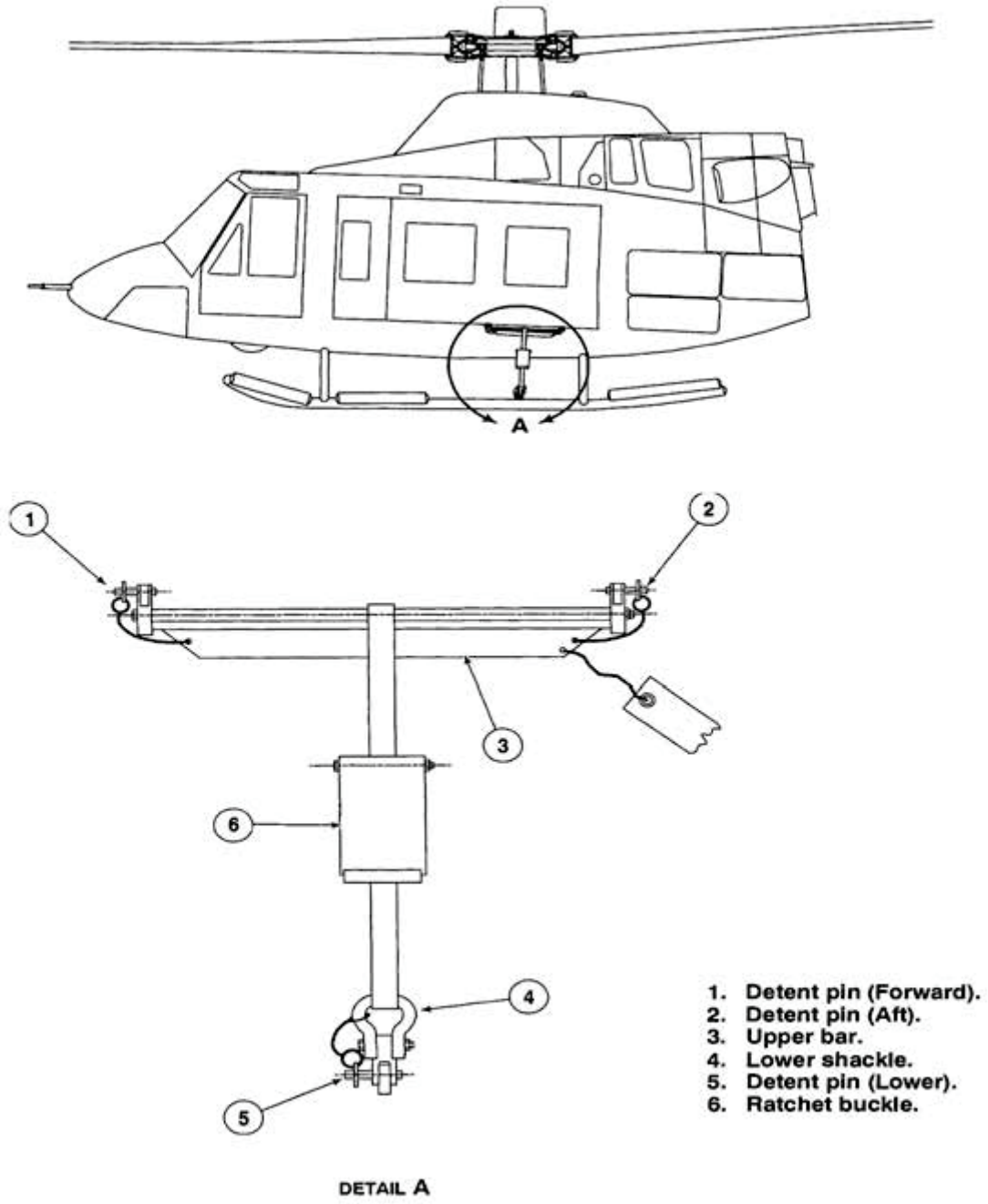
Handling - Maintenance Procedure

Procedure

WARNING

MAINTENANCE ROLL STABILIZER STRAPS SHALL NOT BE INSTALLED WHILE THE ROTOR IS TURNING.

1. For additional stability, for maintenance purposes, maintenance roll stabilizer straps may be installed (Figure 1).



ICN-412-A-021000-B-97499-00007-A-001-01

Figure 1. (MRSS) Maintenance Roll Stabilizer Strap (Sheet 1 of 1)

Roll Stabilizer - Maintenance Procedure

Procedure

1. Function

1.1. Maintenance roll stabilizer may be utilized as necessary to reduce roll of helicopter with respect to vessels deck.

1.2. Intended function of stabilizer is to increase safety of flight operations and maintenance personnel during severe weather operations.

1.3. Roll stabilizers may be used at operator's discretion anytime engines are not running and ground handling wheels are not installed.

1.4. Roll stabilizers fit on left and right side of cabin. They extend from hard points at cabin floor level to wheel handling gear lug on skid tube.

1.5. Stabilizers are to be used only in pairs of one left and one right.

2. Installation

NOTE

Ensure electrically actuated passenger steps are in up position.

2.1. Prepare stabilizers for installation by removing two ball detent pins (1 and 2 Figure 1) in upper bar (3) and one ball detent pin (5) in shackle.

2.2. Release ratchet buckle (6) and extend strap to its full length.

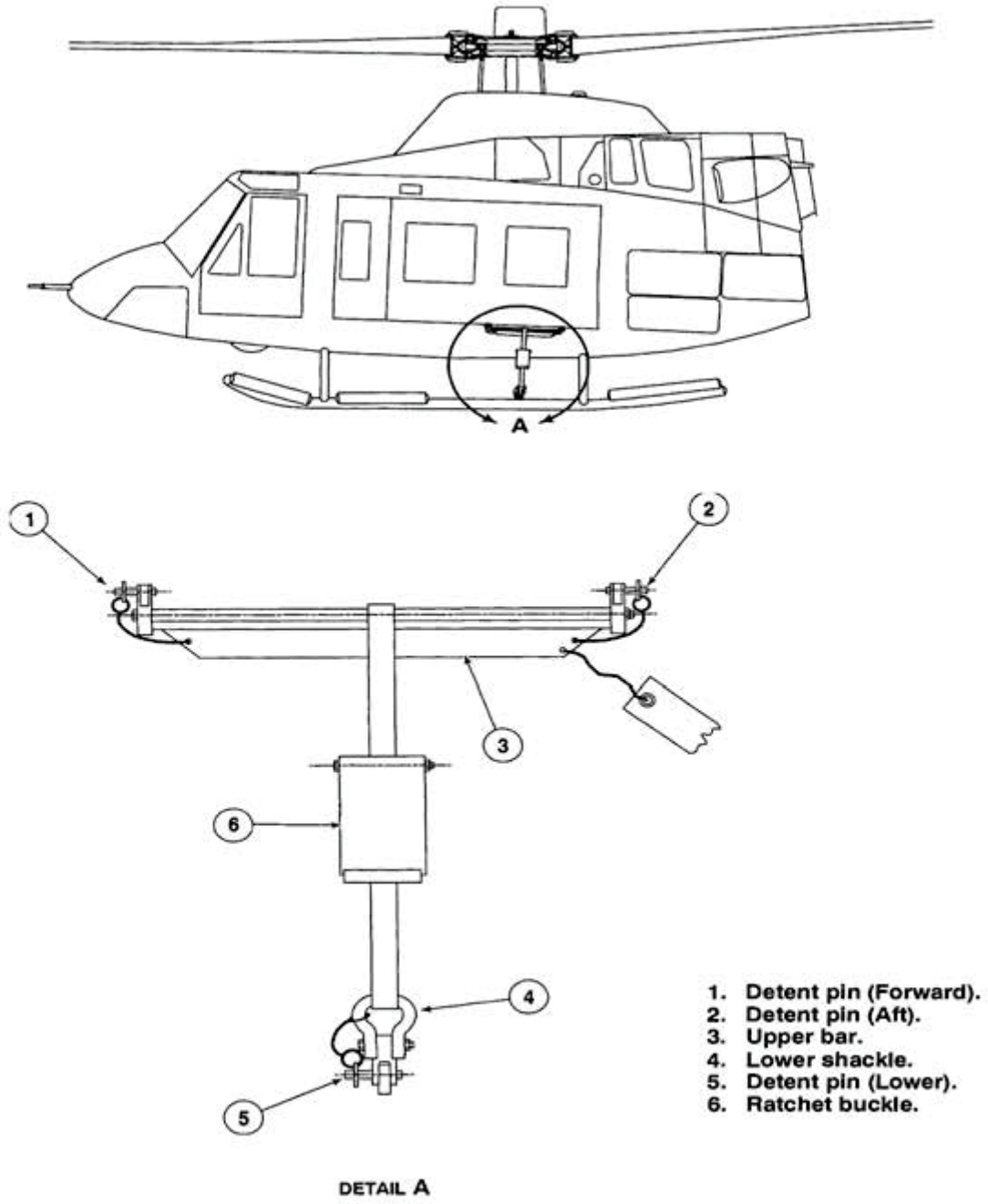
2.3. Locate upper bar (3) over aft cabin floor line fuselage hard points and insert two ball detent pins (1 and 2).

MAINTENANCE ROLL STABILIZER STRAPS SHALL NOT BE IN STALLED WHILE THE ROTOR IS TURNING.

2.4. Locate lower shackle (4) over forward ground handling wheel attach lug and insert ball detent pin (5).

2.5. Using ratchet buckle (6), take up all slack in web strap on left and right units.

2.6. Apply additional tension as evenly as possible to both units as necessary to dampen airframe roll.



ICN-412-A-021000-B-97499-00007-A-001-01

Figure 1. (MRSS) Maintenance Roll Stabilizer Strap (Sheet 1 of 1)

Deck Landing - Maintenance Procedure

Procedure

IF AUTOMATIC FLIGHT CONTROL SYSTEM (AFCS) IS ENGAGED THE ROTORS WILL MOVE IN RESPONSE TO SHIP MOVEMENTS.

1. Pilot lands helicopter on deck landing area and in line with deck markings:

1.1. Orientation: As required to wind direction. Along the ship's centerline or one of the two 25° centerlines.

1.2. Position: Landing gear must be within touchdown circle.

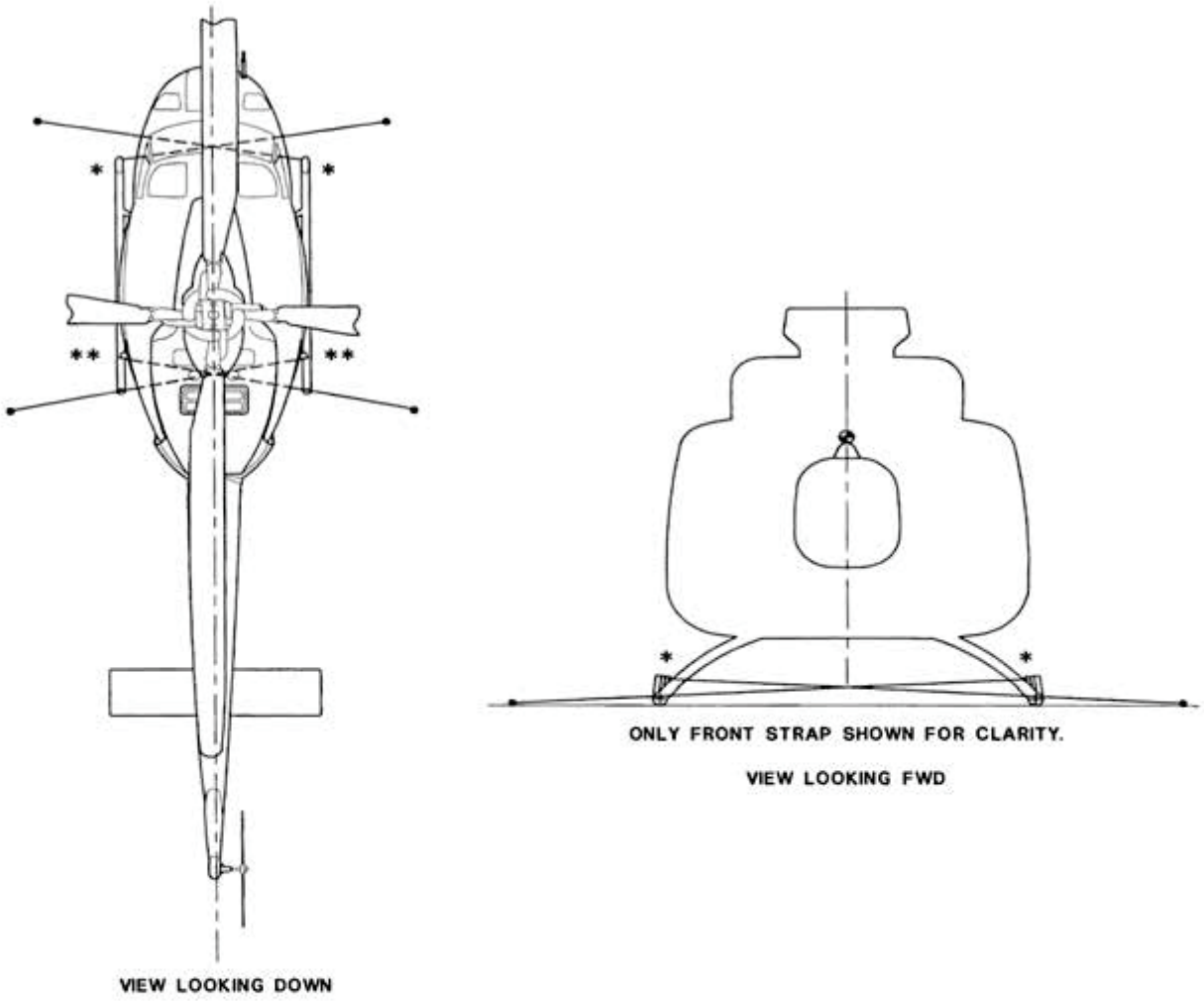
Intentionally blank page

Lashing - Maintenance Procedure

Procedure

TIEDOWN LINES SHALL NOT BE ATTACHED TO ANY FUSELAGE HARD POINT FITTINGS WHEN THE ROTOR IS TURNING.

1. When deck landing has been completed, flight deck crew should attach lashings to skid tube tow rings and to aft crosstube saddles (Figure 1). These lashings will prevent lateral shifting of helicopter as ship rolls. They must remain slack (not taut) with minimum of 1 inch slack, until rotor is stopped. After rotor is stopped, tighten lashings and add primary lines as the sea conditions require (Figure 2 and Figure 3).



NOTES

* Tow skid ring loops

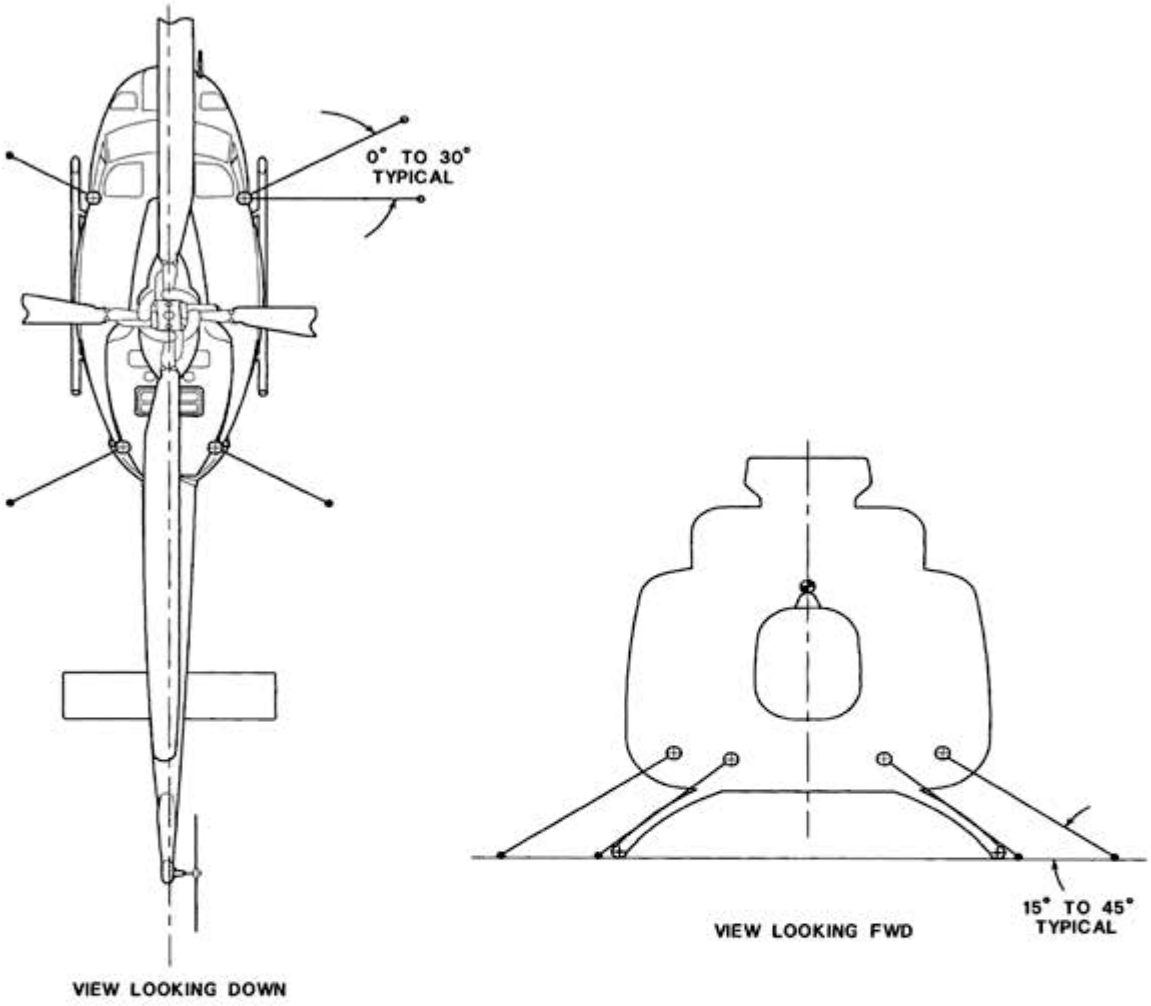
** Aft cross-tube saddles or aft tow rings if installed.

• Nearest available deck tiedown

Lashing lines must be hand tightened only with a minimum of 1.0 inch (25.4 mm) slack.

ICN-412-A-021000-B-97499-00001-A-001-01

Figure 1. Lashing (Rotor Turning) (Sheet 1 of 1)



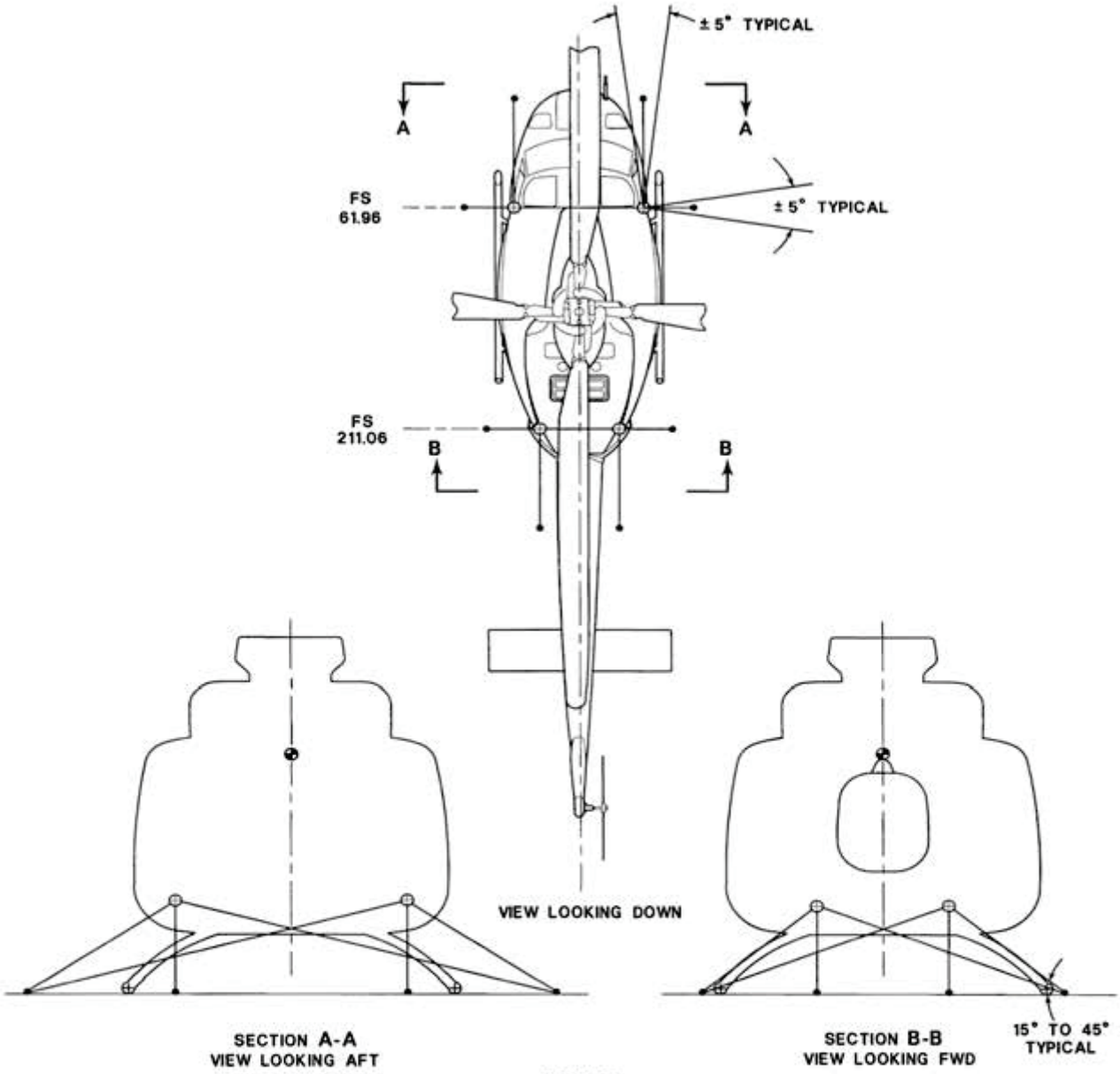
NOTES

Install main rotor blade folding kit and forward crosstube strap at wind speeds above 45 knots (See figure 2-5).

⊖ Fuselage tiedown rings.

ICN-412-A-021000-B-97499-00002-A-001-01

Figure 2. Mooring-Normal Weather (Sea State 3 or Less) on Deck (Rotor Stopped) (Sheet 1 of 1)



ICN-412-A-021000-B-97499-00003-A-001-01

Figure 3. Mooring-Moderate Weather (Sea State 4 or Less) on Deck (Sheet 1 of 1)

Shutdown - Description

Once rotor has stopped, tighten lashings and add primary mooring lines as required.

Intentionally blank page

Mooring - Maintenance Procedure

Procedure

MAINTENANCE ROLL STABILIZER STRAPS SHALL NOT BE INSTALLED WHILE THE ROTOR IS TURNING.

1. Moor helicopter as required.

2. For additional stability, for maintenance purposes, maintenance roll stabilizer straps may be installed.

Intentionally blank page

Fold Main Rotor Blades - Maintenance Procedure

Procedure

1. Refer to 412-A-00-00-00-00A-009A-A / 00001 and BHT-412-SI-59 for main rotor blades folding procedures.

Intentionally blank page

Forward Displacement - Maintenance Procedure

Procedure

1. Preparation for handling:

1.1. If required, attach high gross weight towing strap.

1.2. Attach and secure suitable tow bar.

1.3. Remove lash down and mooring system.

2. Handling:

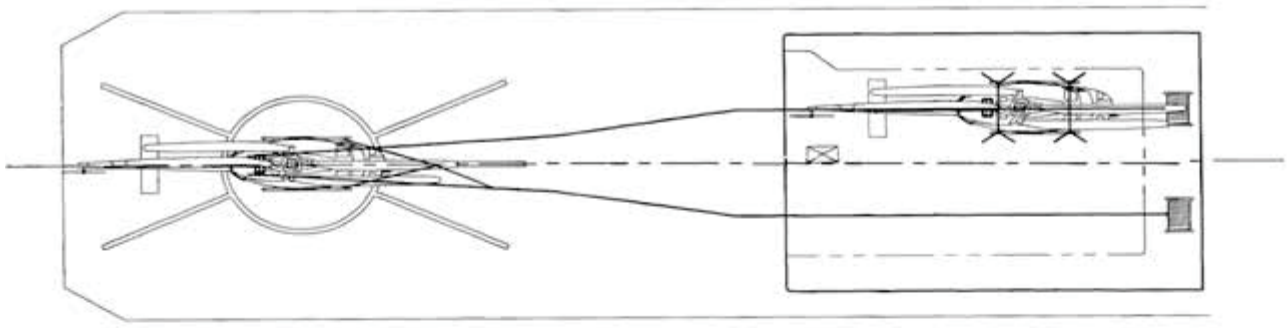
2.1. Move helicopter forward towards its hangar by gradually increasing cable load (Figure 1).

2.2. Helicopter will automatically line up with its rails as it moves forward.

2.3. Move helicopter into hangar to its mooring position (Figure 2).

3. Mooring:

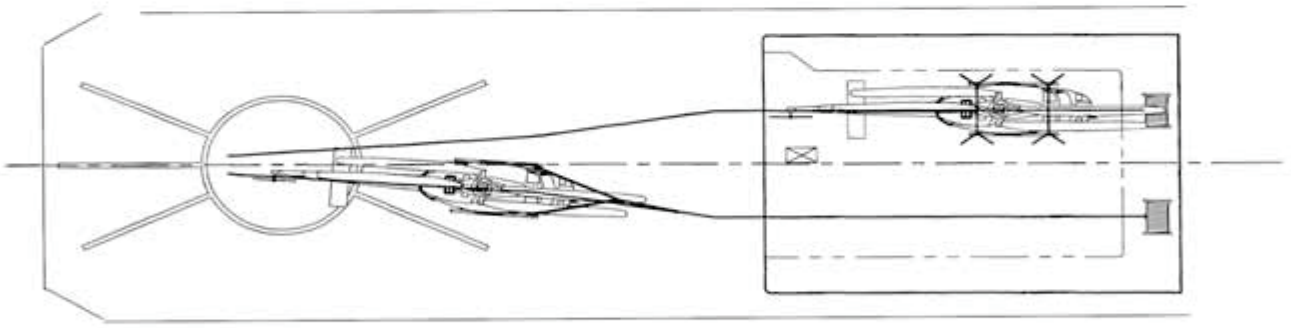
3.1. Moor helicopter (Figure 3).



NOTE ATTACHED TOW BAR TO FWD TOW BAR SUPPORT RINGS
PULLING HELICOPTER TOWARDS HANGAR

ICN-412-A-021000-B-97499-00008-A-001-01

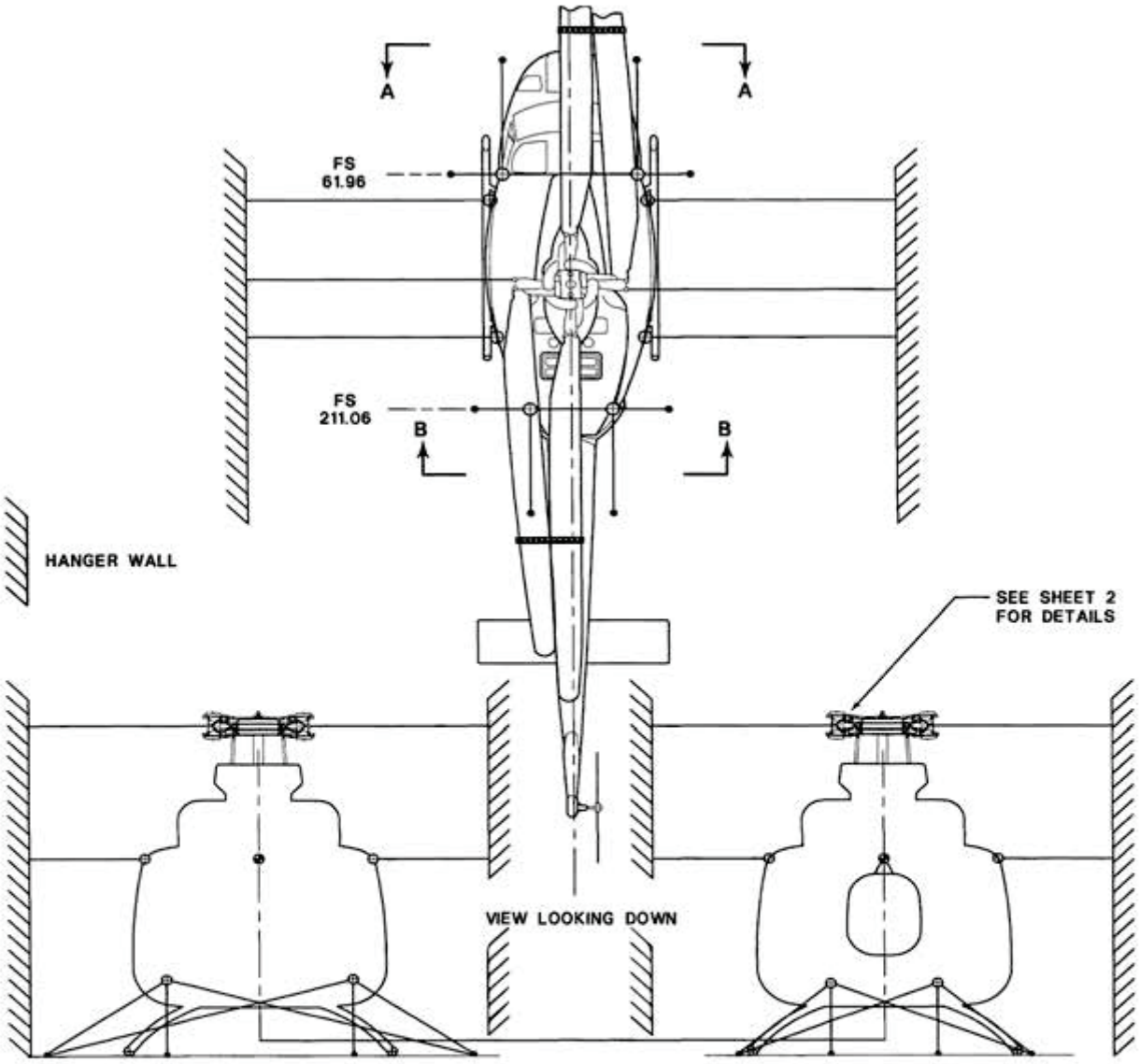
Figure 1. Forward Displacement (Sheet 1 of 1)



HELICOPTER MOVEMENT TO HANGAR

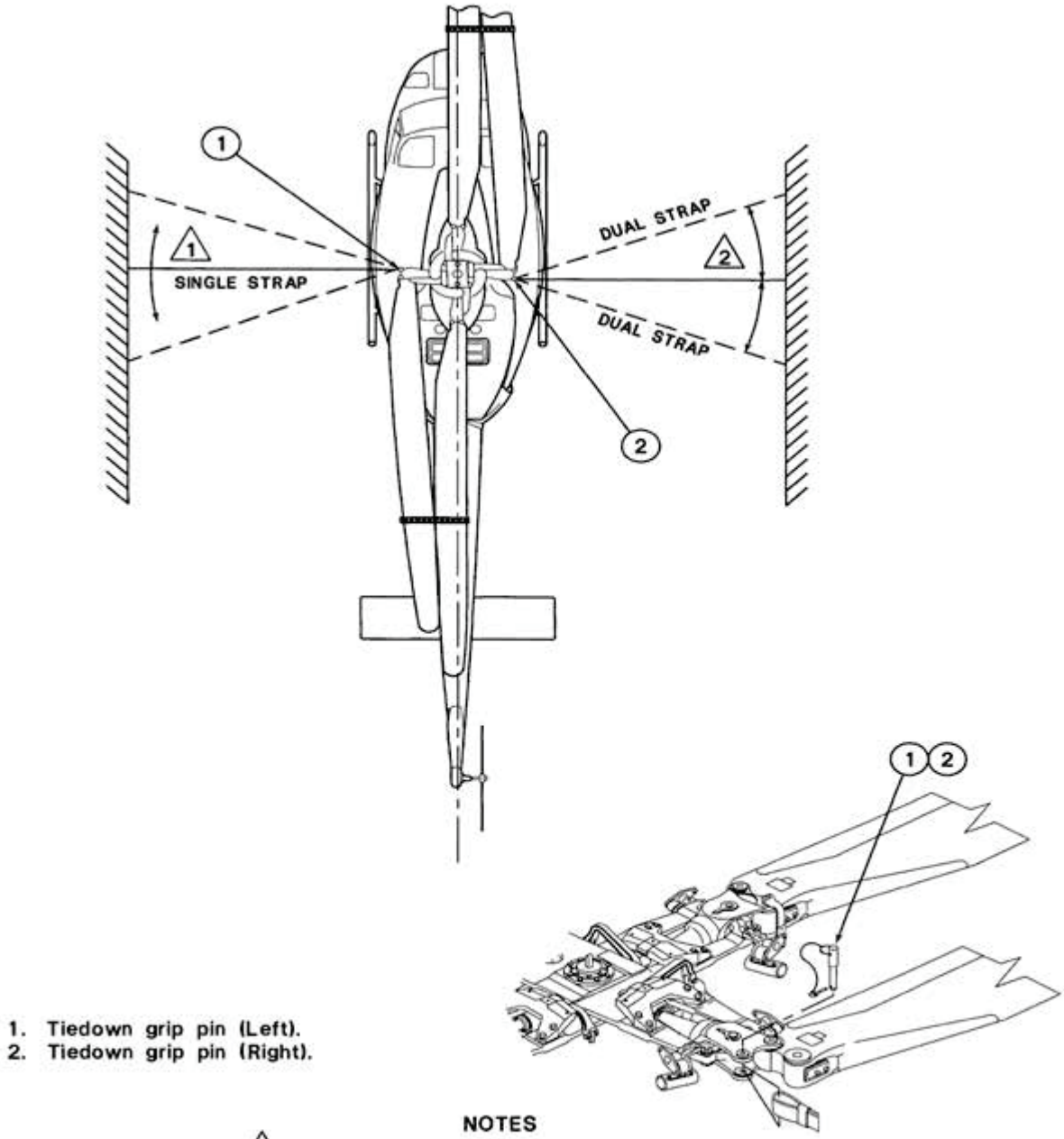
ICN-412-A-021000-B-97499-00009-A-001-01

Figure 2. Forward Displacement (Sheet 1 of 1)



ICN-412-A-021000-B-97499-00004-A-001-01

Figure 3. Mooring-Heavy Weather (Sea State Greater Than 4) in Hangar (Sheet 1 of 2)



1. Tiedown grip pin (Left).
2. Tiedown grip pin (Right).

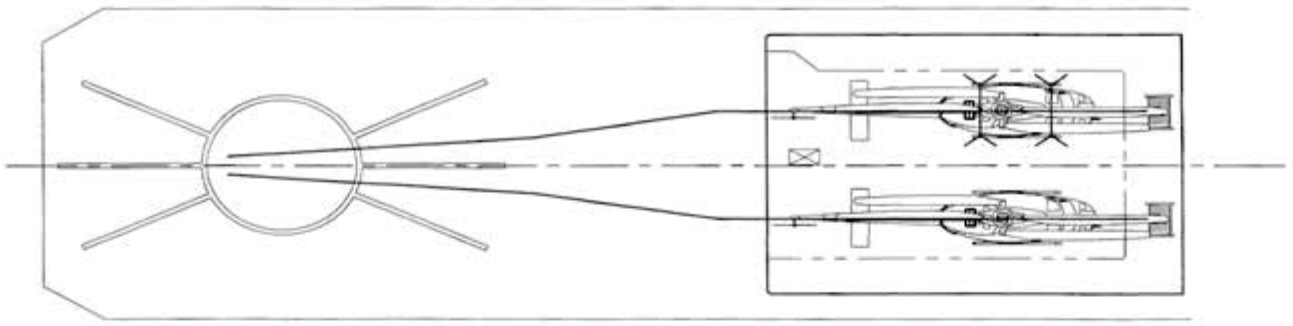
NOTES

1 Single strap, limit ± 5°

2 Dual straps equal angles, 5° to 15° ± 5°.

ICN-412-A-021000-B-97499-00005-A-001-01

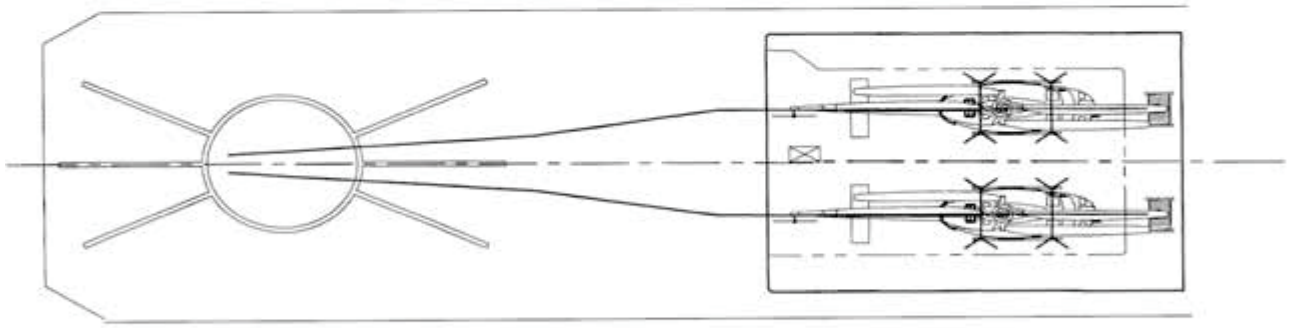
Figure 3. Mooring-Heavy Weather (Sea State Greater Than 4) in Hangar (Sheet 2 of 2)



HELICOPTER IN HANGAR

ICN-412-A-021000-B-97499-00010-A-001-01

Figure 4. Forward Displacement (Sheet 1 of 1)



MOORING THE HELICOPTER

ICN-412-A-021000-B-97499-00011-A-001-01

Figure 5. Forward Displacement (Sheet 1 of 1)

Intentionally blank page

Rearward Displacement - Maintenance Procedure

Procedure

1. Preparation for handling:

1.1. Attach and secure tow bar kit to aft tow bar support rings.

1.2. Remove lashing and mooring system.

2. Handling:

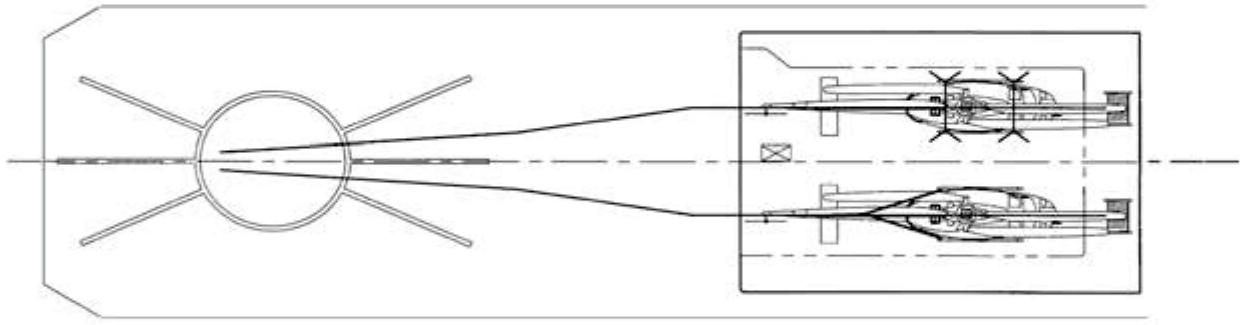
2.1. Move helicopter backward towards takeoff area by increasing cable tension (Figure 1).

2.2. Stop helicopter when it reaches take off area (Figure 2).

3. Moor helicopter as required.

4. Unfold main rotor blades (Figure 3).

Refer to 412-A-00-00-00-00A-009A-A / 00001 and BHT-412-SI-59.

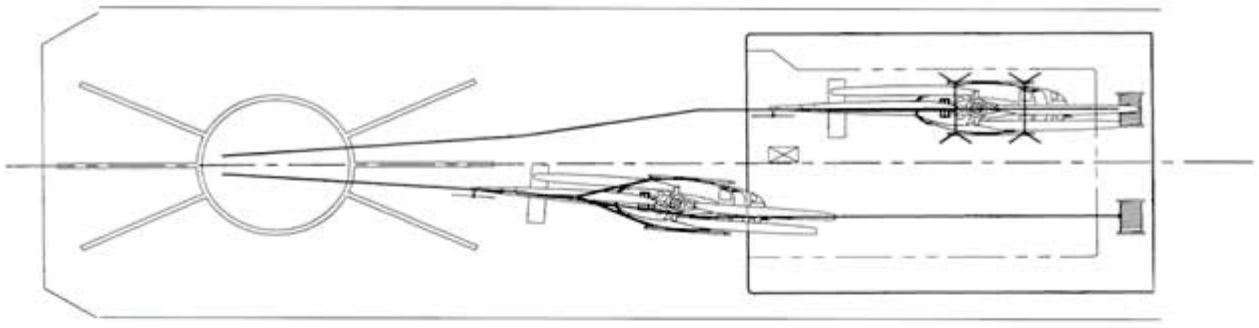


NOTE ATTACHED TOW BAR TO AFT TOW BAR SUPPORT RINGS

PULLING HELICOPTER TOWARDS TAKEOFF AREA

ICN-412-A-021000-B-97499-00012-A-001-01

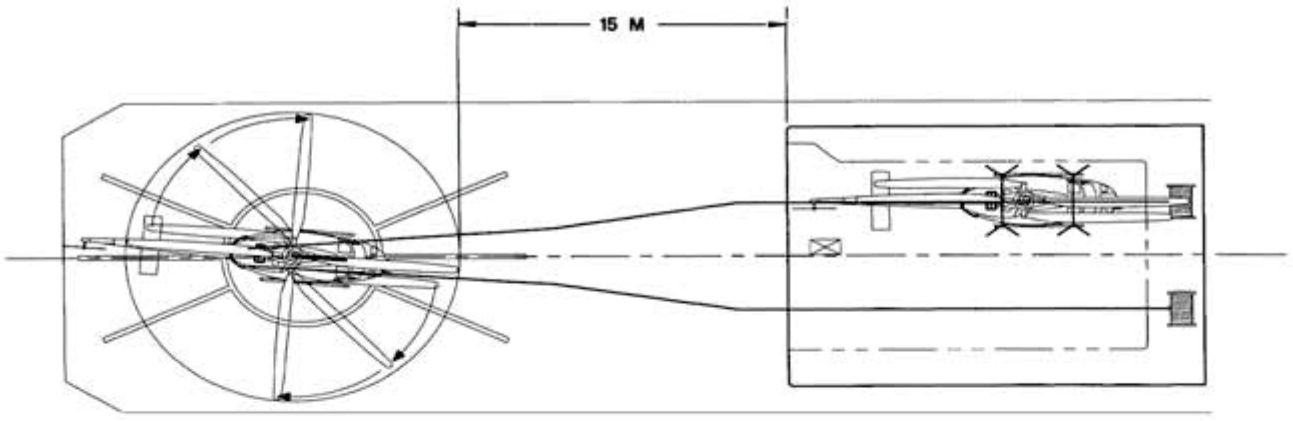
Figure 1. Rearward Displacement (Sheet 1 of 1)



HELICOPTER MOVEMENT OUT OF HANGAR TO TAKEOFF AREA

ICN-412-A-021000-B-97499-00013-A-001-01

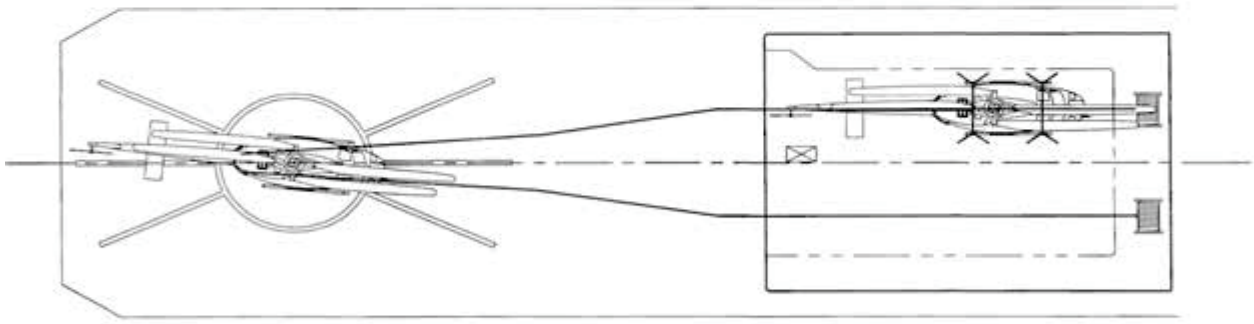
Figure 2. Rearward Displacement (Sheet 1 of 1)



UNFOLDING THE MAIN ROTOR BLADES

ICN-412-A-021000-B-97499-00015-A-001-01

Figure 3. Transfer From Hangar and Take-Off (Sheet 1 of 1)



HELICOPTER AT TAKEOFF AREA

ICN-412-A-021000-B-97499-00014-A-001-01

Figure 4. Rearward Displacement (Sheet 1 of 1)

Intentionally blank page

Launch - Maintenance Procedure

Procedure

1. Before launch helicopter should be positioned on deck landing area and in line with deck marking:

1.1. Orientation: Into wind. Along the ship's centering or one of two 25° centerlines.

1.2. Position: Landing gear must be within touchdown circle.

1.3. Lashings may be attached loosely until shortly before launch.

Intentionally blank page

Engine Start - Maintenance Procedure

Procedure

TIEDOWN LINES SHALL NOT BE ATTACHED TO ANY FUSELAGE HARD POINTS WHEN ROTOR IS TURNING.

DURING FLIGHT CONTROL CHECKS, ROTOR TIP PATH PLANE WILL BE DEFLECTED. IF THE AFCS IS ENGAGED, ROTORS WILL MOVE IN RESPONSE TO SHIP MOVEMENTS.

1. Refer to 412/412EP Flight Manual.

Intentionally blank page

Maintenance Procedures - Description

This supplement contains information and instructions for maintenance of instrument, electrical, avionic, and flight control systems/components used in Search and Rescue (SAR) configured Bell Model 412 helicopters.

Supplemental electrical and avionic systems in SAR configured Model 412 helicopters consists of Digital Automatic Flight Control System (DAFCS) , Electronic Flight Instrument System (EFIS) , radar altimeter navigation system and the weather radar system. Other customizing avionic equipment installed includes Doppler velocity sensor, DATA NAV III, and navigation system (GNS-X).

The DDAFCS consists of dual/autopilot flight director computers, mode selector, autopilot controller, air data sensors and servos. It also comprises the standard flight controls plus added (DAFCS) (SAR) components in the collective, cyclic, and directional control systems. Mechanical linkage systems, actuated by electrical actuators and conventional helicopter controls are used to control flight attitude, altitude and direction.

The system is fully coupled, four axis (pitch, roll, yaw, and collective) system with SAR modes combining autopilot and flight director functions. The system also provides auto trim, heading hold, automatic turn coordination, and auto level.

The flight director computers provide the pilot and copilot with steering information to fly the helicopter manually or the information can be coupled to the autopilot for fully automatic flight path control. SAR specific modes include, radio altitude hold, velocity hold, mark-on-target, approach 1, approach 2, and climb is also coupled to the autopilots.

The EFIS is comprised of pilot and copilot electronic displays Electronic Attitude Director Indicator (EADI) and Electronic Horizontal Situation Indicator (EHSI) , symbol generators, display controllers, and remote instrument controller.

The EFIS displays pitch and roll attitude, heading, course orientation, source annunciators, doppler velocities, weather presentations, navigation waypoints, and flight paths.

The radio altimeter system consists of a receiver/transmitter, transmit and receive antennas, and an indicator. The receiver/transmitter provides radio altitude to the indicator, EFIS, and DAFCS from 0 to 2500 feet. The indicator displays radio altitude from 0 to 1500 feet.

The weather radar system consists of a receiver/transmitter, antenna, indicator, and controller. The system is a high resolution, high power airborne weather/ground-mapping radar optimized for over-water search and rescue missions.

Refer to 412-A-97-00-00-00A-009A-A / 02235 and 412-A-98-00-00-00A-009A-A / 02454 for avionic and electrical wiring diagrams of SAR systems installed.

Intentionally blank page

List of abbreviations

The abbreviations used in this publication are as follows:

Abbreviation	Definition
A.S.B.	Alert Service Bulletins
AAI	Aeronautical Accessories Inc
AD	Airworthiness Directives
ADC	Air Data Computer
ADF	Automatic Direction Finder
ADI	Attitude Director Indicator
ADS	Air Data Sensor
AFCS	Automatic Flight Control System
AFCU	AUTOMATIC FUEL CONTROL UNIT
AHRS	Attitude Heading Reference System
AHRU	ATTITUDE HEADING REFERENCE UNIT
AOG	Aircraft-On-Ground
AOSS	After Over Station Sensing
API	Actuator Position Indicator
ASB	Alert Service Bulletins
ASBs	Alert Service Bulletins
ASCB	Avionics Standard Communications Bus
ASCDB	Aircraft Specific Configuration Database
ASD	European Association of Aerospace Industries
ATA	Airline Transportation Association
ATC	Air Traffic Control
ATCRBS	Air Traffic Control Radar Beacon Site
ATP	Acceptance Test Procedure
BC	Back Course
BHT	Bell Helicopter Textron
BHVM	Bell Helicopter Vibration Monitor
BIT	Built-In-Test
BL	BUTTOCK LINES
BS	Bottom Stations
CADU	Control and Data Unit
CAM	Cockpit Area Microphone
CAS	Crew Alerting System
CDU	CONTROL DISPLAY UNIT
CG	Center of Gravity
CPM	Crash Protected Memory
CSI	Course Select Indicator
CVFDR	Cockpit Voice Flight Data Recorder
CVR	Cockpit Voice Recorder
CYC CTR	Cyclic Centering
DAFCS	Digital Automatic Flight Control System

Abbreviation	Definition
DDAFCS	Dual Digital Automatic Flight Control System
DFC	Digital Flight Control
DG	Directional Gyro
DH	Decision Height
DMC	Data Module Code
DME	Distance Measuring Equipment
EADI	Electronic Attitude Director Indicator
EED	ELECTRO-EXPLOSIVE DEVICE
EEPROM	Electrically Erasable Programmable Read Only Memory
EFIS	Electronic Flight Instrument System
EHSI	Electronic Horizontal Situation Indicator
ES	ELEVATOR STATION
ESDS	Electrostatic Discharge Sensitive
FCC	Flight Control Computer
FDAU	Flight Data Acquisition Unit
FDR	Flight Data Recorder
FDS	Flight Director System
FOD	Foreign Object Damage
FPI	Fluorescent Penetrant Inspection
FPM	Feet Per Minute
FRATS	Flight Recorder Analysis Test Station
FS	Fuselage Stations
FSB	Forced Standby
FT	Feet
FTR	Force Trim Release
GA	Go-Around
GPU	Ground Power Unit
GRE	Ground Readout Equipment
GSE	Ground Support Equipment
HSI	Horizontal Situation Indicator
HSR	Historical Service Records
I/O	Input/Output
IAS	Indicated Airspeed
ICA	Instructions for Continued Airworthiness
ICS	Intercommunication System
IFCS	Integrated Flight Control System
IGS	Integrated Ground Software
ILS	Instrument Landing System
IOGP	IDARS Operational Ground Program
ISA	International Standard Atmosphere
ITT	Inter-Turbine Temperature
IVV	Instantaneous Vertical Velocity
LBL	Left Buttock Line
LED	Light Emitting Diode

Abbreviation	Definition
LOEDM	List of Effective Data Modules
LRU	Line Replaceable Unit
LTF	Lead The Fleet
LVDT	Linear Variable Differential Transformer
MADRAS	Modular Airborne Data Recording & Acquisition System
MFCU	Manual Fuel Control Unit
MM	Maintenance Manual
MMS	MAINTENANCE MANUAL SUPPLEMENT
MTL	Minimum Trigger Level
NDI	Non Destructive Inspection
NMU	NAVIGATION MANAGEMENT UNIT
NOC	Nav On Course
Ni-Cad	Nickel-Cadmium
OAT	Outside Air Temperature
OGP	Operational Ground Program
OMS	On-board Maintenance System
OSS	Over Station Sensing
POD	Pitch Out of Detent
PSE	Product Support Engineering
PUB	Power-up BIT
PVT	Private
RADS AT	Rotor Analysis and Diagnostic System Model AT
RADS AT	Rotor Analysis and Diagnostic System, Model AT
RBL	Right Buttock Line
RCU	Recorder Control Unit
RDF	Recorded Data File
RDU	Recorder Download Unit
RIN	Retirement Index Number
ROD	Roll Out of Detent
SAR	Search and Rescue
SI	SERVICE INSTRUCTION
SNS	Standard Numbering System
SPI	Special Pulse Identification
STA	STATION
STC	Supplement Type Certificate
SYS 2	System 2
T.B.	Technical Bulletins
TAS	True Airspeed
TB	Technical Bulletins
TBO	Time Between Overhaul
TBs	Technical Bulletins
TCAS	Traffic Advisory and Collision Avoidance System
TCU	Torque Control Unit
TIG	Tungsten Inert Gas

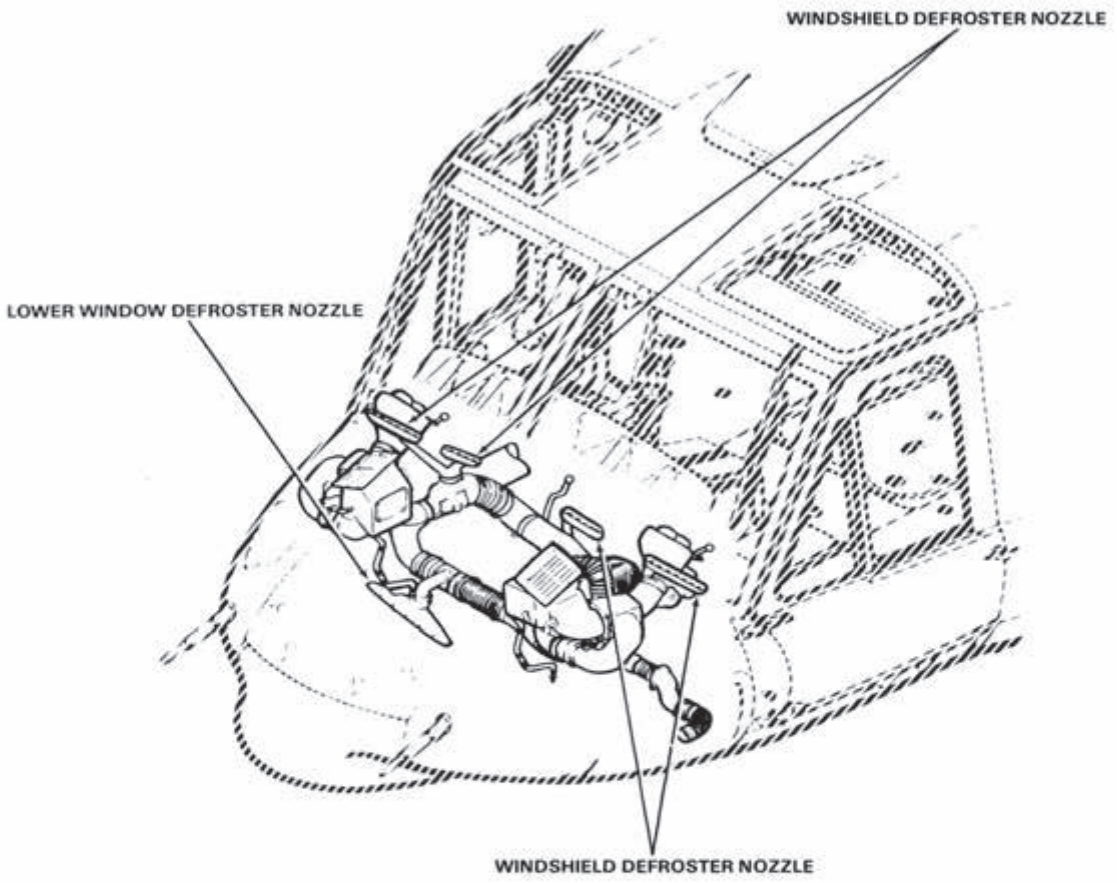
Abbreviation	Definition
TIS	Time-In-Service
TTG	Time-to-Go
TTL	Transistor Logic
ULB	Underwater Location Beacon
VFS	VERTICAL FIN STATION
VPU	Voice Processing Unit
VRLA	Valve-Regulated Lead-Acid
VRO	Vertical Reference Operations
VS	Vertical Speed
VSI	Vertical Speed Indicator
WL	WATER LINE
WOG	Weight-On-Gear
WOW	Weight-On-Wheels
WinDRTTM	Windows Data Review Tool
WinDRT™	Windows Data Review Tool

Heating and Ventilation Air Distribution System - Description

The cabin air distribution system (Figure 1) includes two subsystems which provide optional use of heated or ventilated air from separate sources.

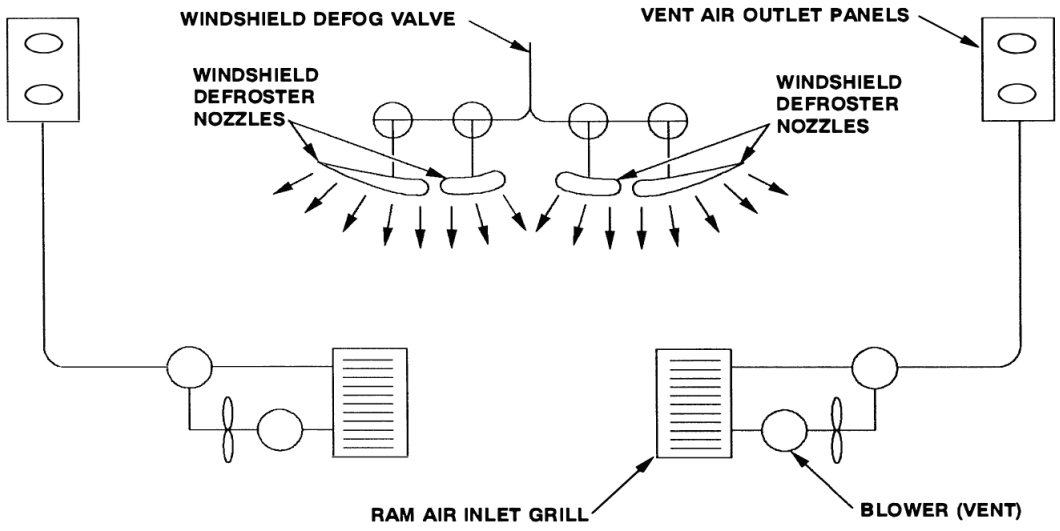
Heated air is supplied to lower window defroster nozzles and to four windshield defroster nozzles for cabin heating-defrosting using bleed air from engine compressors as heat source.

Ventilating system uses ram air as source from ventilating air intakes on cabin nose with optional use of blowers for ventilating air outlets ahead of pilot-copilot stations (Figure 2).



ICN-412-A-212001-B-97499-00002-A-001-01

Figure 1. Heating and Ventilation Air Distribution System (Sheet 1 of 1)



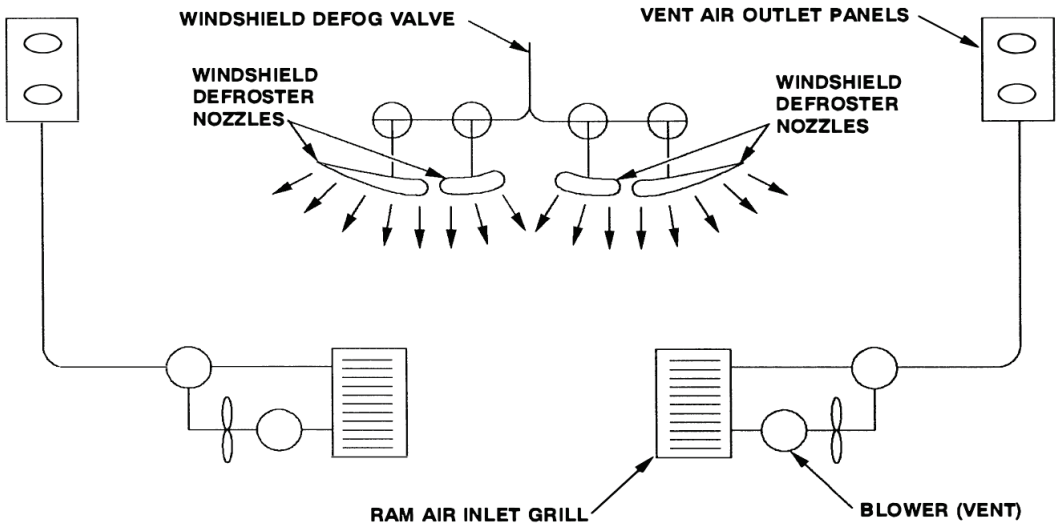
ICN-412-A-212001-B-97499-00003-A-001-01

Figure 2. Heating and Ventilation Air Distribution System Schematic (Sheet 1 of 1)

Intentionally blank page

Ventilation System - Description

Ventilation system (Figure 1) delivers outside air to ventilating air outlets ahead of pilot-copilot stations. Ram air inlet grills are located on cabin nose below each windshield. Ducts extending from air inlet grills direct ram air to ventilating air outlets. Air intake volume is controlled by push/pull knobs near ventilating air outlets. When ram air is low, outside air may be pulled in by turning “ON” the blower which is part of the system.



ICN-412-A-212001-B-97499-00003-A-001-01

Figure 1. Heating and Ventilation Air Distribution System Schematic (Sheet 1 of 1)

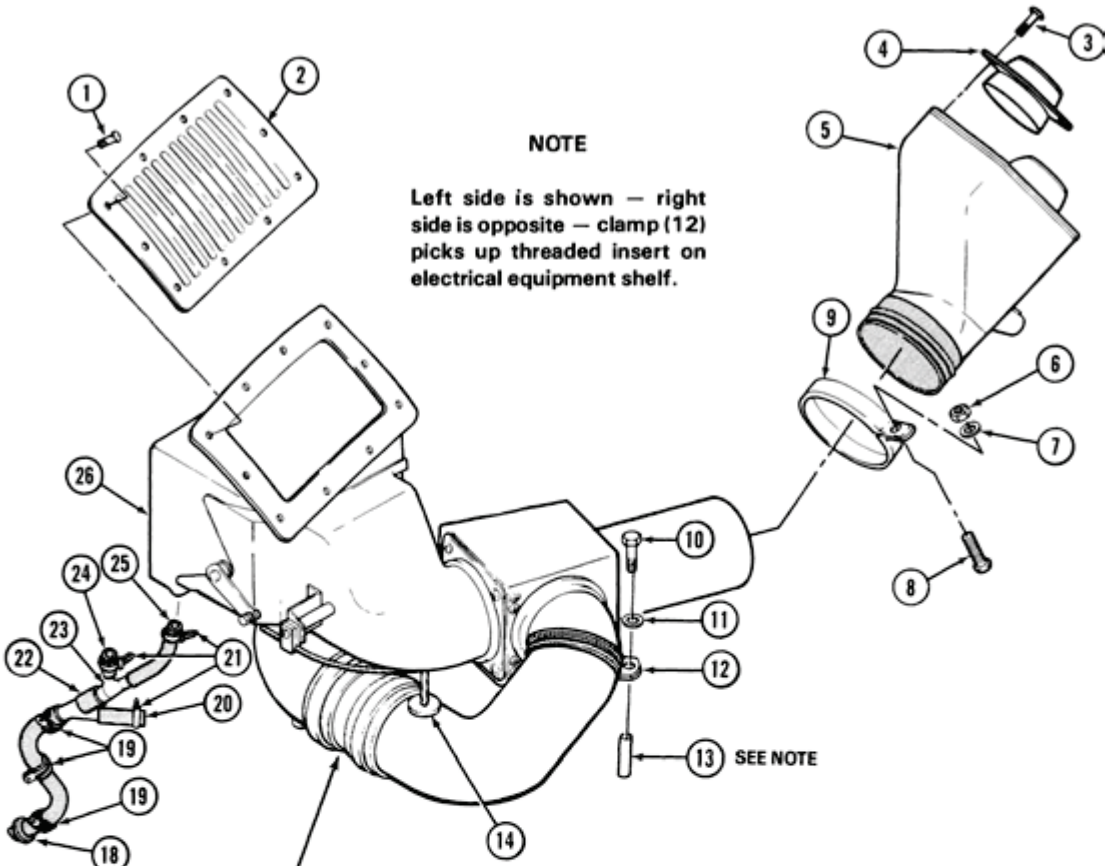
Ventilation System - Removal

Procedure

NOTE

The removal procedure for left plenum and blower is given. The removal procedure for the right plenum and blower is similar.

1. Remove ten screws (1, Figure 1) and remove ram air inlet grill (2) on left side of cabin nose.
2. Remove tie wraps (21) from tubes (20, 24, and 25).
3. Remove three clamps (19).
4. Remove tubes (20, 22, 24, and 25) and adapter (23) between grommet (18) and bottom of plenum (26).
5. Remove clamp (9) and outlet plenum (5).
6. Remove clamp (12) and remove plenum (26) as a unit.
7. Remove clamps (15 and 17) and remove ventilation blower (16).



1. Screw
2. Ram air inlet grill
3. Screw
4. Air valve
5. Outlet plenum
6. Nut
7. Thin aluminum washer
8. Screw
9. Clamp
10. Screw
11. Thin aluminum washer
12. Clamp
13. Spacer (see note)
14. Grommet
15. Clamp
16. Ventilation blower
17. Clamp
18. Grommet
19. Clamp
20. Tube
21. Tie wraps
22. Tube
23. Adapter
24. Tube
25. Tube
26. Plenum

ICN-412-A-212001-B-97499-00004-A-001-01

Figure 1. Ventilation Components Installation (Sheet 1 of 1)

Ventilation System - Installation

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Wraps, Tie	- MS3367-5-9	AR	

Procedure

NOTE

The installation procedure for left plenum and blower is given. The installation procedure for the right plenum and blower is similar.

1. Position plenum (26, Figure 1) in cabin nose and install clamp (12), screw (10), washer (11), and spacer (13). Thread screw (10) into threaded insert on electrical equipment shelf.

2. Position clamp (9) on outlet plenum (5) and plenum (26). Install screw (8), washer (7), and nut (6).

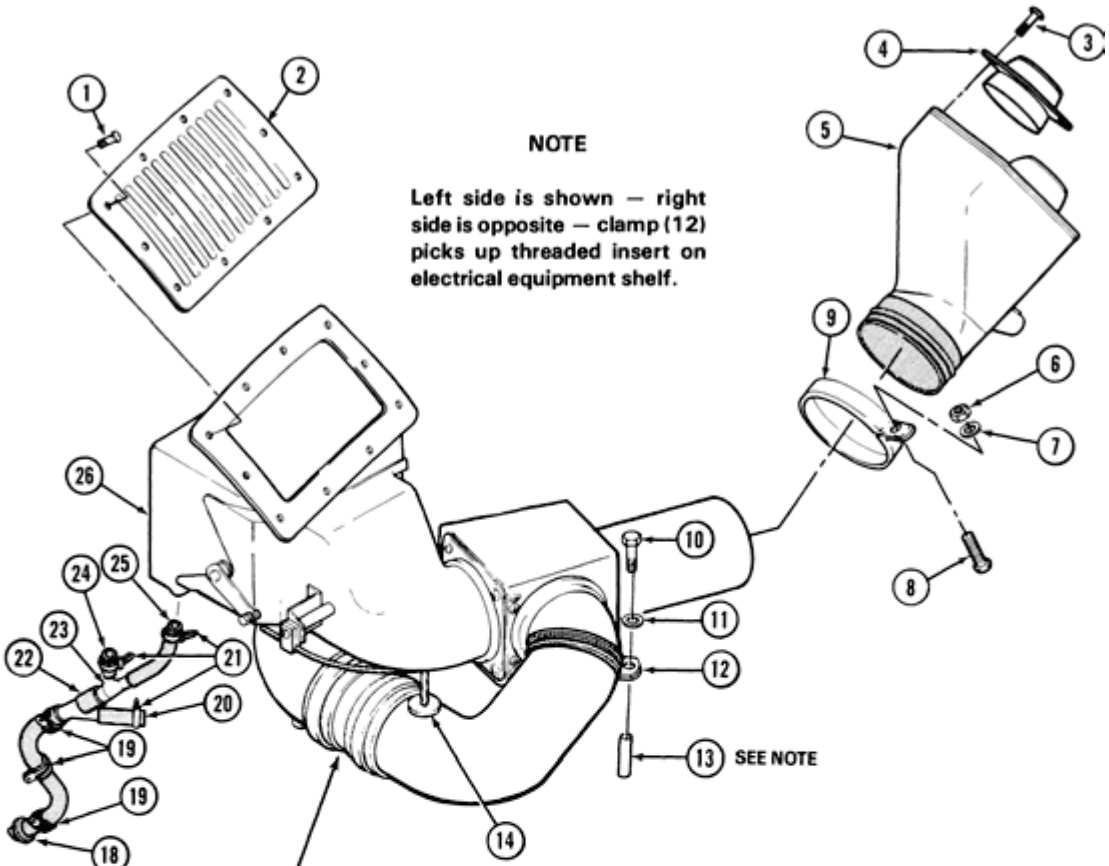
3. If not previously accomplished, position air valve (4) on outlet plenum (5) and install four screws (3).

4. Position tubes (25, 24, 22, and 20) and adapter (23) on plenum (26). Install three clamps (19) and install tiedown wraps (21) on tubes (25, 24, 22, and 20).

5. If not previously accomplished, position ventilation blower (16) on plenum (26), and install clamps (15 and 17).

6. Position ram air inlet grill (2) on top of cabin nose left side. Install ten screws (1) to secure grill to plenum (26).

7. Perform functional check of ventilation blower (16).



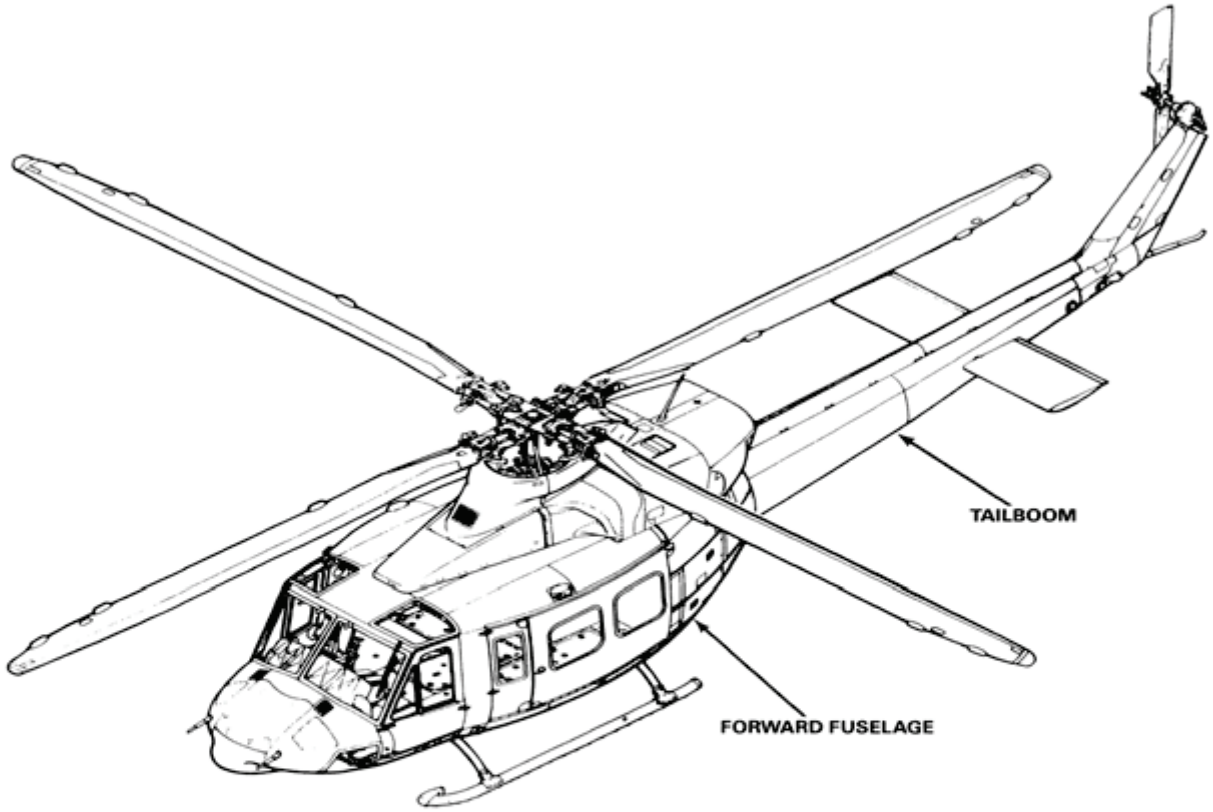
1. Screw
2. Ram air inlet grill
3. Screw
4. Air valve
5. Outlet plenum
6. Nut
7. Thin aluminum washer
8. Screw
9. Clamp
10. Screw
11. Thin aluminum washer
12. Clamp
13. Spacer (see note)
14. Grommet
15. Clamp
16. Ventilation blower
17. Clamp
18. Grommet
19. Clamp
20. Tube
21. Tie wraps
22. Tube
23. Adapter
24. Tube
25. Tube
26. Plenum

ICN-412-A-212001-B-97499-00004-A-001-01

Figure 1. Ventilation Components Installation (Sheet 1 of 1)

Fuselage - Description

The fuselage (Figure 1) consists of forward fuselage and tailboom. Maintenance data for the fuselage and tailboom is contained in 412-A-00-00-00-01A-028A-A / 02630.

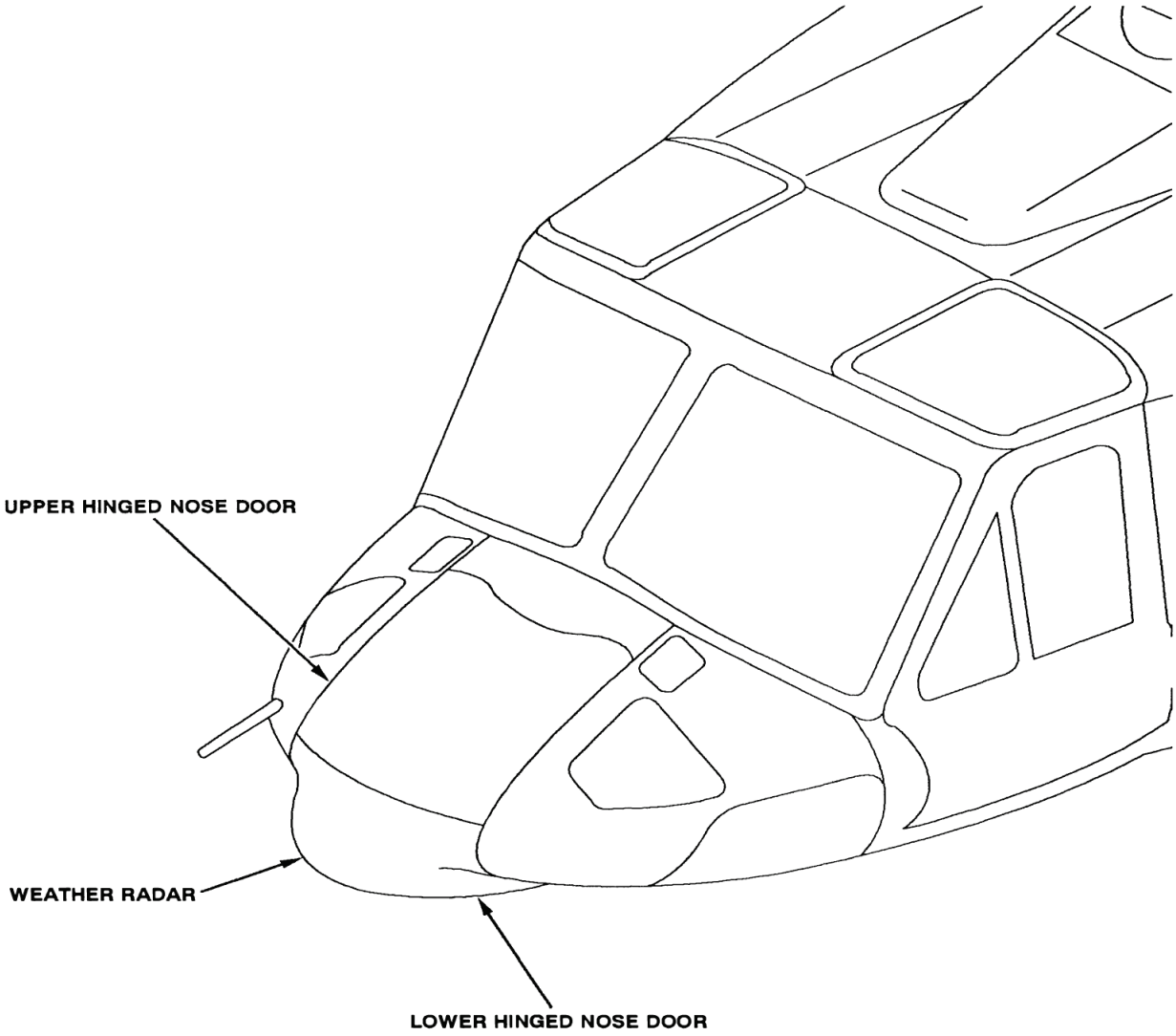


ICN-412-A-530001-B-97499-00001-A-001-01

Figure 1. Model 412 SAR Helicopter (Sheet 1 of 1)

Forward Fuselage - Description

Avionics and weather radar equipment is contained in the nose compartment (Figure 1). Upper and lower hinged nose compartment doors provides access to the electrical compartments.



ICN-412-A-531001-B-97499-00001-A-001-01

Figure 1. Model 412 SAR Forward Fuselage (Sheet 1 of 1)

Digital AFCS Flight Controls - Description

This section contains information and instructions for maintenance of digital automatic flight control system with search and rescue configuration (DAFCS) (SAR) which includes the standard flight controls plus added (DAFCS) (SAR) components in the collective, cyclic, and directional control systems. Mechanical linkage systems, actuated by electrical actuators and conventional helicopter controls are used to control flight attitude, altitude and direction.

Standard flight control systems include dual flight controls, consisting of cyclic control sticks for fore and aft and lateral controls, collective pitch control sticks for vertical control and antitorque pedals for directional control. An aerodynamically actuated elevator provides stability about the longitudinal axis of the helicopter. Force trim systems consisting of a rotary trim actuator and a force gradient are connected in parallel to the cyclic and directional controls.

The cyclic stick centering indicator system consists of two proximity switches and two Cyclic Centering (CYC CTR) caution lights. The lights are located on the pilot and copilots instrument panel. System circuitry is activated whenever the rotor RPM warning light is illuminated. During ground operations below normal operation range, excessive cyclic inputs will illuminate the (CYC CTR) caution lights. Properly positioning the cyclic stick will extinguish the lights.

The cyclic control system incorporates four electrical (SCAS) servo actuators, two connected in series in the pitch system and two connected in series in the roll system. Four transducers, two in roll system and two in pitch system, are used to convert mechanical movement to electrical signals for input of pilot control commands to the AFCS. A force trim system is connected to both cyclic pitch and roll control system.

The antitorque directional system incorporates two electrical (SCAS) servo actuators, a force trim system, an electromechanical rotary actuator and two transducers that are used to convert mechanical movement to electrical signals for input of pilot control commands to the AFCS. The servo actuator is located forward of the directional hydraulic servo actuator. The transducers and force gradient are connected to the bellcrank located aft of pilot directional control pedals.

Forward servo actuator in each cyclic system and directional actuator are controlled by helipilot, HP-1, system. Aft actuators in each system are controlled by helipilot, HP-2 system.

The collective control system incorporates a force trim system consisting of a rotary trim actuator, force gradient, and linkage connected in parallel to the collective controls, and two transducers that are connected to the collective jackshaft.

Refer to 412-A-98-01-01-00A-051A-A / 02974 for (DAFCS) (SAR) systems wiring diagrams.

Intentionally blank page

Digital AFCS Flight Controls - Troubleshooting

Procedure

1. Refer 412-A-67-00-00-00A-422A-A / 01321 for flight control troubleshooting.

Intentionally blank page

Power Lever Controls - Rigging

Procedure

1. Refer 412-A-76-10-00-00A-276A-A / 01725 for rigging power lever controls.

Intentionally blank page

Collective Controls - Rigging

Preliminary requirements

Support equipment

Nomenclature	Identification No.	Qty	Remarks
Spring Scale	- Commercial	1	Range to Measure 4.0 to 4.5 lbs. - 17.79 to 20.02 N

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Corrosion preventive compound	C-101	AR	
Sealant	C-308	AR	
Lockwire	C-405	AR	

Procedure

NOTE

All (DAFCS) (SAR) configurated helicopters will have dual controls installed. Refer to BHT-412-IPC for illustrated parts list.

1. Comply with the following general procedures when rigging collective controls:

1.1. When a rigging procedure requires a specific position to accomplish following steps, manually hold pilot collective in position and secure with collective friction adjustment.

1.2. Tolerance for rigging dimensions is ± 0.03 in. (0.76 mm).

1.3. All adjustable control tubes shall have a maximum of 1.00 in. (25.4 mm) of exposed threads after adjustment. If control tubes have inspection holes, the threads must be engaged sufficiently to cover holes.

1.4. Apply corrosion preventive compound (C-101) to threads of all adjustable end bearings and clevises after adjustment.

1.5. Torque control jamnuts 80 to 100 in. lbs. (9.0 to 11.3 Nm) after adjustment unless other torque is specified.

1.6. Control tubes must be free to rotate several degrees about longitudinal axis. Loosen jamnuts, adjust end fittings, and torque jamnuts if necessary.

2. Disconnect the following control tubes:

2.1. Disconnect extension tube (2, Figure 1) from collective lever (1).

2.2. Disconnect control tube (4) from collective hydraulic cylinder (3).

2.3. Disconnect control tube (7) from lever (8).

2.4. Disconnect force gradient (9) from lever (8).

2.5. Disconnect link (11) from rotary actuator (12).

3. If available, connect hydraulic power cart to helicopter.

4. Position pilot collective stick (17) against ‘up’ stop, hold in position and tighten friction adjuster (16).

5. Position top edge of bellcrank (6) 0.52 in. (13.21 mm) above top of support (5) as shown on detail C.

6. Adjust tube (7) to lever (8). Tighten jamnut on tube (7) clevis. Install tube (7) on lever (8).

7. Align collective hydraulic cylinder (3) using same procedures as flight control cylinders. Refer to 412-A-67-11-00-18A-271A-A / 01395.

8. Ensure pilot collective stick (17) is positioned as described in Step 4..

9. Move actuator valve lever (19) up. Manually pull extension tube (2) up to bottom collective cylinder (3) in the UP position. Then set actuator valve lever (19) 0.070 in. (1.778 mm) above center line neutral valve position as shown on detail B. Hold this position.

10. Adjust tube (4) to fit actuator valve lever (19) then shorten 3 turns of rod end bearing. Tighten jamnut against rod end bearing on tube (4). Install tube (4) on actuator valve lever (19).

11. Position collective lever (1) 4.67 to 4.70 in. (118.61 to 119.38 mm) above base of swashplate support (21) to center of pin (20) as shown on detail A.

12. Apply hydraulic power from hydraulic cart to helicopter.

NOTE

If hydraulic power cart is not available refer to Note preceeding Step 16..

13. Position lever (1) as measured in Step 11.. Position pilot collective stick (17) against low stop. Tighten friction adjuster (16).

14. Adjust extension tube (2) to fit collective lever (1) torque jamnut 1500 to 1800 in. lbs. (168.0 to 201.6 Nm) on extension tube (2), and lockwire (C-405). Apply sealant (C-308) around nut, lockwire, and threads of clevis.

15. Install extension tube (2) to collective lever (1).

NOTE

Alternate method of adjusting extension tube (2) without pressurizing hydraulic system is outlined in Step 16. through Step 20..

16. Position collective lever (1) and pilot collective stick (17) as outlined in Step 13..

17. Exert sufficient downward pressure on hydraulic cylinder (3) to hold actuator valve lever (19) at top of travel as shown on detail B.

COLLECTIVE CONTROLS BELOW COLLECTIVE HYDRAULIC CYLINDER (3) MUST NOT MOVE DURING ACCOMPLISHING THE FOLLOWING STEPS.

18. Adjust extension tube (2) to fit collective lever (1), then shorten by one complete turn of upper clevis in tube (2).

19. Torque jamnut 1500 to 1800 in. lbs. (168.0 to 201.6 Nm) on extension tube (2) and lockwire (C-405). Apply sealant (C-308) around nut, lockwire, and threads of clevis.

20. Install extension tube (2) to collective lever (1).

21. Rig 4-axis collective trim system as follows:

21.1. Position pilots collective stick (17) against its down stop.

21.2. Connect force gradient (9) to lever (8).

21.3. Position output arm of rotary actuator (12) as shown in (detail G).

21.4. Position and hold output arm of rotary actuator (12) against its outboard left hand stop.

21.5. Adjust link (11) to fit between the bellcrank (10) and rotary actuator (12) positioned per Step 21.4., then extend the rod end bearing two and one-half complete turns and secure to output arm of the rotary actuator (12).

NOTE

Dual element control motion transducers (412-074-101-103) are aligned and positioned using calibration tool (412-274-001-101).

NOTE

Refer to cyclic control motion transducer adjustment 412-A-67-02-00-02A-276A-A / 02761, step 2.1, using calibration tool for electrical alignment of collective transducers.

NOTE

This calibration tool (412-274-001-101) is used to electrically align the control motion transducers installed on the helicopter to be used as control position sensors for the flight data recorder. This tool will insure that the control motion transducers are aligned within ± 0.5 percent of the prescribed position. Fore and aft position is set at 50 percent. Lateral position is set at 50 percent. Collective position is set at 12 percent. Pedal position is set at 12 percent.

22. Perform collective transducer adjustment as follows:

22.1. Attach collective transducers (30) to structure (see detail H).

22.2. Move collective transducers (30) through their full travel.

22.3. Position pilot collective stick (17) against the up stick stop, with collective transducers extended but not bottomed. Adjust rod ends on collective transducers (30) to align with hole in lever (31).

22.4. Position pilots collective stick (17) against the down stick stop, with the collective transducers (30) retracted but not bottomed, connect collective transducers (30) to lever (31). Overtravel of collective transducers (30) should be approximately equal.

23. Adjust collective minimum friction adjustment as follows:

23.1. Apply hydraulic power from hydraulic cart to helicopter.

23.2. With hydraulic test cart operating, adjust minimum collective friction clamp (13, detail D) to obtain 4.0 to 4.5 lbs. (17.79 to 20.02 N) up-collective stick force. Record this reading by applying a spring scale with a range that will measure 4.0 to 4.5 lbs. (17.79 to 20.02 N) up-collective stick force at center of upper throttle grip within ± 0.50 in. (12.7 mm).

23.3. Tighten or loosen nut (29) to adjust minimum collective friction clamp (13) during measure of sliding force as shown on detail D.

23.4. After collective friction is adjusted within limits, move collective controls through full throw and check for binding and/or obstruction of control movement.

23.5. Turn off hydraulic power cart and disconnect from helicopter.

24. Ensure that all collective controls bolts are installed and cotter pins and lockwire (C-405) installed, where applicable.

25. Check throttle controls for binding or obstruction as follows:

25.1. Disconnect left and right adjustable throttle control tubes (27) and (28) from each bell crank.

25.2. Check that mark tooth of flexshaft is engaged with 10th tooth space of mating gear sectors (24) and (25) on both pilot and copilot collective elbow.

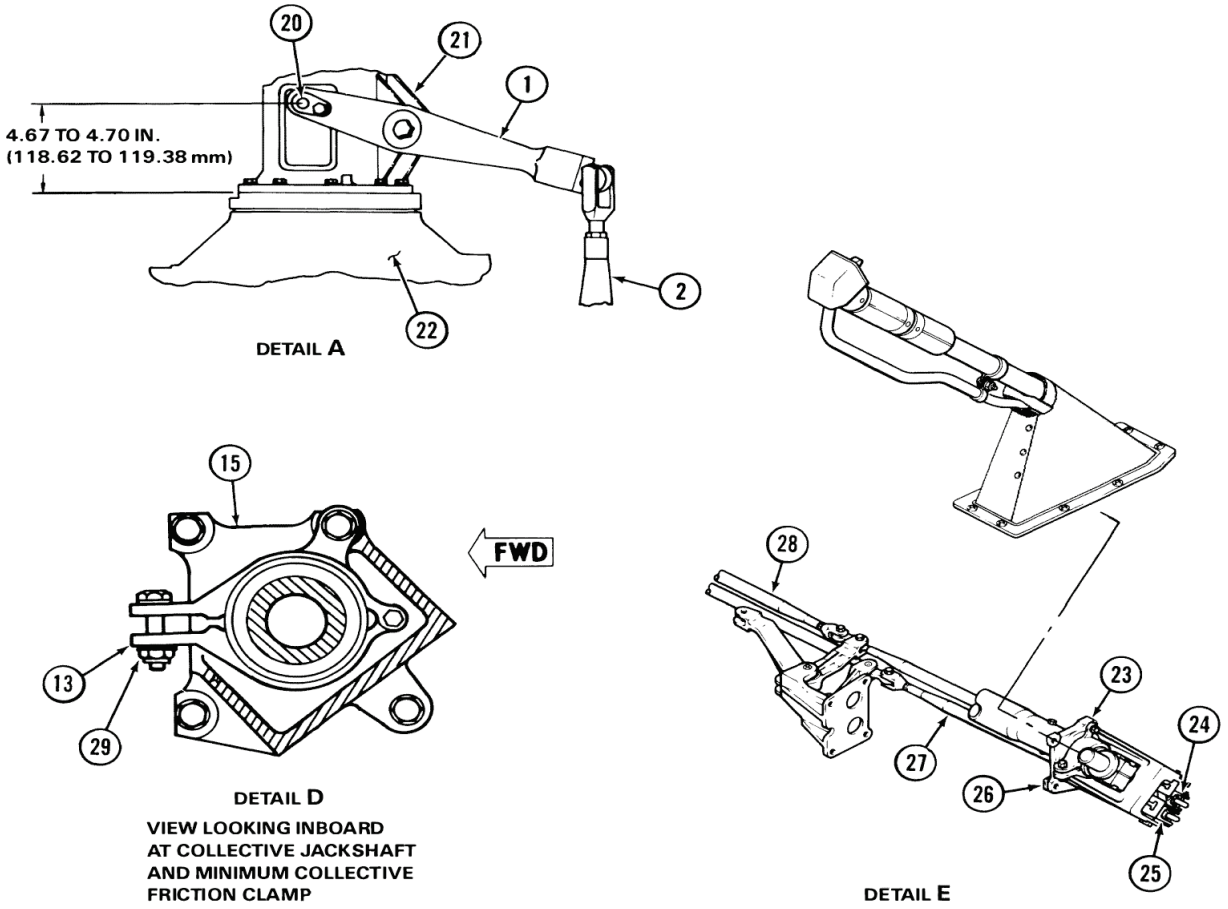
25.3. Roll both engine throttle grips on pilot and copilot collective sticks to full open position against stops.

NOTE

If binding or obstruction occurs in copilot collective stick throttle grips when rotated, adjust shims (26) between airframe structure and elbow support to minimize lateral deflection of elbow assembly flex shafts.

25.4. Adjust control tube (27) and (28) to fit. Torque jamnuts and install control tubes (27) and (28).

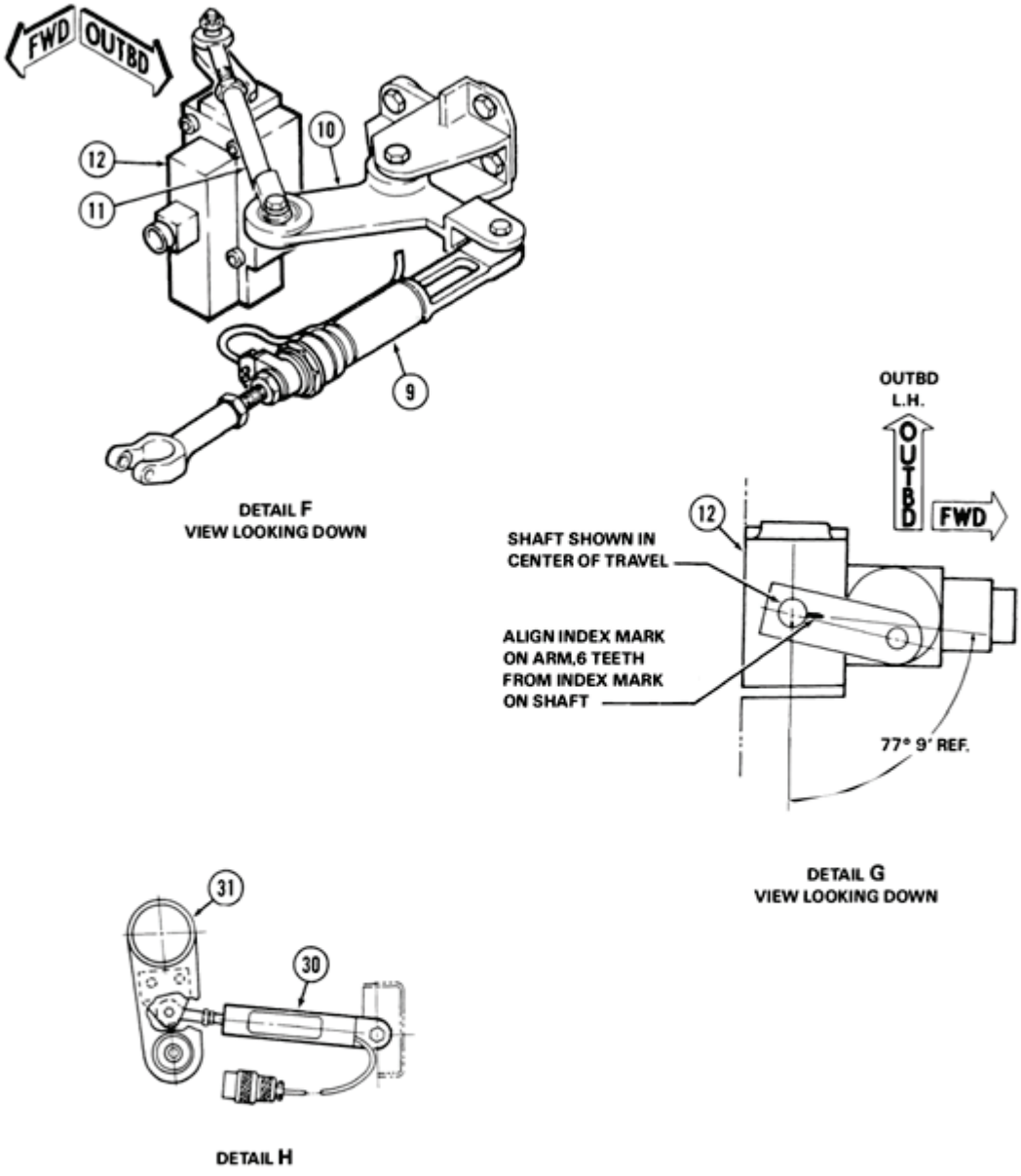
25.5. Cycle both throttle grips through full travel and check to see that each gear sector on both pilot and copilot elbow support is within no less than one tooth of rolling off its mating pinion.



1. Collective lever
2. Extension tube
3. Collective hydraulic cylinder
4. Control tube
5. Support
6. Bellcrank
7. Control tube
8. Lever
9. Force gradient
10. Bellcrank
11. Link
12. Rotary actuator
13. Minimum collective friction clamp
14. Control tube
15. Collective jackshaft
16. Pilot collective friction adjuster
17. Pilot collective stick
18. Piston
19. Actuator valve lever
20. Pin
21. Swashplate support
22. Transmission
23. Elbow support
24. Gear sector
25. Gear sector
26. Shim
27. Control tube
28. Control tube
29. Nut
30. Collective transducers
31. Lever

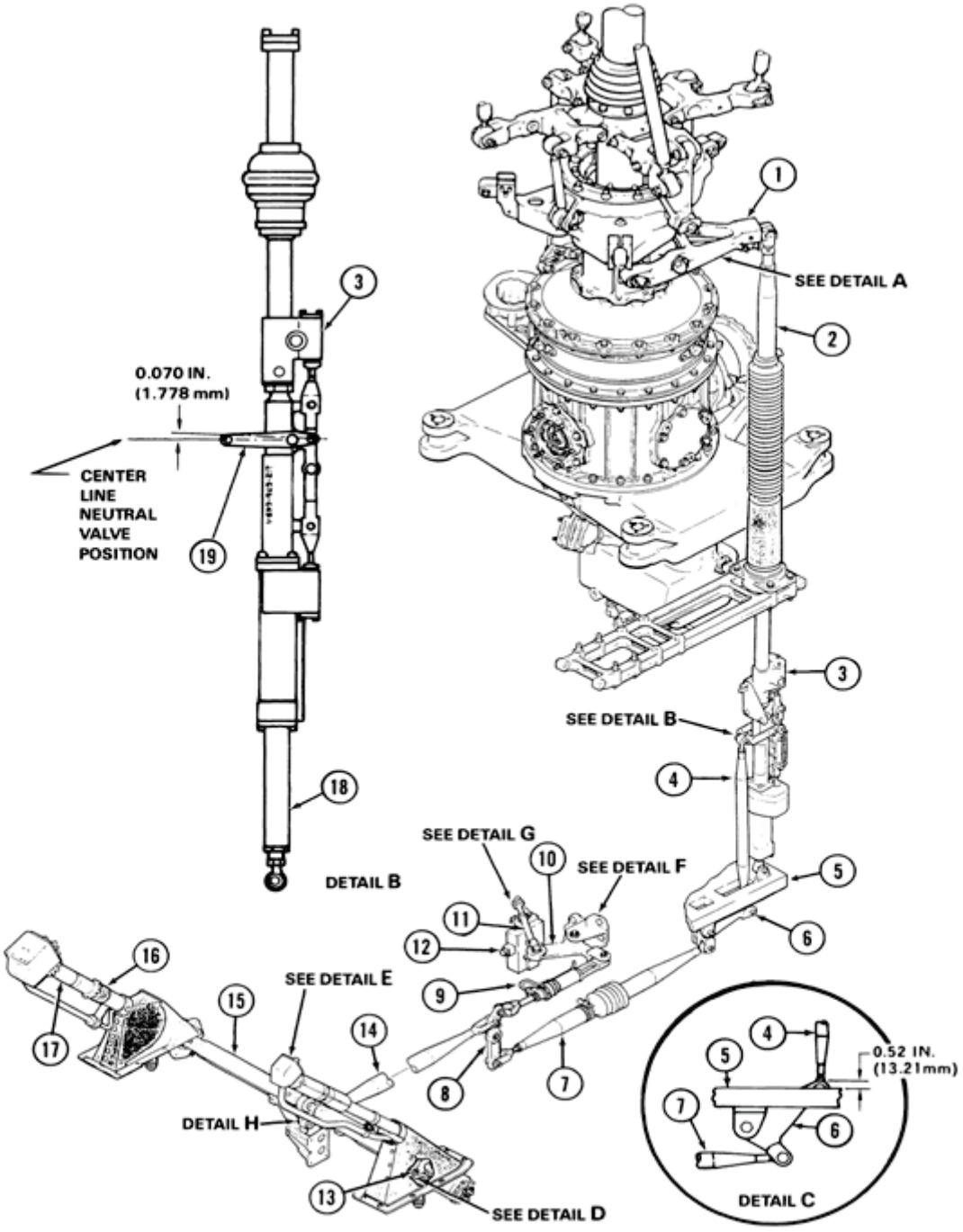
ICN-412-A-670200-B-97499-00005-A-001-01

Figure 1. Collective Controls Rigging (Sheet 1 of 3)



ICN-412-A-670200-B-97499-00006-A-001-01

Figure 1. Collective Controls Rigging (Sheet 2 of 3)



ICN-412-A-670200-B-97499-00004-A-001-01

Figure 1. Collective Controls Rigging (Sheet 3 of 3)

Cyclic Controls - Rigging

Preliminary requirements

Support equipment

Nomenclature	Identification No.	Qty	Remarks
Calibration tool	- 412-274-001-101	1	Pot calibrator

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Corrosion preventive compound	C-101	AR	

Procedure

1. Rigging procedures for the cyclic digital automatic flight control system with the search and rescue configuration (DAFCS SAR) are same as AFCS rigging. Refer to 412-A-67-01-00-04A-276A-A / 01325.

NOTE

For maintenance procedures of cyclic dual controls refer to BHT-412-SI-13 and BHT-412-IPC for illustrated parts list.

1.1. Comply with the following general procedures when rigging cyclic controls (Figure 1).

1.1.1. When a rigging procedure requires a specific position to accomplish following steps, manually hold controls in position and secure with pilot and copilot friction adjustment.

1.1.2. Tolerance for rigging dimensions is ± 0.03 in. (0.76 mm).

1.1.3. Apply corrosion preventive compound (C-101) to threads of all adjustable rod end bearings and clevises after adjustment.

1.1.4. Rod end bearings must be free to rotate several degrees about longitudinal axis. Loosen jamnuts, adjust end fittings, and torque jamnuts if necessary.

1.2. Perform fore and aft cyclic control motion transducer (34, Figure 1, detail A) adjustment as follows:

1.2.1. Position pilot cyclic stick (33) perpendicular to the deck within 1/2 degree longitudinally and laterally. Loosen jam nut (36) on fore and aft transducer (34) so the transducer shaft (37) can rotate freely.

NOTE

Dual element control motion transducers (412-074-101-103) are aligned and positioned using calibration tool (412-274-001-101).

NOTE

This calibration tool (412-274-001-101) is used to electrically align the control motion transducers installed on the helicopter to be used as control position sensors for the flight data recorder. This tool will insure that the control motion transducers are aligned within ± 0.5 percent of the prescribed position. Fore and aft position is set at 50 percent. Lateral position is set at 50 percent. Collective position is set at 12 percent. Pedal position is set at 12 percent.

1.2.2. Using calibration tool (412-274-001-101) (38) turn calibrator power switch (39) to the O (off) position. Connect calibrator box connector (40), to one of the fore and aft cyclic transducers (34) electrical connector (41). Turn selector knob (42) to fore and aft F/A position.

1.2.3. Turn calibrator power switch (39) to I (on) position.

NOTE

Any time the LO BAT lamp comes on, remove lid from calibrator box and replace batteries, use (9v dc alkaline batteries). Low battery is set to illuminate at 12 ± 1 volt.

1.2.4. If a RED lamp comes on, the transducer shaft (37) shall be adjusted in the direction indicated. An ‘extend’ indication means extend the transducer shaft (37) by screwing it into rod end bearing (43) (clockwise looking at rod

end). A ‘retract’ indication means retract the transducer shaft (37) by unscrewing it from the rod end bearing (43) (counterclockwise looking at the rod end). Adjust transducer shaft (37) until GREEN IN RANGE lamp comes on.

1.2.5. Tighten jam nut (36) on transducer shaft (37) and verify that GREEN IN RANGE lamp is on and RED lamps are off.

1.2.6. Move calibrator power switch (39) to O (off) position. Disconnect calibrator box connector (40) from transducer electrical connector (41).

1.2.7. Connect transducer electrical connector (41) to electrical connector on the bulkhead.

1.2.8. Perform calibration adjustment on remaining fore and aft cyclic transducer (34), using Step 1.2.1.. through Step 1.2.7.. above.

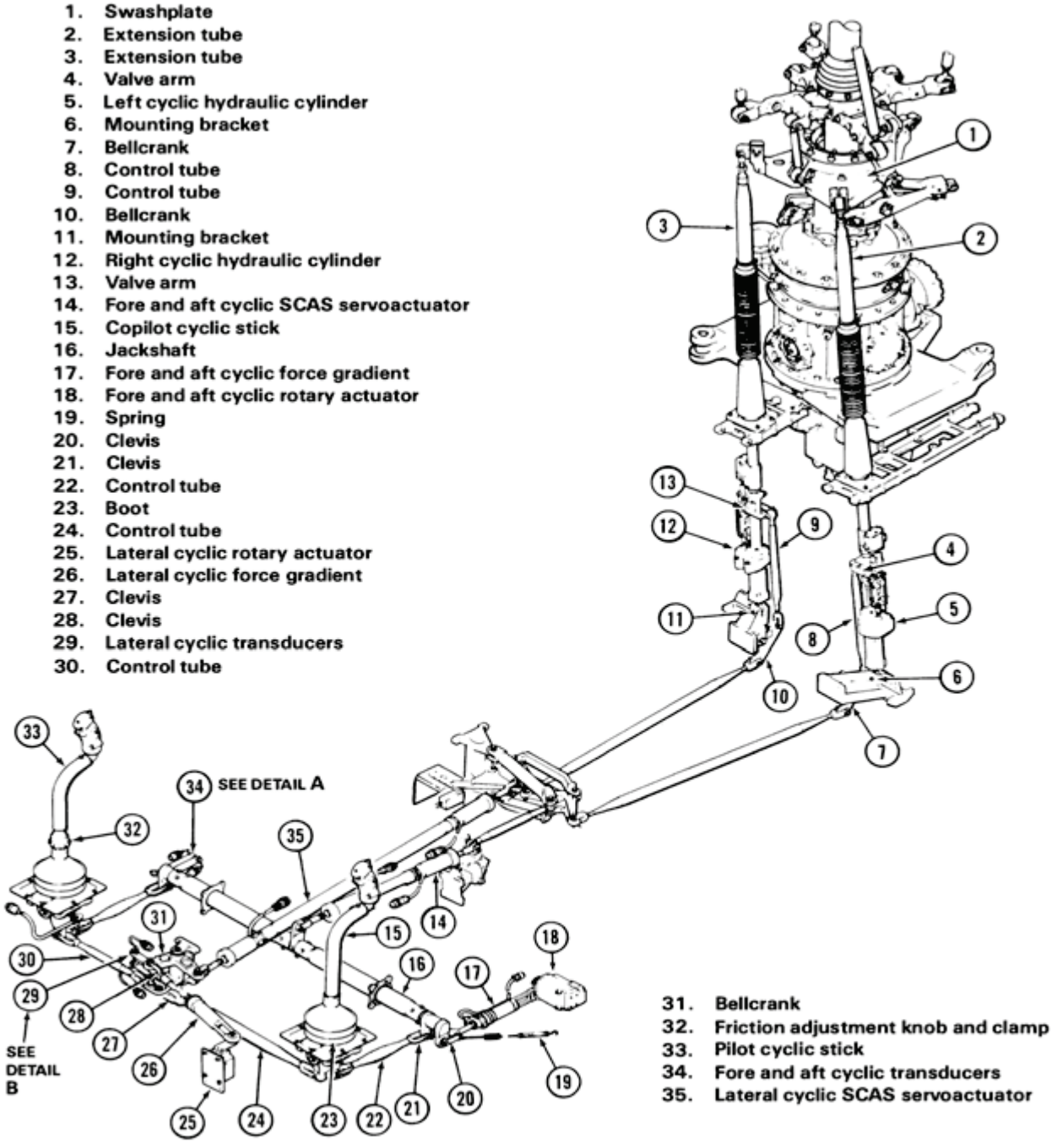
1.3. Perform lateral cyclic control motion transducer (29, Figure 1 detail B) adjustment as follows:

1.3.1. Position pilot cyclic stick (33) perpendicular to the deck within 1/2 degree longitudinally and laterally. Loosen jam nut (36) on lateral transducer (29) so that transducer shaft (37) can rotate freely.

1.3.2. Using calibration tool (412-274-001-101) (38) turn calibrator power switch (39) to O (off) position. Connect calibrator box connector (40), to one of the lateral cyclic transducers (29) electrical connector (41). Turn selector knob (42) to lateral LAT position.

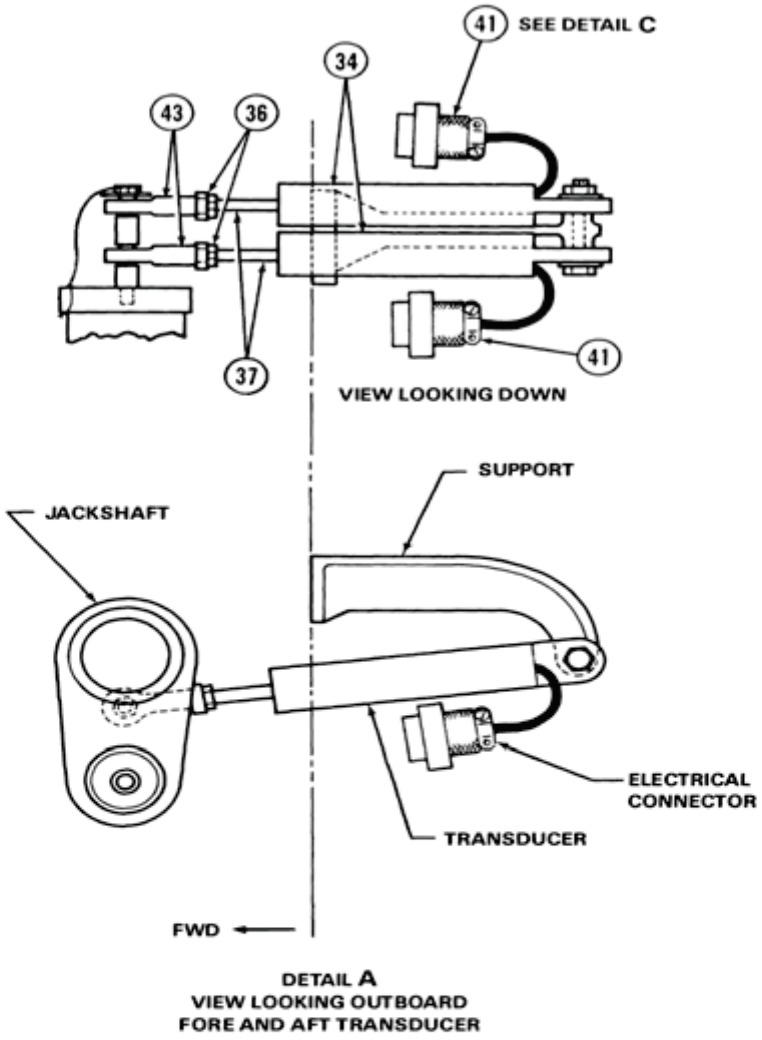
1.3.3. Perform calibration adjustment on lateral cyclic transducer (29), by repeating Step 1.2.3.. through Step 1.2.7.. above.

1.3.4. Perform calibration adjustment on remaining lateral cyclic transducer (29), by repeating Step 1.2.3.. through Step 1.2.7.. above.



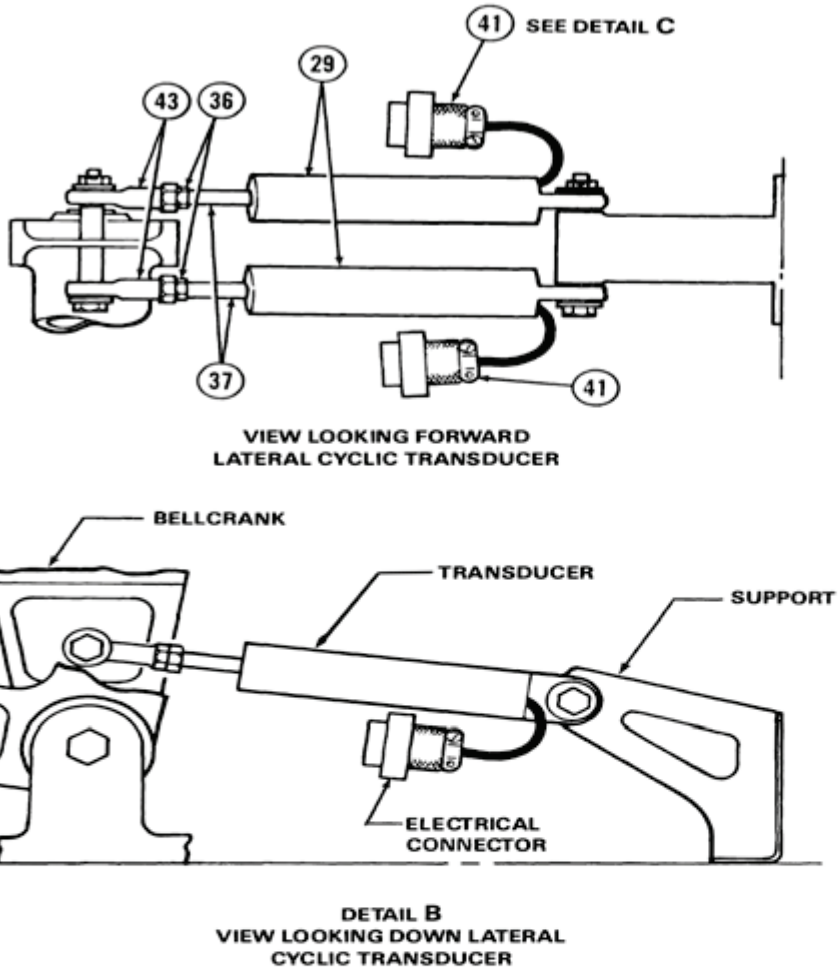
ICN-412-A-670200-B-97499-00007-A-001-01

Figure 1. Cyclic Controls Rigging (Sheet 1 of 4)



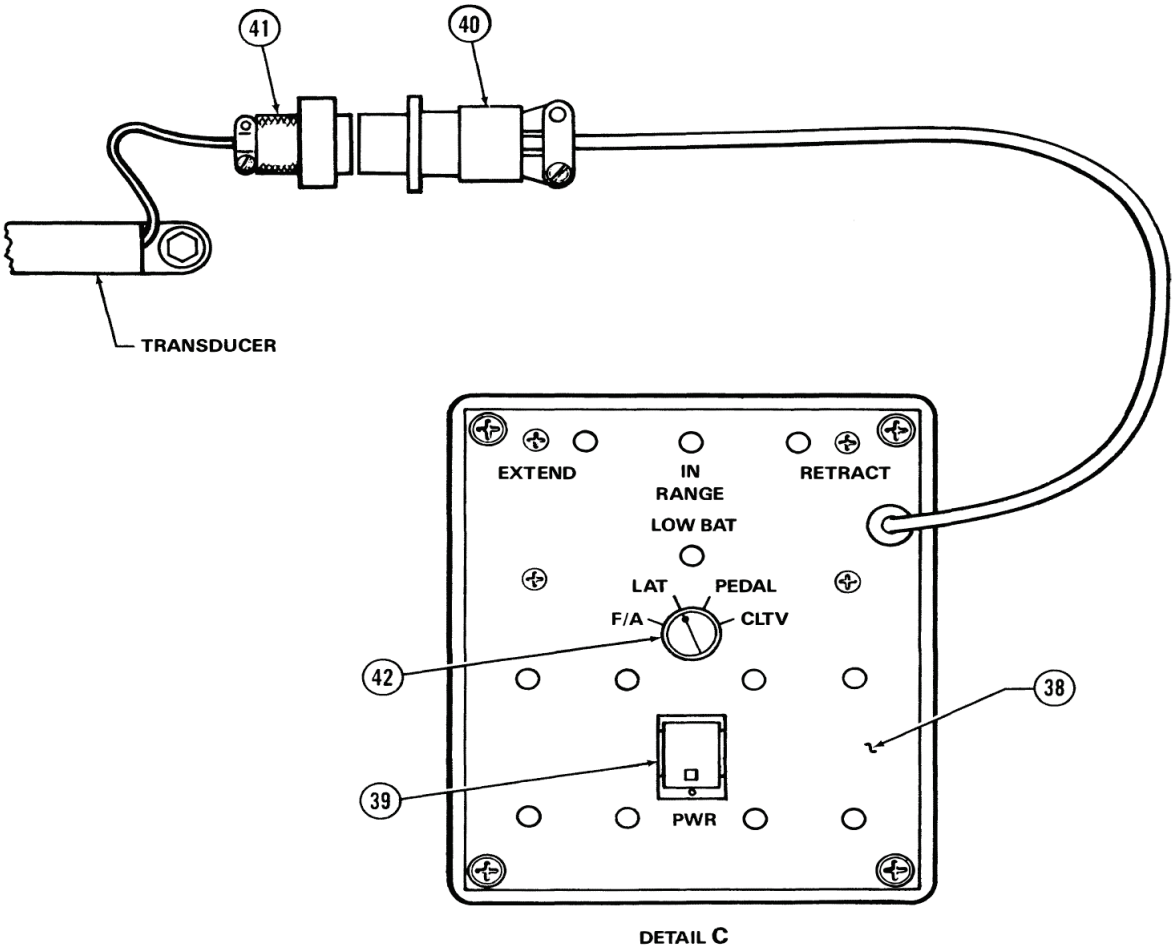
ICN-412-A-670200-B-97499-00008-A-001-01

Figure 1. Cyclic Controls Rigging (Sheet 2 of 4)



ICN-412-A-670200-B-97499-00009-A-001-01

Figure 1. Cyclic Controls Rigging (Sheet 3 of 4)



36. Jamnut
37. Transducer
38. Calibration tool (412-274-001-101)
39. Calibrator power switch
40. Calibrator box connector
41. Electrical connector
42. Selector knob
43. Rod end bearing

ICN-412-A-670200-B-97499-00010-A-001-01

Figure 1. Cyclic Controls Rigging (Sheet 4 of 4)

Antitorque Controls - Rigging

Preliminary requirements

Support equipment

Nomenclature	Identification No.	Qty	Remarks
Tail rotor rigging tool	- T103206	1	
Spring Scale	- Commercial	1	Range to measure 4 to 5 lbs. - 17.80 to 22.24 N)
Tool assembly	- 412-201-700-101	1	
Tool assembly	- 412-201-701-101	1	

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Corrosion preventive compound	C-101	AR	
Sealant	C-308	AR	
Lockwire	C-405	AR	

Procedure

NOTE

Rigging procedures contained herein, pertain to helicopters with (DAFCS) (SAR) with dual controls installed in basic controls systems.

NOTE

For maintenance procedures of dual controls refer to BHT-412-SI-13 and BHT-412-IPC for illustrated parts list.

1. Comply with the following general procedures when rigging antitorque controls:

1.1. When a rigging procedure requires a specific position to accomplish following steps, manually hold controls in position or secure controls in position with pilot friction adjustment.

1.2. Tolerance for rigging dimensions is ± 0.03 in. (0.76 mm).

1.3. All adjustable control tubes shall have a maximum of 1.00 in. (25.4 mm) of exposed threads after adjustment. If control tubes have inspection holes, the threads must be engaged sufficiently to cover holes.

1.4. Apply corrosion preventive compound (C-101) to threads of all adjustable end bearings and clevises after adjustment.

1.5. Torque control jamnuts 80 to 100 in. lbs. (9.0 to 11.3 Nm) after adjustment unless other torque is specified.

1.6. Control tubes must be free to rotate several degrees about longitudinal axis. Loosen jamnuts, adjust end fittings, and torque jamnuts if necessary.

2. Disconnect the following control tubes and force gradient.

2.1. Disconnect force gradient (14, Figure 1) and transducers (20) from pilot pedal adjuster (21).

2.2. Disconnect control tube (16) from copilot pedal adjuster (17).

2.3. Disconnect control tube (24) from pilot pedal adjuster (21).

2.4. Disconnect control tube (26) from bellcrank (25).

2.5. Disconnect link (4) from lever (3).

3. Connect hydraulic cart to helicopter.

4. Verify length of servo actuator (10), 62.865 to 62.915 in. (1596.77 to 1598.04 mm).

5. Apply hydraulic power from hydraulic cart to helicopter.

6. Position both pilot and copilot left directional pedals (18 and 22) against extreme forward stop. Adjust control tube (16) to fit, torque jam nut and install control tube (16).

7. Connect control tube (24) to pilot pedal adjuster (21).

8. Retract two stop bolts on bellcrank (9, detail A).

9. Install tool assembly (412-201-701-101) (27, Detail B) on pilot pedals (22 and 23).

10. Remove bolt (28) which secures servo actuator (10) to bellcrank (9), and install tool assembly (412-201-700-101) (29) and indexing pin (30, detail A).

11. Remove bolt (31) which secures bellcrank (9) to support (8, detail A).

12. Retract both antirotation screws (32) in tool assembly (412-201-700-101) (29, detail A).

13. Position tool assembly (412-201-700-101) (29) against the inboard surface of support (8). Install bolt (NAS6604-30D) through tool assembly (412-201-700-101) (29), support (8), and bellcrank (9) as shown in detail A, but do not completely tighten at this time. Rotate tool assembly (29) body until the indexing pin (30) (installed in Step 10.) engages the center slot of the tool assembly (29).

14. Adjust both tool assembly (29) antirotation screws (32) to make contact with the support (8) as shown in detail A. As the antirotation screws (32) are tightened, ensure indexing pin (30) is free to disengage and reengage the center slot in the tool assembly (29) body, then tighten bolt (NAS6604-30D).

15. Remove tool assembly (412-201-701-101) (27) from the pilots left and right pedals (22 and 23, detail C).

16. Retract indexing pin (30) until it clears tool assembly (412-201-700-101) (29) body.

17. Slowly move pilots left pedal (22) forward until indexing pin (30) engages the forward slot in tool assembly (412-201-700-101) (29) body. Adjust lower stop bolt in bellcrank (9) to the left pedal stop within 0.000 to 0.002 in. (0.000 to 0.0508 mm) as shown in detail A.

18. Retract indexing pin (30) until it clears tool assembly (412-201-700-101) (29) body.

19. Slowly move pilots right pedal (23) forward until indexing pin (30) engages the aft slot in tool assembly (412-201-700-101) (29) body. Adjust upper stop bolt in bellcrank (9) to the right pedal stop within 0.000 to 0.002 in. (0.000 to 0.0508 mm) as shown in detail A.

20. Remove tool assembly (412-201-700-101) (29) and install bolt (28) and bolt (31) that were removed in Step 10.. and Step 11..

21. Position and hold pilots left pedal (22) forward against stop.

22. Check two pitch links (2, detail E) for 6.180 in. ± 0.010 in. length (156.97 mm ± 0.254 mm) between bolt hole centers.

23. Remove bolt (34, detail E) retaining pitch link to crosshead (1).

24. Install tail rotor rigging tool (T103206) (35, detail E) to crosshead (1) and against trunnion (37). Retain with bolt (34).

25. Adjust link (4, detail F) to obtain 0.58 in. (14.73 mm) clearance with bellcrank (6). Torque jamnut on link (4), and install link (4) on bellcrank (6) and lever (3).

26. With pilots left pedal (22) positioned (per Step 21.), and link (4) adjusted (per Step 25.), adjust tube (26) to fit to bellcrank (25). Torque jamnut and install control tube (26).

27. Remove tail rotor rigging tool (T103206) (35, detail E) and install pitch link (2) to crosshead (1).

28. Rig antitorque trim system as follows:

28.1. Install tool assembly (412-201-701-101) (27) on pilot left and right pedals (22 and 23, detail B).

28.2. Position output arm (33) on rotary actuator (15, detail C).

28.3. Place output arm (33) of rotary actuator (15) at center of travel, adjust force gradient (14) to fit to the pilot pedal adjuster (21). Torque jam nut and install force gradient (14) on pilot pedal adjuster (21).

28.4. Remove tool assembly (412-201-701-101) (27) from pilot left and right pedals (22 and 23) as shown in detail B.

28.5. Check clearance between force gradient (14) and structure at extreme position of the output arm (33). Reindex output arm (33) relative to shaft by one serration, if required for clearance, then repeat Step 28.2.. through Step 28.4..

29. Ensure that all antitorque controls bolts are installed and cotter pins, sealant (C-308)), and lockwire (C-405), is installed where applicable.

30. Perform antitorque control motion transducer (20, Figure 1 detail D) adjustment as follows:

30.1.

NOTE

Dual element control motion transducers (412-074-101-103) are aligned and positioned using calibration tool (412-274-001-101).

NOTE

Refer to cyclic control motion transducer adjustment 412-A-67-02-00-02A-276A-A / 02761, step 2.1, using calibration tool for electrical alignment of antitorque transducers

NOTE

This calibration tool (412-274-001-101) is used to electrically align the control motion transducers installed on the helicopter to be used as control position sensors for the flight data recorder. This tool will insure that the control motion transducers are aligned within ± 0.5 percent of the prescribed position. Fore and aft position is set at 50 percent. Lateral position is set at 50 percent. Collective position is set at 12 percent. Pedal position is set at 12 percent.

31. Adjust antitorque minimum friction as follows:

31.1. Disconnect force gradient (14) from pedal adjuster (21)

31.2. Disconnect control tube (12) from walking beam (13).

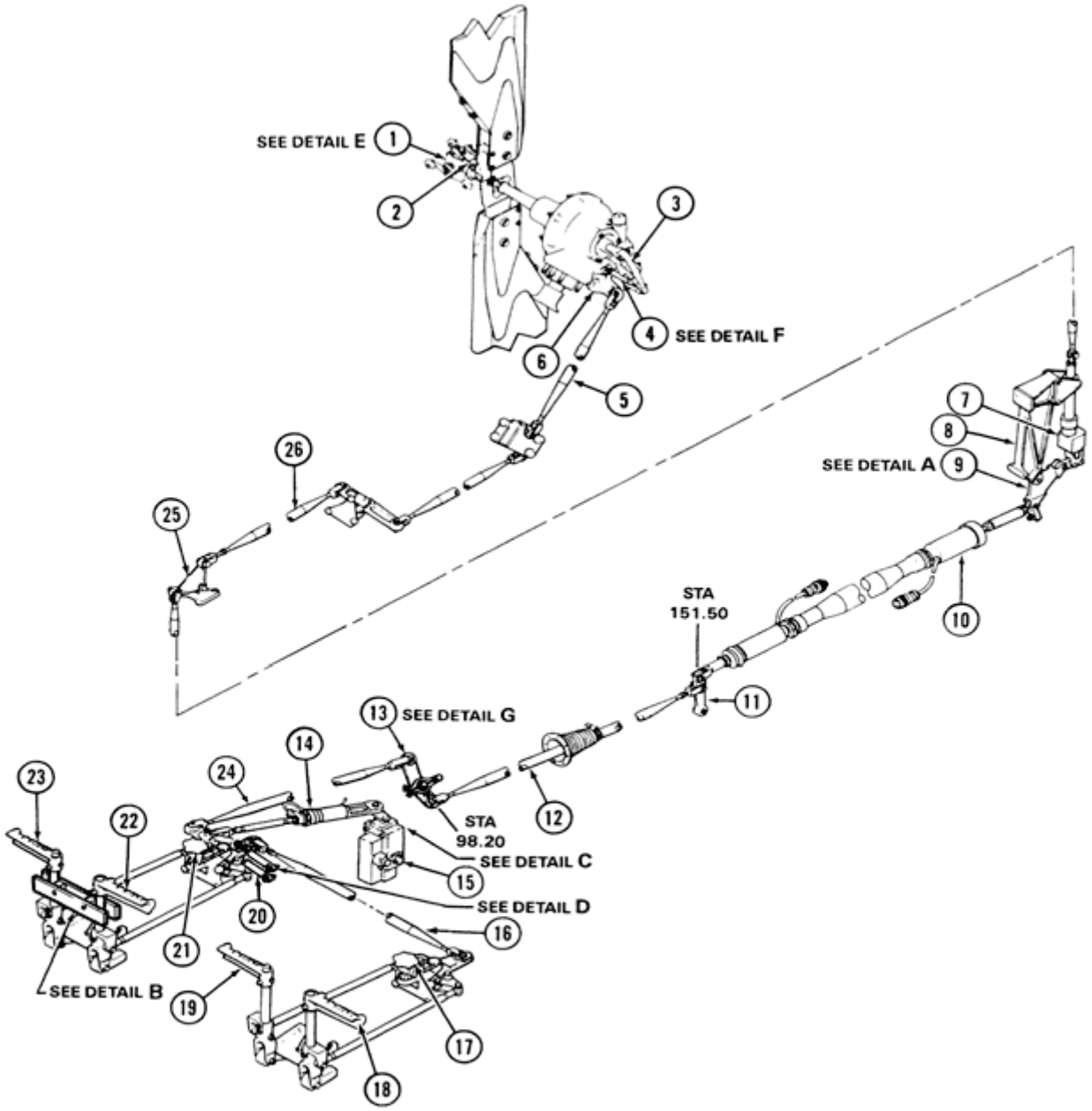
31.3. Attach spring scale to one pilot antitorque pedal. Measure force required to move pedal through neutral. Acceptable force range is 4.25 to 4.75 lbs. (18.91 to 21.13 N). Adjust screw (39, detail G) on clamp (38) to obtain friction within acceptable range.

31.4. Install control tube (12) on walking beam (13).

31.5. Install force gradient (14) on pedal adjuster (21).

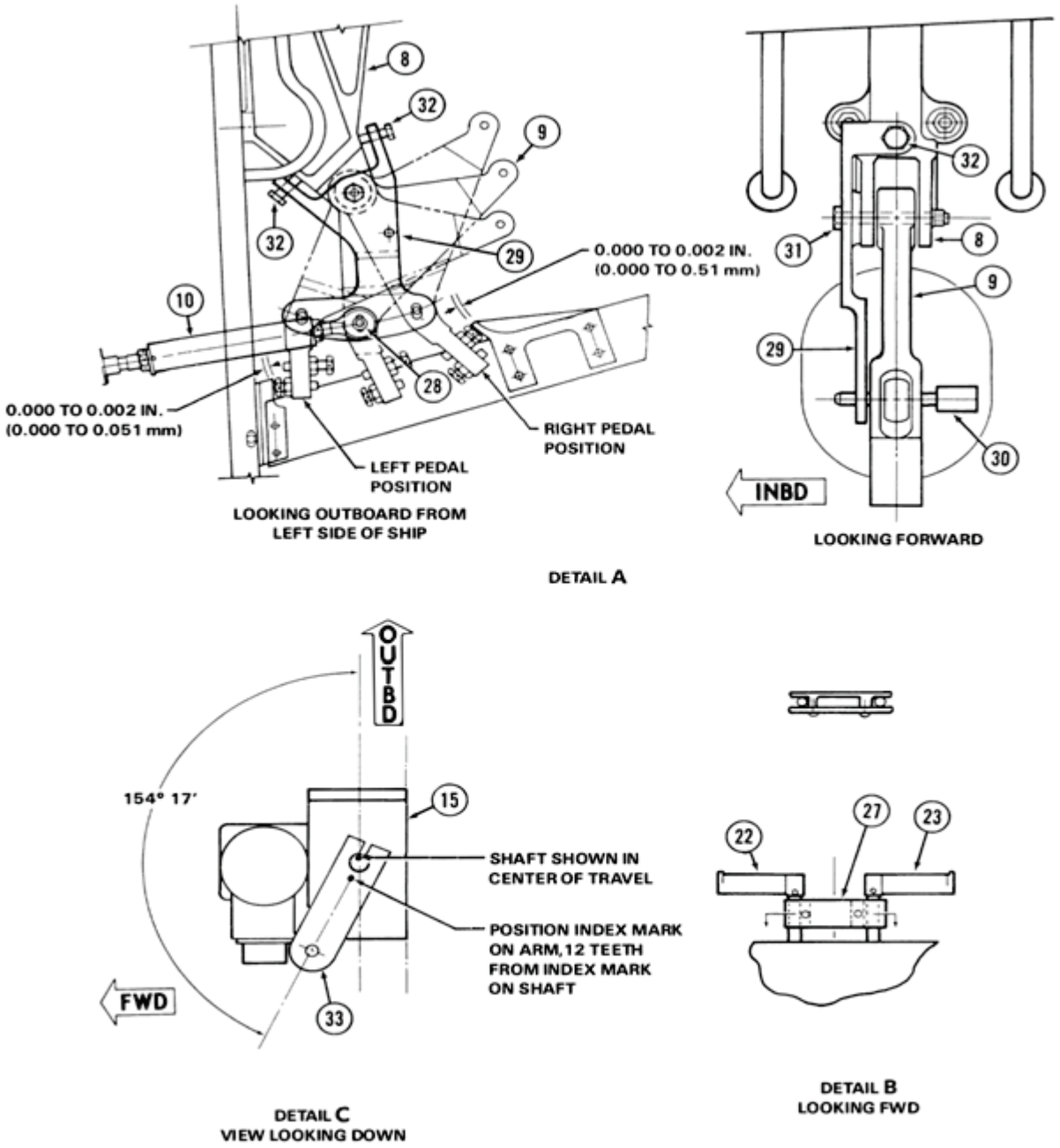
32. Move antitorque controls through full travel and check for binding and/or obstruction of control movement.

33. Ensure that all antitorque controls bolts are installed and cotter pins, sealant (C-308)), lockwire (C-405), installed where applicable.



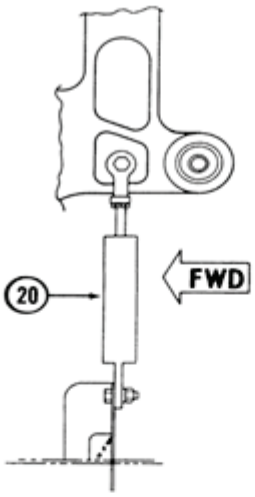
ICN-412-A-670200-B-97499-00011-A-001-01

Figure 1. Antitorque Controls Rigging (Sheet 1 of 4)

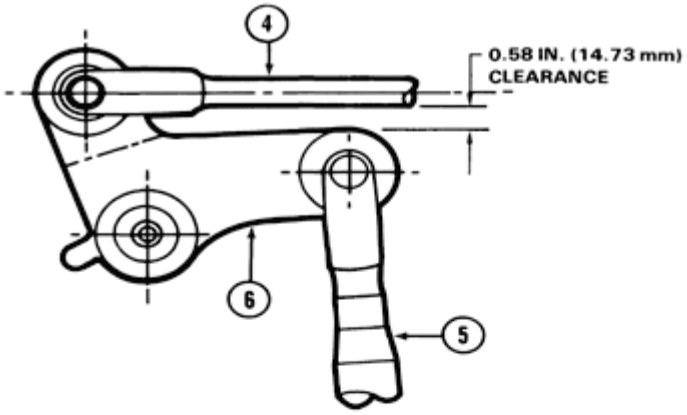


ICN-412-A-670200-B-97499-00012-A-001-01

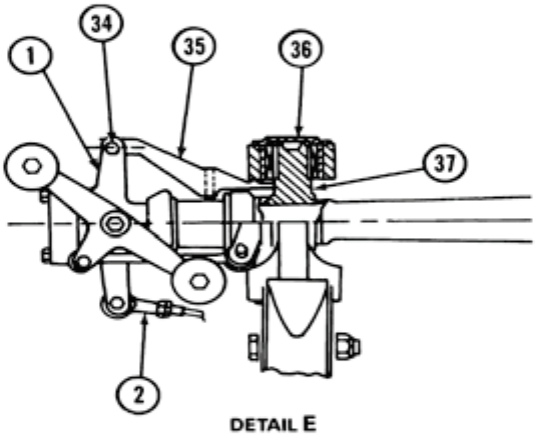
Figure 1. Antitorque Controls Rigging (Sheet 2 of 4)



DETAIL D
VIEW LOOKING DOWN



DETAIL F



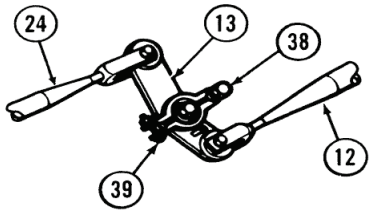
DETAIL E

NOTES

1. Remove one pitch link (2).
2. Position rigging tool (35) on crosshead (1) and trunnion (37) and install bolt (34).
3. Push inboard on crosshead (1) to cause tool to make firm contact with trunnion (37).
4. After rigging is completed, remove rigging tool (35) from tail rotor (36).
5. Install pitch link (2) that was removed in note 1.

ICN-412-A-670200-B-97499-00013-A-001-01

Figure 1. Antitorque Controls Rigging (Sheet 3 of 4)



DETAIL G

1. Crosshead
2. Pitch links
3. Lever
4. Link
5. Control tube
6. Bellcrank
7. Hydraulic cylinder
8. Support
9. Bellcrank
10. Servoactuator
11. Lever
12. Control tube
13. Walking beam
14. Force gradient
15. Rotary actuator
16. Control tube
17. Copilot pedal adjuster
18. Copilot left pedal
19. Copilot right pedal
20. Transducers
21. Pilot pedal adjuster
22. Pilot left pedal
23. Pilot right pedal
24. Control tube
25. Bellcrank
26. Control tube
27. Tool assembly (412-201-701-101)
28. Bolt (20-057-4-18D)
29. Tool assembly (412-201-700-101)
30. Indexing pin
31. Bolt (20-057-4-23D)
32. Antirotation screws
33. Output arm
34. Bolt
35. Tail rotor rigging tool (T103206)
36. Tail rotor
37. Tail rotor hub trunnion
38. Clamp
39. Screw

ICN-412-A-670200-B-97499-00014-A-001-01

Figure 1. Antitorque Controls Rigging (Sheet 4 of 4)

Intentionally blank page

Collective Controls - DAFCS - Description

The collective controls installation (Figure 1) consists of the control linkage and components from pilot and copilot collective sticks (9 and 13) up to but not including collective lever (1).

The collective controls installation consists of a force gradient (18, Figure 1), lever (17), bellcrank (19), link (20), support (21), electromechanical rotary actuator (22), and two transducers (14). The two transducers (14) are connected to the arm of jackshaft tube (11) and support.

The collective trim system consists of an electromechanical rotary actuator (22) and a force gradient (18) with a detent switch connected in parallel to the primary collective controls. The rotary actuator (22) repositions the collective controls in response to a signal from either the pilot or the AFCS system. Actuation of the force gradient (18) detent switch either by the rotary actuator driving to its internal stop or by a pilot control input will disengage the rotary actuator.

Two dual element linear transducers (14) connected in parallel to the collective controls are used to convert mechanical movement to an electrical position signals for the DAFCS and flight data recorder.

The cylinder (5) reduces effort required for control and reduces feedback forces from the main rotor.

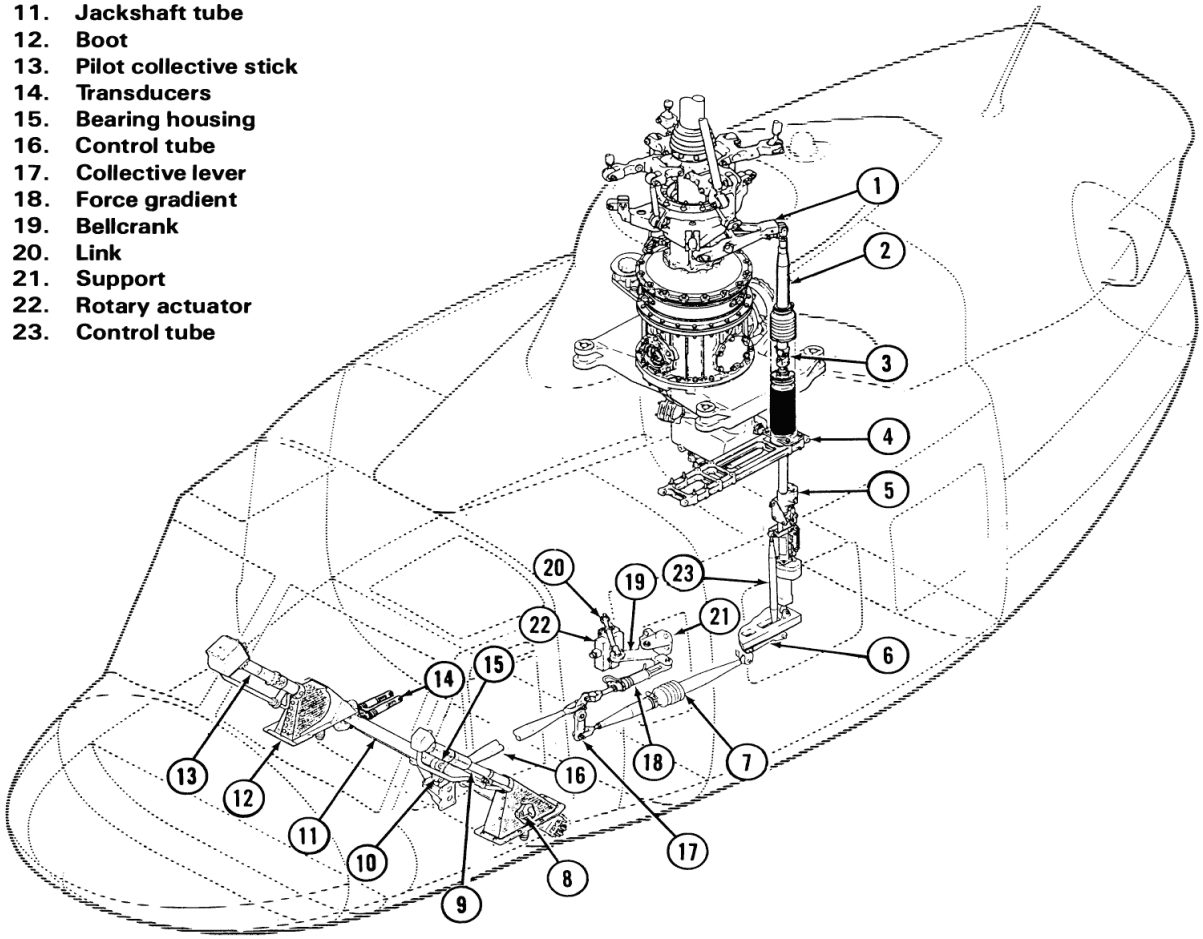
Actuation of the pilot collective stick (13) or copilot collective stick (9) causes hydraulic power assisted collective pitch change of the main rotor blades.

Collective dual controls are installed for copilot control of the helicopter.

Operation of collective dual controls system is basically the same as operation of single system. Refer to Flight Manual for operating procedures.

For removal and installation instructions of copilot collective control system refer to BHT-412-SI-13. Refer to BHT-412-IPC for illustrated parts list.

1. Collective lever
2. Control tube
3. Universal
4. Support
5. Cylinder
6. Bellcrank
7. Adjustable control tube
8. Clamp
9. Copilot collective stick
10. Control arm
11. Jackshaft tube
12. Boot
13. Pilot collective stick
14. Transducers
15. Bearing housing
16. Control tube
17. Collective lever
18. Force gradient
19. Bellcrank
20. Link
21. Support
22. Rotary actuator
23. Control tube



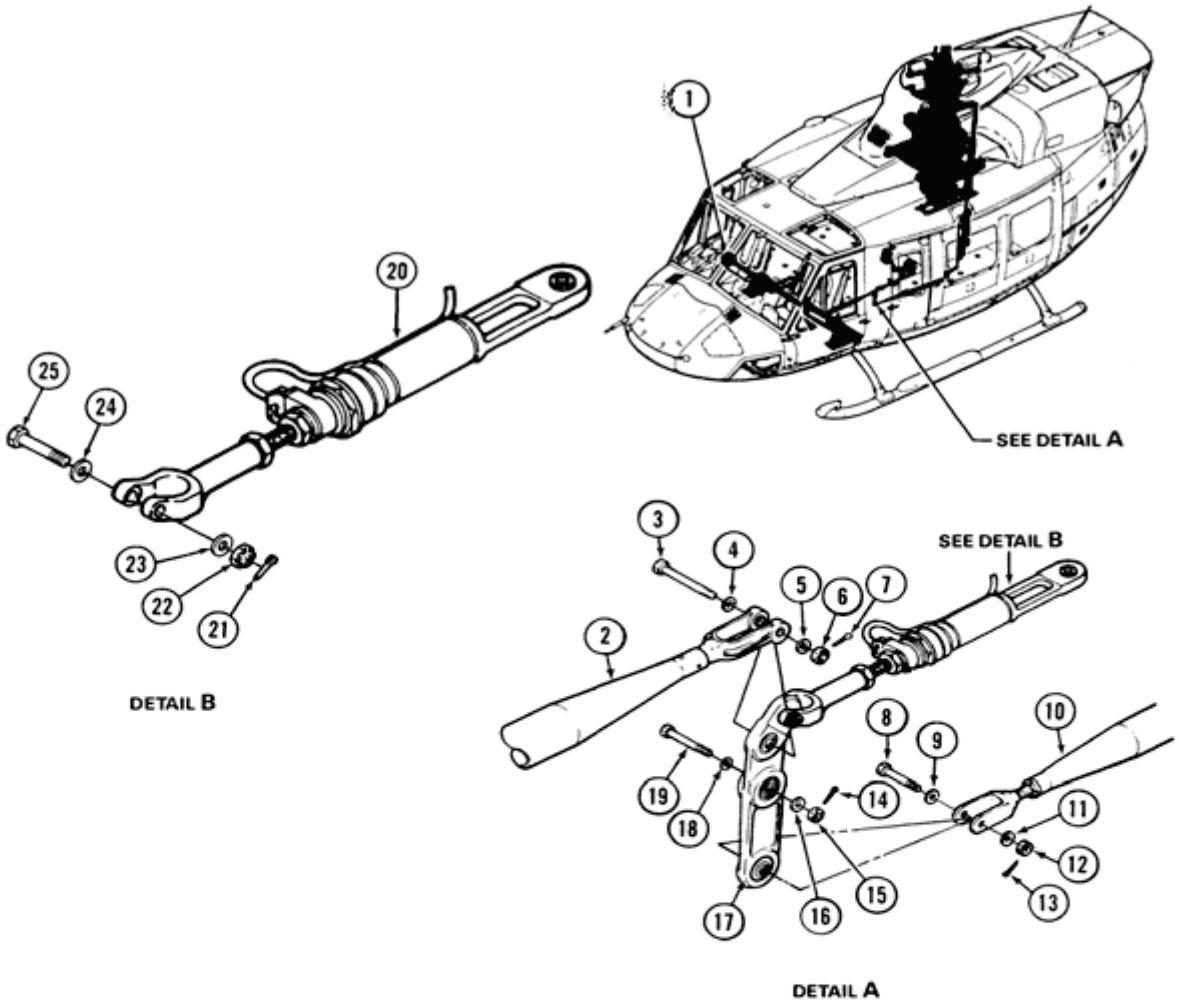
ICN-412-A-671101-B-97499-00001-A-001-01

Figure 1. Collective Controls (Sheet 1 of 1)

Collective Lever - Removal

Procedure

1. Remove access panel from cabin floor forward of pylon.
2. Remove bolt (3, Figure 1).
3. Remove bolt (8).
4. Remove bolt (19).
5. Remove bolt (25).
6. Remove lever (17) from air frame structure.



1. Pilot collective stick
2. Control tube
3. Bolt
4. Aluminum washer
5. Aluminum washer
6. Nut
7. Cotter pin
8. Bolt
9. Steel washer
10. Adjustable control tube
11. Steel washer
12. Nut
13. Cotter pin
14. Cotter pin
15. Nut
16. Aluminum washer
17. Lever
18. Aluminum washer
19. Bolt
20. Force gradient
21. Cotter pin
22. Nut
23. Steel washer
24. Steel washer
25. Bolt

ICN-412-A-671101-B-97499-00002-A-001-01

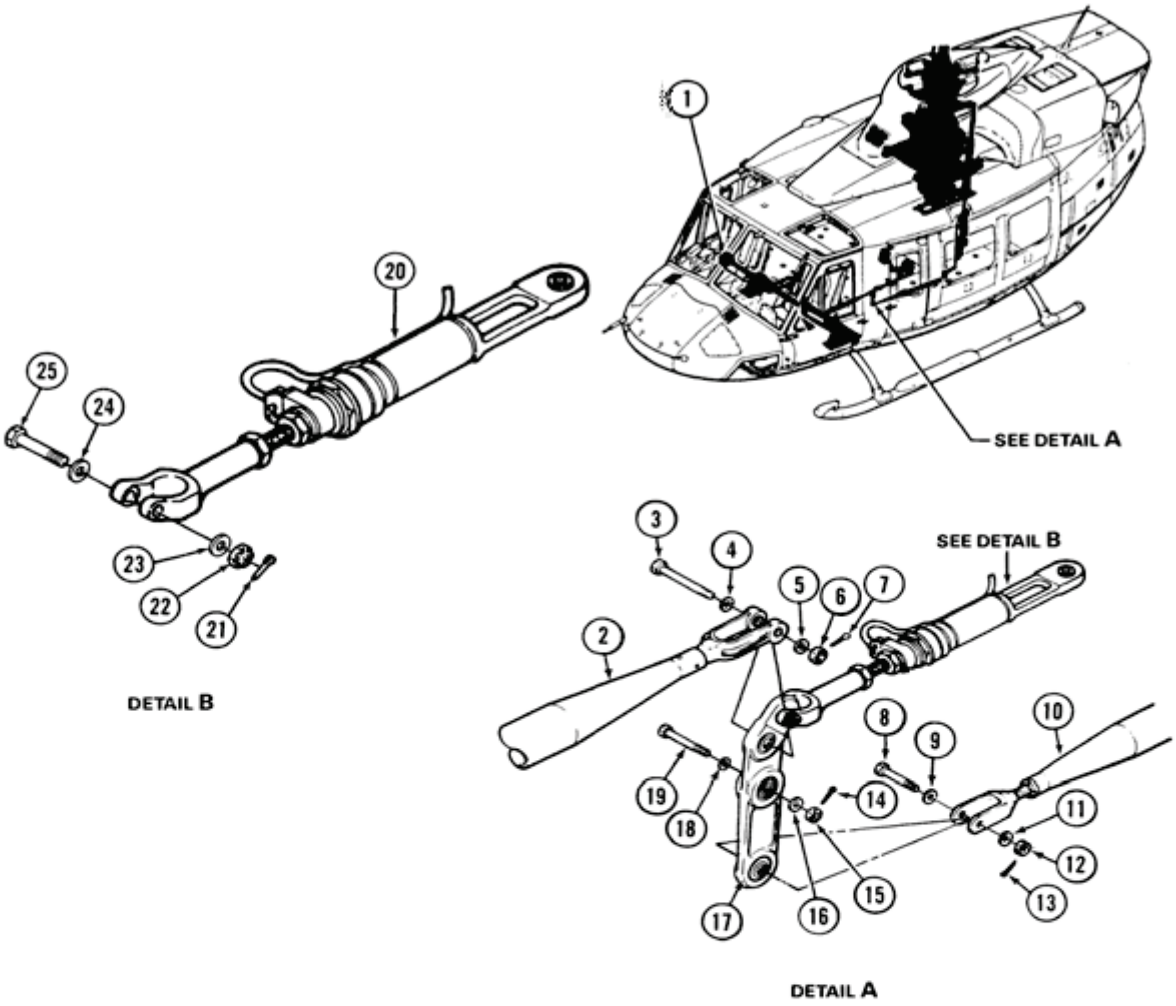
Figure 1. Collective Lever Removal/Installation (Sheet 1 of 1)

Collective Lever - Inspection

Procedure

1. Inspect lever (17, Figure 1) for mechanical and corrosion damage in accordance with 412-CRO - Component Repair and Overhaul Manual.

2. Inspect bearings in lever (17) for secure installation in lever and for wear (BHT-ALL-SPM, Standard Practices Manual).



1. Pilot collective stick
2. Control tube
3. Bolt
4. Aluminum washer
5. Aluminum washer
6. Nut
7. Cotter pin
8. Bolt
9. Steel washer
10. Adjustable control tube
11. Steel washer
12. Nut
13. Cotter pin
14. Cotter pin
15. Nut
16. Aluminum washer
17. Lever
18. Aluminum washer
19. Bolt
20. Force gradient
21. Cotter pin
22. Nut
23. Steel washer
24. Steel washer
25. Bolt

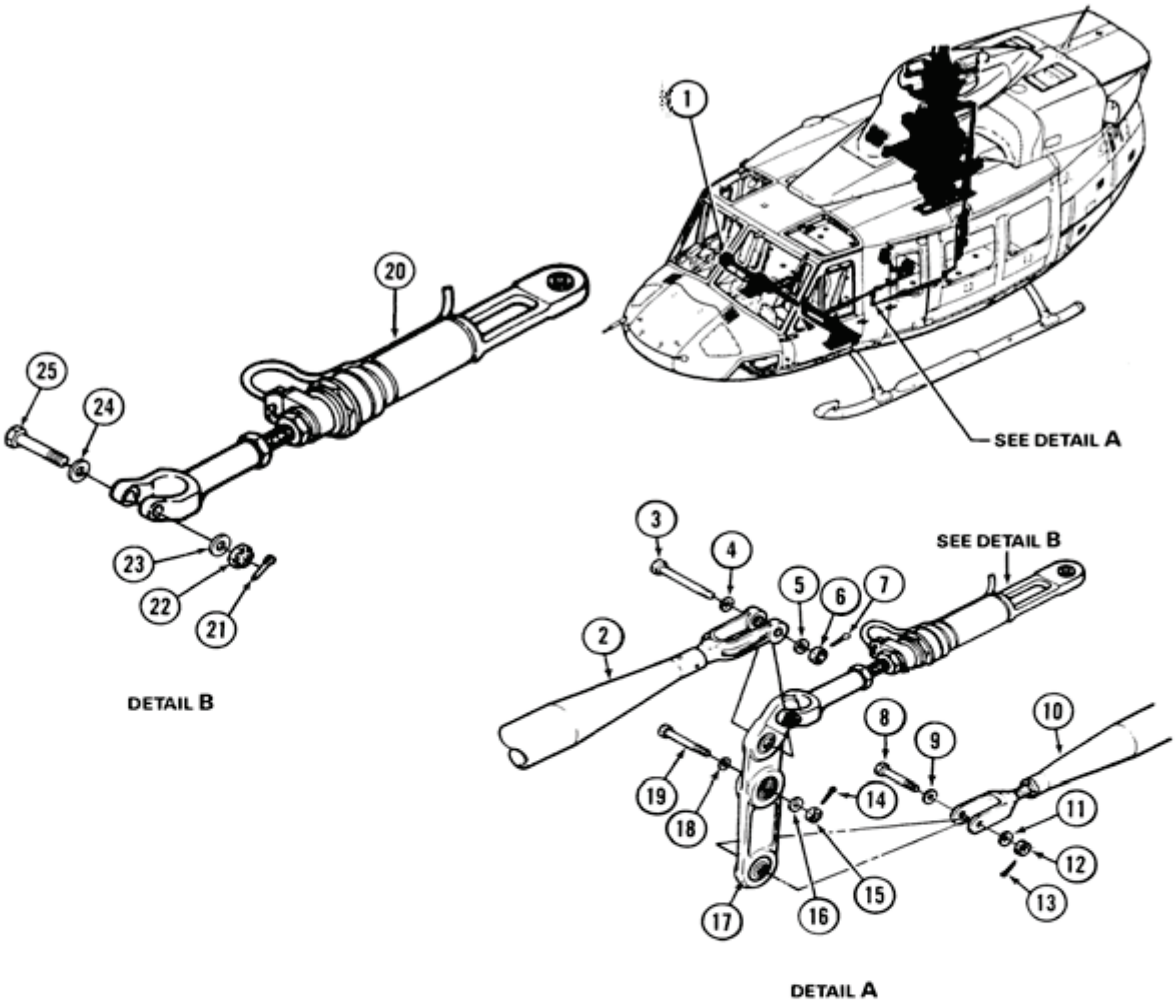
ICN-412-A-671101-B-97499-00002-A-001-01

Figure 1. Collective Lever Removal/Installation (Sheet 1 of 1)

Collective Lever - Installation

Procedure

1. Place lever (17, Figure 1) into position in airframe structure with long end down. Install bolt (19), aluminum washers (18 and 16), nut (15), and cotter pin (14).
2. Position adjustable control tube (10) on lever (17) and install bolt (8), steel washers (9 and 11), nut (12), and cotter pin (13).
3. Position control tube (2) on lever (17) and install bolt (3), steel washers (4 and 5), nut (6) and cotter pin (7).
4. Position force gradient (20) on lever (17) and install bolt (25), aluminum washers (24 and 23), nut (22), and cotter pin (21).
5. Move pilot collective stick (1) through full throw and check for binding and/or obstruction of controls.
6. Confirm that collective controls rigging is within limits. Refer 412-A-67-02-00-01A-276A-A / 02760.
7. Install access panel in cabin floor forward of pylon.



1. Pilot collective stick
2. Control tube
3. Bolt
4. Aluminum washer
5. Aluminum washer
6. Nut
7. Cotter pin
8. Bolt
9. Steel washer
10. Adjustable control tube
11. Steel washer
12. Nut
13. Cotter pin
14. Cotter pin
15. Nut
16. Aluminum washer
17. Lever
18. Aluminum washer
19. Bolt
20. Force gradient
21. Cotter pin
22. Nut
23. Steel washer
24. Steel washer
25. Bolt

ICN-412-A-671101-B-97499-00002-A-001-01

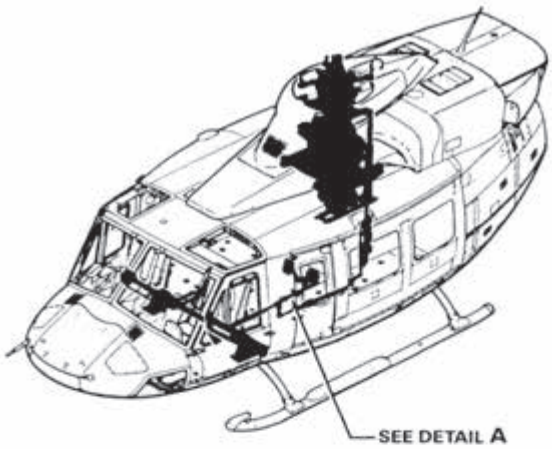
Figure 1. Collective Lever Removal/Installation (Sheet 1 of 1)

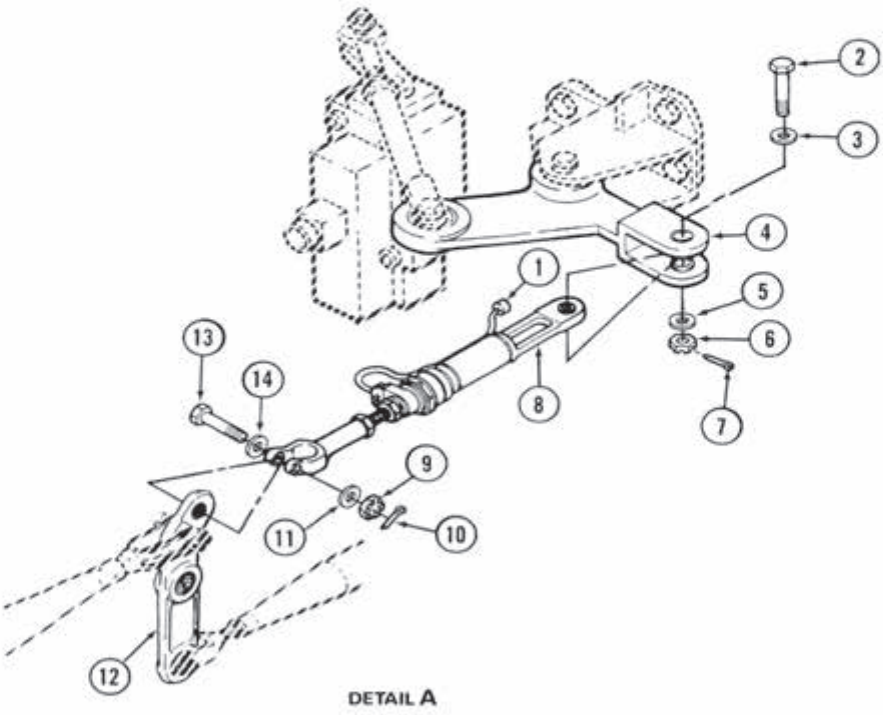
Collective Force Gradient - Removal

Procedure

1. Remove access panel from cabin floor forward of pylon.
2. Disconnect electrical connector (1, Figure 1).
3. Remove bolt (2).
4. Remove bolt (13).
5. Remove force gradient (8).

1. Electrical connector
2. Bolt
3. Aluminum washer
4. Bellcrank
5. Aluminum washer
6. Nut
7. Cotter pin
8. Force gradient
9. Cotter pin
10. Nut
11. Washer
12. Lever
13. Bolt
14. Washer

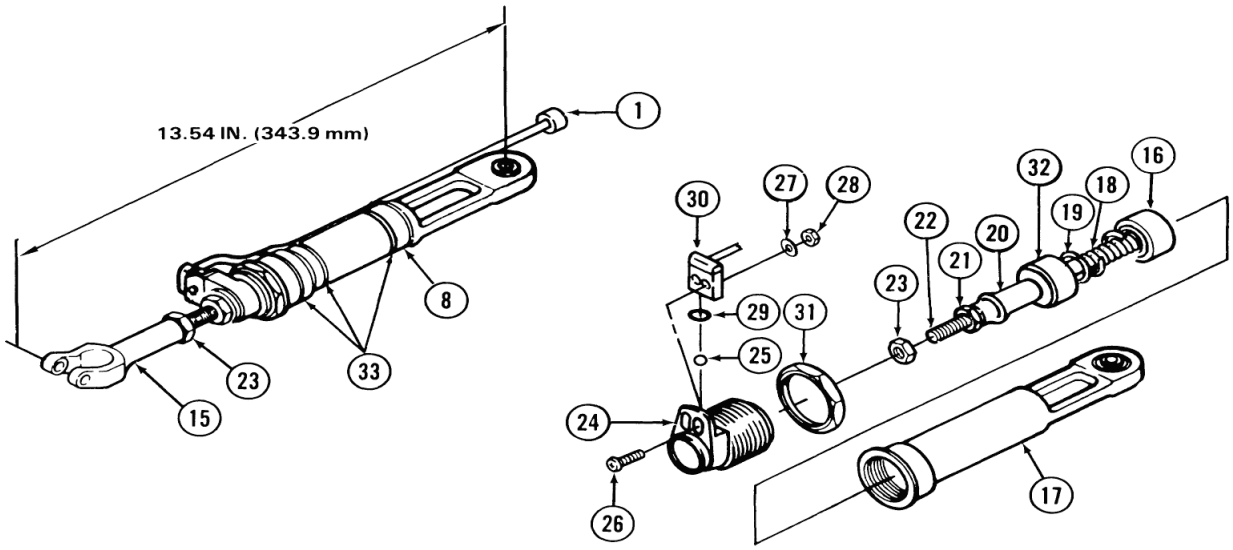




DETAIL A

ICN-412-A-671101-B-97499-00003-A-001-01

Figure 1. Collective Force Gradient Removal/Installation (Sheet 1 of 2)



DETAIL B
VIEW LOOKING DOWN

15. Clevis
16. Guide
17. Cylinder
18. Spring
19. Spring
20. Spacer
21. Jamnut
22. Shaft
23. Jamnut
24. Cap
25. Ball
26. Screw
27. Washer
28. Nut
29. Packing
30. Switch and cable assembly
31. Jamnut
32. Spring assembly
33. Strap

ICN-412-A-671101-B-97499-00004-A-001-01

Figure 1. Collective Force Gradient Removal/Installation (Sheet 2 of 2)

Intentionally blank page

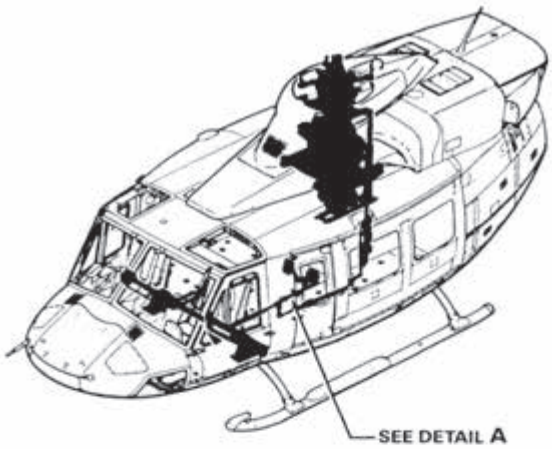
Collective Force Gradient - Inspection

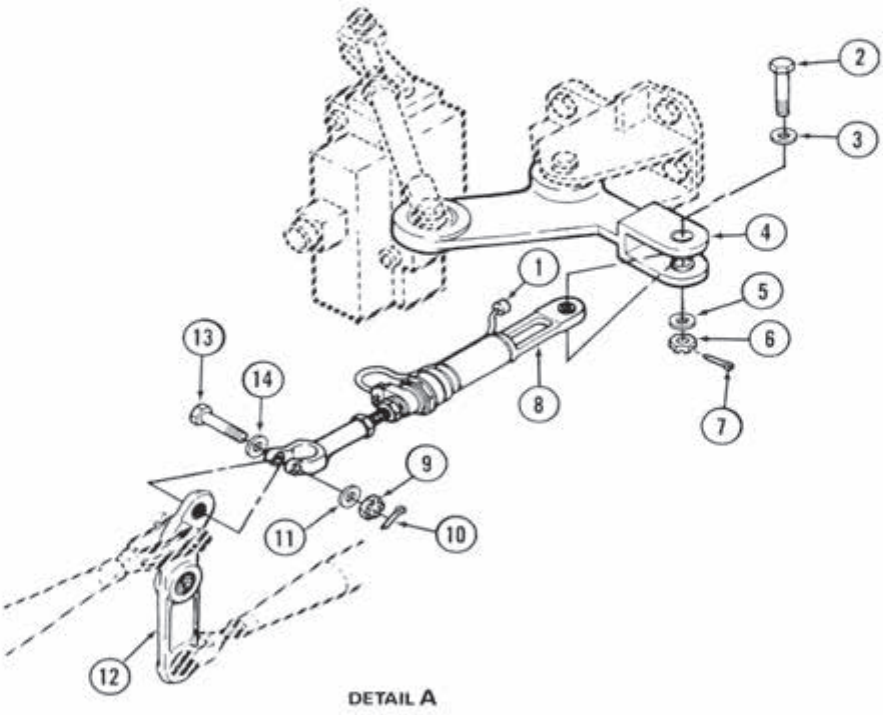
Procedure

1. Inspect force gradient (8, Figure 1) for corrosion and mechanical damage (412-CRO - Component Repair and Overhaul Manual). Superficial mechanical damage is acceptable without repair. Mechanical damage in excess of superficial and/or corrosion damage is not acceptable.

2. Inspect bearing for secure installation in housing and for wear (BHT-ALL-SPM, Standard Practices Manual).

1. Electrical connector
2. Bolt
3. Aluminum washer
4. Bellcrank
5. Aluminum washer
6. Nut
7. Cotter pin
8. Force gradient
9. Cotter pin
10. Nut
11. Washer
12. Lever
13. Bolt
14. Washer

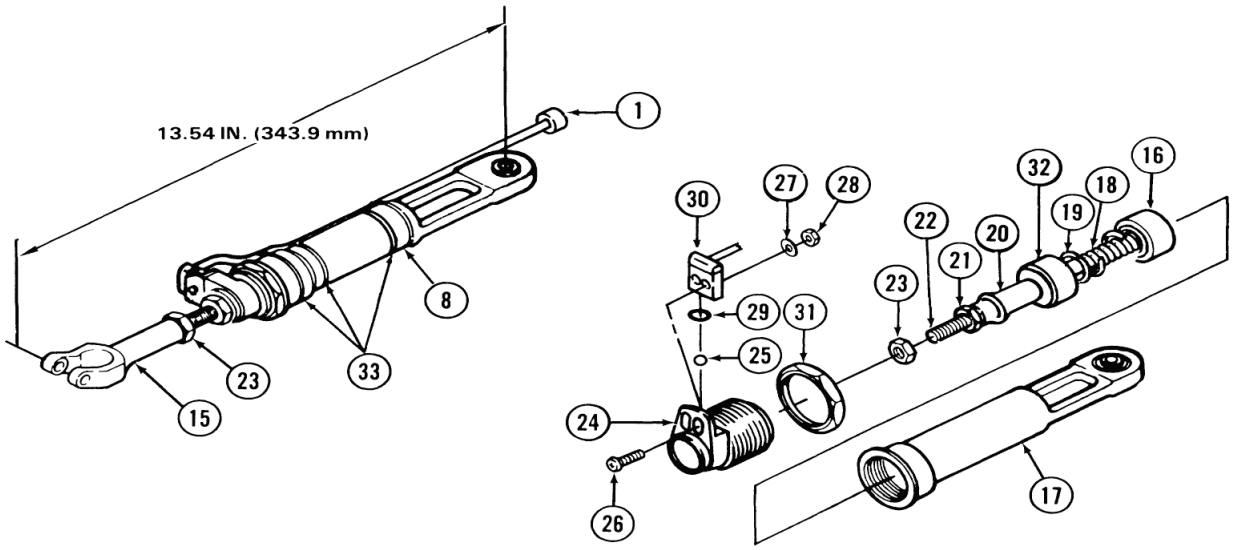




DETAIL A

ICN-412-A-671101-B-97499-00003-A-001-01

Figure 1. Collective Force Gradient Removal/Installation (Sheet 1 of 2)



DETAIL B
VIEW LOOKING DOWN

15. Clevis
16. Guide
17. Cylinder
18. Spring
19. Spring
20. Spacer
21. Jamnut
22. Shaft
23. Jamnut
24. Cap
25. Ball
26. Screw
27. Washer
28. Nut
29. Packing
30. Switch and cable assembly
31. Jamnut
32. Spring assembly
33. Strap

ICN-412-A-671101-B-97499-00004-A-001-01

Figure 1. Collective Force Gradient Removal/Installation (Sheet 2 of 2)

Intentionally blank page

Collective Force Gradient - Repair

Procedure

1. Repair of force gradient consists of replacement of unserviceable components and adjustment of spring preload in accordance with 412-A-67-11-01-02A-720A-A / 02770, step 1 .

Intentionally blank page

Collective Force Gradient - Installation

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Lockwire	C-405	AR	

Procedure

ENSURE THAT FORCE GRADIENT LENGTH IS ADJUSTED PRIOR TO INSTALLATION ON ROTARY ACTUATOR.

NOTE

The collective force gradient is similar in appearance to the force gradients used in the cyclic controls but has a different part number. Ensure that correct part number force gradient is installed for the collective controls system.

1. If not previously accomplished, adjust spring preload on force gradient (8, Figure 1, Sheet 2), 40 to 41 lbs. (177.92 to 182.368 N) as follows:

1.1. Loosen jamnut (23) and remove clevis (15).

1.2. Cut straps (33) holding switch and cable assembly (30) to force gradient (8).

1.3. Remove nut (28), washer (27), screw (26), and packing (29), from cap (24), and switch and cable assembly (30).

1.4. Cut lockwire and loosen jamnut (31), remove cap (24) from cylinder (17).

1.5. Remove spring assembly (32) from cylinder (17).

1.6. Loosen jamnut (21) while holding shaft (22) at wrench flat. Adjust spring preload to 40 to 41 lbs. (177 .92 to 182.368 N) by turning spacer (20), (using wrench flat) clockwise against spring assembly (32). Tighten jamnut (21) against spacer (20).

1.7. Position spring assembly (32) in cylinder (17) and install cap (24) until all end play is eliminated. Torque jamnut (31) to 100 in. lbs. (11.298 Nm) and lockwire (C-405) to prevent turning in either direction. After assembly verify spring preload of 40 to 41 lbs. (177.92 to 182.368 N).

1.8. Install packing (29), and switch and cable assembly (30), on cap (24).

1.9. Install screws (26), washers (27), nuts (28), in cap (24) and switch and cable assembly (30). Do not tighten screws at this time.

1.10. Install clevis (15) on shaft (22). Adjust length of force gradient (8) as shown in Figure 1, Sheet 2. Tighten jamnut (23) to clevis (15).

1.11. Push and hold switch and cable assembly (30), and ball (25) against cap (24). Hold this position to initially adjust switch and cable assembly (30), then adjust switch to actuate within ± 0.050 in. (1.27 mm) linear shaft travel of spring assembly (32). Secure screws (26) mounting switch and cable assembly (30) to cap (24).

1.12. Install straps (33) holding switch and cable assembly (30) to force gradient (8).

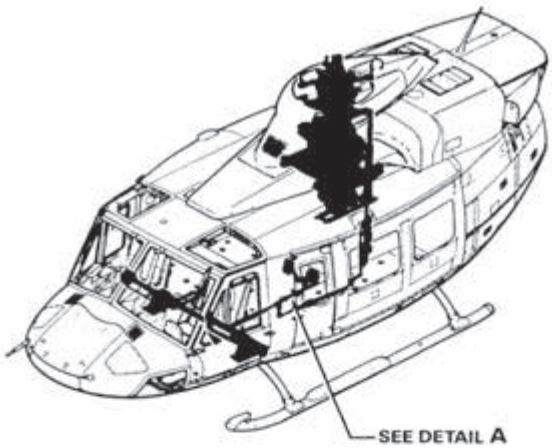
2. Place force gradient (8) into position on bellcrank (3). Install bolt (2), steel washers (3 and 5), nut (6), and cotter pin (7).

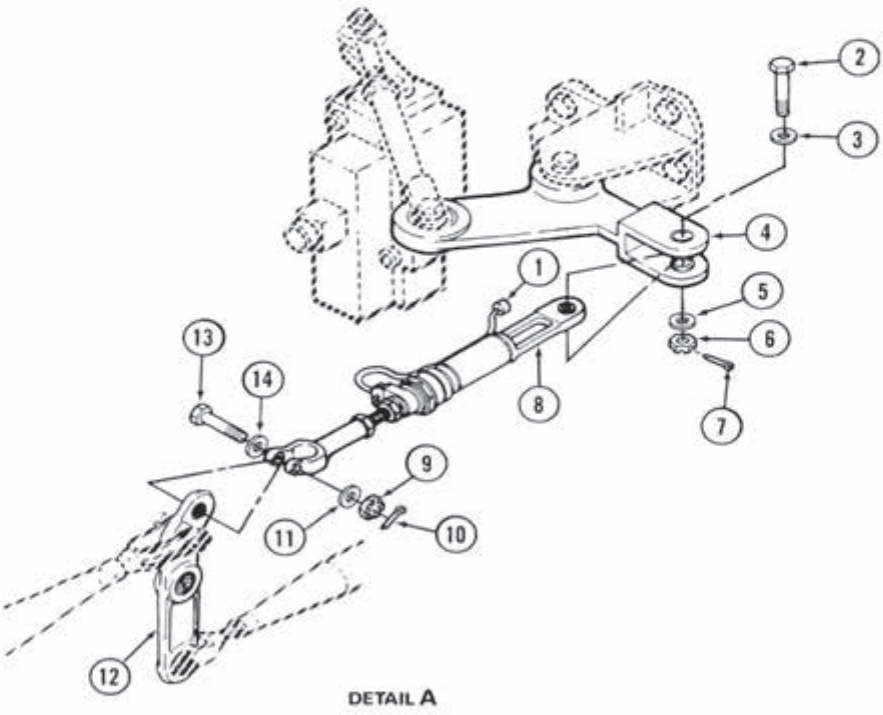
3. Place force gradient (8) into position on lever (12). Install bolt (13), steel washers (14 and 11), nut (10), and cotter pin (9).

4. Connect electrical connector (1) at fuselage bracket.

5. Install access panel in cabin floor forward of pylon.

1. Electrical connector
2. Bolt
3. Aluminum washer
4. Bellcrank
5. Aluminum washer
6. Nut
7. Cotter pin
8. Force gradient
9. Cotter pin
10. Nut
11. Washer
12. Lever
13. Bolt
14. Washer

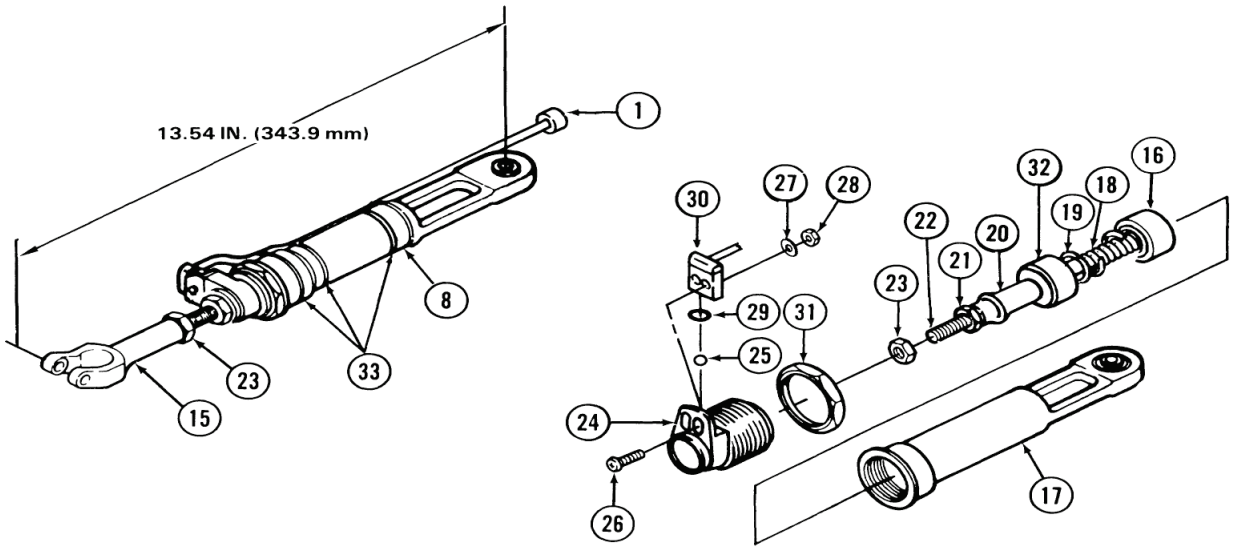




DETAIL A

ICN-412-A-671101-B-97499-00003-A-001-01

Figure 1. Collective Force Gradient Removal/Installation (Sheet 1 of 2)



DETAIL B
VIEW LOOKING DOWN

15. Clevis
16. Guide
17. Cylinder
18. Spring
19. Spring
20. Spacer
21. Jamnut
22. Shaft
23. Jamnut
24. Cap
25. Ball
26. Screw
27. Washer
28. Nut
29. Packing
30. Switch and cable assembly
31. Jamnut
32. Spring assembly
33. Strap

ICN-412-A-671101-B-97499-00004-A-001-01

Figure 1. Collective Force Gradient Removal/Installation (Sheet 2 of 2)

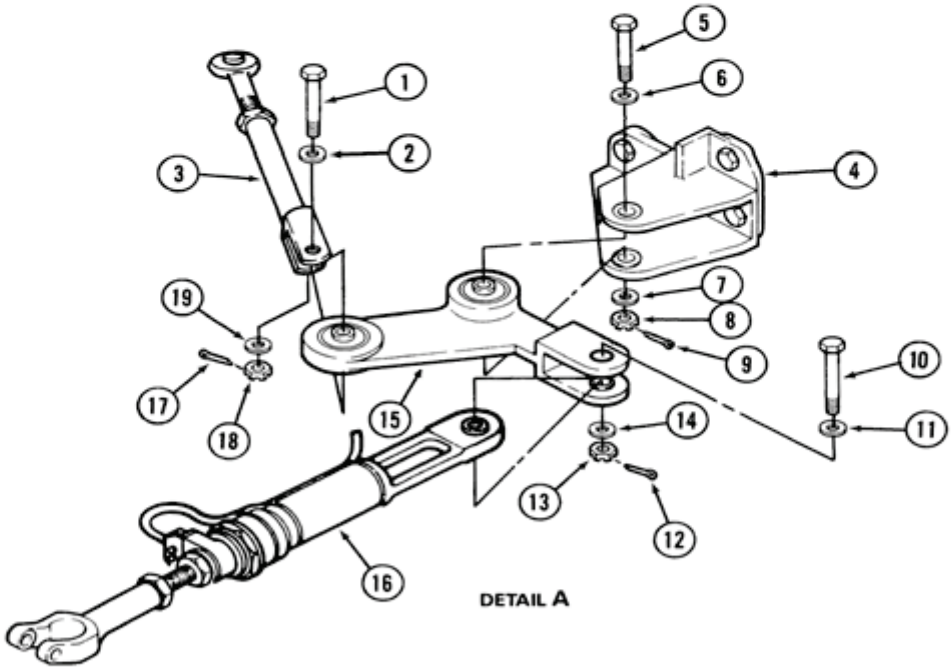
Intentionally blank page

Collective Bellcrank - Removal

Procedure

1. Remove access panel from cabin floor forward of pylon.
2. Remove bolt (1, Figure 1).
3. Remove bolt (10).
4. Remove bolt (5).
5. Remove bellcrank (15) from support (4).

1. Bolt
2. Steel washer
3. Link
4. Support
5. Bolt
6. Aluminum washer
7. Aluminum washer
8. Nut
9. Cotter pin
10. Bolt
11. Aluminum washer
12. Cotter pin
13. Nut
14. Aluminum washer
15. Bellcrank
16. Force gradient
17. Cotter pin
18. Nut
19. Steel washer



ICN-412-A-671101-B-97499-00005-A-001-01

Figure 1. Collective Bellcrank Removal/Installation (Sheet 1 of 1)

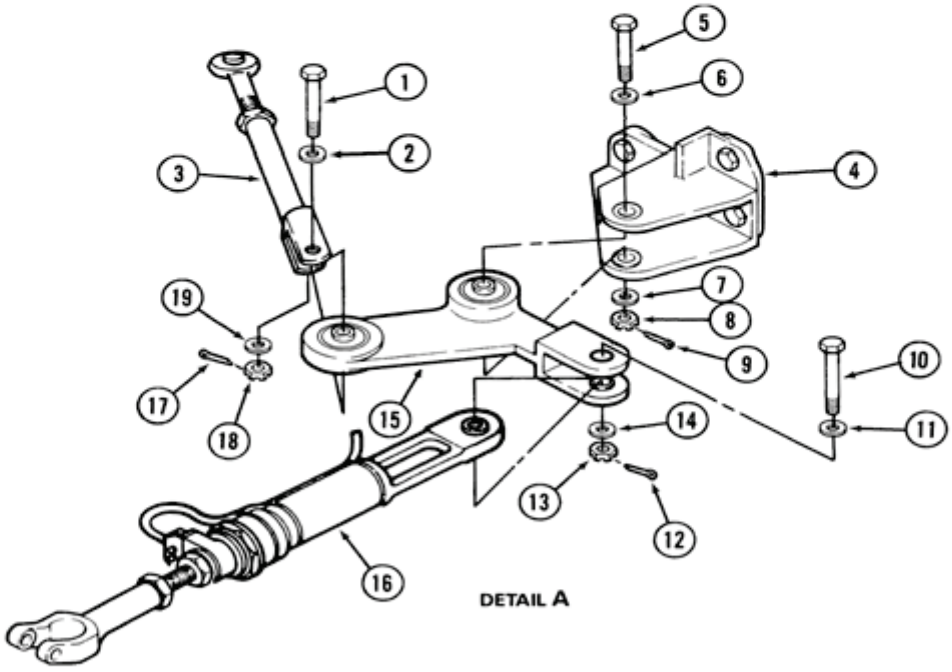
Collective Bellcrank - Inspection

Procedure

1. Inspect bellcrank (15, Figure 1) for mechanical and corrosion damage in accordance with 412-CRO - Component Repair and Overhaul Manual.

2. Inspect bearings in bellcrank (15) for secure installation in bellcrank and for wear (BHT-ALL-SPM, Standard Practices Manual).

1. Bolt
2. Steel washer
3. Link
4. Support
5. Bolt
6. Aluminum washer
7. Aluminum washer
8. Nut
9. Cotter pin
10. Bolt
11. Aluminum washer
12. Cotter pin
13. Nut
14. Aluminum washer
15. Bellcrank
16. Force gradient
17. Cotter pin
18. Nut
19. Steel washer



ICN-412-A-671101-B-97499-00005-A-001-01

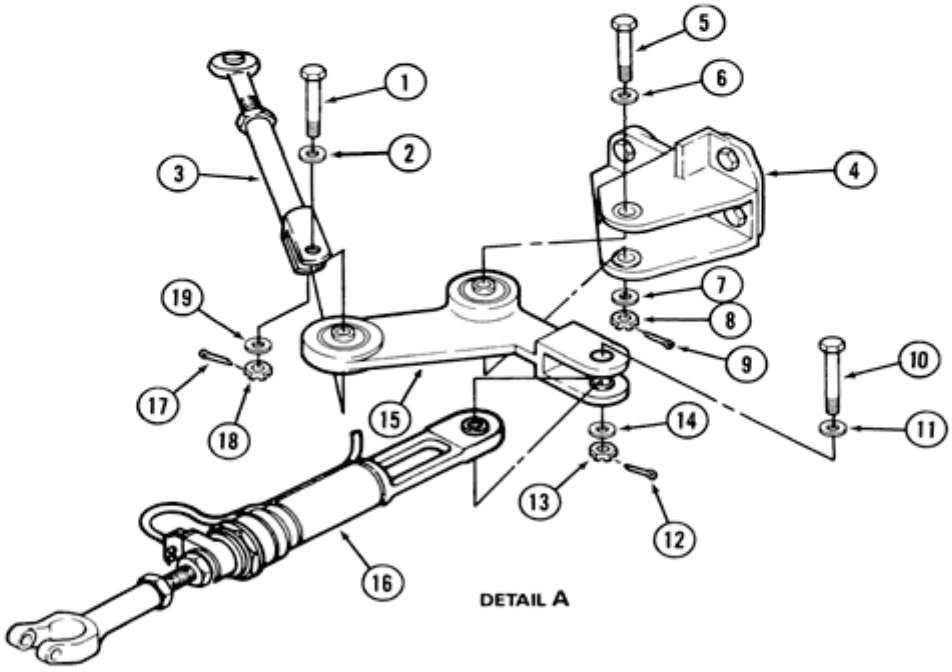
Figure 1. Collective Bellcrank Removal/Installation (Sheet 1 of 1)

Collective Bellcrank - Installation

Procedure

1. Place bellcrank (15, Figure 1) into position on support (4). Install bolt (5), aluminum washers (6 and 7), nut (8), and cotter pin (9).
2. Position force gradient (16) on bellcrank (15) and install bolt (10), aluminum washers (11 and 14), nut (13), and cotter pin (12).
3. Position link (3) on bellcrank (15) and install bolt (1), steel washers (2 and 19), nut (18) and cotter pin (17).
4. Install access panel in cabin floor forward of pylon.

1. Bolt
2. Steel washer
3. Link
4. Support
5. Bolt
6. Aluminum washer
7. Aluminum washer
8. Nut
9. Cotter pin
10. Bolt
11. Aluminum washer
12. Cotter pin
13. Nut
14. Aluminum washer
15. Bellcrank
16. Force gradient
17. Cotter pin
18. Nut
19. Steel washer



ICN-412-A-671101-B-97499-00005-A-001-01

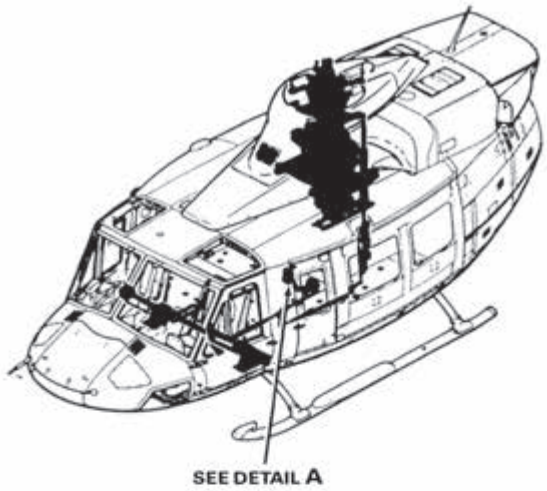
Figure 1. Collective Bellcrank Removal/Installation (Sheet 1 of 1)

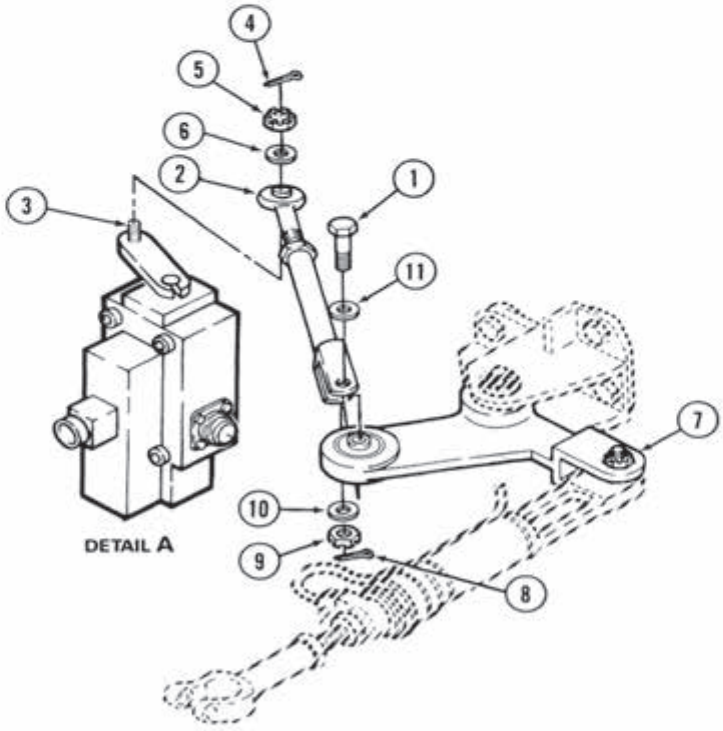
Collective Link - Removal

Procedure

1. Remove access panel from cabin floor forward of pylon.
2. Remove bolt (1, Figure 1).
3. Remove nut (5).
4. Remove link (2) from bellcrank (7) and rotary actuator arm stud (3).

1. Bolt
2. Link
3. Rotary actuator arm stud
4. Cotter pin
5. Nut
6. Steel washer
7. Bellcrank
8. Cotter pin
9. Nut
10. Steel washer
11. Steel washer





ICN-412-A-671101-B-97499-00006-A-001-01

Figure 1. Collective Link Removal/Installation (Sheet 1 of 1)

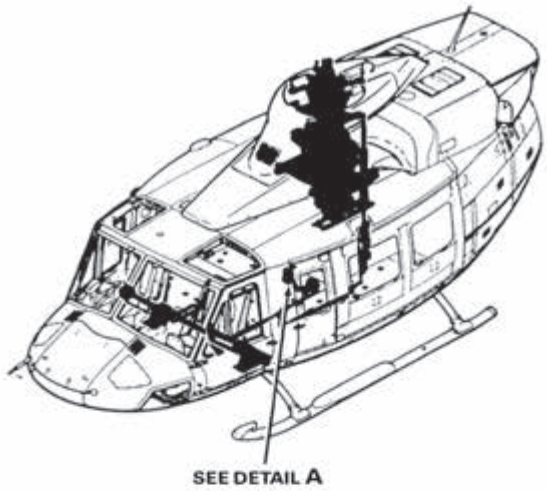
Collective Link - Inspection

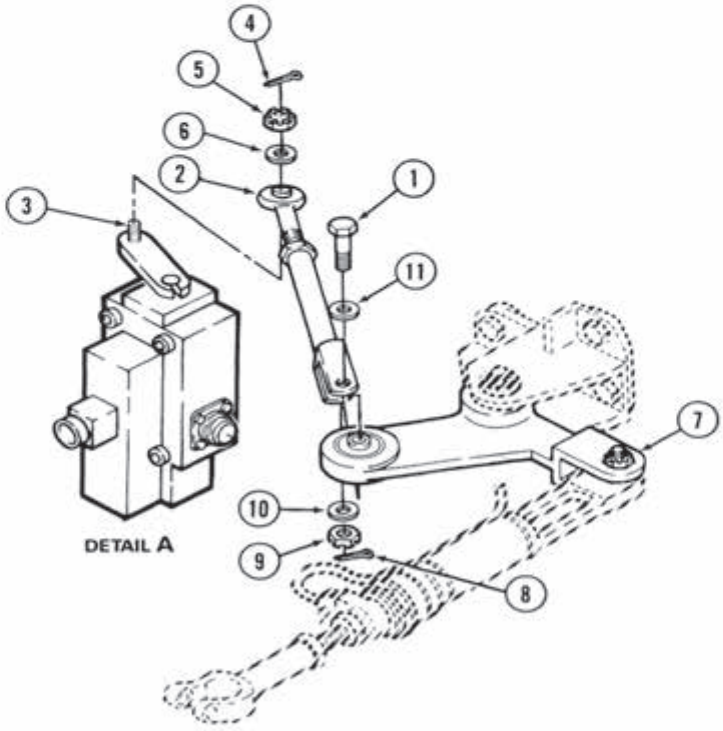
Procedure

1. Inspect link (2, Figure 1) for mechanical and corrosion damage in accordance with 412-CRO - Component Repair and Overhaul Manual.

2. Inspect bearing in link (2) for security and for wear (BHT-ALL-SPM, Standard Practices Manual).

1. Bolt
2. Link
3. Rotary actuator arm stud
4. Cotter pin
5. Nut
6. Steel washer
7. Bellcrank
8. Cotter pin
9. Nut
10. Steel washer
11. Steel washer





ICN-412-A-671101-B-97499-00006-A-001-01

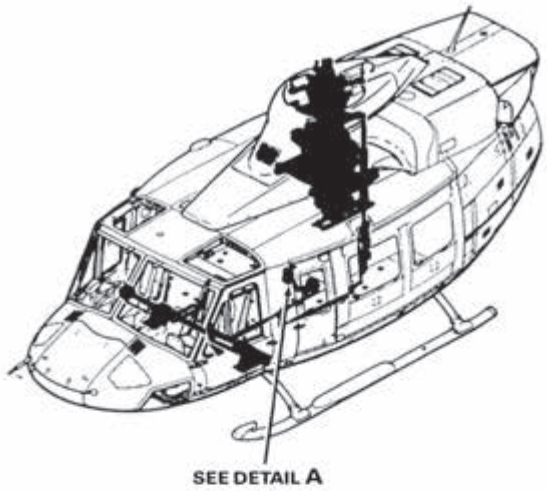
Figure 1. Collective Link Removal/Installation (Sheet 1 of 1)

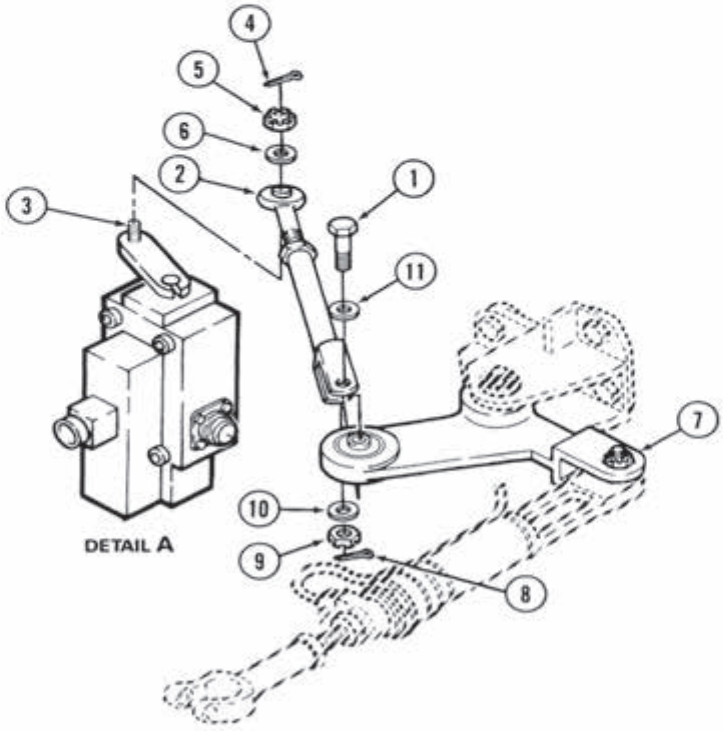
Collective Link - Installation

Procedure

1. Place link (2, Figure 1) into position on bellcrank (7). Install bolt (1), steel washers (10 and 11), nut (9), and cotter pin (8).
2. Position link (2) on rotary actuator arm stud (3). Install steel washer (6), nut (5), and cotter pin (4).
3. Install access panel in cabin floor forward of pylon.

1. Bolt
2. Link
3. Rotary actuator arm stud
4. Cotter pin
5. Nut
6. Steel washer
7. Bellcrank
8. Cotter pin
9. Nut
10. Steel washer
11. Steel washer





ICN-412-A-671101-B-97499-00006-A-001-01

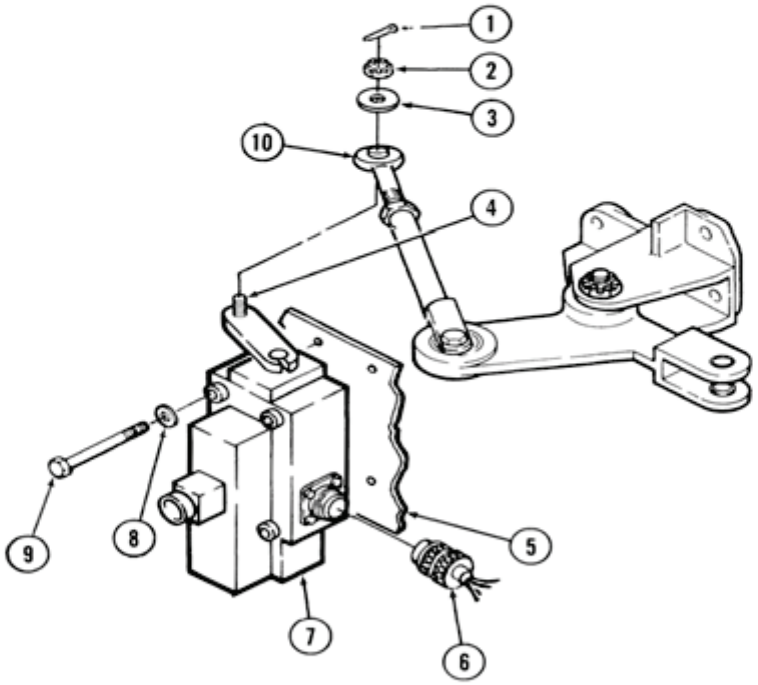
Figure 1. Collective Link Removal/Installation (Sheet 1 of 1)

Collective Rotary Actuator - Removal

Procedure

1. Remove access panel from cabin floor forward of pylon.
2. Remove nut (2, Figure 1).
3. Disconnect electrical connector (6) from rotary actuator (7).
4. Remove bolts (9), four places.
5. Remove rotary actuator (7) from structure (5).

1. Cotter pin
2. Nut
3. Steel washer
4. Rotary actuator arm stud
5. Structure
6. Electrical connector
7. Rotary actuator
8. Aluminum washer
9. Bolt
10. Link



DETAIL A

ICN-412-A-671101-B-97499-00007-A-001-01

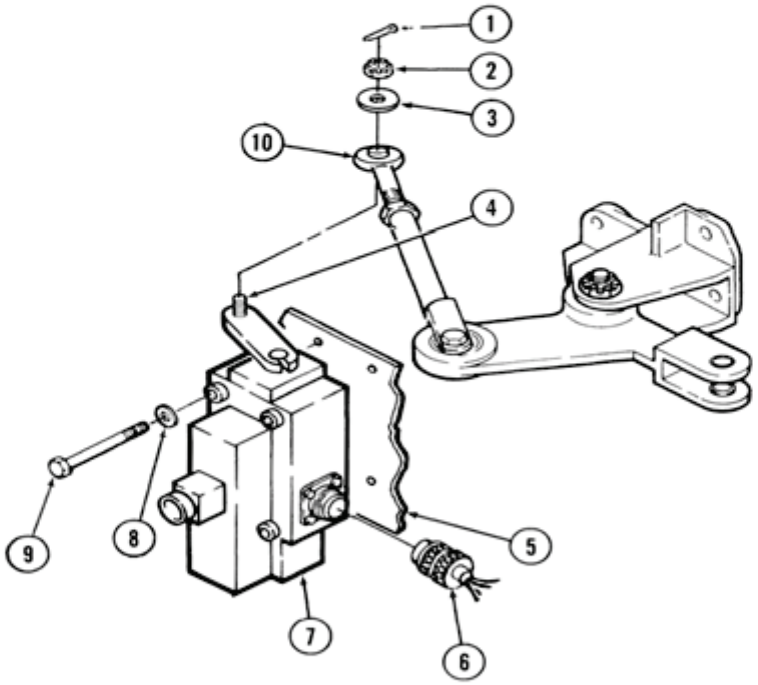
Figure 1. Collective Rotary Actuator Removal/Installation (Sheet 1 of 1)

Collective Rotary Actuator - Inspection

Procedure

1. Inspect rotary actuator (7, Figure 1) for mechanical and corrosion damage and wear (BHT-ALL-SPM, Standard Practices Manual).
2. Inspect rotary actuator arm stud (4) for damaged threads. No damage that can be detected visually is acceptable.
3. Inspect electrical receptacle for bent pins and damaged threads. Superficial damage is acceptable without repair.

1. Cotter pin
2. Nut
3. Steel washer
4. Rotary actuator arm stud
5. Structure
6. Electrical connector
7. Rotary actuator
8. Aluminum washer
9. Bolt
10. Link



DETAIL A

ICN-412-A-671101-B-97499-00007-A-001-01

Figure 1. Collective Rotary Actuator Removal/Installation (Sheet 1 of 1)

Collective Rotary Actuator - Repair

Procedure

1. Refer to vendor manual for rotary actuator repair instructions.

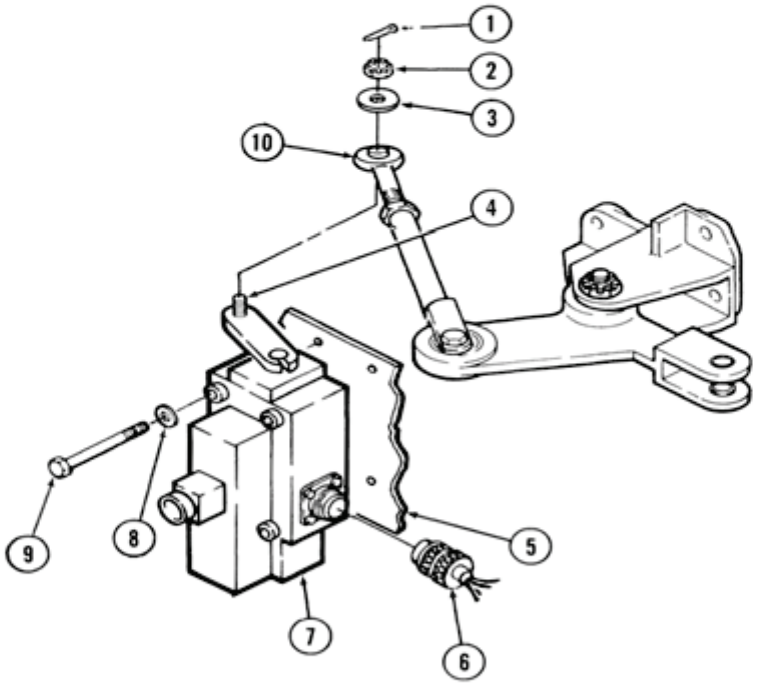
Intentionally blank page

Collective Rotary Actuator - Installation

Procedure

1. Place rotary actuator (7, Figure 1) into position on structure (5). Install four aluminum washers (8), and four bolts (9).
2. Position link (10) on rotary actuator arm stud (4) and install steel washer (3), nut (2), and cotter pin (1).
3. Connect electrical connector (6) to rotary actuator (7).
4. Install access panel in cabin floor forward of pylon.

1. Cotter pin
2. Nut
3. Steel washer
4. Rotary actuator arm stud
5. Structure
6. Electrical connector
7. Rotary actuator
8. Aluminum washer
9. Bolt
10. Link



DETAIL A

ICN-412-A-671101-B-97499-00007-A-001-01

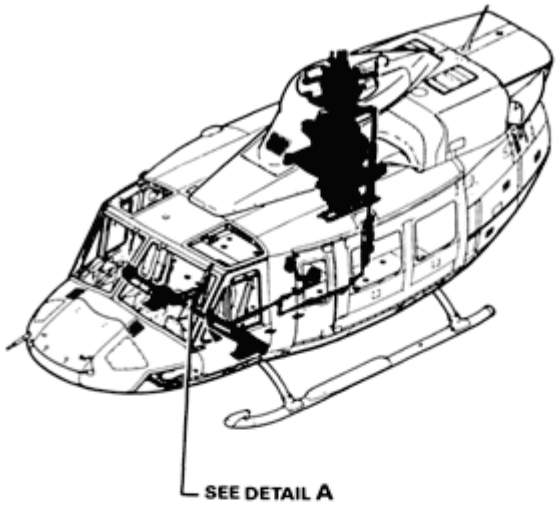
Figure 1. Collective Rotary Actuator Removal/Installation (Sheet 1 of 1)

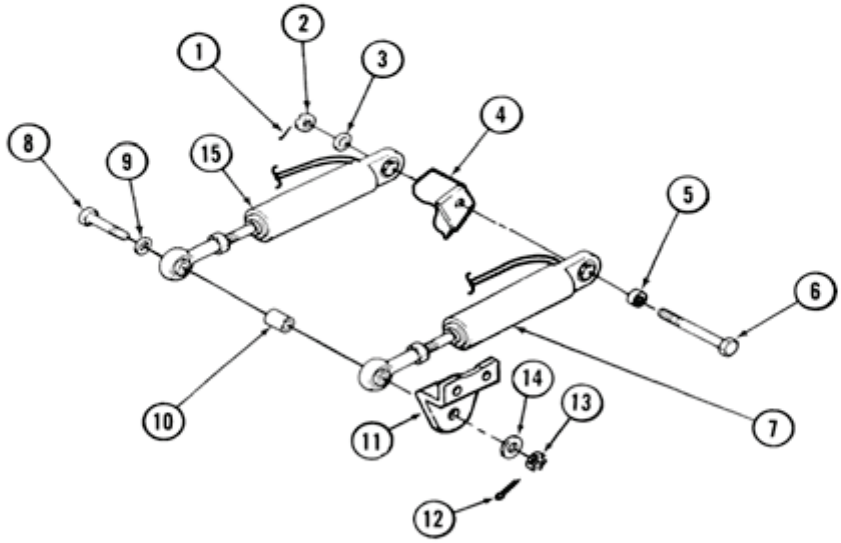
Collective Transducers - Removal

Procedure

1. Remove access panel from cabin floor forward of pylon.
2. Remove bolt (6, Figure 1).
3. Remove bolt (8).
4. Disconnect electrical connectors from both transducers (7 and 15).
5. Remove transducers (7 and 15) from bracket (11) of collective jackshaft tube and support (4).

1. Cotter pin
2. Nut
3. Spacer
4. Support
5. Spacer
6. Bolt
7. Transducer
8. Bolt
9. Spacer
10. Spacer
11. Bracket
12. Cotter pin
13. Nut
14. Spacer
15. Transducer





DETAIL A

ICN-412-A-671101-B-97499-00008-A-001-01

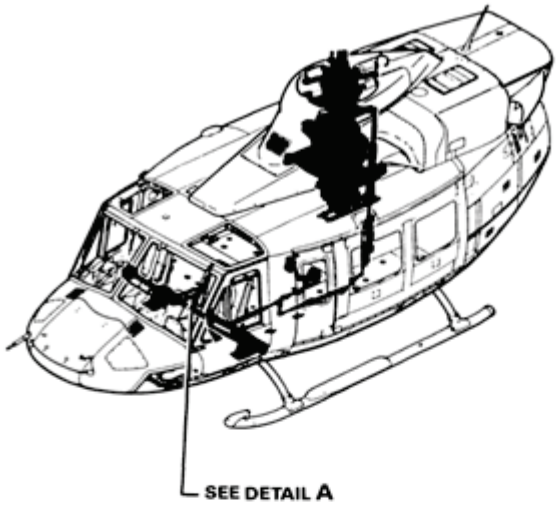
Figure 1. Collective Transducer Removal/Installation (Sheet 1 of 1)

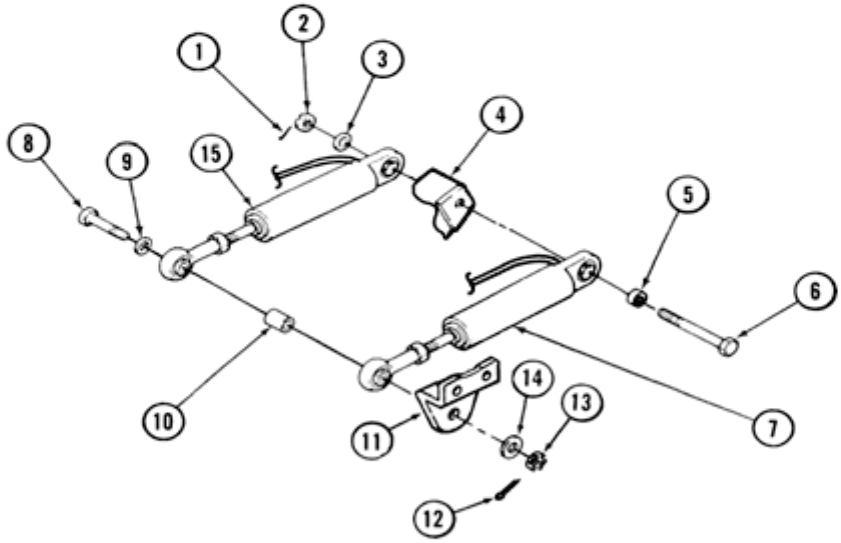
Collective Transducers - Inspection

Procedure

1. Inspect transducers (7 and 15, Figure 1) for mechanical and corrosion damage.
2. Inspect bearings in transducers (7 and 15) for secure installation in transducer and for wear (BHT-ALL-SPM, Standard Practices Manual).
3. Inspect electrical connector for bent pins and damaged threads. Superficial damage is acceptable without repair.

1. Cotter pin
2. Nut
3. Spacer
4. Support
5. Spacer
6. Bolt
7. Transducer
8. Bolt
9. Spacer
10. Spacer
11. Bracket
12. Cotter pin
13. Nut
14. Spacer
15. Transducer





DETAIL A

ICN-412-A-671101-B-97499-00008-A-001-01

Figure 1. Collective Transducer Removal/Installation (Sheet 1 of 1)

Collective Transducers - Repair

Procedure

1. Collective control transducers are nonrepairable and must be replaced when defective.

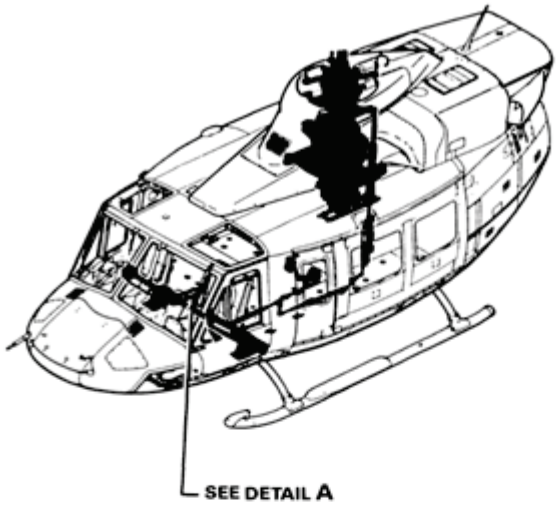
Intentionally blank page

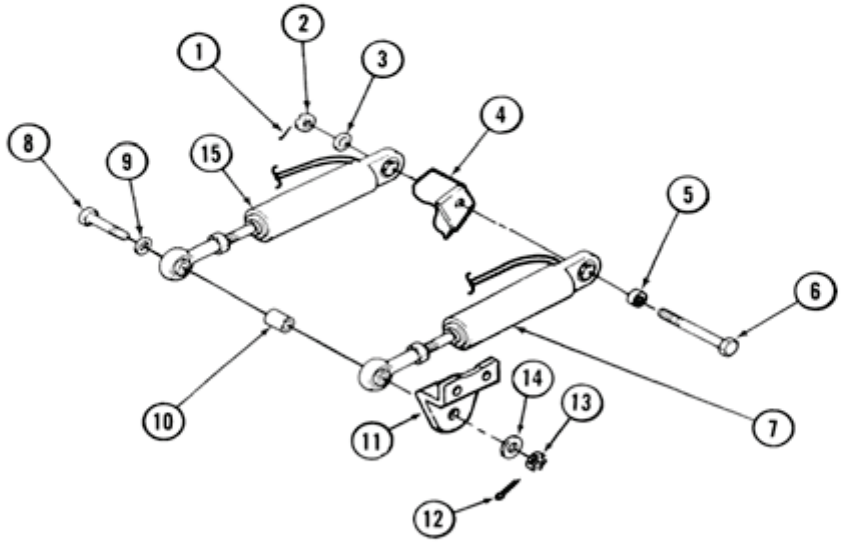
Collective Transducers - Installation

Procedure

1. Position transducers (7 and 15, Figure 1) on support (4). Install bolt (6), spacer (5), spacer (3) under nut (2), and cotter pin (1).
2. Place transducers (7 and 15) into position on bracket (11) of collective jackshaft lever. Install spacer (9) under head of bolt (8), spacer (10), spacer (14), nut (13), and cotter pin (12).
3. Connect electrical connectors to both transducers (7 and 15).
4. Install access panel in cabin floor forward of pylon.

1. Cotter pin
2. Nut
3. Spacer
4. Support
5. Spacer
6. Bolt
7. Transducer
8. Bolt
9. Spacer
10. Spacer
11. Bracket
12. Cotter pin
13. Nut
14. Spacer
15. Transducer





DETAIL A

ICN-412-A-671101-B-97499-00008-A-001-01

Figure 1. Collective Transducer Removal/Installation (Sheet 1 of 1)

Cyclic Controls - DAFCS - Description

The cyclic controls installation (Figure 1) consist of the controls and components from the pilot and copilot cyclic stick (33 and 15) up to but not including the swashplate (1).

The cyclic control system incorporates two servoactuators (14 and 35), one connected in roll system and one connected in pitch system. A control tube and idler (37) are attached to servo actuator (14) in the pitch system.

Four transducers, two lateral cyclic transducers (29) in roll system and two fore and aft cyclic transducers (34) in pitch system, are used to convert mechanical movement to electrical signals for injecting pilot and copilot control commands to the DAFCS.

Two fore and aft cyclic transducers (34) are attached to right end of cyclic jackshaft (16) providing a fore and aft cyclic signal. Two lateral cyclic transducers (29) are attached to a forward cyclic bellcrank (31) providing a lateral cyclic signal.

A force gradient assembly connected to an electromechanical rotary actuator is also installed in each cyclic system. Cyclic roll system force gradient attaches to a bellcrank forward of jackshaft. The cyclic pitch system force gradient attaches directly to the left end of cyclic jackshaft.

A support assembly and idler (37) are used to connect the fore and aft cyclic servo actuator (14) and a pitch control tube which is attached to the mixing lever assembly (39).

Two proximity switches (36 and 38), and two cyclic centering (CYC CTR) caution lights are installed for aid in centering cyclic sticks. The lights are located in the pilots and copilots instrument panel. The two proximity switches (36 and 38) are located underneath the floor near the mixing lever assembly (39).

NOTE

Cyclic control DAFCS system components forward of mixing lever assembly (39) are identified as ‘lateral cyclic’ or ‘fore and aft cyclic’ as applicable, i.e., ‘lateral cyclic force gradient’.

NOTE

Cyclic control system components aft of mixing lever assembly (39) are identified as ‘left cyclic’ or ‘right cyclic’ as applicable, i.e., ‘left cyclic flight control cylinder.

The fore and aft control linkage and the lateral control linkage are separate from the cyclic stick to the mixing lever assembly (39). From the mixing lever assembly (39) to the swashplate (1) the effects of the control linkages are combined and cannot be considered separately as to effect.

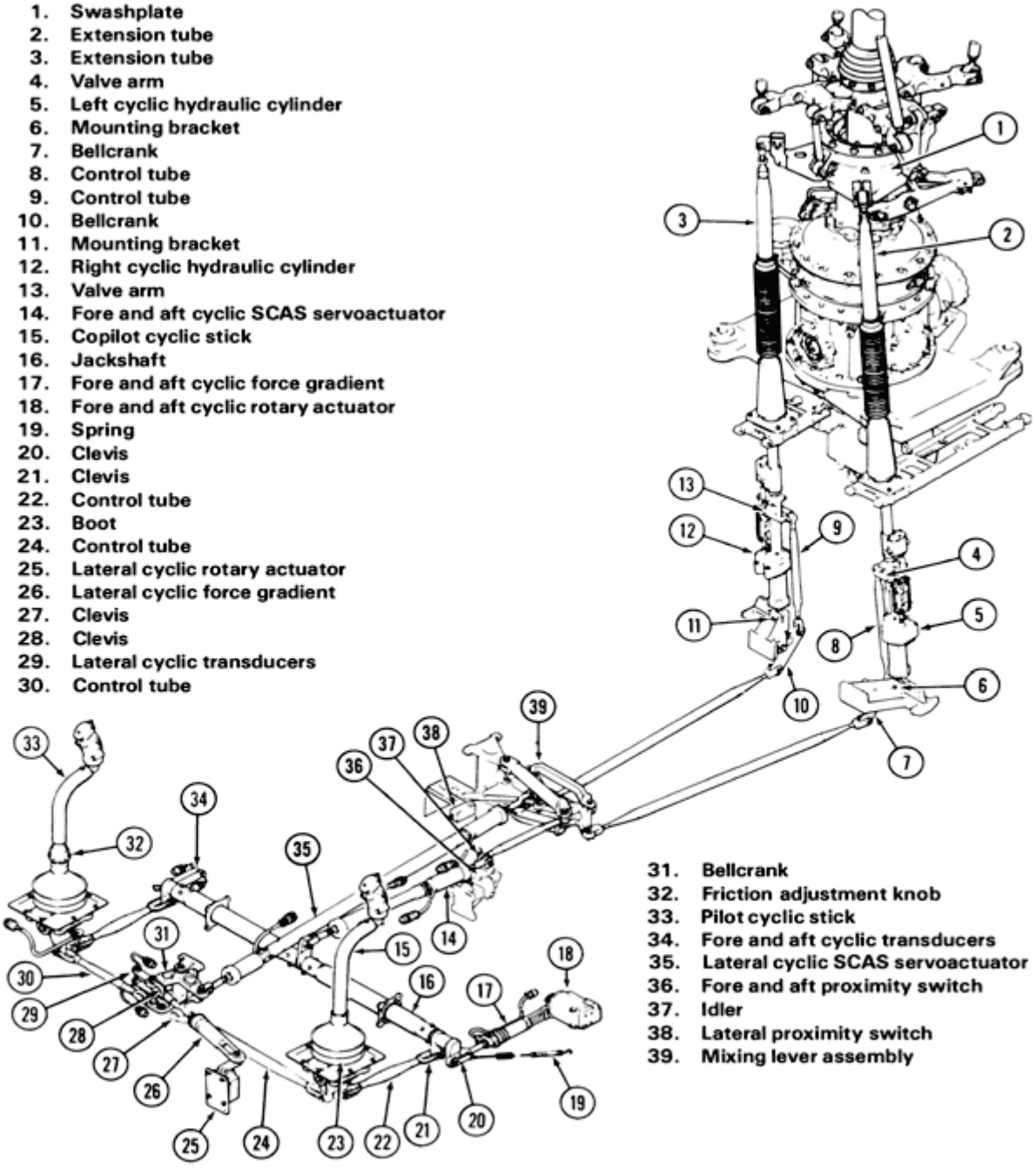
Actuation of the pilot and copilot cyclic sticks (33 and 15) causes hydraulic power assisted cyclic pitch change of the main rotor blades.

The forward actuators in each cyclic system are connected to Helipilot (HP-1) system. Aft actuators in each cyclic system are connected to Helipilot (HP-2) system.

Refer to 412-A-98-01-01-00A-051A-A / 02974, for DAFCS system wiring diagrams.

Dual controls are installed for copilot control of the helicopter.

Operation of dual controls system is basically the same as operation of single system. Refer to Flight Manual for operating procedures.



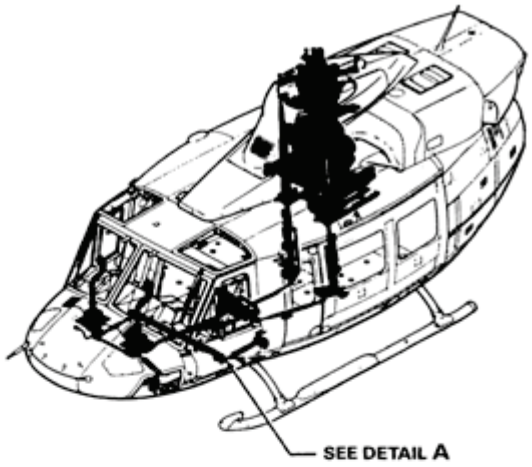
ICN-412-A-671203-B-97499-00001-A-001-01

Figure 1. Cyclic Controls (Sheet 1 of 1)

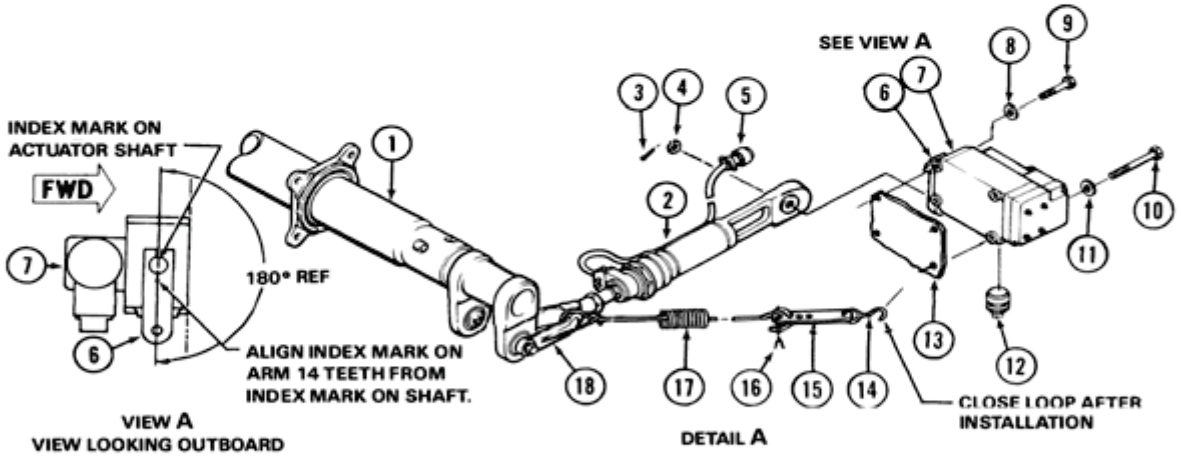
Rotary Actuator (Fore and Aft Cyclic) - Removal

Procedure

1. Remove access panel from cabin floor under left crew seat. Remove access panel from bottom left side of helicopter just forward of forward landing gear crosstube.
2. Disconnect hook (14, Figure 1) from bolt (10).
3. Remove nut (4) and separate force gradient (2) from arm (6).
4. Disconnect electrical connector (12).
5. Remove three bolts (9).
6. Remove one bolt (10).
7. Remove rotary actuator (7).



1. Tube and lever assembly
2. Force gradient
3. Cotter pin
4. Nut
5. Electrical connector
6. Arm
7. Rotary actuator
8. Aluminum washer
9. Bolt
10. Bolt (212-001-315-1)
11. Aluminum washer
12. Electrical connector
13. Structure
14. Hook
15. Plate
16. Lockwire
17. Spring
18. Clevis



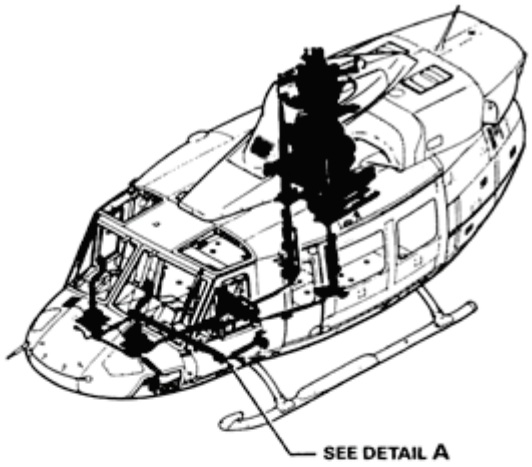
ICN-412-A-671203-B-97499-00002-A-001-01

Figure 1. Fore and Aft Cyclic Rotary Actuator Removal/Installation (Sheet 1 of 1)

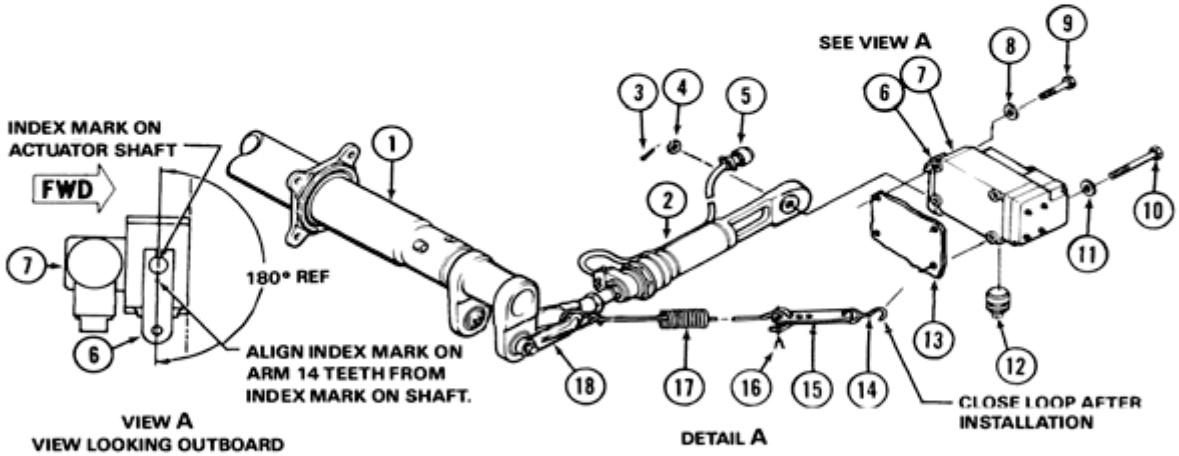
Rotary Actuator (Fore and Aft Cyclic) - Inspection

Procedure

1. Inspect rotary actuator (7, Figure 1) for mechanical and corrosion damage and wear (BHT-ALL-SPM, Standard Practices Manual).
2. Inspect rotary actuator arm (6) stud for damaged threads. No damage that can be detected visually is acceptable.
3. Inspect electrical receptacle for bent pins and damaged threads. Superficial damage is acceptable without repair.



1. Tube and lever assembly
2. Force gradient
3. Cotter pin
4. Nut
5. Electrical connector
6. Arm
7. Rotary actuator
8. Aluminum washer
9. Bolt
10. Bolt (212-001-315-1)
11. Aluminum washer
12. Electrical connector
13. Structure
14. Hook
15. Plate
16. Lockwire
17. Spring
18. Clevis



ICN-412-A-671203-B-97499-00002-A-001-01

Figure 1. Fore and Aft Cyclic Rotary Actuator Removal/Installation (Sheet 1 of 1)

Rotary Actuator (Fore and Aft Cyclic) - Installation

Procedure

1. Inspect rotary actuator (7, Figure 1) to ensure that arm (6) is installed on rotary actuator with index mark on arm positioned 14 teeth from index mark on rotary actuator shaft as shown on view A. Reposition arm (6) on rotary actuator if index marks are not properly located.

2. Position rotary actuator (7) on structure (13). Install special bolt (10) and aluminum washer (11) in lower outboard hole. Install three bolts (9) and aluminum washers (8) in remaining holes.

3. Position force gradient (2) on arm (6) stud. Install nut (4) and cotter pin (3).

4. Install spring (17), plate (15), and hook (14) on bolt (10). Refer to (412-A-67-12-03-03A-720A-A / 02795) for procedure to position spring (17) in correct hole in plate (15).

NOTE

For initial installation, install spring in fourth hole of plate (15). With hydraulic boost on, or during ground run, adjust spring tension to position pilot control stick (longitudinal cyclic only) perpendicular to deck within 2 degrees.

5. Close loop on hook (14) through bolt (10).

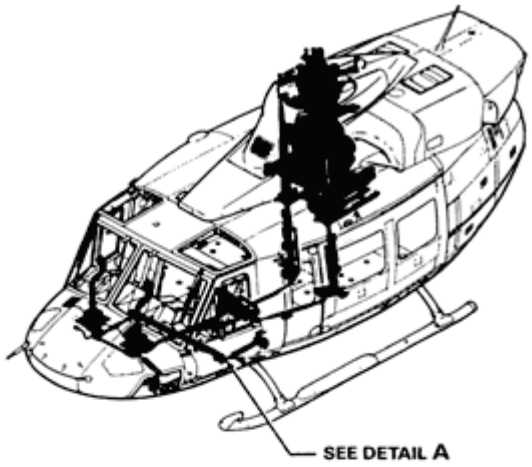
6. Install electrical connector (12).

7. Move cyclic controls through full travel and check for binding and/or obstructions.

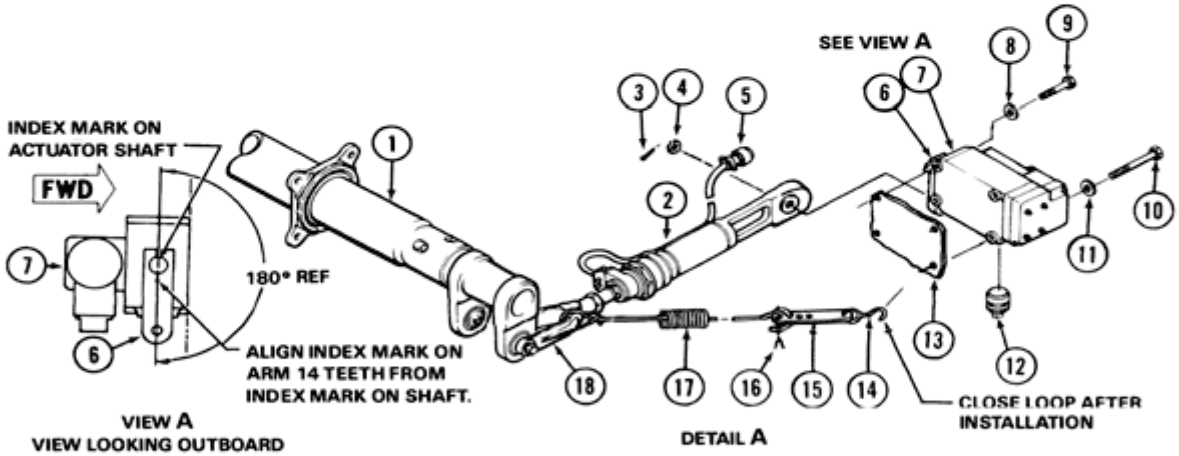
8. Confirm that cyclic control rigging is within limits (412-A-67-01-00-03A-276A-A / 01324).

9. Install access panel on cabin floor under left crew seat.

10. Install access panel on bottom left side of helicopter just forward of forward landing gear crosstube.



1. Tube and lever assembly
2. Force gradient
3. Cotter pin
4. Nut
5. Electrical connector
6. Arm
7. Rotary actuator
8. Aluminum washer
9. Bolt
10. Bolt (212-001-315-1)
11. Aluminum washer
12. Electrical connector
13. Structure
14. Hook
15. Plate
16. Lockwire
17. Spring
18. Clevis



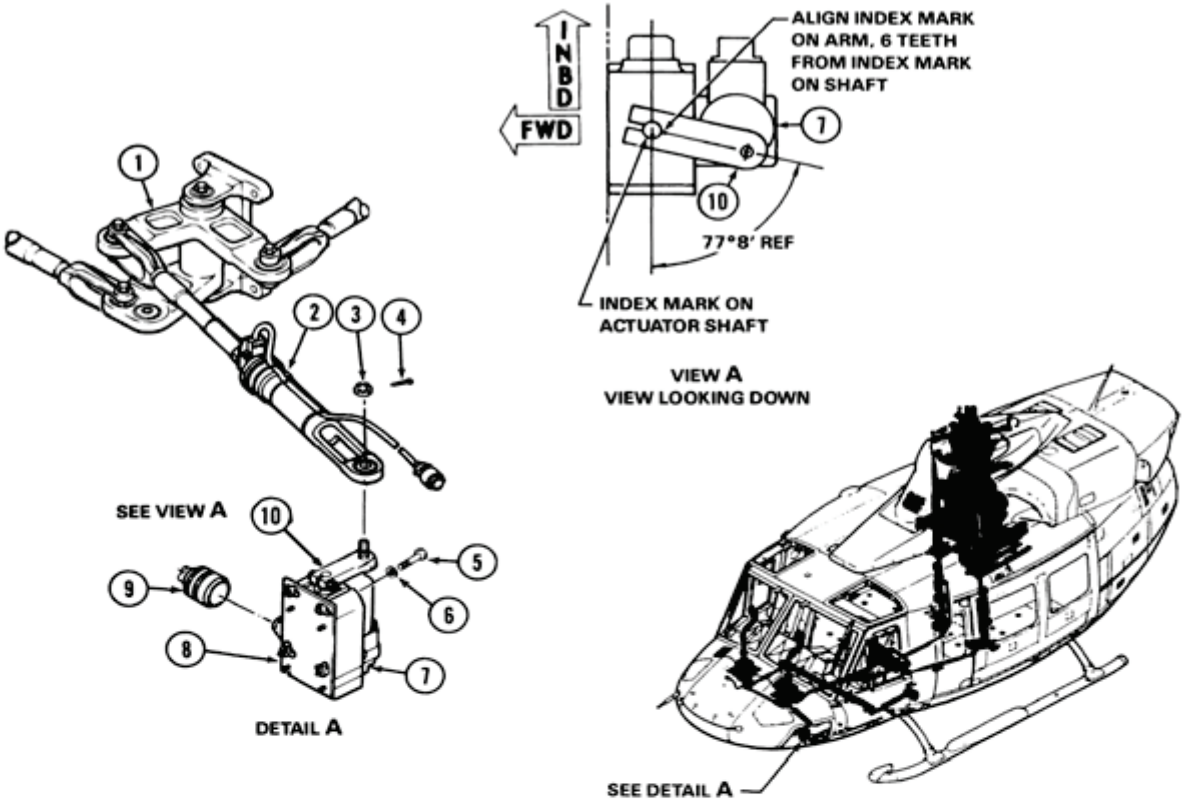
ICN-412-A-671203-B-97499-00002-A-001-01

Figure 1. Fore and Aft Cyclic Rotary Actuator Removal/Installation (Sheet 1 of 1)

Rotary Actuator (Lateral Cyclic) - Removal

Procedure

1. Remove access panel from bottom of helicopter just aft and inboard of left lower nose window.
2. Remove nut (3, Figure 1).
3. Disconnect electrical connector (9) from rotary actuator (7).
4. Remove four bolts (5).
5. Remove rotary actuator (7).



1. Lateral control bellcrank
2. Force gradient
3. Nut
4. Cotter pin
5. Bolt
6. Aluminum washer
7. Rotary actuator
8. Rotary actuator adapter plate (structure)
9. Electrical connector
10. Arm

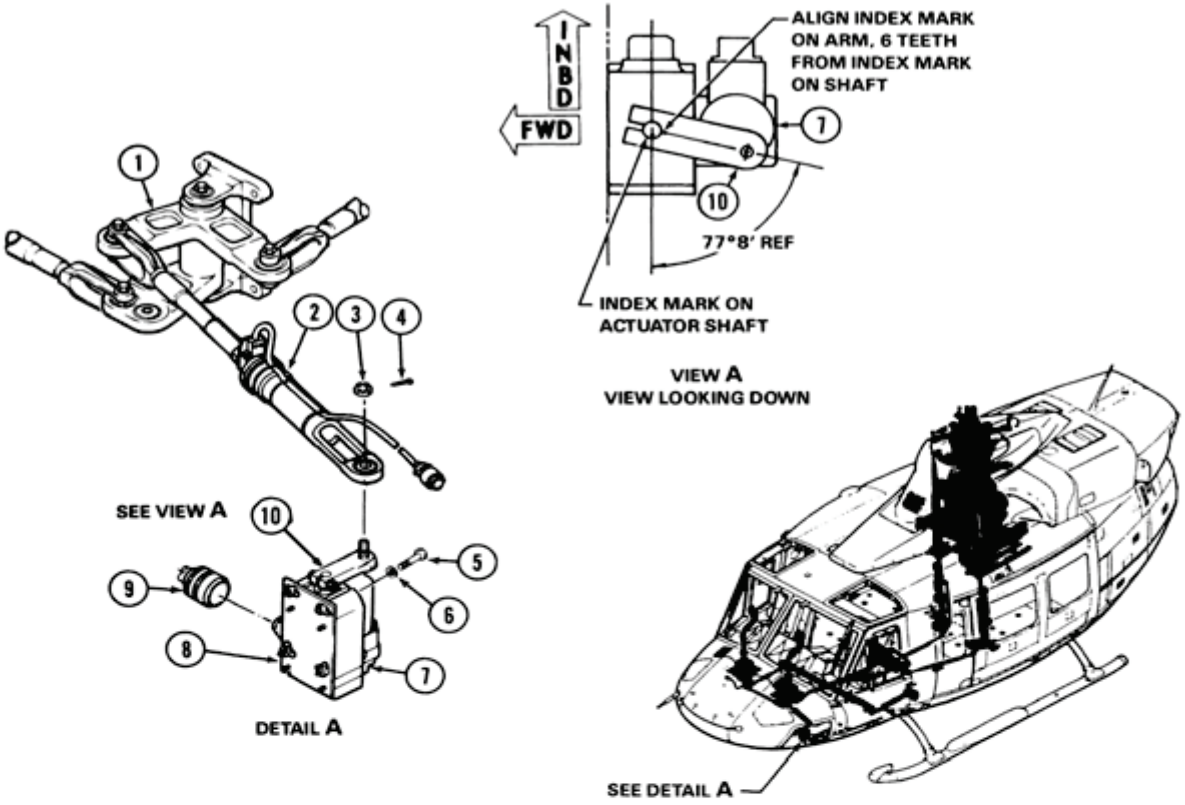
ICN-412-A-671203-B-97499-00003-A-001-01

Figure 1. Lateral Cyclic Rotary Actuator Removal/Installation (Sheet 1 of 1)

Rotary Actuator (Lateral Cyclic) - Inspection

Procedure

1. Inspect rotary actuator (7, Figure 1) for mechanical and corrosion damage and wear (BHT-ALL-SPM, Standard Practices Manual).
2. Inspect rotary actuator arm (10) stud for damaged threads. No damage that can be detected visually is acceptable.
3. Inspect electrical receptacle for bent pins and damaged threads. Superficial damage is acceptable without repair.



1. Lateral control bellcrank
2. Force gradient
3. Nut
4. Cotter pin
5. Bolt
6. Aluminum washer
7. Rotary actuator
8. Rotary actuator adapter plate (structure)
9. Electrical connector
10. Arm

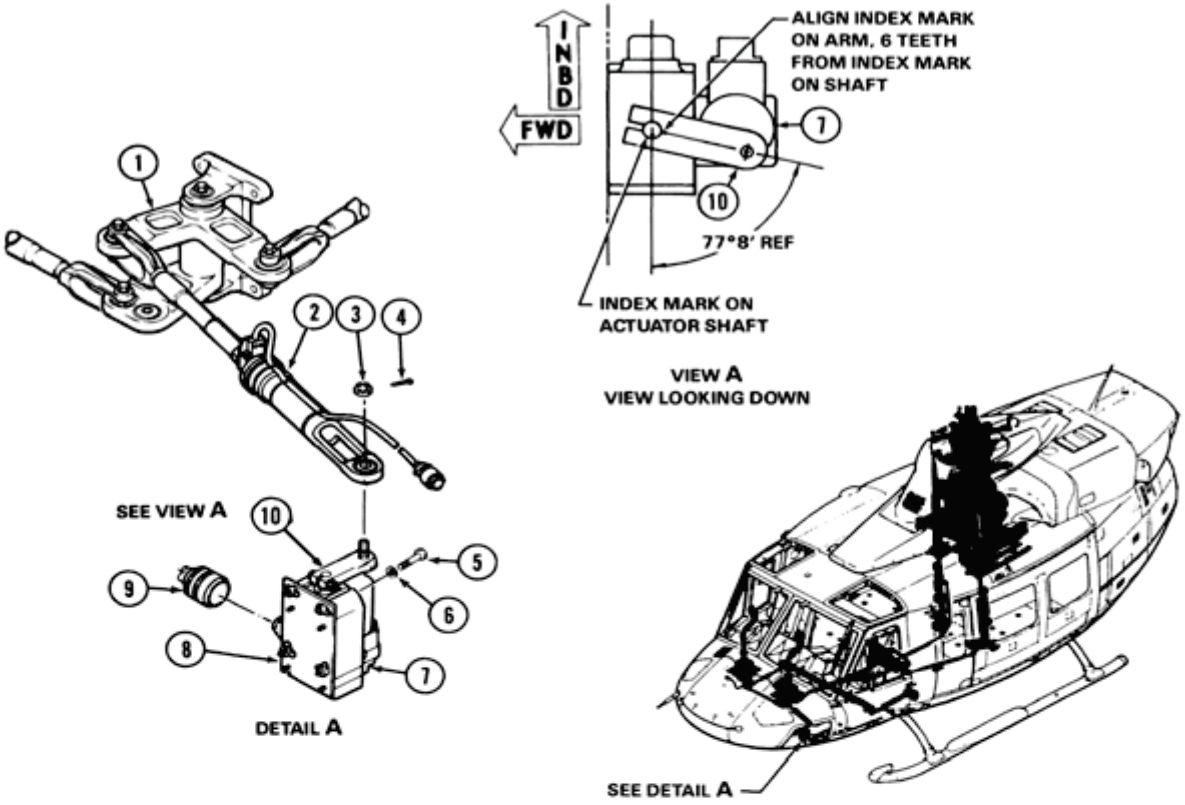
ICN-412-A-671203-B-97499-00003-A-001-01

Figure 1. Lateral Cyclic Rotary Actuator Removal/Installation (Sheet 1 of 1)

Rotary Actuator (Lateral Cyclic) - Installation

Procedure

1. Inspect rotary actuator (7, Figure 1) to ensure that arm (10) is installed on rotary actuator with index mark on arm positioned six teeth from index mark on rotary actuator shaft as shown in view A. Reposition arm (10) on rotary actuator if index marks are not properly located.
2. Position rotary actuator (7) on rotary actuator adapter plate (structure) (8). Install four bolts (5) and four aluminum washers (6).
3. Position force gradient (2) on arm (10). Install nut (3) and cotter pin (4).
4. Install electrical connector (9).
5. Move cyclic controls through full throw and check for binding and/or obstructions.
6. Confirm that cyclic control rigging is within limits (412-A-67-01-00-03A-276A-A / 01324).
7. Install access panel on bottom of helicopter just aft and inboard of left lower nose window.



1. Lateral control bellcrank
2. Force gradient
3. Nut
4. Cotter pin
5. Bolt
6. Aluminum washer
7. Rotary actuator
8. Rotary actuator adapter plate (structure)
9. Electrical connector
10. Arm

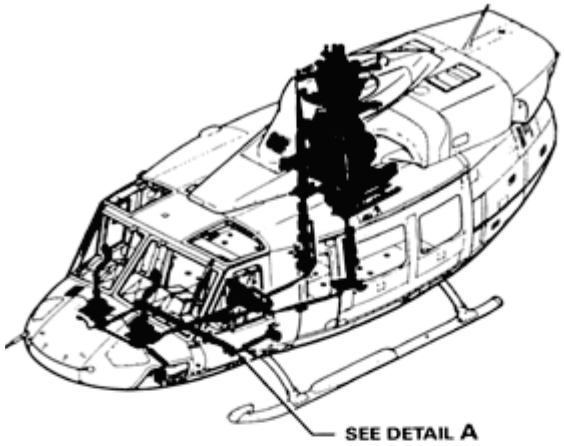
ICN-412-A-671203-B-97499-00003-A-001-01

Figure 1. Lateral Cyclic Rotary Actuator Removal/Installation (Sheet 1 of 1)

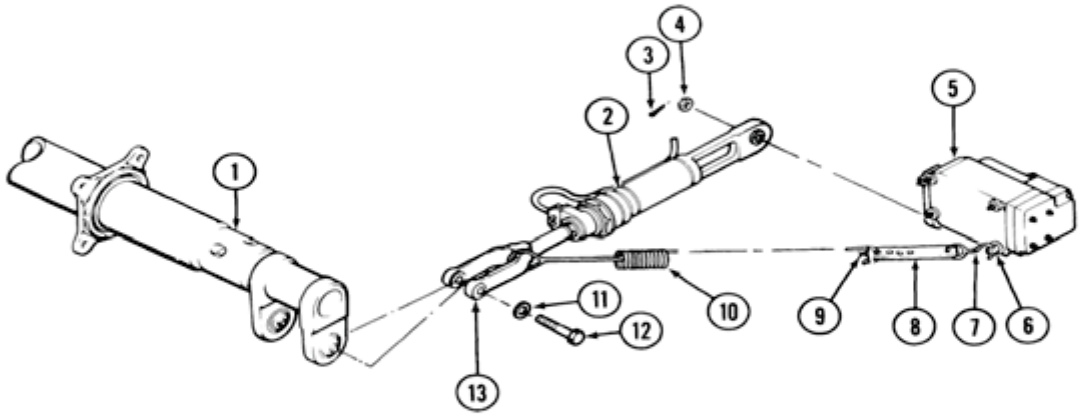
Force Gradient (Fore and Aft Cyclic) - Removal

Procedure

1. Remove access panel from cabin floor under left crew seat.
2. Remove access panel from bottom of helicopter on left side forward of forward landing gear crosstube.
3. Remove hook (7, Figure 1) from bolt (6).
4. Disconnect electrical connector.
5. Remove nut (4).
6. Remove bolt (12).
7. Remove force gradient (2).



1. Tube and lever assembly
2. Fore and aft cyclic controls force gradient
3. Cotter pin
4. Nut
5. Rotary actuator
6. Bolt (212-001-315-1)
7. Hook
8. Plate
9. Lockwire
10. Spring
11. Aluminum washer
12. Bolt
13. Clevis



DETAIL A

ICN-412-A-671203-B-97499-00004-A-001-01

Figure 1. Fore and Aft Cyclic Force Gradient Removal/Installation (Sheet 1 of 1)

Force Gradient (Fore and Aft Cyclic) - Inspection

Procedure

1. Inspect force gradient (2, Figure 1) for corrosion and mechanical damage. Superficial mechanical damage is acceptable without repair. Mechanical damage in excess of superficial and/or corrosion is not acceptable.

2. Inspect clevis (13) for damaged threaded helicoil insert and for elongation hole in bushing opposite the threaded helicoil insert.

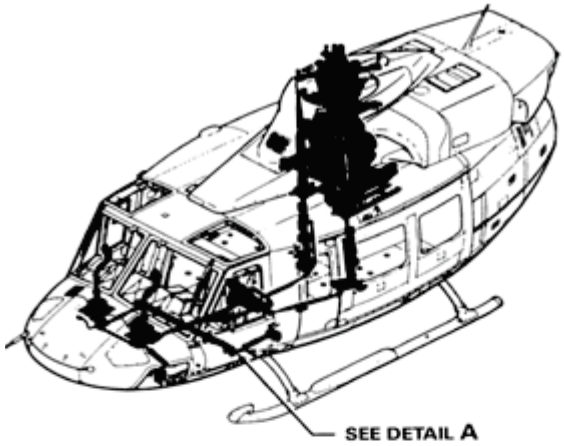
3. Inspect hook (7), plate (8), and spring (10) for obvious mechanical damage and for loss of tension in spring. Spring tension at 10.36 in. (263.1 mm) is 6.4 ± 0.5 lbs. (2.9 ± 0.2 kg).

4. Inspect bearing for wear and for secure installation in cylinder (5, Figure 2). Refer to Bearing Wear Limit Table in BHT-ALL-SPM, Standard Practices Manual for acceptable limits.

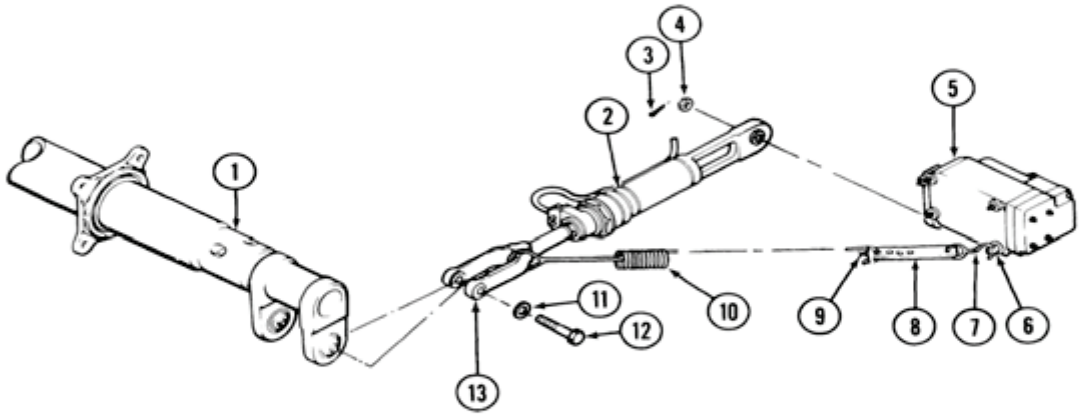
5. Inspect two nuts (6 and 8) for secure installation. If adjustment of nuts (2 and 3) (spring preload) is suspected of being incorrect, adjust spring preload as outlined in 412-A-67-12-03-03A-720A-A / 02795., step 1.

6. Inspect nut (1) and cylinder (5) for correct lockwire installation.

7. Inspect switch and electrical cable assembly (14) for frayed wires and broken pins in connector. Check continuity from connector pins through switch.



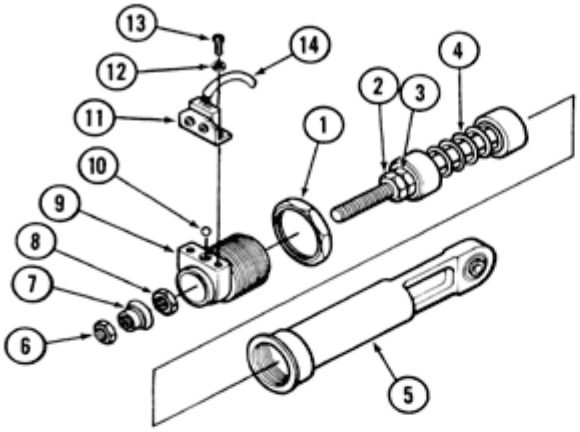
1. Tube and lever assembly
2. Fore and aft cyclic controls force gradient
3. Cotter pin
4. Nut
5. Rotary actuator
6. Bolt (212-001-315-1)
7. Hook
8. Plate
9. Lockwire
10. Spring
11. Aluminum washer
12. Bolt
13. Clevis



DETAIL A

ICN-412-A-671203-B-97499-00004-A-001-01

Figure 1. Fore and Aft Cyclic Force Gradient Removal/Installation (Sheet 1 of 1)



1. Jamnut
2. Outboard nut
3. Inboard nut
4. Spring assembly
5. Cylinder
6. Nut
7. Actuator
8. Nut
9. Cap
10. Ball
11. Retainer
12. Washer
13. Screw
14. Switch and cable assembly

ICN-412-A-671203-B-97499-00005-A-001-01

Figure 2. Fore and Aft Cyclic Force Gradient (Sheet 1 of 1)

Intentionally blank page

Force Gradient (Fore and Aft Cyclic) - Repair

Procedure

1. Repair of force gradient consists of replacement of unservicable components and adjustment of spring preload in accordance with 412-A-67-12-03-03A-720A-A / 02795., step 1.

Intentionally blank page

Force Gradient (Fore and Aft Cyclic) - Installation

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Lockwire	C-405	AR	

Procedure

NOTE

Force gradient (2, Figure 1) in the fore and aft cyclic controls is similar in appearance to the force gradient in the lateral cyclic controls, but the force gradients have different part numbers. Ensure that correct part number force gradient is installed in the fore and aft position.

1. If not previously accomplished, adjust spring preload on force gradient as follows:

1.1. Remove cap (9, Figure 2).

1.2. Remove spring assembly (4) from cylinder (5).

1.3. Adjust inboard nut (3) to provide 9 to 10 lbs. (40.032 to 44.48 N) preload against spring assembly (4). Tighten outboard nut (2) against inboard nut (3).

1.4. Position spring assembly (4) in cylinder (5) and install cap (9) until all end play is eliminated. Install jamnut (1) and lockwire (C-405) to prevent turning in either direction.

2. If not previously accomplished, adjust switch (14, Figure 2) on force gradient as follows:

2.1. Loosen nut (8) on shaft of spring assembly (4).

2.2. Align actuator (7) on shaft of spring assembly (4) with ball (10) and plunger of switch (14). Secure actuator (7) with nut (8).

2.3. Loosen screws (13) mounting switch (14) to retainer (11). Push and hold actuator (7) sideways against cap (9) at point opposite location of switch (14). Hold this position to initially adjust switch (14) then adjust switch to actuate within ± 0.050 in. (1.27 mm) linear travel of spring assembly (4). Secure screws (13) mounting switch (14) to retainer (11).

3. Position spring (10, Figure 1) on clevis (13).

4. Position force gradient (2) on tube and lever (1). Clevis (13) has a threaded helicoil insert in one side. Position the clevis with the threaded insert inboard. Install bolt (12) and aluminum washer (11). lockwire (C-405) bolt head to hole in clevis (13).

ENSURE THAT FORCE GRADIENT LENGTH IS ADJUSTED PRIOR TO INSTALLATION OF FORCE GRADIENT ON ROTARY ACTUATOR.

5. Position force gradient (2) on rotary actuator (5). Install nut (4) and cotter pin (3).

6. Install plate (8) on spring (10) with spring through fourth hole in plate (8). Do not lockwire spring to plate at this time.

7. Attach plate (8) with hook (7) to bolt (6). Do not close loop on hook (7) at this time.

8. Connect electrical connector.

9. Move fore and aft cyclic controls through full throw and check for binding and obstructions.

10. Apply external hydraulic power to helicopter (412-A-29-00-00-00A-028A-A / 00611).

11. Loosen cyclic friction and observe cyclic stick position. If cyclic stick is not perpendicular (longitudinally) within two degrees, move spring (10) to different hole in plate (8) to cause cyclic stick to be vertical (longitudinally).

12. When tension on spring (10) is correctly adjusted, secure spring (10) to plate (8) with lockwire (C-405). Close loop on hook (7) to prevent accidental disengagement of hook.

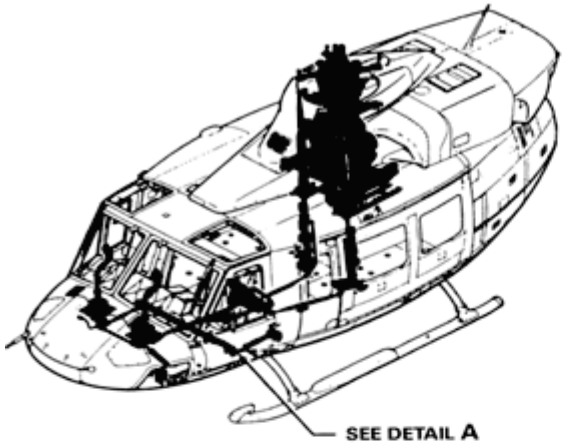
13. Move cyclic controls through full throw and check for binding and obstructions.

14. Confirm that cyclic control rigging is within limits (412-A-67-01-00-03A-276A-A / 01324).

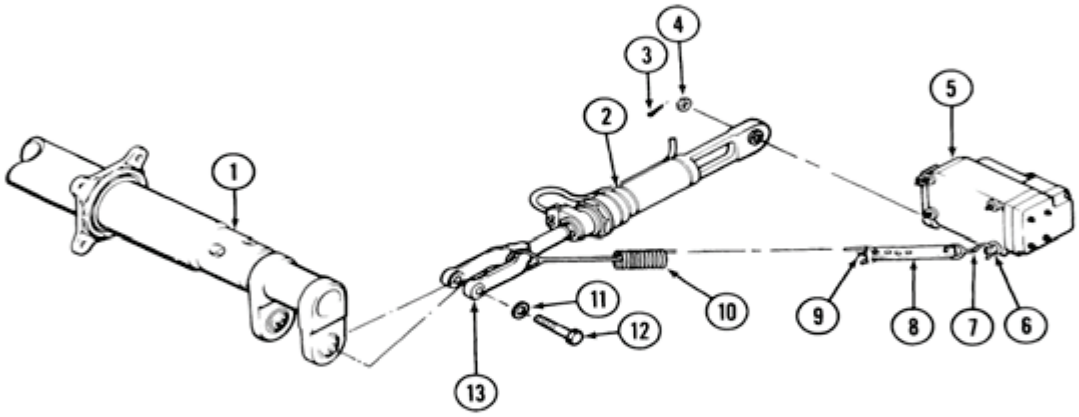
15. Remove external power from helicopter.

16. Install access panel on bottom of helicopter on left side forward of forward landing gear crosstube.

17. Install access panel on cabin floor under left crew seat.



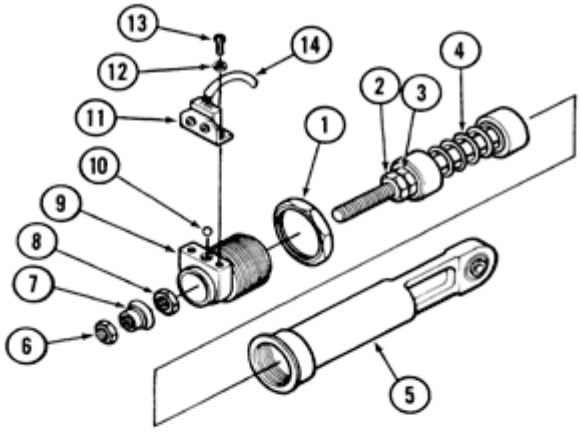
1. Tube and lever assembly
2. Fore and aft cyclic controls force gradient
3. Cotter pin
4. Nut
5. Rotary actuator
6. Bolt (212-001-315-1)
7. Hook
8. Plate
9. Lockwire
10. Spring
11. Aluminum washer
12. Bolt
13. Clevis



DETAIL A

ICN-412-A-671203-B-97499-00004-A-001-01

Figure 1. Fore and Aft Cyclic Force Gradient Removal/Installation (Sheet 1 of 1)



1. Jamnut
2. Outboard nut
3. Inboard nut
4. Spring assembly
5. Cylinder
6. Nut
7. Actuator
8. Nut
9. Cap
10. Ball
11. Retainer
12. Washer
13. Screw
14. Switch and cable assembly

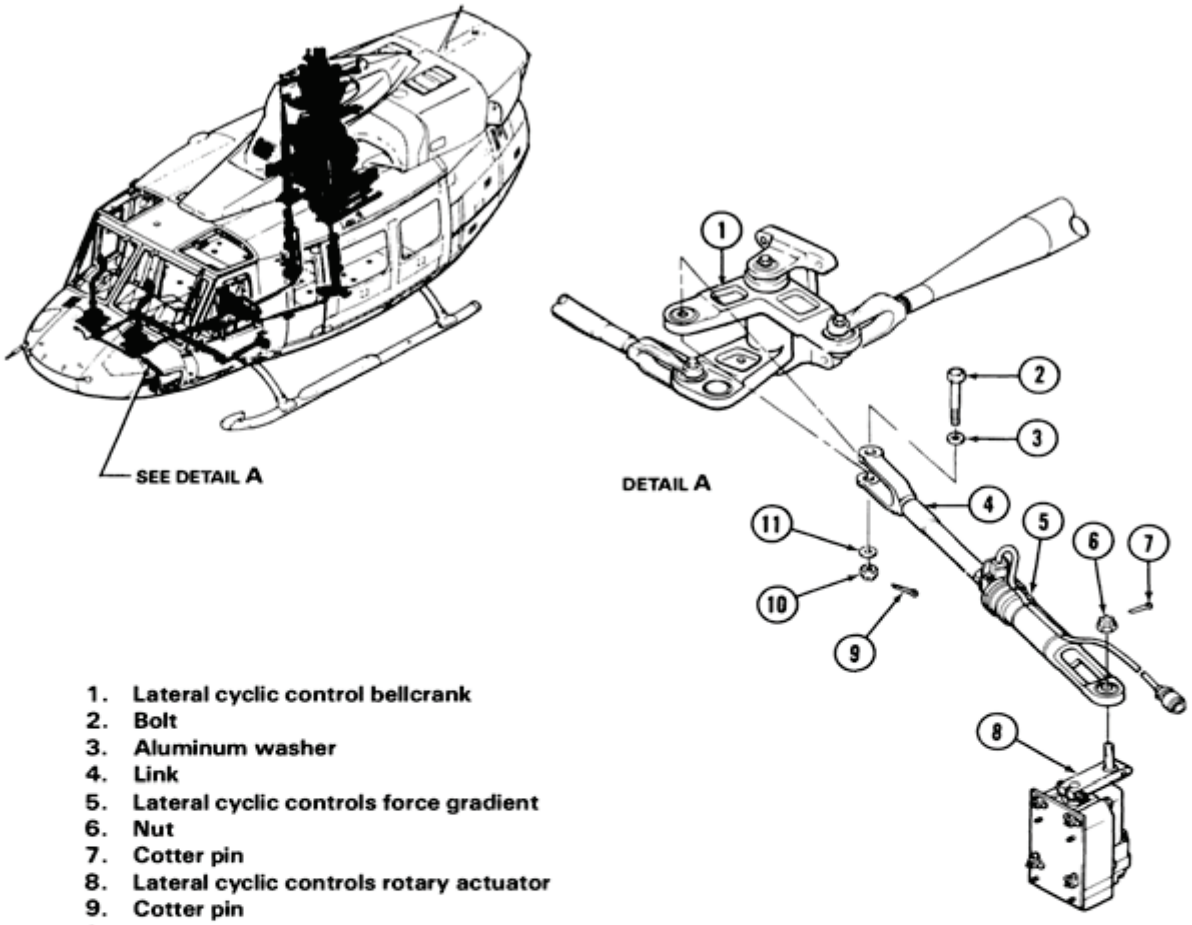
ICN-412-A-671203-B-97499-00005-A-001-01

Figure 2. Fore and Aft Cyclic Force Gradient (Sheet 1 of 1)

Force Gradient (Lateral Cyclic) - Removal

Procedure

1. Remove access panel from bottom of helicopter below force gradient (5, Figure 1).
2. Disconnect electrical connector.
3. Remove nut (6).
4. Remove bolt (2).
5. Remove force gradient (5).



1. Lateral cyclic control bellcrank
2. Bolt
3. Aluminum washer
4. Link
5. Lateral cyclic controls force gradient
6. Nut
7. Cotter pin
8. Lateral cyclic controls rotary actuator
9. Cotter pin
10. Nut
11. Aluminum washer

ICN-412-A-671203-B-97499-00006-A-001-01

Figure 1. Lateral Cyclic Force Gradient Removal/Installation (Sheet 1 of 1)

Force Gradient (Lateral Cyclic) - Inspection

Procedure

1. Inspect force gradient (5, Figure 1) for corrosion and mechanical damage. Superficial mechanical damage is acceptable without repair. Mechanical damage in excess of superficial and/or corrosion is not acceptable.

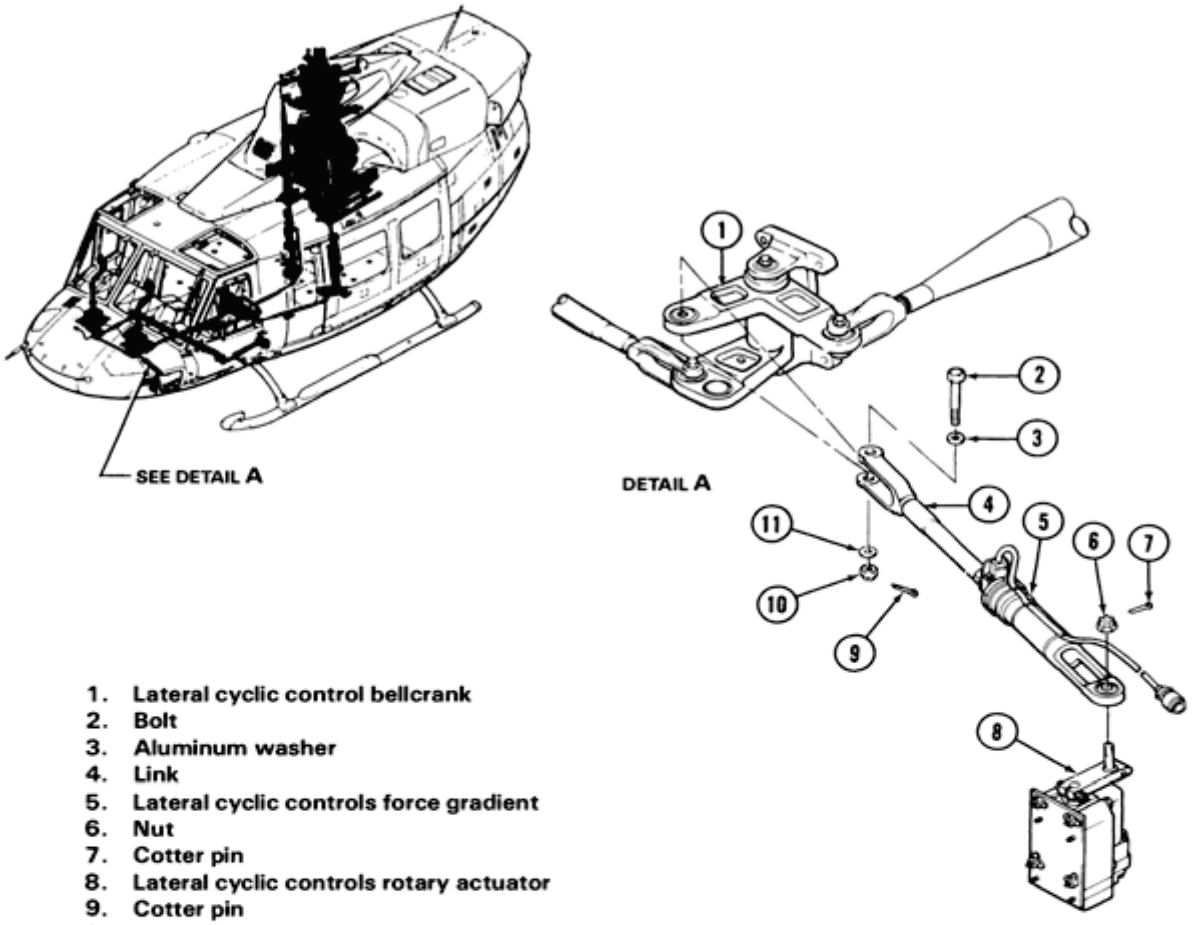
2. Inspect clevis on link (4) for elongated holes.

3. Inspect bearing for wear and for secure installation in cylinder (5, Figure 2). Refer to Bearing Wear Limit Table in BHT-ALL-SPM, Standard Practices Manual for acceptable limits.

4. Inspect two nuts (6 and 8) for secure installation. If adjustment of nuts (2 and 3) (spring preload) is suspected of being incorrect, adjust spring preload as outlined in 412-A-67-12-03-04A-720A-A / 02799., step 1.

5. Inspect nut (1) and cylinder (5) for correct lockwire installation.

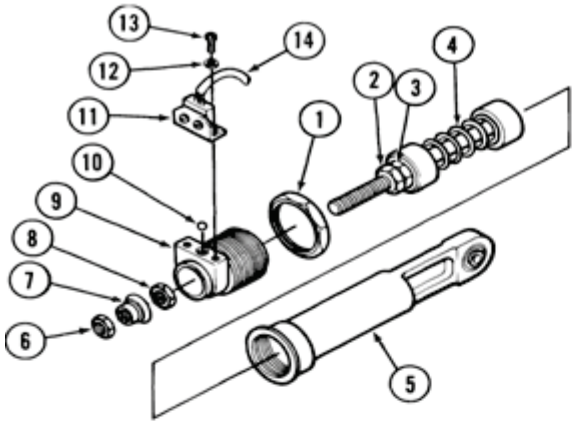
6. Inspect switch and electrical cable assembly (14) for frayed wires and broken pins in connector. Check continuity from connector pins through switch.



1. Lateral cyclic control bellcrank
2. Bolt
3. Aluminum washer
4. Link
5. Lateral cyclic controls force gradient
6. Nut
7. Cotter pin
8. Lateral cyclic controls rotary actuator
9. Cotter pin
10. Nut
11. Aluminum washer

ICN-412-A-671203-B-97499-00006-A-001-01

Figure 1. Lateral Cyclic Force Gradient Removal/Installation (Sheet 1 of 1)



1. Jamnut
2. Outboard nut
3. Inboard nut
4. Spring assembly
5. Cylinder
6. Nut
7. Actuator
8. Nut
9. Cap
10. Ball
11. Retainer
12. Washer
13. Screw
14. Switch and cable assembly

ICN-412-A-671203-B-97499-00007-A-001-01

Figure 2. Lateral Cyclic Force Gradient (Sheet 1 of 1)

Intentionally blank page

Force Gradient (Lateral Cyclic) - Repair

Procedure

1. Repair of force gradient consists of replacement of unservicable components and adjustment of spring preload in accordance with 412-A-67-12-03-04A-720A-A / 02799.

Intentionally blank page

Force Gradient (Lateral Cyclic) - Installation

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Lockwire	C-405	AR	

Procedure

NOTE

Force gradient (5, Figure 1) in the lateral cyclic controls is similar in appearance to the force gradient in the fore and aft cyclic controls, but the force gradients have different part numbers. Ensure that correct part number force gradient is installed in the lateral position.

1. If not previously accomplished, adjust spring preload on force gradient as follows:

1.1. Remove cap (9, Figure 2).

1.2. Remove spring assembly (4) from cylinder (5).

1.3. Adjust inboard nut (3) to provide 6 to 7 lbs. (26.688 to 31.136 N) preload against spring assembly (4). Tighten outboard nut (2) against inboard nut (3).

1.4. Position spring assembly (4) in cylinder (5) and install cap (9) until all end play is eliminated. Install jamnut (1) and lockwire (C-405) to prevent turning in either direction.

2. If not previously accomplished, adjust switch (14, Figure 2) on force gradient as follows:

2.1. Loosen nut (8) on shaft of spring assembly (4).

2.2. Align actuator (7) on shaft of spring assembly (4) with ball (10) and plunger of switch (14). Secure actuator (7) with nut (8).

2.3. Loosen screws (13) mounting switch (14) to retainer (11). Push and hold actuator (7) sideways against cap (9) at point opposite location of switch (14). Hold this position to initially adjust switch (14) then adjust switch to actuate within ± 0.050 in. (1.27 mm) linear travel of spring assembly (4). Secure screws (13) mounting switch (14) to retainer (11).

3. Position force gradient (5, Figure 1) on lateral cyclic control bellcrank (1). Install bolt (2), aluminum washers (3 and 11), nut (10), and cotter pin (9).

ENSURE THAT FORCE GRADIENT LENGTH IS ADJUSTED PRIOR TO INSTALLATION OF FORCE GRADIENT ON ROTARY ACTUATOR.

4. Position force gradient (5) on rotary actuator (8). Install nut (6) and cotter pin (7).

5. Connect electrical connector.

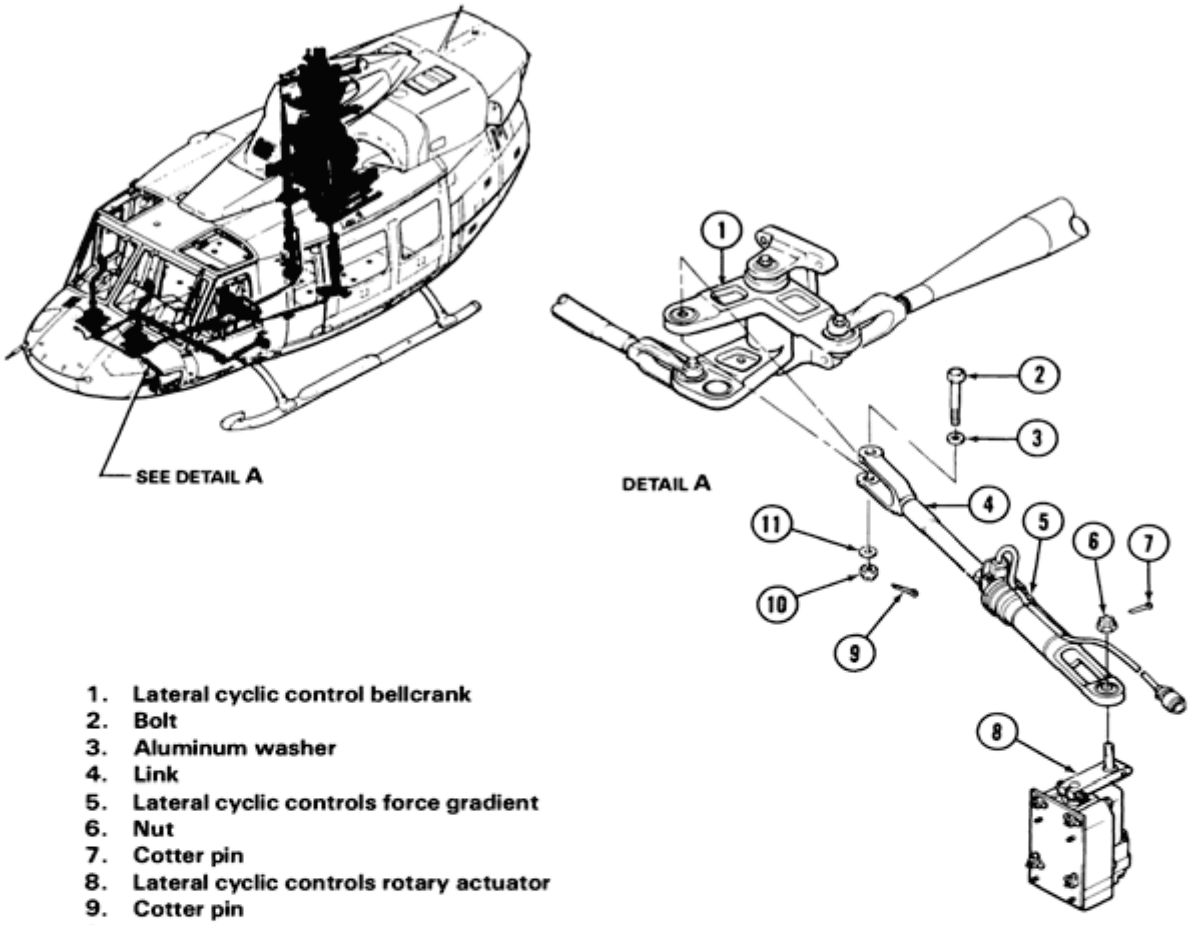
6. Apply external hydraulic power to helicopter (412-A-29-00-00-00A-028A-A / 00611).

7. Move cyclic controls through full throw and check for binding and/or obstructions.

8. Verify that cyclic control rigging is within limits (412-A-67-01-00-03A-276A-A / 01324).

9. Remove external power from helicopter.

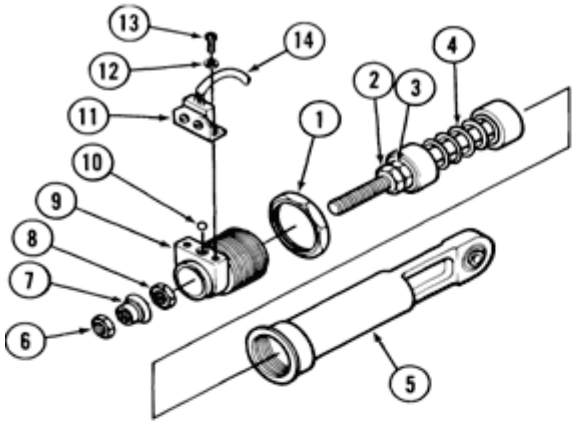
10. Install access panel on bottom of helicopter below force gradient.



1. Lateral cyclic control bellcrank
2. Bolt
3. Aluminum washer
4. Link
5. Lateral cyclic controls force gradient
6. Nut
7. Cotter pin
8. Lateral cyclic controls rotary actuator
9. Cotter pin
10. Nut
11. Aluminum washer

ICN-412-A-671203-B-97499-00006-A-001-01

Figure 1. Lateral Cyclic Force Gradient Removal/Installation (Sheet 1 of 1)



1. Jamnut
2. Outboard nut
3. Inboard nut
4. Spring assembly
5. Cylinder
6. Nut
7. Actuator
8. Nut
9. Cap
10. Ball
11. Retainer
12. Washer
13. Screw
14. Switch and cable assembly

ICN-412-A-671203-B-97499-00007-A-001-01

Figure 2. Lateral Cyclic Force Gradient (Sheet 1 of 1)

Intentionally blank page

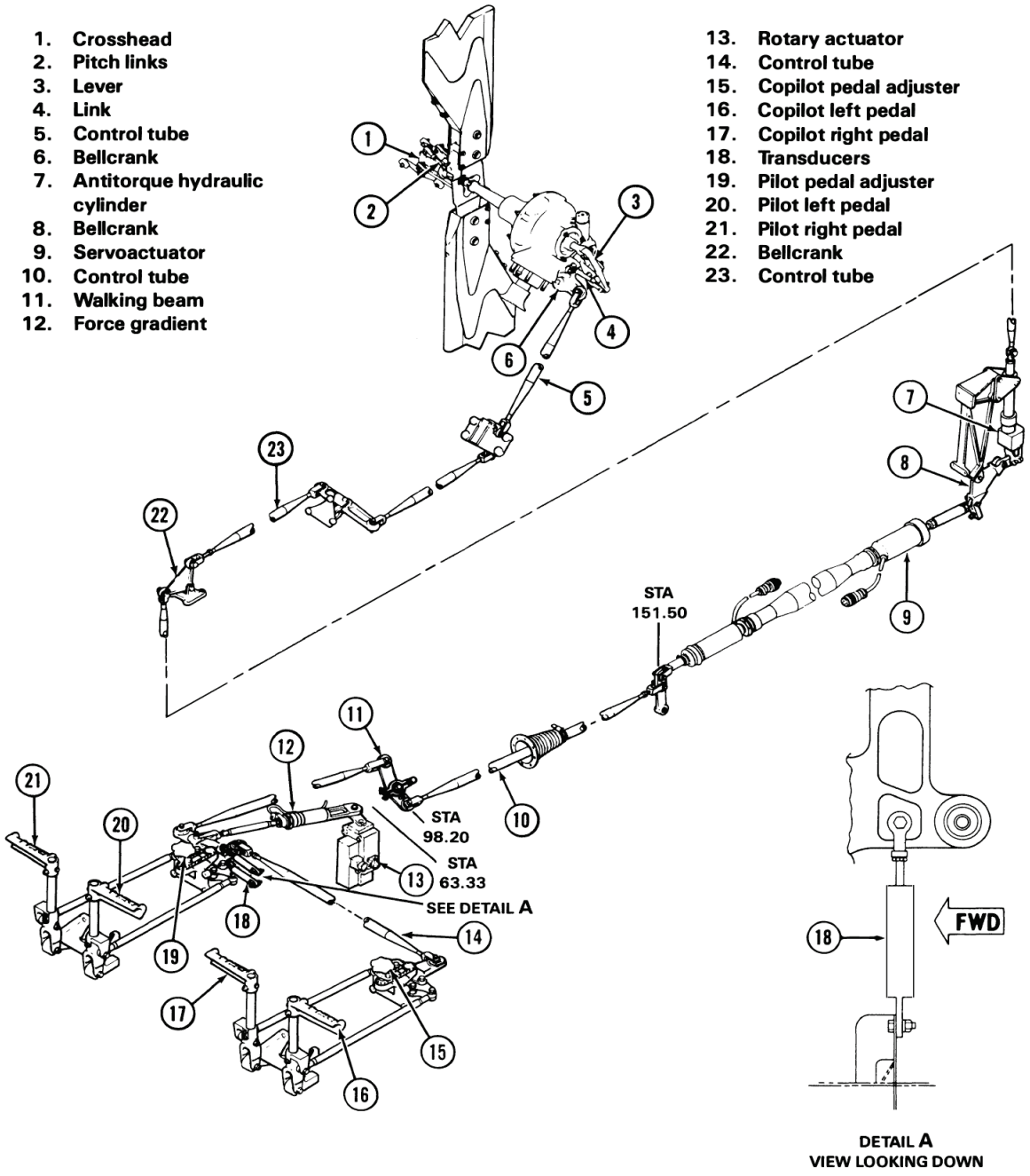
Antitorque Controls - DAFCS - Description

The antitorque control system (Figure 1), consists of control linkage and components from pilot and copilot pedals (20, 21 and 16, 17) aft to crosshead (1) on tail rotor. The DAFCS SAR incorporates an electromechanical servo actuator (9), rotary actuator (13), force gradient (12), and two transducers (18). The servo actuator (9) is located forward of antitorque hydraulic cylinder (7) and support. The rotary actuator (13) is located on main bulkhead under access panel below pilot seat. The force gradient (12) is connected to the rotary actuator (13) and directional control bellcrank. The two transducers (18) are connected to directional control bellcrank located aft of pilot left and right pedals (20 and 21) and to the bracket on main beam of airframe structure.

Actuation of the pilot or copilot pedals causes a hydraulic power assisted pitch change of the tail rotor blades to offset main rotor torque and to control the directional heading of the helicopter.

Dual controls are installed for copilot control of the helicopter.

Operation of antitorque dual controls system is basically the same as operation of single system. Refer to Flight Manual for operating procedures.



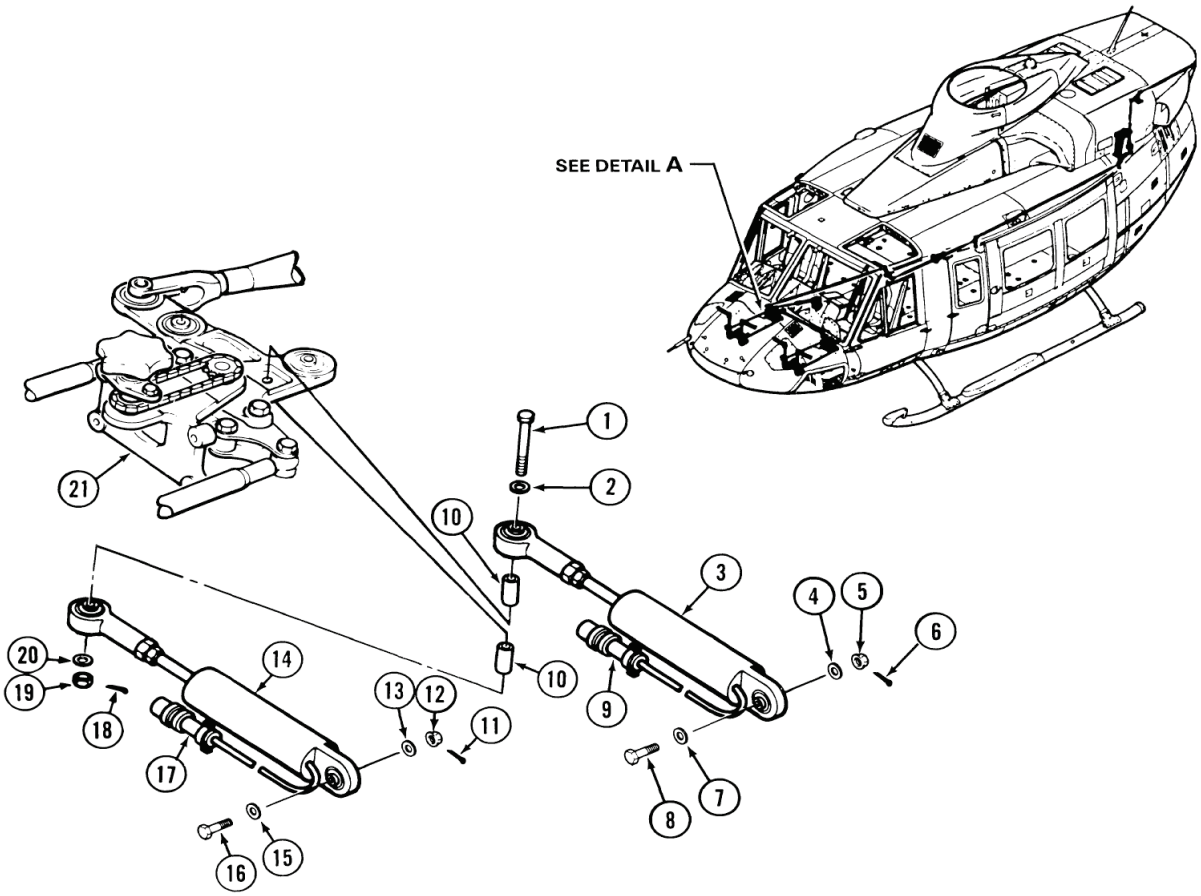
ICN-412-A-671204-B-97499-00001-A-001-01

Figure 1. Antitorque Controls (Sheet 1 of 1)

Antitorque Transducers - Removal

Procedure

1. Remove access panel from cabin floor under pilot seat.
2. Disconnect electrical connectors (9 and 17, Figure 1).
3. Remove bolt (1).
4. Remove bolt (8).
5. Remove bolt (16).
6. Remove transducers (3 and 14).



1. Bolt
2. Aluminum washer
3. Transducer
4. Spacer
5. Nut
6. Cotter pin
7. Aluminum washer
8. Bolt
9. Electrical connector
10. Spacer
11. Cotter pin
12. Nut
13. Spacer
14. Transducer
15. Aluminum washer
16. Bolt
17. Electrical connector
18. Cotter pin
19. Nut
20. Aluminum washer
21. Pedal adjuster

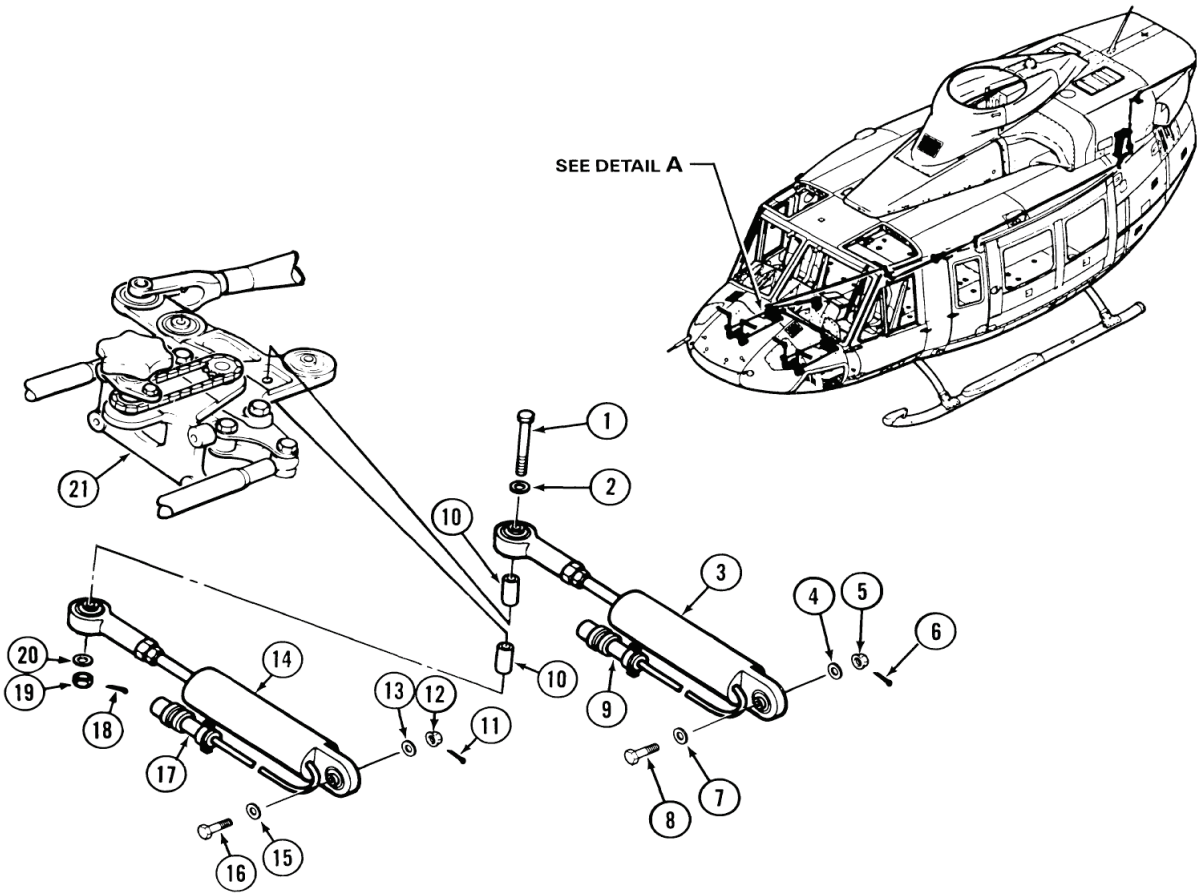
ICN-412-A-671204-B-97499-00002-A-001-01

Figure 1. Antitorque Transducer Removal/Installation (Sheet 1 of 1)

Antitorque Transducers - Inspection

Procedure

1. Inspect transducers (3 and 14, Figure 1) for damage and secured electrical plug.
2. Inspect bearings for looseness (BHT-ALL-SPM, Standard Practices Manual).



1. Bolt	8. Bolt	15. Aluminum washer
2. Aluminum washer	9. Electrical connector	16. Bolt
3. Transducer	10. Spacer	17. Electrical connector
4. Spacer	11. Cotter pin	18. Cotter pin
5. Nut	12. Nut	19. Nut
6. Cotter pin	13. Spacer	20. Aluminum washer
7. Aluminum washer	14. Transducer	21. Pedal adjuster

ICN-412-A-671204-B-97499-00002-A-001-01

Figure 1. Antitorque Transducer Removal/Installation (Sheet 1 of 1)

Antitorque Transducers - Repair

Procedure

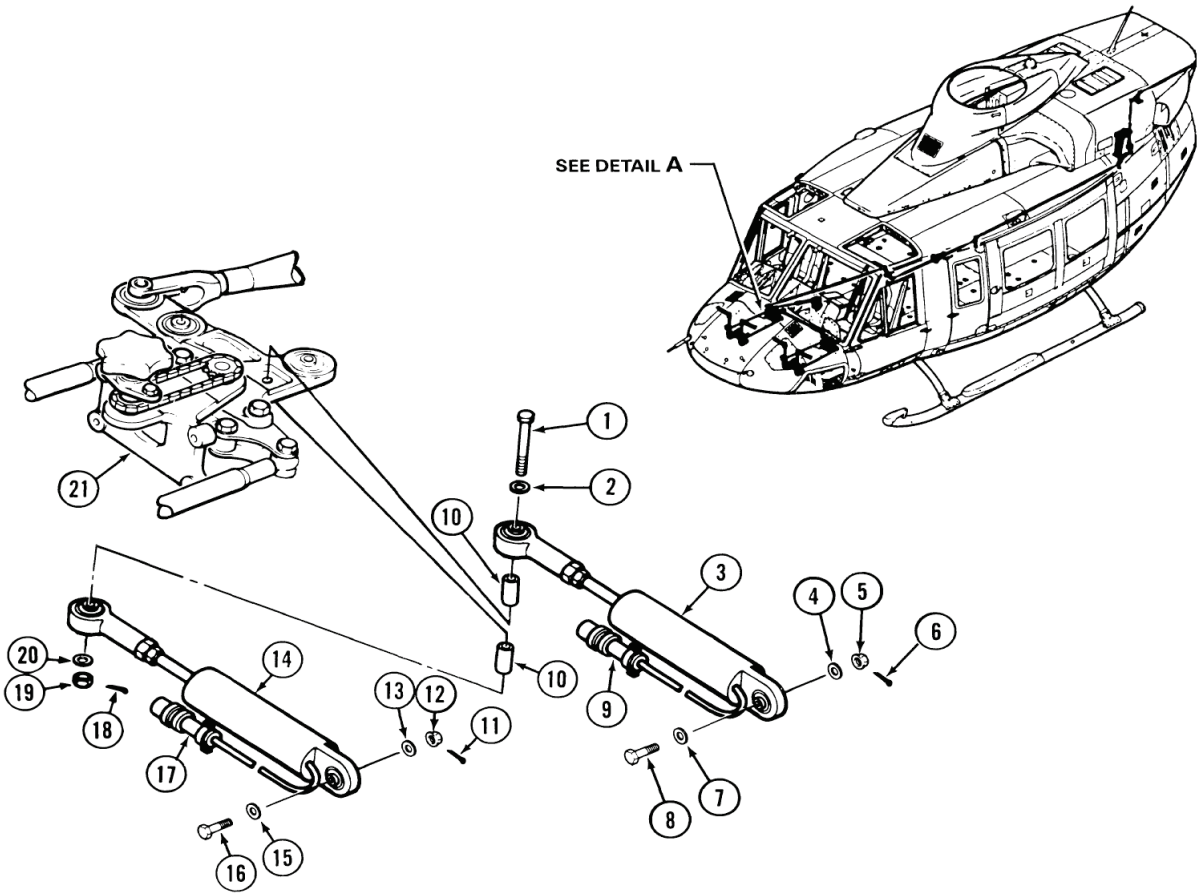
1. Antitorque control transducers are nonrepairable and must be replaced when defective.

Intentionally blank page

Antitorque Transducers - Installation

Procedure

1. Position transducers (3 and 14, Figure 1) on bracket of main beam of airframe structure.
2. Install bolt (8), aluminum washer (7), spacer (4), nut (5), and cotter pin (6).
3. Install bolt (16), aluminum washer (15), spacer (13), nut (12), and cotter pin (11).
4. Position transducers (3 and 14) on pedal adjuster (21) bellcrank.
5. Install bolt (1), aluminum washer (2), spacers (10), aluminum washer (20), nut (19), and cotter pin (18).
6. Connect electrical connectors (9 and 17).
7. Install access panel on floor under pilot seat.



1. Bolt
2. Aluminum washer
3. Transducer
4. Spacer
5. Nut
6. Cotter pin
7. Aluminum washer
8. Bolt
9. Electrical connector
10. Spacer
11. Cotter pin
12. Nut
13. Spacer
14. Transducer
15. Aluminum washer
16. Bolt
17. Electrical connector
18. Cotter pin
19. Nut
20. Aluminum washer
21. Pedal adjuster

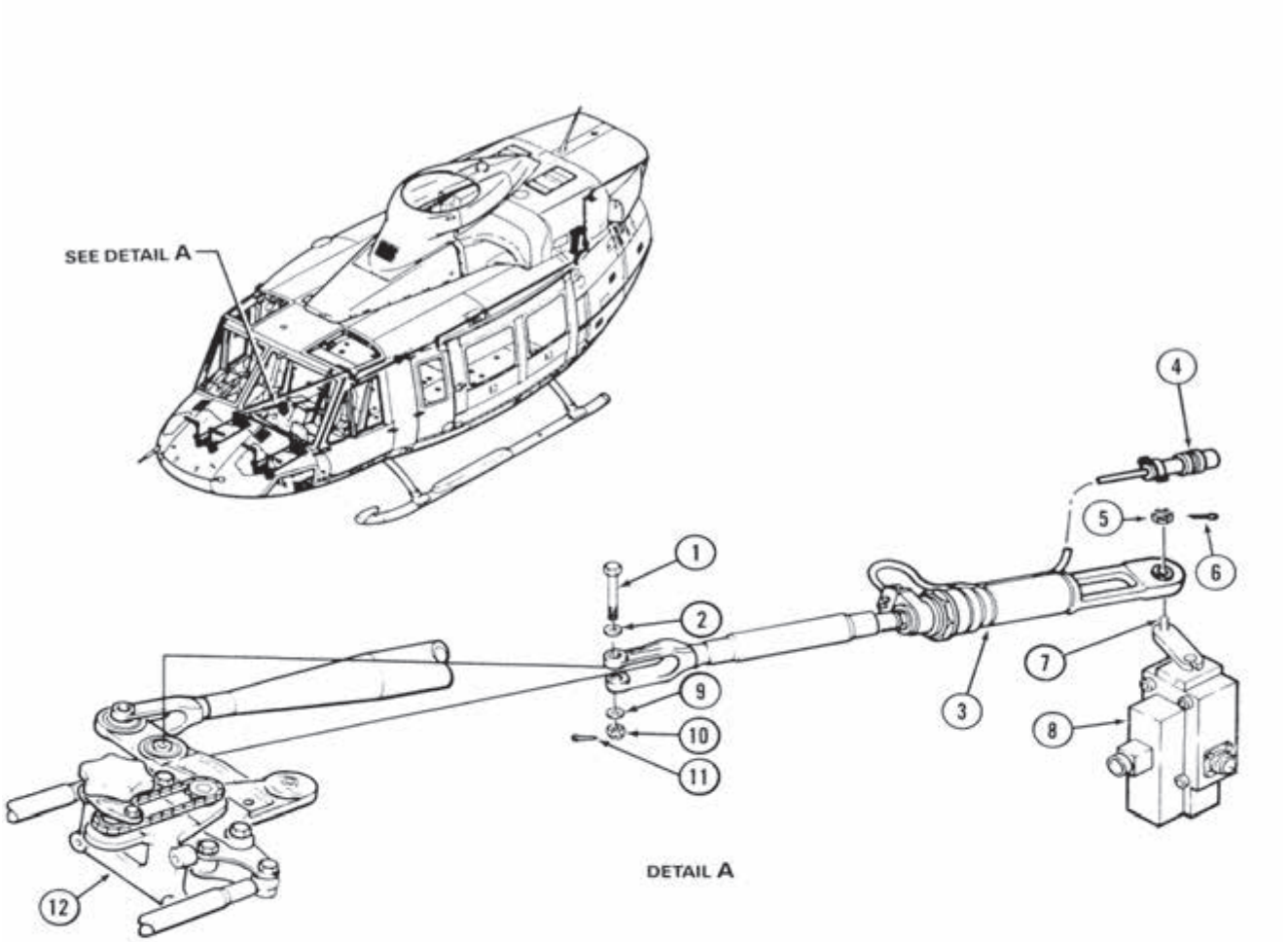
ICN-412-A-671204-B-97499-00002-A-001-01

Figure 1. Antitorque Transducer Removal/Installation (Sheet 1 of 1)

Antitorque Force Gradient - Removal

Procedure

1. Remove access panel from cabin floor under pilot seat.
2. Disconnect electrical connector (4, Figure 1).
3. Remove bolt (1).
4. Remove nut (5).
5. Remove force gradient (3).



1. Bolt
2. Aluminum washer
3. Force gradient
4. Electrical connector
5. Nut
6. Cotter pin
7. Stud
8. Rotary actuator
9. Aluminum washer
10. Nut
11. Cotter pin
12. Pedal adjuster

ICN-412-A-671204-B-97499-00003-A-001-01

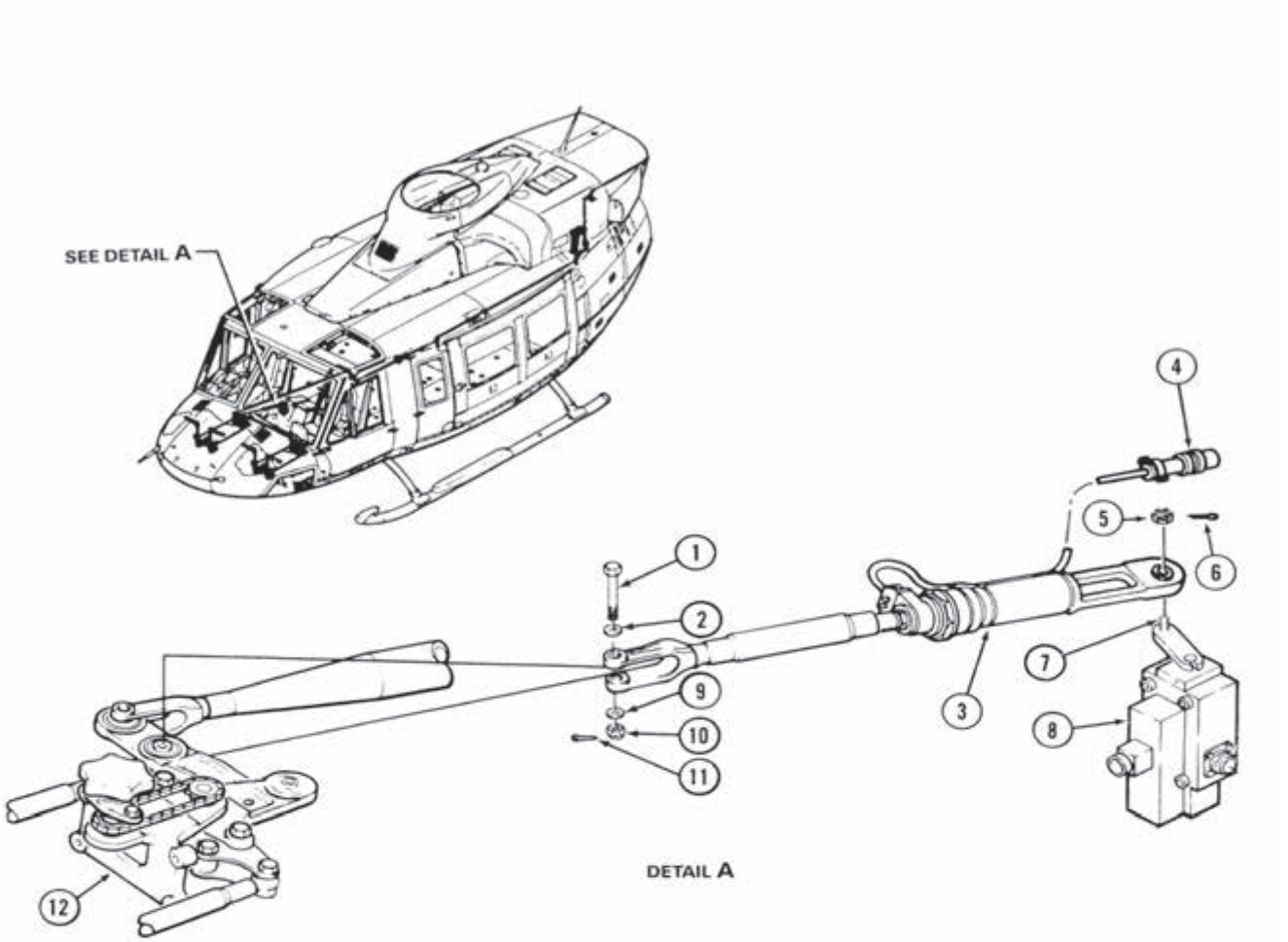
Figure 1. Antitorque Force Gradient Removal/Installation (Sheet 1 of 1)

Antitorque Force Gradient - Inspection

Procedure

1. Inspect force gradient (3, Figure 1) for corrosion and mechanical damage (412-CRO - Component Repair and Overhaul Manual). Superficial mechanical damage is acceptable without repair. Mechanical damage in excess of superficial and/or corrosion damage is not acceptable.

2. Inspect bearing for secure installation in housing and for wear (BHT-ALL-SPM, Standard Practices Manual).



1. Bolt
2. Aluminum washer
3. Force gradient
4. Electrical connector
5. Nut
6. Cotter pin
7. Stud
8. Rotary actuator
9. Aluminum washer
10. Nut
11. Cotter pin
12. Pedal adjuster

ICN-412-A-671204-B-97499-00003-A-001-01

Figure 1. Antitorque Force Gradient Removal/Installation (Sheet 1 of 1)

Antitorque Force Gradient - Repair

Procedure

1. Repair of force gradient consists of replacement of unserviciable components and adjustment of spring preload in accordance with 412-A-67-12-03-04A-720A-A / 02799, step 1.

Intentionally blank page

Antitorque Force Gradient - Installation

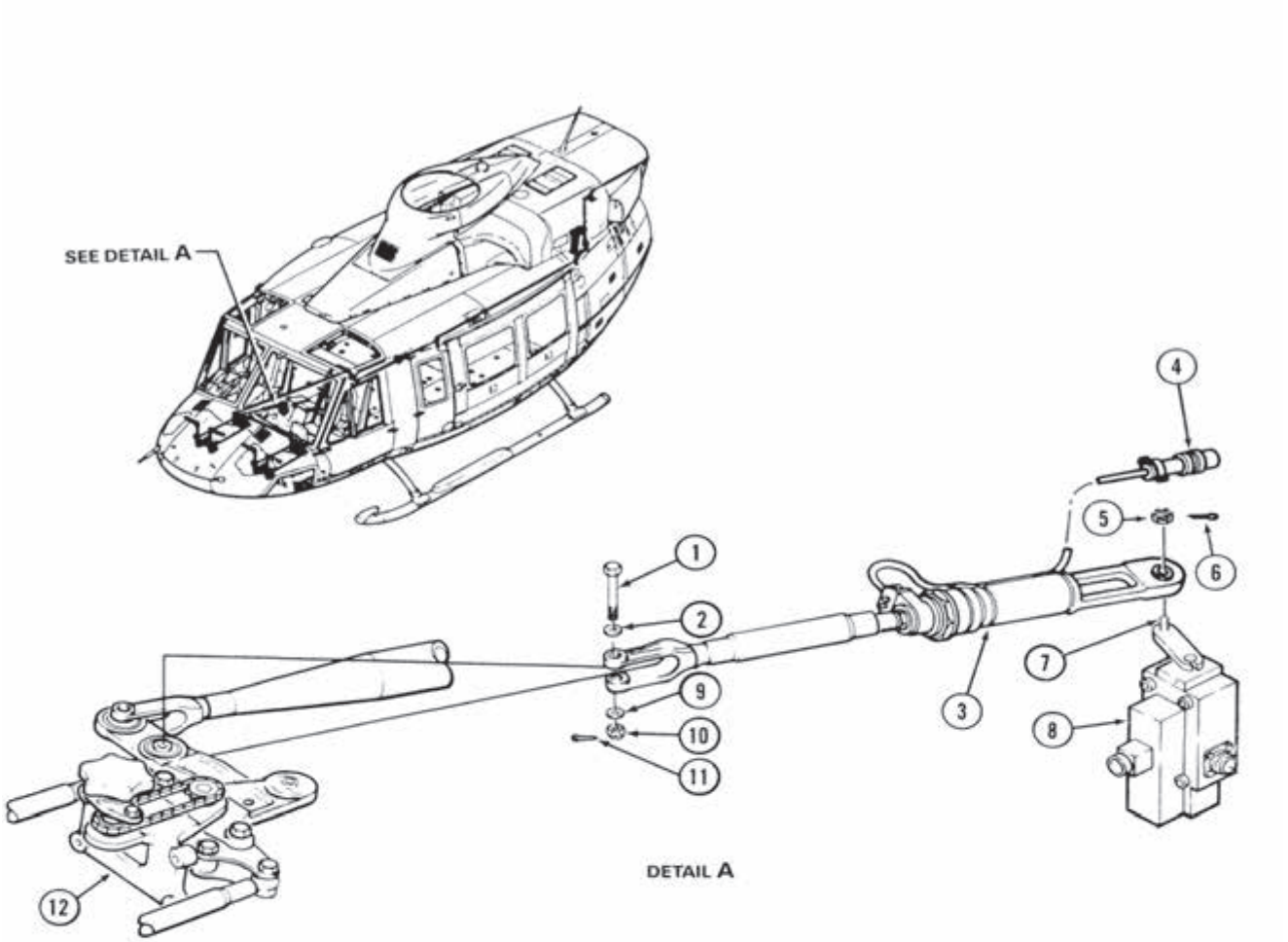
Procedure

ENSURE THAT FORCE GRADIENT LENGTH IS ADJUSTED PRIOR TO INSTALLATION ON ROTARY ACTUATOR.

NOTE

Force gradient is similar in appearance to the force gradients used in the cyclic controls but has a different part number. Ensure that correct part number force gradient is installed for the antitorque control system.

1. If not previously accomplished, adjust spring preload on force gradient (3, Figure 1), 18 to 20 lbs. (80.064 to 88.96 N) in accordance with 412-A-67-12-03-04A-720A-A / 02799, step 1.
2. Position force gradient (3) on pedal adjuster (12).
3. Install bolt (1), aluminum washers (2 and 9), nut (10), and cotter pin (11).
4. Position force gradient (3, Figure 1) on stud (7) of rotary actuator (8).
5. Install nut (5), and cotter pin (6).
6. Connect electrical connector (4).
7. Move antitorque controls through full throw and check for obstructions and binding.
8. Confirm that antitorque control rigging is within limits. Refer 412-A-67-02-00-03A-276A-A / 02762.
9. Install access panel on cabin floor under pilot seat.



1. Bolt
2. Aluminum washer
3. Force gradient
4. Electrical connector
5. Nut
6. Cotter pin
7. Stud
8. Rotary actuator
9. Aluminum washer
10. Nut
11. Cotter pin
12. Pedal adjuster

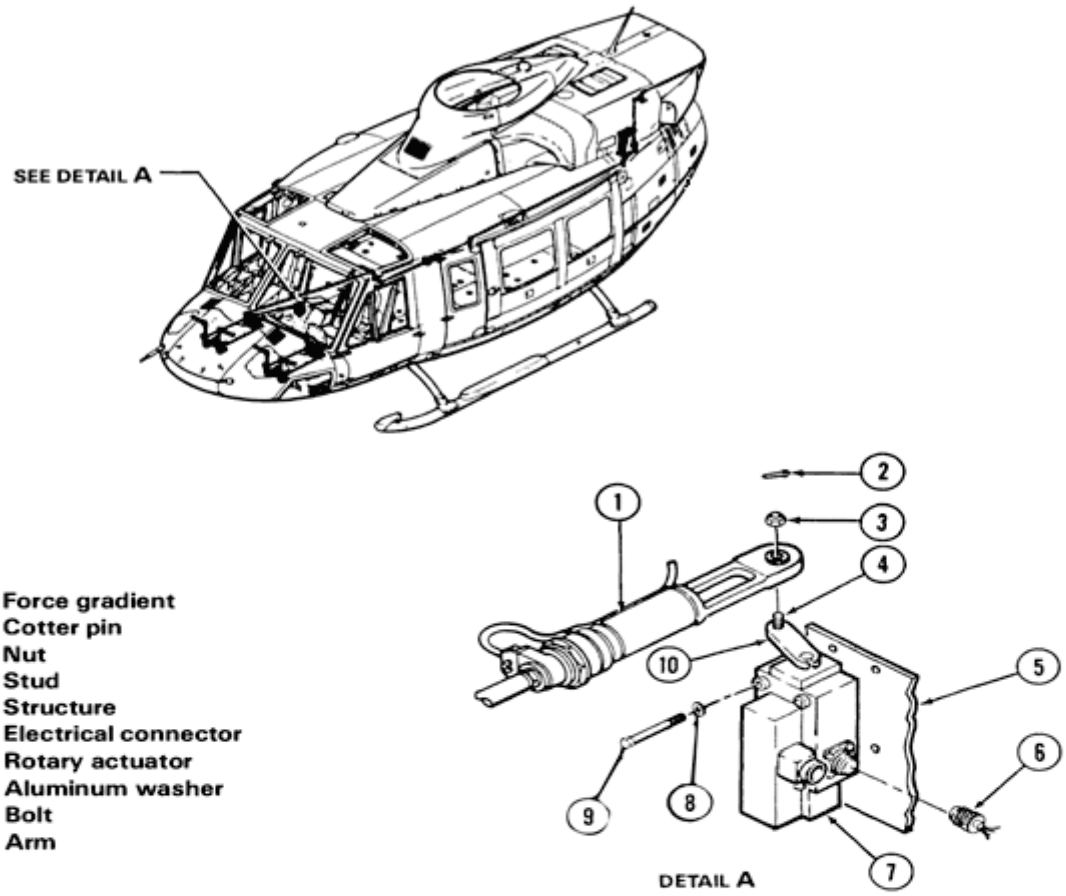
ICN-412-A-671204-B-97499-00003-A-001-01

Figure 1. Antitorque Force Gradient Removal/Installation (Sheet 1 of 1)

Antitorque Rotary Actuator - Removal

Procedure

1. Remove access panel from cabin floor under pilot seat.
2. Disconnect electrical connector (6, Figure 1).
3. Remove nut (3) from stud (4).
4. Remove four bolts (9) and four aluminum washers (8).
5. Remove rotary actuator (7) from structure (5).



ICN-412-A-671204-B-97499-00004-A-001-01

Figure 1. Antitorque Rotary Actuator Removal/Installation (Sheet 1 of 1)

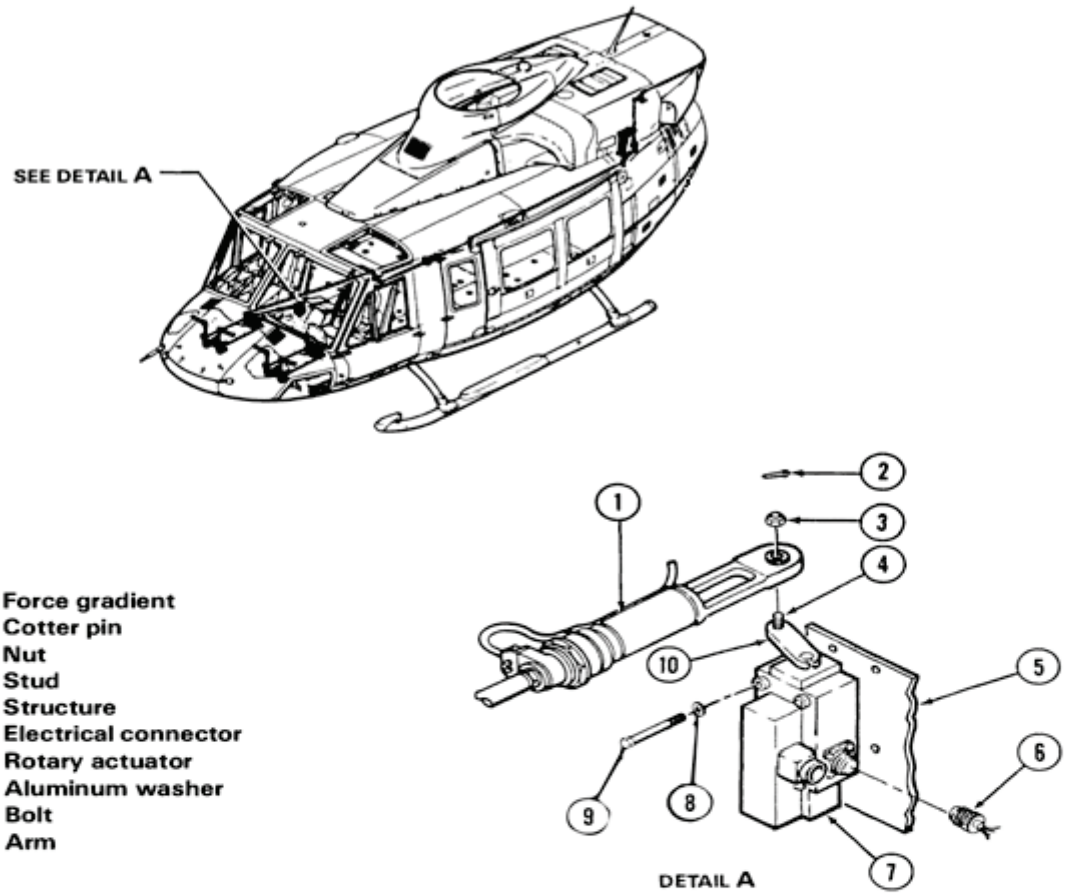
Antitorque Rotary Actuator - Inspection

Procedure

1. Inspect rotary actuator (7, Figure 1) for corrosion and mechanical damage. Superficial mechanical damage is acceptable without repair. Mechanical damage in excess of superficial and/or corrosion damage is not acceptable

2. Inspect stud (4) for damaged threads. No damage that can be detected visually is acceptable.

3. Inspect electrical receptable for bent pins and threads. Superficial damage is acceptable without repair.



ICN-412-A-671204-B-97499-00004-A-001-01

Figure 1. Antitorque Rotary Actuator Removal/Installation (Sheet 1 of 1)

Antitorque Rotary Actuator - Repair

Procedure

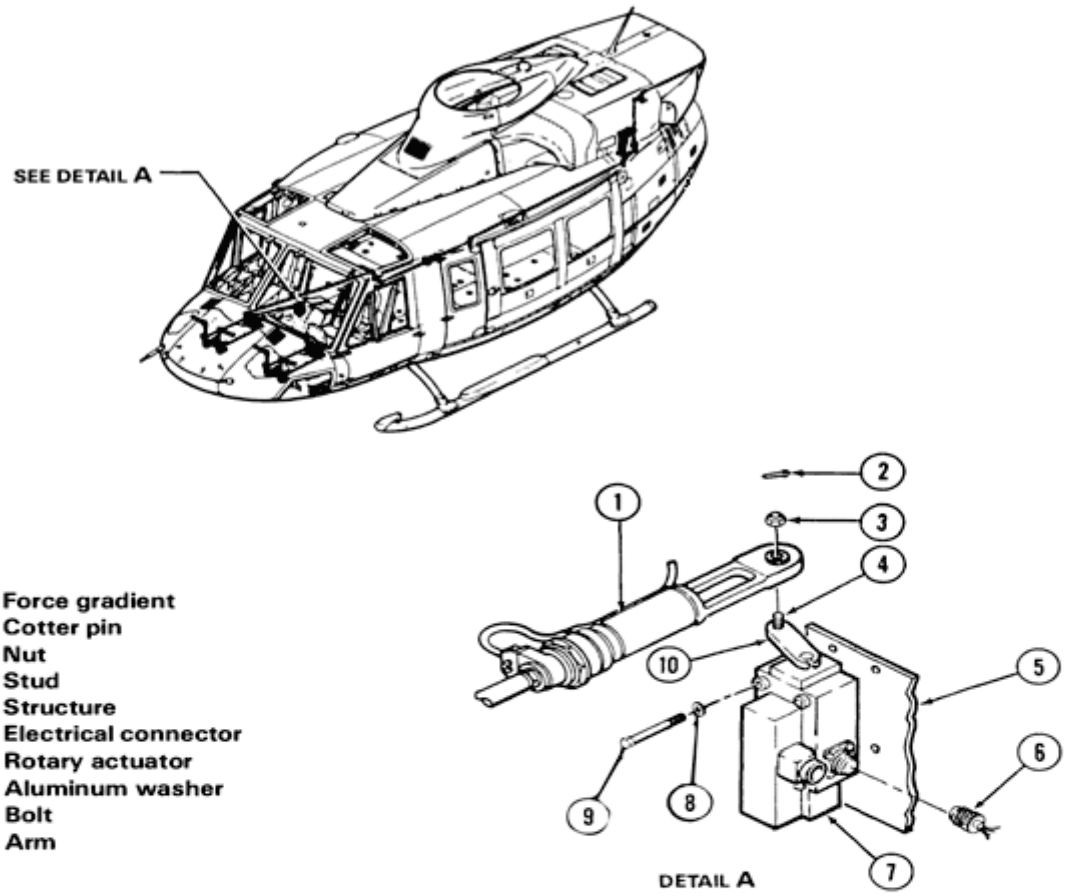
1. Refer to vendor manual for rotary actuator repair instructions.

Intentionally blank page

Antitorque Rotary Actuator - Installation

Procedure

1. Inspect rotary actuator (7, Figure 1) to ensure arm (10) is properly installed on rotary actuator (7) as outlined in 412-A-67-02-00-03A-276A-A / 02762 rigging instructions.
2. Position rotary actuator (7) on structure (5). Install four bolts (9) and four aluminum washers (8) as follows:
3. Install aluminum washer (8) with each bolt (9).
4. Install four bolts (9) of correct length to obtain proper thread engagement. Three different length bolts are required.
5. Install bolts (9) with drilled heads in two inboard holes in rotary actuator (7) and structure (5).
6. Position force gradient (1) on stud (4) of arm (10). Install nut (3) and cotter pin (2).
7. Install electrical connector (6).
8. Move antitorque controls through full throw and check for binding and obstructions.
9. Confirm that antitorque control rigging is within limits 412-A-67-02-00-03A-276A-A / 02762.
10. Install access panel on cabin floor under pilot seat.



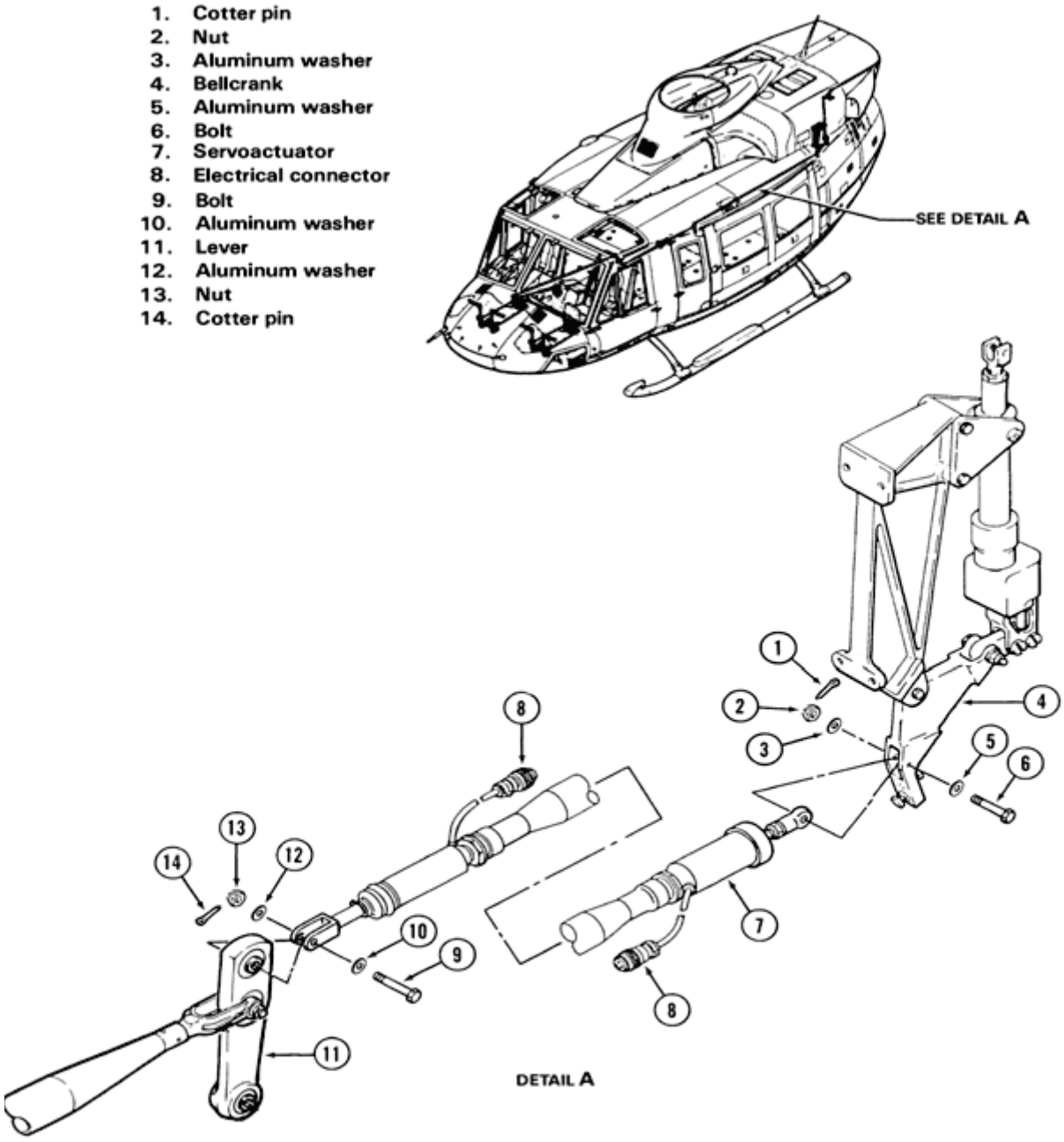
ICN-412-A-671204-B-97499-00004-A-001-01

Figure 1. Antitorque Rotary Actuator Removal/Installation (Sheet 1 of 1)

Antitorque SCAS Servoactuator - Removal

Procedure

1. Remove access panel from bottom of aft fuselage. Open access door on right lower right side of aft fuselage.
2. Disconnect electrical connectors (8, Figure 1).
3. Remove bolt (6).
4. Remove bolt (9).
5. Remove servoactuator (7).



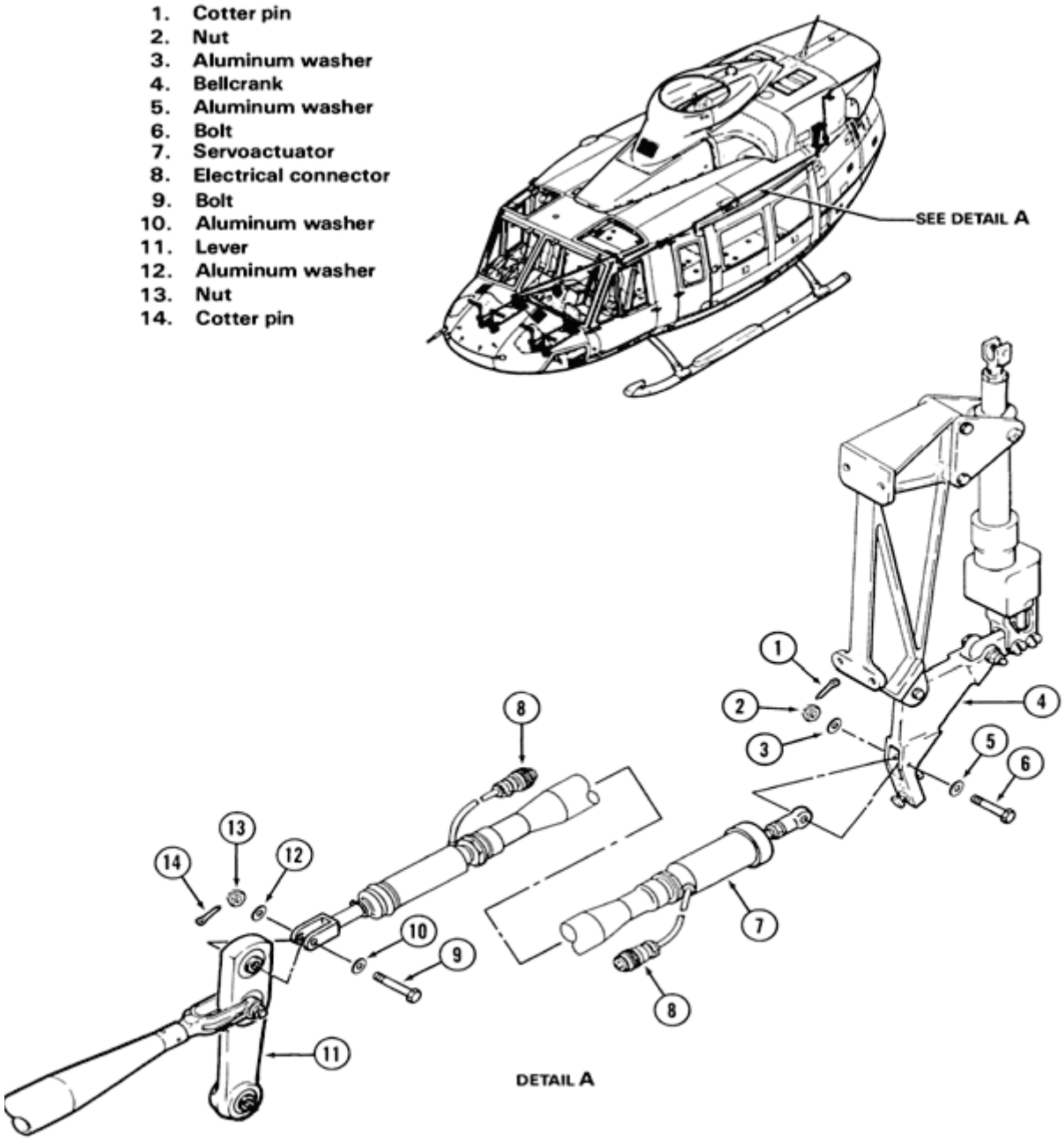
ICN-412-A-671204-B-97499-00005-A-001-01

Figure 1. Antitorque SCAS Servoactuator Removal/Installation (Sheet 1 of 1)

Antitorque SCAS Servoactuator - Inspection

Procedure

1. Inspect servoactuator (7, Figure 1) in accordance with procedures in 412-CRO - Component Repair and Overhaul Manual.



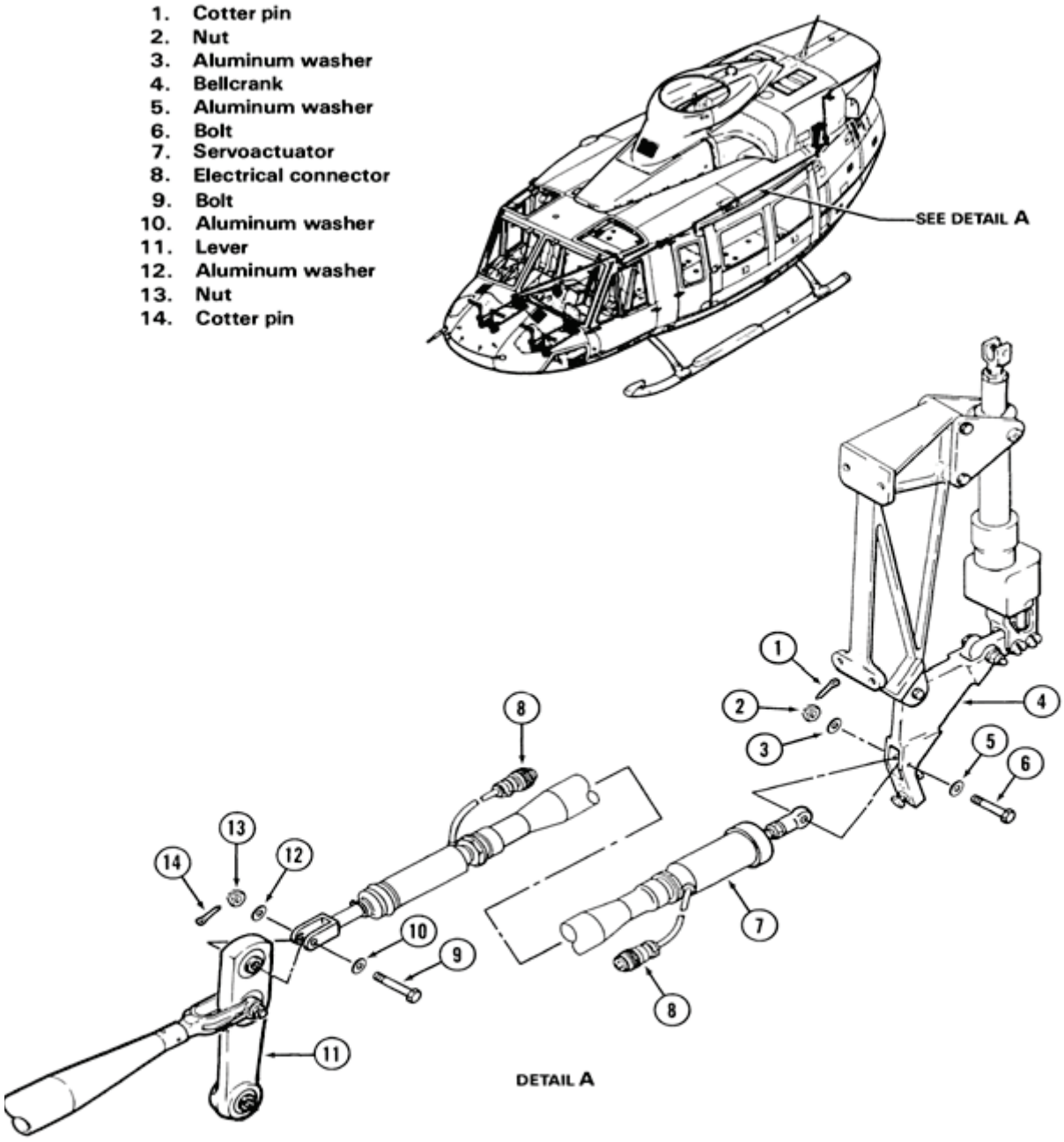
ICN-412-A-671204-B-97499-00005-A-001-01

Figure 1. Antitorque SCAS Servoactuator Removal/Installation (Sheet 1 of 1)

Antitorque SCAS Servoactuator - Installation

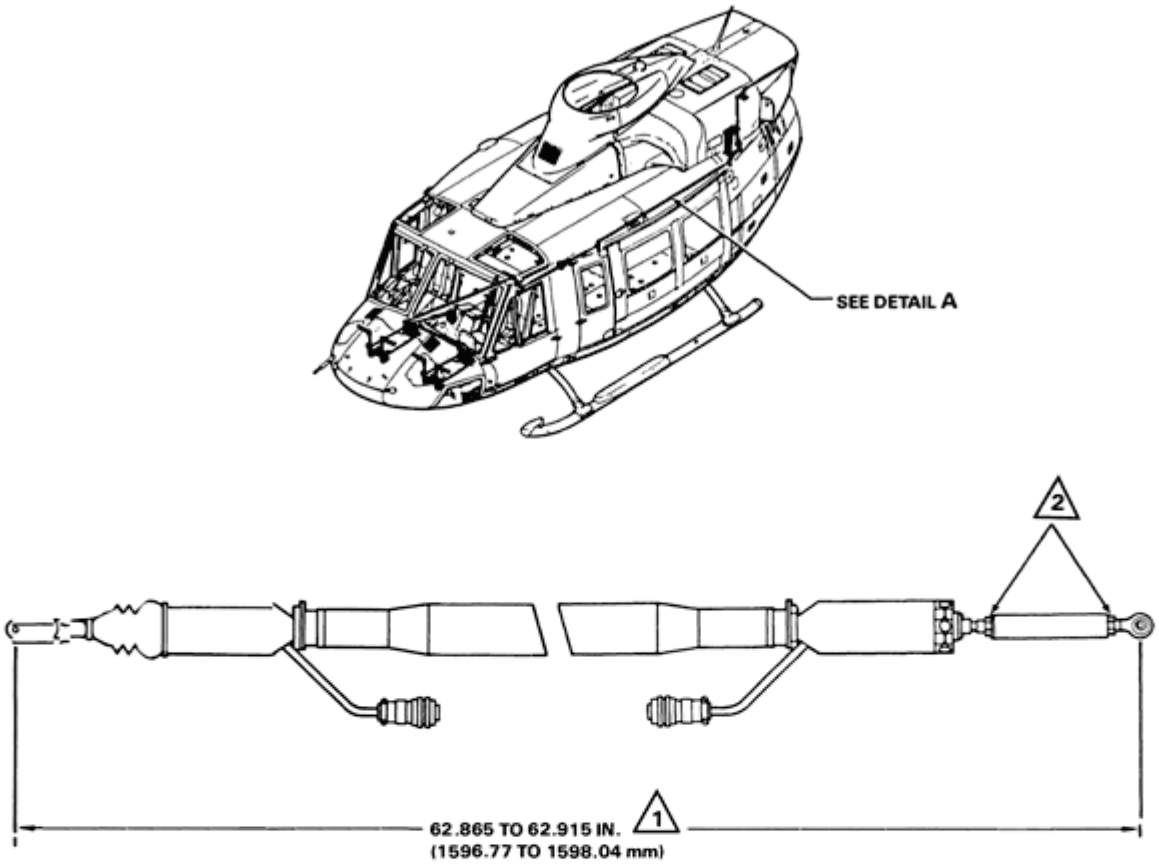
Procedure

1. Establish overall dimensions of servoactuator (7, Figure 1) as shown in Figure 2.
2. Place servoactuator (7, Figure 1) into position with rod end bearing in bellcrank (4).
3. Install bolt (6), aluminum washers (5 and 3), nut (2), and cotter pin (1).
4. Place forward end of servoactuator (7) on lever (11).
5. Install bolt (9), aluminum washers (10 and 12), nut (13), and cotter pin (14).
6. Move antitorque controls through full travel and check for obstructions and binding.
7. Connect electrical connectors (8).
8. Confirm that antitorque rigging is within limits. (Refer to 412-A-67-02-00-03A-276A-A / 02762).
9. Install access panel on bottom of aft fuselage.
10. Close access door on lower right side of aft fuselage.



ICN-412-A-671204-B-97499-00005-A-001-01

Figure 1. Antitorque SCAS Servoactuator Removal/Installation (Sheet 1 of 1)



DETAIL A
412-301-700-101
DIRECTION SCAS
SERVOACTUATOR

NOTES

1. Length shown is with actuator at mid-position.
2. Nut to have full engagement of threads.
3. Actuator length is 63.259 to 63.309 inches (1606.77 to 1608.04 mm) fully extended. Minimum length is 62.471 to 62.521 inches (1586.763 to 1588.033 mm) fully retracted.

ICN-412-A-671204-B-97499-00006-A-001-01

Figure 2. Antitorque SCAS Servoactuator (Sheet 1 of 1)

Intentionally blank page

Instrument System - General Description

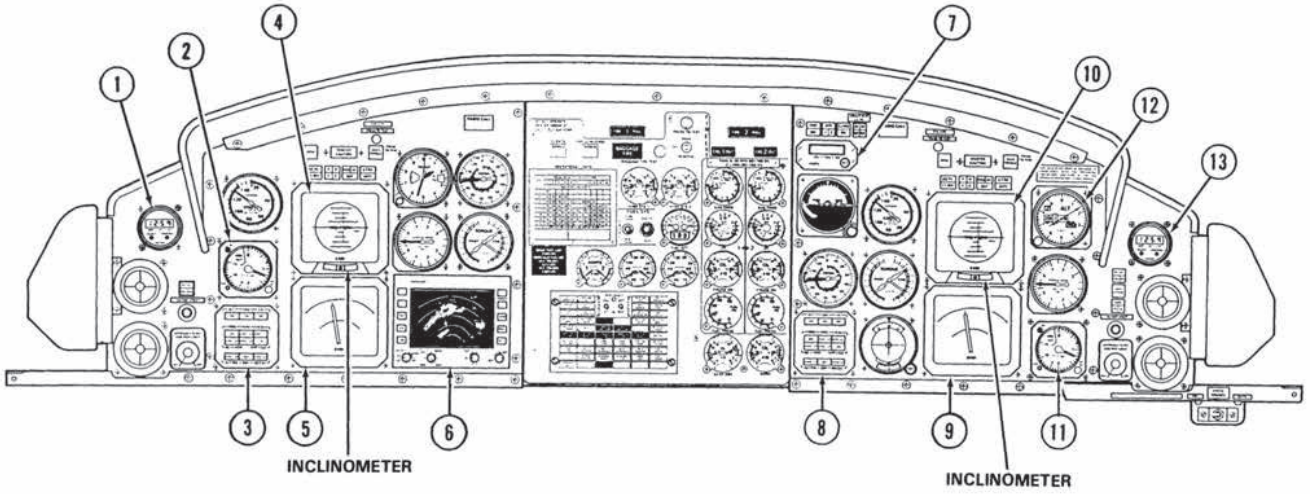
Supplemental maintenance information for flight, navigation, propulsion, hydraulic, and miscellaneous instruments installed in Search And Rescue (SAR) configured model 412 helicopter is presented in this section.

Intentionally blank page

Panel Mounted Instruments - Description

Refer to Figure 1 for SAR instrument panel.

Intentionally blank page



1. Digital clock (copilot)
2. Radio altimeter indicator (copilot)
3. Mode selector (copilot)
4. EADI (copilot)
5. EHSI (copilot)
6. Color indicator (radar indicator)
7. Altitude preselect
8. Mode selector (pilot)
9. EHSI (pilot)
10. EADI (pilot)
11. Radar altimeter indicator (pilot)
12. Encoding altimeter
13. Digital clock (pilot)

ICN-412-A-950101-B-97499-00001-A-001-01

Figure 1. Instrument Panel (Sheet 1 of 1)

Intentionally blank page

Panel Mounted Instruments - Removal/Installation

Procedure

1. Refer to 412-MM, Chapter 95 for removal/installation of instrument panel mounted components.

Intentionally blank page

Panel Mounted Instruments - Inspection/Cleaning

Procedure

1. Refer to 412-A-95-00-01-00A-280A-A / 01793 for inspection/cleaning of Instrument panel mounted components.

Intentionally blank page

Panel Mounted Instruments - Repair or Replacement

Procedure

1. Refer to 412-A-95-00-01-00A-685A-A / 01795 for repair or replacement of instrument panel mounted components.

Intentionally blank page

Instrument Panel - Maintenance Procedures

Procedure

1. Refer to 412-A-95-00-01-00A-042A-A / 01790 for removal/installation/adjustment of instrument panel mounted components.

Intentionally blank page

Flight Instruments - General Description

Flight instruments that are unique to search and rescue configured helicopters include an electronic attitude director indicator.

Intentionally blank page

Electronic Attitude Director Indicator (EADI) - Description

Refer to Honeywell, SPZ-7600 Integrated Flight Control System Bell 412HP System Maintenance Manual, Publication Number A09-1169-024 for component description and operation.

The EADI provides the traditional sphere-type attitude display with lateral and vertical steering signals to assist the pilot in intercepting and maintaining a desired flightpath.

The EADI (Figure 1) displays the following information:

- Attitude
- Pitch and Roll
- Command Bars
- Collective Command Cues
- Glideslope Deviation
- Expanded Localizer Deviation
- Marker Beacon Annunciator
- Radio Altitude
- - Rising Runway
- - Digital Readout
- Decision Height Setting/Warning
- Flight Director and SAR Mode Annunciators
- Rate-of-Turn

In addition to the familiar ADI information the EADI displays the following amber caution and failure annunciations (Figure 2):

- Attitude Source Annunciator
- Flight Director Failure Annunciator
- Decision Height Warning Annunciator
- Flight Management System Message Annunciator
- Radio Altitude Failure Annunciator
- Comparison Monitor Annunciator
- Decision Height Failure Annunciator
- Common Symbol Generator Annunciator

The EADI also displays the following red failure annunciations (Figure 3):

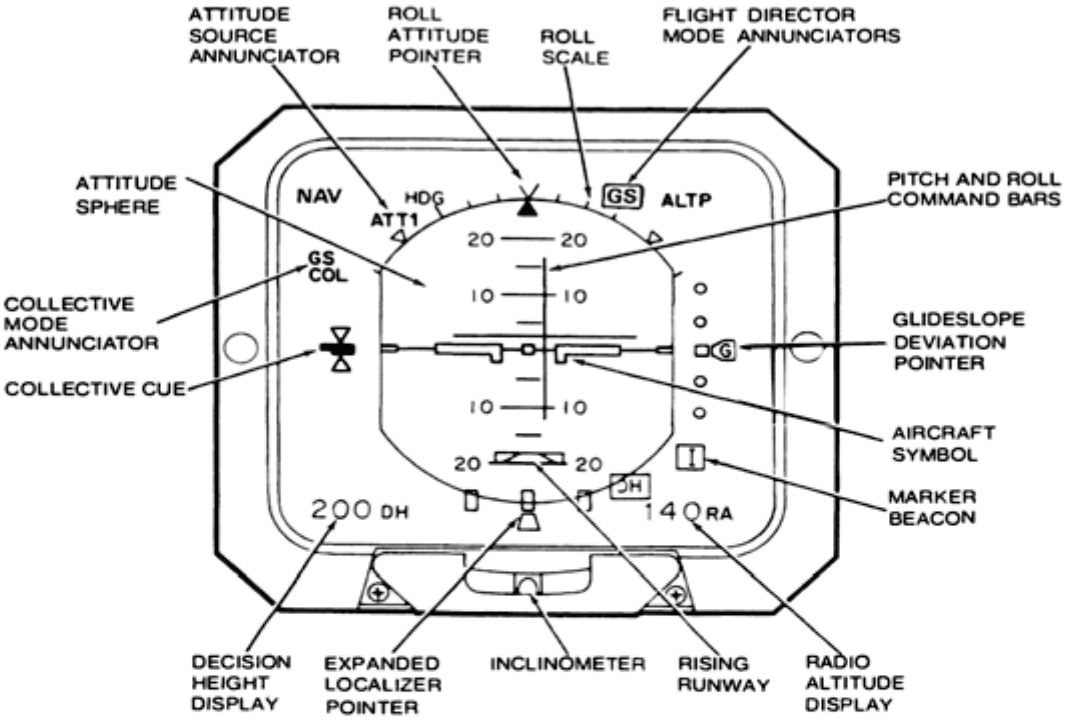
- Attitude Failure
- Glideslope, Glidepath or VNAV, Expanded Localizer, or Rate-of-Turn
- Internal Failure

Failure of the attitude source results in the pitch scale and roll pointer being removed, the sphere will be displayed blue, and a red ATT FAIL annunciator being displayed.

Failure of the glideslope, glidepath or VNAV, expanded localizer or azimuth, or rate-of-turn will result in the pointer being removed and a red X drawn through the scale. The letter G or V at the zero deviation position identifies the invalid information.

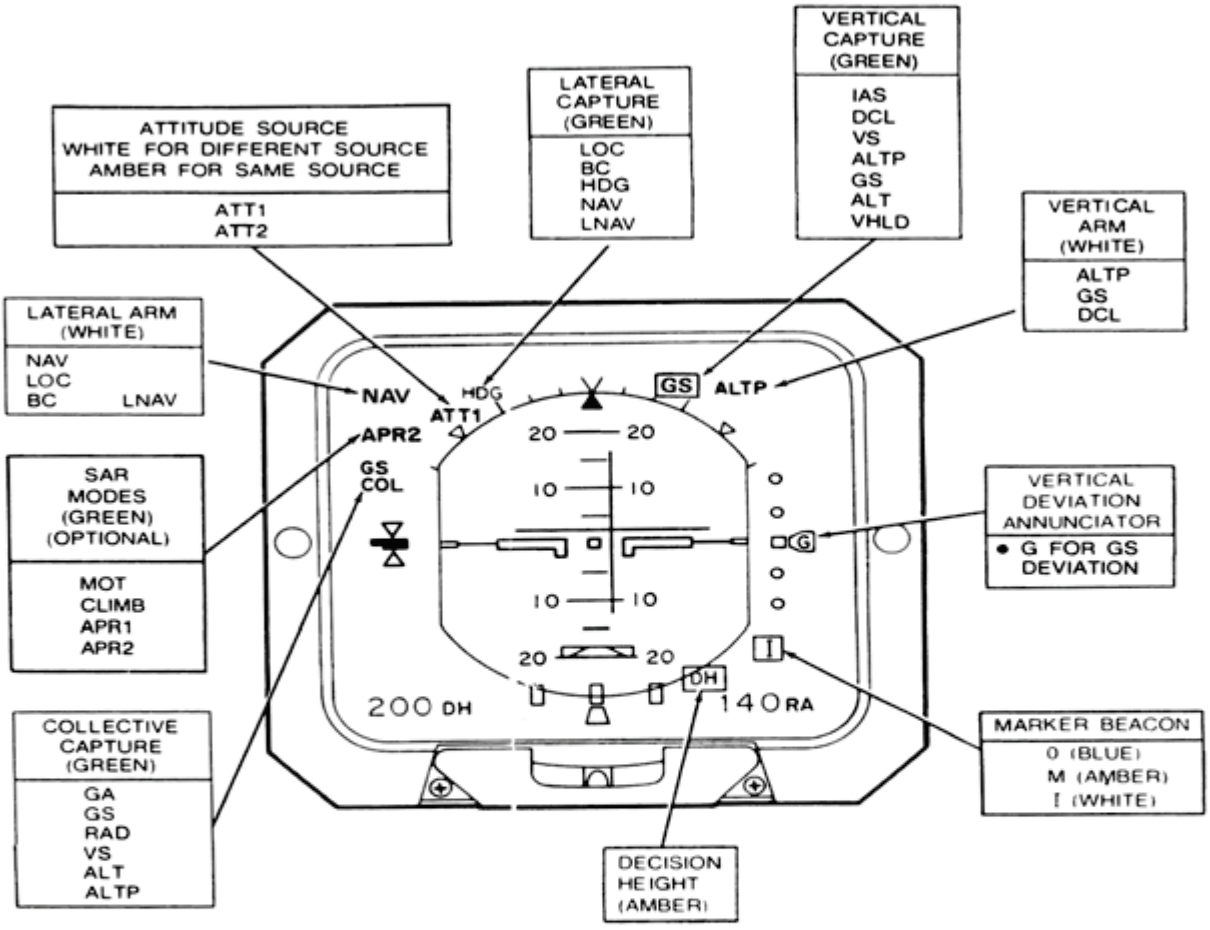
An extended failure of the symbol generator will cause the EADI to go blank. A failure of the input/processor results in a red X centered on the display, with annunciation SG FAIL.

For additional information on component displays/annunciation and operation of the EADI, refer to Honeywell, SPZ-7600 Integrated Flight Control System Bell 412HP System Maintenance, Manual Publication Number A09-1169-024.



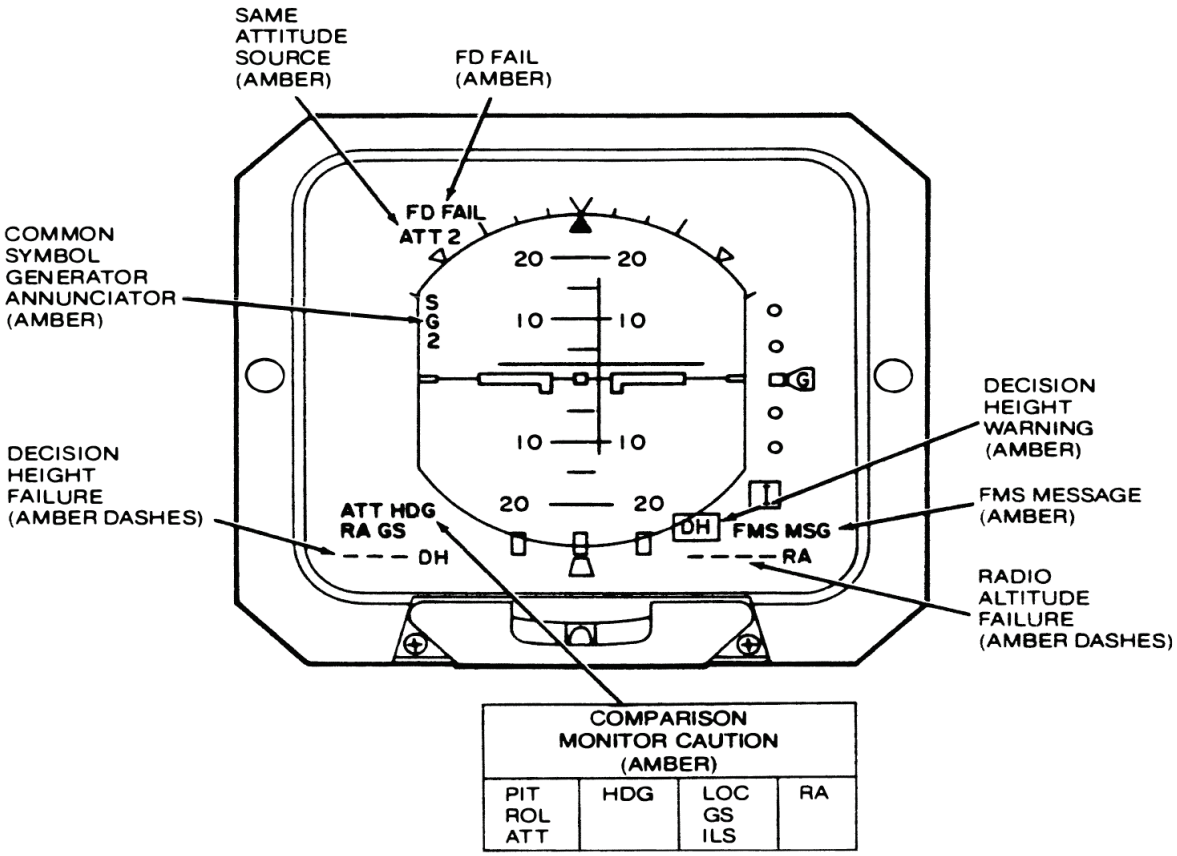
ICN-412-A-951101-B-97499-00001-A-001-01

Figure 1. EADI Displays and Annunciators (Sheet 1 of 2)



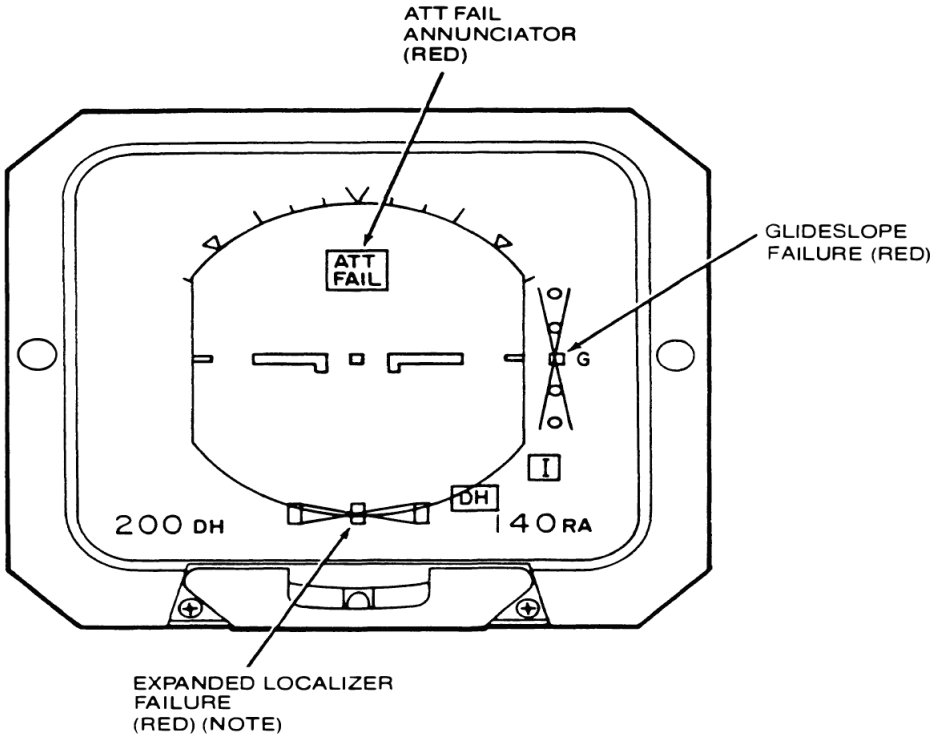
ICN-412-A-951101-B-97499-00002-A-001-01

Figure 1. EADI Displays and Annunciators (Sheet 2 of 2)



ICN-412-A-951101-B-97499-00003-A-001-01

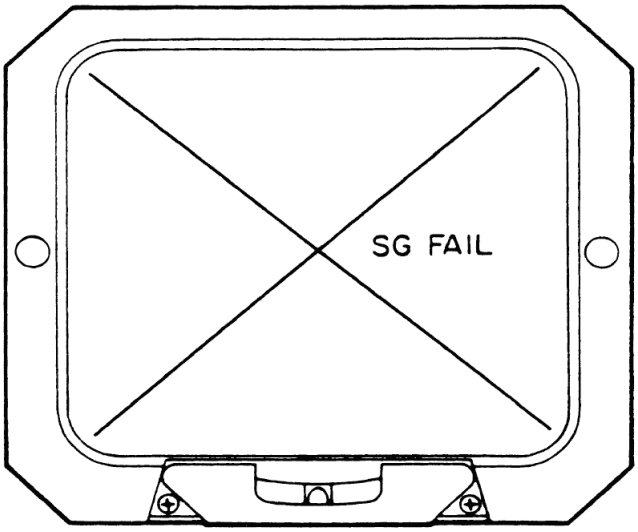
Figure 2. EADI Caution and Failure Annunciations (Amber) (Sheet 1 of 1)



NOTE: RATE-OF-TURN FAILURE IS SIMILAR.

ICN-412-A-951101-B-97499-00004-A-001-01

Figure 3. EADI Failure Annunciations (Red) (Sheet 1 of 1)



NOTE: COMPLETE FAILURE OF A SYMBOL GENERATOR WILL RESULT IN BLANK EADI AND EHSI DISPLAYS.

ICN-412-A-951101-B-97499-00005-A-001-01

Figure 4. Internal System Failure (Red) (Sheet 1 of 1)

EADI - Maintenance Procedures

Procedure

1. Refer to 412-A-95-11-02-00A-042A-A / 01802 for maintenance procedures of panel mounted instrument.

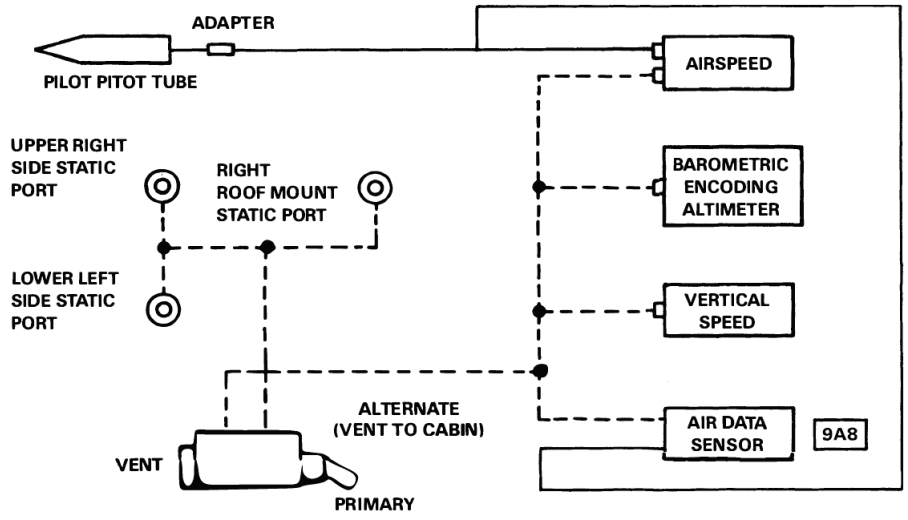
Intentionally blank page

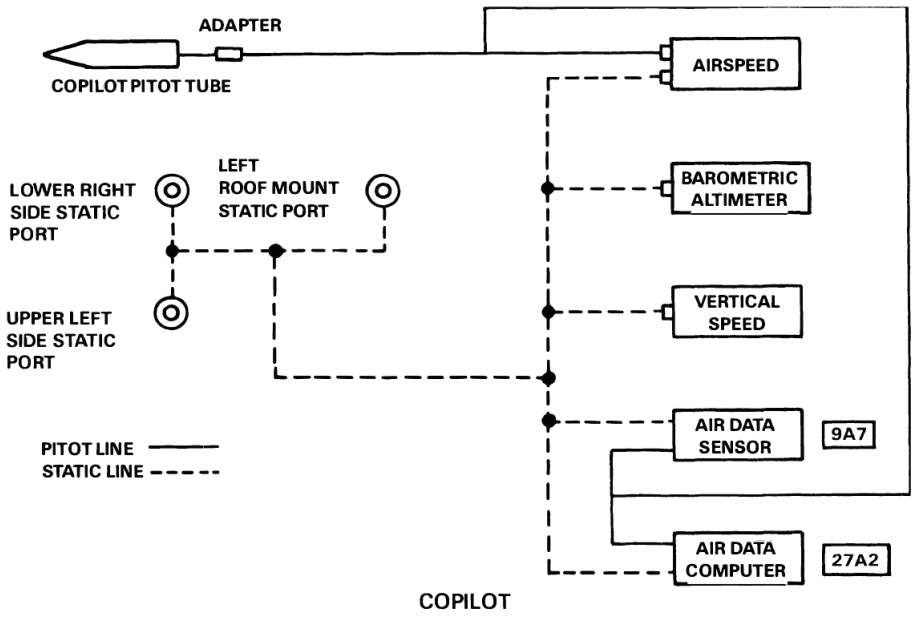
Pilot Pitot-Static System - Description

The pilot pitot-static system (Figure 1) consists of right pitot tube, right and left static ports, right roof-mounted static port, alternate static air source, pilot airspeed indicator, barometric encoding altimeter, vertical speed indicator, air data sensor, and appropriate tubing and hardware for interconnection of system components.

The right pitot tube supplies pitot pressure (impact air) directly to pilot airspeed indicator and air data sensor. Upper right, lower left and right roof-mounted static ports are connected together and supply equalized static pressure to pilot airspeed indicator, barometric encoding altimeter, vertical speed indicator and air data sensor. Pitot tube and static ports are equipped with electric heaters to provide all-weather capability.

An alternate static air source is provided by a pneumatic valve installed below the pilot instrument panel. Tubing and fittings are included to connect pneumatic valve to pilot static ports and instruments. With the toggle of the pneumatic valve set to PRI, static air is supplied to instruments from outside pilot static air ports. With the toggle set to ALTN, an alternate static air source (inside cabin air) is selected and outside static ports are shut off.





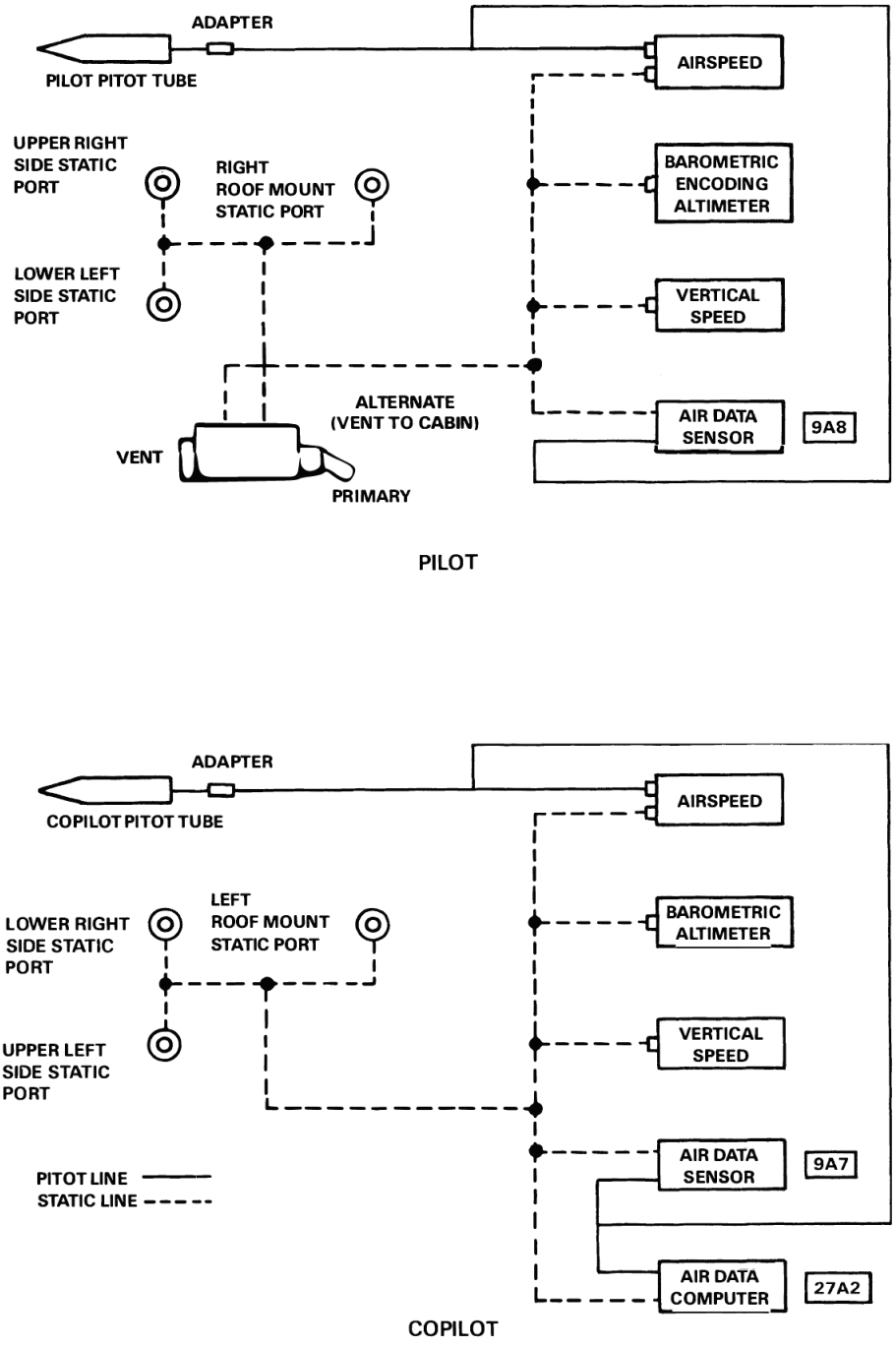
ICN-412-A-951101-B-97499-00006-A-001-01

Figure 1. Pitot-static System Schematic (Sheet 1 of 1)

Copilot Pitot-Static System - Description

The copilot pitot-static system (Figure 1) consists of left pitot tube, right and left static ports, left roof-mounted static port, copilot airspeed indicator, barometric altimeter, vertical airspeed indicator, air data sensor, air data computer, and appropriate tubing and hardware for interconnection of system components.

The left pitot tube supplies pitot pressure (impact air) directly to copilot airspeed indicator, air data sensor, and air data computer. Lower right, upper left and left roof-mounted static ports are connected together and supply equalized static pressure to copilot airspeed indicator, barometric altimeter, vertical speed indicator, air data sensor, and air data computer. Pitot tube and static ports are equipped with electric heaters to provide all-weather capability.



ICN-412-A-951101-B-97499-00006-A-001-01

Figure 1. Pitot-static System Schematic (Sheet 1 of 1)

Navigation Instruments - General Description

Navigation instruments are electrically powered, except for the standby compass.

Intentionally blank page

Electronic Horizontal Situation Indicator (EHSI) - Description

Refer to Honeywell, SPZ-7600 Integrated Flight Control System Bell 412HP System Maintenance Manual, Publication Number A09-1169-024 for component description and operation.

The EHSI provides compass displays and a map-like display of the helicopter position relative to VOR radials, localizer, glideslope beam, or FMS waypoints.

The EHSI (Figure 1) provides the following displays and annunciators:

- Course select/desired
- Drift angle bug
- FMS approach message
- Heading source annunciator
- Heading select bug and heading select display
- Navigation source annunciators
- Distance display
- Preselected Nav source annunciator
- Course preselect pointer
- Compass card and fore and aft lubber lines
- Glideslope deviation pointer
- Groundspeed, time-to-go, or elapsed time display
- Bearing pointers and source annunciators
- Reciprocal course pointer
- Aircraft symbol
- Course deviation bar
- Weather radar mode and tilt annunciator
- To-From arrow
- Course select pointer
- Compass sync annunciator.

In addition to the standard displays and annunciators, the EHSI also displays the following amber caution and failure annunciations (Figure 2):

- Waypoint alert annunciator
- Same heading or navigation source
- DME hold annunciator
- Weather radar (WX) failure
- Displays failures

  - amber dashes replace the digital display for the following systems when they fail.
- - Groundspeed (GSPD)
- - Time-To-Go (TTG)
- - Elapsed Time (ET)
- - Distance Measuring Equipment (DME)
- - Heading Select (HDG)
- - Course Select (CRS)
- - Desired Track (DTRK).

The EHSI also displays the following red failure annunciations (Figure 3):

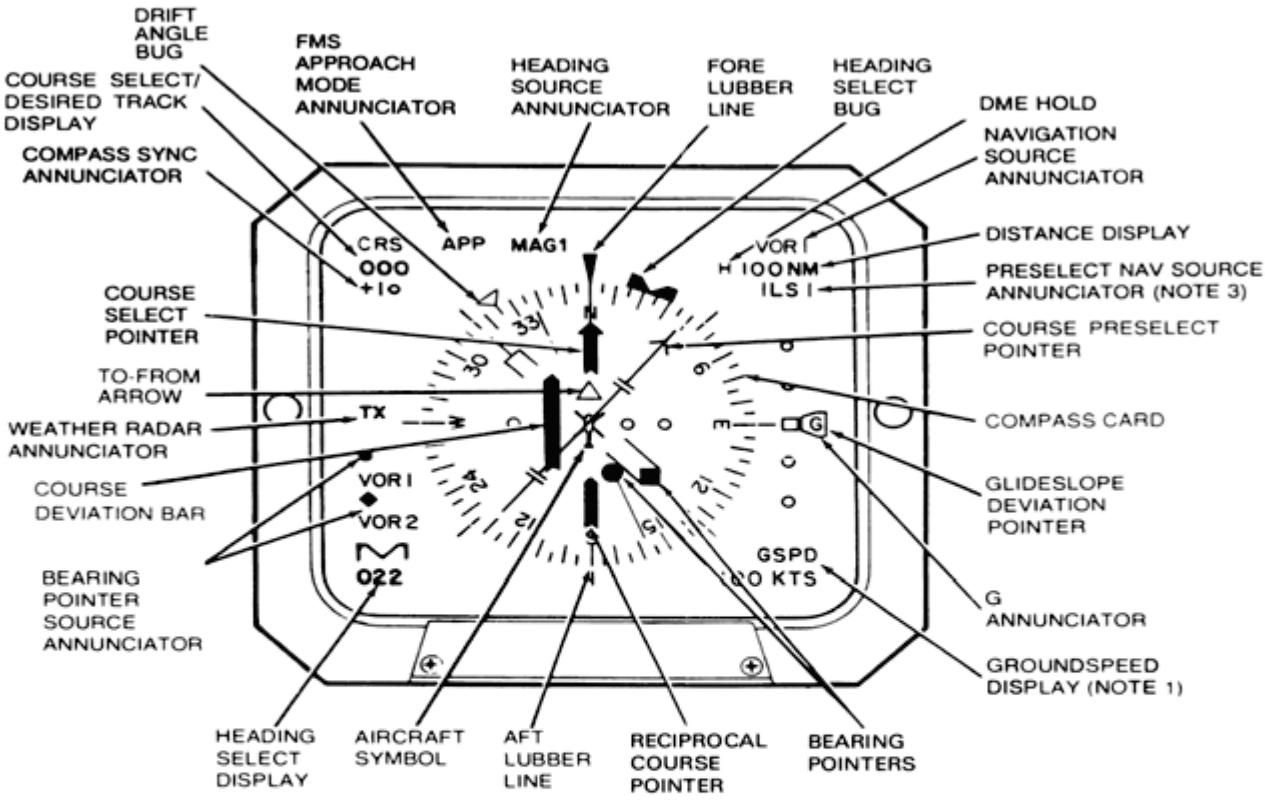
- Heading failure
- Course deviation
- Glideslope deviation.

Failure of the heading system is indicated by the removal of the bearing pointers, TO/FROM arrow, select course pointer, course deviation bar, and course scale. The digital select course and digital heading select readouts are dashed and a red HDG FAIL is displayed. The heading source annunciator changes to a white HDG 1 through HDG 4 to include which heading source has failed.

A failure of the course deviation, or glideslope deviation results in the removal of the deviation pointer, and a red X drawn through the scale.

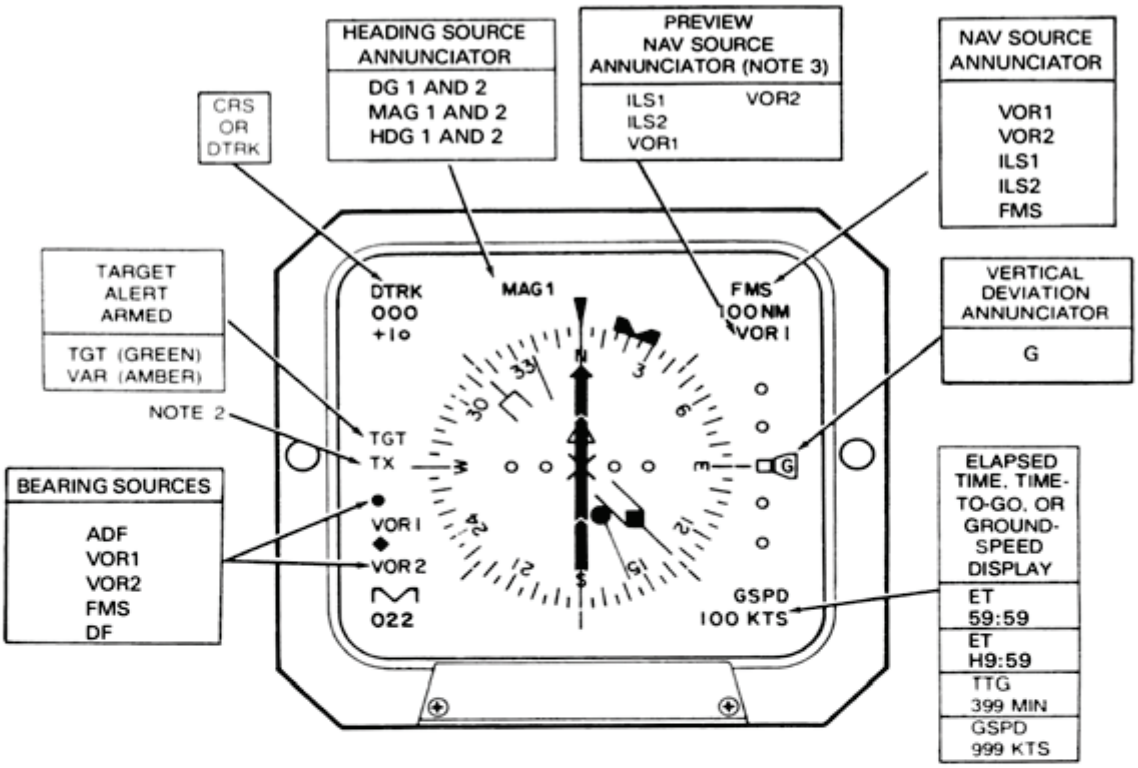
An internal failure of the symbol generator will cause the EHSI to go blank. A failure of the input/processor results in a red X centered on the display, with annunciation SG FAIL.

For additional information on EHSI displays/annunciations and operation refer to Honeywell, SPZ-7600 Integrated Flight Control System Bell 412 HP System Maintenance Manual, Publication Number A09-1169-024.



ICN-412-A-951201-B-97499-00001-A-001-01

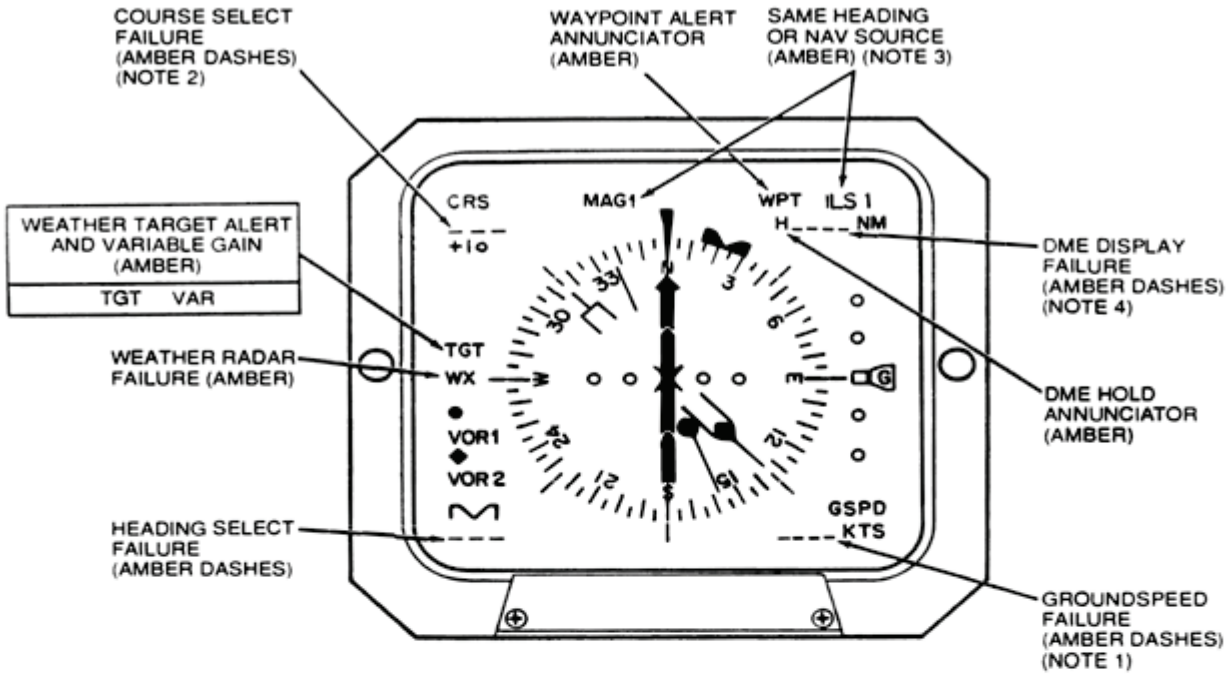
Figure 1. EHSI Full Compass Displays and Annunciators (Sheet 1 of 2)



NOTES:
1. TIME-TO-GO AND ELAPSED TIME ARE ALSO DISPLAYED AT THIS LOCATION.
2. TX APPEARS WHENEVER THE WEATHER RADAR IS TRANSMITTING AND WX IS NOT SELECTED FOR DISPLAY.
3. DME IDENT DISPLAYED WHEN SELECTED AS NAV SOURCE.

ICN-412-A-951201-B-97499-00002-A-001-01

Figure 1. EHSI Full Compass Displays and Annunciators (Sheet 2 of 2)

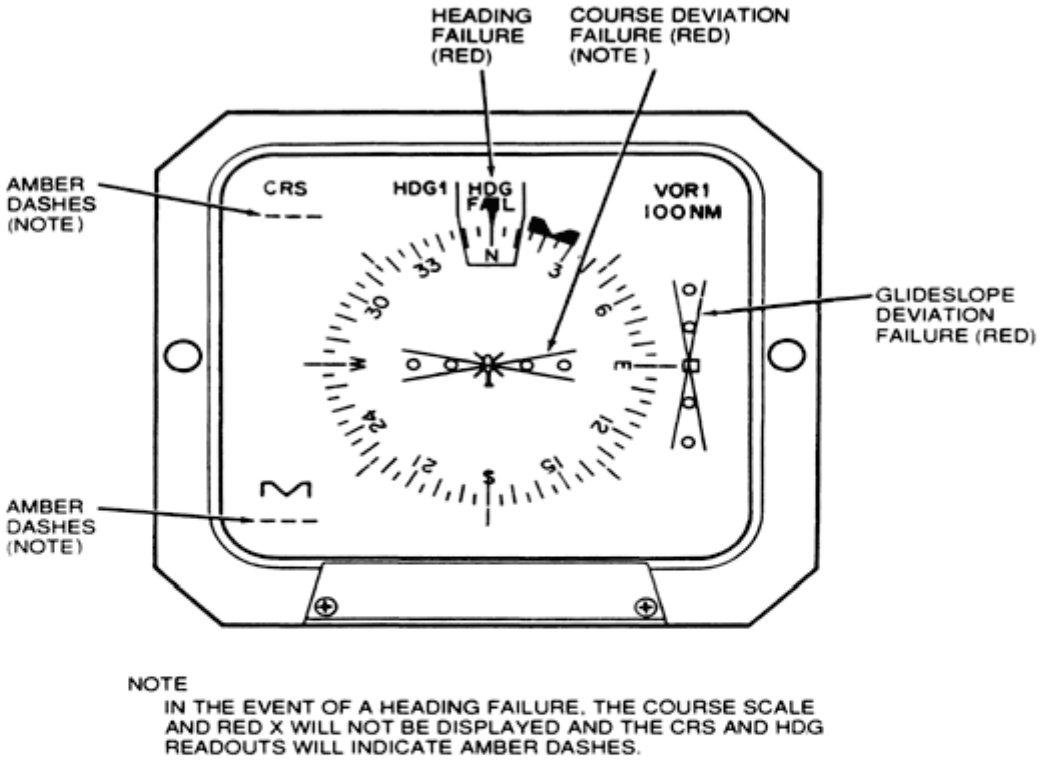


NOTES:

1. TTG AND ET FAILURES ARE SIMILAR
2. DTRK FAILURE IS SIMILAR.
3. IF BOTH THE PILOT AND THE COPILOT SELECT THE CROSS-SIDE NAV SOURCE, THE NAV ANNUNCIATORS ARE AMBER.
4. AN AMBER N/A IS DISPLAYED WHEN SELECTED LRN DISTANCE IS NOT AVAILABLE.
5. THIS DIAGRAM IS FOR ILLUSTRATION PURPOSES ONLY. IT DOES NOT NECESSARILY REPRESENT AN ACTUAL PRESENTATION.

ICN-412-A-951201-B-97499-00003-A-001-01

Figure 2. EHSI Full Compass Cautions and Failure Annunciations (Amber) (Sheet 1 of 1)



NOTE

IN THE EVENT OF A HEADING FAILURE, THE COURSE SCALE AND RED X WILL NOT BE DISPLAYED AND THE CRS AND HDG READOUTS WILL INDICATE AMBER DASHES.

ICN-412-A-951201-B-97499-00004-A-001-01

Figure 3. EHSI Full Compass Display Annunciations (Red) (Sheet 1 of 1)

EHSI - Maintenance Procedures

Procedure

1. Refer to 412-A-95-00-01-00A-042A-A / 01790 for maintenance procedures of panel mounted instrument.

Intentionally blank page

Digital Clock - Description

The clock has a liquid crystal display, lighting system, and ultraviolet filter to provide uniform night illumination and protection from sunlight. The clocks two push buttons control the four displays

- GMT = Greenwich Mean Time (24 hour format)
- LT = Local Time (12 or 24 hour format)
- ET = Elapsed Time Counter (1 second to 99 hours and 59 minutes range)
- ET = Elapsed Time Countdown Timer (1 second to 1 hour range)

Intentionally blank page

Digital Clock - Functional Check

Procedure

1. Press and hold down SELECT button for 3 seconds. The display will read '88.88' and GMT, LT, and ET will be illuminated.

Intentionally blank page

Digital Clock - Operational Check

Procedure

1. SELECT Button. The SELECT button sequentially pressed will illuminate GMT (Greenwich mean time), LT (local time), and ET (elapsed time).

2. CONTROL Button. The CONTROL button sequentially pressed (after pressing the SELECT and CONTROL button simultaneously) will change the display. Pressing the button momentarily will start and reset ET (elapsed time).

Intentionally blank page

GMT (Greenwich Mean Time) Setting - Functional Check

Procedure

1. Press SELECT button momentarily until GMT is illuminated.
2. Press SELECT and CONTROL buttons once simultaneously to enter set mode (tens of hours digit starts flashing).
3. Press CONTROL button once to display first digit to be set.
4. Press CONTROL button momentarily to set digit. Each button push advances the digit.
5. Press SELECT button once to select the next digit to be set.
6. Perform Step 3., Step 4., and Step 5. above until remaining digits are set.
7. Press SELECT button once to exit the set mode (flashing display indicates clock is running).

Intentionally blank page

LT (Local Time) Setting - Functional Check

Procedure

1. Press SELECT button momentarily until LT is illuminated.
2. Press SELECT and CONTROL buttons once simultaneously to enter set mode (tens of hours digit starts flashing).
3. Press CONTROL button once to display first digit to be set.
4. Press CONTROL button momentarily to set digit. Each button push advances the digit.
5. Press SELECT button once to select the next digit to be set.

NOTE

Minutes are synchronized with GMT and can not be changed in the LT (local time) setting mode. If minutes require changing follow GMT procedures for minutes.

6. Perform Step 3., Step 4., and Step 5. above until remaining digits are set.
7. Press SELECT button once to exit the set mode (flashing display indicates clock is running).

Intentionally blank page

ET (Elapsed Time) Count Up Setting - Functional Check

Procedure

1. Press SELECT button momentarily until ET is illuminated.
2. Press CONTROL button once to start count up. Counter will start counting in minutes and seconds up to 59 minutes and 59 seconds. Then the counter will start counting in hours and minutes up to 99 hours and 59 minutes.
3. Press CONTROL button once to reset the ET count up to zero.

Intentionally blank page

ET (Elapsed Time) Countdown Setting - Functional Check

Procedure

1. Press SELECT button momentarily until ET is illuminated.
2. Press SELECT and CONTROL buttons once simultaneously to enter set mode.
3. Press CONTROL button once to display first digit to be set.
4. Press CONTROL button momentarily to set digit. Each button push advances the digit up from 1 second to 59 minutes and 59 seconds.
5. Press SELECT button once to select the next digit to be set.
6. Perform Step 3., Step 4., and Step 5. above until remaining digits are set.
7. Press SELECT button once to exit the set mode.
8. Press CONTROL button once to start the countdown. Display will flash and an external alarm will be activated when the counter reaches zero.
9. Press SELECT or CONTROL button to deactivate the flashing display and external alarm.

Intentionally blank page

Electrical System - Description

The supplemental electrical equipment installed in the Search and Rescue (SAR) configured model 412 helicopter consists of the Dual Digital Automatic Flight Control System (DDAFCS) , and the Electronic Flight Instrument System (EFIS) .

Components that are part of the electrical configuration on SAR helicopters are illustrated on Figure 1 through Figure 7. Refer to 412-A-98-01-01-00A-051A-A / 02974 for schematic and wiring diagrams. For code designation correlation, refer to Electrical Equipment List (Table 1).

Table 1. Electrical Equipment List

REF DES	NOMENCLATURE
10DS1	LIGHT, WINDSHIELD HEATER
10DS1P1	CONN, WINDSHIELD HEATER LIGHT
22WT12	TERMINAL BOARD, GROUND
22WT14	TERMINAL BOARD, GROUND
22WT8	TERMINAL BOARD, GROUND
2300J1	CONN
24A1	CONTROL PANEL, MISCELLANEOUS
24A1P1	CONN, MISCELLANEOUS CONTROL PANEL
24A3	CONTROL UNIT, HOIST PENDANT
24A3J2	CONN, HOIST PENDANT CONTROL
24A3J3	CONN, HOIST PENDANT CONTROL
24A3P2	CONN, HOIST PENDANT CONTROL
24A3P3	CONN, HOIST PENDANT CONTROL
24A3S1	SWITCH, BOOM
24A3S3	SWITCH, HOIST UP-DOWN
24P1	CONN, HOIST REMOTE BEEP - CABIN ROOF DISCONNECT
27A2	COMPARATOR, AIR DATA
27A2P1	CONNECTOR, AIR DATA COMPARATOR
27TB1	TERMINAL BOARD, FLIGHT DATA RECORDER
3400CB3	CIRCUIT BREAKER, HSI
3400CP30	NAV CONTROL
3400CP30P1	CONN, NAV CONTROL
3400P13B	CONN
3400TB30	TERMINAL BOARD
3400WT2	TERMINAL BOARD, GROUND
3400WT8	TERMINAL BOARD, GROUND
3407CP1	PANEL, COPILOT COMPASS CONTROL
3407CP1P1	CONN, COPILOT COMPASS CONTROL PANEL
3407CP2	PANEL, PILOT COMPASS CONTROL
3407CP2P1	CONN, PILOT COMPASS CONTROL PANEL
3407F1	FUSE, VERTICAL GYRO
3407F2	FUSE, VERTICAL GYRO
3407L1	COMPENSATOR, DUAL REMOTE
3407L1P1	CONN, DUAL REMOTE COMPENSATOR
3407L1P2	CONN, DUAL REMOTE COMPENSATOR
3407MP1	SENSOR, COPILOT 3-AXIS REFERENCE

Table 1. Electrical Equipment List (continued)

REF DES	NOMENCLATURE
3407MP1P1	CONN, COPILOT 3-AXIS REFERENCE SENSOR
3407MP1P2	CONN, COPILOT 3-AXIS REFERENCE SENSOR
3407MP2	SENSOR, PILOT 3-AXIS REFERENCE
3407MP2P1	CONN, PILOT 3-AXIS REFERENCE SENSOR
3407MP2P2	CONN, PILOT 3-AXIS REFERENCE SENSOR
3440A1	SYMBOL GENERATOR, COPILOT
3440A10	FAN, EFIS
3440A10	TERMINATOR, RESISTIVE LOAD
3440A10J1	CONN, RESISTIVE LOAD TERMINATOR
3440A11	FAN, EFIS
3440A11	TERMINATOR, RESISTIVE LOAD
3440A11J1	CONN, RESISTIVE LOAD TERMINATOR
3440A12	MOD, COMM
3440A12	TERMINATOR, RESISTIVE LOAD
3440A12J1	CONN, RESISTIVE LOAD TERMINATOR
3440A13	MOD, COMM
3440A1P1A	CONN, COPILOT SYMBOL GENERATOR
3440A1P1B	CONN, COPILOT SYMBOL GENERATOR
3440A1P2A	CONN, COPILOT SYMBOL GENERATOR
3440A1P2B	CONN, COPILOT SYMBOL GENERATOR
3440A2	SYMBOL GENERATOR, PILOT
3440A2P1A	CONN, PILOT SYMBOL GENERATOR
3440A2P1B	CONN, PILOT SYMBOL GENERATOR
3440A2P2A	CONN, PILOT SYMBOL GENERATOR
3440A2P2B	CONN, PILOT SYMBOL GENERATOR
3440A3	SWITCH PANEL, COPILOT EFIS AUXILIARY
3440A3P1	CONN, COPILOT EFIS AUXILIARY SWITCH PANEL
3440A4	SWITCH PANEL, PILOT EFIS AUXILIARY
3440A4P1	CONN, PILOT EFIS AUXILIARY SWITCH PANEL
3440A5	TERMINATOR, RESISTIVE LOAD
3440A5J1	CONN, RESISTIVE LOAD TERMINATOR
3440A6	TERMINATOR, RESISTIVE LOAD
3440A6J1	CONN, RESISTIVE LOAD TERMINATOR
3440A7	TERMINAL, RESISTIVE LOAD
3440A7J1	CONN, RESISTIVE LOAD TERMINATOR
3440A8	TERMINATOR, RESISTIVE LOAD
3440A8J1	CONN, RESISTIVE LOAD TERMINATOR
3440A9	TERMINATOR, RESISTIVE LOAD
3440A9J1	CONN, RESISTIVE LOAD TERMINAL
3440CB1	CIRCUIT BREAKER, EFIS NO. 1 ADI
3440CB10	CIRCUIT BREAKER, EFIS BLOWER
3440CB2	CIRCUIT BREAKER, EFIS NO. 2 ADI
3440CB3	CIRCUIT BREAKER, EFIS NO. 1 HSI

Table 1. Electrical Equipment List (continued)

REF DES	NOMENCLATURE
3440CB4	CIRCUIT BREAKER, EFIS NO. 2 HSI
3440CB5	CIRCUIT BREAKER, EFIS NO. 1 SG/DC
3440CB6	CIRCUIT BREAKER, EFIS NO. 2 SG/DC
3440CB7	CIRCUIT BREAKER, 26V NO. 1
3440CB8	CIRCUIT BREAKER, 26V NO. 2
3440CB9	CIRCUIT BREAKER, EFIS BLOWER
3440CP1	INSTRUMENT CONTROLLER
3440CP1P1	CONN, INSTRUMENT CONTROLLER
3440CP1P2	CONN, INSTRUMENT CONTROLLER
3440CP3	DISPLAY CONTROLLER, COPILOT
3440CP3P1	CONN, COPILOT DISPLAY CONTROLLER
3440CP4	DISPLAY CONTROLLER, PILOT
3440CP4P1	CONN, PILOT DISPLAY CONTROLLER
3440CR1	DIODE
3440DS1	EADI, COPILOT
3440DS1P1	CONN, COPILOT EADI
3440DS10	SWITCH, EFIS MASTER
3440DS2	PILOT, EADI
3440DS2P1	CONN, PILOT EADI
3440DS3	EHSI, COPILOT
3440DS3P1	CONN, COPILOT EHSI
3440DS4	EHSI, PILOT
3440DS4P1	CONN, PILOT EHSI
3440J1	CONN
3440J10	CONN, TEE
3440J10C	CONN
3440J11	CONN, TEE
3440J11C	CONN
3440J12	CONN, TEE
3440J12C	CONN
3440J13	CONN, TEE
3440J13C	CONN
3440J14	CONN, TEE
3440J14C	CONN
3440J15	CONN, TEE
3440J15C	CONN
3440J16	CONN, TEE
3440J16C	CONN
3440J17	CONN, TEE
3440J17C	CONN
3440J2	CONN
3440J3	CONN
3440J4	CONN

Table 1. Electrical Equipment List (continued)

REF DES	NOMENCLATURE
3440K1	RELAY
3440K2	RELAY
3440K3	RELAY
3440K4	RELAY
3440K5	RELAY
3440K6	RELAY
3440P1	CONN
3440P10A	CONN
3440P10B	CONN
3440P11A	CONN
3440P11B	CONN
3440P12A	CONN
3440P12B	CONN
3440P13A	CONN
3440P14A	CONN
3440P14B	CONN
3440P15A	CONN
3440P15B	CONN
3440P16A	CONN
3440P16B	CONN
3440P17A	CONN
3440P17B	CONN
3440P2	CONN
3440P3	CONN
3440P4	CONN
3440TB1	TERMINAL BOARD
3440TB2	TERMINAL BOARD
3440TB3	TERMINAL BOARD
3440WT1	TERMINAL BOARD, GROUND
3440WT2	TERMINAL BOARD, GROUND
4A2	COLLECTIVE STICK, PILOT
4A2J1	CONN, PILOT COLLECTIVE STICK
4M5	CLOCK, PILOT DIGITAL
4M5P1	CONN, PILOT DIGITAL CLOCK
5A2	COLLECTIVE STICK, COPILOT
5A2J1	CONN, COPILOT COLLECTIVE STICK
5M5	CLOCK, COPILOT DIGITAL
5M5P1	CONN, COPILOT DIGITAL CLOCK
8CR52	DIODE
8CR53	DIODE
8A1	PANEL, CAUTION
8A1P1	CONN
8A1P2	CONN

Table 1. Electrical Equipment List (continued)

REF DES	NOMENCLATURE
8K20	RELAY, SWITCHING
8Q20	TRANSISTOR
8Q21	TRANSISTOR
8TB1	TERMINAL BOARD, LIGHTING
8TB2	TERMINAL BOARD, RPM WARNING
8TB3	TERMINAL BOARD, 5 VOLT LIGHT POWER
9A1	COMPUTER, COPILOT DFC NO. 1
9A10	SWITCH, PITCH DETENT
9A10J1	CONN, PITCH DETENT SWITCH
9A10P1	CONN, PITCH DETENT SWITCH
9A11	INDICATOR PANEL AFCS ACTUATOR
9A11P1	CONN, AFCS ACTUATOR PANEL
9A1P1A	CONN, COPILOT DFC COMPUTER NO. 1
9A1P1B	CONN, COPILOT DFC COMPUTER NO. 1
9A1P1C	CONN, COPILOT DFC COMPUTER NO. 1
9A2	COMPUTER, PILOT DFC NO. 2
9A2P1A	CONN, PILOT DFC COMPUTER NO. 2
9A2P1B	CONN, PILOT DFC COMPUTER NO. 2
9A2P1C	CONN, PILOT DFC COMPUTER NO. 2
9A3	MODE SELECTOR, COPILOT
9A3P1	CONN, COPILOT MODE SELECTOR
9A30	CONTROL BOX, AFCS RELAY
9A30P1	CONN, AFCS RELAY CONTROL BOX
9A30P1	CONN, AFCS RELAY CONTROL BOX
9A31	GYRO NO. 1 YAW RATE
9A31P1	CONN, YAW RATE GYRO NO. 1
9A32	GYRO NO. 2, YAW RATE
9A32P1	CONN, YAW RATE GYRO NO. 2
9A33	SWITCH, YAW DETENT
9A33J1	CONN, YAW DETENT SWITCH
9A33P1	CONN, YAW DETENT SWITCH
9A34	SWITCH, COLLECTIVE DETENT
9A34J1	CONN, COLLECTIVE DETENT SWITCH
9A34P1	CONN, COLLECTIVE DETENT SWITCH
9A35	GYRO ASSEMBLY, YAW RATE
9A35J1	CONN, YAW RATE GYRO ASSEMBLY
9A35J2	CONN, YAW RATE GYRO ASSEMBLY
9A35P1	CONN, YAW RATE GYRO ASSEMBLY
9A35P2	CONN, YAW RATE GYRO ASSEMBLY
9A3P1	CONN, COPILOT MODE SELECTOR
9A3P1	CONN, COPILOT MODE SELECTOR
9A4	MODE SELECTOR, PILOT
9A4P1	CONN, PILOT MODE SELECTOR

Table 1. Electrical Equipment List (continued)

REF DES	NOMENCLATURE
9A5	CONTROLLER, AUTOPILOT
9A5P1	CONN, AUTOPILOT CONTROLLER
9A5P2	CONN, AUTOPILOT CONTROLLER
9A6	PRESELECT, ALTITUDE
9A6P1	CONN, ALTITUDE PRESELECT
9A7	SENSOR, AIR DATA NO. 1
9A7P1	CONN, AIR DATA SENSOR NO. 1
9A8	SENSOR, AIR DATA NO. 2
9A8P1	CONN, AIR DATA SENSOR NO. 2
9A9	SWITCH, ROLL DETENT
9A9J1	CONN, ROLL DETENT SWITCH
9A9P1	CONN, ROLL DETENT SWITCH
9B1	ACTUATOR, PILOT LINEAR NO. 1
9B1J1	CONN, PITCH LINEAR ACTUATOR NO. 1
9B2	ACTUATOR, ROLL LINEAR NO. 1
9B2J1	CONN, ROLL LINEAR ACTUATOR NO. 1
9B3	ACTUATOR, YAW LINEAR NO. 1
9B30	ACTUATOR, YAW TRIM
9B30P1	CONN, YAW TRIM ACTUATOR
9B31	ACTUATOR, COLLECTIVE TRIM
9B31P1	CONN, COLLECTIVE TRIM ACTUATOR
9B32	LINEAR ACTUATOR, YAW NO. 2
9B32J1	CONN, YAW LINEAR ACTUATOR NO. 2
9B3J1	CONN, YAW LINEAR ACTUATOR NO. 1
9B4	LINEAR ACTUATOR, PITCH NO. 2
9B4J1	CONN, PITCH LINEAR ACTUATOR NO. 2
9B5	LINEAR ACTUATOR, ROLL NO. 2
9B5J1	CONN, ROLL LINEAR ACTUATOR NO. 2
9B6	ACTUATOR, PITCH TRIM
9B6P1	CONN, PITCH TRIM ACTUATOR
9B7	ACTUATOR, YAW TRIM
9B7P1	CONN, YAW TRIM ACTUATOR
9CB2	CIRCUIT BREAKER, AFCS NO. 2
9CB3	CIRCUIT BREAKER, FORCE TRIM
9CB30	CIRCUIT BREAKER, RATE GYRO NO. 2
9CB31	CIRCUIT BREAKER, RATE GYRO NO. 1
9CB32	CIRCUIT BREAKER, AFCS NO. 1 ADD/CAUTION
9CB33	CIRCUIT BREAKER, AFCS NO. 2 CAUT
9CB4	CIRCUIT BREAKER, AFCS 26V NO. 1
9CB7	CIRCUIT BREAKER, AFCS NO. 2
9CR30	DIODE
9CR31	DIODE
9CR32	DIODE

Table 1. Electrical Equipment List (continued)

REF DES	NOMENCLATURE
9CR33	DIODE
9CR34	DIODE
9CR35	DIODE
9CR36	DIODE
9CR37	DIODE
9CR38	DIODE
9CR39	DIODE
9CR40	DIODE
9CR41	DIODE
9CR42	DIODE
9CR43	DIODE
9CR44	DIODE
9CR45	DIODE
9CR46	DIODE
9CR47	DIODE
9CR48	DIODE
9CR49	DIODE
9CR50	DIODE
9CR51	DIODE
9CR52	DIODE
9CR53	DIODE
9CR54	DIODE
9CR55	DIODE
9CR56	DIODE
9CR57	DIODE
9CR58	DIODE
9CR59	DIODE
9DS30	LIGHT, COPILOT AUTOPILOT
9DS30P1	CONN, COPILOT AUTOPILOT LIGHT
9DS31	LIGHT COPILOT PITCH, ROLL, YAW
9DS31P1	CONN, COPILOT PITCH, ROLL, YAW LIGHT
9DS32	LIGHT, COPILOT FORCE TRIM DECOUPLER
9DS32P1	CONN, COPILOT FORCE TRIM DECOUPLER LIGHT
9DS33	LIGHT, PILOT AUTOPILOT
9DS33P1	CONN, PILOT AUTOPILOT LIGHT
9DS34	LIGHT, PILOT PITCH, ROLL, YAW
9DS34P1	CONN, PILOT PITCH, ROLL, YAW LIGHT
9DS35	LIGHT, PILOT FORCE TRIM DECOUPLER
9DS35P1	CONN, PILOT FORCE TRIM DECOUPLER LIGHT
9DS36	LIGHT, MOT/ON
9DS36P1	CONN, MOT/ON LIGHT
9DS37	LIGHT, CLIMB/ON
9DS37P1	CONN, CLIMB/ON LIGHT

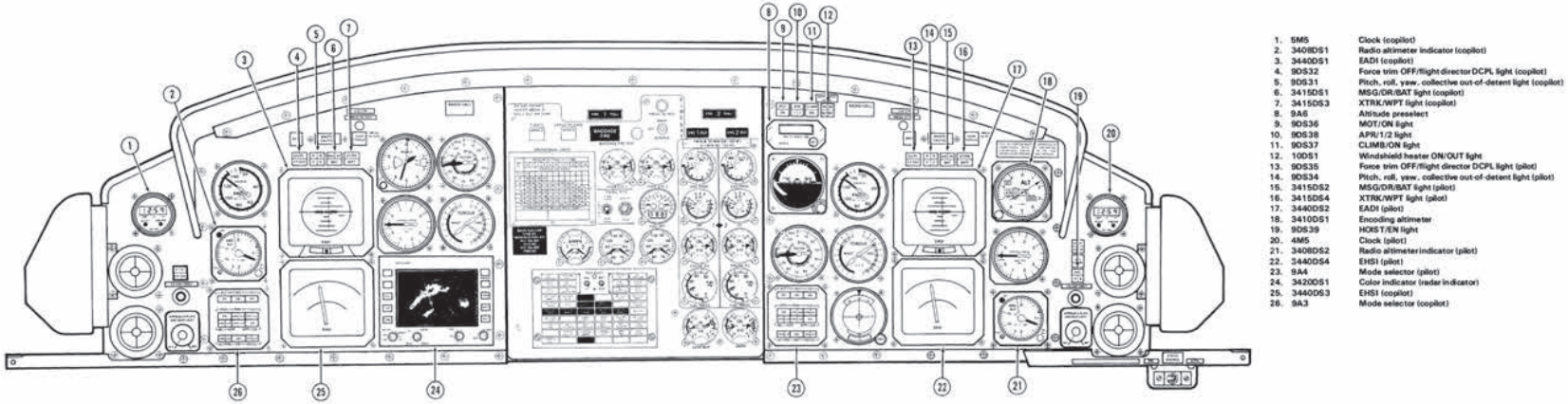
Table 1. Electrical Equipment List (continued)

REF DES	NOMENCLATURE
9DS38	LIGHT, APR/1/2
9DS38P1	CONN, APR/1/2 LIGHT
9DS39	LIGHT, HOIST/EN
9DS39P1	CONN, HOIST/EN LIGHT
9J30	CONN
9J31	CONN
9K2	RELAY, TRIM FAIL ANNUN
9K20	RELAY
9K21	RELAY
9K22	RELAY
9K23	RELAY
9K24	RELAY
9K25	RELAY
9K3	RELAY, CLTV WARN ANNUN
9MT1	DUAL CMT, COPILOT CYCLIC PITCH
9MT1J1	CONN, COPILOT CYCLIC PITCH DUAL CMT
9MT2	DUAL CMT, COPILOT CYCLIC ROLL
9MT2J1	CONN, COPILOT CYCLIC ROLL DUAL CMT
9MT3	DUAL CMT, COPILOT PEDALS YAW
9MT30	DUAL CMT, PILOT COLLECTIVE
9MT30J1	CONN, PILOT COLLECTIVE DUAL CMT
9MT31	DUAL CMT, PILOT PEDALS YAW
9MT31J1	CONN, PILOT PEDALS YAW DUAL CMT
9MT32	ACCELEROMETER, LONGITUDE NO. 1
9MT32P1	CONN, LONGITUDE NO. 1 ACCELEROMETER
9MT33	ACCELEROMETER, LONGITUDE
9MT33P1	CONN, LONGITUDE ACCELEROMETER
9MT34	ACCELEROMETER, LATITUDE NO. 1
9MT34P1	CONN, LATITUDE NO. 1 ACCELEROMETER
9MT35	ACCELEROMETER, LATITUDE
9MT35P1	CONN, LATITUDE ACCELEROMETER
9MT36	ACCELEROMETER, VERTICAL NO. 1
9MT36P1	CONN, VERTICAL NO. 1 ACCELEROMETER
9MT37	ACCELEROMETER, VERTICAL
9MT37P1	CONN, VERTICAL ACCELEROMETER
9MT3J1	CONN, COPILOT PEDALS YAW DUAL CMT
9MT4	DUAL CMT, PILOT PITCH CYCLIC
9MT4J1	CONN, PILOT PITCH CYCLIC DUAL CMT
9MT5	DUAL CMT, PILOT ROLL CYCLIC
9MT5J1	CONN, PILOT ROLL CYCLIC DUAL CMT
9MT6	DUAL CMT, COPILOT COLLECTIVE
9MT6J1	CONN, COPILOT COLLECTIVE DUAL CMT
9P30	CONN

Table 1. Electrical Equipment List (continued)

REF DES	NOMENCLATURE
9P31	CONN
9S34	SWITCH, READ EEPROM
9S35	SWITCH, ERASE EEPROM
9S36	SWITCH, HOIST REMOTE BEEP
9TB1	TERMINAL BOARD
9TB3	TERMINAL BOARD
9TB30	TERMINAL BOARD
9TB31	TERMINAL BLOCK
9TB32	TERMINAL BOARD
9TB33	TERMINAL BOARD
9TB34	TERMINAL BOARD
9TB4	TERMINAL BOARD
9TB6	TERMINAL BOARD
9WT30	TERMINAL BOARD, GROUND
9WT31	TERMINAL BOARD, GROUND
9WT32	TERMINAL BOARD, GROUND
9WT33	TERMINAL BOARD, GROUND
9WT34	TERMINAL BOARD, GROUND
9WT35	TERMINAL BOARD, GROUND
J14	CONN, GLARE SHIELD FLOOD LIGHT ASSEMBLY
P14	CONN, GLARE SHIELD FLOOD LIGHT ASSEMBLY

Intentionally blank page

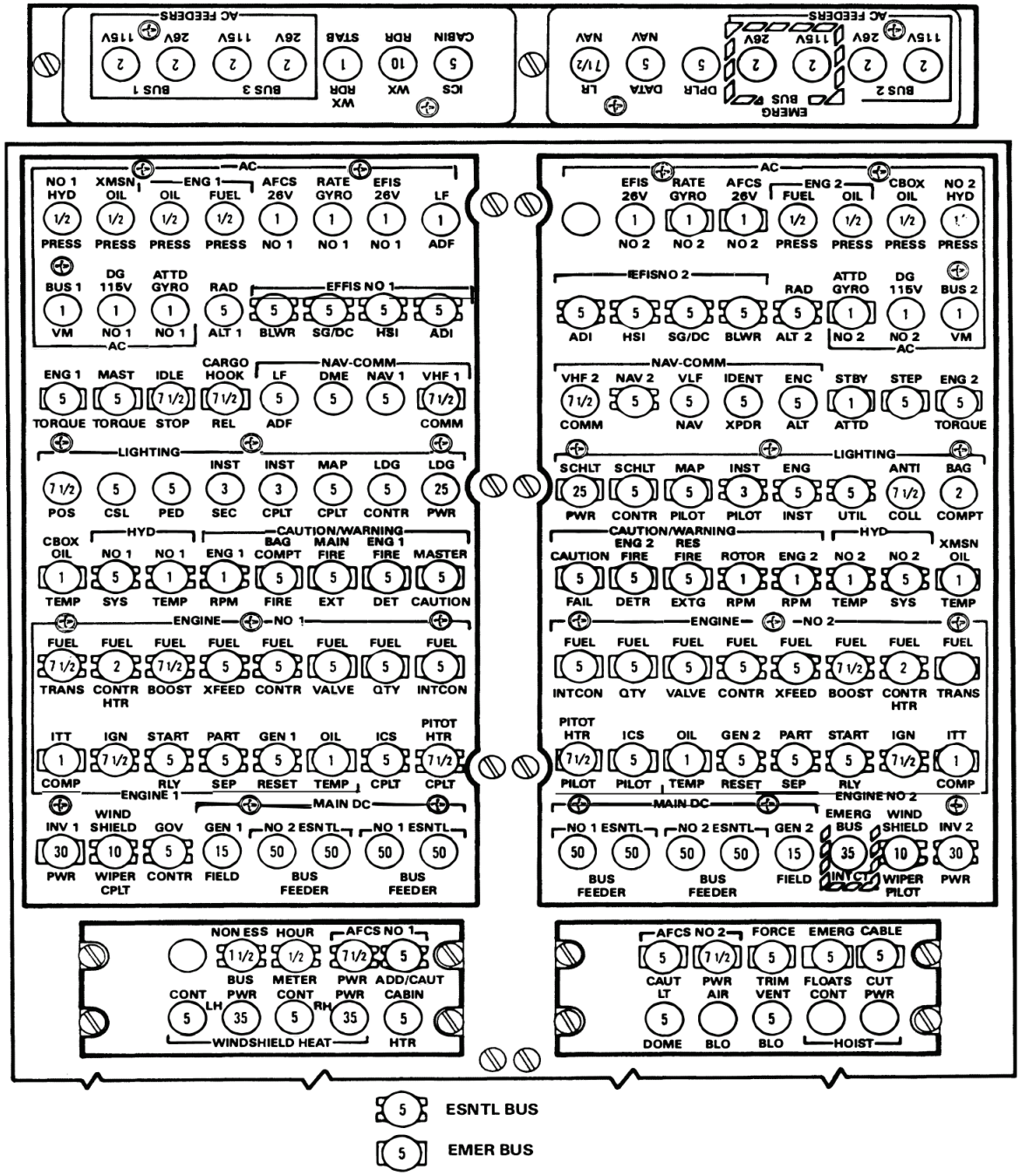


1.	5M5	Clock (copilot)
2.	3408DS1	Radio altimeter indicator (copilot)
3.	3440DS1	EADI (copilot)
4.	9DS32	Force trim OFF/flight director DCPL light (copilot)
5.	9DS31	Pitch, roll, yaw, collective out-of-detent light (copilot)
6.	3415DS1	MSG/DR/BAT light (copilot)
7.	3415DS3	XTRK/WPT light (copilot)
8.	9A6	Altitude preselect
9.	9DS36	MOT/ON light
10.	9DS38	APR/1/2 light
11.	9DS37	CLIMB/ON light
12.	10DS1	Windshield heater ON/OUT light
13.	9DS35	Force trim OFF/flight director DCPL light (pilot)
14.	9DS34	Pitch, roll, yaw, collective out-of-detent light (pilot)
15.	3415DS2	MSG/DR/BAT light (pilot)
16.	3415DS4	XTRK/WPT light (pilot)
17.	3440DS2	EADI (pilot)
18.	3410DS1	Encoding altimeter
19.	9DS39	HOIST/E/N light
20.	4M5	Clock (pilot)
21.	3408DS2	Radio altimeter indicator (pilot)
22.	3440DS4	EHSI (pilot)
23.	9A4	Mode selector (pilot)
24.	3420DS1	Color indicator (radar indicator)
25.	3440DS3	EHSI (copilot)
26.	9A3	Mode selector (copilot)

ICN-412-A-960001-B-97499-00001-A-001-01

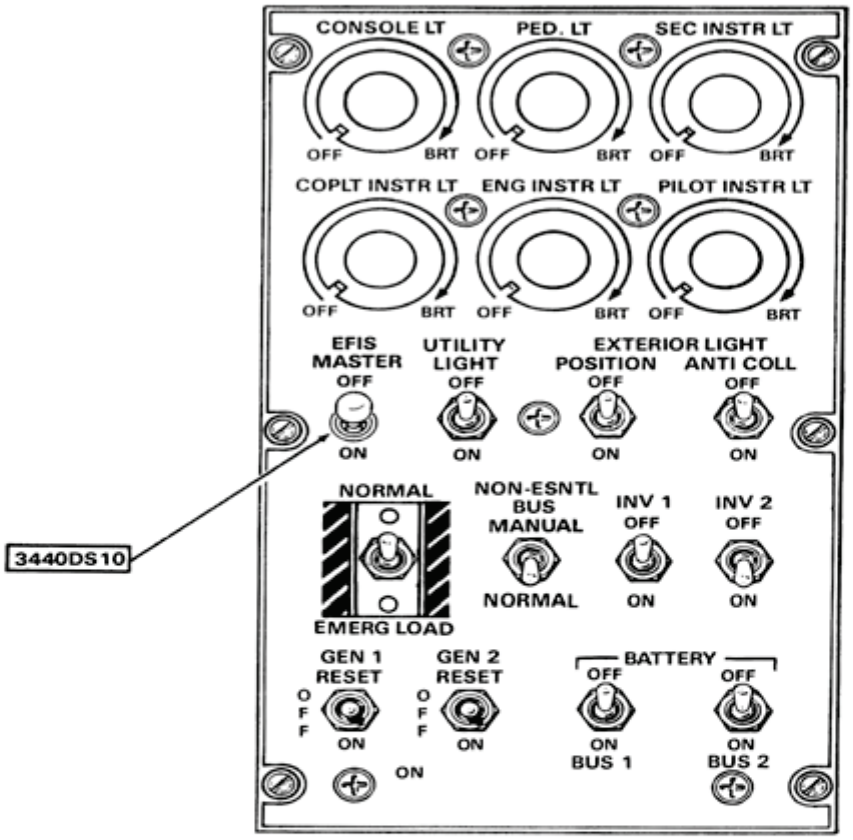
Figure 1. Instrument Panel (Sheet 1 of 1)

Intentionally blank page



ICN-412-A-960001-B-97499-00002-A-001-01

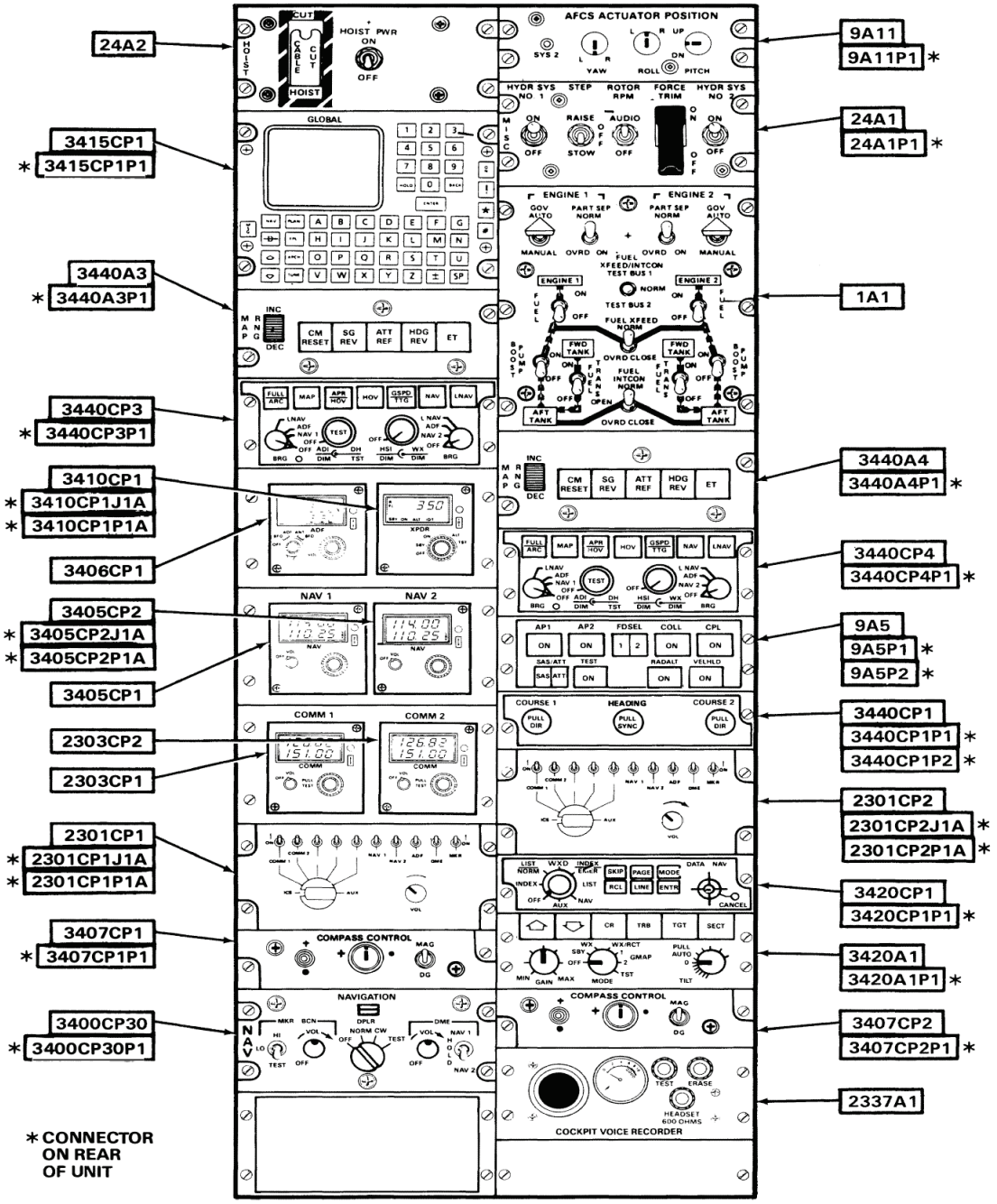
Figure 2. Console (Sheet 1 of 2)



OVERHEAD CONSOLE
PILOT SIDE

ICN-412-A-960001-B-97499-00003-A-001-01

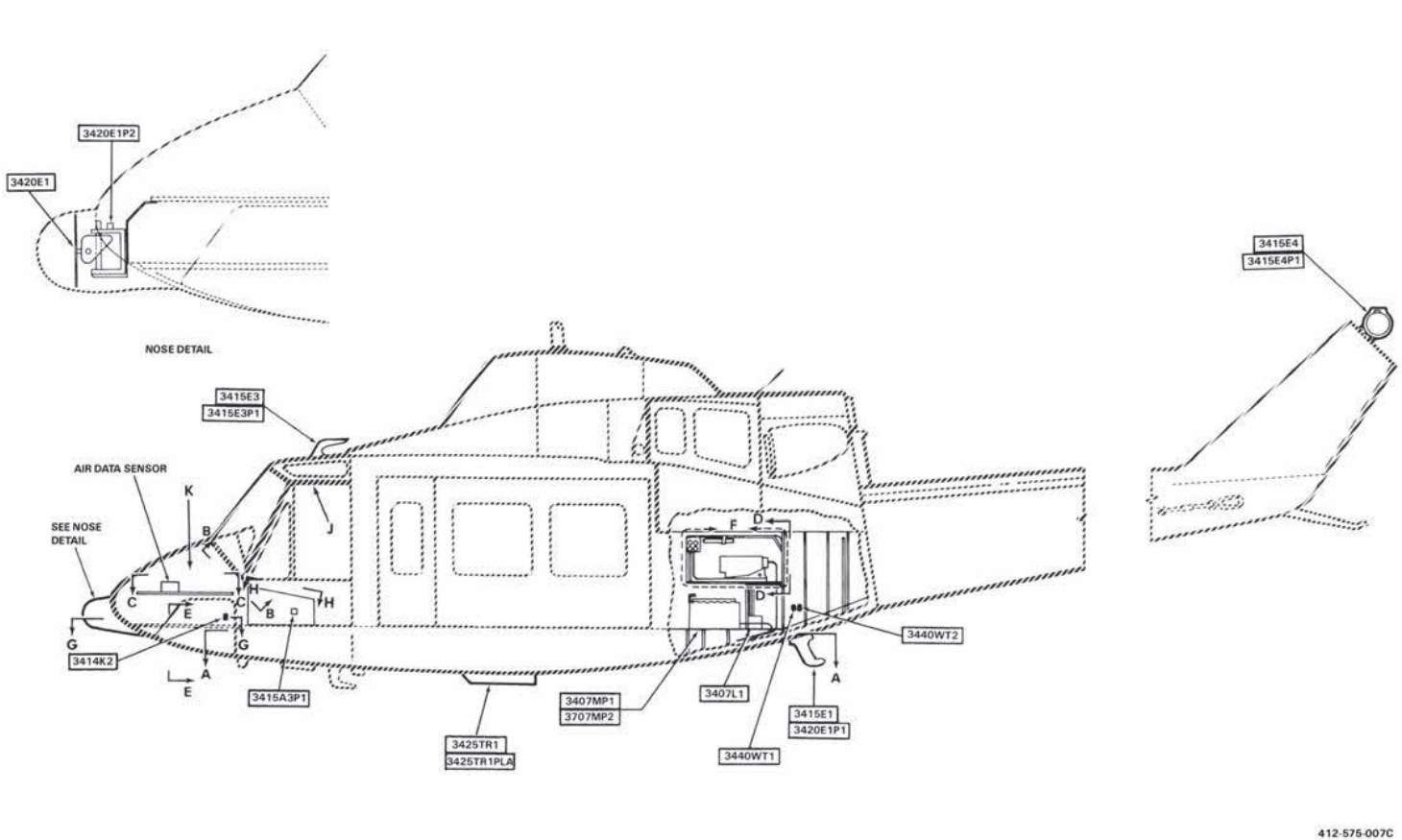
Figure 2. Console (Sheet 2 of 2)



ICN-412-A-960001-B-97499-00004-A-001-01

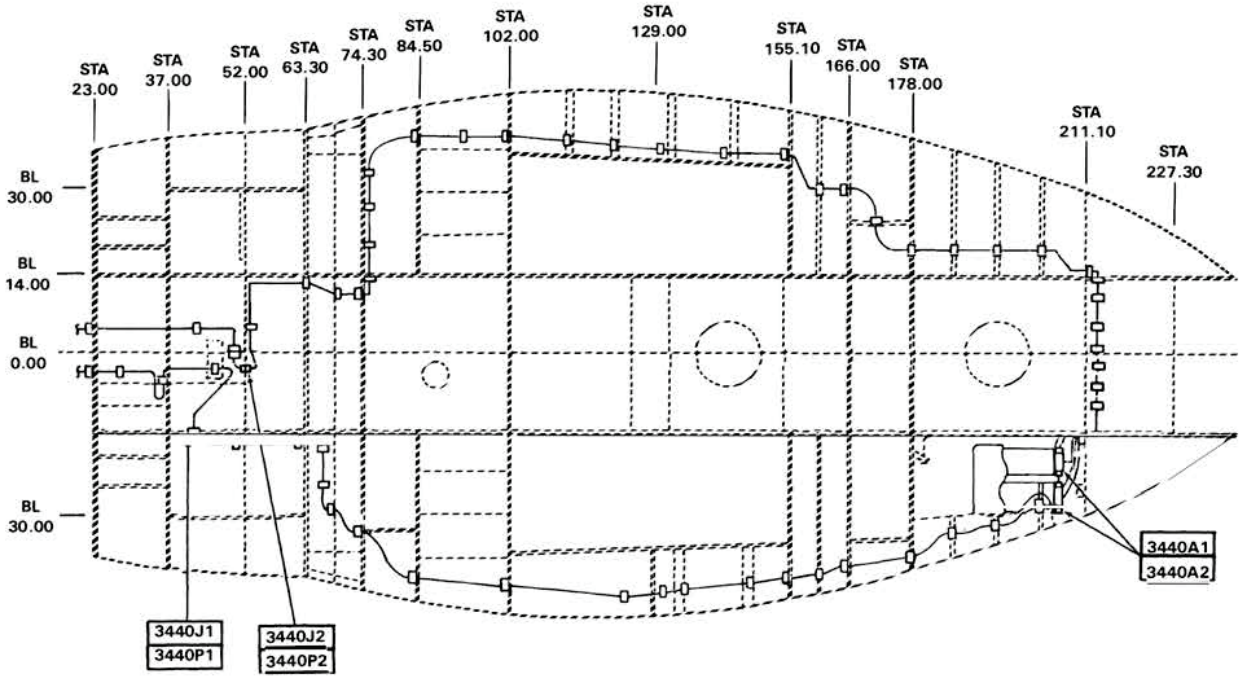
Figure 3. Pedestal (Sheet 1 of 1)

Intentionally blank page



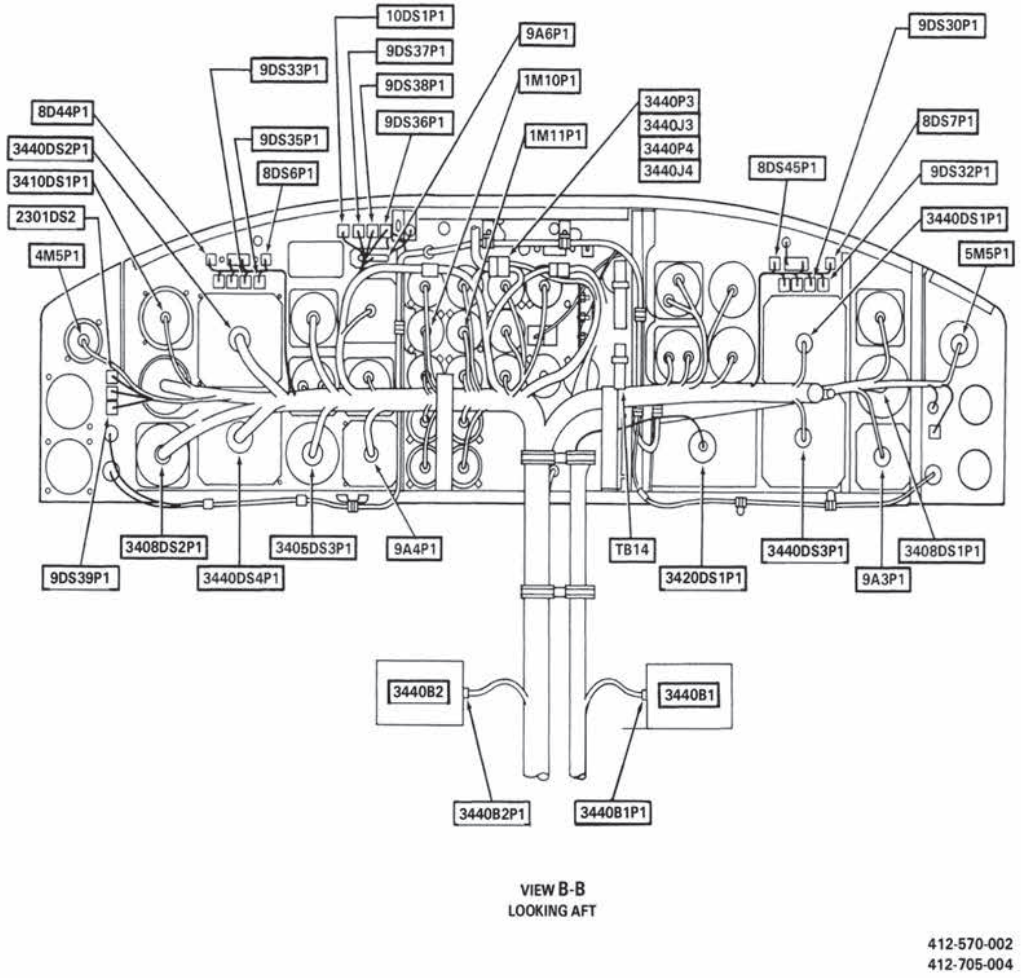
ICN-412-A-960001-B-97499-00005-A-001-01

Figure 4. EFIS Related Component Location (Sheet 1 of 8)



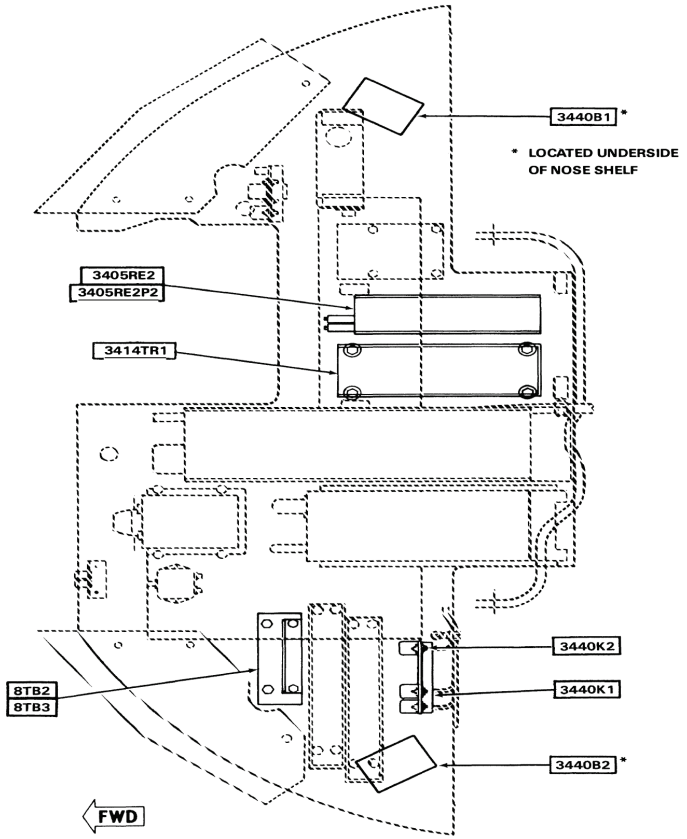
ICN-412-A-960001-B-97499-00006-A-001-01

Figure 4. EFIS Related Component Location (Sheet 2 of 8)



ICN-412-A-960001-B-97499-00007-A-001-01

Figure 4. EFIS Related Component Location (Sheet 3 of 8)



ICN-412-A-960001-B-97499-00008-A-001-01

Figure 4. EFIS Related Component Location (Sheet 4 of 8)